

replace the actual problem by a set of effective sources uniformly distributed over the open space beyond the object.

We imagine a large number of very closely spaced antennas, and we want the intensity at some point P . That looks just like what we have been doing. Not quite; because our screen is not at infinity. We do not want the intensity at infinity, but at a finite point. To calculate the intensity at some particular place, we have to add the contributions from all the antennas. First there is an antenna at D , exactly opposite P ; if we go up a little bit in angle, let us say a height h , then there is an increase in delay (there is also a change in amplitude because of the change in distance, but this is a very small effect if we are at all far away, and is much less important than the difference in the phases). Now the path difference $EP - DP$ is $h^2/2s$, so that the phase difference is proportional to the *square* of how far we go from D , while in our previous work s was infinite, and the phase difference was *linearly* proportional to h . When the phases are linearly proportional, each vector adds at a constant angle to the next vector. What we now need is a curve which is made by adding a lot of infinitesimal vectors with the requirement that the angle they make shall increase, not linearly, but as the *square* of the length of the curve. To construct that curve involves slightly advanced mathematics, but we can always construct it by actually drawing the arrows and measuring the angles. In any case, we get the marvelous curve (called Cornu's spiral) shown in Fig. 30-8. Now how do we use this curve?

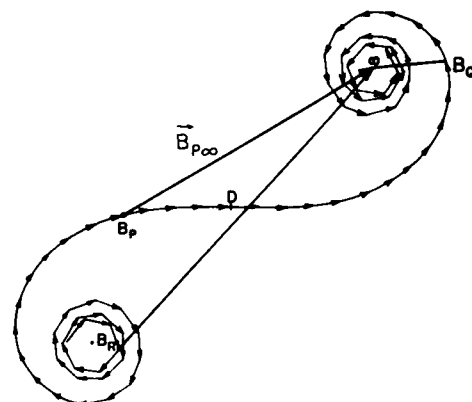


Fig. 30-8. The addition of amplitudes for many in-phase oscillators whose phase delays vary as the square of the distance from point D of the previous figure.

If we want the intensity, let us say, at point P , we add a lot of contributions of different phases from point D on up to infinity, and from D down only to point B_P . So we start at B_P in Fig. 30-8, and draw a series of arrows of ever-increasing angle. Therefore the total contribution above point B_P all goes along the spiraling curve. If we were to stop integrating at some place, then the total amplitude would be a vector from B to that point; in this particular problem we are going to infinity, so the total answer is the vector $B_{P\infty}$. Now the position on the curve which corresponds to point B_P on the object depends upon where point P is located, since point D , the inflection point, always corresponds to the position of point P . Thus, depending upon where P is located above B , the beginning point will fall at various positions on the lower left part of the curve, and the resultant vector $B_{P\infty}$ will have many maxima and minima (Fig. 30-9).

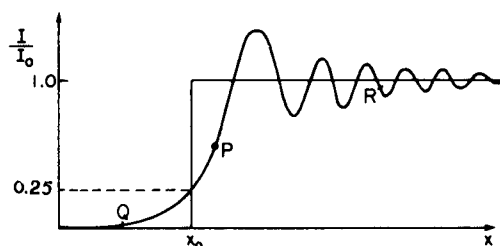


Fig. 30-9. The intensity near the edge of a shadow. The geometrical shadow edge is at x_0 .

On the other hand, if we are at Q , on the other side of P , then we are using only one end of the spiral curve, and not the other end. In other words, we do not even start at D , but at B_Q , so on this side we get an intensity which continuously falls off as Q goes farther into the shadow.

One point that we can immediately calculate with ease, to show that we really understand it, is the intensity exactly opposite the edge. The intensity here is $1/4$ that of the incident light. Reason: Exactly at the edge (so the endpoint B of the arrow is at D in Fig. 30-8) we have half the curve that we would have had if we were far into the bright region. If our point R is far into the light we go from one end of the curve to the other, that is, one full unit vector; but if we are at the edge of the shadow, we have only half the amplitude— $1/4$ the intensity.

In this chapter we have been finding the intensity produced in various directions from various distributions of sources. As a final example we shall derive a formula which we shall need for the next chapter on the theory of the index of refraction. Up to this point relative intensities have been sufficient for our purpose, but this time we shall find the complete formula for the field in the following situation.

30-7 The field of a plane of oscillating charges

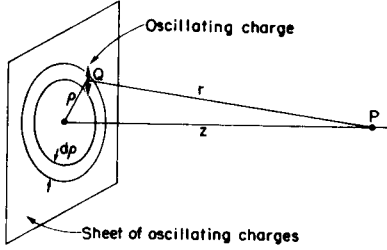


Fig. 30-10. Radiation field of a sheet of oscillating charges.

Suppose that we have a plane full of sources, all oscillating together, with their motion in the plane and all having the same amplitude and phase. What is the field at a finite, but very large, distance away from the plane? (We cannot get very close, of course, because we do not have the right formulas for the field close to the sources.) If we let the plane of the charges be the XY -plane, then we want to find the field at the point P far out on the Z -axis (Fig. 30-10). We suppose that there are η charges per unit area of the plane, and that each one of them has a charge q . All of the charges move with simple harmonic motion, with the same direction, amplitude, and phase. We let the motion of each charge, with respect to its own average position, be $x_0 \cos \omega t$. Or, using the complex notation and remembering that the real part represents the actual motion, the motion can be described by $x_0 e^{i\omega t}$.

Now we find the field at the point P from all of the charges by finding the field there from each charge q , and then adding the contributions from all the charges. We know that the radiation field is proportional to the acceleration of the charge, which is $-\omega^2 x_0 e^{i\omega t}$ (and is the same for every charge). The electric field that we want at the point P due to a charge at the point Q is proportional to the acceleration of the charge q , but we have to remember that the field at the point P at the instant t is given by the acceleration of the charge at the earlier time $t' = t - r/c$, where r/c is the time it takes the waves to travel the distance r from Q to P . Therefore the field at P is proportional to

$$-\omega^2 x_0 e^{i\omega(t-r/c)}. \quad (30.10)$$

Using this value for the acceleration as seen from P in our formula for the electric field at large distances from a radiating charge, we get

$$\left(\text{Electric field at } P \right)_{\text{from charge at } Q} = \frac{q}{4\pi\epsilon_0 c^2} \frac{\omega^2 x_0 e^{i\omega(t-r/c)}}{r} \quad (\text{approx.}). \quad (30.11)$$

Now this formula is not quite right, because we should have used *not* the acceleration of the charge but *its component* perpendicular to the line QP . We shall suppose, however, that the point P is so far away, compared with the distance of the point Q from the axis (the distance ρ in Fig. 30-9), for those changes that we need to take into account, that we can leave out the cosine factor (which would be nearly equal to 1 anyway).

To get the total field at P , we now add the effects of all the charges in the plane. We should, of course, make a *vector* sum. But since the direction of the electric field is nearly the same for all the charges, we may, in keeping with the approximation we have already made, just add the magnitudes of the fields. To our approximation the field at P depends only on the distance r , so all charges at the same r produce equal fields. So we add, first, the fields of those charges in a ring of width $d\rho$ and radius ρ . Then, by taking the integral over all ρ , we will obtain the total field.

The number of charges in the ring is the product of the surface area of the ring, $2\pi\rho d\rho$, and η , the number of charges per unit area. We have, then,

$$\text{Total field at } P = \int \frac{q}{4\pi\epsilon_0 c^2} \frac{\omega^2 x_0 e^{i\omega(t-r/c)}}{r} \cdot \eta \cdot 2\pi\rho d\rho. \quad (30.12)$$

We wish to evaluate this integral from $\rho = 0$ to $\rho = \infty$. The variable t , of course, is to be held fixed while we do the integral, so the only varying quantities are ρ and r . Leaving out all the constant factors, including the factor $e^{i\omega t}$, for the moment, the integral we wish is

$$\int_{\rho=0}^{\rho=\infty} \frac{e^{-i\omega r/c}}{r} \rho d\rho. \quad (30.13)$$

To do this integral we need to use the relation between r and ρ :

$$r^2 = \rho^2 + z^2. \quad (30.14)$$

Since z is independent of ρ , when we take the differential of this equation, we get

$$2r dr = 2\rho d\rho,$$

which is lucky, since in our integral we can replace $\rho d\rho$ by $r dr$ and the r will cancel the one in the denominator. The integral we want is then the simpler one

$$\int_{r=z}^{r=\infty} e^{-i\omega r/c} dr. \quad (30.15)$$

To integrate an exponential is very easy. We divide by the coefficient of r in the exponent and evaluate the exponential at the limits. But the limits of r are not the same as the limits of ρ . When $\rho = 0$, we have $r = z$, so the limits of r are z to infinity. We get for the integral

$$-\frac{c}{i\omega} [e^{-i\infty} - e^{-(i\omega/c)z}], \quad (30.16)$$

where we have written ∞ for $(r/c)\infty$, since they both just mean a very large number!

Now $e^{-i\infty}$ is a mysterious quantity. Its real part, for example, is $\cos(-\infty)$, which, mathematically speaking, is completely indefinite (although we would expect it to be somewhere—or everywhere (?)—between $+1$ and -1 !). But in a *physical* situation, it can mean something quite reasonable, and usually can just be taken to be zero. To see that this is so in our case, we go back to consider again the original integral (30.15).

We can understand (30.15) as a sum of many small complex numbers, each of magnitude Δr , and with the angle $\theta = -\omega r/c$ in the complex plane. We can try to evaluate the sum by a graphical method. In Fig. 30-11 we have drawn the first five pieces of the sum. Each segment of the curve has the length Δr and is placed at the angle $\Delta\theta = -\omega \Delta r/c$ with respect to the preceding piece. The sum for these first five pieces is represented by the arrow from the starting point to the end of the fifth segment. As we continue to add pieces we shall trace out a polygon until we get back to the starting point (approximately) and then start around once more. Adding more pieces, we just go round and round, staying close to a circle whose radius is easily shown to be c/ω . We can see now why the integral does not give a definite answer!

But now we have to go back to the *physics* of the situation. In any real situation the plane of charges *cannot* be infinite in extent, but must sometime stop. If it stopped suddenly, and was exactly circular in shape, our integral would have some value on the circle in Fig. 30-11. If, however, we let the number of charges in the plane gradually taper off at some large distance from the center (or else stop suddenly but in an irregular shape so for larger ρ the entire ring of width $d\rho$ no longer contributes), then the coefficient η in the exact integral would decrease toward zero. Since we are adding smaller pieces but still turning through the same angle, the graph of our integral would then become a curve which is a spiral. The spiral would eventually end up at the center of our original circle, as drawn in Fig. 30-12. The *physically* correct integral is the complex number A in the figure represented by the interval from the starting point to the center of the circle, which is just equal to

$$\frac{c}{i\omega} e^{-i\omega z/c}, \quad (30.17)$$

as you can work out for yourself. This is the same result we would get from Eq. (30.16) if we set $e^{-i\infty} = 0$.

(There is also another reason why the contribution to the integral tapers off for large values of r , and that is the factor we have omitted for the projection of the acceleration on the plane perpendicular to the line PQ .)

We are, of course, interested only in physical situations, so we will take $e^{-i\infty}$ equal to zero. Returning to our original formula (30.12) for the field and putting

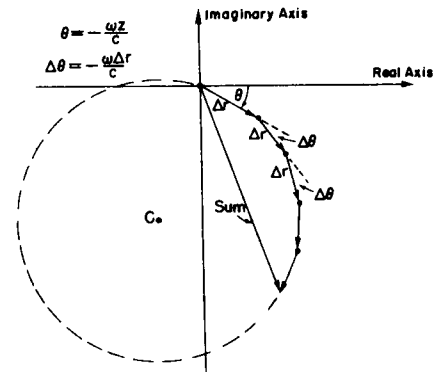


Fig. 30-11. Graphical solution of $\int_z^\infty e^{-i\omega r/c} dr$.

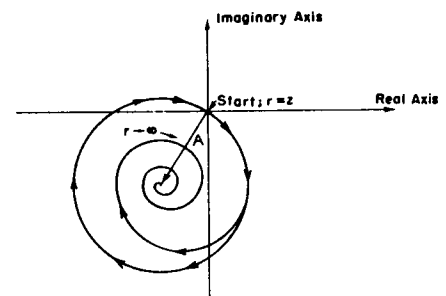


Fig. 30-12. Graphical solution of $\int_z^\infty \eta e^{-i\omega r/c} dr$.

back all of the factors that go with the integral, we have the result

$$\text{Total field at } P = - \frac{\eta q}{2\epsilon_0 c} i\omega x_0 e^{i\omega(t-z/c)} \quad (30.18)$$

(remembering that $1/i = -i$).

It is interesting to note that $(i\omega x_0 e^{i\omega t})$ is just equal to the *velocity* of the charges, so that we can also write the equation for the field as

$$\text{Total field at } P = - \frac{\eta q}{2\epsilon_0 c} [\text{velocity of charges}]_{\text{at } t-z/c}, \quad (30.19)$$

which is a little strange, because the retardation is just by the distance z , which is the shortest distance from P to the plane of charges. But that is the way it comes out—fortunately a rather simple formula. (We may add, by the way, that although our derivation is valid only for distances far from the plane of oscillatory charges, it turns out that the formula (30.18) or (30.19) is correct at any distance z , even for $z < \lambda$.)

The Origin of the Refractive Index

31-1 The index of refraction

We have said before that light goes slower in water than in air, and slower, slightly, in air than in vacuum. This effect is described by the index of refraction n . Now we would like to understand how such a slower velocity could come about. In particular, we should try to see what the relation is to some physical assumptions, or statements, we made earlier, which were the following:

- (a) That the total electric field in any physical circumstance can always be represented by the sum of the fields from all the charges in the universe.
- (b) That the field from a single charge is given by its acceleration evaluated with a retardation at the speed c , *always* (for the *radiation* field).

But, for a piece of glass, you might think: "Oh, no, you should modify all this. You should say it is retarded at the speed c/n ." That, however, is not right, and we have to understand why it is not.

It is approximately true that light or any electrical wave *does appear* to travel at the speed c/n through a material whose index of refraction is n , but the fields are still produced by the motions of *all* the charges—including the charges moving in the material—and with these basic contributions of the field travelling at the ultimate velocity c . Our problem is to understand how the *apparently* slower velocity comes about.

We shall try to understand the effect in a very simple case. A source which we shall call "the *external source*" is placed a large distance away from a thin plate of transparent material, say glass. We inquire about the field at a large distance on the opposite side of the plate. The situation is illustrated by the diagram of Fig. 31-1, where S and P are imagined to be very far away from the plate. According to the principles we have stated earlier, an electric field anywhere that is far from all moving charges is the (vector) sum of the fields produced by the external source (at S) and the fields produced by *each* of the charges in the plate of glass, *every one with its proper retardation at the velocity c* . Remember that the contribution of each charge is not changed by the presence of the other charges. These are our basic principles. The field at P can be written thus:

$$\mathbf{E} = \sum_{\text{all charges}} \mathbf{E}_{\text{each charge}} \quad (31.1)$$

or

$$\mathbf{E} = \mathbf{E}_s + \sum_{\text{all other charges}} \mathbf{E}_{\text{each charge}}, \quad (31.2)$$

where $\mathbf{E}_s$ is the field due to the source alone and would be precisely the field at P if there were no material present. We expect the field at P to be different from $\mathbf{E}_s$ if there are any other moving charges.

Why should there be charges moving in the glass? We know that all material consists of atoms which contain electrons. When the electric field of the source acts on these atoms it drives the electrons up and down, because it exerts a force on the electrons. And moving electrons generate a field—they constitute new radiators. These new radiators are related to the source S , because they are driven by the field of the source. The total field is not just the field of the source S , but it is modified by the additional contribution from the other moving charges. This means that the field is not the same as the one which was there before the glass was there, but is modified, and it turns out that it is modified in such a way that

31-1 The index of refraction

31-2 The field due to the material

31-3 Dispersion

31-4 Absorption

31-5 The energy carried by an electric wave

31-6 Diffraction of light by a screen

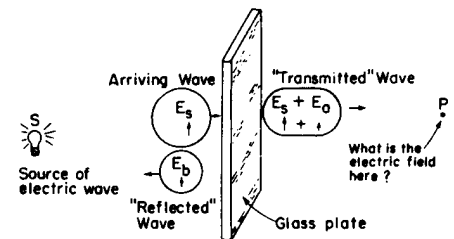


Fig. 31-1. Electric waves passing through a layer of transparent material.

the field inside the glass appears to be moving at a different speed. That is the idea which we would like to work out quantitatively.

Now this is, in the exact case, pretty complicated, because although we have said that all the other moving charges are driven by the source field, that is not quite true. If we think of a particular charge, it feels not only the source, but like anything else in the world, it feels *all* of the charges that are moving. It feels, in particular, the charges that are moving somewhere else in the glass. So the total field which is acting on a *particular charge* is a combination of the fields from the other charges, *whose motions depend on what this particular charge is doing!* You can see that it would take a complicated set of equations to get the complete and exact formula. It is so complicated that we postpone this problem until next year.

Instead we shall work out a very simple case in order to understand all the physical principles very clearly. We take a circumstance in which the effects from the other atoms is very small relative to the effects from the source. In other words, we take a material in which the total field is not modified very much by the motion of the other charges. That corresponds to a material in which the index of refraction is very close to 1, which will happen, for example, if the density of the atoms is very low. Our calculation will be valid for any case in which the index is for any reason very close to 1. In this way we shall avoid the complications of the most general, complete solution.

Incidentally, you should notice that there is another effect caused by the motion of the charges in the plate. These charges will also radiate waves back toward the source S . This backward-going field is the light we see reflected from the surfaces of transparent materials. It does not come from just the surface. The backward radiation comes from everywhere in the interior, but it turns out that the total effect is equivalent to a reflection from the surfaces. These reflection effects are beyond our approximation at the moment because we shall be limited to a calculation for a material with an index so close to 1 that very little light is reflected.

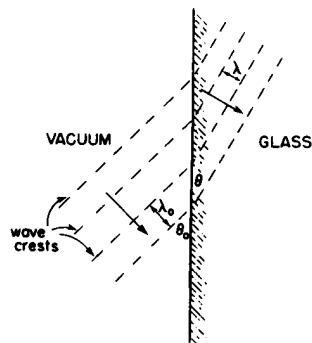


Fig. 31-2. Relation between refraction and velocity change.

Before we proceed with our study of how the index of refraction comes about, we should understand that all that is required to understand refraction is to understand why the apparent wave *velocity* is different in different materials. The *bending* of light rays comes about just *because* the effective speed of the waves is different in the materials. To remind you how that comes about we have drawn in Fig. 31-2 several successive crests of an electric wave which arrives from a vacuum onto the surface of a block of glass. The arrow perpendicular to the wave crests indicates the direction of travel of the wave. Now all oscillations in the wave must have the same *frequency*. (We have seen that driven oscillations have the same frequency as the driving source.) This means, also, that the wave crests for the waves on both sides of the surface must have the *same spacing along the surface* because they must travel together, so that a charge sitting at the boundary will feel only one frequency. The *shortest* distance between crests of the wave, however, is the wavelength which is the velocity divided by the frequency. On the vacuum side it is $\lambda_0 = 2\pi c/\omega$, and on the other side it is $\lambda = 2\pi v/\omega$ or $2\pi c/\omega n$, if $v = c/n$ is the velocity of the wave. From the figure we can see that the only way for the waves to “fit” properly at the boundary is for the waves in the material to be travelling at a different angle with respect to the surface. From the geometry of the figure you can see that for a “fit” we must have $\lambda_0/\sin \theta_0 = \lambda/\sin \theta$, or $\sin \theta_0/\sin \theta = n$, which is Snell’s law. We shall, for the rest of our discussion, consider only why light has an effective speed of c/n in material of index n , and no longer worry, in this chapter, about the bending of the light direction.

We go back now to the situation shown in Fig. 31-1. We see that what we have to do is to calculate the field produced at P by all the oscillating charges in the glass plate. We shall call this part of the field E_a , and it is just the sum written as the second term in Eq. (31.2). When we add it to the term E_s , due to the source, we will have the total field at P .

This is probably the most complicated thing that we are going to do this year, but it is complicated only in that there are many pieces that have to be put together; each piece, however, is very simple. Unlike other derivations where we say, "Forget the derivation, just look at the answer!" in this case we do not need the answer so much as the derivation. In other words, the thing to understand now is the physical machinery for the production of the index.

To see where we are going, let us first find out what the "correction field" E_a would have to be if the total field at P is going to look like radiation from the source that is slowed down while passing through the thin plate. If the plate had no effect on it, the field of a wave travelling to the right (along the z -axis) would be

$$E_s = E_0 \cos \omega(t - z/c) \quad (31.3)$$

or, using the exponential notation,

$$E_s = E_0 e^{i\omega(t-z/c)}. \quad (31.4)$$

Now what would happen if the wave travelled more slowly in going through the plate? Let us call the thickness of the plate Δz . If the plate were not there the wave would travel the distance Δz in the time $\Delta z/c$. But if it appears to travel at the speed c/n then it should take the longer time $n \Delta z/c$ or the *additional* time $\Delta t = (n - 1) \Delta z/c$. After that it would continue to travel at the speed c again. We can take into account the extra delay in getting through the plate by replacing t in Eq. (31.4) by $(t - \Delta t)$ or by $[t - (n - 1) \Delta z/c]$. So the wave after insertion of the plate should be written

$$E_{\text{after plate}} = E_0 e^{i\omega[t - (n-1)\Delta z/c - z/c]}. \quad (31.5)$$

We can also write this equation as

$$E_{\text{after plate}} = e^{-i\omega(n-1)\Delta z/c} E_0 e^{i\omega(t-z/c)}, \quad (31.6)$$

which says that the wave after the plate is obtained from the wave which could exist without the plate, i. e., from E_s , by multiplying by the factor $e^{-i\omega(n-1)\Delta z/c}$. Now we know that multiplying an oscillating function like $e^{i\omega t}$ by a factor $e^{i\theta}$ just says that we change the phase of the oscillation by the angle θ , which is, of course, what the extra delay in passing through the thickness Δz has done. It has retarded the phase by the amount $\omega(n - 1) \Delta z/c$ (retarded, because of the minus sign in the exponent).

We have said earlier that the plate should *add* a field E_a to the original field $E_s = E_0 e^{i\omega(t-z/c)}$, but we have found instead that the effect of the plate is to *multiply* the field by a factor which shifts its phase. However, that is really all right because we can get the same result by adding a suitable complex number. It is particularly easy to find the right number to add in the case that Δz is small, for you will remember that if x is a small number then e^x is nearly equal to $(1 + x)$. We can write, therefore,

$$e^{-i\omega(n-1)\Delta z/c} = 1 - i\omega(n - 1) \Delta z/c. \quad (31.7)$$

Using this equality in Eq. (31.6), we have

$$E_{\text{after plate}} = \underbrace{E_0 e^{i\omega(t-z/c)}}_{E_s} - \underbrace{\frac{i\omega(n-1)\Delta z}{c} E_0 e^{i\omega(t-z/c)}}_{E_a}. \quad (31.8)$$

The first term is just the field from the source, and the second term must just be equal to E_a , the field produced to the right of the plate by the oscillating charges of the plate—expressed here in terms of the index of refraction n , and depending, of course, on the strength of the wave from the source.

What we have been doing is easily visualized if we look at the complex number diagram in Fig. 31-3. We first draw the number E_s (we chose some values for z and t so that E_s comes out horizontal, but this is not necessary). The delay due to

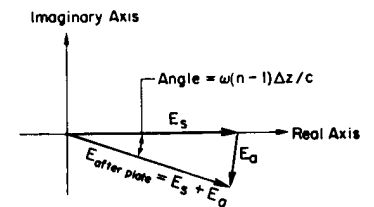


Fig. 31-3. Diagram for the transmitted wave at a particular t and z .

slowing down in the plate would delay the phase of this number, that is, it would rotate E_s through a negative angle. But this is equivalent to adding the small vector E_a at roughly right angles to E_s . But that is just what the factor $-i$ means in the second term of Eq. (31.8). It says that if E_s is real, then E_a is negative imaginary or that, in general, E_s and E_a make a right angle.

31-2 The field due to the material

We now have to ask: Is the field E_a obtained in the second term of Eq. (31.8) the kind we would expect from oscillating charges in the plate? If we can show that it is, we will then have calculated what the index n should be! [Since n is the only nonfundamental number in Eq. (31.8).] We turn now to calculating what field E_a the charges in the material will produce. (To help you keep track of the many symbols we have used up to now, and will be using in the rest of our calculation, we have put them all together in Table 31-1.)

Table 31-1

Symbols used in the calculations

E_s	= field from the source
E_a	= field produced by charges in the plate
Δz	= thickness of the plate
z	= perpendicular distance from the plate
n	= index of refraction
ω	= frequency (angular) of the radiation
N	= number of charges per unit volume in the plate
η	= number of charges per unit area of the plate
q_e	= charge on an electron
m	= mass of an electron
ω_0	= resonant frequency of an electron bound in an atom

If the source S (of Fig. 31-1) is far off to the left, then the field E_s will have the same phase everywhere on the plate, so we can write that in the neighborhood of the plate

$$E_s = E_0 e^{i(\omega t - z/c)} \quad (31.9)$$

Right at the plate, where $z = 0$, we will have

$$E_s = E_0 e^{i\omega t} \text{ (at the plate)} \quad (31.10)$$

Each of the electrons in the atoms of the plate will feel this electric field and will be driven up and down (we assume the direction of E_0 is vertical) by the electric force qE . To find what motion we expect for the electrons, we will assume that the atoms are little oscillators, that is, that the electrons are fastened elastically to the atoms, which means that if a force is applied to an electron its displacement from its normal position will be proportional to the force.

You may think that this is a funny model of an atom if you have heard about electrons whirling around in orbits. But that is just an oversimplified picture. The correct picture of an atom, which is given by the theory of wave mechanics, says that, *so far as problems involving light are concerned*, the electrons behave as though they were held by springs. So we shall suppose that the electrons have a linear restoring force which, together with their mass m , makes them behave like little oscillators, with a resonant frequency ω_0 . We have already studied such oscillators, and we know that the equation of their motion is written this way:

$$m \left(\frac{d^2 x}{dt^2} + \omega_0^2 x \right) = F, \quad (31.11)$$

where F is the driving force.

For our problem, the driving force comes from the electric field of the wave from the source, so we should use

$$F = q_e E_s = q_e E_0 e^{i\omega t}, \quad (31.12)$$

where q_e is the electric charge on the electron and for E_s we use the expression $E_s = E_0 e^{i\omega t}$ from (31.10). Our equation of motion for the electron is then

$$m \left(\frac{d^2 x}{dt^2} + \omega_0^2 x \right) = q_e E_0 e^{i\omega t}. \quad (31.13)$$

We have solved this equation before, and we know that the solution is

$$x = x_0 e^{i\omega t}, \quad (31.14)$$

where, by substituting in (31.13), we find that

$$x_0 = \frac{q_e E_0}{m(\omega_0^2 - \omega^2)}, \quad (31.15)$$

so that

$$x = \frac{q_e E_0}{m(\omega_0^2 - \omega^2)} e^{i\omega t}. \quad (31.16)$$

We have what we needed to know—the motion of the electrons in the plate. And it is the same for every electron, except that the mean position (the “zero” of the motion) is, of course, different for each electron.

Now we are ready to find the field E_a that these atoms produce at the point P , because we have already worked out (at the end of Chapter 30) what field is produced by a sheet of charges that all move together. Referring back to Eq. (30.19), we see that the field E_a at P is just a negative constant times the velocity of the charges retarded in time the amount z/c . Differentiating x in Eq. (31.16) to get the velocity, and sticking in the retardation [or just putting x_0 from (31.15) into (30.18)] yields

$$E_a = - \frac{\eta q_e}{2\epsilon_0 c} \left[i\omega \frac{q_e E_0}{m(\omega_0^2 - \omega^2)} e^{i\omega(t-z/c)} \right]. \quad (31.17)$$

Just as we expected, the driven motion of the electrons produced an extra wave which travels to the right (that is what the factor $e^{i\omega(t-z/c)}$ says), and the amplitude of this wave is proportional to the number of atoms per unit area in the plate (the factor η) and also proportional to the strength of the source field (the factor E_0). Then there are some factors which depend on the atomic properties (q_e , m , and ω_0), as we should expect.

The most important thing, however, is that this formula (31.17) for E_a looks very much like the expression for E_a that we got in Eq. (31.8) by saying that the original wave was delayed in passing through a material with an index of refraction n . The two expressions will, in fact, be identical if

$$(n - 1) \Delta z = \frac{\eta q_e^2}{2\epsilon_0 m(\omega_0^2 - \omega^2)}. \quad (31.18)$$

Notice that both sides are proportional to Δz , since η , which is the number of atoms *per unit area*, is equal to $N \Delta z$, where N is the number of atoms *per unit volume* of the plate. Substituting $N \Delta z$ for η and cancelling the Δz , we get our main result, a formula for the index of refraction in terms of the properties of the atoms of the material—and of the frequency of the light:

$$n = 1 + \frac{N q_e^2}{2\epsilon_0 m(\omega_0^2 - \omega^2)}. \quad (31.19)$$

This equation gives the “explanation” of the index of refraction that we wished to obtain.

31-3 Dispersion

Notice that in the above process we have obtained something very interesting. For we have not only a number for the index of refraction which can be computed from the basic atomic quantities, but we have also learned how the index of refraction should vary with the frequency ω of the light. This is something we would never understand from the simple statement that "light travels slower in a transparent material." We still have the problem, of course, of knowing how many atoms per unit volume there are, and what is their natural frequency ω_0 . We do not know this just yet, because it is different for every different material, and we cannot get a general theory of that now. Formulation of a general theory of the properties of different substances—their natural frequencies, and so on—is possible only with quantum atomic mechanics. Also, different materials have different properties and different indexes, so we cannot expect, anyway, to get a general formula for the index which will apply to all substances.

However, we shall discuss the formula we have obtained, in various possible circumstances. First of all, for most ordinary gases (for instance, for air, most colorless gases, hydrogen, helium, and so on) the natural frequencies of the electron oscillators correspond to ultraviolet light. These frequencies are higher than the frequencies of visible light, that is, ω_0 is much larger than ω of visible light, and to a first approximation, we can disregard ω^2 in comparison with ω_0^2 . Then we find that the index is nearly constant. So for a gas, the index is nearly constant. This is also true for most other transparent substances, like glass. If we look at our expression a little more closely, however, we notice that as ω rises, taking a little bit more away from the denominator, the index also rises. So n rises slowly with frequency. The index is higher for blue light than for red light. That is the reason why a prism bends the light more in the blue than in the red.

The phenomenon that the index depends upon the frequency is called the phenomenon of *dispersion*, because it is the basis of the fact that light is "dispersed" by a prism into a spectrum. The equation for the index of refraction as a function of frequency is called a *dispersion equation*. So we have obtained a dispersion equation. (In the past few years "dispersion equations" have been finding a new use in the theory of elementary particles.)

Our dispersion equation suggests other interesting effects. If we have a natural frequency ω_0 which lies in the visible region, or if we measure the index of refraction of a material like glass in the ultraviolet, where ω gets near ω_0 , we see that at frequencies very close to the natural frequency the index can get enormously large, because the denominator can go to zero. Next, suppose that ω is greater than ω_0 . This would occur, for example, if we take a material like glass, say, and shine x-ray radiation on it. In fact, since many materials which are opaque to visible light, like graphite for instance, are transparent to x-rays, we can also talk about the index of refraction of carbon for x-rays. All the natural frequencies of the carbon atoms would be much lower than the frequency we are using in the x-rays, since x-ray radiation has a very high frequency. The index of refraction is that given by our dispersion equation if we set ω_0 equal to zero (we neglect ω_0^2 in comparison with ω^2).

A similar situation would occur if we beam radiowaves (or light) on a gas of free electrons. In the upper atmosphere electrons are liberated from their atoms by ultraviolet light from the sun and they sit up there as free electrons. For free electrons $\omega_0 = 0$ (there is no elastic restoring force). Setting $\omega_0 = 0$ in our dispersion equation yields the correct formula for the index of refraction for radiowaves in the stratosphere, where N is now to represent the density of free electrons (number per unit volume) in the stratosphere. But let us look again at the equation, if we beam x-rays on matter, or radiowaves (or any electric waves) on free electrons the term $(\omega_0^2 - \omega^2)$ becomes *negative*, and we obtain the result that n is *less than one*. That means that the effective speed of the waves in the substance is *faster* than c ! Can that be correct?

It is correct. In spite of the fact that it is said that you cannot send signals any faster than the speed of light, it is nevertheless true that the index of refraction of materials at a particular frequency can be either greater or less than 1. This

just means that the *phase shift* which is produced by the scattered light can be either positive or negative. It can be shown, however, that the speed at which you can send a *signal* is not determined by the index at one frequency, but depends on what the index is at *many* frequencies. What the index tells us is the speed at which the *nodes* (or crests) of the wave travel. The *node* of a wave is not a signal by itself. In a perfect wave, which has no modulations of any kind, i.e., which is a steady oscillation, you cannot really say when it “starts,” so you cannot use it for a timing signal. In order to send a *signal* you have to change the wave somehow, make a notch in it, make it a little bit fatter or thinner. That means that you have to have more than one frequency in the wave, and it can be shown that the speed at which *signals* travel is not dependent upon the index alone, but upon the way that the index changes with the frequency. This subject we must also delay (until Chapter 48). Then we will calculate for you the actual speed of *signals* through such a piece of glass, and you will see that it will not be faster than the speed of light, although the nodes, which are mathematical points, do travel faster than the speed of light.

Just to give a slight hint as to how that happens, you will note that the real difficulty has to do with the fact that the responses of the charges are opposite to the field, i.e., the sign has gotten reversed. Thus in our expression for x (Eq. 31.16) the displacement of the charge is in the direction opposite to the driving field, because $(\omega_0^2 - \omega^2)$ is negative for small ω_0 . The formula says that when the electric field is pulling in one direction, the charge is moving in the opposite direction.

How does the charge happen to be going in the opposite direction? It certainly does not start off in the opposite direction when the field is first turned on. When the motion first starts there is a transient, which settles down after awhile, and only *then* is the phase of the oscillation of the charge opposite to the driving field. And it is then that the *phase* of the transmitted field can appear to be *advanced* with respect to the source wave. It is this *advance in phase* which is meant when we say that the “phase velocity” or velocity of the nodes is greater than c . In Fig. 31-4 we give a schematic idea of how the waves might look for a case where the wave is suddenly turned on (to make a signal). You will see from the diagram that the *signal* (i.e., the *start* of the wave) is *not* earlier for the wave which ends up with an advance in phase.

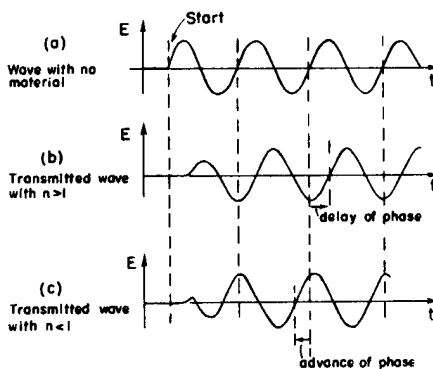


Fig. 31-4. Wave “signals.”

Let us now look again at our dispersion equation. We should remark that our analysis of the refractive index gives a result that is somewhat simpler than you would actually find in nature. To be completely accurate we must add some refinements. First, we should expect that our model of the atomic oscillator should have some damping force (otherwise once started it would oscillate forever, and we do not expect that to happen). We have worked out before (Eq. 23.8) the motion of a damped oscillator and the result is that the denominator in Eq. (31.16), and therefore in (31.19), is changed from $(\omega_0^2 - \omega^2)$ to $(\omega_0^2 - \omega^2 + i\gamma\omega)$, where γ is the damping coefficient.

We need a second modification to take into account the fact that there are several resonant frequencies for a particular kind of atom. It is easy to fix up our

dispersion equation by imagining that there are several different kinds of oscillators, but that each oscillator acts separately, and so we simply add the contributions of all the oscillators. Let us say that there are N_k electrons per unit of volume, whose natural frequency is ω_k and whose damping factor is γ_k . We would then have for our dispersion equation

$$n = 1 + \frac{q_e^2}{2\epsilon_0 m} \sum_k \frac{N_k}{\omega_k^2 - \omega^2 + i\gamma_k \omega}. \quad (31.20)$$

We have, finally, a complete expression which describes the index of refraction that is observed for many substances.* The index described by this formula varies with frequency roughly like the curve shown in Fig. 31-5.

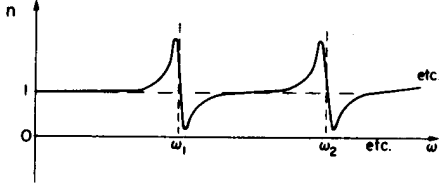


Fig. 31-5. The index of refraction as a function of frequency.

You will note that so long as ω is not too close to one of the resonant frequencies, the slope of the curve is positive. Such a positive slope is called “normal” dispersion (because it is clearly the most common occurrence). Very near the resonant frequencies, however, there is a small range of ω 's for which the slope is negative. Such a negative slope is often referred to as “anomalous” (meaning abnormal) dispersion, because it seemed unusual when it was first observed, long before anyone even knew there were such things as electrons. From our point of view both slopes are quite “normal”!

31-4 Absorption

Perhaps you have noticed something a little strange about the last form (Eq. 31.20) we obtained for our dispersion equation. Because of the term $i\gamma$ we put in to take account of damping, the index of refraction is now a *complex number*! What does *that* mean? By working out what the real and imaginary parts of n are we could write

$$n = n' - in'', \quad (31.21)$$

where n' and n'' are real numbers. (We use the minus sign in front of the in'' because then n'' will turn out to be a positive number, as you can show for yourself.)

We can see what such a complex index means by going back to Eq. (31.6), which is the equation of the wave after it goes through a plate of material with an index n . If we put our complex n into this equation, and do some rearranging, we get

$$E_{\text{after plate}} = \underbrace{e^{-\omega n'' \Delta z / c}}_A \underbrace{e^{-i\omega(n' - 1)\Delta z / c} E_0 e^{i\omega(t - z/c)}}_B \quad (31.22)$$

The last factors, marked B in Eq. (31.22), are just the form we had before, and again describe a wave whose phase has been delayed by the angle $\omega(n' - 1)\Delta z / c$ in traversing the material. The first term (A) is new and is an exponential factor with a *real* exponent, because there were two i 's that cancelled. Also, the exponent is negative, so the factor is a real number less than one. It describes a *decrease* in the magnitude of the field and, as we should expect, by an amount which is more the larger Δz is. As the wave goes through the material, it is weakened. The material is “absorbing” part of the wave. The wave comes out the other side with less energy. We should not be surprised at this, because the damping we put in for the oscillators is indeed a friction force and must be expected to cause a loss of energy. We see that the imaginary part n'' of a complex index of refraction represents an absorption (or “attenuation”) of the wave. In fact, n'' is sometimes referred to as the “absorption index.”

We may also point out that an imaginary part to the index n corresponds to bending the arrow E_a in Fig. 31-3 toward the origin. It is clear why the transmitted field is then decreased.

* Actually, although in quantum mechanics Eq. (31.20) is still valid, its interpretation is somewhat different. In quantum mechanics even an atom with one electron, like hydrogen, has several resonant frequencies. Therefore N_k is not really the number of electrons having the frequency ω_k , but is replaced instead by Nf_k , where N is the number of atoms per unit volume and f_k (called the oscillator strength) is a factor that tells how strongly the atom exhibits each of its resonant frequencies ω_k .

Normally, for instance as in glass, the absorption of light is very small. This is to be expected from our Eq. (31.20), because the imaginary part of the denominator, $i\gamma_k\omega$, is much smaller than the term $(\omega_k^2 - \omega^2)$. But if the light frequency ω is very close to ω_k then the resonance term $(\omega_k^2 - \omega^2)$ can become small compared with $i\gamma_k\omega$ and the index becomes almost completely imaginary. The absorption of the light becomes the dominant effect. It is just this effect that gives the dark lines in the spectrum of light which we receive from the sun. The light from the solar surface has passed through the sun's atmosphere (as well as the earth's), and the light has been strongly absorbed at the resonant frequencies of the atoms in the solar atmosphere.

The observation of such spectral lines in the sunlight allows us to tell the resonant frequencies of the atoms and hence the chemical composition of the sun's atmosphere. The same kind of observations tell us about the materials in the stars. From such measurements we know that the chemical elements in the sun and in the stars are the same as those we find on the earth.

31-5 The energy carried by an electric wave

We have seen that the imaginary part of the index means absorption. We shall now use this knowledge to find out how much energy is carried by a light wave. We have given earlier an argument that the energy carried by light is proportional to $\overline{E^2}$, the time average of the square of the electric field in the wave. The decrease in E due to absorption must mean a loss of energy, which would go into some friction of the electrons and, we might guess, would end up as heat in the material.

If we consider the light arriving on a unit area, say one square centimeter, of our plate in Fig. 31-1, then we can write the following energy equation (if we assume that energy is conserved, as we *do!*):

$$\text{Energy in per sec} = \text{energy out per sec} + \text{work done per sec.} \quad (31.23)$$

For the first term we can write $\alpha\overline{E_s^2}$, where α is the as yet unknown constant of proportionality which relates the average value of E^2 to the energy being carried. For the second term we must include the part from the radiating atoms of the material, so we should use $\alpha(\overline{E_s + E_a})^2$, or (evaluating the square) $\alpha(\overline{E_s^2} + 2\overline{E_s E_a} + \overline{E_a^2})$.

All of our calculations have been made for a thin layer of material whose index is not too far from 1, so that E_a would always be much less than E_s (just to make the calculations easier). In keeping with our approximations, we should, therefore, leave out the term $\overline{E_a^2}$, because it is much smaller than $\overline{E_s E_a}$. You may say: "Then you should leave out $\overline{E_s E_a}$ also, because it is much smaller than $\overline{E_s^2}$." It is true that $\overline{E_s E_a}$ is much smaller than $\overline{E_s^2}$, but we must keep $\overline{E_s E_a}$ or our approximation will be the one that would apply if we neglected the presence of the material completely! One way of checking that our calculations are consistent is to see that we always keep terms which are proportional to $N\Delta z$, the area density of atoms in the material, but we leave out terms which are proportional to $(N\Delta z)^2$ or any higher power of $N\Delta z$. Ours is what should be called a "low-density approximation."

In the same spirit, we might remark that our energy equation has neglected the energy in the reflected wave. But that is OK because this term, too, is proportional to $(N\Delta z)^2$, since the amplitude of the reflected wave is proportional to $N\Delta z$.

For the last term in Eq. (31.23) we wish to compute the rate at which the incoming wave is doing work on the electrons. We know that work is force times distance, so the *rate* of doing work (also called power) is the force times the velocity. It is really $\mathbf{F} \cdot \mathbf{V}$, but we do not need to worry about the dot product when the velocity and force are along the same direction as they are here (except for a possible minus sign). So for each atom we take $q_e \overline{E_s} v$ for the average rate of

doing work. Since there are $N \Delta z$ atoms in a unit area, the last term in Eq. (31.23) should be $N \Delta z q_e \overline{E_s v}$. Our energy equation now looks like

$$\alpha \overline{E_s^2} = \alpha \overline{E_s^2} + 2\alpha \overline{E_s E_a} + N \Delta z q_e \overline{E_s v}. \quad (31.24)$$

The $\overline{E_s^2}$ terms cancel, and we have

$$2\alpha \overline{E_s E_a} = N \Delta z q_e \overline{E_s v}. \quad (31.25)$$

We now go back to Eq. (30.19), which tells us that for large z

$$E_a = \frac{N \Delta z q_e}{2\epsilon_0 c} v(\text{ret by } z/c) \quad (31.26)$$

(recalling that $\eta = N \Delta z$). Putting Eq. (31.26) into the left-hand side of (31.25), we get

$$2\alpha \frac{N \Delta z q_e}{2\epsilon_0 c} \overline{E_s(\text{at } z) \cdot v(\text{ret by } z/c)}.$$

However, $E_s(\text{at } z)$ is E_s (at atoms) retarded by z/c . Since the average is independent of time, it is the same now as retarded by z/c , or is $\overline{E_s}(\text{at atom}) \cdot v$, the same average that appears on the right-hand side of (31.25). The two sides are therefore equal if

$$\frac{\alpha}{\epsilon_0 c} = 1, \quad \text{or} \quad \alpha = \epsilon_0 c. \quad (31.27)$$

We have discovered that if energy is to be conserved, the energy carried in an electric wave per unit area and per unit time (or what we have called the *intensity*) must be given by $\epsilon_0 c \overline{E^2}$. If we call the intensity $\overline{S}$, we have

$$\overline{S} = \left\{ \begin{array}{c} \text{intensity} \\ \text{or} \\ \text{energy/area/time} \end{array} \right\} = \epsilon_0 c \overline{E^2}, \quad (31.28)$$

where the *bar* means the *time average*. We have a nice bonus result from our theory of the refractive index!

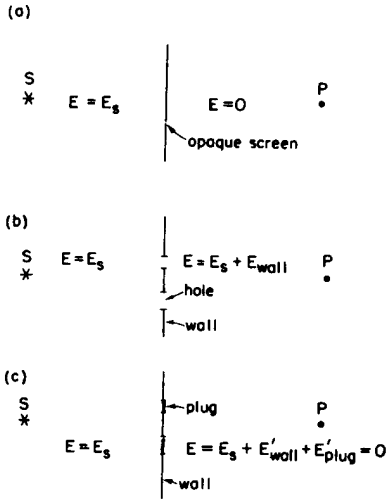


Fig. 31-6. Diffraction by a screen.

31-6 Diffraction of light by a screen

It is now a good time to take up a somewhat different matter which we can handle with the machinery of this chapter. In the last chapter we said that when you have an opaque screen and the light can come through some holes, the distribution of intensity—the diffraction pattern—could be obtained by imagining instead that the holes are replaced by sources (oscillators) uniformly distributed over the hole. In other words, the diffracted wave is the same as though the hole were a new source. We have to explain the reason for that, because the hole is, of course, just where there are *no* sources, where there are *no* accelerating charges.

Let us first ask: “What is an opaque screen?” Suppose we have a completely opaque screen between a source S and an observer at P , as in Fig. 31-6(a). If the screen is “opaque” there is no field at P . Why is there no field there? According to the basic principles we should obtain the field at P as the field E_s of the source delayed, plus the field from all the other charges around. But, as we have seen above, the charges in the screen will be set in motion by the field E_s , and these motions generate a new field which, if the screen is opaque, must *exactly* cancel the field E_s on the back side of the screen. You say: “What a miracle that it balances *exactly*! Suppose it was not exactly right!” If it were not exactly right (remember that this opaque screen has some thickness), the field toward the rear part of the screen would not be exactly zero. So, not being zero, it would set into motion some other charges in the material of the screen, and thus make a little more field, trying to get the total balanced out. So if we make the screen thick enough, there is no residual field, because there is enough opportunity to finally get the thing quieted down. In terms of our formulas above we would say that the

screen has a large and imaginary index, so the wave is absorbed exponentially as it goes through. You know, of course, that a thin enough sheet of the most opaque material, even gold, is transparent.

Now let us see what happens with an opaque screen which has holes in it, as in Fig. 31-6(b). What do we expect for the field at P ? The field at P can be represented as a sum of two parts—the field due to the source S plus the field due to the wall, i.e., due to the motions of the charges in the walls. We might expect the motions of the charges in the walls to be complicated, but we can find out *what fields they produce* in a rather simple way.

Suppose that we were to take the same screen, but plug up the holes, as indicated in part (c) of the figure. We imagine that the plugs are of exactly the same material as the wall. Mind you, the plugs go where the holes were in case (b). Now let us calculate the field at P . The field at P is certainly zero in case (c), but it is *also* equal to the field from the source plus the field due to all the motions of the atoms in the walls and in the plugs. We can write the following equations:

$$\text{Case (b): } E_{\text{at } P} = E_s + E_{\text{wall}},$$

$$\text{Case (c): } E'_{\text{at } P} = 0 = E_s + E'_{\text{wall}} + E'_{\text{plug}},$$

where the primes refer to the case where the plugs are in place, but E_s is, of course, the same in both cases. Now if we subtract the two equations, we get

$$E_{\text{at } P} = (E_{\text{wall}} - E'_{\text{wall}}) - E'_{\text{plug}}.$$

Now if the holes are not too small (say many wavelengths across), we would not expect the presence of the plugs to change the fields which arrive at the walls except possibly for a little bit around the edges of the holes. Neglecting this small effect, we can set $E_{\text{wall}} = E'_{\text{wall}}$ and obtain that

$$E_{\text{at } P} = -E'_{\text{plug}}.$$

We have the result that the field at P *when there are holes* in a screen (case b) is the same (except for sign) as the field that is produced by *that part* of a complete opaque wall which is *located where the holes are!* (The sign is not too interesting, since we are usually interested in intensity which is proportional to the square of the field.) It seems like an amazing backwards-forwards argument. It is, however, not only true (approximately for not too small holes), but useful, and is the justification for the usual theory of diffraction.

The field E'_{plug} is computed in any particular case by remembering that the motion of the charges *everywhere* in the screen is just that which will cancel out the field E_s on the back of the screen. Once we know these motions, we add the radiation fields at P due just to the charges in the plugs.

We remark again that this theory of diffraction is only approximate, and will be good only if the holes are not too small. For holes which are too small the E'_{plug} term will be small and then the difference between E'_{wall} and E_{wall} (which difference we have taken to be zero) may be comparable to or larger than the small E'_{plug} term, and our approximation will no longer be valid.

Radiation Damping. Light Scattering

32-1 Radiation resistance

In the last chapter we learned that when a system is oscillating, energy is carried away, and we deduced a formula for the energy which is radiated by an oscillating system. If we know the electric field, then the average of the square of the field times $\epsilon_0 c$ is the amount of energy that passes per square meter per second through a surface normal to the direction in which the radiation is going:

$$S = \epsilon_0 c \langle E^2 \rangle. \quad (32.1)$$

Any oscillating charge radiates energy; for instance, a driven antenna radiates energy. If the system radiates energy, then in order to account for the conservation of energy we must find that power is being delivered along the wires which lead into the antenna. That is, to the driving circuit the antenna acts like a *resistance*, or a place where energy can be “lost” (the energy is not really lost, it is really radiated out, but so far as the circuit is concerned, the energy is lost). In an ordinary resistance, the energy which is “lost” passes into heat; in this case the energy which is “lost” goes out into space. But from the standpoint of circuit theory, without considering *where* the energy goes, the net effect on the circuit is the same—energy is “lost” from that circuit. Therefore the antenna appears to the generator as having a resistance, even though it may be made with perfectly good copper. In fact, if it is well built it will appear as almost a pure resistance, with very little inductance or capacitance, because we would like to radiate as much energy as possible out of the antenna. This resistance that an antenna shows is called the *radiation resistance*.

If a current I is going to the antenna, then the average rate at which power is delivered to the antenna is the average of the square of the current times the resistance. The rate at which power is *radiated* by the antenna is proportional to the square of the current in the antenna, of course, because all the fields are proportional to the currents, and the energy liberated is proportional to the square of the field. The coefficient of proportionality between radiated power and $\langle I^2 \rangle$ is the radiation resistance.

An interesting question is, what is this radiation resistance due to? Let us take a simple example: let us say that currents are driven up and down in an antenna. We find that we have to put work in, if the antenna is to radiate energy. If we take a charged body and accelerate it up and down it radiates energy; if it were not charged it would not radiate energy. It is one thing to calculate from the conservation of energy that energy is lost, but another thing to answer the question, *against what force* are we doing the work? That is an interesting and very difficult question which has never been completely and satisfactorily answered for electrons, although it has been for antennas. What happens is this: in an antenna, the fields produced by the moving charges in one part of the antenna react on the moving charges in another part of the antenna. We can calculate these forces and find out how much work they do, and so find the right rule for the radiation resistance. When we say “We can calculate—” that is not quite right—we cannot, because we have not yet studied the laws of electricity at short distances; only at large distances do we know what the electric field is. We saw the formula (28.3), but at present it is too complicated for us to calculate the fields inside the wave zone. Of course, since conservation of energy is valid, we can calculate the result all right without knowing the fields at short distances. (As a matter of fact, by using this argument backwards it turns out that one can find the formula for the forces at short

32-1 Radiation resistance

32-2 The rate of radiation of energy

32-3 Radiation damping

32-4 Independent sources

32-5 Scattering of light

distances only by knowing the field at very large distances, by using the laws of conservation of energy, but we shall not go into that here.)

The problem in the case of a single electron is this: if there is only one charge, what can the force act on? It has been proposed, in the old classical theory, that the charge was a little ball, and that one part of the charge acted on the other part. Because of the delay in the action across the tiny electron, the force is not exactly in phase with the motion. That is, if we have the electron standing still, we know that "action equals reaction." So the various internal forces are equal, and there is no net force. But if the electron is accelerating, then because of the time delay across it, the force which is acting on the front from the back is not exactly the same as the force on the back from the front, because of the delay in the effect. This delay in the timing makes for a lack of balance, so, as a net effect, the thing holds itself back by its bootstraps! This model of the origin of the resistance to acceleration, the radiation resistance of a moving charge, has run into many difficulties, because our present view of the electron is that it is *not* a "little ball"; this problem has never been solved. Nevertheless we can calculate exactly, of course, what the net radiation resistance force must be, i.e., how much loss there must be when we accelerate a charge, in spite of not knowing directly the mechanism of how that force works.

32-2 The rate of radiation of energy

Now we shall calculate the total energy radiated by an accelerating charge. To keep the discussion general, we shall take the case of a charge accelerating any which way, but nonrelativistically. At a moment when the acceleration is, say, vertical, we know that the electric field that is generated is the charge multiplied by the projection of the retarded acceleration, divided by the distance. So we know the electric field at any point, and we therefore know the square of the electric field and thus the energy $\epsilon_0 c E^2$ leaving through a unit area per second.

The quantity $\epsilon_0 c$ appears quite often in expressions involving radiowave propagation. Its reciprocal is called the *impedance of a vacuum*, and it is an easy number to remember: it has the value $1/\epsilon_0 c = 377$ ohms. So the power in watts per square meter is equal to the average of the field squared, divided by 377.

Using our expression (29.1) for the electric field, we find that

$$S = \frac{q^2 a'^2 \sin^2 \theta}{16\pi^2 \epsilon_0 r^2 c^3} \quad (32.2)$$

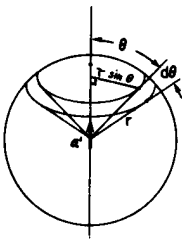


Fig. 32-1. The area of a spherical segment is $2\pi r \sin \theta \cdot r d\theta$.

is the power per square meter radiated in the direction θ . We notice that it goes inversely as the square of the distance, as we said before. Now suppose we wanted the total energy radiated in all directions: then we must integrate (32.2) over all directions. First we multiply by the area, to find the amount that flows within a little angle $d\theta$ (Fig. 32-1). We need the area of a spherical section. The way to think of it is this; if r is the radius, then the width of the annular segment is $r d\theta$, and the circumference is $2\pi r \sin \theta$, because $r \sin \theta$ is the radius of the circle. So the area of the little piece of the sphere is $2\pi r \sin \theta$ times $r d\theta$:

$$dA = 2\pi r^2 \sin \theta d\theta. \quad (32.3)$$

By multiplying the flux [(32.2), the power per square meter] by the area in square meters included in the small angle $d\theta$, we find the amount of energy that is liberated in this direction between θ and $\theta + d\theta$; then we integrate that over all the angles θ from 0 to 180° :

$$P = \int S dA = \frac{q^2 a'^2}{8\pi \epsilon_0 c^3} \int_0^\pi \sin^3 \theta d\theta. \quad (32.4)$$

By writing $\sin^3 \theta = (1 - \cos^2 \theta) \sin \theta$ it is not hard to show that $\int_0^\pi \sin^3 \theta d\theta = 4/3$. Using that fact, we finally get

$$P = \frac{q^2 a'^2}{6\pi \epsilon_0 c^3}. \quad (32.5)$$

This expression deserves some remarks. First of all, since the vector $\mathbf{a}'$ had a certain direction, the a'^2 in (32.5) would be the square of the vector $\mathbf{a}'$, that is, $\mathbf{a}' \cdot \mathbf{a}'$, the length of the vector, squared. Secondly, the flux (32.2) was calculated using the retarded acceleration; that is, the acceleration at the time at which the energy now passing through the sphere was radiated. We might like to say that this energy was in fact liberated at this earlier time. This is not exactly true; it is only an approximate idea. The exact time when the energy is liberated cannot be defined precisely. All we can really calculate precisely is what happens in a complete motion, like an oscillation or something, where the acceleration finally ceases. Then what we find is that the total energy flux per cycle is the average of acceleration squared, for a complete cycle. This is what should really appear in (32.5). Or, if it is a motion with an acceleration that is initially and finally zero, then the total energy that has flown out is the time integral of (32.5).

To illustrate the consequences of formula (32.5) when we have an oscillating system, let us see what happens if the displacement x of the charge is oscillating so that the acceleration a is $-\omega^2 x_0 e^{i\omega t}$. The average of the acceleration squared over a cycle (remember that we have to be very careful when we square things that are written in complex notation—it really is the cosine, and the average of $\cos^2 \omega t$ is one-half) thus is

$$\langle a'^2 \rangle = \frac{1}{2} \omega^4 x_0^2.$$

Therefore

$$P = \frac{q^2 \omega^4 x_0^2}{12 \pi \epsilon_0 c^3}. \quad (32.6)$$

The formulas we are now discussing are relatively advanced and more or less modern; they date from the beginning of the twentieth century, and they are very famous. Because of their historical value, it is important for us to be able to read about them in older books. In fact, the older books also used a system of units different from our present mks system. However, all these complications can be straightened out in the final formulas dealing with electrons by the following rule: The quantity $q_e^2/4\pi\epsilon_0$, where q_e is the electronic charge (in coulombs), has, historically, been written as e^2 . It is very easy to calculate that e in the mks system is numerically equal to 1.5188×10^{-14} , because we know that, numerically, $q_e = 1.60206 \times 10^{-19}$ and $1/4\pi\epsilon_0 = 8.98748 \times 10^9$. Therefore we shall often use the convenient abbreviation

$$e^2 = \frac{q_e^2}{4\pi\epsilon_0}. \quad (32.7)$$

If we use the above numerical value of e in the older formulas and treat them as though they were written in mks units, we will get the right numerical results. For example, the older form of (32.5) is $P = \frac{2}{3} e^2 a^2 / c^3$. Again, the potential energy of a proton and an electron at distance r is $q_e^2/4\pi\epsilon_0 r$ or e^2/r , with $e = 1.5188 \times 10^{-14}$ mks.

32-3 Radiation damping

Now the fact that an oscillator loses a certain energy would mean that if we had a charge on the end of a spring (or an electron in an atom) which has a natural frequency ω_0 , and we start it oscillating and let it go, it will not oscillate forever, even if it is in empty space millions of miles from anything. There is no oil, no resistance, in an ordinary sense; no “viscosity.” But nevertheless it will not oscillate, as we might once have said, “forever,” because if it is charged it is radiating energy, and therefore the oscillation will slowly die out. How slowly? What is the Q of such an oscillator, caused by the electromagnetic effects, the so-called radiation resistance or radiation damping of the oscillator? The Q of any oscillating system is the total energy content of the oscillator at any time divided by the energy loss per radian:

$$Q = \frac{W}{dW/d\phi}.$$

Or (another way to write it), since $dW/d\phi = (dW/dt)/(d\phi/dt) = (dW/dt)/\omega$,

$$Q = \frac{\omega W}{dW/dt}. \quad (32.8)$$

If for a given Q this tells us how the energy of the oscillation dies out, $dW/dt = -(\omega/Q)W$, which has the solution $W = W_0 e^{-\omega t/Q}$ if W_0 is the initial energy (at $t = 0$).

To find the Q for a radiator, we go back to (32.8) and use (32.6) for dW/dt .

Now what do we use for the energy W of the oscillator? The kinetic energy of the oscillator is $\frac{1}{2}mv^2$, and the *mean* kinetic energy is $m\omega^2 x_0^2/4$. But we remember that for the total energy of an oscillator, on the average half is kinetic and half is potential energy, and so we double our result, and find for the total energy of the oscillator

$$W = \frac{1}{2}m\omega^2 x_0^2. \quad (32.9)$$

What do we use for the frequency in our formulas? We use the natural frequency ω_0 because, for all practical purposes, that is the frequency at which our atom is radiating, and for m we use the electron mass m_e . Then, making the necessary divisions and cancellations, the formula comes down to

$$\frac{1}{Q} = \frac{4\pi e^2}{3\lambda m_e c^2}. \quad (32.10)$$

(In order to see it better and in a more historical form we write it using our abbreviation $q_e^2/4\pi\epsilon_0 = e^2$, and the factor ω_0/c which was left over has been written as $2\pi/\lambda$.) Since Q is dimensionless, the combination $e^2/m_e c^2$ must be a property only of the electron charge and mass, an intrinsic property of the electron, and it must be a *length*. It has been given a name, the *classical electron radius*, because the early atomic models, which were invented to explain the radiation resistance on the basis of the force of one part of the electron acting on the other parts, all needed to have an electron whose dimensions were of this general order of magnitude. However, this quantity no longer has the significance that we believe that the electron really *has* such a radius. Numerically, the magnitude of the radius is

$$r_0 = \frac{e^2}{m_e c^2} = 2.82 \times 10^{-15} \text{ m}. \quad (32.11)$$

Now let us actually calculate the Q of an atom that is emitting light—let us say a sodium atom. For a sodium atom, the wavelength is roughly 6000 angstroms, in the yellow part of the visible spectrum, and this is a typical wavelength. Thus

$$Q = \frac{3\lambda}{4\pi r_0} \approx 5 \times 10^7, \quad (32.12)$$

so the Q of an atom is of the order 10^8 . This means that an atomic oscillator will oscillate for 10^8 radians or about 10^7 oscillations, before its energy falls by a factor $1/e$. The frequency of oscillation of light corresponding to 6000 angstroms, $\nu = c/\lambda$, is on the order of 10^{15} cycles/sec, and therefore the lifetime, the time it takes for the energy of a radiating atom to die out by a factor $1/e$, is on the order of 10^{-8} sec. In ordinary circumstances, freely emitting atoms usually take about this long to radiate. This is valid only for atoms which are in empty space, not being disturbed in any way. If the electron is in a solid and it has to hit other atoms or other electrons, then there are additional resistances and different damping.

The effective resistance term γ in the resistance law for the oscillator can be found from the relation $1/Q = \gamma/\omega_0$, and we remember that the size of γ determines how wide the resonance curve is (Fig. 23-2). Thus we have just computed the *widths of spectral lines* for freely radiating atoms! Since $\lambda = 2\pi c/\omega$, we find that

$$\begin{aligned} \Delta\lambda &= 2\pi c \Delta\omega/\omega^2 = 2\pi c \gamma/\omega_0^2 = 2\pi c/Q\omega_0 \\ &= \lambda/Q = 4\pi r_0/3 = 1.18 \times 10^{-14} \text{ m}. \end{aligned} \quad (32.13)$$

32-4 Independent sources

In preparation for our second topic, the scattering of light, we must now discuss a certain feature of the phenomenon of interference that we neglected to discuss previously. This is the question of when interference does *not* occur. If we have two sources S_1 and S_2 , with amplitudes A_1 and A_2 , and we make an observation in a certain direction in which the phases of arrival of the two signals are ϕ_1 and ϕ_2 (a combination of the actual time of oscillation and the delayed time, depending on the position of observation), then the energy that we receive can be found by compounding the two complex number vectors A_1 and A_2 , one at angle ϕ_1 and the other at angle ϕ_2 (as we did in Chapter 30) and we find that the resultant energy is proportional to

$$A_R^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos(\phi_1 - \phi_2). \quad (32.14)$$

Now if the cross term $2A_1A_2 \cos(\phi_1 - \phi_2)$ were not there, then the total energy that would be received in a given direction would simply be the sum of the energies, $A_1^2 + A_2^2$, that would be liberated by each source separately, which is what we usually expect. That is, the combined intensity of light shining on something from two sources is the sum of the intensities of the two lights. On the other hand, if we have things set just right and we have a cross term, it is not such a sum, because there is also some interference. If there are circumstances in which this term is of no importance, then we would say the interference is apparently lost. Of course, in nature it is always there, but we may not be able to detect it.

Let us consider some examples. Suppose, first, that the two sources are 7,000,000,000 wavelengths apart, not an impossible arrangement. Then in a given direction it is true that there is a very definite value of these phase differences. But, on the other hand, if we move just a hair in one direction, a few wavelengths, which is no distance at all (our eye already has a hole in it that is so large that we are averaging the effects over a range very wide compared with one wavelength) then we change the relative phase, and the cosine changes very rapidly. If we take the *average* of the intensity over a little region, then the cosine, which goes plus, minus, plus, minus, as we move around, averages to zero.

So if we average over regions where the phase varies very rapidly with position, we get no interference.

Another example. Suppose that the two sources are two independent radio oscillators—not a single oscillator being fed by two wires, which guarantees that the phases are kept together, but two independent sources—and that they are not *precisely* tuned at the same frequency (it is very hard to make them at exactly the same frequency without actually wiring them together). In this case we have what we call two *independent* sources. Of course, since the frequencies are not exactly equal, although they started in phase, one of them begins to get a little ahead of the other, and pretty soon they are out of phase, and then it gets still further ahead, and pretty soon they are in phase again. So the phase difference between the two is gradually drifting with time, but if our observation is so crude that we cannot see that little time, if we average over a much longer time, then although the intensity swells and falls like what we call “beats” in sound, if these swellings and fallings are too rapid for our equipment to follow, then again this term averages out.

In other words, in any circumstance in which the phase shift averages out, we get no interference!

One finds many books which say that two distinct light sources never interfere. This is not a statement of physics, but is merely a statement of the degree of sensitivity of the technique of the experiments at the time the book was written. What happens in a light source is that first one atom radiates, then another atom radiates, and so forth, and we have just seen that atoms radiate a train of waves only for about 10^{-8} sec; after 10^{-8} sec, some atom has probably taken over, then another atom takes over, and so on. So the phases can really only stay the same for about 10^{-8} sec. Therefore, if we average for very much more than 10^{-8} sec, we do not see an interference from two different sources, because they cannot hold their phases steady for longer than 10^{-8} sec. With photocells, very high-speed detection

is possible, and one can show that there is an interference which varies with time, up and down, in about 10^{-8} sec. But most detection equipment, of course, does not look at such fine time intervals, and thus sees no interference. Certainly with the eye, which has a tenth-of-a-second averaging time, there is no chance whatever of seeing an interference between two different ordinary sources.

Recently it has become possible to make light sources which get around this effect by making all the atoms emit *together* in time. The device which does this is a very complicated thing, and has to be understood in a quantum-mechanical way. It is called a *laser*, and it is possible to produce from a laser a source in which the interference frequency, the time at which the phase is kept constant, is very much longer than 10^{-8} sec. It can be of the order of a hundredth, a tenth, or even one second, and so, with ordinary photocells, one can pick up the frequency between two different lasers. One can easily detect the pulsing of the beats between two laser sources. Soon, no doubt, someone will be able to demonstrate two sources shining on a wall, in which the beats are so slow that one can *see* the wall get bright and dark!

Another case in which the interference averages out is that in which, instead of having only *two* sources, we have *many*. In this case, we would write the expression for A_R^2 as the sum of a whole lot of amplitudes, complex numbers, squared, and we would get the square of each one, all added together, plus cross terms between every pair, and if the circumstances are such that the latter average out, then there will be no effects of interference. It may be that the various sources are located in such random positions that, although the phase difference between A_2 and A_3 is also definite, it is very different from that between A_1 and A_2 , etc. So we would get a whole lot of cosines, many plus, many minus, all averaging out.

So it is that in many circumstances we do not see the effects of interference, but see only a collective, total intensity equal to the sum of all the intensities.

32-5 Scattering of light

The above leads us to an effect which occurs in air as a consequence of the irregular positions of the atoms. When we were discussing the index of refraction, we saw that an incoming beam of light will make the atoms radiate again. The electric field of the incoming beam drives the electrons up and down, and they radiate because of their acceleration. This scattered radiation combines to give a beam in the same direction as the incoming beam, but of somewhat different phase, and this is the origin of the index of refraction.

But what can we say about the amount of re-radiated light in some other direction? Ordinarily, if the atoms are very beautifully located in a nice pattern, it is easy to show that we get nothing in other directions, because we are adding a lot of vectors with their phases always changing, and the result comes to zero. But if the objects are *randomly* located, then the total intensity in any direction is the *sum* of the intensities that are scattered by each atom, as we have just discussed. Furthermore, the atoms in a gas are in actual motion, so that although the relative phase of two atoms is a definite amount now, later the phase would be quite different, and therefore *each* cosine term will average out. Therefore, to find out how much light is scattered in a given direction by a gas, we merely study the effects of *one atom* and multiply the intensity it radiates by the number of atoms.

Earlier, we remarked that the phenomenon of scattering of light of this nature is the origin of the blue of the sky. The sunlight goes through the air, and when we look to one side of the sun—say at 90° to the beam—we see blue light; what we now have to calculate is *how much* light we see and *why it is blue*.

If the incident beam has the electric field $\mathbf{E} = \mathbf{E}_0 e^{i\omega t}$ at the point where the atom is located, we know that an electron in the atom will vibrate up and down in response to this $\mathbf{E}$ (Fig. 32-2). From Eq. (23.8), the amplitude will be

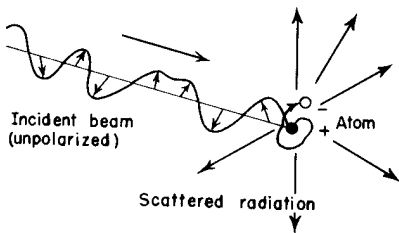


Fig. 32-2. A beam of radiation falls on an atom and causes the charges (electrons) in the atom to move. The moving electrons in turn radiate in various directions.

$$\hat{x} = \frac{q_e \mathbf{E}_0}{m(\omega_0^2 - \omega^2 + i\omega\gamma)}. \quad (32.15)$$

We could include the damping and the possibility that the atom acts like several oscillators of different frequency and sum over the various frequencies, but for simplicity let us just take one oscillator and neglect the damping. Then the response to the external electric field, which we have already used in the calculation of the index of refraction, is simply

$$\hat{\mathbf{x}} = \frac{q_e \hat{\mathbf{E}}_0}{m(\omega_0^2 - \omega^2)}. \quad (32.16)$$

We could now easily calculate the intensity of light that is emitted in various directions, using formula (32.2) and the acceleration corresponding to the above $\hat{\mathbf{x}}$.

Rather than do this, however, we shall simply calculate the *total amount* of light scattered in *all* directions, just to save time. The total amount of light energy per second, scattered in all directions by the single atom, is of course given by Eq. (32.7). So, putting together the various pieces and regrouping them, we get

$$\begin{aligned} P &= [(q_e^2 \omega^4 / 12 \pi \epsilon_0 c^3) q_e^2 E_0^2 / m^2 (\omega^2 - \omega_0^2)^2] \\ &= (\frac{1}{2} \epsilon_0 c E_0^2) (8 \pi / 3) (q_e^4 / 16 \pi^2 \epsilon_0^2 m^2 c^4) [\omega^4 / (\omega^2 - \omega_0^2)^2] \\ &= (\frac{1}{2} \epsilon_0 c E_0^2) (8 \pi r_0^2 / 3) [\omega^4 / (\omega^2 - \omega_0^2)^2] \end{aligned} \quad (32.17)$$

for the total scattered power, radiated in all directions.

We have written the result in the above form because it is then easy to remember: First, the total energy that is scattered is proportional to the square of the incident field. What does that mean? Obviously, the square of the incident field is proportional to the energy which is coming in per second. In fact, the energy incident per square meter per second is $\epsilon_0 c$ times the average $\langle E^2 \rangle$ of the square of the electric field, and if E_0 is the maximum value of E , then $\langle E^2 \rangle = \frac{1}{2} E_0^2$. In other words, the total energy scattered is proportional to the energy per square meter that comes in; the brighter the sunlight that is shining in the sky, the brighter the sky is going to look.

Next, what *fraction* of the incoming light is scattered? Let us imagine a “target” with a certain area, let us say σ , in the beam (not a real, material target, because this would diffract light, and so on; we mean an imaginary area drawn in space). The total amount of energy that would pass through this surface σ in a given circumstance is proportional both to the incoming intensity and to σ , and would be

$$P = (\frac{1}{2} \epsilon_0 c E_0^2) \sigma. \quad (32.18)$$

Now we invent an idea: we say that the atom scatters a total amount of intensity which is the amount which would fall on a certain geometrical area, and we give the answer by giving that area. That answer, then, is independent of the incident intensity; it gives the ratio of the energy scattered to the energy incident per square meter. In other words, the ratio

$$\frac{\text{total energy scattered per second}}{\text{energy incident per square meter per second}} \text{ is an } \textit{area}.$$

The significance of this area is that, if all the energy that impinged on that area were to be spewed in all directions, then that is the amount of energy that would be scattered by the atom.

This area is called a *cross section for scattering*; the idea of cross section is used constantly, whenever some phenomenon occurs in proportion to the intensity of a beam. In such cases one always describes the amount of the phenomenon by saying what the effective area would have to be to pick up that much of the beam. It does not mean in any way that this oscillator actually *has* such an area. If there were nothing present but a free electron shaking up and down there would be no area directly associated with it, physically. It is merely a way of expressing the answer to a certain kind of problem; it tells us what area the incident beam would

have to hit in order to account for that much energy coming off. Thus, for our case,

$$\sigma_s = \frac{8\pi r_0^2}{3} \cdot \frac{\omega^4}{(\omega^2 - \omega_0^2)^2} \quad (32.19)$$

(the subscript s is for “scattering”).

Let us look at some examples. First, if we go to a very low natural frequency ω_0 , or to completely unbound electrons, for which $\omega_0 = 0$, then the frequency ω cancels out and the cross section is a constant. This low-frequency limit, or the free electron cross section, is known as the *Thompson scattering cross section*. It is an area whose dimensions are approximately 10^{-15} meter, more or less, on a side, i.e., 10^{-30} square meter, which is rather small!

On the other hand, if we take the case of light in the air, we remember that for air the natural frequencies of the oscillators are higher than the frequency of the light that we use. This means that, to a first approximation, we can disregard ω^2 in the denominator, and we find that the scattering is proportional to the *fourth power* of the frequency. That is to say, light which is of higher frequency by, say, a factor of two, is *sixteen times* more intensely scattered, which is a quite sizable difference. This means that blue light, which has about twice the frequency of the reddish end of the spectrum, is scattered to a far greater extent than red light. Thus when we look at the sky it looks that glorious blue that we see all the time!

There are several points to be made about the above results. One interesting question is, why do we ever see the *clouds*? Where do the clouds come from? Everybody knows it is the condensation of water vapor. But, of course, the water vapor is already in the atmosphere *before* it condenses, so why don't we see it then? After it condenses it is perfectly obvious. It wasn't there, now it *is* there. So the mystery of where the clouds come from is not really such a childish mystery as “Where does the water come from, Daddy?,” but has to be explained.

We have just explained that every atom scatters light, and of course the water vapor will scatter light, too. The mystery is why, when the water is condensed into clouds, does it scatter such a *tremendously greater amount* of light?

Consider what would happen if, instead of a single atom, we had an agglomerate of atoms, say two, very close together compared with the wavelength of the light. Remember, atoms are only an angstrom or so across, while the wavelength of light is some 5000 angstroms, so when they form a clump, a few atoms together, they can *be* very close together compared with the wavelength of light. Then when the electric field acts, *both of the atoms will move together*. The electric field that is scattered will then be the sum of the two electric fields in phase, i.e., double the amplitude that there was with a single atom, and the *energy* which is scattered is therefore *four times* what it is with a single atom, not twice! So lumps of atoms radiate or scatter more energy than they do as single atoms. Our argument that the phases are independent is based on the assumption that there is a real and large difference in phase between any two atoms, which is true only if they are several wavelengths apart and randomly spaced, or moving. But if they are right next to each other, they necessarily scatter in phase, and they have a coherent interference which produces an increase in the scattering.

If we have N atoms in a lump, which is a tiny droplet of water, then each one will be driven by the electric field in about the same way as before (the effect of one atom on the other is not important; it is just to get the idea anyway) and the amplitude of scattering from each one is the same, so the total field which is scattered is N -fold increased. The *intensity* of the light which is scattered is then the square, or N^2 -fold, increased. We would have expected, if the atoms were spread out in space, only N times as much as 1, whereas we get N^2 times as much as 1! That is to say, the scattering of water in lumps of N molecules each is N times more intense than the scattering of the single atoms. So as the water agglomerates the scattering increases. Does it increase *ad infinitum*? No! When does this analysis begin to fail? How many atoms can we put together before we cannot drive this argument any further? *Answer*: If the water drop gets so big that from one end to the other is a wavelength or so, then the atoms are no longer all in

phase because they are too far apart. So as we keep increasing the size of the drop-lets we get more and more scattering, until such a time that a drop gets about the size of a wavelength, and then the scattering does not increase anywhere nearly as rapidly as the drop gets bigger. Furthermore, the blue disappears, because for long wavelengths the drops can be bigger, before this limit is reached, than they can be for short wavelengths. Although the short waves scatter more per atom than the long waves, there is a bigger enhancement for the red end of the spectrum than for the blue end when all the drops are bigger than the wavelength, so the color is shifted from the blue toward the red.

Now we can make an experiment that demonstrates this. We can make particles that are very small at first, and then gradually grow in size. We use a solution of sodium thiosulfate (hypo) with sulphuric acid, which precipitates very fine grains of sulphur. As the sulphur precipitates, the grains first start very small, and the scattering is a little bluish. As it precipitates more it gets more intense, and then it will get whitish as the particles get bigger. In addition, the light which goes straight through will have the blue taken out. That is why the sunset is red, of course, because the light that comes through a lot of air to the eye has had a lot of blue light scattered out, so it is yellow-red.

Finally, there is one other important feature which really belongs in the next chapter, on polarization, but it is so interesting that we point it out now. This is that the electric field of the scattered light tends to vibrate in a particular direction. The electric field in the incoming light is oscillating in some way, and the driven oscillator goes in this same direction, and if we are situated about at right angles to the beam, we will see *polarized* light, that is to say, light in which the electric field is going only one way. In general, the atoms can vibrate in any direction at right angles to the beam, but if they are driven directly toward or away from us, we do not see it. So if the incoming light has an electric field which changes and oscillates in any direction, which we call unpolarized light, then the light which is coming out at 90° to the beam vibrates in only one direction! (See Fig. 32-3.)

There is a substance called polaroid which has the property that when light goes through it, only the piece of the electric field which is along one particular axis can get through. We can use this to test for polarization, and indeed we find the light scattered by the hypo solution to be strongly polarized.

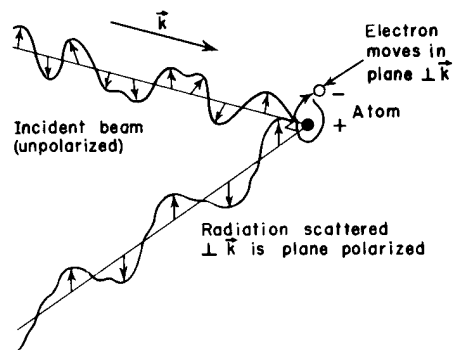


Fig. 32-3. Illustration of the origin of the polarization of radiation scattered at right angles to the incident beam.

Polarization

33-1 The electric vector of light

In this chapter we shall consider those phenomena which depend on the fact that the electric field that describes the light is a vector. In previous chapters we have not been concerned with the direction of oscillation of the electric field, except to note that the electric vector lies in a plane perpendicular to the direction of propagation. The particular direction in this plane has not concerned us. We now consider those phenomena whose central feature is the particular direction of oscillation of the electric field.

In ideally monochromatic light, the electric field must oscillate at a definite frequency, but since the x -component and the y -component can oscillate independently at a definite frequency, we must first consider the resultant effect produced by superposing two independent oscillations at right angles to each other. What kind of electric field is made up of an x -component and a y -component which oscillate at the same frequency? If one adds to an x -vibration a certain amount of y -vibration at the same phase, the result is a vibration in a new direction in the xy -plane. Figure 33-1 illustrates the superposition of different amplitudes for the x -vibration and the y -vibration. But the resultants shown in Fig. 33-1 are not the only possibilities; in all of these cases we have assumed that the x -vibration and the y -vibration are *in phase*, but it does not have to be that way. It could be that the x -vibration and the y -vibration are out of phase.

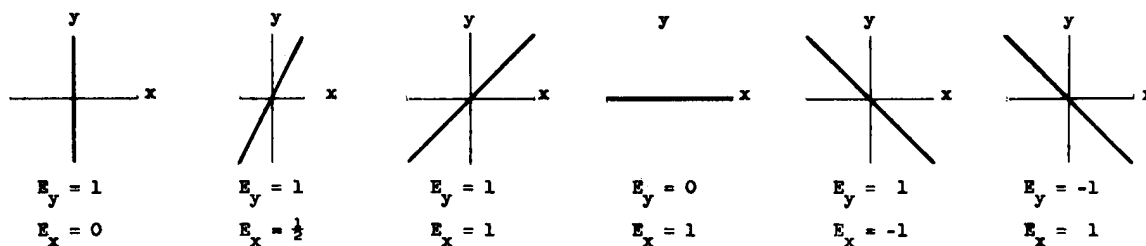


Fig. 33-1. Superposition of x -vibrations and y -vibrations in phase.

When the x -vibration and the y -vibration are not in phase, the electric field vector moves around in an ellipse, and we can illustrate this in a familiar way. If we hang a ball from a support by a long string, so that it can swing freely in a horizontal plane, it will execute sinusoidal oscillations. If we imagine horizontal x - and y -coordinates with their origin at the rest position of the ball, the ball can swing in either the x - or y -direction with the same pendulum frequency. By selecting the proper initial displacement and initial velocity, we can set the ball in oscillation along either the x -axis or the y -axis, or along any straight line in the xy -plane. These motions of the ball are analogous to the oscillations of the electric field vector illustrated in Fig. 33-1. In each instance, since the x -vibrations and the y -vibrations reach their maxima and minima at the same time, the x - and y -oscillations are in phase. But we know that the most general motion of the ball is motion in an ellipse, which corresponds to oscillations in which the x - and y -directions are *not* in the same phase. The superposition of x - and y -vibrations which are not in phase is illustrated in Fig. 33-2 for a variety of angles between the phase of the x -vibration and that of the y -vibration. The general result is that the electric vector moves around an ellipse. The motion in a straight line is a particular

33-1 The electric vector of light

33-2 Polarization of scattered light

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case corresponding to a phase difference of zero (or an integral multiple of π); motion in a circle corresponds to equal amplitudes with a phase difference of 90° (or any odd integral multiple of $\pi/2$).

In Fig. 33-2 we have labeled the electric field vectors in the x - and y -directions with complex numbers, which are a convenient representation in which to express the phase difference. Do not confuse the real and imaginary components of the complex electric vector in this notation with the x - and y -coordinates of the field. The x - and y -coordinates plotted in Fig. 33-1 and Fig. 33-2 are actual electric fields that we can measure. The real and imaginary components of a complex electric field vector are only a mathematical convenience and have no physical significance.

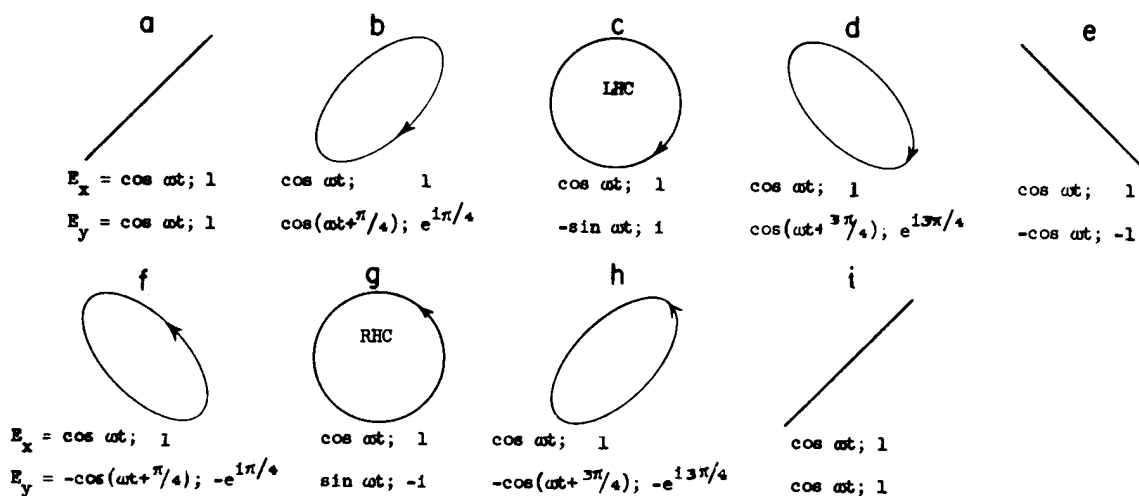


Fig. 33-2. Superposition of x -vibrations and y -vibrations with equal amplitudes but various relative phases. The components E_x and E_y are expressed in both real and complex notations.

Now for some terminology. Light is *linearly polarized* (sometimes called plane polarized) when the electric field oscillates on a straight line; Fig. 33-1 illustrates linear polarization. When the end of the electric field vector travels in an ellipse, the light is *elliptically polarized*. When the end of the electric field vector travels around a circle, we have *circular polarization*. If the end of the electric vector, when we look at it as the light comes straight toward us, goes around in a counterclockwise direction, we call it right-hand circular polarization. Figure 33-2(g) illustrates right-hand circular polarization, and Fig. 33-2(c) shows left-hand circular polarization. In both cases the light is coming out of the paper. Our convention for labeling left-hand and right-hand circular polarization is consistent with that which is used today for all the other particles in physics which exhibit polarization (e.g., electrons). However, in some books on optics the opposite conventions are used, so one must be careful.

We have considered linearly, circularly, and elliptically polarized light, which covers everything except for the case of *unpolarized* light. Now how can the light be unpolarized when we know that it must vibrate in one or another of these ellipses? If the light is not absolutely monochromatic, or if the x - and y -phases are not kept perfectly together, so that the electric vector first vibrates in one direction, then in another, the polarization is constantly changing. Remember that one atom emits during 10^{-8} sec, and if one atom emits a certain polarization, and then another atom emits light with a different polarization, the polarizations will change every 10^{-8} sec. If the polarization changes more rapidly than we can detect it, then we call the light unpolarized, because all the effects of the polarization average out. None of the interference effects of polarization would show up with unpolarized light. But as we see from the definition, light is unpolarized only if we are unable to find out whether the light is polarized or not.

33-2 Polarization of scattered light

The first example of the polarization effect that we have already discussed is the scattering of light. Consider a beam of light, for example from the sun, shining on the air. The electric field will produce oscillations of charges in the air, and motion of these charges will radiate light with its maximum intensity in a plane normal to the direction of vibration of the charges. The beam from the sun is unpolarized, so the direction of polarization changes constantly, and the direction of vibration of the charges in the air changes constantly. If we consider light scattered at 90° , the vibration of the charged particles radiates to the observer only when the vibration is perpendicular to the observer's line of sight, and then light will be polarized along the direction of vibration. So scattering is an example of one means of producing polarization.

33-3 Birefringence

Another interesting effect of polarization is the fact that there are substances for which the index of refraction is different for light linearly polarized in one direction and linearly polarized in another. Suppose that we had some material which consisted of long, nonspherical molecules, longer than they are wide, and suppose that these molecules were arranged in the substance with their long axes parallel. Then what happens when the oscillating electric field passes through this substance? Suppose that because of the structure of the molecule, the electrons in the substance respond more easily to oscillations in the direction parallel to the axes of the molecules than they would respond if the electric field tries to push them at right angles to the molecular axis. In this way we expect a different response for polarization in one direction than for polarization at right angles to that direction. Let us call the direction of the axes of the molecules the *optic axis*. When the polarization is in the direction of the optic axis the index of refraction is different than it would be if the direction of polarization were at right angles to it. Such a substance is called *birefringent*. It has two refrangibilities, i.e., two indexes of refraction, depending on the direction of the polarization inside the substance. What kind of a substance can be birefringent? In a birefringent substance there must be a certain amount of lining up, for one reason or another, of unsymmetrical molecules. Certainly a cubic crystal, which has the symmetry of a cube, cannot be birefringent. But long needlelike crystals undoubtedly contain molecules that are asymmetric, and one observes this effect very easily.

Let us see what effects we would expect if we were to shine polarized light through a plate of a birefringent substance. If the polarization is parallel to the optic axis, the light will go through with one velocity; if the polarization is perpendicular to the axis, the light is transmitted with a different velocity. An interesting situation arises when, say, light is linearly polarized at 45° to the optic axis. Now the 45° polarization, we have already noticed, can be represented as a superposition of the x - and the y -polarizations of equal amplitude and in phase, as shown in Fig. 33-2(a). Since the x - and y -polarizations travel with different velocities, their phases change at a different rate as the light passes through the substance. So, although at the start the x - and y -vibrations are in phase, inside the material the phase difference between x - and y -vibrations is proportional to the depth in the substance. As the light proceeds through the material the polarization changes as shown in the series of diagrams in Fig. 33-2. If the thickness of the plate is just right to introduce a 90° phase shift between the x - and y -polarizations, as in Fig. 33-2(c), the light will come out circularly polarized. Such a thickness is called a quarter-wave plate, because it introduces a quarter-cycle phase difference between the x - and the y -polarizations. If linearly polarized light is sent through two quarter-wave plates, it will come out plane-polarized again, but at right angles to the original direction, as we can see from Fig. 33-2(e).

One can easily illustrate this phenomenon with a piece of cellophane. Cellophane is made of long, fibrous molecules, and is not isotropic, since the fibers lie preferentially in a certain direction. To demonstrate birefringence we need a

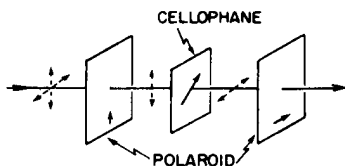


Fig. 33-3. An experimental demonstration of the birefringence of cellophane. The electric vectors in the light are indicated by the dotted lines. The pass axes of the polaroid sheets and optic axes of the cellophane are indicated by arrows. The incident beam is unpolarized.

beam of linearly polarized light, and we can obtain this conveniently by passing unpolarized light through a sheet of polaroid. Polaroid, which we will discuss later in more detail, has the useful property that it transmits light that is linearly polarized parallel to the axis of the polaroid with very little absorption, but light polarized in a direction perpendicular to the axis of the polaroid is strongly absorbed. When we pass unpolarized light through a sheet of polaroid, only that part of the unpolarized beam which is vibrating parallel to the axis of the polaroid gets through, so that the transmitted beam is linearly polarized. This same property of polaroid is also useful in detecting the direction of polarization of a linearly polarized beam, or in determining whether a beam is linearly polarized or not. One simply passes the beam of light through the polaroid sheet and rotates the polaroid in the plane normal to the beam. If the beam is linearly polarized, it will not be transmitted through the sheet when the axis of the polaroid is normal to the direction of polarization. The transmitted beam is only slightly attenuated when the axis of the polaroid sheet is rotated through 90° . If the transmitted intensity is independent of the orientation of the polaroid, the beam is not linearly polarized.

To demonstrate the birefringence of cellophane, we use two sheets of polaroid, as shown in Fig. 33-3. The first gives us a linearly polarized beam which we pass through the cellophane and then through the second polaroid sheet, which serves to detect any effect the cellophane may have had on the polarized light passing through it. If we first set the axes of the two polaroid sheets perpendicular to each other and remove the cellophane, no light will be transmitted through the second polaroid. If we now introduce the cellophane between the two polaroid sheets, and rotate the sheet about the beam axis, we observe that in general the cellophane makes it possible for some light to pass through the second polaroid. However, there are two orientations of the cellophane sheet, at right angles to each other, which permit no light to pass through the second polaroid. These orientations in which linearly polarized light is transmitted through the cellophane with no effect on the direction of polarization must be the directions parallel and perpendicular to the optic axis of the cellophane sheet.

We suppose that the light passes through the cellophane with two different velocities in these two different orientations, but it is transmitted without changing the direction of polarization. When the cellophane is turned halfway between these two orientations, as shown in Fig. 33-3, we see that the light transmitted through the second polaroid is bright.

It just happens that ordinary cellophane used in commercial packaging is very close to a half-wave thickness for most of the colors in white light. Such a sheet will turn the axis of linearly polarized light through 90° if the incident linearly polarized beam makes an angle of 45° with the optic axis, so that the beam emerging from the cellophane is then vibrating in the right direction to pass through the second polaroid sheet.

If we use white light in our demonstration, the cellophane sheet will be of the proper half-wave thickness only for a particular component of the white light, and the transmitted beam will have the color of this component. The color transmitted depends on the thickness of the cellophane sheet, and we can vary the effective thickness of the cellophane by tilting it so that the light passes through the cellophane at an angle, consequently through a longer path in the cellophane. As the sheet is tilted the transmitted color changes. With cellophane of different thicknesses one can construct filters that will transmit different colors. These filters have the interesting property that they transmit one color when the two polaroid sheets have their axes perpendicular, and the complementary color when the axes of the two polaroid sheets are parallel.

Another interesting application of aligned molecules is quite practical. Certain plastics are composed of very long and complicated molecules all twisted together. When the plastic is solidified very carefully, the molecules are all twisted in a mass, so that there are as many aligned in one direction as another, and so the plastic is not particularly birefringent. Usually there are strains and stresses introduced when the material is solidified, so the material is not perfectly homo-

geneous. However, if we apply tension to a piece of this plastic material, it is as if we were pulling a whole tangle of strings, and there will be more strings preferentially aligned parallel to the tension than in any other direction. So when a stress is applied to certain plastics, they become birefringent, and one can see the effects of the birefringence by passing polarized light through the plastic. If we examine the transmitted light through a polaroid sheet, patterns of light and dark fringes will be observed (in color, if white light is used). The patterns move as stress is applied to the sample, and by counting the fringes and seeing where most of them are, one can determine what the stress is. Engineers use this phenomenon as a means of finding the stresses in odd-shaped pieces that are difficult to calculate.

Another interesting example of a way of obtaining birefringence is by means of a liquid substance. Consider a liquid composed of long asymmetric molecules which carry a plus or minus average charge near the ends of the molecule, so that the molecule is an electric dipole. In the collisions in the liquid the molecules will ordinarily be randomly oriented, with as many molecules pointed in one direction as in another. If we apply an electric field the molecules will tend to line up, and the moment they line up the liquid becomes birefringent. With two polaroid sheets and a transparent cell containing such a polar liquid, we can devise an arrangement with the property that light is transmitted only when the electric field is applied. So we have an electrical switch for light, which is called a *Kerr cell*. This effect, that an electric field can produce birefringence in certain liquids, is called the Kerr effect.

33-4 Polarizers

So far we have considered substances in which the refractive index is different for light polarized in different directions. Of very practical value are those crystals and other substances in which not only the index, but also the coefficient of absorption, is different for light polarized in different directions. By the same arguments which supported the idea of birefringence, it is understandable that absorption can vary with the direction in which the charges are forced to vibrate in an anisotropic substance. Tourmaline is an old, famous example and polaroid is another. Polaroid consists of a thin layer of small crystals of herapathite (a salt of iodine and quinine), all aligned with their axes parallel. These crystals absorb light when the oscillations are in one direction, and they do not absorb appreciably when the oscillations are in the other direction.

Suppose that we send light into a polaroid sheet polarized linearly at an angle θ to the passing direction. What intensity will come through? This incident light can be resolved into a component perpendicular to the pass direction which is proportional to $\sin \theta$, and a component along the pass direction which is proportional to $\cos \theta$. The amplitude which comes out of the polaroid is only the cosine θ part; the $\sin \theta$ component is absorbed. The amplitude which passes through the polaroid is smaller than the amplitude which entered, by a factor $\cos \theta$. The energy which passes through the polaroid, i.e., the intensity of the light, is proportional to the square of $\cos \theta$. $\cos^2 \theta$, then, is the intensity transmitted when the light enters polarized at an angle θ to the pass direction. The absorbed intensity, of course, is $\sin^2 \theta$.

An interesting paradox is presented by the following situation. We know that it is not possible to send a beam of light through two polaroid sheets with their axes crossed at right angles. But if we place a third polaroid sheet *between* the first two, with its pass axis at 45° to the crossed axes, some light is transmitted. We know that polaroid absorbs light, it does not create anything. Nevertheless, the addition of a third polaroid at 45° allows more light to get through. The analysis of this phenomenon is left as an exercise for the student.

One of the most interesting examples of polarization is not in complicated crystals or difficult substances, but in one of the simplest and most familiar of situations—the reflection of light from a surface. Believe it or not, when light is reflected from a glass surface it may be polarized, and the physical explanation of this is very simple. It was discovered empirically by Brewster that light reflected

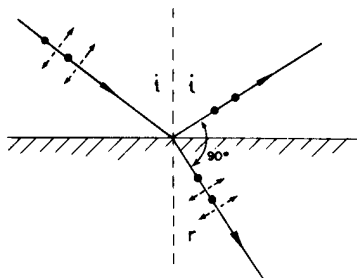


Fig. 33-4. Reflection of linearly polarized light at Brewster's angle. The polarization direction is indicated by dashed arrows; round dots indicate polarization normal to the paper.

from a surface is completely polarized if the reflected beam and the beam refracted into the material form a right angle. The situation is illustrated in Fig. 33-4. If the incident beam is polarized in the plane of incidence, there will be no reflection at all. Only if the incident beam is polarized normal to the plane of incidence will it be reflected. The reason is very easy to understand. In the reflecting material the light is polarized transversely, and we know that it is the motion of the charges in the material which generates the emergent beam, which we call the reflected beam. The source of this so-called reflected light is not simply that the incident beam is reflected; our deeper understanding of this phenomenon tells us that the incident beam drives an oscillation of the charges in the material, which in turn generates the reflected beam. From Fig. 33-4 it is clear that only oscillations normal to the paper can radiate in the direction of reflection, and consequently the reflected beam will be polarized normal to the plane of incidence. If the incident beam is polarized in the plane of incidence, there will be no reflected light.

This phenomenon is readily demonstrated by reflecting a linearly polarized beam from a flat piece of glass. If the glass is turned to present different angles of incidence to the polarized beam, sharp attenuation of the reflected intensity is observed when the angle of incidence passes through Brewster's angle. This attenuation is observed only if the plane of polarization lies in the plane of incidence. If the plane of polarization is normal to the plane of incidence, the usual reflected intensity is observed at all angles.

33-5 Optical activity

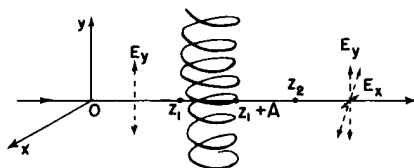


Fig. 33-5. A molecule with a shape that is not symmetric when reflected in a mirror. A beam of light, linearly polarized in the y -direction, falls on the molecule.

Another most remarkable effect of polarization is observed in materials composed of molecules which do not have reflection symmetry: molecules shaped something like a corkscrew, or like a gloved hand, or any shape which, if viewed through a mirror, would be reversed in the same way that a left-hand glove reflects as a right-hand glove. Suppose all of the molecules in the substance are the same, i.e., none is a mirror image of any other. Such a substance may show an interesting effect called optical activity, whereby as linearly polarized light passes through the substance, the direction of polarization rotates about the beam axis.

To understand the phenomenon of optical activity requires some calculation, but we can see qualitatively how the effect might come about, without actually carrying out the calculations. Consider an asymmetric molecule in the shape of a spiral, as shown in Fig. 33-5. Molecules need not actually be shaped like a corkscrew in order to exhibit optical activity, but this is a simple shape which we shall take as a typical example of those that do not have reflection symmetry. When a light beam linearly polarized along the y -direction falls on this molecule, the electric field will drive charges up and down the helix, thereby generating a current in the y -direction and radiating an electric field E_y polarized in the y -direction. However, if the electrons are constrained to move along the spiral, they must also move in the x -direction as they are driven up and down. When a current is flowing up the spiral, it is also flowing into the paper at $z = z_1$ and out of the paper at $z = z_1 + A$, if A is the diameter of our molecular spiral. One might suppose that the current in the x -direction would produce no net radiation, since the currents are in opposite directions on opposite sides of the spiral. However, if we consider the x -components of the electric field arriving at $z = z_2$, we see that the field radiated by the current at $z = z_1 + A$ and the field radiated from $z = z_1$ arrive at z_2 separated in time by the amount A/c , and thus separated in phase by $\pi + \omega A/c$. Since the phase difference is not exactly π , the two fields do not cancel exactly, and we are left with a small x -component in the electric field generated by the motion of the electrons in the molecule, whereas the driving electric field had only a y -component. This small x -component, added to the large y -component, produces a resultant field that is tilted slightly with respect to the y -axis, the original direction of polarization. As the light moves through the material, the direction of polarization rotates about the beam axis. By drawing a few examples and considering the currents that will be set in motion by an incident

electric field, one can convince himself that the existence of optical activity and the sign of the rotation are independent of the orientation of the molecules.

Corn syrup is a common substance which possesses optical activity. The phenomenon is easily demonstrated with a polaroid sheet to produce a linearly polarized beam, a transmission cell containing corn syrup, and a second polaroid sheet to detect the rotation of the direction of polarization as the light passes through the corn syrup.

33-6 The intensity of reflected light

Let us now consider quantitatively the reflection coefficient as a function of angle. Figure 33-6(a) shows a beam of light striking a glass surface, where it is partly reflected and partly refracted into the glass. Let us suppose that the incident beam, of unit amplitude, is linearly polarized normal to the plane of the paper. We will call the amplitude of the reflected wave b , and the amplitude of the refracted wave a . The refracted and reflected waves will, of course, be linearly polarized, and the electric field vectors of the incident, reflected, and refracted waves are all parallel to each other. Figure 33-6(b) shows the same situation, but now we suppose that the incident wave, of unit amplitude, is polarized in the plane of the paper. Now let us call the amplitude of the reflected and refracted wave B and A , respectively.

We wish to calculate how strong the reflection is in the two situations illustrated in Fig. 33-6(a) and 33-6(b). We already know that when the angle between the reflected beam and refracted beam is a right angle, there will be no reflected wave in Fig. 33-6(b), but let us see if we cannot get a quantitative answer—an exact formula for B and b as a function of the angle of incidence, i .

The principle that we must understand is as follows. The currents that are generated in the glass produce two waves. First, they produce the reflected wave. Moreover, we know that if there were no currents generated in the glass, the incident wave would continue straight into the glass. Remember that all the sources in the world make the net field. The source of the incident light beam produces a field of unit amplitude which would move into the glass along the dotted line in the figure. This field is not observed, and therefore the currents generated in the glass must produce a field of amplitude -1 which moves along the dotted line. Using this fact, we will calculate the amplitude of the refracted waves, a and A .

In Fig. 33-6(a) we see that the field of amplitude b is radiated by the motion of charges inside the glass which are responding to a field a inside the glass, and that therefore b is proportional to a . We might suppose that since our two figures are exactly the same, except for the direction of polarization, the ratio B/A would be the same as the ratio b/a . This is not quite true, however, because in Fig. 33-6(b) the polarization directions are not all parallel to each other, as they are in Fig. 33-6(a). It is only the component of A which is perpendicular to B , $A \cos(i + r)$, which is effective in producing B . The correct expression for the proportionality is then

$$\frac{b}{a} = \frac{B}{A \cos(i + r)}. \quad (33.1)$$

Now we use a trick. We know that in both (a) and (b) of Fig. 33-6 the electric field in the glass must produce oscillations of the charges which generate a field of amplitude -1 , polarized parallel to the incident beam, and moving in the direction of the dotted line. But we see from part (b) of the figure that only the component of A that is normal to the dashed line has the right polarization to produce this field, whereas in Fig. 33-6(a) the full amplitude a is effective, since the polarization of wave a is parallel to the polarization of the wave of amplitude -1 . Therefore we can write

$$\frac{A \cos(i - r)}{a} = \frac{-1}{-1}, \quad (33.2)$$

since the two amplitudes on the left side of Eq. (33.2) each produce the wave of amplitude -1 .

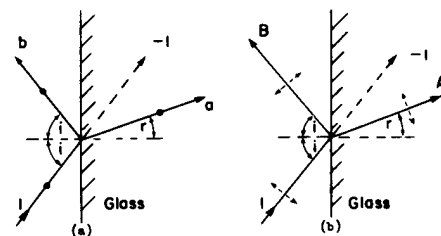


Fig. 33-6. An incident wave of unit amplitude is reflected and refracted at a glass surface. In (a) the incident wave is linearly polarized normal to the plane of the paper. In (b) the incident wave is linearly polarized in the direction shown by the dotted electric vector.

Dividing Eq. (33.1) by Eq. (33.2), we obtain

$$\frac{B}{b} = \frac{\cos(i+r)}{\cos(i-r)}, \quad (33.3)$$

a result which we can check against what we already know. If we set $(i+r) = 90^\circ$, Eq. (33.3) gives $B = 0$, as Brewster says it should be, so our results so far are at least not obviously wrong.

We have assumed unit amplitudes for the incident waves, so that $|B|^2/1^2$ is the reflection coefficient for waves polarized in the plane of incidence, and $|b|^2/1^2$ is the reflection coefficient for waves polarized normal to the plane of incidence. The ratio of these two reflection coefficients is determined by Eq. (33.3).

Now we perform a miracle, and compute not just the ratio, but each coefficient $|B|^2$ and $|b|^2$ individually! We know from the conservation of energy that the energy in the refracted wave must be equal to the incident energy minus the energy in the reflected wave, $1 - |B|^2$ in one case, $1 - |b|^2$ in the other. Furthermore, the energy which passes into the glass in Fig. 33-6(b) is to the energy which passes into the glass in Fig. 33-6(a) as the ratio of the squares of the refracted amplitudes, $|A|^2/|a|^2$. One might ask whether we really know how to compute the energy inside the glass, because, after all, there are energies of motion of the atoms in addition to the energy in the electric field. But it is obvious that all of the various contributions to the total energy will be proportional to the square of the amplitude of the electric field. Therefore we can write

$$\frac{1 - |B|^2}{1 - |b|^2} = \frac{|A|^2}{|a|^2}. \quad (33.4)$$

We now substitute Eq. (33.2) to eliminate A/a from the expression above, and express B in terms of b by means of Eq. (33.3):

$$\frac{1 - |b|^2 \frac{\cos^2(i+r)}{\cos^2(i-r)}}{1 - |b|^2} = \frac{1}{\cos^2(i-r)}. \quad (33.5)$$

This equation contains only one unknown amplitude, b . Solving for $|b|^2$, we obtain

$$|b|^2 = \frac{\sin^2(i-r)}{\sin^2(i+r)} \quad (33.6)$$

and, with the aid of (33.3),

$$|B|^2 = \frac{\tan^2(i-r)}{\tan^2(i+r)}. \quad (33.7)$$

So we have found the reflection coefficient $|b|^2$ for an incident wave polarized perpendicular to the plane of incidence, and also the reflection coefficient $|B|^2$ for an incident wave polarized in the plane of incidence!

It is possible to go on with arguments of this nature and deduce that b is real. To prove this, one must consider a case where light is coming from both sides of the glass surface at the same time, a situation not easy to arrange experimentally, but fun to analyze theoretically. If we analyze this general case, we can prove that b must be real, and therefore, in fact, that $b = \pm \sin(i-r)/\sin(i+r)$. It is even possible to determine the sign by considering the case of a very, very thin layer in which there is reflection from the front and from the back surfaces, and calculating how much light is reflected. We know how much light should be reflected by a thin layer, because we know how much current is generated, and we have even worked out the fields produced by such currents.

One can show by these arguments that

$$b = -\frac{\sin(i-r)}{\sin(i+r)}, \quad B = -\frac{\tan(i-r)}{\tan(i+r)}. \quad (33.8)$$

These expressions for the reflection coefficients as a function of the angles of incidence and refraction are called Fresnel's reflection formulas.

If we consider the limit as the angles i and r go to zero, we find, for the case of normal incidence, that $B^2 \approx b^2 \approx (i-r)^2/(i+r)^2$ for both polarizations,

since the sines are practically equal to the angles, as are also the tangents. But we know that $\sin i / \sin r = n$, and when the angles are small, $i/r \approx n$. It is thus easy to show that the coefficient of reflection for normal incidence is

$$B^2 = b^2 = \frac{(n - 1)^2}{(n + 1)^2}.$$

It is interesting to find out how much light is reflected at normal incidence from the surface of water, for example. For water, n is $4/3$, so that the reflection coefficient is $(1/7)^2 \approx 2\%$. At normal incidence, only two percent of the light is reflected from the surface of water.

33-7 Anomalous refraction

The last polarization effect we shall consider was actually one of the first to be discovered: anomalous refraction. Sailors visiting Iceland brought back to Europe crystals of Iceland spar (CaCO_3) which had the amusing property of making anything seen through the crystal appear doubled, i.e., as two images. This came to the attention of Huygens, and played an important role in the discovery of polarization. As is often the case, the phenomena which are discovered first are the hardest, ultimately, to explain. It is only after we understand a physical concept thoroughly that we can carefully select those phenomena which most clearly and simply demonstrate the concept.

Anomalous refraction is a particular case of the same birefringence that we considered earlier. Anomalous refraction comes about when the optic axis, the long axis of our asymmetric molecules, is *not* parallel to the surface of the crystal. In Fig. 33-7 are drawn two pieces of birefringent material, with the optic axis as shown. In the upper figure, the incident beam falling on the material is linearly polarized in a direction perpendicular to the optic axis of the material. When this beam strikes the surface of the material, each point on the surface acts as a source of a wave which travels into the crystal with velocity $v_{\perp}$, the velocity of light in the crystal when the plane of polarization is normal to the optic axis. The wave-front is just the envelope or locus of all these little spherical waves, and this wave-front moves straight through the crystal and out the other side. This is just the ordinary behavior we would expect, and this ray is called the *ordinary ray*.

In the lower figure the linearly polarized light falling on the crystal has its direction of polarization turned through 90° , so that the optic axis lies in the plane of polarization. When we now consider the little waves originating at any point on the surface of the crystal, we see that they do not spread out as spherical waves. Light travelling along the optic axis travels with velocity $v_{\perp}$ because the polarization is perpendicular to the optic axis, whereas the light travelling perpendicular to the optic axis travels with velocity $v_{\parallel}$ because the polarization is parallel to the optic axis. In a birefringent material $v_{\parallel} \neq v_{\perp}$, and in the figure $v_{\parallel} < v_{\perp}$. A more complete analysis will show that the waves spread out on the surface of an ellipsoid, with the optic axis as major axis of the ellipsoid. The envelope of all these elliptical waves is the wavefront which proceeds through the crystal in the direction shown. Again, at the back surface the beam will be deflected just as it was at the front surface, so that the light emerges parallel to the incident beam, but displaced from it. Clearly, this beam does not follow Snell's law, but goes in an extraordinary direction. It is therefore called the *extraordinary ray*.

When an unpolarized beam strikes an anomalously refracting crystal, it is separated into an ordinary ray, which travels straight through in the normal manner, and an extraordinary ray which is displaced as it passes through the crystal. These two emergent rays are linearly polarized at right angles to each other. That this is true can be readily demonstrated with a sheet of polaroid to analyze the polarization of the emergent rays. We can also demonstrate that our interpretation of this phenomenon is correct by sending linearly polarized light into the crystal. By properly orienting the direction of polarization of the incident beam, we can make this light go straight through without splitting, or we can make it go through without splitting but with a displacement.

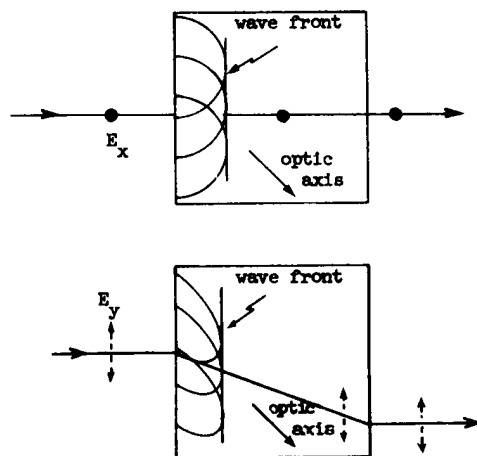


Fig. 33-7. The upper diagram shows the path of the ordinary ray through a doubly refracting crystal. The extraordinary ray is shown in the lower diagram. The optic axis lies in the plane of the paper.

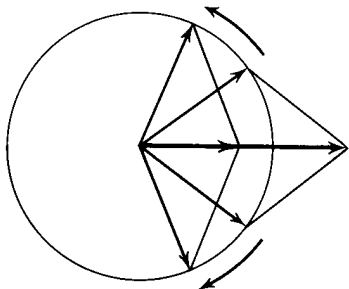


Fig. 33-8. Two oppositely rotating vectors of equal amplitude add to produce a vector in a fixed direction, but with an oscillating amplitude.

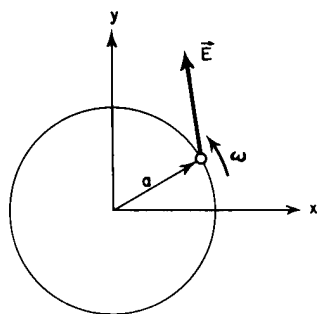


Fig. 33-9. A charge moving in a circle in response to circularly polarized light.

We have represented all the various polarization cases in Figs. 33-1 and 33-2 as superpositions of two special polarization cases, namely x and y in various amounts and phases. Other pairs could equally well have been used. Polarization along any two perpendicular axes x' , y' inclined to x and y would serve as well [for example, any polarization can be made up of superpositions of cases (a) and (e) of Fig. 33-2]. It is interesting, however, that this idea can be extended to other cases also. For example, any *linear* polarization can be made up by superposing suitable amounts at suitable phases of right and left *circular* polarizations [cases (c) and (g) of Fig. 33-2], since two equal vectors rotating in opposite directions add to give a single vector oscillating in a straight line (Fig. 33-8). If the phase of one is shifted relative to the other, the line is inclined. Thus all the pictures of Fig. 33-1 could be labeled “the superposition of equal amounts of right and left circularly polarized light at various relative phases.” As the left slips behind the right in phase, the direction of the linear polarization changes. Therefore optically active materials are, in a sense, birefringent. Their properties can be described by saying that they have different indexes for right- and left-hand circularly polarized light. Superposition of right and left circularly polarized light of different intensities produces elliptically polarized light.

Circularly polarized light has another interesting property—it carries *angular momentum* (about the direction of propagation). To illustrate this, suppose that such light falls on an atom represented by a harmonic oscillator that can be displaced equally well in any direction in the plane xy . Then the x -displacement of the electron will respond to the E_x component of the field, while the y -component responds, equally, to the equal E_y component of the field but 90° behind in phase. That is, the responding electron goes around in a circle, with angular velocity ω , in response to the rotating electric field of the light (Fig. 33-9). Depending on the damping characteristics of the response of the oscillator, the direction of the displacement $\mathbf{a}$ of the electron, and the direction of the force $q_e\mathbf{E}$ on it need not be the same but they rotate around together. The $\mathbf{E}$ may have a component at right angles to $\mathbf{a}$, so work is done on the system and a torque τ is exerted. The work done per second is $\tau\omega$. Over a period of time T the energy absorbed is $\tau\omega T$, while τT is the angular momentum delivered to the matter absorbing the energy. We see therefore that a beam of right circularly polarized light containing a total energy $\mathcal{E}$ carries an angular momentum (with vector directed along the direction of propagation) $\mathcal{E}/\omega$. For when this beam is absorbed that angular momentum is delivered to the absorber. Left-hand circular light carries angular momentum of the opposite sign, $-\mathcal{E}/\omega$.

Relativistic Effects in Radiation

34-1 Moving sources

In the present chapter we shall describe a number of miscellaneous effects in connection with radiation, and then we shall be finished with the classical theory of light propagation. In our analysis of light, we have gone rather far and into considerable detail. The only phenomena of any consequence associated with electromagnetic radiation that we have not discussed is what happens if radiowaves are contained in a box with reflecting walls, the size of the box being comparable to a wavelength, or are transmitted down a long tube. The phenomena of so-called *cavity resonators* and *waveguides* we shall discuss later; we shall first use another physical example—sound—and then we shall return to this subject. Except for this, the present chapter is our last consideration of the classical theory of light.

We can summarize all the effects that we shall now discuss by remarking that they have to do with the effects of *moving sources*. We no longer assume that the source is localized, with all its motion being at a relatively low speed near a fixed point.

We recall that the fundamental laws of electrodynamics say that, at large distances from a moving charge, the electric field is given by the formula

$$\mathbf{E} = -\frac{q}{4\pi\epsilon_0 c^2} \frac{d^2 \mathbf{e}_{R'}}{dt^2}. \quad (34.1)$$

The second derivative of the unit vector $\mathbf{e}_{R'}$ which points in the apparent direction of the charge, is the determining feature of the electric field. This unit vector does not point toward the *present* position of the charge, of course, but rather in the direction that the charge would seem to be, if the information travels only at the finite speed c from the charge to the observer.

Associated with the electric field is a magnetic field, always at right angles to the electric field and at right angles to the apparent direction of the source, given by the formula

$$\mathbf{B} = -\mathbf{e}_{R'} \times \mathbf{E}/c. \quad (34.2)$$

Until now we have considered only the case in which motions are nonrelativistic in speed, so that there is no appreciable motion in the direction of the source to be considered. Now we shall be more general and study the case where the motion is at an arbitrary velocity, and see what different effects may be expected in those circumstances. We shall let the motion be at an arbitrary speed, but of course we shall still assume that the detector is very far from the source.

We already know from our discussion in Chapter 28 that the only things that count in $d^2 \mathbf{e}_{R'}/dt^2$ are the changes in the *direction* of $\mathbf{e}_{R'}$. Let the coordinates of the charge be (x, y, z) , with z measured along the direction of observation (Fig. 34-1). At a given moment in time, say the moment τ , the three components of the position are $x(\tau)$, $y(\tau)$, and $z(\tau)$. The distance R is very nearly equal to $R(\tau) = R_0 + z(\tau)$. Now the direction of the vector $\mathbf{e}_{R'}$ depends mainly on x and y , but hardly at all upon z : the transverse components of the unit vector are x/R and y/R , and when we differentiate these components we get things like R^2 in the denominator:

$$\frac{d(x/R)}{dt} = \frac{dx/dt}{R} - \frac{dz}{dt} \frac{x}{R^2}.$$

34-1 Moving sources

34-2 Finding the “apparent” motion

34-3 Synchrotron radiation

34-4 Cosmic synchrotron radiation

34-5 Bremsstrahlung

34-6 The Doppler effect

34-7 The $\omega, \mathbf{k}$ four-vector

34-8 Aberration

34-9 The momentum of light

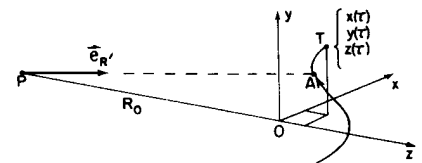


Fig. 34-1. The path of a moving charge. The true position at the time τ is at T , but the retarded position is at A .

So, when we are far enough away the only terms we have to worry about are the variations of x and y . Thus we take out the factor R_0 and get

$$\begin{aligned} E_x &= -\frac{q}{4\pi\epsilon_0 c^2 R_0} \frac{d^2 x'}{dt^2}, \\ E_y &= -\frac{q}{4\pi\epsilon_0 c^2 R_0} \frac{d^2 y'}{dt^2}, \end{aligned} \quad (34.3)$$

where R_0 is the distance, more or less, to q ; let us take it as the distance OP to the origin of the coordinates (x, y, z) . Thus the electric field is a constant multiplied by a very simple thing, the second derivatives of the x - and y -coordinates. (We could put it more mathematically by calling x and y the *transverse* components of the position vector $\mathbf{r}$ of the charge, but this would not add to the clarity.)

Of course, we realize that the coordinates must be measured at the retarded time. Here we find that $z(\tau)$ *does* affect the retardation. What time is the retarded time? If the time of observation is called t (the time at P) then the time τ to which this corresponds at A is not the time t , but is delayed by the total distance that the light has to go, divided by the speed of light. In the first approximation, this delay is R_0/c , a constant (an uninteresting feature), but in the next approximation we must include the effects of the position in the z -direction at the time τ , because if q is a little farther back, there is a little more retardation. This is an effect that we have neglected before, and it is the only change needed in order to make our results valid for all speeds.

What we must now do is to choose a certain value of t and calculate the value of τ from it, and thus find out where x and y are at that τ . These are then the retarded x and y , which we call x' and y' , whose second derivatives determine the field. Thus τ is determined by

$$t = \tau + \frac{R_0}{c} + \frac{z(\tau)}{c}$$

and

$$x'(t) = x(\tau), \quad y'(t) = y(\tau). \quad (34.4)$$

Now these are complicated equations, but it is easy enough to make a geometrical picture to describe their solution. This picture will give us a good qualitative feeling for how things work, but it still takes a lot of detailed mathematics to deduce the precise results of a complicated problem.

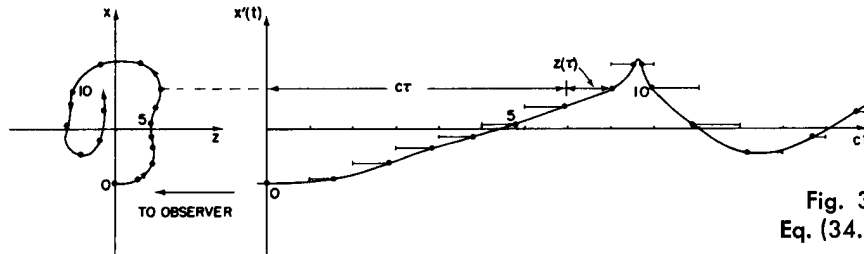


Fig. 34-2. A geometrical solution of Eq. (34.5) to find $x'(t)$.

34-2 Finding the “apparent” motion

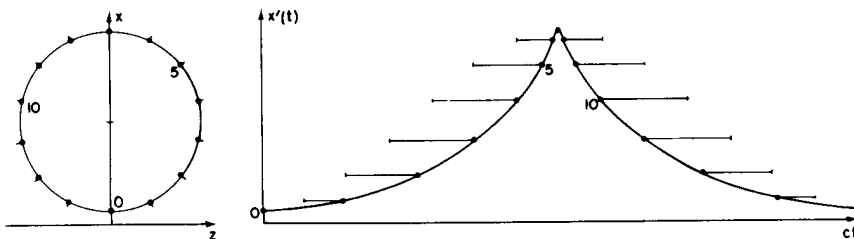
The above equation has an interesting simplification. If we disregard the uninteresting constant delay R_0/c , which just means that we must change the origin of t by a constant, then it says that

$$ct = c\tau + z(\tau), \quad x' = x(\tau), \quad y' = y(\tau). \quad (34.5)$$

Now we need to find x' and y' as functions of t , not τ , and we can do this in the following way: Eq. (34.5) says that we should take the actual motion and add a constant (the speed of light) times τ . What that turns out to mean is shown in Fig. 34-2. We take the actual motion of the charge (shown at left) and imagine that as it is going around it is being swept away from the point P at the speed c (there are no contractions from relativity or anything like that; this is just a mathematical addition of the $c\tau$). In this way we get a new motion, in which the line-

of-sight coordinate is ct , as shown at the right. (The figure shows the result for a rather complicated motion in a plane, but of course the motion may not be in one plane—it may be even more complicated than motion in a plane.) The point is that the horizontal (i.e., line-of-sight) distance now is no longer the old z , but is $z + c\tau$, and therefore is ct . Thus we have found a picture of the curve, x' (and y') against t ! All we have to do to find the field is to look at the acceleration of this curve, i.e., to differentiate it twice. So the final answer is: in order to find the electric field for a moving charge, take the motion of the charge and translate it back at the speed c to “open it out”; then the curve, so drawn, is a curve of the x' and y' positions of the function of t . The acceleration of this curve gives the electric field as a function of t . Or, if we wish, we can now imagine that this whole “rigid” curve moves forward at the speed c through the plane of sight, so that the point of intersection with the plane of sight has the coordinates x' and y' . The acceleration of this point makes the electric field. This solution is just as exact as the formula we started with—it is simply a geometrical representation.

Fig. 34-3. The $x'(t)$ curve for a particle moving at constant speed $v = 0.94c$, a circle.



If the motion is relatively slow, for instance if we have an oscillator just going up and down slowly, then when we shoot that motion away at the speed of light, we would get, of course, a simple cosine curve, and that gives a formula we have been looking at for a long time: it gives the field produced by an oscillating charge. A more interesting example is an electron moving rapidly, very nearly at the speed of light, in a circle. If we look in the plane of the circle, the retarded $x'(t)$ appears as shown in Fig. 34-3. What is this curve? If we imagine a radius vector from the center of the circle to the charge, and if we extend this radial line a little bit past the charge, just a shade if it is going fast, then we come to a point on the line that goes at the speed of light. Therefore, when we translate the motion back at the speed of light, that corresponds to having a wheel with a charge on it rolling backward (without slipping) at the speed c ; thus we find a curve which is very close to a cycloid—it is called a *hypocyloid*. If the charge is going very nearly at the speed of light, the “cusps” are very sharp indeed; if it went at exactly the speed of light, they would be actual cusps, infinitely sharp. “Infinitely sharp” is interesting; it means that near a cusp the second derivative is enormous. Once in each cycle we get a sharp pulse of electric field. This is not at all what we would get from a nonrelativistic motion, where each time the charge goes around there is an oscillation which is of about the same “strength” all the time. Instead, there are very sharp pulses of electric field spaced at time intervals $1/T_0$ apart, where T_0 is the period of revolution. These strong electric fields are emitted in a narrow cone in the direction of motion of the charge. When the charge is moving away from P , there is very little curvature and there is very little radiated field in the direction of P .

34-3 Synchrotron radiation

We have very fast electrons moving in circular paths in the synchrotron; they are travelling at very nearly the speed c , and it is possible to see the above radiation as actual *light*! Let us discuss this in more detail.

In the synchrotron we have electrons which go around in circles in a uniform magnetic field. First, let us see why they go in circles. From Eq. (12.10), we know that the force on a particle in a magnetic field is given by

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}, \quad (34.6)$$

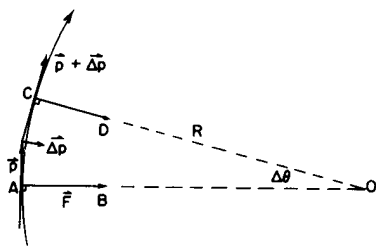


Fig. 34-4. A charged particle moves in a circular (or helical) path in a uniform magnetic field.

and it is at right angles both to the field and to the velocity. As usual, the force is equal to the rate of change of momentum with time. If the field is directed upward out of the paper, the momentum of the particle and the force on it are as shown in Fig. 34-4. Since the force is at right angles to the velocity, the kinetic energy, and therefore the speed, remains *constant*. All the magnetic field does is to change the *direction of motion*. In a short time Δt , the momentum vector changes at right angles to itself by an amount $\Delta \mathbf{p} = \mathbf{F} \Delta t$, and therefore $\mathbf{p}$ turns through an angle $\Delta \theta = \Delta p/p = qvB \Delta t/p$, since $|F| = qvB$. But in this same time the particle has gone a distance $\Delta s = v \Delta t$. Evidently, the two lines AB and CD will intersect at a point O such that $OA = OC = R$, where $\Delta s = R \Delta \theta$. Combining this with the previous expressions, we find $R \Delta \theta / \Delta t = R \omega = v = qvBR/p$, from which we find

$$p = qBR \quad (34.7)$$

and

$$\omega = qvB/p. \quad (34.8)$$

Since this same argument can be applied during the next instant, the next, and so on, we conclude that the particle must be moving in a *circle* of radius R , with angular velocity ω .

The result that the momentum of the particle is equal to a charge times the radius times the magnetic field is a very important law that is used a great deal. It is important for practical purposes because if we have elementary particles which all have the same charge and we observe them in a magnetic field, we can measure the radii of curvature of their orbits and, knowing the magnetic field, thus determine the momenta of the particles. If we multiply both sides of Eq. (34.7) by c , and express q in terms of the electronic charge, we can measure the momentum in units of the *electron volt*. In those units our formula is

$$pc(\text{ev}) = 3 \times 10^8 (q/q_e) BR, \quad (34.9)$$

where B , R , and the speed of light are all expressed in the mks system, the latter being 3×10^8 , numerically.

The mks unit of magnetic field is called a *weber per square meter*. There is an older unit which is still in common use, called a *gauss*. One weber/m^2 is equal to 10^4 gauss. To give an idea of how big magnetic fields are, the strongest magnetic field that one can usually make in iron is about 1.5×10^4 gauss; beyond that, the advantage of using iron disappears. Today, electromagnets wound with superconducting wire are able to produce steady fields of over 10^5 gauss strength—that is, 10 mks units. The field of the earth is a few tenths of a gauss at the equator.

Returning to Eq. (34.9), we could imagine the synchrotron running at a billion electron volts, so pc would be 10^9 for a billion electron volts. (We shall come back to the energy in just a moment.) Then, if we had a B corresponding to, say, 10,000 gauss, which is a good substantial field, one mks unit, then we see that R would have to be 3.3 meters. The actual radius of the Caltech synchrotron is 3.7 meters, the field is a little bigger, and the energy is 1.5 billion, but it is the same idea. So now we have a feeling for why the synchrotron has the size it has.

We have calculated the momentum, but we know that the total energy, including the rest energy, is given by $W = \sqrt{p^2 c^2 + m^2 c^4}$, and for an electron the rest energy corresponding to mc^2 is 0.511×10^6 ev, so when pc is 10^9 ev we can neglect mc^2 , and so for all practical purposes $W = pc$ when the speeds are relativistic. It is practically the same to say the energy of an electron is a billion electron volts as to say the momentum times c is a billion electron volts. If $W = 10^9$ ev, it is easy to show that the speed differs from the speed of light by but one part in eight million!

We turn now to the radiation emitted by such a particle. A particle moving on a circle of radius 3.3 meters, or 20 meters circumference, goes around once in roughly the time it takes light to go 20 meters. So the wavelength that should be emitted by such a particle would be 20 meters—in the shortwave radio region. But because of the piling up effect that we have been discussing (Fig. 34-3), and because the distance by which we must extend the radius to reach the speed c is

only one part in eight million of the radius, the cusps of the hypocycloid are enormously sharp compared with the distance between them. The acceleration, which involves a second derivative with respect to time, gets twice the “compression factor” of 8×10^6 because the time scale is reduced by eight million twice in the neighborhood of the cusp. Thus we might expect the effective wavelength to be much shorter, to the extent of 64 times 10^{12} smaller than 20 meters, and that corresponds to the x-ray region. (Actually, the cusp itself is not the entire determining factor; one must also include a certain region about the cusp. This changes the factor to the $3/2$ power instead of the square, but still leaves us above the optical region.) Thus, even though a slowly moving electron would have radiated 20-meter radiowaves, the relativistic effect cuts down the wavelength so much that we can see it! Clearly, the light should be *polarized*, with the electric field perpendicular to the uniform magnetic field.

To further appreciate what we would observe, suppose that we were to take such light (to simplify things, because these pulses are so far apart in time, we shall just take one pulse) and direct it onto a diffraction grating, which is a lot of scattering wires. After this pulse comes away from the grating, what do we see? (We should see red light, blue light, and so on, if we see any light at all.) What *do* we see? The pulse strikes the grating head-on, and all the oscillators in the grating, together, are violently moved up and then back down again, just once. They then produce effects in various directions, as shown in Fig. 34-5. But the point *P* is closer to one end of the grating than to the other, so at this point the electric field arrives first from wire *A*, next from *B*, and so on; finally, the pulse from the last wire arrives. In short, the sum of the reflections from all the successive wires is as shown in Fig. 34-6(a); it is an electric field which is a series of pulses, and it is very like a sine wave whose wavelength is the distance between the pulses, just as it would be for monochromatic light striking the grating! So, we get colored light all right. But, by the same argument, will we not get light from any kind of a “pulse”? No. Suppose that the curve were much smoother; then we would add all the scattered waves together, separated by a small time between them (Fig. 34-6b). Then we see that the field would not shake at all, it would be a very smooth curve, because each pulse does not vary much in the time interval between pulses.

The electromagnetic radiation emitted by relativistic charged particles circulating in a magnetic field is called *synchrotron radiation*. It is so named for obvious reasons, but it is not limited specifically to synchrotrons, or even to earthbound laboratories. It is exciting and interesting that it also occurs in nature!

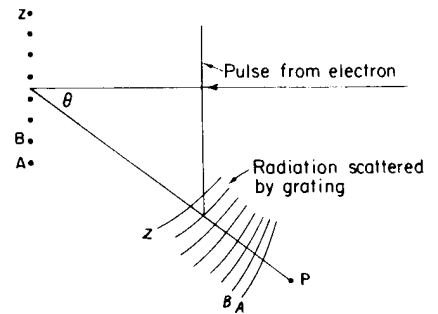


Fig. 34-5. The light which strikes a grating as a single, sharp pulse is scattered in various directions as different colors.

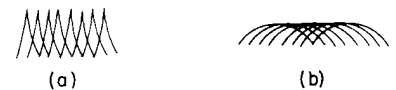


Fig. 34-6. The total electric field due to a series of (a) sharp pulses and (b) smooth pulses.



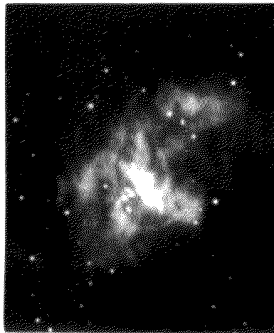
Fig. 34-7. The crab nebula as seen in all colors (no filter).

34-4 Cosmic synchrotron radiation

In the year 1054 the Chinese and Japanese civilizations were among the most advanced in the world; they were conscious of the external universe, and they recorded, most remarkably, an explosive bright star in that year. (It is amazing that none of the European monks, writing all the books of the middle ages, even bothered to write that a star exploded in the sky, but they did not.) Today we may take a picture of that star, and what we see is shown in Fig. 34-7. On the outside is a big mass of red filaments, which is produced by the atoms of the thin gas “ringing” at their natural frequencies; this makes a bright line spectrum with different frequencies in it. The red happens in this case to be due to nitrogen. On the other hand, in the central region is a mysterious, fuzzy patch of light in a *continuous* distribution of frequency, i.e., there are no special frequencies associated with particular atoms. Yet this is not dust “lit up” by nearby stars, which is one way by which one can get a continuous spectrum. We can see stars through it, so it is transparent, but it is *emitting* light.



(a)



(b)

Fig. 34-8. The crab nebula as seen through a blue filter and a polaroid. (a) Electric vector vertical. (b) Electric vector horizontal.

In Fig. 34-8 we look at the same object, using light in a region of the spectrum which has no bright spectral line, so that we see only the central region. But in this case, also, polarizers have been put on the telescope, and the two views correspond to two orientations 90° apart. We see that the pictures are different! That is to say, the light is polarized. The reason, presumably, is that there is a local magnetic field, and many very energetic electrons are going around in that magnetic field.

We have just illustrated how the electrons could go around the field in a circle. We can add to this, of course, any uniform motion in the direction of the field, since the force, $q\mathbf{v} \times \mathbf{B}$, has no component in this direction and, as we have already remarked, the synchrotron radiation is evidently polarized in a direction at right angles to the projection of the magnetic field onto the plane of sight.

Putting these two facts together, we see that in a region where one picture is bright and the other one is black, the light must have its electric field completely polarized in one direction. This means that there is a magnetic field at right angles to this direction, while in other regions, where there is a strong emission in the other picture, the magnetic field must be the other way. If we look carefully at Fig. 34-8, we may notice that there is, roughly speaking, a general set of “lines” that go one way in one picture and at right angles to this in the other. The pictures show a kind of fibrous structure. Presumably, the magnetic field lines will tend to extend relatively long distances in their own direction, and so, presumably, there are long regions of magnetic field with all the electrons spiralling one way, while in another region the field is the other way and the electrons are also spiralling that way.

What keeps the electron energy so high for so long a time? After all, it is 900 years since the explosion—how can they keep going so fast? How they maintain their energy and how this whole thing keeps going is still not thoroughly understood.

34-5 Bremsstrahlung

We shall next remark briefly on one other interesting effect of a very fast-moving particle that radiates energy. The idea is very similar to the one we have just discussed. Suppose that there are charged particles in a piece of matter and a very fast electron, say, comes by (Fig. 34-9). Then, because of the electric field around the atomic nucleus the electron is pulled, accelerated, so that the curve of its motion has a slight kink or bend in it. If the electron is travelling at very nearly the speed of light, what is the electric field produced in the direction c ? Remember our rule: we take the actual motion, translate it backwards at speed c , and that gives us a curve whose curvature measures the electric field. It was coming toward us at the speed v , so we get a backward motion, with the whole picture compressed into a smaller distance in proportion as $c - v$ is smaller than c . So, if $1 - v/c \ll 1$, there is a very sharp and rapid curvature at B' , and when we take the second derivative of that we get a very high field in the direction of the motion. So when very energetic electrons move through matter they spit radiation in a forward direction. This is called *bremsstrahlung*. As a matter of fact, the synchrotron

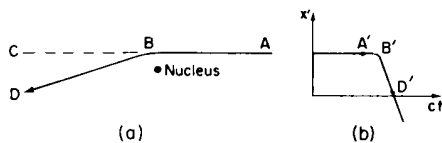


Fig. 34-9. A fast electron passing near a nucleus radiates energy in the direction of its motion.

is used, not so much to make high-energy electrons (actually if we could get them out of the machine more conveniently we would not say this) as to make very energetic photons—gamma rays—by passing the energetic electrons through a solid tungsten “target,” and letting them radiate photons from this bremsstrahlung effect.

34-6 The Doppler effect

Now we go on to consider some other examples of the effects of moving sources. Let us suppose that the source is a stationary atom which is oscillating at one of its natural frequencies, ω_0 . Then we know that the frequency of the light we would observe is ω_0 . But now let us take another example, in which we have a similar oscillator oscillating with a frequency ω_1 , and at the same time the whole atom, the whole oscillator, is moving along in a direction toward the observer at velocity v . Then the actual motion in space, of course, is as shown in Fig. 34-10(a). Now we play our usual game, we add $c\tau$; that is to say, we translate the whole curve backward and we find then that it oscillates as in Fig. 34-10(b). In a given amount of time τ , when the oscillator would have gone a distance $v\tau$, on the x' vs. ct diagram it goes a distance $(c - v)\tau$. So all the oscillations of frequency ω_1 in the time $\Delta\tau$ are now found in the interval $\Delta\tau = (1 - v/c)\Delta\tau$; they are squashed together, and as this curve comes by us at speed c , we will see light of a *higher frequency*, higher by just the compression factor $(1 - v/c)$. Thus we observe

$$\omega = \frac{\omega_1}{1 - v/c}. \quad (34.10)$$

We can, of course, analyze this situation in various other ways. Suppose that the atom were emitting, instead of sine waves, a series of pulses, pip, pip, pip, pip, at a certain frequency ω_1 . At what frequency would they be received by us? The first one that arrives has a certain delay, but the next one is delayed less because in the meantime the atom moves closer to the receiver. Therefore, the time between the “pips” is decreased by the motion. If we analyze the geometry of the situation, we find that the frequency of the pips is increased by the factor $1/(1 - v/c)$.

Is $\omega = \omega_0/(1 - v/c)$, then, the frequency that would be observed if we took an ordinary atom, which had a natural frequency ω_0 , and moved it toward the receiver at speed v ? No; as we well know, the natural frequency ω_1 of a moving atom is not the same as that measured when it is standing still, because of the relativistic dilation in the rate of passage of time. Thus if ω_0 were the true natural frequency, then the modified natural frequency ω_1 would be

$$\omega_1 = \omega_0 \sqrt{1 - v^2/c^2}. \quad (34.11)$$

Therefore the observed frequency ω is

$$\omega = \frac{\omega_0 \sqrt{1 - v^2/c^2}}{1 - v/c}. \quad (34.12)$$

The shift in frequency observed in the above situation is called the *Doppler effect*: if something moves toward us the light it emits appears more violet, and if it moves away it appears more red.

We shall now give two more derivations of this same interesting and important result. Suppose, now, that the *source* is standing still and is emitting waves at frequency ω_0 , while the *observer* is moving with speed v toward the source. After a certain period of time t the observer will have moved to a new position, a distance vt from where he was at $t = 0$. How many radians of phase will he have seen go by? A certain number, $\omega_0 t$, went past any fixed point, and in addition the observer has swept past some more by his own motion, namely a number $vt k_0$ (the number of radians per meter times the distance). So the total number of radians in the time t , or the observed frequency, would be $\omega_1 = \omega_0 + k_0 v$. We have made this analysis from the point of view of a man at rest; we would like to know how it would look to the man who is moving. Here we have to worry again about the difference in clock rate for the two observers, and this time that means that we have to *divide* by $\sqrt{1 - v^2/c^2}$. So if k_0 is the wave number, the number of radians

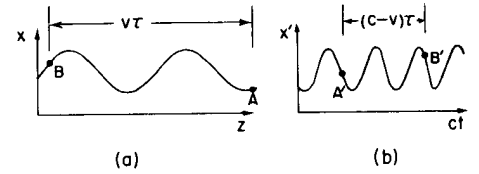


Fig. 34-10. The $x - z$ and $x' - t$ curves of a moving oscillator.

per meter in the direction of motion, and ω_0 is the frequency, then the observed frequency for a moving man is

$$\omega = \frac{\omega_0 + k_0 v}{\sqrt{1 - v^2/c^2}}. \quad (34.13)$$

For the case of light, we know that $k_0 = \omega_0/c$. So, in this particular problem, the equation would read

$$\omega = \frac{\omega_0(1 + v/c)}{\sqrt{1 - v^2/c^2}}, \quad (34.14)$$

which looks completely unlike formula (34.12)! Is the frequency that we would observe if we move toward a source different than the frequency that we would see if the source moved toward us? Of course not! The theory of relativity says that these two must be *exactly equal*. If we were expert enough mathematicians we would probably recognize that these two mathematical expressions *are* exactly equal! In fact, the *necessary* equality of the two expressions is one of the ways by which some people like to demonstrate that relativity requires a time dilation, because if we did not put those square-root factors in, they would no longer be equal.

Since we know about relativity, let us analyze it in still a third way, which may appear a little more general. (It is really the same thing, since it makes no difference *how* we do it!) According to the relativity theory there is a relationship between position and time as observed by one man and position and time as seen by another who is moving relative to him. We wrote down those relationships long ago (Chapter 16). This is the *Lorentz transformation* and its inverse:

$$\begin{aligned} x' &= \frac{x + vt}{\sqrt{1 - v^2/c^2}}, & x &= \frac{x' - vt'}{\sqrt{1 - v^2/c^2}}, \\ t' &= \frac{t + vx/c^2}{\sqrt{1 - v^2/c^2}}, & t &= \frac{t' - vx'/c^2}{\sqrt{1 - v^2/c^2}}. \end{aligned} \quad (34.15)$$

If we were standing still on the ground, the form of a wave would be $\cos(\omega t - kx)$; all the nodes and maxima and minima would follow this form. But what would a man in motion, observing the same physical wave, see? Where the field is zero, the positions of all the nodes are the same (when the field is *zero*, *everyone* measures the field as zero); that is a relativistic invariant. So the form is the same for the other man too, except that we must transform it into his frame of reference:

$$\cos(\omega t - kx) = \cos \left[\omega \frac{t' - vx'/c^2}{\sqrt{1 - v^2/c^2}} - k \frac{x' - vt'}{\sqrt{1 - v^2/c^2}} \right].$$

If we regroup the terms inside the brackets, we get

$$\begin{aligned} \cos(\omega t - kx) &= \cos \left[\underbrace{\frac{\omega + kv}{\sqrt{1 - v^2/c^2}}}_{\omega'} t' - \underbrace{\frac{k + v\omega/c^2}{\sqrt{1 - v^2/c^2}}}_{k'} x' \right] \\ &= \cos \left[\omega' t' - k' x' \right]. \end{aligned} \quad (34.16)$$

This is again a wave, a cosine wave, in which there is a certain frequency ω' , a constant multiplying t' , and some other constant, k' , multiplying x' . We call k' the wave number, or the number of waves per meter, for the other man. Therefore the other man will see a new frequency and a new wave number given by

$$\omega' = \frac{\omega + kv}{\sqrt{1 - v^2/c^2}}, \quad (34.17)$$

$$k' = \frac{k + v\omega/c^2}{\sqrt{1 - v^2/c^2}}. \quad (34.18)$$

If we look at (34.17), we see that it is the same formula (34.13), that we obtained by a more physical argument.

34-7 The ω, k four-vector

The relationships indicated in Eqs. (34.17) and (34.18) are very interesting, because these say that the new frequency ω' is a combination of the old frequency ω and the old wave number k , and that the new wave number is a combination of the old wave number and frequency. Now the wave number is the rate of change of phase with distance, and the frequency is the rate of change of phase with time, and in these expressions we see a close analogy with the Lorentz transformation of the position and time: if ω is thought of as being like t , and k is thought of as being like x divided by c^2 , then the new ω' will be like t' , and the new k' will be like x'/c^2 . That is to say, *under the Lorentz transformation ω and k transform the same way as do t and x* . They constitute what we call a *four-vector*; when a quantity has four components transforming like time and space, it is a four-vector. Everything seems all right, then, except for one little thing: we said that a four-vector has to have *four components*; where are the other two components? We have seen that ω and k are like time and space in one space direction, but not in all directions, and so we must next study the problem of the propagation of light in three space dimensions, not just in one direction, as we have been doing up until now.

Suppose that we have a coordinate system, x, y, z , and a wave which is travelling along and whose wavefronts are as shown in Fig. 34-11. The wavelength of the wave is λ , but the direction of motion of the wave does not happen to be in the direction of one of the axes. What is the formula for such a wave? The answer is clearly $\cos(\omega t - ks)$, where $k = 2\pi/\lambda$ and s is the distance along the direction of motion of the wave—the component of the spatial position in the direction of motion. Let us put it this way: if $\mathbf{r}$ is the vector position of a point in space, then s is $\mathbf{r} \cdot \mathbf{e}_k$, where $\mathbf{e}_k$ is a unit vector in the direction of motion. That is, s is just $r \cos(\mathbf{r}, \mathbf{e}_k)$, the component of distance in the direction of motion. Therefore our wave is $\cos(\omega t - k\mathbf{e}_k \cdot \mathbf{r})$.

Now it turns out to be very convenient to define a vector $\mathbf{k}$, which is called the *wave vector*, which has a magnitude equal to the wave number, $2\pi/\lambda$, and is pointed in the direction of propagation of the waves:

$$\mathbf{k} = 2\pi\mathbf{e}_k/\lambda = k\mathbf{e}_k. \quad (34.19)$$

Using this vector, our wave can be written as $\cos(\omega t - \mathbf{k} \cdot \mathbf{r})$, or as $\cos(\omega t - k_x x - k_y y - k_z z)$. What is the significance of a component of k , say k_x ? Clearly, k_x is the rate of change of phase with respect to x . Referring to Fig. 34-11, we see that the phase changes as we change x , just as if there were a wave along x , *but of a longer wavelength*. The “wavelength in the x -direction” is longer than a natural, true wavelength by the secant of the angle α between the actual direction of propagation and the x -axis:

$$\lambda_x = \lambda / \cos \alpha. \quad (34.20)$$

Therefore the rate of change of phase, which is proportional to the *reciprocal* of λ_x , is *smaller* by the factor $\cos \alpha$; that is just how k_x would vary—it would be the magnitude of $\mathbf{k}$, times the cosine of the angle between $\mathbf{k}$ and the x -axis!

That, then, is the nature of the wave vector that we use to represent a wave in three dimensions. The four quantities ω, k_x, k_y, k_z transform in relativity as a four-vector, where ω corresponds to the time, and k_x, k_y, k_z correspond to the x -, y -, and z -components of the four-vector.

In our previous discussion of special relativity (Chapter 17), we learned that there are ways of making relativistic dot products with four-vectors. If we use the position vector x_μ , where μ stands for the four components (time and three space ones), and if we call the wave vector k_μ , where the index μ again has four values, time and three space ones, then the dot product of x_μ and k_μ is written $\Sigma' k_\mu x_\mu$ (see Chapter 17). This dot product is an invariant, independent of the coordinate system; what is it equal to? By the definition of this dot product in four dimensions, it is

$$\Sigma' k_\mu x_\mu = \omega t - k_x x - k_y y - k_z z. \quad (34.21)$$

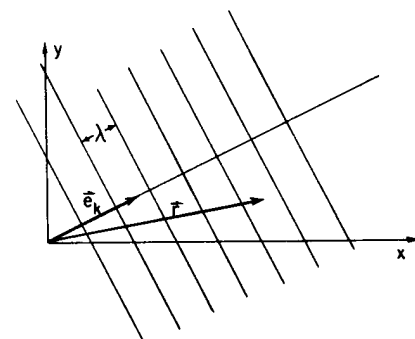


Fig. 34-11. A plane wave travelling in an oblique direction.

We know from our study of vectors that $\Sigma' k_\mu x_\mu$ is invariant under the Lorentz transformation, since k_μ is a four-vector. But this quantity is precisely what appears inside the cosine for a plane wave, and it *ought* to be invariant under a Lorentz transformation. We cannot have a formula with something that changes inside the cosine, since we know that the phase of the wave cannot change when we change the coordinate system.

34-8 Aberration

In deriving Eqs. (34.17) and (34.18), we have taken a simple example where $\mathbf{k}$ happened to be in a direction of motion, but of course we can generalize it to other cases also. For example, suppose there is a source sending out light in a certain direction from the point of view of a man at rest, but we are moving along on the earth, say (Fig. 34-12). From which direction does the light appear to come? To find out, we will have to write down the four components of k_μ and apply the Lorentz transformation. The answer, however, can be found by the following argument: we have to point our telescope at an angle to see the light. Why? Because light is coming down at the speed c , and we are moving sidewise at the speed v , so the telescope has to be tilted forward so that as the light comes down it goes "straight" down the tube. It is very easy to see that the horizontal distance is vt when the vertical distance is ct , and therefore, if θ' is the angle of tilt, $\tan \theta' = v/c$. How nice! How nice, indeed—except for one little thing: θ' is *not* the angle at which we would have to set the telescope *relative to the earth*, because we made our analysis from the point of view of a "fixed" observer. When we said the horizontal distance is vt , the man on the earth would have found a different distance, since he measured with a "squashed" ruler. It turns out that, because of that contraction effect,

$$\tan \theta = \frac{v/c}{\sqrt{1 - v^2/c^2}}, \quad (34.22)$$

which is equivalent to

$$\sin \theta = v/c. \quad (34.23)$$

It will be instructive for the student to derive this result, using the Lorentz transformation.

This effect, that a telescope has to be tilted, is called *aberration*, and it has been observed. *How* can we observe it? Who can say where a given star *should* be? Suppose we *do* have to look in the wrong direction to see a star; how do we know it is the wrong direction? Because the earth goes around the sun. Today we have to point the telescope one way; six months later we have to tilt the telescope the other way. That is how we can tell that there is such an effect.

34-9 The momentum of light

Now we turn to a different topic. We have never, in all our discussion of the past few chapters, said anything about the effects of the *magnetic* field that is associated with light. Ordinarily, the effects of the magnetic field are very small, but there is one interesting and important effect which is a consequence of the magnetic field. Suppose that light is coming from a source and is acting on a charge and driving that charge up and down. We will suppose that the electric field is in the x -direction, so the motion of the charge is also in the x -direction: it has a position x and a velocity v , as shown in Fig. 34-13. The magnetic field is at right angles to the electric field. Now as the electric field acts on the charge and moves it up and down, what does the magnetic field do? The magnetic field acts on the charge (say an electron) only when it is moving; but the electron *is* moving, it is driven by the electric field, so the two of them work together: While the thing is going up and down it has a velocity and there is a force on it, B times v times q ; but in which *direction* is this force? *It is in the direction of the propagation of light.* Therefore, when light is shining on a charge and it is oscillating in response to that

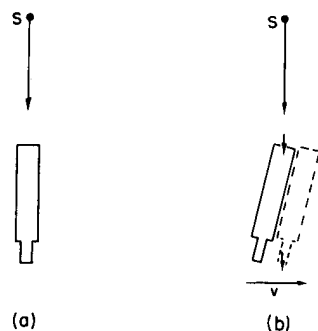


Fig. 34-12. A distant source S is viewed by (a) a stationary telescope, and (b) a laterally moving telescope.

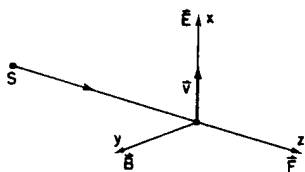


Fig. 34-13. The magnetic force on a charge which is driven by the electric field is in the direction of the light beam.

charge, there is a driving force in the direction of the light beam. This is called *radiation pressure* or light pressure.

Let us determine how strong the radiation pressure is. Evidently it is $F = qvB$ or, since everything is oscillating, it is the *time average* of this, $\langle F \rangle$. From (34.2) the strength of the magnetic field is the same as the strength of the electric field divided by c , so we need to find the average of the electric field, times the velocity, times the charge, times $1/c$: $\langle F \rangle = q\langle vE \rangle/c$. But the charge q times the field E is the electric force on a charge, and the force on the charge times the velocity is the work dW/dt being done on the charge! Therefore the force, the “pushing momentum,” that is delivered per second by the light, is equal to $1/c$ times the *energy absorbed* from the light per second! That is a general rule, since we did not say how strong the oscillator was, or whether some of the charges cancel out. *In any circumstance where light is being absorbed, there is a pressure.* The momentum that the light delivers is always equal to the energy that is absorbed, divided by c :

$$\langle F \rangle = \frac{dW/dt}{c}. \quad (34.24)$$

That light carries energy we already know. We now understand that it also carries *momentum*, and further, that the momentum carried is always $1/c$ times the energy.

When light is emitted from a source there is a recoil effect: the same thing in reverse. If an atom is emitting an energy W in some direction, then there is a recoil momentum $p = W/c$. If light is *reflected* normally from a mirror, we get twice the force.

That is as far as we shall go using the classical theory of light. Of course we know that there is a quantum theory, and that in many respects light acts like a particle. The energy of a light-particle is a constant times the frequency:

$$W = h\nu = \hbar\omega \quad (34.25)$$

We now appreciate that light also carries a momentum equal to the energy divided by c , so it is also true that these effective particles, these *photons*, carry a momentum

$$p = W/c = \hbar\omega/c = \hbar k. \quad (34.26)$$

The *direction* of the momentum is, of course, the direction of propagation of the light. So, to put it in vector form,

$$W = \hbar\omega, \quad \mathbf{p} = \hbar\mathbf{k}. \quad (34.27)$$

We also know, of course, that the energy and momentum of a particle should form a four-vector. We have just discovered that ω and $\mathbf{k}$ form a four-vector. Therefore it is a good thing that (34.27) has the same constant in both cases; it means that the quantum theory and the theory of relativity are mutually consistent.

Equation (34.27) can be written more elegantly as $p_\mu = \hbar k_\mu$, a relativistic equation, for a particle associated with a wave. Although we have discussed this only for photons, for which k (the magnitude of $\mathbf{k}$) equals ω/c and $p = W/c$, the relation is much more general. In quantum mechanics *all* particles, not only photons, exhibit wavelike properties, but the frequency and wave number of the waves is related to the energy and momentum of particles by (34.27) (called the deBroglie relations) even when p is not equal to W/c .

In the last chapter we saw that a beam of right or left circularly polarized light also carries *angular momentum* in an amount proportional to the energy $\mathcal{E}$ of the wave. In the quantum picture, a beam of circularly polarized light is regarded as a stream of photons, each carrying an angular momentum $\pm\hbar$ along the direction of propagation. That is what becomes of polarization in the corpuscular point of view—the photons carry angular momentum like spinning rifle bullets. But this “bullet” picture is really as incomplete as the “wave” picture, and we shall have to discuss these ideas more fully in a later chapter on Quantum Behavior.

Color Vision

35-1 The human eye

The phenomenon of colors depends partly on the physical world. We discuss the colors of soap films and so on as being produced by interference. But also, of course, it depends on the eye, or what happens behind the eye, in the brain. Physics characterizes the light that enters the eye, but after that, our sensations are the result of photochemical-neural processes and psychological responses.

There are many interesting phenomena associated with vision which involve a mixture of physical phenomena and physiological processes, and the full appreciation of natural phenomena, as we *see* them, must go beyond physics in the usual sense. We make no apologies for making these excursions into other fields, because the separation of fields, as we have emphasized, is merely a human convenience, and an unnatural thing. Nature is not interested in our separations, and many of the interesting phenomena bridge the gaps between fields.

In Chapter 3 we have already discussed the relation of physics to the other sciences in general terms, but now we are going to look in some detail at a specific field in which physics and other sciences are very, very closely interrelated. That area is *vision*. In particular, we shall discuss *color vision*. In the present chapter we shall discuss mainly the observable phenomena of human vision, and in the next chapter we shall consider the physiological aspects of vision, both in man and in other animals.

It all begins with the eye; so, in order to understand what phenomena we see, some knowledge of the eye is required. In the next chapter we shall discuss in some detail how the various parts of the eye work, and how they are interconnected with the nervous system. For the present, we shall describe only briefly how the eye functions (Fig. 35-1).

Light enters the eye through the *cornea*; we have already discussed how it is bent and is imaged on a layer called the *retina* in the back of the eye, so that different parts of the retina receive light from different parts of the visual field outside. The retina is not absolutely uniform: there is a place, a spot, in the center of our field of view which we use when we are trying to see things very carefully, and at which we have the greatest acuity of vision; it is called the *fovea* or *macula*. The side parts of the eye, as we can immediately appreciate from our experience in looking at things, are not as effective for seeing detail as is the center of the eye. There is also a spot in the retina where the nerves carrying all the information run out; that is a blind spot. There is no sensitive part of the retina here, and it is possible to demonstrate that if we close, say, the left eye and look straight at something, and then move a finger or another small object slowly out of the field of view it suddenly disappears somewhere. The only practical use of this fact that we know of is that some physiologist became quite a favorite in the court of a king of France by pointing this out to him; in the boring sessions that he had with his courtiers, the king could amuse himself by "cutting off their heads" by looking at one and watching another's head disappear.

Figure 35-2 shows a magnified view of the inside of the retina in somewhat schematic form. In different parts of the retina there are different kinds of structures. The objects that occur more densely near the periphery of the retina are called *rods*. Closer to the fovea, we find, besides these rod cells, also *cone* cells. We shall describe the structure of these cells later. As we get close to the fovea, the number of cones increases, and in the fovea itself there are in fact nothing but cone cells, packed very tightly, so tightly that the cone cells are much finer, or narrower

35-1 The human eye

35-2 Color depends on intensity

35-3 Measuring the color sensation

35-4 The chromaticity diagram

35-5 The mechanism of color vision

35-6 Physiochemistry of color vision

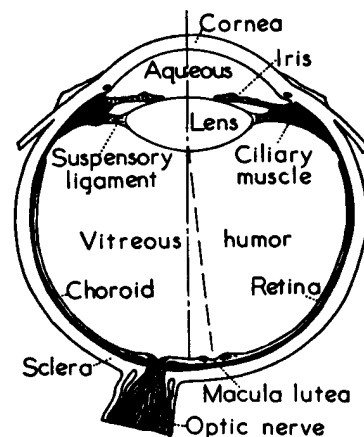


Fig. 35-1. The eye.

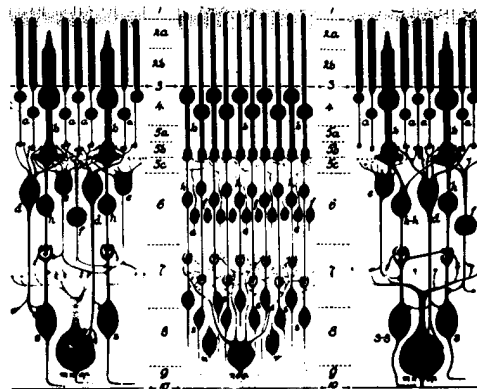


Fig. 35-2. The structure of the retina.
(Light enters from below.)

here than anywhere else. So we must appreciate that we see with the cones right in the middle of the field of view, but as we go to the periphery we have the other cells, the rods. Now the interesting thing is that in the retina each of the cells which is sensitive to light is not connected by a fiber directly to the optic nerve, but is connected to many other cells, which are themselves connected to each other. There are several kinds of cells: there are cells that carry the information toward the optic nerve, but there are others that are mainly interconnected "horizontally." There are essentially four kinds of cells, but we shall not go into these details now. The main thing we emphasize is that the light signal is already being "thought about." That is to say, the information from the various cells does not immediately go to the brain, spot for spot, but in the retina a certain amount of the information has already been digested, by a combining of the information from several visual receptors. It is important to understand that some brain-function phenomena occur in the eye itself.

35-2 Color depends on intensity

One of the most striking phenomena of vision is the dark adaptation of the eye. If we go into the dark from a brightly lighted room, we cannot see very well for a while, but gradually things become more and more apparent, and eventually we can see something where we could see nothing before. If the intensity of the light is very low, the things that we see have *no color*. It is known that this dark-adapted vision is almost entirely due to the rods, while the vision in bright light is due to the cones. As a result, there are a number of phenomena that we can easily appreciate because of this transfer of function from the cones and rods together, to just the rods.

There are many situations in which, if the light intensity were stronger, we could see color, and we would find these things quite beautiful. One example is that through a telescope we nearly always see "black and white" images of faint nebulae, but W. C. Miller of the Mt. Wilson and Palomar Observatories had the patience to make *color* pictures of some of these objects. Nobody has ever really seen these colors with the eye, but they are not artificial colors, it is merely that the light intensity is not strong enough for the cones in our eye to see them. Among the more spectacular such objects are the ring nebula and the Crab nebula. The former shows a beautiful blue inner part, with a bright red outer halo, and the latter shows a general bluish haze permeated by bright red-orange filaments.

In the bright light, apparently, the rods are at very low sensitivity but, in the dark, as time goes on they pick up their ability to see light. The variations in light intensity for which one can adapt is over a million to one. Nature does not do all this with just one kind of cell, but she passes her job from bright-light-seeing cells, the color-seeing cells, the cones, to low-intensity, dark-adapted cells, the rods. Among the interesting consequences of this shift is, first, that there is no color, and second, that there is a difference in the relative brightness of differently colored objects. It turns out that the rods see better toward the blue than the cones do, and the cones can see, for example, deep red light, while the rods find that absolutely impossible to see. So red light is black so far as the rods are concerned. Thus two pieces of colored paper, say blue and red, in which the red might be even brighter than the blue in good light, will, in the dark, appear completely reversed. It is a very striking effect. If we are in the dark and can find a magazine or something that has colors and, before we know for sure what the colors are, we judge the lighter and darker areas, and if we then carry the magazine into the light, we may see this very remarkable shift between which was the brightest color and which was not. The phenomenon is called the *Purkinje effect*.

In Fig. 35-3, the dashed curve represents the sensitivity of the eye in the dark, i.e., using the rods, while the solid curve represents it in the light. We see that the peak sensitivity of the rods is in the green region and that of the cones is more in the yellow region. If there is a red-colored page (red is about $650\text{ m}\mu$) we can see it if it is brightly lighted, but in the dark it is almost invisible.

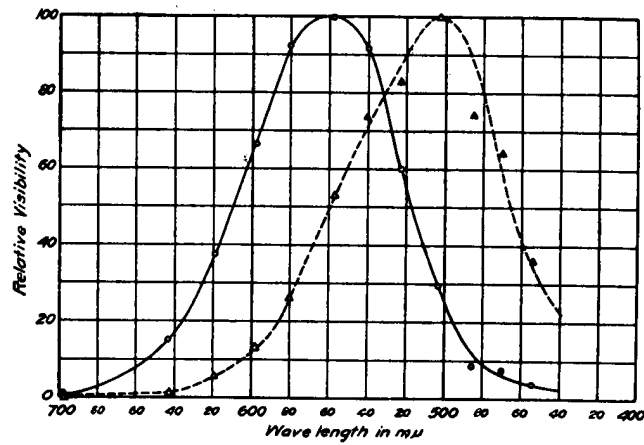


Fig. 35-3. The spectral sensitivity of the eye. Dashed curve, rods; solid curve, cones.

Another effect of the fact that rods take over in the dark, and that there are no rods in the fovea, is that when we look straight at something in the dark, our vision is not quite as acute as when we look to one side. A faint star or nebula can sometimes be seen better by looking a little to one side than directly at it, because we do not have sensitive rods in the middle of the fovea.

Another interesting effect of the fact that the number of cones decreases as we go farther to the side of the field of view is that even in a bright light color disappears as the object goes far to one side. The way to test that is to look in some particular fixed direction, let a friend walk in from one side with colored cards, and try to decide what color they are before they are right in front of you. One finds that he can see that the cards are there long before he can determine the color. When doing this, it is advisable to come in from the side opposite the blind spot, because it is otherwise rather confusing to almost see the color, then not see anything, then to see the color again.

Another interesting phenomenon is that the periphery of the retina is very sensitive to motion. Although we cannot see very well from the corner of our eye, if a little bug moves and we do not expect anything to be moving over there, we are immediately sensitive to it. We are all “wired up” to look for something jiggling to the side of the field.

35-3 Measuring the color sensation

Now we go to the cone vision, to the brighter vision, and we come to the question which is most characteristic of cone vision, and that is color. As we know, white light can be split by a prism into a whole spectrum of wavelengths which appear to us to have different colors; that is what colors are, of course: appearances. Any source of light can be analyzed by a grating or a prism, and one can determine the spectral distribution, i.e., the “amount” of each wavelength. A certain light may have a lot of blue, considerable red, very little yellow, and so on. That is all very precise in the sense of physics, but the question is, what *color* will it appear to be? It is evident that the different colors depend somehow upon the spectral distribution of the light, but the problem is to find what characteristics of the spectral distribution produce the various sensations. For example, what do we have to do to get a green color? We all know that we can simply take a piece of the spectrum which is green. But is that the *only* way to get green, or orange, or any other color?

Is there more than one spectral distribution which produces the same apparent visual effect? The answer is, definitely *yes*. There is a very limited number of visual effects, in fact just a three-dimensional manifold of them, as we shall shortly see, but there is an infinite number of different curves that we can draw for the light that comes from different sources. Now the question we have to discuss is, under what conditions do different distributions of light appear as exactly the same color to the eye?

The most powerful psycho-physical technique in color judgment is to use the eye as a *null instrument*. That is, we do not try to define what constitutes a green sensation, or to measure in what circumstances we get a green sensation, because it turns out that this is extremely complicated. Instead, we study the conditions under which two stimuli are *indistinguishable*. Then we do not have to decide whether two people see the same sensation in different circumstances, but only whether, if for one person two sensations are the same, they are also the same for another. We do not have to decide whether, when one sees something green, what it feels like inside is the same as what it feels like inside someone else when he sees something green; we do not know anything about that.

To illustrate the possibilities, we may use a series of four projector lamps which have filters on them, and whose brightnesses are continuously adjustable over a wide range: one has a red filter and makes a spot of red light on the screen, the next one has a green filter and makes a green spot, the third one has a blue filter, and the fourth one is a white circle with a black spot in the middle of it. Now if we turn on some red light, and next to it put some green, we see that in the area of overlap it produces a sensation which is not what we call reddish green, but a new color, yellow in this particular case. By changing the proportions of the red and the green, we can go through various shades of orange and so forth. If we have set it for a certain yellow, we can also obtain that same yellow, not by mixing these two colors but by mixing some other ones, perhaps a yellow filter with white light, or something like that, to get the same sensation. In other words, it is possible to make various colors in more than one way by mixing the lights from various filters.

What we have just discovered may be expressed analytically as follows. A particular yellow, for example, can be represented by a certain symbol Y , which is the "sum" of certain amounts of red-filtered light (R) and green-filtered light (G). By using two numbers, say r and g , to describe how bright the (R) and (G) are, we can write a formula for this yellow:

$$Y = rR + gG. \quad (35.1)$$

The question is, can we make *all* the different colors by adding together two or three lights of different, fixed colors? Let us see what can be done in that connection. We certainly cannot get all the different colors by mixing only red and green, because, for instance, blue never appears in such a mixture. However, by putting in some blue the central region, where all three spots overlap, may be made to appear to be a fairly nice white. By mixing the various colors and looking at this central region, we find that we can get a considerable range of colors in that region by changing the proportions, and so it is not impossible that *all* the colors can be made by mixing these three colored lights. We shall discuss to what extent this is true; it is in fact essentially correct, and we shall shortly see how to define the proposition better.

In order to illustrate our point, we move the spots on the screen so that they all fall on top of each other, and then we try to match a particular color which appears in the annular ring made by the fourth lamp. What we once thought was "white" coming from the fourth lamp now appears yellowish. We may try to match that by adjusting the red and green and blue as best we can by a kind of trial and error, and we find that we can approach rather closely this particular shade of "cream" color. So it is not hard to believe that we can make all colors. We shall try to make yellow in a moment, but before we do that, there is one color that might be very hard to make. People who give lectures on color make all the "bright" colors, but they never make *brown*, and it is hard to recall ever having seen brown light. As a matter of fact, this color is never used for any stage effect, one never sees a spotlight with brown light; so we think it might be impossible to make brown. In order to find out whether it is possible to make brown, we point out that brown light is merely something that we are not used to seeing without its background. As a matter of fact, we can make it by mixing some red and yellow. To prove that we are looking at brown light, we merely increase the brightness of the annular

background against which we see the very same light, and we see that that is, in fact, what we call brown! Brown is always a dark color next to a lighter background. We can easily change the character of the brown. For example, if we take some green out we get a reddish brown, apparently a chocolatey reddish brown, and if we put more green into it, in proportion, we get that horrible color which all the uniforms of the Army are made of, but the light from that color is not so horrible by itself; it is of yellowish green, but seen against a light background.

Now we put a yellow filter in front of the fourth light and try to match that. (The intensity must of course be within the range of the various lamps; we cannot match something which is too bright, because we do not have enough power in the lamp.) But we *can* match the yellow; we use a green and red mixture, and put in a touch of blue to make it even more perfect. Perhaps we are ready to believe that, under good conditions, we can make a perfect match of any given color.

Now let us discuss the laws of color mixture. In the first place, we found that different spectral distributions can produce the same color; next, we saw that "any" color can be made by adding together three special colors, red, blue, and green. The most interesting feature of color mixing is this: if we have a certain light, which we may call X , and if it appears indistinguishable from Y , to the eye (it may be a different spectral distribution, but it *appears* indistinguishable), we call these colors "equal," in the sense that the eye sees them as equal, and we write

$$X = Y. \quad (35.2)$$

Here is one of the great laws of color: if two spectral distributions are indistinguishable, and we *add to each one* a certain light, say Z (if we write $X + Z$, this means that we shine both lights on the same patch), and then we take Y and add the same amount of the same other light, Z , *the new mixtures are also indistinguishable*:

$$X + Z = Y + Z. \quad (35.3)$$

We have just matched our yellow; if we now shine pink light on the whole thing, it will still match. So adding any other light to the matched lights leaves a match. In other words, we can summarize all these color phenomena by saying that once we have a match between two colored lights, seen next to each other in the same circumstances, then this match will remain, and one light can be substituted for the other light in any other color mixing situation. In fact, it turns out, and it is very important and interesting, that this matching of the color of lights is not dependent upon the characteristics of the eye at the moment of observation: we know that if we look for a long time at a bright red surface, or a bright red light, and then look at a white paper, it looks greenish, and other colors are also distorted by our having looked so long at the bright red. If we now have a match between, say, two yellows, and we look at them and make them match, then we look at a bright red surface for a long time, and then turn back to the yellow, it may not look yellow any more; I do not know what color it will look, but it will not look yellow. Nevertheless *the yellows will still look matched*, and so, as the eye adapts to various levels of intensity, the color match still works, with the obvious exception of when we go into the region where the intensity of the light gets so low that we have shifted from cones to rods; then the color match is no longer a color match, because we are using a different system.

The second principle of color mixing of lights is this: *any color at all can be made from three different colors*, in our case, red, green, and blue lights. By suitably mixing the three together we can make anything at all, as we demonstrated with our two examples. Further, these laws are very interesting mathematically. For those who are interested in the mathematics of the thing, it turns out as follows. Suppose that we take our three colors, which were red, green, and blue, but label them A , B , and C , and call them our *primary* colors. Then any color could be made by certain amounts of these three: say an amount a of color A , an amount b of color B , and an amount c of color C makes X :

$$X = aA + bB + cC. \quad (35.4)$$

Now suppose another color Y is made from the same three colors:

$$Y = a'A + b'B + c'C. \quad (35.5)$$

Then it turns out that the mixture of the two lights (it is one of the consequences of the laws that we have already mentioned) is obtained by taking the sum of the components of X and Y :

$$Z = X + Y = (a + a')A + (b + b')B + (c + c')C. \quad (35.6)$$

It is just like the mathematics of the addition of vectors, where (a, b, c) are the components of one vector, and (a', b', c') are those of another vector, and the new light Z is then the “sum” of the vectors. This subject has always appealed to physicists and mathematicians. In fact, Schrödinger wrote a wonderful paper on color vision in which he developed this theory of vector analysis as applied to the mixing of colors.

Now a question is, what are the correct primary colors to use? There is no such thing as “the” correct primary colors for the mixing of lights. There may be, for practical purposes, three paints that are more useful than others for getting a greater variety of mixed pigments, but we are not discussing that matter now. *Any three differently colored lights whatsoever** can always be mixed in the correct proportion to produce *any color whatsoever*. Can we demonstrate this fantastic fact? Instead of using red, green, and blue, let us use red, blue, and yellow in our projector. Can we use red, blue, and yellow to make, say, green?

By mixing these three colors in various proportions, we get quite an array of different colors, ranging over quite a spectrum. But as a matter of fact, after a lot of trial and error, we find that nothing ever looks like green. The question is, *can* we make green? The answer is yes. How? *By projecting some red onto the green*, then we can make a match with a certain mixture of yellow and blue! So we have matched them, except that we had to cheat by putting the red on the other side. But since we have some mathematical sophistication, we can appreciate that what we really showed was not that X could always be made, say, of red, blue, and yellow, but by putting the red on the other side we found that red plus X could be made out of blue and yellow. Putting it on the other side of the equation, we can interpret that as a *negative amount*, so if we will allow that the coefficients in equations like (35.4) can be both positive and negative, and if we interpret negative amounts to mean that we have to *add* those to the *other side*, then any color can be matched by any three, and there is no such thing as “the” fundamental primaries.

We may ask whether there are three colors that come only with positive amounts for all mixings. The answer is no. Every set of three primaries requires negative amounts for some colors, and therefore there is no unique way to define a primary. In elementary books they are said to be red, green, and blue, but that is merely because with these a *wider range* of colors is available without minus signs for some of the combinations.

35-4 The chromaticity diagram

Now let us discuss the combination of colors on a mathematical level as a geometrical proposition. If any one color is represented by Eq. (35.4), we can plot it as a vector in space by plotting along three axes the amounts a , b , and c , and then a certain color is a point. If another color is a' , b' , c' , that color is located somewhere else. The sum of the two, as we know, is the color which comes from adding these as vectors. We can simplify this diagram and represent everything on a plane by the following observation: if we had a certain color light, and merely doubled a and b and c , that is, if we make them all stronger in the same ratio, it is the same color, but brighter. So if we agree to reduce everything to the *same light intensity*, then we can project everything onto a plane, and this has been done in Fig. 35-4. It follows that any color obtained by mixing a given two in some pro-

* Except, of course, if one of the three can be matched by mixing the other two.

portion will lie somewhere on a line drawn between the two points. For instance, a fifty-fifty mixture would appear halfway between them, and $1/4$ of one and $3/4$ of the other would appear $1/4$ of the way from one point to the other, and so on. If we use a blue and a green and a red, as primaries, we see that all the colors that we can make with positive coefficients are inside the dotted triangle, which contains almost all of the colors that we can ever see, because all the colors that we can ever see are enclosed in the oddly shaped area bounded by the curve. Where did this area come from? Once somebody made a very careful match of all the colors that we can see against three special ones. But we do not have to check *all* colors that we can see, we only have to check the pure spectral colors, the lines of the spectrum. Any light can be considered as a sum of various positive amounts of various pure spectral colors—pure from the physical standpoint. A given light will have a certain amount of red, yellow, blue, and so on—spectral colors. So if we know how much of each of our three chosen primaries is needed to make each of these pure components, we can calculate how much of each is needed to make our given color. So, if we find out what the *color coefficients* of all the spectral colors are for any given three primary colors, then we can work out the whole color mixing table.

An example of such experimental results for mixing three lights together is given in Fig. 35-5. This figure shows the amount of each of three different particular primaries, red, green and blue, which is required to make each of the spectral colors. Red is at the left end of the spectrum, yellow is next, and so on, all the way to blue. Notice that at some points minus signs are necessary. It is from such data that it is possible to locate the position of all of the colors on a chart, where the x - and the y -coordinates are related to the amounts of the different primaries that are used. That is the way that the curved boundary line has been found. It is the locus of the pure spectral colors. Now any other color can be made by adding spectral lines, of course, and so we find that anything that can be produced by connecting one part of this curve to another is a color that is available in nature. The straight line connects the extreme violet end of the spectrum with the extreme red end. It is the locus of the purples. Inside the boundary are colors that can be made with lights, and outside it are colors that cannot be made with lights, and nobody has ever seen them (except, possibly, in after-images!).

35-5 The mechanism of color vision

Now the next aspect of the matter is the question, *why* do colors behave in this way? The simplest theory, proposed by Young and Helmholtz, supposes that in the eye there are three different pigments which receive the light and that these have different absorption spectra, so that one pigment absorbs strongly, say, in the red, another absorbs strongly in the blue, another absorbs in the green. Then when we shine a light on them we will get different amounts of absorptions in the three regions, and these three pieces of information are somehow maneuvered in the brain or in the eye, or somewhere, to decide what the color is. It is easy to demonstrate that all of the rules of color mixing would be a consequence of this proposition. There has been considerable debate about the thing because the next problem, of course, is to find the absorption characteristics of each of the three pigments. It turns out, unfortunately, that because we can transform the color coordinates in any manner we want to, we can only find all kinds of linear combinations of absorption curves by the color-mixing experiments, but not the curves for the individual pigments. People have tried in various ways to obtain a specific curve which does describe some particular physical property of the eye. One such curve is called a *brightness curve*, demonstrated in Fig. 35-3. In this figure are two curves, one for eyes in the dark, the other for eyes in the light; the latter is the cone brightness curve. This is measured by finding what is the smallest amount of colored light we need in order to be able to just see it. This measures how sensitive the eye is in different spectral regions. There is another very interesting way to measure this. If we take two colors and make them appear in an area, by flickering back and forth from one to the other, we see a flicker if the frequency is too low. However, as the frequency increases,

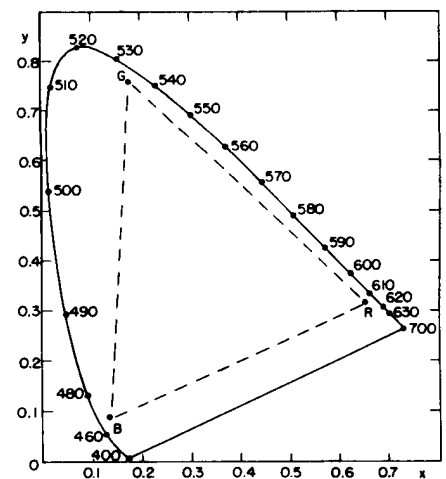


Fig. 35-4. The standard chromaticity diagram.

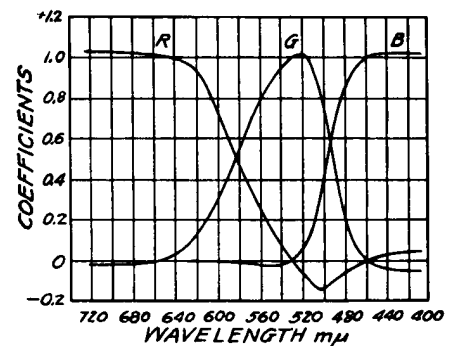


Fig. 35-5. The color coefficients of pure spectral colors in terms of a certain set of standard primary colors.

the flicker will ultimately disappear at a certain frequency that depends on the brightness of the light, let us say at 16 repetitions per second. Now if we adjust the brightness or the intensity of one color against the other, there comes an intensity where the flicker at 16 cycles disappears. To get flicker with the brightness so adjusted, we have to go to a much lower frequency in order to see a flicker of the color. So, we get what we call a flicker of the brightness at a higher frequency and, at a lower frequency, a flicker of the color. It is possible to match two colors for "equal brightness" by this flicker technique. The results are almost, but not exactly, the same as those obtained by measuring the threshold sensitivity of the eye for seeing weak light by the cones. Most workers use the flicker system as a definition of the brightness curve.

Now, if there are three color-sensitive pigments in the eye, the problem is to determine the shape of the absorption spectrum of each one. How? We know there are people who are color blind—eight percent of the male population, and one-half of one percent of the female population. Most of the people who are color blind or abnormal in color vision have a different degree of sensitivity than others to a variation of color, but they still need three colors to match. However, there are some who are called *dichromats*, for whom any color can be matched using only *two* primary colors. The obvious suggestion, then, is to say that they are missing one of the three pigments. If we can find three kinds of color-blind dichromats who have different color-mixing rules, one kind should be missing the *red*, another the *green*, and another the *blue* pigmentation. By measuring all these types we can determine the three curves! It turns out that there *are* three types of dichromatic color blindness; there are two common types and a third very rare type, and from these three it has been possible to deduce the pigment absorption spectra.

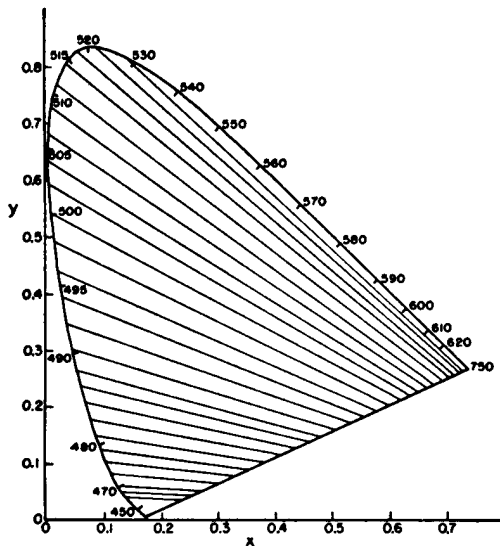


Fig. 35-6. Loci of colors confused by deuteranopes.

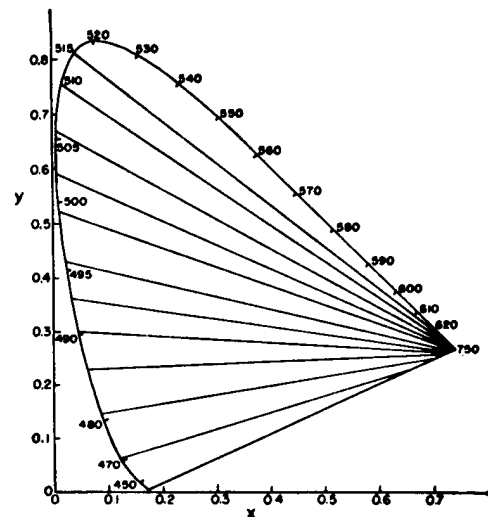


Fig. 35-7. Loci of colors confused by protanopes.

Figure 35-6 shows the color mixing of a particular type of color-blind person called a deuteranope. For him, the loci of constant colors are not points, but certain lines, along each of which the color appears to him to be the same. If the theory that he is missing one of the three pieces of information is right, all these lines should intersect at a point. If we carefully measure on this graph, they *do* intersect perfectly. Obviously, therefore, this has been made by a mathematician and does not represent real data! As a matter of fact, if we look at the latest paper with real data, it turns out that in the graph of Fig. 35-6, the point of focus of all the lines is not exactly at the right place. Using the lines in the above figure, we cannot find reasonable spectra; we need negative and positive absorptions in different regions. But using the new data of Yustova, it turns out that each of the absorption curves is everywhere positive.

Figure 35-7 shows a different kind of color blindness, that of the protanope, which has a focus near the red end of the boundary curve. Yustova gets approximately the same position in this case. Using the three different kinds of color blindness, the three pigment response curves have finally been determined, and are shown in Fig. 35-8. Finally? Perhaps. There is a question as to whether the three-pigment idea is right, whether color blindness results from lack of one pigment, and even whether the color-mix data on color blindness are right. Different workers get different results. This field is still very much under development.

35-6 Physiochemistry of color vision

Now, what about checking these curves against actual pigments in the eye? The pigments that can be obtained from a retina consist mainly of a pigment called *visual purple*. The most remarkable features of this are, first, that it is in the eye of almost every vertebrate animal, and second, that its response curve fits beautifully with the sensitivity of the eye, as seen in Fig. 35-9, in which are plotted on the same scale the absorption of visual purple and the sensitivity of the dark-adapted eye. This pigment is evidently the pigment that we see with in the dark: visual purple is the pigment for the rods, and it has nothing to do with color vision. This fact was discovered in 1877. Even today it can be said that the color pigments of the cones have never been obtained in a test tube. In 1958 it could be said that the color pigments had never been seen at all. But since that time, two of them have been detected by Rushton by a very simple and beautiful technique.

The trouble is, presumably, that since the eye is so weakly sensitive to bright light compared with light of low intensity, it needs a lot of visual purple to see with, but not much of the color pigments for seeing colors. Rushton's idea is to *leave the pigment in the eye*, and measure it anyway. What he does is this. There is an instrument called an ophthalmoscope for sending light into the eye through the lens and then focusing the light that comes back out. With it one can measure how much is reflected. So one measures the reflection coefficient of light which has gone *twice* through the pigment (reflected by a back layer in the eyeball, and coming out through the pigment of the cone again). Nature is not always so beautifully designed. The cones are interestingly designed so that the light that comes into the cone bounces around and works its way down into the little sensitive points at the apex. The light goes right down into the sensitive point, bounces at the bottom and comes back out again, having traversed a considerable amount of the color-vision pigment; also, by looking at the fovea, where there are no rods, one is not confused by visual purple. But the color of the retina has been seen a long time ago: it is a sort of orangey pink; then there are all the blood vessels, and the color of the material at the back, and so on. How do we know when we are looking at the pigment? *Answer:* First we take a color-blind person, who has fewer pigments and for whom it is therefore easier to make the analysis. Second, the various pigments, like visual purple, have an intensity change when they are bleached by light; when we shine light on them they change their concentration. So, while looking at the absorption spectrum of the eye, Rushton put *another* beam in the whole eye, which changes the concentration of the pigment, and he measured the *change* in the spectrum, and the difference, of course, has nothing to do with the amount of blood or the color of the reflecting layers, and so on, but only the pigment, and in this manner Rushton obtained a curve for the pigment of the protanope eye, which is given in Fig. 35-10.

The second curve in Fig. 35-10 is a curve obtained with a normal eye. This was obtained by taking a normal eye and, having already determined what one pigment was, bleaching the other one in the red where the first one is insensitive. Red light has no effect on the protanope eye, but does in the normal eye, and thus one can obtain the curve for the missing pigment. The shape of one curve fits beautifully with Yustova's green curve, but the red curve is a little bit displaced. So perhaps we are getting on the right track. Or perhaps not—the latest work with deuteranopes does not show any definite pigment missing.

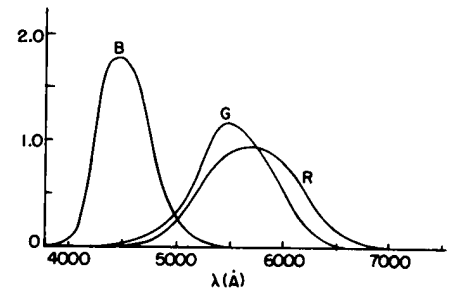


Fig. 35-8. The spectral sensitivity curves of a normal trichromat's receptors.

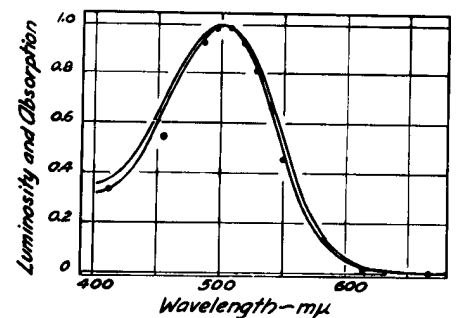


Fig. 35-9. The sensitivity curve of the dark-adapted eye, compared with the absorption curve of visual purple.

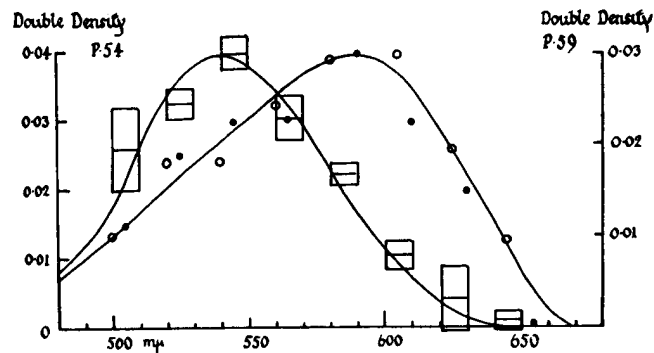


Fig. 35-10. Absorption spectrum of the color pigment of a protanope color-blind eye (squares) and a normal eye (dots).

Color is not a question of the physics of the light itself. Color is a *sensation*, and the sensation for different colors is different in different circumstances. For instance, if we have a pink light, made by superimposing crossing beams of white light and red light (all we can make with white and red is pink, obviously), we may show that white light may appear blue. If we place an object in the beams, it casts two shadows—one illuminated by the white light alone and the other by the red. For most people the “white” shadow of an object looks blue, but if we keep expanding this shadow until it covers the entire screen, we see that it suddenly appears white, not blue! We can get other effects of the same nature by mixing red, yellow, and white light. Red, yellow, and white light can produce only orangey yellows, and so on. So if we mix such lights roughly equally, we get only orange light. Nevertheless, by casting different kinds of shadows in the light, with various overlaps of colors, one gets quite a series of beautiful colors which are not in the light themselves (that is only orange), but in our *sensations*. We clearly *see* many different colors that are quite unlike the “physical” ones in the beam. It is very important to appreciate that a retina is already “thinking” about the light; it is comparing what it sees in one region with what it sees in another, although not consciously. What we know of how it does that is the subject of the next chapter.

BIBLIOGRAPHY

- Committee on Colorimetry, Optical Society of America, *The Science of Color*, Thomas Y. Crowell Company, New York, 1953.
- HECHT, S., S. SHLAER, and M. H. PIRENNE, “Energy, Quanta, and Vision,” *Journal of General Physiology*, 1942, **25**, 819–840.
- MORGAN, CLIFFORD and ELIOT STELLAR, *Physiological Psychology*, 2nd ed., McGraw-Hill Book Company, Inc., 1950.
- NUBERG, N. D. and E. N. YUSTOVA, “Researches on Dichromatic Vision and the Spectral Sensitivity of the Receptors of Trichromats,” presented at Symposium No. 8, *Visual Problems of Colour*, Vol. II, National Physical Laboratory, Teddington, England, September 1957. Published by Her Majesty’s Stationery Office, London, 1958.
- RUSHTON, W. A., “The Cone Pigments of the Human Fovea in Colour Blind and Normal,” presented at Symposium No. 8, *Visual Problems of Colour*, Vol. I, National Physical Laboratory, Teddington, England, September 1957. Published by Her Majesty’s Stationery Office, London, 1958.
- WOODWORTH, ROBERT S., *Experimental Psychology*, Henry Holt and Company, New York, 1938. Revised edition, 1954, by Robert S. Woodworth and H. Schlosberg.

Mechanisms of Seeing

36-1 The sensation of color

In discussing the sense of sight, we have to realize that (outside of a gallery of modern art!) one does not see random spots of color or spots of light. When we look at an object we see a *man* or a *thing*; in other words, the brain interprets what we see. How it does that, no one knows, and it does it, of course, at a very high level. Although we evidently do learn to recognize what a man looks like after much experience, there are a number of features of vision which are more elementary but which also involve combining information from different parts of what we see. To help us understand how we make an interpretation of an entire image, it is worth while to study the earliest stages of the putting together of information from the different retinal cells. In the present chapter we shall concentrate mainly on that aspect of vision, although we shall also mention a number of side issues as we go along.

An example of the fact that we have an accumulation, at a very elementary level, of information from several parts of the eye at the same time, beyond our voluntary control or ability to learn, was that blue shadow which was produced by white light when both white and red were shining on the same screen. This effect at least involves the knowledge that the background of the screen is pink, even though, when we are looking at the blue shadow, it is only "white" light coming into a particular spot in the eye; somewhere, pieces of information have been put together. The more complete and familiar the context is, the more the eye will make corrections for peculiarities. In fact, Land has shown that if we mix that apparent blue and the red in various proportions, by using two photographic transparencies with absorption in front of the red and the white in different proportions, it can be made to represent a real scene, with real objects, rather faithfully. In this case we get a lot of intermediate apparent colors too, analogous to what we would get by mixing red and blue-green; it seems to be an almost complete set of colors, but if we look very hard at them, they are not so very good. Even so, it is surprising how much can be obtained from just red and white. The more the scene looks like a real situation, the more one is able to compensate for the fact that all the light is actually nothing but pink!

Another example is the appearance of "colors" in a black-and-white rotating disc, whose black and white areas are as shown in Fig. 36-1. When the disc is rotated, the variations of light and dark at any one radius are exactly the same; it is only the background that is different for the two kinds of "stripes." Yet one of the "rings" appears colored with one color and the other with another.* No one yet understands the reason for those colors, but it is clear that information is being put together at a very elementary level, in the eye itself, most likely.

Almost all present-day theories of color vision agree that the color-mixing data indicate that there are only three pigments in the cones of the eye, and that it is the spectral absorption in those three pigments that fundamentally produces the color sense. But the total sensation that is associated with the absorption characteristics of the three pigments acting together is not necessarily the sum of the individual sensations. We all agree that yellow simply *does not* seem to be reddish green; in fact it might be a tremendous surprise to most people to discover that light is, in fact, a mixture of colors, because presumably the sensation of light

36-1 The sensation of color

36-2 The physiology of the eye

36-3 The rod cells

36-4 The compound (insect) eye

36-5 Other eyes

36-6 Neurology of vision

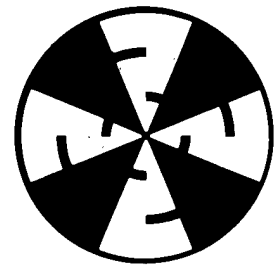


Fig. 36-1. When a disc like the above is spun, colors appear in only one of the two darker "rings." If the spin direction is reversed, the colors appear in the other ring.

* The colors depend on speed of rotation, on the brightness of illumination, and to some extent on who looks at them and how intently he stares at them.

is due to some other process than a simple mixture like a chord in music, where the three notes are there at the same time and if we listen hard we can hear them individually. We cannot look hard and see the red and the green.

The earliest theories of vision said that there are three pigments and three kinds of cones, each kind containing one pigment; that a nerve runs from each cone to the brain, so that the three pieces of information are carried to the brain; and then in the brain, anything can happen. This, of course, is an incomplete idea: it does no good to discover that the information is carried along the optic nerve to the brain, because we have not even started to solve the problem. We must ask more basic questions: Does it make any difference *where* the information is put together? Is it important that it be carried right up into the brain in the optic nerve, or could the retina do some analysis first? We have seen a picture of the retina as an extremely complicated thing with lots of interconnections (Fig. 35-2) and it might make some analyses.

As a matter of fact, people who study anatomy and the development of the eye have shown that the retina is, in fact, the brain: in the development of the embryo, a piece of the brain comes out in front, and long fibers grow back, connecting the eyes to the brain. The retina is organized in just the way the brain is organized and, as someone has beautifully put it, "The brain has developed a way to look out upon the world." The eye is a piece of brain that is touching light, so to speak, on the outside. So it is not at all unlikely that some analysis of the color has already been made in the retina.

This gives us a very interesting opportunity. None of the other senses involves such a large amount of calculation, so to speak, before the signal gets into a nerve that one can make measurements on. The calculations for all the rest of the senses usually happen in the brain itself, where it is very difficult to get at specific places to make measurements, because there are so many interconnections. Here, with the visual sense, we have the light, three layers of cells making calculations, and the results of the calculations being transmitted through the optic nerve. So we have the first chance to observe physiologically how, perhaps, the first layers of the brain work in their first steps. It is thus of double interest, not simply interesting for vision, but interesting to the whole problem of physiology.

The fact that there are three pigments does not mean that there must be three kinds of sensations. One of the other theories of color vision has it that there are really opposing color schemes (Fig. 36-2). That is, one of the nerve fibers carries a lot of impulses if there is yellow being seen, and less than usual for blue. Another nerve fiber carries green and red information in the same way, and another, white and black. In other words, in this theory someone has already started to make a guess as to the system of wiring, the method of calculation.

The problems we are trying to solve by guessing at these first calculations are questions about the apparent colors that are seen on a pink background, what happens when the eye is adapted to different colors, and also the so-called psychological phenomena. The psychological phenomena are of the nature, for instance, that white does not "feel" like red and yellow and blue, and this theory was advanced because the psychologists say that there are *four* apparent pure colors: "There are four stimuli which have a remarkable capacity to evoke psychologically simple blue, yellow, green, and red hues respectively. Unlike sienna, magenta, purple, or most of the discriminable colors, these simple hues are unmixed in the sense that none partakes of the nature of the other; specifically, blue is not yellowish, reddish, or greenish, and so on; these are psychologically primary hues." That is a psychological fact, so-called. To find out from what evidence this psychological fact was deduced, we must search very hard indeed through all the literature. In the modern literature all we find on the subject are repeats of the same statement, or of one by a German psychologist, who uses as one of his authorities Leonardo da Vinci, who, of course, we all know was a great artist. He says, "Leonardo thought there were five colors." Then, looking still further, we find, in a still older book, the evidence for the subject. The book says something like this: "Purple is reddish-blue, orange is reddish-yellow, but can red be seen as purplish-orange? Are not red and yellow more unitary than purple or orange? The average person, asked to state which

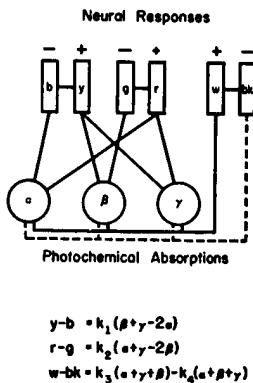


Fig. 36-2. Neural connections according to an "opponent" theory of color vision.

colors are unitary, names red, yellow, and blue, these three, and some observers add a fourth, green. Psychologists are accustomed to accept the four as salient hues." So that is the situation in the psychological analysis of this matter: if everybody says there are three, and somebody says there are four, and they want it to be four, it will be four. That shows the difficulty with psychological researches. It is clear that we have such feelings, but it is very difficult to obtain much information about them.

So the other direction to go is the physiological direction, to find out experimentally what actually happens in the brain, the eye, the retina, or wherever, and perhaps to discover that some combinations of impulses from various cells move along certain nerve fibers. Incidentally, primary pigments do not have to be in separate cells; one could have cells in which are mixtures of the various pigments, cells with the red and the green pigments, cells with all three (the information of all three is then white information), and so on. There are many ways of hooking the system up, and we have to find out which way nature has used. It would be hoped, ultimately, that when we understand the physiological connections we will have a little bit of understanding of some of those aspects of the psychology, so we look in that direction.

36-2 The physiology of the eye

We begin by talking not only about color vision, but about vision in general, just to remind ourselves about the interconnections in the retina, shown in Fig. 35-2. The retina is really like the surface of the brain. Although the actual picture through a microscope is a little more complicated looking than this somewhat schematized drawing, by careful analysis one can see all these interconnections. There is no question that one part of the surface of the retina is connected to other parts, and that the information that comes out on the long axons, which produce the optic nerve, are combinations of information from many cells. There are three layers of cells in the succession of function: there are retinal cells, which are the ones that the light affects, an intermediate cell which takes information from a single or a few retinal cells and gives it out again to several cells in a third layer of cells and carries it to the brain. There are all kinds of cross connections between cells in the layers.

We now turn to some aspects of the structure and performance of the eye (see Fig. 35-1). The focusing of the light is accomplished mainly by the cornea, by the fact that it has a curved surface which "bends" the light. This is why we cannot see clearly under water, because we then do not have enough difference between the index of the cornea, which is 1.37, and that of the water, which is 1.33. Behind the cornea is water, practically, with an index of 1.33, and behind that is a lens which has a very interesting structure: it is a series of layers, like an onion, except that it is all transparent, and it has an index of 1.40 in the middle and 1.38 at the outside. (It would be nice if we could make optical glass in which we could adjust the index throughout, for then we would not have to curve it as much as we do when we have a uniform index.) Furthermore, the shape of the cornea is not that of a sphere. A spherical lens has a certain amount of spherical aberration. The cornea is "flatter" at the outside than is a sphere, in just such a manner that the spherical aberration is less for the cornea than it would be if we put a spherical lens in there! The light is focused by the cornea-lens system onto the retina. As we look at things that are closer and farther away, the lens tightens and loosens and changes the focus to adjust for the different distances. To adjust for the total amount of light there is the iris, which is what we call the color of the eye, brown or blue, depending on who it is; as the amount of light increases and decreases, the iris moves in and out.

Let us now look at the neural machinery for controlling the accommodation of the lens, the motion of the eye, the muscles which turn the eye in the socket, and the iris, shown schematically in Fig. 36-3. Of all the information that comes out of the optic nerve *A*, the great majority is divided into one of two bundles (which we will talk about later) and thence to the brain. But there are a few fibers, of

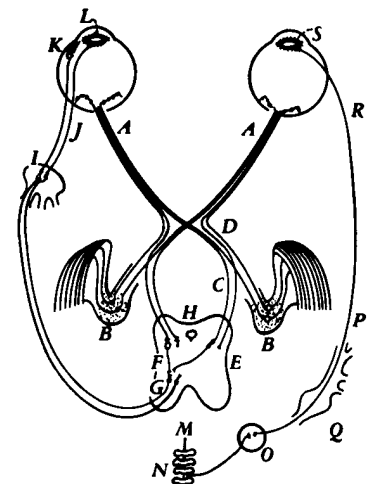


Fig. 36-3. The neural interconnections for the mechanical operation of the eyes.

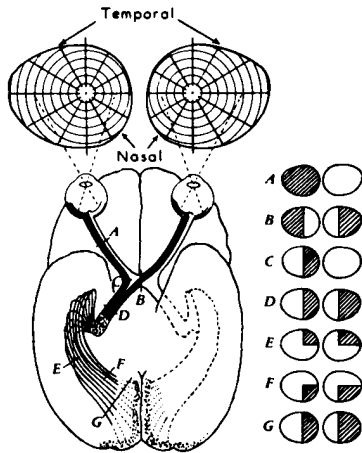


Fig. 36-4. The neural connections from the eyes to the visual cortex.

interest to us now, which do not run directly to the visual cortex of the brain where we "see" the images, but instead go into the mid-brain *H*. These are the fibers which measure the average light and make adjustment for the iris; or, if the image looks foggy, they try to correct the lens; or, if there is a double image, they try to adjust the eye for binocular vision. At any rate, they go through the mid-brain and feed back into the eye. At *K* are the muscles which run the accommodation of the lens, and at *L* another one that runs into the iris. The iris has two muscle systems. One is a circular muscle *L* which, when it is excited, pulls in and closes down the iris; it acts very rapidly and the nerves are directly connected from the brain through short axons into the iris. The opposite muscles are radial muscles, so that, when the things get dark and the circular muscle relaxes, these radial muscles pull out. Here we have, as in many places in the body, a pair of muscles which work in opposite directions, and in almost every such case the nerve systems which control the two are very delicately adjusted, so that when signals are sent in to tighten one, signals are automatically sent in to loosen the other. The iris is a peculiar exception: the nerves which make the iris contract are the ones we have already described, but the nerves which make the iris *expand* come out from no one knows exactly where, go down into the spinal cord back of the chest, into the thoracic sections, out of the spinal cord, up through the neck ganglia, and all the way around and back up into the head in order to run the other end of the iris. In fact, the signal goes through a completely different nervous system, not the central nervous system at all, but the sympathetic nervous system, so it is a very strange way of making things go.

We have already emphasized another strange thing about the eye, that the light-sensitive cells are on the wrong side, so that the light has to go through several layers of other cells before it gets to the receptors—it is built inside out! So some of the features are wonderful and some are apparently stupid.

Figure 36-4 shows the connections of the eye to the part of the brain which is most directly concerned with the visual process. The optic nerve fibers run into a certain area just beyond *D*, called the lateral geniculate, whereupon they run out to a section of the brain called the visual cortex. Notice that some of the fibers from each eye are sent over to the other side of the brain, so the picture formed is incomplete. The optic nerves from the left side of the right eye run across the optic chiasma *B*, while the ones on the left side of the left eye come around and go this same way. So the left side of the brain receives all the information which comes from the left side of the eyeball of each eye, i.e., on the right side of the visual field, while the right side of the brain sees the left side of the visual field. This is the manner in which the information from each of the two eyes is put together in order to tell how far away things are. This is the system of binocular vision.

The connections between the retina and the visual cortex are interesting. If a spot in the retina is excised or destroyed in any way, then the whole fiber will die, and we can thereby find out where it is connected. It turns out that, essentially, the connections are one to one—for each spot in the retina there is one spot in the visual cortex—and spots that are very close together in the retina are very close together in the visual cortex. So the visual cortex still represents the spatial arrangement of the rods and cones, but of course much distorted. Things which are in the center of the field, which occupy a very small part of the retina, are expanded over many, many cells in the visual cortex. It is clear that it is useful to have things which are originally close together, still close together. The most remarkable aspect of the matter, however, is the following. The place where one would think it would be most important to have things close together would be right in the middle of the visual field. Believe it or not, the up-and-down line in our visual field as we look at something is of such a nature that the information from all the points on the right side of that line is going into the left side of the brain, and information from the points on the left side is going into the right side of the brain, and the way this area is made, there is a cut right down through the middle, so that the things that are very close together right in the middle are very far apart in the brain! Somehow, the information has to go from one side of the brain to the other through some other channels, which is quite surprising.

The question of how this network ever gets “wired” together is very interesting. The problem of how much is already wired and how much is learned is an old one. It used to be thought long ago that perhaps it does not have to be wired carefully at all, it is only just roughly interconnected, and then, by experience, the young child learns that when a thing is “up there” it produces some sensation in the brain. (Doctors always tell us what the young child “feels,” but how do *they* know what a child feels at the age of one?) The child, at the age of one, supposedly sees that an object is “up there,” gets a certain sensation, and learns to reach “there,” because when he reaches “here,” it does not work. That approach probably is not correct, because we already see that in many cases there are these special detailed interconnections. More illuminating are some most remarkable experiments done with a salamander. (Incidentally, with the salamander there is a direct crossover connection, without the optic chiasma, because the eyes are on each side of the head and have no common area. Salamanders do not have binocular vision.) The experiment is this. We can cut the optic nerve in a salamander and the nerve will grow out from the eyes again. Thousands and thousands of cell fibers will thus re-establish themselves. Now, in the optic nerve the fibers do not stay adjacent to each other—it is like a great, sloppily made telephone cable, all the fibers twisting and turning, but when it gets to the brain they are all sorted out again. When we cut the optic nerve of the salamander, the interesting question is, will it ever get straightened out? The answer is remarkable: yes. If we cut the optic nerve of the salamander and it grows back, the salamander has good visual acuity again. However, if we cut the optic nerve and *turn the eye upside down* and let it grow back again, it has good visual acuity all right, but it has a terrible error: when the salamander sees a fly “up here,” it jumps at it “down there,” and it never learns. Therefore there is some mysterious way by which the thousands and thousands of fibers find their right places in the brain.

This problem of how much is wired in, and how much is not, is an important problem in the theory of the development of creatures. The answer is not known, but is being studied intensively.

The same experiment in the case of a goldfish shows that there is a terrible knot, like a great scar or complication, in the optic nerve where we cut it, but in spite of all this the fibers grow back to their right places in the brain.

In order to do this, as they grow into the old channels of the optic nerve they must make several decisions about the direction in which they should grow. How do they do this? There seem to be chemical clues that different fibers respond to differently. Think of the enormous number of growing fibers, each of which is an individual differing in some way from its neighbors; in responding to whatever the chemical clues are, it responds in a unique enough way to find its proper place for ultimate connection in the brain! This is an interesting—a fantastic—thing. It is one of the great recently discovered phenomena of biology and is undoubtedly connected to many older unsolved problems of growth, organization, and development of organisms, and particularly of embryos.

One other interesting phenomenon has to do with the motion of the eye. The eyes must be moved in order to make the two images coincide in different circumstances. These motions are of different kinds: one is to follow something, which requires that both eyes must go in the same direction, right or left, and the other is to point them toward the same place at various distances away, which requires that they must move oppositely. The nerves going into the muscles of the eye are already wired up for just such purposes. There is one set of nerves which will pull the muscles on the inside of one eye and the outside of the other, and relax the opposite muscles, so that the two eyes move together. There is another center where an excitation will cause the eyes to move in toward each other from parallel. Either eye can be turned out to the corner if the other eye moves toward the nose, but it is impossible consciously or unconsciously to turn both eyes *out* at the same time, not because there are no *muscles*, but because there is no way to send a signal to turn both eyes out, unless we have had an accident or there is something the matter, for instance if a nerve has been cut. Although the muscles of one eye can certainly steer that eye about, not even a Yogi is able to move *both* eyes out freely

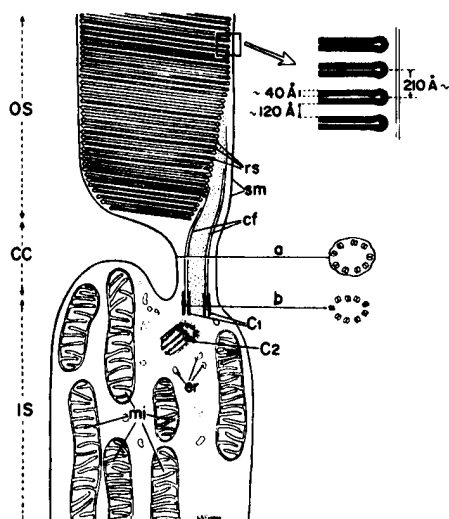


Fig. 36-5. Electron micrograph of a rod cell.

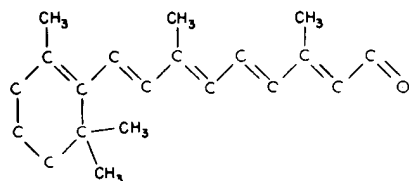


Fig. 36-6. The structure of retinene.

under voluntary control, because there does not seem to be any way to do it. We are already wired to a certain extent. This is an important point, because most of the earlier books on anatomy and psychology, and so on, do not appreciate or do not emphasize the fact that we are so completely wired already—they say that everything is just learned.

36-3 The rod cells

Let us now examine in more detail what happens in the rod cells. Figure 36-5 shows an electron micrograph of the middle of a rod cell (the rod cell keeps going up out of the field). There are layer after layer of plane structures, shown magnified at the right, which contain the substance rhodopsin (visual purple), the dye, or pigment, which produces the effects of vision in the rods. The rhodopsin, which is the pigment, is a big protein which contains a special group called retinene, which can be taken off the protein, and which is, undoubtedly, the main cause of the absorption of light. We do not understand the reason for the planes, but it is very likely that there is some reason for holding all the rhodopsin molecules parallel. The chemistry of the thing has been worked out to a large extent, but there might be some physics to it. It may be that all of the molecules are arranged in some kind of a row so that when one is excited an electron which is generated, say, may run all the way down to some place at the end to get the signal out, or something. This subject is very important, and has not been worked out. It is a field in which both biochemistry and solid state physics, or something like it, will ultimately be used.

This kind of a structure, with layers, appears in other circumstances where light is important, for example in the chloroplast in plants, where the light causes photosynthesis. If we magnify those, we find the same thing with almost the same kind of layers, but there we have chlorophyll, of course, instead of retinene. The chemical form of retinene is shown in Fig. 36-6. It has a series of alternate double bonds along the side chain, which is characteristic of nearly all strongly absorbing organic substances, like chlorophyll, blood, and so on. This substance is impossible for human beings to manufacture in their own cells—we have to eat it. So we eat it in the form of a special substance, which is exactly the same as retinene except that there is a hydrogen tied on the right end; it is called vitamin A, and if we do not eat enough of it, we do not get a supply of retinene, and the eye becomes what we call *night blind*, because there is then not enough pigment in the rhodopsin to see with the rods at night.

The reason why such a series of double bonds absorbs light very strongly is also known. We may just give a hint: The alternating series of double bonds is called a *conjugated* double bond; a double bond means that there is an extra electron there, and this extra electron is easily shifted to the right or left. When light strikes this molecule, the electron of each double bond is shifted over by one step. All the electrons in the whole chain shift, like a string of dominoes falling over, and though each one moves only a little distance (we would expect that, in a single atom, we could move the electron only a little distance), the net effect is the same as though the one at the end was moved over to the other end! It is the same as though one electron went the whole distance back and forth, and so, in this manner, we get a much stronger absorption under the influence of the electric field, than if we could only move the electron a distance which is associated with one atom. So, since it is easy to move the electrons back and forth, retinene absorbs light very strongly; that is the machinery of the physical-chemical end of it.

36-4 The compound (insect) eye

Let us now return to biology. The human eye is not the only kind of eye. In the vertebrates, almost all eyes are essentially like the human eye. However, in the lower animals there are many other kinds of eyes: eye spots, various eye cups, and other less sensitive things, which we have no time to discuss. But there is one other highly developed eye among the invertebrates, the *compound* eye of the insect. (Most insects having large compound eyes also have various additional simpler eyes as well.) A bee is an insect whose vision has been studied very carefully. It is

easy to study the properties of the vision of bees because they are attracted to honey, and we can make experiments in which we identify the honey by putting it on blue paper or red paper, and see which one they come to. By this method some very interesting things have been discovered about the vision of the bee.

In the first place, in trying to measure how acutely bees could see the color difference between two pieces of "white" paper, some researchers found they were not very good, and others found they were fantastically good. Even if the two pieces of white paper were almost exactly the same, the bees could still tell the difference. The experimenters used zinc white for one piece of paper and lead white for the other, and although these look exactly the same to us, the bee could easily distinguish them, because they reflect a different amount in the ultraviolet. In this way it was discovered that the bee's eye is sensitive over a wider range of the spectrum than is our own. Our eye works from 7000 angstroms to 4000 angstroms, from red to violet, but the bee's can see down to 3000 angstroms into the ultraviolet! This makes for a number of different interesting effects. In the first place, bees can distinguish between many flowers which to us look alike. Of course, we must realize that the colors of flowers are not designed for *our* eyes, but for the bee; they are signals to attract the bees to a specific flower. We all know that there are many "white" flowers. Apparently white is not very interesting to the bees, because it turns out that all of the white flowers have different proportions of reflection in the *ultraviolet*; they do not reflect one hundred percent of the ultraviolet as would a true white. All the light is not coming back, the ultraviolet is missing, and that is a color, just as, for us, if the blue is missing, it comes out yellow. So, all the flowers are colored for the bees. However, we also know that red cannot be seen by bees. Thus we might expect that all red flowers should look black to the bee. Not so! A careful study of red flowers shows, first, that even with our own eye we can see that a great majority of red flowers have a bluish tinge because they are mainly reflecting an additional amount in the blue, which is the part that the bee sees. Furthermore, experiments also show that flowers vary in their reflection of the ultraviolet over different parts of the petals, and so on. So if we could see the flowers as bees see them they would be even more beautiful and varied!

It has been shown, however, that there are a few red flowers which do *not* reflect in the blue or in the ultraviolet, and *would*, therefore, appear black to the bee! This was of quite some concern to the people who worry about this matter, because black does not seem like an interesting color, since it is hard to tell from a dirty old shadow. It actually turned out that these flowers were *not* visited by bees, these are the flowers that are visited by *hummingbirds*, and hummingbirds *can* see the red!

Another interesting aspect of the vision of the bee is that bees can apparently tell the direction of the sun by looking at a patch of blue sky, without seeing the sun itself. We cannot easily do this. If we look out the window at the sky and see that it is blue, in which direction is the sun? The bee can tell, because the bee is quite sensitive to the *polarization* of light, and the scattered light of the sky is polarized.* There is still some debate about how this sensitivity operates. Whether it is because the reflections of the light are different in different circumstances, or the bee's eye is directly sensitive, is not yet known.†

It is also said that the bee can notice flicker up to 200 oscillations per second, while we see it only up to 20. The motions of bees in the hives are very quick; the feet move and the wings vibrate, but it is very hard for us to see these motions with our eye. However, if we could see more rapidly we would be able to see the motion. It is probably very important to the bee that its eye has such a rapid response.

* The human eye also has a slight sensitivity to the polarization of light, and one *can* learn to tell the direction of the sun! The phenomenon that is involved here is called *Haidinger's brush*; it is a faint, yellowish hourglass-like pattern seen at the center of the visual field when one looks at a broad, featureless expanse using polarizing glasses. It can also be seen in the blue sky without polarizing glasses if one rotates his head back and forth about the axis of vision.

† Evidence obtained since this lecture was given indicates that the eye is directly sensitive.



Fig. 36-7. The structure of an ommatidium (a single cell of a compound eye).

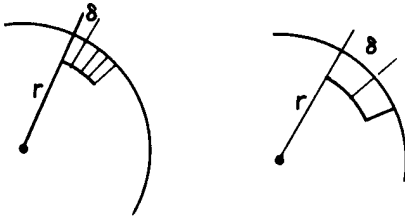


Fig. 36-8. Schematic view of packing of ommatidia in the eye of a bee.

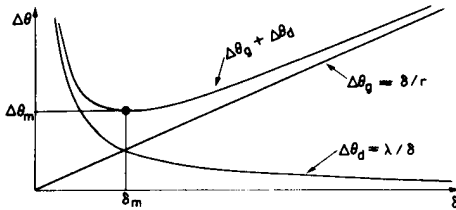


Fig. 36-9. The optimum size for an ommatidium is δ_m .

Now let us discuss the visual acuity we could expect from the bee. The eye of a bee is a compound eye, and it is made of a large number of special cells called *ommatidia*, which are arranged conically on the surface of a sphere (roughly) on the outside of the bee's head. Figure 36-7 shows a picture of one such ommatidium. At the top there is a transparent area, a kind of "lens," but actually it is more like a filter or light pipe to make the light come down along the narrow fiber, which is where the absorption presumably occurs. Out of the other end of it comes the nerve fiber. The central fiber is surrounded on its sides by six cells which, in fact, have secreted the fiber. That is enough description for our purposes; the point is that it is a conical thing and many can fit next to each other all over the surface of the eye of the bee.

Now let us discuss the resolution of the eye of the bee. If we draw lines (Fig. 36-8) to represent the ommatidia on the surface, which we suppose is a sphere of radius r , we may actually *calculate* how wide each ommatidium is by using our brains, and assuming that evolution is as clever as we are! If we have a very large ommatidium we do not have much resolution. That is, one cell gets a piece of information from one direction, and the adjacent cell gets a piece of information from another direction, and so on, and the bee cannot see things in between very well. So the uncertainty of visual acuity in the eye will surely correspond to an angle, the angle of the end of the ommatidium relative to the center of curvature of the eye. (The eye cells, of course, exist only at the surface of the sphere; inside that is the head of the bee.) This angle, from one ommatidium to the next, is, of course, the diameter of the ommatidia divided by the radius of the eye surface:

$$\Delta\theta_g = \delta/r. \quad (36.1)$$

So, we may say, "The finer we make the δ , the more the visual acuity. So why doesn't the bee just use very, very fine ommatidia?" *Answer:* We know enough physics to realize that if we are trying to get light down into a narrow slot, we cannot see accurately in a given direction because of the diffraction effect. The light that comes from several directions can enter and, due to diffraction, we will get light coming in at angle $\Delta\theta_d$ such that

$$\Delta\theta_d = \lambda/\delta. \quad (36.2)$$

Now we see that if we make the δ too small, then each ommatidium does not look in only one direction, because of diffraction! If we make them too big, each one sees in a definite direction, but there are not enough of them to get a good view of the scene. So we adjust the distance d in order to make minimal the total effect of these two. If we add the two together, and find the place where the sum has a minimum (Fig. 36-9), we find that

$$\frac{d(\Delta\theta_g + \Delta\theta_d)}{d\delta} = 0 = \frac{1}{r} - \frac{\lambda}{\delta^2}, \quad (36.3)$$

which gives us a distance

$$\delta = \sqrt{\lambda r}. \quad (36.4)$$

If we guess that r is about 3 millimeters, take the light that the bee sees as 4000 angstroms, and put the two together and take the square root, we find

$$\begin{aligned} \delta &= (3 \times 10^{-3} \times 4 \times 10^{-7})^{1/2} \text{ m} \\ &= 3.5 \times 10^{-5} \text{ m} = 35 \mu. \end{aligned} \quad (36.5)$$

The book says the diameter is 30μ , so that is rather good agreement! So, apparently, it really works, and we can understand what determines the size of the bee's eye! It is also easy to put the above number back in and find out how good the bee's eye actually is in angular resolution; it is very poor relative to our own. We can see things that are thirty times smaller in apparent size than the bee; the bee has a rather fuzzy out-of-focus image relative to what we can see. Nevertheless it is all right, and it is the best they can do. We might ask why the bees do not develop

a good eye like our own, with a lens and so on. There are several interesting reasons. In the first place, the bee is too small; if it had an eye like ours, but on his scale, the opening would be about $30\ \mu$ in size and diffraction would be so important that it would not be able to see very well anyway. The eye is not good if it is too small. Secondly, if it were as big as the bee's head, then the eye would occupy the whole head of the bee. The beauty of the compound eye is that it takes up no space, it is just a very thin layer on the surface of the bee. So when we argue that they should have done it our way, we must remember that they had their own problems!

36-5 Other eyes

Besides the bees, many other animals can see color. Fish, butterflies, birds, and reptiles can see color, but it is believed that most mammals cannot. The primates can see color. The birds certainly see color, and that accounts for the colors of birds. There would be no point in having such brilliantly colored males if the females could not notice it! That is, the evolution of the sexual "whatever it is" that the birds have is a result of the female being able to see color. So next time we look at a peacock and think of what a brilliant display of gorgeous color it is, and how delicate all the colors are, and what a wonderful aesthetic sense it takes to appreciate all that, we should not compliment the peacock, but should compliment the visual acuity and aesthetic sense of the *peahen*, because that is what has generated the beautiful scene!

All invertebrates have poorly developed eyes or compound eyes, but all the vertebrates have eyes very similar to our own, with one exception. If we consider the highest form of animal, we usually say, "Here we are!," but if we take a less prejudiced point of view and restrict ourselves to the invertebrates, so that we cannot include ourselves, and ask what is the highest invertebrate animal, most zoologists agree that the *octopus* is the highest animal! It is very interesting that, besides the development of its brain and its reactions and so on, which are rather good for an invertebrate, it has also developed, independently, a different eye. It is not a compound eye or an eye spot—it has a cornea, it has lids, it has an iris, it has a lens, it has two regions of water, it has a retina behind. It is essentially the same as the eye of the vertebrates! It is a remarkable example of a coincidence in evolution where nature has twice discovered the same solution to a problem, with one slight improvement. In the octopus it also turns out, amazingly, that the retina is a piece of the brain that has come out in the same way in its embryonic development as is true for vertebrates, but the interesting thing which is different is that the cells which are sensitive to light are on the *inside*, and the cells which do the calculation are in back of them, rather than "inside out," as in our eye. So we see, at least, that there is no good reason for its being inside out. The other time nature tried it, she got it straightened out! (See Fig. 36-10.) The biggest eyes in the world are those of the giant squid; they have been found up to 15 inches in diameter!

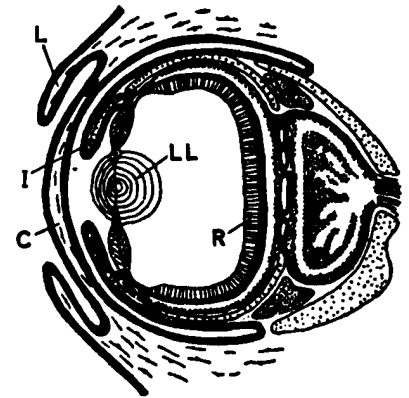


Fig. 36-10. The eye of an octopus.

36-6 Neurology of vision

One of the main points of our subject is the interconnection of information from one part of the eye to the other. Let us consider the compound eye of the horseshoe crab, on which considerable experimentation has been done. First of all, we must appreciate what kind of information can come along nerves. A nerve carries a kind of disturbance which has an electrical effect that is easy to detect, a kind of wavelike disturbance which runs down the nerve and produces an effect at the other end: a long piece of the nerve cell, called the axon, carries the information along, and a certain kind of impulse, called a "spike," goes along if it is excited at one end. When one spike goes down the nerve, another cannot immediately follow. All the spikes are of the same size, so it is not that we get *higher* spikes when the thing is more strongly excited, but that we get *more spikes per second*. The *size* of the spike is determined by the fiber. It is important to appreciate this in order to see what happens next.

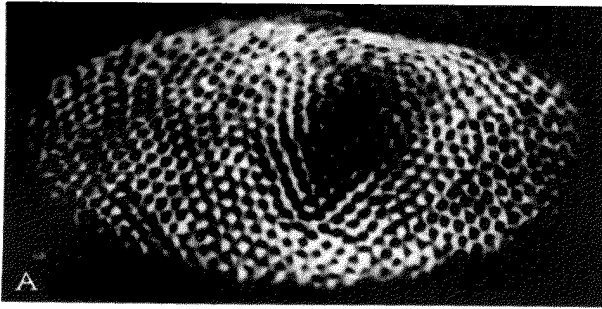


Fig. 36-11. The compound eye of the horseshoe crab. (a) Normal view. (b) Cross section.

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Figure 36-11(a) shows the compound eye of the horseshoe crab; it is not very much of an eye, it has only about a thousand ommatidia. Figure 36-11(b) is a cross section through the system; one can see the ommatidia, with the nerve fibers that run out of them and go into the brain. But note that even in a horseshoe crab there are little interconnections. They are much less elaborate than in the human eye, and it gives us a chance to study a simpler example.

Let us now look at the experiments which have been done by putting fine electrodes into the optic nerve of the horseshoe crab, and shining light on only one of the ommatidia, which is easy to do with lenses. If we turn a light on at some instant t_0 , and measure the electrical pulses that come out, we find that there is a slight delay and then a rapid series of discharges which gradually slow down to a uniform rate, as shown in Fig. 36-12(a). When the light goes out, the discharge stops. Now it is very interesting that if, while our amplifier is connected to this same nerve fiber, we shine light on a *different* ommatidium nothing happens; no signal.

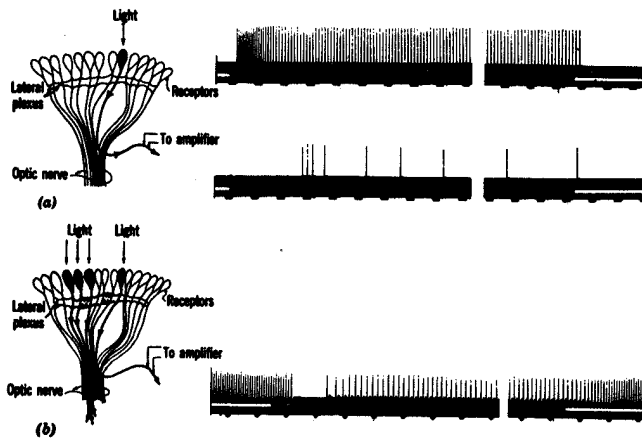


Fig. 36-12. The response to light of the nerve fibers of the eye of the horseshoe crab.

Now we do another experiment: we shine the light on the original ommatidium and get the same response, but if we now turn light on another one nearby as well, the pulses are interrupted briefly and then run at a much lower rate (Fig. 36-12b). The rate of one is inhibited by the impulses which are coming out of the other! In other words, each nerve fiber carries the information from one ommatidium, but the amount that it carries is inhibited by the signals from the others. So, for example, if the whole eye is more or less uniformly illuminated, the information coming from any one ommatidium will be relatively weak, because it is inhibited by so many. In fact the inhibition is additive—if we shine light on several nearby ommatidia the inhibition is very great. The inhibition is greater when the ommatidia are closer, and if the ommatidia are far enough away from one another, inhibition is practically zero. So it is additive and depends on the distance; here is a first example of information from different parts of the eye being combined in the eye itself. We can see, perhaps, if we think about it awhile, that this is a device to *enhance contrast* at the edges of objects, because if a part of the scene is light and a part is black, then the ommatidia in the lighted area give impulses that are

inhibited by all the other light in the neighborhood, so it is relatively weak. On the other hand, an ommatidium at the boundary which is given a “white” impulse is also inhibited by others in the neighborhood, but there are not as many of them, since some are black; the net signal is therefore stronger. The result would be a curve, something like that of Fig. 36-13. The crab will see an enhancement of the contour.

The fact that there is an enhancement of contours has long been known; in fact it is a remarkable thing that has been commented on by psychologists many times. In order to draw an object, we have only to draw its outline. How used we are to looking at pictures that have only the outline! What is the outline? The outline is only the edge difference between light and dark or one color and another. It is not something definite. It is not, believe it or not, that every object has a line around it! There is no such line. It is only in our own psychological makeup that there is a line; we are beginning to understand the reasons why the “line” is enough of a clue to get the whole thing. Presumably our own eye works in some similar manner—much more complicated, but similar.

Finally, we shall briefly describe the more elaborate work, the beautiful, advanced work that has been done on the frog. Doing a corresponding experiment on a frog, by putting very fine, beautifully built needlelike probes into the optic nerve of a frog, one can obtain the signals that are going along one particular axon and, just as in the case of the horseshoe crab, we find that the information does not depend on just one spot in the eye, but is a sum of information over several spots.

The most recent picture of the operation of the frog’s eye is the following. One can find four different kinds of optic nerve fibers, in the sense that there are four different kinds of responses. These experiments were not done by shining on-and-off impulses of light, because that is not what a frog sees. A frog just sits there and his eyes never move, unless the lily pad is flopping back and forth, and in that case his eyes wobble just right so that the image stays put. He does not turn his eyes. If anything moves in his field of vision, like a little bug (he has to be able to see something small moving in the fixed background), it turns out that there are four different kinds of fibers which discharge, whose properties are summarized in Table 36-1. Sustained edge detection, nonerasable, means that if we bring an object with an edge into the field of view of the frog, then there are a lot of impulses in this particular fiber while the object is moving, but they die down to a sustained impulse that continues as long as the edge is there, even if it is standing still. If we turn out the light, the impulses stop. If we turn it on again while the edge is still in view, they start again. They are not erasable. Another kind of fiber is very similar, except that if the edge is straight it does not work. It must be a convex edge with dark behind it! How complicated must be the system of interconnections in the retina of the eye of the frog in order for it to understand that a convex surface has moved in! Furthermore, although this fiber does sustain somewhat, it does not sustain as long as the other, and if we turn out the light and turn it on again it does *not* build up again. It depends on the moving in of the convex

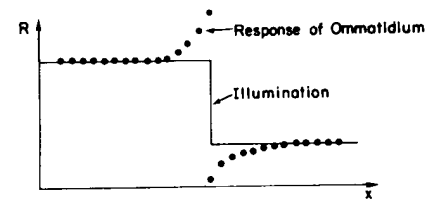


Fig. 36-13. The net response of horseshoe crab ommatidia near a sharp change in illumination.

Table 36-1

Types of response in optic nerve fibers of a frog

Type	Speed	Angular field
1. Sustained edge detection (nonerasable)	0.2–0.5 m/sec	1°
2. Convex edge detection (erasable)	0.5 m/sec	2°–3°
3. Changing contrast detection	1–2 m/sec	7°–10°
4. Dimming detection	Up to $\frac{1}{2}$ m/sec	Up to 15°
5. Darkness detection	?	Very large

surface. The eye sees it move in and remembers that it is there, but if we merely turn out the light for a moment, it simply forgets it and no longer sees it.

Another example is change-in-contrast detection. If there is an edge moving in or out there are pulses, but if the thing stands still there are no pulses at all.

Then there is a dimming detector. If the light intensity is going down it creates pulses, but if it stays down or stays up, the impulse stops; it only works while the light is dimming.

Then, finally, there are a few fibers which are dark detectors—a most amazing thing—they fire all the time! If we increase the light, they fire less rapidly, but all the time. If we decrease the light, they fire more rapidly, all the time. In the dark they fire like mad, perpetually saying, “It is dark! It is dark! It is dark!”

Now these responses seem to be rather complicated to classify, and we might wonder whether perhaps the experiments are being misinterpreted. But it is very interesting that these same classes are very clearly separated in the anatomy of the frog! By other measurements, after these responses had been classified (*afterwards*, that is what is important about this), it was discovered that the *speed* of the signals on the different fibers was not the same, so here was another, independent way to check which kind of a fiber we have found!

Another interesting question is from how big an area is one particular fiber making its calculations? The answer is different for the different classes.

Figure 36-14 shows the surface of the so-called tectum of a frog, where the nerves come into the brain from the optic nerve. All the nerve fibers coming in from the optic nerve make connections in various layers of the tectum. This layered structure is analogous to the retina; that is partly why we know that the brain and retina are very similar. Now, by taking an electrode and moving it down in succession through the layers, we can find out which kinds of optic nerves end where, and the beautiful and wonderful result is that the different kinds of fibers end in different layers! The first ones end in number 1 type, the second in number 2, the threes and fives end in the same place, and deepest of all is number four. (What a coincidence, they got the numbers almost in the right order! No, that is why they numbered them that way, the first paper had the numbers in a different order!)

We may briefly summarize what we have just learned this way: There are three pigments, presumably. There may be many different kinds of receptor cells containing the three pigments in different proportions, but there are many cross connections which may permit additions and subtractions through addition and reinforcement in the nervous system. So before we really understand color vision, we will have to understand the final sensation. This subject is still an open one, but these researches with microelectrodes and so on will perhaps ultimately give us more information on how we see color.



Fig. 36-14. The tectum of a frog.

BIBLIOGRAPHY

Committee on Colorimetry, Optical Society of America, *The Science of Color*, Thomas Y. Crowell Company, New York, 1953.

“Mechanisms of Vision,” 2nd Supplement to *Journal of General Physiology*, Vol. 43, No. 6, Part 2, July 1960, Rockefeller Institute Press.

SPECIFIC ARTICLES:

DEROBERTIS, E., “Some Observations on the Ultrastructure and Morphogenesis of Photoreceptors,” pp. 1-15.

HURVICH, L. M. and D. JAMESON, “Perceived Color, Induction Effects, and Opponent-Response Mechanisms,” pp. 63-80.

ROSENBLITH, W. A., ed., *Sensory Communication*, Massachusetts Institute of Technology Press, Cambridge, Mass., 1961.

“Sight, Sense of,” *Encyclopaedia Britannica*, Vol. 20, 1957, pp. 628-635.

Quantum Behavior

37-1 Atomic mechanics

In the last few chapters we have treated the essential ideas necessary for an understanding of most of the important phenomena of light—or electromagnetic radiation in general. (We have left a few special topics for next year. Specifically, the theory of the index of dense materials and total internal reflection.) What we have dealt with is called the “classical theory” of electric waves, which turns out to be a completely adequate description of nature for a large number of effects. We have not had to worry yet about the fact that light energy comes in lumps or “photons.”

We would like to take up as our next subject the problem of the behavior of relatively large pieces of matter—their mechanical and thermal properties, for instance. In discussing these, we will find that the “classical” (or older) theory fails almost immediately, because matter is really made up of atomic-sized particles. Still, we will deal only with the classical part, because that is the only part that we can understand using the classical mechanics we have been learning. But we shall not be very successful. We shall find that in the case of matter, unlike the case of light, we shall be in difficulty relatively soon. We could, of course, continuously skirt away from the atomic effects, but we shall instead interpose here a short excursion in which we will describe the basic ideas of the quantum properties of matter, i.e., the quantum ideas of atomic physics, so that you will have some feeling for what it is we are leaving out. For we will have to leave out some important subjects that we cannot avoid coming close to.

So we will give now the *introduction* to the subject of quantum mechanics, but will not be able actually to get into the subject until much later.

“Quantum mechanics” is the description of the behavior of matter in all its details and, in particular, of the happenings on an atomic scale. Things on a very small scale behave like nothing that you have any direct experience about. They do not behave like waves, they do not behave like particles, they do not behave like clouds, or billiard balls, or weights on springs, or like anything that you have ever seen.

Newton thought that light was made up of particles, but then it was discovered, as we have seen here, that it behaves like a wave. Later, however (in the beginning of the twentieth century) it was found that light did indeed sometimes behave like a particle. Historically, the electron, for example, was thought to behave like a particle, and then it was found that in many respects it behaved like a wave. So it really behaves like neither. Now we have given up. We say: “It is like *neither*.”

There is one lucky break, however—electrons behave just like light. The quantum behavior of atomic objects (electrons, protons, neutrons, photons, and so on) is the same for all, they are all “particle waves,” or whatever you want to call them. So what we learn about the properties of electrons (which we shall use for our examples) will apply also to all “particles,” including photons of light.

The gradual accumulation of information about atomic and small-scale behavior during the first quarter of this century, which gave some indications about how small things do behave, produced an increasing confusion which was finally resolved in 1926 and 1927 by Schrödinger, Heisenberg, and Born. They finally obtained a consistent description of the behavior of matter on a small scale. We take up the main features of that description in this chapter.

Because atomic behavior is so unlike ordinary experience, it is very difficult to get used to and it appears peculiar and mysterious to everyone, both to the

37-1 Atomic mechanics

37-2 An experiment with bullets

37-3 An experiment with waves

37-4 An experiment with electrons

37-5 The interference of electron waves

37-6 Watching the electrons

37-7 First principles of quantum mechanics

37-8 The uncertainty principle

novice and to the experienced physicist. Even the experts do not understand it the way they would like to, and it is perfectly reasonable that they should not, because all of direct, human experience and of human intuition applies to large objects. We know how large objects will act, but things on a small scale just do not act that way. So we have to learn about them in a sort of abstract or imaginative fashion and not by connection with our direct experience.

In this chapter we shall tackle immediately the basic element of the mysterious behavior in its most strange form. We choose to examine a phenomenon which is impossible, *absolutely* impossible, to explain in any classical way, and which has in it the heart of quantum mechanics. In reality, it contains the *only* mystery. We cannot explain the mystery in the sense of “explaining” how it works. We will *tell* you how it works. In telling you how it works we will have told you about the basic peculiarities of all quantum mechanics.

37-2 An experiment with bullets

To try to understand the quantum behavior of electrons, we shall compare and contrast their behavior, in a particular experimental setup, with the more familiar behavior of particles like bullets, and with the behavior of waves like water waves. We consider first the behavior of bullets in the experimental setup shown diagrammatically in Fig. 37-1. We have a machine gun that shoots a stream of bullets. It is not a very good gun, in that it sprays the bullets (randomly) over a fairly large angular spread, as indicated in the figure. In front of the gun we have a wall (made of armor plate) that has in it two holes just about big enough to let a bullet through. Beyond the wall is a backstop (say a thick wall of wood) which will “absorb” the bullets when they hit it. In front of the wall we have an object which we shall call a “detector” of bullets. It might be a box containing sand. Any bullet that enters the detector will be stopped and accumulated. When we wish, we can empty the box and count the number of bullets that have been caught. The detector can be moved back and forth (in what we will call the x -direction). With this apparatus, we can find out experimentally the answer to the question: “What is the probability that a bullet which passes through the holes in the wall will arrive at the backstop at the distance x from the center?” First, you should realize that we should talk about probability, because we cannot say definitely where any particular bullet will go. A bullet which happens to hit one of the holes may bounce off the edges of the hole, and may end up anywhere at all. By “probability” we mean the chance that the bullet will arrive at the detector, which we can measure by counting the number which arrive at the detector in a certain time and then taking the ratio of this number to the *total* number that hit the backstop during that time. Or, if we assume that the gun always shoots at the same rate during the measurements, the probability we want is just proportional to the number that reach the detector in some standard time interval.

For our present purposes we would like to imagine a somewhat idealized experiment in which the bullets are not real bullets, but are *indestructible* bullets—they cannot break in half. In our experiment we find that bullets always arrive in lumps, and when we find something in the detector, it is always one whole bullet. If the rate at which the machine gun fires is made very low, we find that at any given moment either nothing arrives, or one and only one—exactly one—bullet arrives at the backstop. Also, the size of the lump certainly does not depend on the rate of firing of the gun. We shall say: “Bullets *always* arrive in identical lumps.” What we measure with our detector is the probability of arrival of a lump. And we measure the probability as a function of x . The result of such measurements with this apparatus (we have not yet done the experiment, so we are really imagining the result) are plotted in the graph drawn in part (c) of Fig. 37-1. In the graph we plot the probability to the right and x vertically, so that the x -scale fits the diagram of the apparatus. We call the probability P_{12} because the bullets may have come either through hole 1 or through hole 2. You will not be surprised that P_{12} is large near the middle of the graph but gets small if x is very large. You may wonder, however, why P_{12} has its maximum value at $x = 0$. We can understand

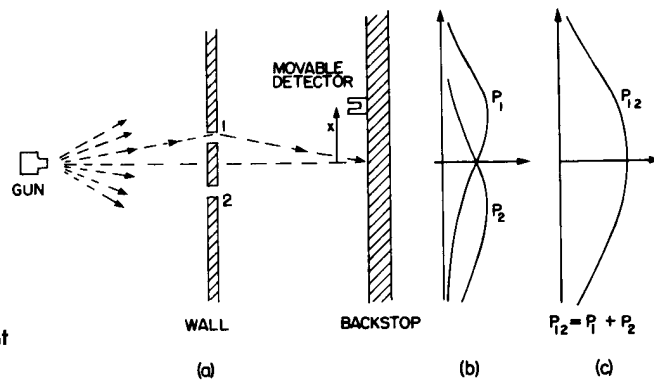


Fig. 37-1. Interference experiment with bullets.

this fact if we do our experiment again after covering up hole 2, and once more while covering up hole 1. When hole 2 is covered, bullets can pass only through hole 1, and we get the curve marked P_1 in part (b) of the figure. As you would expect, the maximum of P_1 occurs at the value of x which is on a straight line with the gun and hole 1. When hole 1 is closed, we get the symmetric curve P_2 drawn in the figure. P_2 is the probability distribution for bullets that pass through hole 2. Comparing parts (b) and (c) of Fig. 37-1, we find the important result that

$$P_{12} = P_1 + P_2. \quad (37.1)$$

The probabilities just add together. The effect with both holes open is the sum of the effects with each hole open alone. We shall call this result an observation of “no interference,” for a reason that you will see later. So much for bullets. They come in lumps, and their probability of arrival shows no interference.

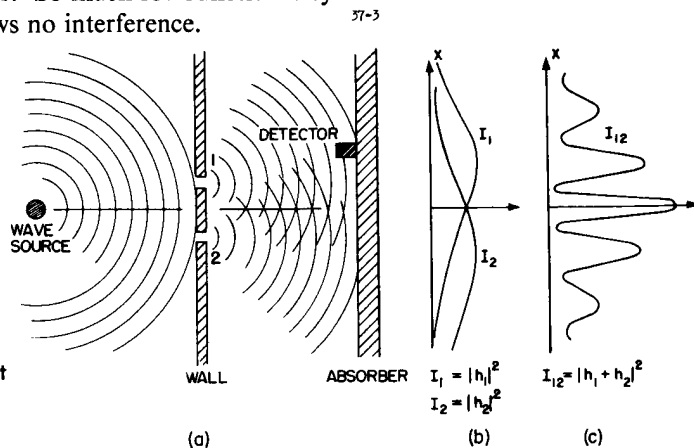


Fig. 37-2. Interference experiment with water waves.

37-3 An experiment with waves

Now we wish to consider an experiment with water waves. The apparatus is shown diagrammatically in Fig. 37-2. We have a shallow trough of water. A small object labeled the “wave source” is jiggled up and down by a motor and makes circular waves. To the right of the source we have again a wall with two holes, and beyond that is a second wall, which, to keep things simple, is an “absorber,” so that there is no reflection of the waves that arrive there. This can be done by building a gradual sand “beach.” In front of the beach we place a detector which can be moved back and forth in the x -direction, as before. The detector is now a device which measures the “intensity” of the wave motion. You can imagine a gadget which measures the height of the wave motion, but whose scale is calibrated in proportion to the *square* of the actual height, so that the reading is proportional to the intensity of the wave. Our detector reads, then, in proportion to the *energy* being carried by the wave—or rather, the rate at which energy is carried to the detector.

With our wave apparatus, the first thing to notice is that the intensity can have *any* size. If the source just moves a very small amount, then there is just a little bit of wave motion at the detector. When there is more motion at the source,

there is more intensity at the detector. The intensity of the wave can have any value at all. We would *not* say that there was any “lumpiness” in the wave intensity.

Now let us measure the wave intensity for various values of x (keeping the wave source operating always in the same way). We get the interesting-looking curve marked I_{12} in part (c) of the figure.

We have already worked out how such patterns can come about when we studied the interference of electric waves. In this case we would observe that the original wave is diffracted at the holes, and new circular waves spread out from each hole. If we cover one hole at a time and measure the intensity distribution at the absorber we find the rather simple intensity curves shown in part (b) of the figure. I_1 is the intensity of the wave from hole 1 (which we find by measuring when hole 2 is blocked off) and I_2 is the intensity of the wave from hole 2 (seen when hole 1 is blocked).

The intensity I_{12} observed when both holes are open is certainly *not* the sum of I_1 and I_2 . We say that there is “interference” of the two waves. At some places (where the curve I_{12} has its maxima) the waves are “in phase” and the wave peaks add together to give a large amplitude and, therefore, a large intensity. We say that the two waves are “interfering constructively” at such places. There will be such constructive interference wherever the distance from the detector to one hole is a whole number of wavelengths larger (or shorter) than the distance from the detector to the other hole.

At those places where the two waves arrive at the detector with a phase difference of π (where they are “out of phase”) the resulting wave motion at the detector will be the difference of the two amplitudes. The waves “interfere destructively,” and we get a low value for the wave intensity. We expect such low values wherever the distance between hole 1 and the detector is different from the distance between hole 2 and the detector by an odd number of half-wavelengths. The low values of I_{12} in Fig. 37-2 correspond to the places where the two waves interfere destructively.

You will remember that the quantitative relationship between I_1 , I_2 , and I_{12} can be expressed in the following way: The instantaneous height of the water wave at the detector for the wave from hole 1 can be written as (the real part of) $\hat{h}_1 e^{i\omega t}$, where the “amplitude” $\hat{h}_1$ is, in general, a complex number. The intensity is proportional to the mean squared height or, when we use the complex numbers, to $|\hat{h}_1|^2$. Similarly, for hole 2 the height is $\hat{h}_2 e^{i\omega t}$ and the intensity is proportional to $|\hat{h}_2|^2$. When both holes are open, the wave heights add to give the height $(\hat{h}_1 + \hat{h}_2)e^{i\omega t}$ and the intensity $|\hat{h}_1 + \hat{h}_2|^2$. Omitting the constant of proportionality for our present purposes, the proper relations for *interfering waves* are

$$I_1 = |\hat{h}_1|^2, \quad I_2 = |\hat{h}_2|^2, \quad I_{12} = |\hat{h}_1 + \hat{h}_2|^2. \quad (37.2)$$

You will notice that the result is quite different from that obtained with bullets (Eq. 37.1). If we expand $|\hat{h}_1 + \hat{h}_2|^2$ we see that

$$|\hat{h}_1 + \hat{h}_2|^2 = |\hat{h}_1|^2 + |\hat{h}_2|^2 + 2|\hat{h}_1||\hat{h}_2| \cos \delta, \quad (37.3)$$

where δ is the phase difference between $\hat{h}_1$ and $\hat{h}_2$. In terms of the intensities, we could write

$$I_{12} = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta. \quad (37.4)$$

The last term in (37.4) is the “interference term.” So much for water waves. The intensity can have any value, and it shows interference.

37-4 An experiment with electrons

Now we imagine a similar experiment with electrons. It is shown diagrammatically in Fig. 37-3. We make an electron gun which consists of a tungsten wire heated by an electric current and surrounded by a metal box with a hole in it. If the wire is at a negative voltage with respect to the box, electrons emitted by the wire will be accelerated toward the walls and some will pass through the hole. All the electrons which come out of the gun will have (nearly) the same energy. In front of the gun is again a wall (just a thin metal plate) with two holes in it.

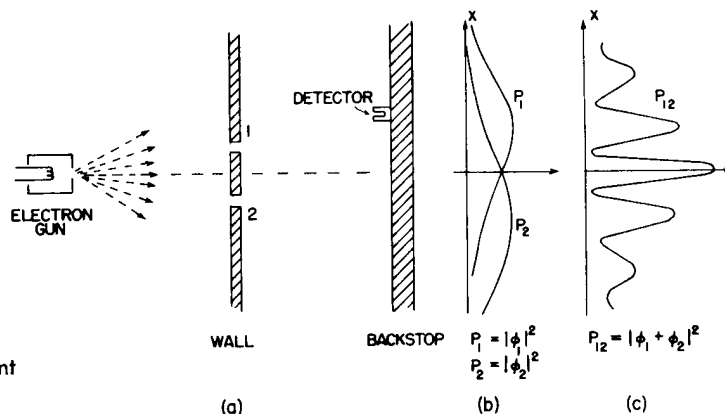


Fig. 37-3. Interference experiment with electrons.

Beyond the wall is another plate which will serve as a “backstop.” In front of the backstop we place a movable detector. The detector might be a geiger counter or, perhaps better, an electron multiplier, which is connected to a loudspeaker.

We should say right away that you should not try to set up this experiment (as you could have done with the two we have already described). This experiment has never been done in just this way. The trouble is that the apparatus would have to be made on an impossibly small scale to show the effects we are interested in. We are doing a “thought experiment,” which we have chosen because it is easy to think about. We know the results that *would* be obtained because there *are* many experiments that have been done, in which the scale and the proportions have been chosen to show the effects we shall describe.

The first thing we notice with our electron experiment is that we hear sharp “clicks” from the detector (that is, from the loudspeaker). And all “clicks” are the same. There are *no* “half-clicks.”

We would also notice that the “clicks” come very erratically. Something like: click click-click . . . click click click-click click . . . , etc., just as you have, no doubt, heard a geiger counter operating. If we count the clicks which arrive in a sufficiently long time—say for many minutes—and then count again for another equal period, we find that the two numbers are very nearly the same. So we can speak of the *average rate* at which the clicks are heard (so-and-so-many clicks per minute on the average).

As we move the detector around, the *rate* at which the clicks appear is faster or slower, but the size (loudness) of each click is always the same. If we lower the temperature of the wire in the gun the rate of clicking slows down, but still each click sounds the same. We would notice also that if we put two separate detectors at the backstop, one *or* the other would click, but never both at once. (Except that once in a while, if there were two clicks very close together in time, our ear might not sense the separation.) We conclude, therefore, that whatever arrives at the backstop arrives in “lumps.” All the “lumps” are the same size: only whole “lumps” arrive, and they arrive one at a time at the backstop. We shall say: “Electrons always arrive in identical lumps.”

Just as for our experiment with bullets, we can now proceed to find experimentally the answer to the question: “What is the relative probability that an electron ‘lump’ will arrive at the backstop at various distances x from the center?” As before, we obtain the relative probability by observing the rate of clicks, holding the operation of the gun constant. The probability that lumps will arrive at a particular x is proportional to the average rate of clicks at that x .

The result of our experiment is the interesting curve marked P_{12} in part (c) of Fig. 37-3. Yes! That is the way electrons go.

37-5 The interference of electron waves

Now let us try to analyze the curve of Fig. 37-3 to see whether we can understand the behavior of the electrons. The first thing we would say is that since they come in lumps, each lump, which we may as well call an electron, has come either through hole 1 or through hole 2. Let us write this in the form of a “Proposition”:

Proposition A: Each electron *either* goes through hole 1 *or* it goes through hole 2.

Assuming Proposition A, all electrons that arrive at the backstop can be divided into two classes: (1) those that come through hole 1, and (2) those that come through hole 2. So our observed curve must be the sum of the effects of the electrons which come through hole 1 and the electrons which come through hole 2. Let us check this idea by experiment. First, we will make a measurement for those electrons that come through hole 1. We block off hole 2 and make our counts of the clicks from the detector. From the clicking rate, we get P_1 . The result of the measurement is shown by the curve marked P_1 in part (b) of Fig. 37-3. The result seems quite reasonable. In a similar way, we measure P_2 , the probability distribution for the electrons that come through hole 2. The result of this measurement is also drawn in the figure.

The result P_{12} obtained with *both* holes open is clearly not the sum of P_1 and P_2 , the probabilities for each hole alone. In analogy with our water-wave experiment, we say: "There is interference."

$$\text{For electrons: } P_{12} \neq P_1 + P_2. \quad (37.5)$$

How can such an interference come about? Perhaps we should say: "Well, that means, presumably, that it is *not true* that the lumps go either through hole 1 or hole 2, because if they did, the probabilities should add. Perhaps they go in a more complicated way. They split in half and . . ." But no! They cannot, they always arrive in lumps . . . "Well, perhaps some of them go through 1, and then they go around through 2, and then around a few more times, or by some other complicated path . . . then by closing hole 2, we changed the chance that an electron that *started out* through hole 1 would finally get to the backstop . . ." But notice! There are some points at which very few electrons arrive when *both* holes are open, but which receive many electrons if we close one hole, so *closing* one hole *increased* the number from the other. Notice, however, that at the center of the pattern, P_{12} is more than twice as large as $P_1 + P_2$. It is as though closing one hole *decreased* the number of electrons which come through the other hole. It seems hard to explain *both* effects by proposing that the electrons travel in complicated paths.

It is all quite mysterious. And the more you look at it the more mysterious it seems. Many ideas have been concocted to try to explain the curve for P_{12} in terms of individual electrons going around in complicated ways through the holes. None of them has succeeded. None of them can get the right curve for P_{12} in terms of P_1 and P_2 .

Yet, surprisingly enough, the *mathematics* for relating P_1 and P_2 to P_{12} is extremely simple. For P_{12} is just like the curve I_{12} of Fig. 37-2, and *that* was simple. What is going on at the backstop can be described by two complex numbers that we can call $\hat{\phi}_1$ and $\hat{\phi}_2$ (they are functions of x , of course). The absolute square of $\hat{\phi}_1$ gives the effect with only hole 1 open. That is, $P_1 = |\hat{\phi}_1|^2$. The effect with only hole 2 open is given by $\hat{\phi}_2$ in the same way. That is, $P_2 = |\hat{\phi}_2|^2$. And the combined effect of the two holes is just $P_{12} = |\hat{\phi}_1 + \hat{\phi}_2|^2$. The *mathematics* is the same as that we had for the water waves! (It is hard to see how one could get such a simple result from a complicated game of electrons going back and forth through the plate on some strange trajectory.)

We conclude the following: The electrons arrive in lumps, like particles, and the probability of arrival of these lumps is distributed like the distribution of intensity of a wave. It is in this sense that an electron behaves "sometimes like a particle and sometimes like a wave."

Incidentally, when we were dealing with classical waves we defined the intensity as the mean over time of the square of the wave amplitude, and we used complex numbers as a mathematical trick to simplify the analysis. But in quantum mechanics it turns out that the amplitudes *must* be represented by complex numbers. The real parts alone will not do. That is a technical point, for the moment, because the formulas look just the same.

Since the probability of arrival through both holes is given so simply, although it is not equal to $(P_1 + P_2)$, that is really all there is to say. But there are a large number of subtleties involved in the fact that nature does work this way. We would like to illustrate some of these subtleties for you now. First, since the number that arrives at a particular point is *not* equal to the number that arrives through 1 plus the number that arrives through 2, as we would have concluded from Proposition A, undoubtedly we should conclude that *Proposition A is false*. It is *not* true that the electrons go *either* through hole 1 or hole 2. But that conclusion can be tested by another experiment.

37-6 Watching the electrons

We shall now try the following experiment. To our electron apparatus we add a very strong light source, placed behind the wall and between the two holes, as shown in Fig. 37-4. We know that electric charges scatter light. So when an electron passes, however it does pass, on its way to the detector, it will scatter some light to our eye, and we can *see* where the electron goes. If, for instance, an electron were to take the path via hole 2 that is sketched in Fig. 37-4, we should see a flash of light coming from the vicinity of the place marked *A* in the figure. If an electron passes through hole 1 we would expect to see a flash from the vicinity of the upper hole. If it should happen that we get light from both places at the same time, because the electron divides in half . . . Let us just do the experiment!

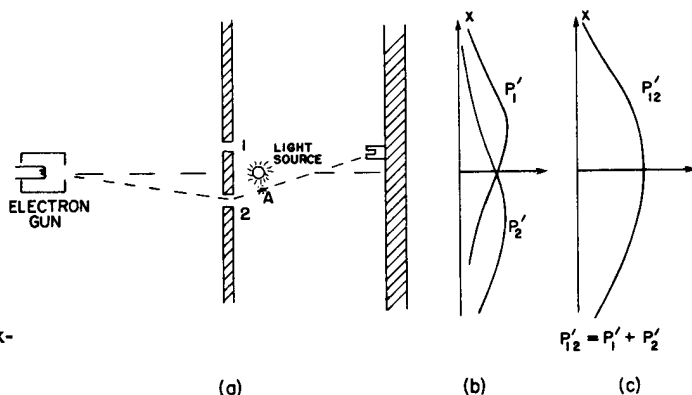


Fig. 37-4. A different electron experiment.

Here is what we see: *every* time that we hear a “click” from our electron detector (at the backstop), we *also* see a flash of light *either* near hole 1 *or* near hole 2, but *never* both at once! And we observe the same result no matter where we put the detector. From this observation we conclude that when we look at the electrons we find that the electrons go either through one hole or the other. Experimentally, Proposition A is necessarily true.

What, then, is wrong with our argument *against* Proposition A? Why *isn't* P_{12} just equal to $P_1 + P_2$? Back to experiment! Let us keep track of the electrons and find out what they are doing. For each position (x -location) of the detector we will count the electrons that arrive and *also* keep track of which hole they went through, by watching for the flashes. We can keep track of things this way: whenever we hear a “click” we will put a count in Column 1 if we see the flash near hole 1, and if we see the flash near hole 2, we will record a count in Column 2. Every electron which arrives is recorded in one of two classes: those which come through 1 and those which come through 2. From the number recorded in Column 1 we get the probability P'_1 that an electron will arrive at the detector via hole 1; and from the number recorded in Column 2 we get P'_2 , the probability that an electron will arrive at the detector via hole 2. If we now repeat such a measurement for many values of x , we get the curves for P'_1 and P'_2 shown in part (b) of Fig. 37-4.

Well, that is not too surprising! We get for P'_1 something quite similar to what we got before for P_1 by blocking off hole 2; and P'_2 is similar to what we got by blocking hole 1. So there is *not* any complicated business like going through both holes. When we watch them, the electrons come through just as we would

expect them to come through. Whether the holes are closed or open, those which we see come through hole 1 are distributed in the same way whether hole 2 is open or closed.

But wait! What do we have *now* for the *total* probability, the probability that an electron will arrive at the detector by any route? We already have that information. We just pretend that we never looked at the light flashes, and we lump together the detector clicks which we have separated into the two columns. We *must* just *add* the numbers. For the probability that an electron will arrive at the backstop by passing through *either* hole, we do find $P'_{12} = P_1 + P_2$. That is, although we succeeded in watching which hole our electrons come through, we no longer get the old interference curve P_{12} , but a new one, P'_{12} , showing no interference! If we turn out the light P_{12} is restored.

We must conclude that *when we look at the electrons* the distribution of them on the screen is different than when we do not look. Perhaps it is turning on our light source that disturbs things? It must be that the electrons are very delicate, and the light, when it scatters off the electrons, gives them a jolt that changes their motion. We know that the electric field of the light acting on a charge will exert a force on it. So perhaps we *should* expect the motion to be changed. Anyway, the light exerts a big influence on the electrons. By trying to “watch” the electrons we have changed their motions. That is, the jolt given to the electron when the photon is scattered by it is such as to change the electron’s motion enough so that if it *might* have gone to where P_{12} was at a maximum it will instead land where P_{12} was a minimum; that is why we no longer see the wavy interference effects.

You may be thinking: “Don’t use such a bright source! Turn the brightness down! The light waves will then be weaker and will not disturb the electrons so much. Surely, by making the light dimmer and dimmer, eventually the wave will be weak enough that it will have a negligible effect.” O.K. Let’s try it. The first thing we observe is that the flashes of light scattered from the electrons as they pass by does *not* get weaker. *It is always the same-sized flash.* The only thing that happens as the light is made dimmer is that sometimes we hear a “click” from the detector but see *no flash at all*. The electron has gone by without being “seen.” What we are observing is that light *also* acts like electrons, we *knew* that it was “wavy,” but now we find that it is also “lumpy.” It always arrives—or is scattered—in lumps that we call “photons.” As we turn down the *intensity* of the light source we do not change the *size* of the photons, only the *rate* at which they are emitted. *That* explains why, when our source is dim, some electrons get by without being seen. There did not happen to be a photon around at the time the electron went through.

This is all a little discouraging. If it is true that whenever we “see” the electron we see the same-sized flash, then those electrons we see are *always* the disturbed ones. Let us try the experiment with a dim light anyway. Now whenever we hear a click in the detector we will keep a count in three columns: in Column (1) those electrons seen by hole 1, in Column (2) those electrons seen by hole 2, and in Column (3) those electrons not seen at all. When we work up our data (computing the probabilities) we find these results: Those “seen by hole 1” have a distribution like P'_1 ; those “seen by hole 2” have a distribution like P'_2 (so that those “seen by either hole 1 or 2” have a distribution like P'_{12}); and those “not seen at all” have a “wavy” distribution just like P_{12} of Fig. 37-3! *If the electrons are not seen, we have interference!*

That is understandable. When we do not see the electron, no photon disturbs it, and when we do see it, a photon has disturbed it. There is always the same amount of disturbance because the light photons all produce the same-sized effects and the effect of the photons being scattered is enough to smear out any interference effect.

Is there not *some* way we can see the electrons without disturbing them? We learned in an earlier chapter that the momentum carried by a “photon” is inversely proportional to its wavelength ($p = h/\lambda$). Certainly the jolt given to the electron when the photon is scattered toward our eye depends on the momentum that photon carries. Aha! If we want to disturb the electrons only

slightly we should not have lowered the *intensity* of the light, we should have lowered its *frequency* (the same as increasing its wavelength). Let us use light of a redder color. We could even use infrared light, or radiowaves (like radar), and “see” where the electron went with the help of some equipment that can “see” light of these longer wavelengths. If we use “gentler” light perhaps we can avoid disturbing the electrons so much.

Let us try the experiment with longer waves. We shall keep repeating our experiment, each time with light of a longer wavelength. At first, nothing seems to change. The results are the same. Then a terrible thing happens. You remember that when we discussed the microscope we pointed out that, due to the *wave nature* of the light, there is a limitation on how close two spots can be and still be seen as two separate spots. This distance is of the order of the wavelength of light. So now, when we make the wavelength longer than the distance between our holes, we see a *big* fuzzy flash when the light is scattered by the electrons. We can no longer tell which hole the electron went through! We just know it went somewhere! And it is just with light of this color that we find that the jolts given to the electron are small enough so that P'_{12} begins to look like P_{12} —that we begin to get some interference effect. And it is only for wavelengths much longer than the separation of the two holes (when we have no chance at all of telling where the electron went) that the disturbance due to the light gets sufficiently small that we again get the curve P_{12} shown in Fig. 37-3.

In our experiment we find that it is impossible to arrange the light in such a way that one can tell which hole the electron went through, and at the same time not disturb the pattern. It was suggested by Heisenberg that the then new laws of nature could only be consistent if there were some basic limitation on our experimental capabilities not previously recognized. He proposed, as a general principle, his *uncertainty principle*, which we can state in terms of our experiment as follows: “It is impossible to design an apparatus to determine which hole the electron passes through, that will not at the same time disturb the electrons enough to destroy the interference pattern.” If an apparatus is capable of determining which hole the electron goes through, it *cannot* be so delicate that it does not disturb the pattern in an essential way. No one has ever found (or even thought of) a way around the uncertainty principle. So we must assume that it describes a basic characteristic of nature.

The complete theory of quantum mechanics which we now use to describe atoms and, in fact, all matter depends on the correctness of the uncertainty principle. Since quantum mechanics is such a successful theory, our belief in the uncertainty principle is reinforced. But if a way to “beat” the uncertainty principle were ever discovered, quantum mechanics would give inconsistent results and would have to be discarded as a valid theory of nature.

“Well,” you say, “what about Proposition A? It is true, or is it *not* true, that the electron either goes through hole 1 or it goes through hole 2?” The only answer that can be given is that we have found from experiment that there is a certain special way that we have to think in order that we do not get into inconsistencies. What we must say (to avoid making wrong predictions) is the following. If one looks at the holes or, more accurately, if one has a piece of apparatus which is capable of determining whether the electrons go through hole 1 or hole 2, then one *can* say that it goes either through hole 1 or hole 2. *But*, when one does *not* try to tell which way the electron goes, when there is nothing in the experiment to disturb the electrons, then one may *not* say that an electron goes either through hole 1 or hole 2. If one does say that, and starts to make any deductions from the statement, he will make errors in the analysis. This is the logical tightrope on which we must walk if we wish to describe nature successfully.

If the motion of all matter—as well as electrons—must be described in terms of waves, what about the bullets in our first experiment? Why didn’t we see an interference pattern there? It turns out that for the bullets the wavelengths were so tiny that the interference patterns became very fine. So fine, in fact, that with any

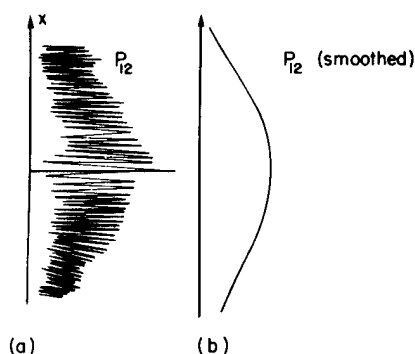


Fig. 37-5. Interference pattern with bullets: (a) actual (schematic), (b) observed.

detector of finite size one could not distinguish the separate maxima and minima. What we saw was only a kind of average, which is the classical curve. In Fig. 37-5 we have tried to indicate schematically what happens with large-scale objects. Part (a) of the figure shows the probability distribution one might predict for bullets, using quantum mechanics. The rapid wiggles are supposed to represent the interference pattern one gets for waves of very short wavelength. Any physical detector, however, straddles several wiggles of the probability curve, so that the measurements show the smooth curve drawn in part (b) of the figure.

37-7 First principles of quantum mechanics

We will now write a summary of the main conclusions of our experiments. We will, however, put the results in a form which makes them true for a general class of such experiments. We can write our summary more simply if we first define an “ideal experiment” as one in which there are no uncertain external influences, i.e., no jiggling or other things going on that we cannot take into account. We would be quite precise if we said: “An ideal experiment is one in which all of the initial and final conditions of the experiment are completely specified.” What we will call “an event” is, in general, just a specific set of initial and final conditions. (For example: “an electron leaves the gun, arrives at the detector, and nothing else happens.”) Now for our summary.

SUMMARY

- (1) The probability of an event in an ideal experiment is given by the square of the absolute value of a complex number ϕ which is called the probability amplitude.

$$\begin{aligned} P &= \text{probability,} \\ \phi &= \text{probability amplitude,} \\ P &= |\phi|^2. \end{aligned} \tag{37.6}$$

- (2) When an event can occur in several alternative ways, the probability amplitude for the event is the sum of the probability amplitudes for each way considered separately. There is interference.

$$\begin{aligned} \phi &= \phi_1 + \phi_2, \\ P &= |\phi_1 + \phi_2|^2. \end{aligned} \tag{37.7}$$

- (3) If an experiment is performed which is capable of determining whether one or another alternative is actually taken, the probability of the event is the sum of the probabilities for each alternative. The interference is lost.

$$P = P_1 + P_2. \tag{37.8}$$

One might still like to ask: “How does it work? What is the machinery behind the law?” No one has found any machinery behind the law. No one can “explain” any more than we have just “explained.” No one will give you any deeper representation of the situation. We have no ideas about a more basic mechanism from which these results can be deduced.

We would like to emphasize a very important difference between classical and quantum mechanics. We have been talking about the probability that an electron will arrive in a given circumstance. We have implied that in our experimental arrangement (or even in the best possible one) it would be impossible to predict exactly what would happen. We can only predict the odds! This would mean, if it were true, that physics has given up on the problem of trying to predict exactly what will happen in a definite circumstance. Yes! physics *has* given up. *We do not know how to predict what would happen in a given circumstance*, and we believe now that it is impossible, that the only thing that can be predicted is the probability of different events. It must be recognized that this is a retrenchment in our earlier ideal of understanding nature. It may be a backward step, but no one has seen a way to avoid it.

We make now a few remarks on a suggestion that has sometimes been made to try to avoid the description we have given: "Perhaps the electron has some kind of internal works—some inner variables—that we do not yet know about. Perhaps that is why we cannot predict what will happen. If we could look more closely at the electron we could be able to tell where it would end up." So far as we know, that is impossible. We would still be in difficulty. Suppose we were to assume that inside the electron there is some kind of machinery that determines where it is going to end up. That machine must *also* determine which hole it is going to go through on its way. But we must not forget that what is inside the electron should not be dependent on what *we* do, and in particular upon whether we open or close one of the holes. So if an electron, before it starts, has already made up its mind (a) which hole it is going to use, and (b) where it is going to land, we should find P_1 for those electrons that have chosen hole 1, P_2 for those that have chosen hole 2, and *necessarily* the sum $P_1 + P_2$ for those that arrive through the two holes. There seems to be no way around this. But we have verified experimentally that that is not the case. And no one has figured a way out of this puzzle. So at the present time we must limit ourselves to computing probabilities. We say "at the present time," but we suspect very strongly that it is something that will be with us forever—that it is impossible to beat that puzzle—that this is the way nature really *is*.

37-8 The uncertainty principle

This is the way Heisenberg stated the uncertainty principle originally: If you make the measurement on any object, and you can determine the x -component of its momentum with an uncertainty Δp , you cannot, at the same time, know its x -position more accurately than $\Delta x = h/\Delta p$. The uncertainties in the position and momentum at any instant must have their product greater than Planck's constant. This is a special case of the uncertainty principle that was stated above more generally. The more general statement was that one cannot design equipment in any way to determine which of two alternatives is taken, without, at the same time, destroying the pattern of interference.

Let us show for one particular case that the kind of relation given by Heisenberg must be true in order to keep from getting into trouble. We imagine a modification of the experiment of Fig. 37-3, in which the wall with the holes consists of a plate mounted on rollers so that it can move freely up and down (in the x -direction), as shown in Fig. 37-6. By watching the motion of the plate carefully we can try to tell which hole an electron goes through. Imagine what happens when the detector is placed at $x = 0$. We would expect that an electron which passes through hole 1 must be deflected downward by the plate to reach the detector. Since the vertical component of the electron momentum is changed, the plate must recoil with an equal momentum in the opposite direction. The plate will get an upward kick. If the electron goes through the lower hole, the plate should feel a downward kick. It is clear that for every position of the detector, the momentum received by the plate will have a different value for a traversal via hole 1 than for a traversal via hole 2. So! Without disturbing the electrons *at all*, but just by watching the *plate*, we can tell which path the electron used.

Now in order to do this it is necessary to know what the momentum of the screen is, before the electron goes through. So when we measure the momentum after the electron goes by, we can figure out how much the plate's momentum has changed. But remember, according to the uncertainty principle we cannot at the same time know the position of the plate with an arbitrary accuracy. But if we do not know exactly *where* the plate is we cannot say precisely where the two holes are. They will be in a different place for every electron that goes through. This means that the center of our interference pattern will have a different location for each electron. The wiggles of the interference pattern will be smeared out. We shall show quantitatively in the next chapter that if we determine the momentum of the plate sufficiently accurately to determine from the recoil measurement which hole was used, then the uncertainty in the x -position of the plate will, according to the un-

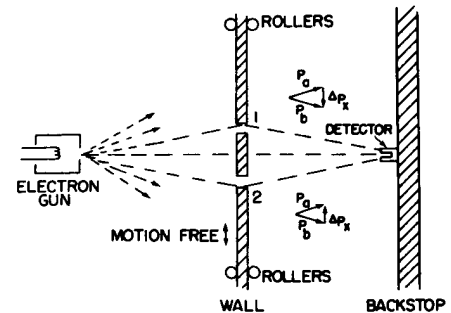


Fig. 37-6. An experiment in which the recoil of the wall is measured.

certainty principle, be enough to shift the pattern observed at the detector up and down in the x -direction about the distance from a maximum to its nearest minimum. Such a random shift is just enough to smear out the pattern so that no interference is observed.

The uncertainty principle “protects” quantum mechanics. Heisenberg recognized that if it were possible to measure the momentum and the position simultaneously with a greater accuracy, the quantum mechanics would collapse. So he proposed that it must be impossible. Then people sat down and tried to figure out ways of doing it, and nobody could figure out a way to measure the position and the momentum of anything—a screen, an electron, a billiard ball, anything—with any greater accuracy. Quantum mechanics maintains its perilous but accurate existence.

The Relation of Wave and Particle Viewpoints

38-1 Probability wave amplitudes

In this chapter we shall discuss the relationship of the wave and particle viewpoints. We already know, from the last chapter, that neither the wave viewpoint nor the particle viewpoint is correct. Usually we have tried to present things accurately, or at least precisely enough that they will not have to be changed when we learn more—it may be extended, but it will not be changed! But when we try to talk about the wave picture or the particle picture, both are approximate, and both will change. Therefore what we learn in this chapter will not be accurate in a certain sense; it is a kind of half-intuitive argument that will be made more precise later, but certain things will be changed a little bit when we interpret them correctly in quantum mechanics. The reason for doing such a thing, of course, is that we are not going to go directly into quantum mechanics, but we want to have at least some idea of the kinds of effects that we will find. Furthermore, all our experiences are with waves and with particles, and so it is rather handy to use the wave and particle ideas to get some understanding of what happens in given circumstances before we know the complete mathematics of the quantum-mechanical amplitudes. We shall try to illustrate the weakest places as we go along, but most of it is very nearly correct—it is just a matter of interpretation.

First of all, we know that the new way of representing the world in quantum mechanics—the new framework—is to give an amplitude for every event that can occur, and if the event involves the reception of one particle then we can give the amplitude to find that one particle at different places and at different times. The probability of finding the particle is then proportional to the absolute square of the amplitude. In general, the amplitude to find a particle in different places at different times varies with position and time.

In a special case the amplitude varies sinusoidally in space and time like $e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}$ (do not forget that these amplitudes are complex numbers, not real numbers) and involves a definite frequency ω and wave number $\mathbf{k}$. Then it turns out that this corresponds to a classical limiting situation where we would have believed that we have a particle whose energy E was known and is related to the frequency by

$$E = \hbar\omega, \quad (38.1)$$

and whose momentum $\mathbf{p}$ is also known and is related to the wave number by

$$\mathbf{p} = \hbar\mathbf{k}. \quad (38.2)$$

This means that the idea of a particle is limited. The idea of a particle—its location, its momentum, etc.—which we use so much, is in certain ways unsatisfactory. For instance, if an amplitude to find a particle at different places is given by $e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}$, whose absolute square is a constant, that would mean that the probability of finding a particle is the same at all points. That means we do not know where it is—it can be anywhere—there is a great uncertainty in its location.

On the other hand, if the position of a particle is more or less well known and we can predict it fairly accurately, then the probability of finding it in different places must be confined to a certain region, whose length we call Δx . Outside this region, the probability is zero. Now this probability is the absolute square of an amplitude, and if the absolute square is zero, the amplitude is also zero, so that

38-1 Probability wave amplitudes

38-2 Measurement of position and momentum

38-3 Crystal diffraction

38-4 The size of an atom

38-5 Energy levels

38-6 Philosophical implications

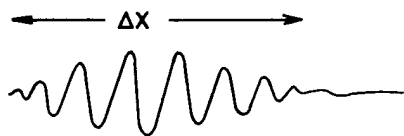


Fig. 38-1. A wave packet of length Δx .

we have a wave train whose length is Δx (Fig. 38-1), and the wavelength (the distance between nodes of the waves in the train) of that wave train is what corresponds to the particle momentum.

Here we encounter a strange thing about waves; a very simple thing which has nothing to do with quantum mechanics strictly. It is something that anybody who works with waves, even if he knows no quantum mechanics, knows: namely, *we cannot define a unique wavelength for a short wave train*. Such a wave train does not have a definite wavelength; there is an indefiniteness in the wave number that is related to the finite length of the train, and thus there is an indefiniteness in the momentum.

38-2 Measurement of position and momentum

Let us consider two examples of this idea—to see the reason why there is an uncertainty in the position and/or the momentum, if quantum mechanics is right. We have also seen before that if there were not such a thing—if it were possible to measure the position and the momentum of anything simultaneously—we would have a paradox; it is fortunate that we do not have such a paradox, and the fact that such an uncertainty comes naturally from the wave picture shows that everything is mutually consistent.

Here is one example which shows the relationship between the position and the momentum in a circumstance that is easy to understand. Suppose we have a single slit, and particles are coming from very far away with a certain energy—so that they are all coming essentially horizontally (Fig. 38-2). We are going to concentrate on the vertical components of momentum. All of these particles have a certain horizontal momentum p_0 , say, in a classical sense. So, in the classical sense, the vertical momentum p_y , before the particle goes through the hole, is definitely known. The particle is moving neither up nor down, because it came from a source that is far away—and so the vertical momentum is of course zero. But now let us suppose that it goes through a hole whose width is B . Then after it has come out through the hole, we know the position vertically—the y position—with considerable accuracy—namely $\pm B$. That is, the uncertainty in position, Δy , is of order B . Now we might also want to say, since we know the momentum is absolutely horizontal, that Δp_y is zero; but that is wrong. We *once* knew the momentum was horizontal, but we do not know it any more. Before the particles passed through the hole, we did not know their vertical positions. Now that we have found the vertical position by having the particle come through the hole, we have lost our information on the vertical momentum! Why? According to the wave theory, there is a spreading out, or diffraction, of the waves after they go through the slit, just as for light. Therefore there is a certain probability that particles coming out of the slit are not coming exactly straight. The pattern is spread out by the diffraction effect, and the angle of spread, which we can define as the angle of the first minimum, is a measure of the uncertainty in the final angle.

How does the pattern become spread? To say it is spread means that there is some chance for the particle to be moving up or down, that is, to have a component of momentum up or down. We say *chance* and *particle* because we can detect this diffraction pattern with a particle counter, and when the counter receives the particle, say at C in Fig. 38-2, it receives the *entire* particle, so that, in a classical sense, the particle has a vertical momentum, in order to get from the slit up to C .

To get a rough idea of the spread of the momentum, the vertical momentum p_y has a spread which is equal to $p_0 \Delta\theta$, where p_0 is the horizontal momentum. And how big is $\Delta\theta$ in the spread-out pattern? We know that the first minimum occurs at an angle $\Delta\theta$ such that the waves from one edge of the slit have to travel one wavelength farther than the waves from the other side—we worked that out before (Chapter 30). Therefore $\Delta\theta$ is λ/B , and so Δp_y in this experiment is $p_0 \lambda/B$. Note that if we make B smaller and make a more accurate measurement of the position of the particle, the diffraction pattern gets wider. Remember, when we closed the slits on the experiment with the microwaves, we had more intensity farther out. So the narrower we make the slit, the wider the pattern gets, and the

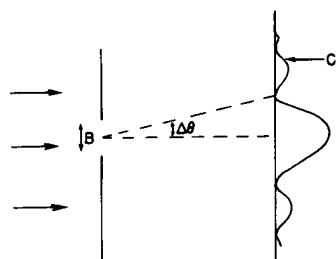


Fig. 38-2. Diffraction of particles passing through a slit.

more is the likelihood that we would find that the particle has sidewise momentum. Thus the uncertainty in the vertical momentum is inversely proportional to the uncertainty of y . In fact, we see that the product of the two is equal to $p_0\lambda$. But λ is the wavelength and p_0 is the momentum, and in accordance with quantum mechanics, the wavelength times the momentum is Planck's constant h . So we obtain the rule that the uncertainties in the vertical momentum and in the vertical position have a product of the order h :

$$\Delta y \Delta p_y \approx h. \quad (38.3)$$

We cannot prepare a system in which we know the vertical position of a particle and can predict how it will move vertically with greater certainty than given by (38.3). That is, the uncertainty in the vertical momentum must exceed $h/\Delta y$, where Δy is the uncertainty in our knowledge of the position.

Sometimes people say quantum mechanics is all wrong. When the particle arrived from the left, its vertical momentum was zero. And now that it has gone through the slit, its position is known. Both position and momentum seem to be known with arbitrary accuracy. It is quite true that we can receive a particle, and on reception determine what its position is and what its momentum would have had to have been to have gotten there. That is true, but that is not what the uncertainty relation (38.3) refers to. Equation (38.3) refers to the *predictability* of a situation, not remarks about the *past*. It does no good to say "I knew what the momentum was before it went through the slit, and now I know the position," because now the momentum knowledge is lost. The fact that it went through the slit no longer permits us to predict the vertical momentum. We are talking about a predictive theory, not just measurements after the fact. So we must talk about what we can predict.

Now let us take the thing the other way around. Let us take another example of the same phenomenon, a little more quantitatively. In the previous example we measured the momentum by a classical method. Namely, we considered the direction and the velocity and the angles, etc., so we got the momentum by classical analysis. But since momentum is related to wave number, there exists in nature still another way to measure the momentum of a particle—photon or otherwise—which has no classical analog, because it uses Eq. (38.2). We measure the *wavelengths of the waves*. Let us try to measure momentum in this way.

Suppose we have a grating with a large number of lines (Fig. 38-3), and send a beam of particles at the grating. We have often discussed this problem: if the particles have a definite momentum, then we get a very sharp pattern in a certain direction, because of the interference. And we have also talked about how accurately we can determine that momentum, that is to say, what the resolving power of such a grating is. Rather than derive it again, we refer to Chapter 30, where we found that the relative uncertainty in the wavelength that can be measured with a given grating is $1/Nm$, where N is the number of lines on the grating and m is the order of the diffraction pattern. That is,

$$\Delta\lambda/\lambda = 1/Nm. \quad (38.4)$$

Now formula (38.4) can be rewritten as

$$\Delta\lambda/\lambda^2 = 1/Nm\lambda = 1/L, \quad (38.5)$$

where L is the distance shown in Fig. 38-3. This distance is the difference between the total distance that the particle or wave or whatever it is has to travel if it is reflected from the bottom of the grating, and the distance that it has to travel if it is reflected from the top of the grating. That is, the waves which form the diffraction pattern are waves which come from different parts of the grating. The first ones that arrive come from the bottom end of the grating, from the beginning of the wave train, and the rest of them come from later parts of the wave train, coming from different parts of the grating, until the last one finally arrives, and that involves a point in the wave train a distance L behind the first point. So in order that we shall have a sharp line in our spectrum corresponding to a definite momentum,

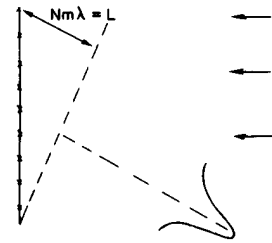


Fig. 38-3. Determination of momentum by using a diffraction grating.

with an uncertainty given by (38.4), we have to have a wave train of at least length L . If the wave train is too short we are not using the entire grating. The waves which form the spectrum are being reflected from only a very short sector of the grating if the wave train is too short, and the grating will not work right—we will find a big angular spread. In order to get a narrower one, we need to use the whole grating, so that at least at some moment the whole wave train is scattering simultaneously from all parts of the grating. Thus the wave train must be of length L in order to have an uncertainty in the wavelength less than that given by (38.5). Incidentally,

$$\Delta\lambda/\lambda^2 = \Delta(1/\lambda) = \Delta k/2\pi. \quad (38.6)$$

Therefore

$$\Delta k = 2\pi/L, \quad (38.7)$$

where L is the length of the wave train.

This means that if we have a wave train whose length is less than L , the uncertainty in the wave number must exceed $2\pi/L$. Or the uncertainty in a wave number times the length of the wave train—we will call that for a moment Δx —exceeds 2π . We call it Δx because that is the uncertainty in the location of the particle. If the wave train exists only in a finite length, then that is where we could find the particle, within an uncertainty Δx . Now this property of waves, that the length of the wave train times the uncertainty of the wave number associated with it is at least 2π , is a property that is known to everyone who studies them. It has nothing to do with quantum mechanics. It is simply that if we have a finite train, we cannot count the waves in it very precisely. Let us try another way to see the reason for that.

Suppose that we have a finite train of length L ; then because of the way it has to decrease at the ends, as in Fig. 38-1, the number of waves in the length L is uncertain by something like ± 1 . But the number of waves in L is $kL/2\pi$. Thus k is uncertain, and we again get the result (38.7), a property merely of waves. The same thing works whether the waves are in space and k is the number of radians per centimeter and L is the length of the train, or the waves are in time and ω is the number of oscillations per second and T is the “length” in time that the wave train comes in. That is, if we have a wave train lasting only for a certain finite time T , then the uncertainty in the frequency is given by

$$\Delta\omega = 2\pi/T. \quad (38.8)$$

We have tried to emphasize that these are properties of waves alone, and they are well known, for example, in the theory of sound.

The point is that in quantum mechanics we interpret the wave number as being a measure of the momentum of a particle, with the rule that $p = \hbar k$, so that relation (38.7) tells us that $\Delta p \approx \hbar/\Delta x$. This, then, is a limitation of the classical idea of momentum. (Naturally, it has to be limited in some ways if we are going to represent particles by waves!) It is nice that we have found a rule that gives us some idea of when there is a failure of classical ideas.

38-3 Crystal diffraction

Next let us consider the reflection of particle waves from a crystal. A crystal is a thick thing which has a whole lot of similar atoms—we will include some complications later—in a nice array. The question is how to set the array so that we get a strong reflected maximum in a given direction for a given beam of, say, light (x-rays), electrons, neutrons, or anything else. In order to obtain a strong reflection, the scattering from all of the atoms must be in phase. There cannot be equal numbers in phase and out of phase, or the waves will cancel out. The way to arrange things is to find the regions of constant phase, as we have already explained; they are planes which make equal angles with the initial and final directions (Fig. 38-4).

If we consider two parallel planes, as in Fig. 38-4, the waves scattered from the two planes will be in phase provided the difference in distance travelled by a wave-

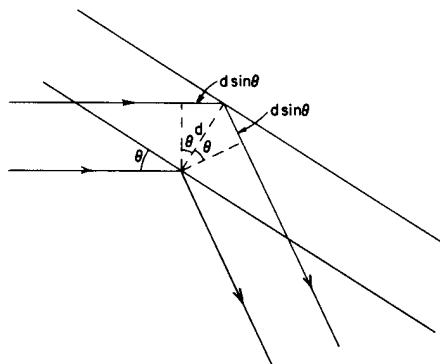


Fig. 38-4. Scattering of waves by crystal planes.

front is an integral number of wavelengths. This difference can be seen to be $2d \sin \theta$, where d is the perpendicular distance between the planes. Thus the condition for coherent reflection is

$$2d \sin \theta = n\lambda \quad (n = 1, 2, \dots). \quad (38.9)$$

If, for example, the crystal is such that the atoms happen to lie on planes obeying condition (38.9) with $n = 1$, then there will be a strong reflection. If, on the other hand, there are other atoms of the same nature (equal in density) halfway between, then the intermediate planes will also scatter equally strongly and will interfere with the others and produce no effect. So d in (38.9) must refer to *adjacent* planes; we cannot take a plane five layers farther back and use this formula!

As a matter of interest, actual crystals are not usually as simple as a single kind of atom repeated in a certain way. Instead, if we make a two-dimensional analog, they are much like wallpaper, in which there is some kind of figure which repeats all over the wallpaper. By "figure" we mean, in the case of atoms, some arrangement—calcium and a carbon and three oxygens, etc., for calcium carbonate, and so on—which may involve a relatively large number of atoms. But whatever it is, the figure is repeated in a pattern. This basic figure is called a *unit cell*.

The basic pattern of repetition defines what we call the *lattice type*; the lattice type can be immediately determined by looking at the reflections and seeing what their symmetry is. In other words, where we find any reflections *at all* determines the lattice type, but in order to determine what is in each of the elements of the lattice one must take into account the *intensity* of the scattering at the various directions. Which directions scatter depends on the type of lattice, but *how strongly* each scatters is determined by what is inside each unit cell, and in that way the structure of crystals is worked out.

Two photographs of x-ray diffraction patterns are shown in Figs. 38-5 and 38-6; they illustrate scattering from rock salt and myoglobin, respectively.

Incidentally, an interesting thing happens if the spacings of the nearest planes are less than $\lambda/2$. In this case (38.9) has no solution for n . Thus if λ is bigger than twice the distance between adjacent planes then there is no side diffraction pattern, and the light—or whatever it is—will go right through the material without bouncing off or getting lost. So in the case of light, where λ is much bigger than the spacing, of course it does go through and there is no pattern of reflection from the planes of the crystal.

This fact also has an interesting consequence in the case of piles which make neutrons (these are obviously particles, for anybody's money!). If we take these neutrons and let them into a long block of graphite, the neutrons diffuse and work their way along (Fig. 38-7). They diffuse because they are bounced by the atoms, but strictly, in the wave theory, they are bounced by the atoms because of diffraction from the crystal planes. It turns out that if we take a very long piece of graphite, the neutrons that come out the far end are all of long wavelength! In fact, if one plots the intensity as a function of wavelength, we get nothing except for wavelengths longer than a certain minimum (Fig. 38-8). In other words, we can get very slow neutrons that way. Only the slowest neutrons come through; they are not diffracted or scattered by the crystal planes of the graphite, but keep going right through like light through glass, and are not scattered out the sides. There are many other demonstrations of the reality of neutron waves and waves of other particles.

38-4 The size of an atom

We now consider another application of the uncertainty relation, Eq. (38.3). It must not be taken too seriously; the idea is right but the analysis is not very accurate. The idea has to do with the determination of the size of atoms, and the fact that, classically, the electrons would radiate light and spiral in until they settle down right on top of the nucleus. But that cannot be right quantum-mechanically because then we would know where each electron was and how fast it was moving.

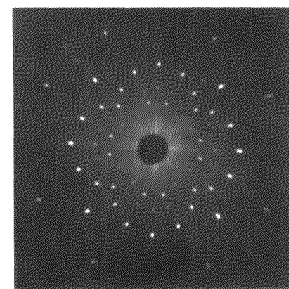


Figure 38-5

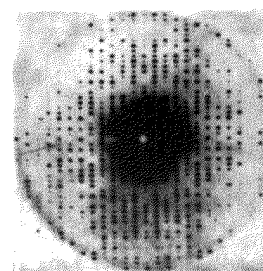


Figure 38-6

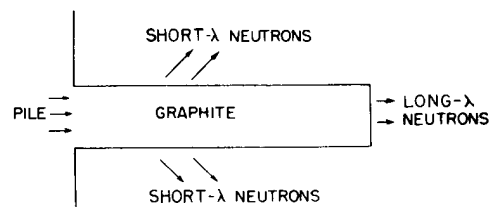


Fig. 38-7. Diffusion of pile neutrons through graphite block.

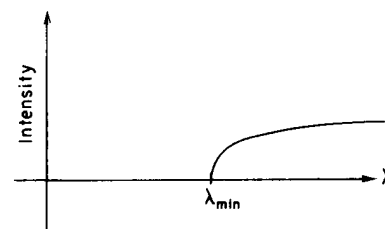


Fig. 38-8. Intensity of neutrons out of graphite rod as function of wavelength.

Suppose we have a hydrogen atom, and measure the position of the electron; we must not be able to predict exactly where the electron will be, or the momentum spread will then turn out to be infinite. Every time we look at the electron, it is somewhere, but it has an amplitude to be in different places so there is a probability of it being found in different places. These places cannot all be at the nucleus; we shall suppose there is a spread in position of order a . That is, the distance of the electron from the nucleus is usually about a . We shall determine a by minimizing the total energy of the atom.

The spread in momentum is roughly h/a because of the uncertainty relation, so that if we try to measure the momentum of the electron in some manner, such as by scattering x-rays off it and looking for the Doppler effect from a moving scatterer, we would expect not to get zero every time—the electron is not standing still—but the momenta must be of the order $p \approx h/a$. Then the kinetic energy is roughly $\frac{1}{2}mv^2 = p^2/2m = h^2/2ma^2$. (In a sense, this is a kind of dimensional analysis to find out in what way the kinetic energy depends upon Planck's constant, upon m , and upon the size of the atom. We need not trust our answer to within factors like 2, π , etc. We have not even defined a very precisely.) Now the potential energy is minus e^2 over the distance from the center, say $-e^2/a$, where, we remember, e^2 is the charge of an electron squared, divided by $4\pi\epsilon_0$. Now the point is that the potential energy is reduced if a gets smaller, but the smaller a is, the higher the momentum required, because of the uncertainty principle, and therefore the higher the kinetic energy. The total energy is

$$E = h^2/2ma^2 - e^2/a. \quad (38.10)$$

We do not know what a is, but we know that the atom is going to arrange itself to make some kind of compromise so that the energy is as little as possible. In order to minimize E , we differentiate with respect to a , set the derivative equal to zero, and solve for a . The derivative of E is

$$dE/da = -h^2/ma^3 + e^2/a^2, \quad (38.11)$$

and setting $dE/da = 0$ gives for a the value

$$\begin{aligned} a_0 &= h^2/me^2 = 0.528 \text{ angstrom} \\ &= 0.528 \times 10^{-10} \text{ meter.} \end{aligned} \quad (38.12)$$

This particular distance is called the *Bohr radius*, and we have thus learned that atomic dimensions are of the order of angstroms, which is right: This is pretty good—in fact, it is amazing, since until now we have had no basis for understanding the size of atoms! Atoms are completely impossible from the classical point of view, since the electrons would spiral into the nucleus.

Now if we put the value (38.12) for a_0 into (38.10) to find the energy, it comes out

$$E_0 = -e^2/2a_0 = -me^4/2h^2 = -13.6 \text{ ev.} \quad (38.13)$$

What does a negative energy mean? It means that the electron has less energy when it is in the atom than when it is free. It means it is bound. It means it takes energy to kick the electron out; it takes energy of the order of 13.6 ev to ionize a hydrogen atom. We have no reason to think that it is not two or three times this—or half of this—or $(1/\pi)$ times this, because we have used such a sloppy argument. However, we have cheated, we have used all the constants in such a way that it happens to come out the right number! This number, 13.6 electron volts, is called a Rydberg of energy; it is the ionization energy of hydrogen.

So we now understand why we do not fall through the floor. As we walk, our shoes with their masses of atoms push against the floor with *its* mass of atoms. In order to squash the atoms closer together, the electrons would be confined to a smaller space and, by the uncertainty principle, their momenta would have to be higher on the average, and that means high energy; the resistance to atomic compression is a quantum-mechanical effect and not a classical effect. Classically,

we would expect that if we were to draw all the electrons and protons closer together, the energy would be reduced still further, and the best arrangement of positive and negative charges in classical physics is all on top of each other. This was well known in classical physics and was a puzzle because of the existence of the atom. Of course, the early scientists invented some ways out of the trouble—but never mind, we have the *right* way out, now! (Maybe.)

Incidentally, although we have no reason to understand it at the moment, in a situation where there are many electrons it turns out that they try to keep away from each other. If one electron is occupying a certain space, then another does not occupy the same space. More precisely, there are two spin cases, so that two can sit on top of each other, one spinning one way and one the other way. But after that we cannot put any more there. We have to put others in another place, and that is the real reason that matter has strength. If we could put all the electrons in the same place it would condense even more than it does. It is the fact that the electrons cannot all get on top of each other that makes tables and everything else solid.

Obviously, in order to understand the properties of matter, we will have to use quantum mechanics and not be satisfied with classical mechanics.

38-5 Energy levels

We have talked about the atom in its lowest possible energy condition, but it turns out that the electron can do other things. It can jiggle and wiggle in a more energetic manner, and so there are many different possible motions for the atom. According to quantum mechanics, in a stationary condition there can only be definite energies for an atom. We make a diagram (Fig. 38-9) in which we plot the energy vertically, and we make a horizontal line for each allowed value of the energy. When the electron is free, i.e., when its energy is positive, it can have any energy; it can be moving at any speed. But bound energies are not arbitrary. The atom must have one or another out of a set of allowed values, such as those in Fig. 38-9.

Now let us call the allowed values of the energy E_0, E_1, E_2, E_3 . If an atom is initially in one of these “excited states,” E_1, E_2 , etc., it does not remain in that state forever. Sooner or later it drops to a lower state and radiates energy in the form of light. The frequency of the light that is emitted is determined by conservation of energy plus the quantum-mechanical understanding that the frequency of the light is related to the energy of the light by (38.1). Therefore the frequency of the light which is liberated in a transition from energy E_3 to energy E_1 (for example) is

$$\omega_{31} = (E_3 - E_1)/\hbar. \quad (38.14)$$

This, then, is a characteristic frequency of the atom and defines a spectral emission line. Another possible transition would be from E_3 to E_0 . That would have a different frequency

$$\omega_{30} = (E_3 - E_0)/\hbar. \quad (38.15)$$

Another possibility is that if the atom were excited to the state E_1 it could drop to the ground state E_0 , emitting a photon of frequency

$$\omega_{10} = (E_1 - E_0)/\hbar. \quad (38.16)$$

The reason we bring up three transitions is to point out an interesting relationship. It is easy to see from (38.14), (38.15), and (38.16) that

$$\omega_{30} = \omega_{31} + \omega_{10}. \quad (38.17)$$

In general, if we find two spectral lines, we shall expect to find another line at the sum of the frequencies (or the difference in the frequencies), and that all the lines can be understood by finding a series of levels such that every line corresponds to the difference in energy of some pair of levels. This remarkable coincidence in

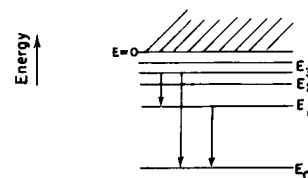


Fig. 38-9. Energy diagram for an atom, showing several possible transitions.

spectral frequencies was noted before quantum mechanics was discovered, and it is called the *Ritz combination principle*. This is again a mystery from the point of view of classical mechanics. Let us not belabor the point that classical mechanics is a failure in the atomic domain; we seem to have demonstrated that pretty well.

We have already talked about quantum mechanics as being represented by amplitudes which behave like waves, with certain frequencies and wave numbers. Let us observe how it comes about from the point of view of amplitudes that the atom has definite energy states. This is something we cannot understand from what has been said so far, but we are all familiar with the fact that confined waves have definite frequencies. For instance, if sound is confined to an organ pipe, or anything like that, then there is more than one way that the sound can vibrate, but for each such way there is a definite frequency. Thus an object in which the waves are confined has certain resonance frequencies. It is therefore a property of waves in a confined space—a subject which we will discuss in detail with formulas later on—that they exist only at definite frequencies. And since the general relation exists between frequencies of the amplitude and energy, we are not surprised to find definite energies associated with electrons bound in atoms.

38-6 Philosophical implications

Let us consider briefly some philosophical implications of quantum mechanics. As always, there are two aspects of the problem: one is the philosophical implication for physics, and the other is the extrapolation of philosophical matters to other fields. When philosophical ideas associated with science are dragged into another field, they are usually completely distorted. Therefore we shall confine our remarks as much as possible to physics itself.

First of all, the most interesting aspect is the idea of the uncertainty principle; making an observation affects the phenomenon. It has always been known that making observations affects a phenomenon, but the point is that the effect cannot be disregarded or minimized or decreased arbitrarily by rearranging the apparatus. When we look for a certain phenomenon we cannot help but disturb it in a certain minimum way, and *the disturbance is necessary for the consistency of the viewpoint*. The observer was sometimes important in prequantum physics, but only in a rather trivial sense. The problem has been raised: if a tree falls in a forest and there is nobody there to hear it, does it make a noise? A *real* tree falling in a *real* forest makes a sound, of course, even if nobody is there. Even if no one is present to hear it, there are other traces left. The sound will shake some leaves, and if we were careful enough we might find somewhere that some thorn had rubbed against a leaf and made a tiny scratch that could not be explained unless we assumed the leaf were vibrating. So in a certain sense we would have to admit that there is a sound made. We might ask: was there a *sensation* of sound? No, sensations have to do, presumably, with consciousness. And whether ants are conscious and whether there were ants in the forest, or whether the tree was conscious, we do not know. Let us leave the problem in that form.

Another thing that people have emphasized since quantum mechanics was developed is the idea that we should not speak about those things which we cannot measure. (Actually relativity theory also said this.) Unless a thing can be defined by measurement, it has no place in a theory. And since an accurate value of the momentum of a localized particle cannot be defined by measurement it therefore has no place in the theory. The idea that this is what was the matter with classical theory is a *false position*. It is a careless analysis of the situation. Just because we cannot *measure* position and momentum precisely does not *a priori* mean that we *cannot* talk about them. It only means that we *need* not talk about them. The situation in the sciences is this: A concept or an idea which cannot be measured or cannot be referred directly to experiment may or may not be useful. It need not exist in a theory. In other words, suppose we compare the classical theory of the world with the quantum theory of the world, and suppose that it is true experimentally that we can measure position and momentum only imprecisely. The question is whether the *ideas* of the exact position of a particle and the exact

momentum of a particle are valid or not. The classical theory admits the ideas; the quantum theory does not. This does not in itself mean that classical physics is wrong. When the new quantum mechanics was discovered, the classical people—which included everybody except Heisenberg, Schrödinger, and Born—said: “Look, your theory is not any good because you cannot answer certain questions like: what is the exact position of a particle?, which hole does it go through?, and some others.” Heisenberg’s answer was: “I do not need to answer such questions because you cannot ask such a question experimentally.” It is that we do not *have* to. Consider two theories (a) and (b); (a) contains an idea that cannot be checked directly but which is used in the analysis, and the other, (b), does not contain the idea. If they disagree in their predictions, one could not claim that (b) is false because it cannot explain this idea that is in (a), because that idea is one of the things that cannot be checked directly. It is always good to know which ideas cannot be checked directly, but it is not necessary to remove them all. It is not true that we can pursue science completely by using only those concepts which are directly subject to experiment.

In quantum mechanics itself there is a wave function amplitude, there is a potential, and there are many constructs that we cannot measure directly. The basis of a science is its ability to *predict*. To predict means to tell what will happen in an experiment that has never been done. How can we do that? By assuming that we know what is there, independent of the experiment. We must extrapolate the experiments to a region where they have not been done. We must take our concepts and extend them to places where they have not yet been checked. If we do not do that, we have no prediction. So it was perfectly sensible for the classical physicists to go happily along and suppose that the position—which obviously means something for a baseball—meant something also for an electron. It was not stupidity. It was a sensible procedure. Today we say that the law of relativity is supposed to be true at all energies, but someday somebody may come along and say how stupid we were. We do not know where we are “stupid” until we “stick our neck out,” and so the whole idea is to put our neck out. And the only way to find out that we are wrong is to find out *what* our predictions are. It is absolutely necessary to make constructs.

We have already made a few remarks about the indeterminacy of quantum mechanics. That is, that we are unable now to predict what will happen in physics in a given physical circumstance which is arranged as carefully as possible. If we have an atom that is in an excited state and so is going to emit a photon, we cannot say *when* it will emit the photon. It has a certain amplitude to emit the photon at any time, and we can predict only a probability for emission; we cannot predict the future exactly. This has given rise to all kinds of nonsense and questions on the meaning of freedom of will, and of the idea that the world is uncertain.

Of course we must emphasize that classical physics is also indeterminate, in a sense. It is usually thought that this indeterminacy, that we cannot predict the future, is an important quantum-mechanical thing, and this is said to explain the behavior of the mind, feelings of free will, etc. But if the world *were* classical—if the laws of mechanics were classical—it is not quite obvious that the mind would not feel more or less the same. It is true classically that if we knew the position and the velocity of every particle in the world, or in a box of gas, we could predict exactly what would happen. And therefore the classical world is deterministic. Suppose, however, that we have a finite accuracy and do not know *exactly* where just one atom is, say to one part in a billion. Then as it goes along it hits another atom, and because we did not know the position better than to one part in a billion, we find an even larger error in the position after the collision. And that is amplified, of course, in the next collision, so that if we start with only a tiny error it rapidly magnifies to a very great uncertainty. To give an example: if water falls over a dam, it splashes. If we stand nearby, every now and then a drop will land on our nose. This appears to be completely random, yet such a behavior would be predicted by purely classical laws. The exact position of all the drops depends upon the precise wiggles of the water before it goes over the dam. How? The tiniest irregularities are magnified in falling, so that we get complete randomness. Ob-

viously, we cannot really predict the position of the drops unless we know the motion of the water *absolutely exactly*.

Speaking more precisely, given an arbitrary accuracy, no matter how precise, one can find a time long enough that we cannot make predictions valid for that long a time. Now the point is that this length of time is not very large. It is not that the time is millions of years if the accuracy is one part in a billion. The time goes, in fact, only logarithmically with the error, and it turns out that in only a very, very tiny time we lose all our information. If the accuracy is taken to be one part in billions and billions and billions—no matter how many billions we wish, provided we do stop somewhere—then we can find a time less than the time it took to state the accuracy—after which we can no longer predict what is going to happen! It is therefore not fair to say that from the apparent freedom and indeterminacy of the human mind, we should have realized that classical “deterministic” physics could not ever hope to understand it, and to welcome quantum mechanics as a release from a “completely mechanistic” universe. For already in classical mechanics there was indeterminability from a practical point of view.

The Kinetic Theory of Gases

39-1 Properties of matter

With this chapter we begin a new subject which will occupy us for some time. It is the first part of the analysis of the properties of matter from the physical point of view, in which, recognizing that matter is made out of a great many atoms, or elementary parts, which interact electrically and obey the laws of mechanics, we try to understand why various aggregates of atoms behave the way they do.

It is obvious that this is a difficult subject, and we emphasize at the beginning that it is in fact an *extremely* difficult subject, and that we have to deal with it differently than we have dealt with the other subjects so far. In the case of mechanics and in the case of light, we were able to begin with a precise statement of some laws, like Newton's laws, or the formula for the field produced by an accelerating charge, from which a whole host of phenomena could be essentially understood, and which would produce a basis for our understanding of mechanics and of light from that time on. That is, we may learn more later, but we do not learn different physics, we only learn better methods of mathematical analysis to deal with the situation.

We cannot use this approach effectively in studying the properties of matter. We can discuss matter only in a most elementary way; it is much too complicated a subject to analyze directly from its specific basic laws, which are none other than the laws of mechanics and electricity. But these are a bit too far away from the properties we wish to study; it takes too many steps to get from Newton's laws to the properties of matter, and these steps are, in themselves, fairly complicated. We will now start to take some of these steps, but while many of our analyses will be quite accurate, they will eventually get less and less accurate. We will have only a rough understanding of the properties of matter.

One of the reasons that we have to perform the analysis so imperfectly is that the mathematics of it requires a deep understanding of the theory of probability; we are not going to want to know where every atom is actually moving, but rather, how many move here and there on the average, and what the odds are for different effects. So this subject involves a knowledge of the theory of probability, and our mathematics is not yet quite ready and we do not want to strain it too hard.

Secondly, and more important from a physical standpoint, the actual behavior of the atoms is not according to classical mechanics, but according to quantum mechanics, and a correct understanding of the subject cannot be attained until we understand quantum mechanics. Here, unlike the case of billiard balls and automobiles, the difference between the classical mechanical laws and the quantum-mechanical laws is very important and very significant, so that many things that we will deduce by classical physics will be fundamentally incorrect. Therefore there will be certain things to be partially unlearned; however, we shall indicate in every case when a result is incorrect, so that we will know just where the "edges" are. One of the reasons for discussing quantum mechanics in the preceding chapters was to give an idea as to why, more or less, classical mechanics is incorrect in the various directions.

Why do we deal with the subject now at all? Why not wait a half a year, or a year, until we know the mathematics of probability better, and we learn a little quantum mechanics, and then we can do it in a more fundamental way? The answer is that it is a difficult subject, and the best way to learn is to do it slowly! The first thing to do is to get some idea, more or less, of what ought to happen in

39-1 Properties of matter

39-2 The pressure of a gas

39-3 Compressibility of radiation

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different circumstances, and then, later, when we know the laws better, we will formulate them better.

Anyone who wants to analyze the properties of matter in a real problem might want to start by writing down the fundamental equations and then try to solve them mathematically. Although there are people who try to use such an approach, these people are the failures in this field; the real successes come to those who start from a *physical* point of view, people who have a rough idea where they are going and then begin by making the right kind of approximations, knowing what is big and what is small in a given complicated situation. These problems are so complicated that even an elementary understanding, although inaccurate and incomplete, is worth while having, and so the subject will be one that we shall go over again and again, each time with more and more accuracy, as we go through our course in physics.

Another reason for beginning the subject right now is that we have already used many of these ideas in, for example, chemistry, and we have even heard of some of them in high school. It is interesting to know the physical basis for these things.

As an interesting example, we all know that equal volumes of gases, at the same pressure and temperature, contain the same number of molecules. The law of multiple proportions, that when two gases combine in a chemical reaction the volumes needed always stand in simple integral proportions, was understood ultimately by Avogadro to mean that equal volumes have equal numbers of atoms. Now *why* do they have equal numbers of atoms? Can we deduce from Newton's laws that the number of atoms should be equal? We shall address ourselves to that specific matter in this chapter. In succeeding chapters, we shall discuss various other phenomena involving pressures, volumes, temperature, and heat.

We shall also find that the subject can be attacked from a nonatomic point of view, and that there are many interrelationships of the properties of substances. For instance, when we compress something, it heats; if we heat it, it expands. There is a relationship between these two facts which can be deduced independently of the machinery underneath. This subject is called *thermodynamics*. The deepest understanding of thermodynamics comes, of course, from understanding the actual machinery underneath, and that is what we shall do: we shall take the atomic viewpoint from the beginning and use it to understand the various properties of matter and the laws of thermodynamics.

Let us, then, discuss the properties of gases from the standpoint of Newton's laws of mechanics.

39-2 The pressure of a gas

First, we know that a gas exerts a pressure, and we must clearly understand what this is due to. If our ears were a few times more sensitive, we would hear a perpetual rushing noise. Evolution has not developed the ear to that point, because it would be useless if it were so much more sensitive—we would hear a perpetual racket. The reason is that the eardrum is in contact with the air, and air is a lot of molecules in perpetual motion and these bang against the eardrums. In banging against the eardrums they make an irregular tattoo—boom, boom, boom—which we do not hear because the atoms are so small, and the sensitivity of the ear is not quite enough to notice it. The result of this perpetual bombardment is to push the drum away, but of course there is an equal perpetual bombardment of atoms on the other side of the eardrum, so the net force on it is zero. If we were to take the air away from one side, or change the relative amounts of air on the two sides, the eardrum would then be pushed one way or the other, because the amount of bombardment on one side would be greater than on the other. We sometimes feel this uncomfortable effect when we go up too fast in an elevator or an airplane, especially if we also have a bad cold (when we have a cold, inflammation closes the tube which connects the air on the inside of the eardrum with the outside air through the throat, so that the two pressures cannot readily equalize).

In considering how to analyze the situation quantitatively, we imagine that we have a volume of gas in a box, at one end of which is a piston which can be moved (Fig. 39-1). We would like to find out what force on the piston results from the fact that there are atoms in this box. The volume of the box is V , and as the atoms move around inside the box with various velocities they bang against the piston. Suppose there is nothing, a vacuum, on the outside of the piston. What of it? If the piston were left alone, and nobody held onto it, each time it got banged it would pick up a little momentum and it would gradually get pushed out of the box. So in order to keep it from being pushed out of the box, we have to hold it with a force F . The problem is, how much force? One way of expressing the force is to talk about the force per unit area: if A is the area of the piston, then the force on the piston will be written as a number times the area. We define the pressure, then, as equal to the force that we have to apply on a piston, divided by the area of the piston:

$$P = F/A. \quad (39.1)$$

To make sure we understand the idea (we have to derive it for another purpose anyway), the differential work dW done on the gas in compressing it by moving the piston in a differential amount $-dx$ would be the force times the distance that we compress it, which, according to (39.1), would be the pressure times the area, times the distance, which is equal to minus the pressure times the change in the volume:

$$dW = F(-dx) = -PA dx = -P dV. \quad (39.2)$$

(The area A times the distance dx is the volume change.) The minus sign is there because, as we compress it, we *decrease* the volume; if we think about it we can see that if a gas is compressed, work is done *on* it.

How much force do we have to apply to balance the banging of the molecules? The piston receives from each collision a certain amount of momentum. A certain amount of momentum per second will pour into the piston, and it will start to move. To keep it from moving, we must pour back into it the same amount of momentum per second from our force. Of course, the force *is* the amount of momentum per second that we must pour in. There is another way to put it: if we let go of the piston it will pick up speed because of the bombardments; with each collision we get a little more speed, and the speed thus accelerates. The rate at which the piston picks up speed, or accelerates, is proportional to the force on it. So we see that the force, which we already have said is the pressure times the area, is equal to the momentum per second delivered to the piston by the colliding molecules.

To calculate the momentum per second is easy—we can do it in two parts: first, we find the momentum delivered to the piston by one particular atom in a collision with the piston, then we have to multiply by the number of collisions per second that the atoms have with the wall. The force will be the product of these two factors. Now let us see what the two factors are: In the first place, we shall suppose that the piston is a perfect “reflector” for the atoms. If it is not, the whole theory is wrong, and the piston will start to heat up and things will change, but eventually, when equilibrium has set in, the net result is that the collisions are effectively perfectly elastic. On the average, every particle that comes in leaves with the same energy. So we shall imagine that the gas is in a steady condition, and we lose no energy to the piston because the piston is standing still. In those circumstances, if a particle comes in with a certain speed, it comes out with the same speed and, we will say, with the same mass.

If $\mathbf{v}$ is the velocity of an atom, and v_x is the x -component of $\mathbf{v}$, then mv_x is the x -component of momentum “in”; but we also have an equal component of momentum “out,” and so the total momentum delivered to the piston by the particle, in one collision, is $2mv_x$, because it is “reflected.”

Now, we need the number of collisions made by the atoms in a second, or in a certain amount of time dt ; then we divide by dt . How many atoms are hitting? Let us suppose that there are N atoms in the volume V , or $n = N/V$ in each unit volume. To find how many atoms hit the piston, we note that, given a certain

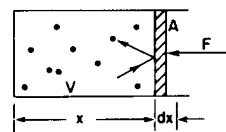


Fig. 39-1. Atoms of a gas in a box with a frictionless piston.

amount of time t , if a particle has a certain velocity toward the piston it will hit during the time t , provided it is close enough. If it is too far away, it goes only part way toward the piston in the time t , but does not reach the piston. Therefore it is clear that only those molecules which are within a distance $v_x t$ from the piston are going to hit the piston in the time t . Thus the number of collisions in a time t is equal to the number of atoms which are in the region within a distance $v_x t$, and since the area of the piston is A , the volume occupied by the atoms which are going to hit the piston is $v_x t A$. But the number of atoms that are going to hit the piston is that volume times the number of atoms per unit volume, $n v_x t A$. Of course we do not want the number that hit in a time t , we want the number that hit per second, so we divide by the time t , to get $n v_x A$. (This time t could be made very short; if we feel we want to be more elegant, we call it dt , then differentiate, but it is the same thing.)

So we find that the force is

$$F = n v_x A \cdot 2 m v_x. \quad (39.3)$$

See, the force is proportional to the area, if we keep the particle density fixed as we change the area! The pressure is then

$$P = 2 n m v_x^2. \quad (39.4)$$

Now we notice a little trouble with this analysis: First, all the molecules do not have the same velocity, and they do not move in the same direction. So, all the v_x^2 's are different! So what we must do, of course, is to take an *average* of the v_x^2 's, since each one makes its own contribution. What we want is the square of v_x , averaged over all the molecules:

$$P = n m \langle v_x^2 \rangle. \quad (39.5)$$

Did we forget to include the factor 2? No; of all the atoms, only half are headed toward the piston. The other half are headed the other way, and if we take $\langle v_x^2 \rangle$, we are averaging the *negative* v_x 's squared, as well as the positive v_x 's. So when we just take $\langle v_x^2 \rangle$, without looking, we are getting twice as much as we want. The average of v_x^2 , for positive v_x , is equal to the average of v_x^2 for *all* v_x , times one half.

Now as the atoms bounce around, it is clear that there is nothing special about the "x-direction"; the atoms may also be moving up and down, back and forth, in and out. Therefore it is going to be true that $\langle v_x^2 \rangle$, the average motion of the atoms in one direction, and the average in the other two directions, are all going to be equal:

$$\langle v_x^2 \rangle = \langle v_y^2 \rangle = \langle v_z^2 \rangle. \quad (39.6)$$

It is only a matter of rather tricky mathematics to notice, therefore, that they are each equal to one-third of their sum, which is of course the square of the magnitude of the velocity:

$$\langle v_x^2 \rangle = \frac{1}{3} \langle v_x^2 + v_y^2 + v_z^2 \rangle = \langle v^2 \rangle / 3. \quad (39.7)$$

This has the advantage that we do not have to worry about any particular direction, and so we write our pressure formula again in this form:

$$P = \left(\frac{2}{3}\right) n \langle m v^2 / 2 \rangle. \quad (39.8)$$

The reason we wrote the last factor as $\langle m v^2 / 2 \rangle$ is that this is the *kinetic energy* of the center-of-mass motion of the molecule. We find, therefore, that

$$P V = N \left(\frac{2}{3}\right) \langle m v^2 / 2 \rangle. \quad (39.9)$$

With this equation we can calculate how much the pressure is, if we know the speeds.

As a very simple example let us take helium gas, or any other gas, like mercury vapor, or potassium vapor of high enough temperature, or argon, in which all the molecules are single atoms, for which we may suppose that there is no internal

motion in the atom. If we had a complex molecule, there might be some internal motion, mutual vibrations, or something. We suppose that we may disregard that; this is actually a serious matter that we will have to come back to, but it turns out to be all right. We suppose that the internal motion of the atoms can be disregarded, and therefore, for this purpose, that the kinetic energy of the center-of-mass motion is all the energy there is. So for a monatomic gas, the kinetic energy is the total energy. In general, we are going to call U the total energy (it is sometimes called the total *internal* energy—we may wonder why, since there is no *external* energy to a gas), i.e., all the energy of all the molecules in the gas, or the object, whatever it is.

For a monatomic gas we will suppose that the total energy U is equal to a number of atoms times the average kinetic energy of each, because we are disregarding any possibility of excitation or motion inside the atoms themselves. Then, in these circumstances, we would have

$$PV = \frac{2}{3}U. \quad (39.10)$$

Incidentally, we can stop here and find the answer to the following question: Suppose that we take a can of gas and compress the gas slowly, how much pressure do we need to squeeze the volume down? It is easy to find out, since the pressure is $\frac{2}{3}$ the energy divided by V . As we squeeze it down, we do work on the gas and we thereby increase the energy U . So we are going to have some kind of a differential equation: If we start out in a given circumstance with a certain energy and a certain volume, we then know the pressure. Now we start to squeeze, but the moment we do, the energy U increases and the volume V decreases, so the pressure goes up.

So, we have to solve a differential equation, and we will solve it in a moment. We must first emphasize, however, that as we are compressing this gas, we are supposing that all the work goes into increasing the energy of the atoms inside. We may ask, "Isn't that necessary? Where else could it go?" It turns out that it *can* go another place. There are what we call "heat leaks" through the walls: the hot (i.e., fast-moving) atoms that bombard the walls, heat the walls, and energy goes away. We shall suppose for the present that this is not the case.

For somewhat wider generality, although we are still making some very special assumptions about our gas, we shall write, not $PV = \frac{2}{3}U$, but

$$PV = (\gamma - 1)U. \quad (39.11)$$

It is written $(\gamma - 1)$ times U for conventional reasons, because we will deal with a few other cases later where the number in front of U will not be $\frac{2}{3}$, but will be a different number. So, in order to do the thing in general, we call it $\gamma - 1$, because people have been calling it that for almost one hundred years. This γ , then, is $\frac{5}{3}$, because $\frac{5}{3} - 1$ is $\frac{2}{3}$ for a monatomic gas like helium.

We have already noticed that when we compress a gas the work done is $-P dV$. A compression in which there is no heat energy added or removed is called an *adiabatic* compression, from the Greek *a* (not) + *dia* (through) + *bainein* (to go). (The word adiabatic is used in physics in several ways, and it is sometimes hard to see what is common about them.) That is, for an adiabatic compression all the work done goes into changing the internal energy. That is the key—that there are no other losses of energy—for then we have $P dV = -dU$. But since $U = PV/(\gamma - 1)$, we may write

$$dU = (P dV + V dP)/(\gamma - 1). \quad (39.12)$$

So we have $P dV = -(P dV + V dP)/(\gamma - 1)$, or, rearranging the terms, $\gamma P dV = -V dP$, or

$$(\gamma dV/V) + (dP/P) = 0. \quad (39.13)$$

Fortunately, assuming that γ is constant, as it is for a monatomic gas, we can integrate this: it gives $\gamma \ln V + \ln P = \ln C$, where $\ln C$ is the constant of integration. If we take the exponential of both sides, we get the law

$$PV^\gamma = C \text{ (a constant)}. \quad (39.14)$$

In other words, under adiabatic conditions, where the temperature rises as we compress because no heat is being lost, the pressure times the volume to the $\frac{5}{3}$ power is a constant for a monatomic gas! Although we derived it theoretically, this *is*, in fact, the way monatomic gases behave experimentally.

39-3 Compressibility of radiation

We may give one other example of the kinetic theory of a gas, one which is not used in chemistry so much, but is used in astronomy. We have a large number of photons in a box in which the temperature is very high. (The box is, of course, the gas in a very hot star. The sun is not hot enough; there are still too many atoms, but at still higher temperatures in certain very hot stars, we may neglect the atoms and suppose that the only objects that we have in the box are photons.) Now then, a photon has a certain momentum $\mathbf{p}$. (We always find that we are in terrible trouble when we do kinetic theory: p is the pressure, but p is the momentum; v is the volume, but v is the velocity; T is the temperature, but T is the kinetic energy or the time or the torque; one must keep one's wits about one!) This $\mathbf{p}$ is momentum, it is a vector. Going through the same analysis as before, it is the x -component of the vector $\mathbf{p}$ which generates the "kick," and twice the x -component of the vector $\mathbf{p}$ is the momentum which is given in the kick. Thus $2p_x$ replaces $2mv_x$, and in evaluating the number of collisions, v_x is still v_x , so when we get all the way through, we find that the pressure in Eq. (39.4) is, instead,

$$P = 2np_xv_x. \quad (39.15)$$

Then, in the averaging, it becomes n times the average of p_xv_x (the same factor of 2) and, finally, putting in the other two directions, we find

$$PV = N\langle \mathbf{p} \cdot \mathbf{v} \rangle / 3. \quad (39.16)$$

This checks with the formula (39.9), because the momentum is $m\mathbf{v}$; it is a little more general, that is all. The pressure times the volume is the total number of atoms times $\frac{1}{3}\langle \mathbf{p} \cdot \mathbf{v} \rangle$, averaged.

Now, for photons, what is $\mathbf{p} \cdot \mathbf{v}$? The momentum and the velocity are in the same direction, and the velocity is the speed of light, so this is the momentum of each of the objects, times the speed of light. The momentum times the speed of light of every photon is its energy: $E = pc$, so these terms are the *energies* of each of the photons, and we should, of course, take an average energy, times the number of photons. So we have $\frac{1}{3}$ of the energy inside the gas:

$$PV = U/3 \text{ (photon gas)}. \quad (39.17)$$

For photons, then, since we have $\frac{1}{3}$ in front, $(\gamma - 1)$ in (39.11) is $\frac{1}{3}$, or $\gamma = \frac{4}{3}$, and we have discovered that radiation in a box obeys the law

$$PV^{4/3} = C. \quad (39.18)$$

So we know the compressibility of radiation! That is what is used in an analysis of the contribution of radiation pressure in a star, that is how we calculate it, and how it changes when we compress it. What wonderful things are already within our power!

39-4 Temperature and kinetic energy

So far we have not dealt with *temperature*; we have purposely been avoiding the temperature. As we compress a gas, we know that the energy of the molecules increases, and we are used to saying that the gas gets hotter; we would like to understand what this has to do with the temperature. If we try to do the experiment, not adiabatically but at what we call *constant temperature*, what are we doing? We know that if we take two boxes of gas and let them sit next to each other long enough, even if at the start they were at what we call different temperatures, they

will in the end come to the same temperature. Now what does that mean? That means that they get to a condition that they would get to if we left them alone long enough! What we mean by equal temperature is just that—the final condition when things have been sitting around interacting with each other long enough.

Let us consider, now, what happens if we have two gases in containers separated by a movable piston as in Fig. 39-2 (just for simplicity we shall take two monatomic gases, say helium and neon). In container (1) the atoms have mass m_1 , velocity v_1 , and there are n_1 per unit volume, and in the other container the atoms have mass m_2 , velocity v_2 , there are n_2 atoms per unit volume. What are the conditions for equilibrium?

Obviously, the bombardment from the left side must be such that it moves the piston to the right and compresses the other gas until its pressure builds up, and the thing will thus slosh back and forth, and will gradually come to rest at a place where the pressures are equal on both sides. So we can arrange that the pressures are equal; that just means that the internal energies per unit volume are equal, or that the numbers n times the average kinetic energies on each side are equal. What we have to try to prove, eventually, is that the *numbers themselves* are equal. So far, all we know is that the numbers times the kinetic energies are equal,

$$n_1 \langle m_1 v_1^2 / 2 \rangle = n_2 \langle m_2 v_2^2 / 2 \rangle,$$

from (39.8), because the pressures are equal. We must realize that this is not the only condition over the long run, but something else must happen more slowly as the true complete equilibrium corresponding to equal temperatures sets in.

To see the idea, suppose that the pressure on the left side were developed by having a very high density but a low velocity. By having a large n and a small v , we can get the same pressure as by having a small n and a large v . The atoms may be moving slowly but be packed nearly solidly, or there may be fewer but they are hitting harder. Will it stay like that forever? At first we might think so, but then we think again and find we have forgotten one important point. That is, that the intermediate piston does not receive a steady pressure; it wiggles, just like the eardrum that we were first talking about, because the bangings are not absolutely uniform. There is not a perpetual, steady pressure, but a tattoo—the pressure varies, and so the thing jiggles. Suppose that the atoms on the right side are not jiggling much, but those on the left are few and far between and very energetic. The piston will, now and then, get a big impulse from the left, and will be driven against the slow atoms on the right, giving them more speed. (As each atom collides with the piston, it either gains or loses energy, depending upon whether the piston is moving one way or the other when the atom strikes it.) So, as a result of the collisions, the piston finds itself jiggling, jiggling, jiggling, and this shakes the other gas—it gives energy to the other atoms, and they build up faster motions, until they balance the jiggling that the piston is giving to them. The system comes to some equilibrium where the piston is moving at such a mean square speed that it picks up energy from the atoms at about the same rate as it puts energy back into them. So the piston picks up a certain mean irregularity in speed, and it is our problem to find it. When we do find it, we can solve our problem better, because the gases will adjust their velocities until the rate at which they are trying to pour energy into each other through the piston will become equal.

It is quite difficult to figure out the details of the piston in this particular circumstance; although it is ideally simple to understand, it turns out to be a little harder to analyze. Before we analyze that, let us analyze another problem in which we have a box of gas but now we have two different kinds of molecules in it, having masses m_1 and m_2 , velocities v_1 and v_2 , and so forth; there is now a much more intimate relationship. If all of the No. 2 molecules are standing still, that condition is not going to last, because they get kicked by the No. 1 molecules and so pick up speed. If they are all going much faster than the No. 1 molecules, then maybe that will not last either—they will pass the energy back to the No. 1 molecules. So when both gases are in the same box, the problem is to find the rule that determines the relative speeds of the two.

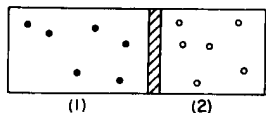


Fig. 39-2. Atoms of two different monatomic gases are separated by a movable piston.

This is still a very difficult problem, but we will solve it as follows. First we consider the following sub-problem (again this is one of those cases where—never mind the derivation—in the end the result is very simple to remember, but the derivation is just ingenious). Let us suppose that we have two molecules, of different mass, colliding, and that the collision is viewed on the center-of-mass (CM) system. In order to remove a complication, we look at the collision in the CM. As we know from the laws of collision, by the conservation of momentum and energy, after the molecules collide the only way they can move is such that each maintains its own original speed—and they just change their *direction*. So we have an average collision that looks like that in Fig. 39-3. Suppose, for a moment, that we watch all the collisions with the CM at rest. Suppose we imagine that they are all initially moving horizontally. Of course, after the first collision some of them are moving at an angle. In other words, if they were all going horizontally, then at least some would later be moving vertically. Now in some other collision, they would be coming in from another direction, and then they would be twisted at still another angle. So even if they were completely organized in the beginning, they would get sprayed around at all angles, and then the sprayed ones would get sprayed some more, and sprayed some more, and sprayed some more. Ultimately, what will be the distribution? *Answer: It will be equally likely to find any pair moving in any direction in space.* After that further collisions could not change the distribution.

They are equally likely to go in all directions, but how do we say that? There is of course *no* likelihood that they will go in any specific direction, because a specific direction is too exact, so we have to talk about per unit “something.” The idea is that any area on a sphere centered at a collision point will have just as many molecules going through it as go through any other equal area on the sphere. So the result of the collisions will be to distribute the directions so that equal areas on a sphere will have equal probabilities.

Incidentally, if we just want to discuss the original direction and some other direction an angle θ from it, it is an interesting property that the differential area of a sphere of unit radius is $\sin \theta d\theta$ times 2π , and that is the same as the differential of $\cos \theta$. So what it means is that the cosine of the angle θ between any two directions is equally likely to be anything from -1 to $+1$.

Next, we have to worry about the actual case, where we do not have the collision in the CM system, but we have two atoms which are coming together with vector velocities $\mathbf{v}_1$ and $\mathbf{v}_2$. What happens now? We can analyze this collision with the vector velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ in the following way: We first say that there is a certain CM; the velocity of the CM is given by the “average” velocity, with weights proportional to the masses, so the velocity of the CM is $\mathbf{v}_{CM} = (m_1\mathbf{v}_1 + m_2\mathbf{v}_2)/(m_1 + m_2)$. If we watch this collision in the CM system, then we see a collision just like that in Fig. 39-3, with a certain relative velocity $\mathbf{w}$ coming in. The relative velocity is just $\mathbf{v}_1 - \mathbf{v}_2$. Now the idea is that, first, the whole CM is moving, and in the CM there is a relative velocity $\mathbf{w}$, and the molecules collide and come off in some new direction. All this happens while the CM keeps right on moving, without any change.

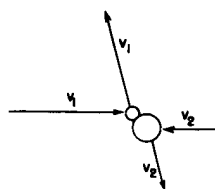


Fig. 39-3. A collision between unequal molecules, viewed in the CM system.

Now then, what is the distribution resulting from this? From our previous argument we conclude this: that at equilibrium, *all directions for $\mathbf{w}$ are equally likely, relative to the direction of the motion of the CM.** There will be no particular correlation, in the end, between the direction of the motion of the relative velocity and that of the motion of the CM. Of course, if there were, the collisions would spray it about, so it is all sprayed around. So the cosine of the angle between $\mathbf{w}$ and $\mathbf{v}_{CM}$ is zero on the average. That is,

$$\langle \mathbf{w} \cdot \mathbf{v}_{CM} \rangle = 0. \quad (39.19)$$

* This argument, which was the one used by Maxwell, involves some subtleties. Although the conclusion is correct, the result does *not* follow purely from the considerations of symmetry that we used before, since, by going to a reference frame moving through the gas, we may find a distorted velocity distribution. We have not found a simple proof of this result.

But $\mathbf{w} \cdot \mathbf{v}_{CM}$ can be expressed in terms of $\mathbf{v}_1$ and $\mathbf{v}_2$ as well:

$$\begin{aligned}\mathbf{w} \cdot \mathbf{v}_{CM} &= \frac{(\mathbf{v}_1 - \mathbf{v}_2) \cdot (m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2)}{m_1 + m_2} \\ &= \frac{(m_1 v_1^2 - m_2 v_2^2) + (m_2 - m_1)(\mathbf{v}_1 \cdot \mathbf{v}_2)}{m_1 + m_2}.\end{aligned}\quad (39.20)$$

First, let us look at the $\mathbf{v}_1 \cdot \mathbf{v}_2$; what is the average of $\mathbf{v}_1 \cdot \mathbf{v}_2$? That is, what is the average of the component of velocity of one molecule in the direction of another? Surely there is just as much likelihood of finding any given molecule moving one way as another. *The average of the velocity $\mathbf{v}_2$ in any direction is zero.* Certainly, then, in the direction of $\mathbf{v}_1$, $\mathbf{v}_2$ has zero average. So, the average of $\mathbf{v}_1 \cdot \mathbf{v}_2$ is zero! Therefore, we conclude that the average of $m_1 v_1^2$ must be equal to the average of $m_2 v_2^2$. That is, *the average kinetic energy of the two must be equal:*

$$\frac{1}{2} m_1 v_1^2 = \frac{1}{2} m_2 v_2^2. \quad (39.21)$$

If we have two kinds of atoms in a gas, it can be shown, and we presume to have shown it, that the average of the kinetic energy of one is the same as the average of the kinetic energy of the other, when they are both in the same gas in the same box in equilibrium. That means that the heavy ones will move slower than the light ones; this is easily shown by experimentation with “atoms” of different masses in an air trough.

Now we would like to go one step further, and say that if we have two different gases *separated* in a box, they will also have equal average kinetic energy when they have finally come to equilibrium, even though they are not in the same box. We can make the argument in a number of ways. One way is to argue that if we have a fixed partition with a tiny hole in it (Fig. 39-4) so that one gas could leak out through the holes while the other could not, because the molecules are too big, and these had attained equilibrium, then we know that in one part, where they are mixed, they have the same average kinetic energy, but some come through the hole without loss of kinetic energy, so the average kinetic energy in the pure gas and in the mixture must be the same. That is not too satisfactory, because maybe there are no holes, for this kind of molecule, that separate one kind from the other.

Let us now go back to the piston problem. We can give an argument which shows that the kinetic energy of this piston must also be $\frac{1}{2} m_2 v_2^2$. Actually, that would be the kinetic energy due to the purely horizontal motion of the piston, so, forgetting its up and down motion, it will have to be the same as $\frac{1}{2} m_2 v_{2x}^2$. Likewise, from the equilibrium on the other side, we can prove that the kinetic energy of the piston is $\frac{1}{2} m_1 v_{1x}^2$. Although this is not in the middle of the gas, but is on one side of the gas, we can still make the argument, although it is a little more difficult, that the average kinetic energy of the piston and of the gas molecules are equal as a result of all the collisions.

If this still does not satisfy us, we may make an artificial example by which the equilibrium is generated by an object which can be hit on all sides. Suppose that we have a short rod with a ball on each end sticking through the piston, on a frictionless sliding universal joint. Each ball is round, like one of the molecules, and can be hit on all sides. This whole object has a certain total mass, m . Now, we have the gas molecules with mass m_1 and mass m_2 as before. The result of the collisions, by the analysis that was made before, is that the kinetic energy of m because of collisions with the molecules on one side must be $\frac{1}{2} m_1 v_1^2$, on the average. Likewise, because of the collisions with molecules on the other side, it has to be $\frac{1}{2} m_2 v_2^2$ on the average. So, therefore, both sides have to have the *same* kinetic energy when they are in thermal equilibrium. So, although we only proved it for a mixture of gases, it is easily extended to the case where there are two different, separate gases at the same temperature.

Thus *when we have two gases at the same temperature, the mean kinetic energy of the CM motions are equal.*

The mean molecular kinetic energy is a property only of the “temperature.” Being a property of the “temperature,” *and not of the gas*, we can use it as a *definition* of the temperature. The mean kinetic energy of a molecule is thus some

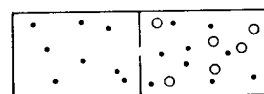


Fig. 39-4. Two gases in a box with a semipermeable membrane.

function of the temperature. But who is to tell us what scale to use for the temperature? We may arbitrarily *define* the scale of temperature so that the mean energy is linearly proportional to the temperature. The best way to do it would be to call the mean energy itself “the temperature.” That would be the simplest possible function. Unfortunately, the scale of temperature has been chosen differently, so instead of calling it temperature directly we use a constant conversion factor between the energy of a molecule and a degree of absolute temperature called a degree Kelvin. The constant of proportionality is $k = 1.38 \times 10^{-23}$ joule for every degree Kelvin.* So if T is absolute temperature, our definition says that the mean molecular kinetic energy is $\frac{3}{2} kT$. (The $\frac{3}{2}$ is put in as a matter of convenience, so as to get rid of it somewhere else.)

We point out that the kinetic energy associated with the component of motion in any particular direction is only $\frac{1}{2} kT$. The three independent directions that are involved make it $\frac{3}{2} kT$.

39-5 The ideal gas law

Now, of course, we can put our definition of temperature into Eq. (39.9) and so find the law for the pressure of gases as a function of the temperature: it is that the pressure times the volume is equal to the total number of atoms times the universal constant k , times the temperature:

$$PV = NkT. \quad (39.22)$$

Furthermore, at the same temperature and pressure and volume, the *number of atoms* is determined; it too is a universal constant! So equal volumes of different gases, at the same pressure and temperature, have the same number of molecules, because of Newton’s laws. That is an amazing conclusion!

In practice, when dealing with molecules, because the numbers are so large, the chemists have artificially chosen a specific number, a very large number, and called it something else. They have a number which they call a *mole*. A mole is merely a handy number. Why they did not choose 10^{24} objects, so it would come out even, is a historical question. They happened to choose, for the convenient number of objects on which they standardize, $N_0 = 6.02 \times 10^{23}$ objects, and this is called a mole of objects. So instead of measuring the number of molecules in units, they measure in terms of numbers of moles.† In terms of N_0 we can write the number of moles, times the number of atoms in a mole, times kT , and if we want to, we can take the number of atoms in a mole times k , which is a mole’s worth of k , and call it something else, and we do—we call it R . A mole’s worth of k is 8.317 joules: $R = N_0 k = 8.317 \text{ j} \cdot \text{mole}^{-1} \cdot ^\circ\text{K}^{-1}$. Thus we also find the gas law written as the number of moles (also called N) times RT , or the number of atoms, times kT :

$$PV = NRT. \quad (39.23)$$

It is the same thing, just a different scale for measuring numbers. We use 1 as a unit, and chemists use 6×10^{23} as a unit!

We now make one more remark about our gas law, and that has to do with the law for objects other than monatomic molecules. We have dealt only with the CM motion of the atoms of a monatomic gas. What happens if there are forces present? First, consider the case that the piston is held by a horizontal spring, and there are forces on it. The exchange of jiggling motion between atoms and piston at any moment does not depend on where the piston is at that moment, of course. The equilibrium conditions are the same. No matter where the piston is, its speed of motion must be such that it passes energy to the molecules in just

* The centigrade scale is just this Kelvin scale with a zero chosen at 273.16 °K, so $T = 273.16 + \text{centigrade temperature}$.

† What the chemists call molecular weights are the masses in grams of a mole of a molecule. The mole is defined so that the mass of a mole of carbon atoms of isotope 12 (i.e., having 6 protons and 6 neutrons in the nucleus) is exactly 12 grams.

the right way. So it makes no difference about the spring. The *speed* at which the piston has to move, on the average, is the same. So our theorem, that the mean value of the kinetic energy in one direction is $\frac{1}{2}kT$, is true whether there are forces present or not.

Consider, for example, a diatomic molecule composed of atoms m_A and m_B . What we have proved is that the motion of the CM of part A and that of part B are such that $\langle \frac{1}{2}m_A v_A^2 \rangle = \langle \frac{1}{2}m_B v_B^2 \rangle = \frac{3}{2}kT$. How can this be, if they are held together? Although they are held together, when they are spinning and turning in there, when something hits them, exchanging energy with them, *the only thing that counts is how fast they are moving*. That alone determines how fast they exchange energy in collisions. At the particular instant, the force is not an essential point. Therefore the same principle is right, even when there are forces.

Let us prove, finally, that the gas law is consistent also with a disregard of the internal motion. We did not really include the internal motions before; we just treated a monatomic gas. But we shall now show that an entire object, considered as a single body of total mass M , has a velocity of the CM such that

$$\frac{1}{2}Mv_{CM}^2 = \frac{3}{2}kT. \quad (39.24)$$

In other words, we can consider either the separate pieces or the whole thing! Let us see the reason for that: The mass of the diatomic molecule is $M = m_A + m_B$, and the velocity of the center of mass is equal to $\mathbf{v}_{CM} = (m_A \mathbf{v}_A + m_B \mathbf{v}_B)/M$. Now we need $\langle v_{CM}^2 \rangle$. If we square $\mathbf{v}_{CM}$, we get

$$v_{CM}^2 = \frac{m_A^2 v_A^2 + 2m_A m_B \mathbf{v}_A \cdot \mathbf{v}_B + m_B^2 v_B^2}{M^2}.$$

Now we multiply $\frac{1}{2}M$ and take the average, and thus we get

$$\begin{aligned} \frac{1}{2}Mv_{CM}^2 &= \frac{m_A \frac{3}{2}kT + 2m_A m_B \langle \mathbf{v}_A \cdot \mathbf{v}_B \rangle + m_B \frac{3}{2}kT}{M} \\ &= \frac{3}{2}kT + \frac{2m_A m_B \langle \mathbf{v}_A \cdot \mathbf{v}_B \rangle}{M}. \end{aligned}$$

(We have used the fact that $(m_A + m_B)/M = 1$.) Now what is $\langle \mathbf{v}_A \cdot \mathbf{v}_B \rangle$? (It had better be zero!) To find out, let us use our assumption that the relative velocity, $\mathbf{w} = \mathbf{v}_A - \mathbf{v}_B$ is not any more likely to point in one direction than in another—that is, that its average component in any direction is zero. Thus we assume that

$$\langle \mathbf{w} \cdot \mathbf{v}_{CM} \rangle = 0.$$

But what is $\mathbf{w} \cdot \mathbf{v}_{CM}$? It is

$$\begin{aligned} \mathbf{w} \cdot \mathbf{v}_{CM} &= \frac{(\mathbf{v}_A - \mathbf{v}_B) \cdot (m_A \mathbf{v}_A + m_B \mathbf{v}_B)}{M} \\ &= \frac{m_A v_A^2 + (m_B - m_A)(\mathbf{v}_A \cdot \mathbf{v}_B) - m_B v_B^2}{M}. \end{aligned}$$

Therefore, since $\langle m_A v_A^2 \rangle = \langle m_B v_B^2 \rangle$, the first and last terms cancel out on the average, and we are left with

$$(m_B - m_A) \langle \mathbf{v}_A \cdot \mathbf{v}_B \rangle = 0.$$

Thus if $m_A \neq m_B$, we find that $\langle \mathbf{v}_A \cdot \mathbf{v}_B \rangle = 0$, and therefore that the bodily motion of the entire molecule, regarded as a single particle of mass M , has a kinetic energy, on the average, equal to $\frac{3}{2}kT$.

Incidentally, we have also proved at the same time that the average kinetic energy of the *internal* motions of the diatomic molecule, disregarding the bodily motion of the CM, is $\frac{3}{2}kT$! For, the total kinetic energy of the parts of the molecule is $\frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2$, whose average is $\frac{3}{2}kT + \frac{3}{2}kT$, or $3kT$. The kinetic energy of the center-of-mass motion is $\frac{3}{2}kT$, so the average kinetic energy of the rotational and vibratory motions of the two atoms inside the molecule is the difference, $\frac{3}{2}kT$.

The theorem concerning the average energy of the CM motion is general: for any object considered as a whole, with forces present or no, for every independent direction of motion that there is, the average kinetic energy in that motion is $\frac{1}{2}kT$. These “independent directions of motion” are sometimes called the *degrees of freedom* of the system. The number of degrees of freedom of a molecule composed of r atoms is $3r$, since each atom needs three coordinates to define its position. The entire kinetic energy of the molecule can be expressed either as the sum of the kinetic energies of the separate atoms, or as the sum of the kinetic energy of the CM motion plus the kinetic energy of the internal motions. The latter can sometimes be expressed as a sum of rotational kinetic energy of the molecule and vibrational energy, but this is an approximation. Our theorem, applied to the r -atom molecule, says that the molecule will have, on the average, $3rkT/2$ joules of kinetic energy, of which $\frac{3}{2}kT$ is kinetic energy of the center-of-mass motion of the entire molecule, and the rest, $\frac{3}{2}(r - 1)kT$, is internal vibrational and rotational kinetic energy.

The Principles of Statistical Mechanics

40-1 The exponential atmosphere

We have discussed some of the properties of large numbers of intercolliding atoms. The subject is called kinetic theory, a description of matter from the point of view of collisions between the atoms. Fundamentally, we assert that the gross properties of matter should be explainable in terms of the motion of its parts.

We limit ourselves for the present to conditions of thermal equilibrium, that is, to a subclass of all the phenomena of nature. The laws of mechanics which apply just to thermal equilibrium are called *statistical mechanics*, and in this section we want to become acquainted with some of the central theorems of this subject.

We already have one of the theorems of statistical mechanics, namely, the mean value of the kinetic energy for any motion at the absolute temperature T is $\frac{1}{2}kT$ for each independent motion, i.e., for each degree of freedom. That tells us something about the mean square velocities of the atoms. Our objective now is to learn more about the positions of the atoms, to discover how many of them are going to be in different places at thermal equilibrium, and also to go into a little more detail on the distribution of the velocities. Although we have the mean square velocity, we do not know how to answer a question such as how many of them are going three times faster than the root mean square, or how many of them are going one-quarter of the root mean square speed. Or have they all the same speed exactly?

So, these are the two questions that we shall try to answer: How are the molecules distributed in space when there are forces acting on them, and how are they distributed in velocity?

It turns out that the two questions are completely independent, and that the distribution of velocities is always the same. We already received a hint of the latter fact when we found that the average kinetic energy is the same, $\frac{1}{2}kT$ per degree of freedom, no matter what forces are acting on the molecules. The distribution of the velocities of the molecules is independent of the forces, because the collision rates do not depend upon the forces.

Let us begin with an example: the distribution of the molecules in an atmosphere like our own, but without the winds and other kinds of disturbance. Suppose that we have a column of gas extending to a great height, and at thermal equilibrium—unlike our atmosphere, which as we know gets colder as we go up. We could remark that if the temperature differed at different heights, we could demonstrate lack of equilibrium by connecting a rod to some balls at the bottom (Fig. 40-1), where they would pick up $\frac{1}{2}kT$ from the molecules there and would shake, via the rod, the balls at the top and those would shake the molecules at the top. So, ultimately, of course, the temperature becomes the same at all heights in a gravitational field.

If the temperature is the same at all heights, the problem is to discover by what law the atmosphere becomes tenuous as we go up. If N is the total number of molecules in a volume V of gas at pressure P , then we know $PV = NkT$, or $P = nkT$, where $n = N/V$ is the number of molecules per unit volume. In other words, if we know the number of molecules per unit volume, we know the pressure, and vice versa: they are proportional to each other, since the temperature is constant in this problem. But the pressure is not constant, it must increase as the altitude is reduced, because it has to hold, so to speak, the weight of all the gas above it. That is the clue by which we may determine how the pressure changes with height. If we take a unit area at height h , then the vertical force from below,

40-1 The exponential atmosphere

40-2 The Boltzmann law

40-3 Evaporation of a liquid

40-4 The distribution of molecular speeds

40-5 The specific heats of gases

40-6 The failure of classical physics

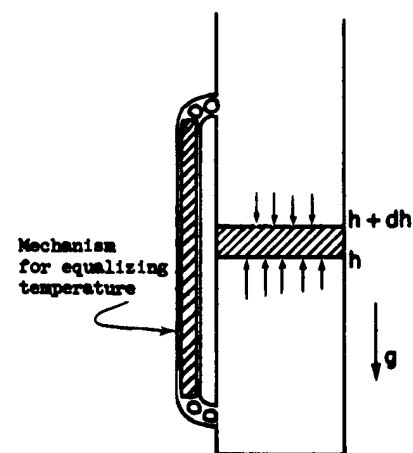


Fig. 40-1. The pressure at height h must exceed that at $h + dh$ by the weight of the intervening gas.

on this unit area, is the pressure P . The vertical force per unit area pushing down at a height $h + dh$ would be the same, in the absence of gravity, but here it is not, because the force from below must exceed the force from above by the weight of gas in the section between h and $h + dh$. Now mg is the force of gravity on each molecule, where g is the acceleration due to gravity, and $n dh$ is the total number of molecules in the unit section. So this gives us the differential equation $P_{h+dh} - P_h = dP = -mgn dh$. Since $P = nkT$, and T is constant, we can eliminate either P or n , say P , and get

$$\frac{dn}{dh} = -\frac{mg}{kT} n$$

for the differential equation, which tells us how the density goes down as we go up in energy.

We thus have an equation for the particle density n , which varies with height, but which has a derivative which is proportional to itself. Now a function which has a derivative proportional to itself is an exponential, and the solution of this differential equation is

$$n = n_0 e^{-mgh/kT}. \quad (40.1)$$

Here the constant of integration, n_0 , is obviously the density at $h = 0$ (which can be chosen anywhere), and the density goes down exponentially with height.

Note that if we have different kinds of molecules with different masses, they go down with different exponentials. The ones which were heavier would decrease with altitude faster than the light ones. Therefore we would expect that because oxygen is heavier than nitrogen, as we go higher and higher in an atmosphere with nitrogen and oxygen the proportion of nitrogen would increase. This does not really happen in our own atmosphere, at least at reasonable heights, because there is so much agitation which mixes the gases back together again. It is not an isothermal atmosphere. Nevertheless, there is a tendency for lighter materials, like hydrogen, to dominate at very great heights in the atmosphere, because the lowest masses continue to exist, while the other exponentials have all died out (Fig. 40-2).

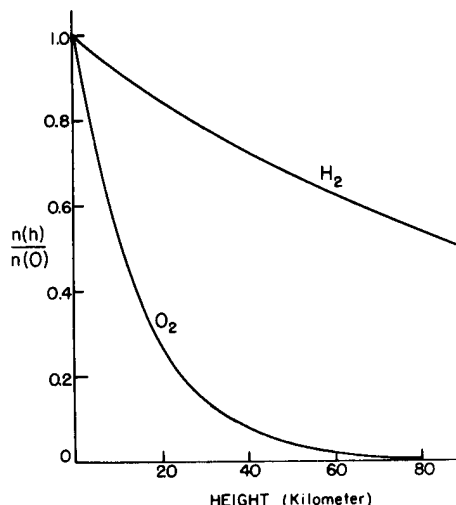


Fig. 40-2. The normalized density as a function of height in the earth's gravitational field for oxygen and for hydrogen, at constant temperature.

40-2 The Boltzmann law

Here we note the interesting fact that the numerator in the exponent of Eq. (40.1) is the *potential energy* of an atom. Therefore we can also state this particular law as: the density at any point is proportional to

$$e^{-(\text{the potential energy of each atom}/kT)}.$$

That may be an accident, i.e., may be true only for this particular case of a uniform gravitational field. However, we can show that it is a more general proposition. Suppose that there were some kind of force other than gravity acting on the molecules in a gas. For example, the molecules may be charged electrically, and may be acted on by an electric field or another charge that attracts them. Or, because of the mutual attractions of the atoms for each other, or for the wall, or for a solid, or something, there is some force of attraction which varies with position and which acts on all the molecules. Now suppose, for simplicity, that the molecules are all the same, and that the force acts on each individual one, so that the total force on a piece of gas would be simply the number of molecules times the force on each one. To avoid unnecessary complication, let us choose a coordinate system with the x -axis in the direction of the force, F .

In the same manner as before, if we take two parallel planes in the gas, separated by a distance dx , then the force on each atom, times the n atoms per cm^3 (the generalization of the previous nmg), times dx , must be balanced by the pressure change: $Fn dx = dP = kT dn$. Or, to put this law in a form which will be useful to us later,

$$F = kT \frac{d}{dx} (\ln n). \quad (40.2)$$

For the present, observe that $-F dx$ is the work we would do in taking a molecule from x to $x + dx$, and if F comes from a potential, i.e., if the work done can be represented by a potential energy at all, then this would also be the difference in the potential energy (P.E.). The negative differential of potential energy is the work done, $F dx$, and we find that $d(\ln n) = -d(\text{P.E.})/kT$, or, after integrating,

$$n = (\text{constant})e^{-\text{P.E.}/kT}. \quad (40.3)$$

Therefore what we noticed in a special case turns out to be true in general. (What if F does not come from a potential? Then (40.2) has no solution at all. Energy can be generated, or lost by the atoms running around in cyclic paths for which the work done is not zero, and no equilibrium can be maintained at all. Thermal equilibrium cannot exist if the external forces on the atoms are not conservative.) Equation (40.3), known as *Boltzmann's law*, is another of the principles of statistical mechanics: that the probability of finding molecules in a given spatial arrangement varies exponentially with the negative of the potential energy of that arrangement, divided by kT .

This, then, could tell us the distribution of molecules: Suppose that we had a positive ion in a liquid, attracting negative ions around it, how many of them would be at different distances? If the potential energy is known as a function of distance, then the proportion of them at different distances is given by this law, and so on, through many applications.

40-3 Evaporation of a liquid

In more advanced statistical mechanics one tries to solve the following important problem. Consider an assembly of molecules which attract each other, and suppose that the force between any two, say i and j , depends only on their separation r_{ij} , and can be represented as the derivative of a potential function $V(r_{ij})$. Figure 40-3 shows a form such a function might have. For $r > r_0$, the energy decreases as the molecules come together, because they attract, and then the energy increases very sharply as they come still closer together, because they repel strongly, which is characteristic of the way molecules behave, roughly speaking.

Now suppose we have a whole box full of such molecules, and we would like to know how they arrange themselves on the average. The answer is $e^{-\text{P.E.}/kT}$. The total potential energy in this case would be the sum over all the pairs, supposing that the forces are all in pairs (there may be three-body forces in more complicated things, but in electricity, for example, the potential energy is all in pairs). Then the probability for finding molecules in any particular combination of r_{ij} 's will be proportional to

$$\exp \left[- \sum_{i,j} V(r_{ij})/kT \right].$$

Now, if the temperature is very high, so that $kT \gg |V(r_0)|$, the exponent is relatively small almost everywhere, and the probability of finding a molecule is almost independent of position. Let us take the case of just two molecules: the $e^{-\text{P.E.}/kT}$ would be the probability of finding them at various mutual distances r . Clearly, where the potential goes most negative, the probability is largest, and where the potential goes toward infinity, the probability is almost zero, which occurs for very small distances. That means that for such atoms in a gas, there is no chance that they are on top of each other, since they repel so strongly. But there is a greater chance of finding them *per unit volume* at the point r_0 than at any other point. How much greater, depends on the temperature. If the temperature is very large compared with the difference in energy between $r = r_0$ and $r = \infty$, the exponential is always nearly unity. In this case, where the mean kinetic energy (about kT) greatly exceeds the potential energy, the forces do not make much difference. But as the temperature falls, the probability of finding the molecules at the preferred distance r_0 gradually increases relative to the probability of finding them elsewhere and, in fact, if kT is much less than $|V(r_0)|$, we have a relatively

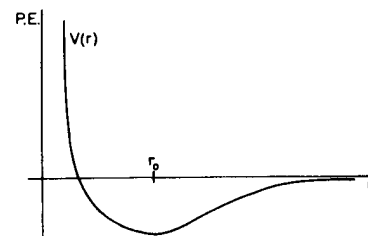


Fig. 40-3. A potential-energy function for two molecules, which depends only on their separation.

large positive exponent in that neighborhood. In other words, in a given volume they are *much more* likely to be at the distance of minimum energy than far apart. As the temperature falls, the atoms fall together, clump in lumps, and reduce to liquids, and solids, and molecules, and as you heat them up they evaporate.

The requirements for the determination of exactly how things evaporate, exactly how things should happen in a given circumstance, involve the following. First, to discover the correct molecular-force law $V(r)$, which must come from something else, quantum mechanics, say, or experiment. But, given the law of force between the molecules, to discover what a billion molecules are going to do merely consists of studying the function $e^{-\sum V_{ij}/kT}$. Surprisingly enough, since it is such a simple function and such an easy idea, given the potential, the labor is *enormously complicated*; the difficulty is the tremendous number of variables.

In spite of such difficulties, the subject is quite exciting and interesting. It is often called an example of a "many-body problem," and it really has been a very interesting thing. In that single formula must be contained all the details, for example, about the solidification of gas, or the forms of the crystals that the solid can take, and people have been trying to squeeze it out, but the mathematical difficulties are very great, not in writing the law, but in dealing with so enormous a number of variables.

That then, is the distribution of particles in space. That is the end of classical statistical mechanics, practically speaking, because if we know the forces, we can, in principle, find the distribution in space, and the distribution of velocities is something that we can work out once and for all, and is not something that is different for the different cases. The great problems are in getting particular information out of our formal solution, and that is the main subject of classical statistical mechanics.

40-4 The distribution of molecular speeds

Now we go on to discuss the distribution of velocities, because sometimes it is interesting or useful to know how many of them are moving at different speeds. In order to do that, we may make use of the facts which we discovered with regard to the gas in the atmosphere. We take it to be a perfect gas, as we have already assumed in writing the potential energy, disregarding the energy of mutual attraction of the atoms. The only potential energy that we included in our first example was gravity. We would, of course, have something more complicated if there were forces between the atoms. Thus we assume that there are no forces between the atoms and, for a moment, disregard collisions also, returning later to the justification of this. Now we saw that there are fewer molecules at the height h than there are at the height 0; according to formula (40.1), they decrease exponentially with height. How can there be fewer at greater heights? After all, do not all the molecules which are moving up at height 0 arrive at h ? No!, because some of those which are moving up at 0 are going too slowly, and cannot climb the potential hill to h . With that clue, we can calculate how many must be moving at various speeds, because from (40.1) we know *how many* are moving with less than enough speed to climb a given distance h . Those are just the ones that account for the fact that the density at h is lower than at 0.

Now let us put that idea a little more precisely: let us count how many molecules are passing from below to above the plane $h = 0$ (by calling it height = 0, we do not mean that there is a floor there; it is just a convenient label, and there is gas at negative h). These gas molecules are moving around in every direction, but some of them are moving through the plane, and at any moment a certain number per second of them are passing through the plane from below to above with different velocities. Now we note the following: if we call u the velocity which is just needed to get up to the height h (kinetic energy $mu^2/2 = mgh$), then the number of molecules per second which are passing upward through the lower plane in a vertical direction with velocity component greater than u is exactly the same as the number which pass through the upper plane with *any* upward velocity. Those molecules whose vertical velocity does not exceed u cannot get through the upper

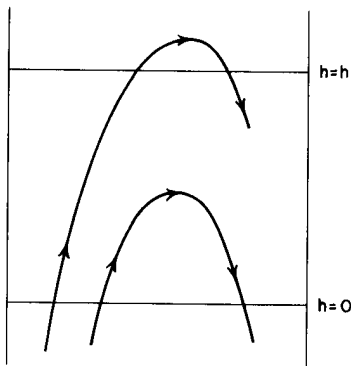


Fig. 40-4. Only those molecules moving up at $h = 0$ with sufficient velocity can arrive at height h .

plane. So therefore we see that

$$\text{Number passing } h = 0 \text{ with } v_z > u = \text{number passing } h = h \text{ with } v_z > 0.$$

But the number which pass through h with any velocity greater than 0 is less than the number which pass through the lower height with any velocity greater than 0, because the number of atoms is greater; that is all we need. We know already that the distribution of velocities is the same, after the argument we made earlier about the temperature being constant all the way through the atmosphere. So, since the velocity distributions are the same, and it is just that there are *more atoms* lower down, clearly the number $n_{>0}(h)$, passing with positive velocity at height h , and the number $n_{>0}(0)$, passing with positive velocity at height 0, are in the same ratio as the densities at the two heights, which is $e^{-mgh/kT}$. But $n_{>0}(h) = n_{>u}(0)$, and therefore we find that

$$\frac{n_{>u}(0)}{n_{>0}(0)} = e^{-mgh/kT} = e^{-mu^2/2kT},$$

since $\frac{1}{2}mu^2 = mgh$. Thus, in words, the number of molecules per unit area per second passing the height 0 with a z -component of velocity greater than u is $e^{-mu^2/2kT}$ times the total number that are passing through the plane with velocity greater than zero.

Now this is not only true at the arbitrarily chosen height 0, but of course it is true at any other height, and thus the distributions of velocities *are* all the same! (The final statement does not involve the height h , which appeared only in the intermediate argument.) The result is a general proposition that gives us the distribution of velocities. It tells us that if we drill a little hole in the side of a gas pipe, a very tiny hole, so that the collisions are few and far between, i.e., are farther apart than the diameter of the hole, then the particles which are coming out will have different velocities, but the fraction of particles which come out at a velocity greater than u is $e^{-mu^2/2kT}$.

Now we return to the question about the neglect of collisions: Why does it not make any difference? We could have pursued the same argument, not with a finite height h , but with an infinitesimal height h , which is so small that there would be no room for collisions between 0 and h . But that was not necessary: the argument is evidently based on an analysis of the energies involved, the conservation of energy, and in the collisions that occur there is an exchange of energies among the molecules. However, we do not really care whether we follow the same molecule if energy is merely exchanged with another molecule. So it turns out that even if the problem is analyzed more carefully (and it is more difficult, naturally, to do a rigorous job), it still makes no difference in the result.

It is interesting that the velocity distribution we have found is just

$$n_{>u} \propto e^{-\text{kinetic energy}/kT}. \quad (40.4)$$

This way of describing the distribution of velocities, by giving the number of molecules that pass a given area with a certain minimum z -component, is not the most convenient way of giving the velocity distribution. For instance, inside the gas, one more often wants to know how many molecules are moving with a z -component of velocity between two given values, and that, of course, is not directly given by Eq. (40.4). We would like to state our result in the more conventional form, even though what we already have written is quite general. *Note that it is not possible to say that any molecule has exactly some stated velocity; none of them has a velocity exactly equal to 1.7962899173 meters per second.* So in order to make a meaningful statement, we have to ask how many are to be found in some *range* of velocities. We have to say how many have velocities between 1.796 and 1.797, and so on. On mathematical terms, let $f(u) du$ be the fraction of all the molecules which have velocities between u and $u + du$ or, what is the same thing (if du is infinitesimal), all that have a velocity u with a range du . Figure 40-5 shows a possible form for the function $f(u)$, and the shaded part, of width du and mean height $f(u)$, represents this fraction $f(u) du$. That is, the ratio of the shaded

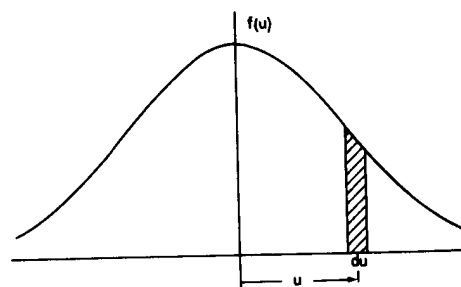


Fig. 40-5. A velocity distribution function. The shaded area is $f(u) du$, the fraction of particles having velocities within a range du about u .

area to the total area of the curve is the relative proportion of molecules with velocity u within du . If we define $f(u)$ so that the fraction having a velocity in this range is given directly by the shaded area, then the total area must be 100 percent of them, that is,

$$\int_{-\infty}^{\infty} f(u) du = 1. \quad (40.5)$$

Now we have only to get this distribution by comparing it with the theorem we derived before. First we ask, what is the number of molecules passing through an area per second with a velocity greater than u , expressed in terms of $f(u)$? At first we might think it is merely the integral of $\int_u^{\infty} f(u) du$, but it is not, because we want the number that are passing the area *per second*. The faster ones pass more often, so to speak, than the slower ones, and in order to express how many pass, you have to multiply by the velocity. (We discussed that in the previous chapter when we talked about the number of collisions.) In a given time t the total number which pass through the surface is all of those which have been able to arrive at the surface, and the number which arrive come from a distance ut . So the number of molecules which arrive is not simply the number which are there, but the number that are there per unit volume, multiplied by the distance that they sweep through in racing for the area through which they are supposed to go, and that distance is proportional to u . Thus we need the integral of u times $f(u) du$, an infinite integral with a lower limit u , and this must be the same as we found before, namely $e^{-mu^2/2kT}$, with a proportionality constant which we will get later:

$$\int_u^{\infty} uf(u) du = \text{const} \cdot e^{-mu^2/2kT}. \quad (40.6)$$

Now if we differentiate the integral with respect to u , we get the thing that is inside the integral, i.e., the integrand (with a minus sign, since u is the lower limit), and if we differentiate the other side, we get u times the same exponential (and some constants). The u 's cancel and we find

$$f(u) du = Ce^{-mu^2/2kT} du. \quad (40.7)$$

We retain the du on both sides as a reminder that it is a *distribution*, and it tells what the proportion is for velocity between u and $u + du$.

The constant C must be so determined that the integral is unity, according to Eq. (40.5). Now we can prove* that

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Using this fact, it is easy to find that $C = \sqrt{m/2\pi kT}$.

Since velocity and momentum are proportional, we may say that the distribution of momenta is also proportional to $e^{-K.E./kT}$ per unit momentum range. It turns out that this theorem is true in relativity too, if it is in terms of momentum, while if it is in velocity it is not, so it is best to learn it in momentum instead of in velocity:

$$f(p) dp = Ce^{-K.E./kT} dp. \quad (40.8)$$

So we find that the probabilities of different conditions of energy, kinetic and potential, are both given by $e^{-\text{energy}/kT}$, a very easy thing to remember and a rather beautiful proposition.

* To get the value of the integral, let

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx.$$

Then

$$I^2 = \int_{-\infty}^{\infty} e^{-x^2} dx \cdot \int_{-\infty}^{\infty} e^{-y^2} dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dy dx,$$

which is a double integral over the whole xy -plane. But this can also be written in polar coordinates as

$$I^2 = \int_0^{\infty} e^{-r^2} \cdot 2\pi r dr = \pi \int_0^{\infty} e^{-t} dt = \pi.$$

So far we have, of course, only the distribution of the velocities “vertically.” We might want to ask, what is the probability that a molecule is moving in another direction? Of course these distributions are connected, and one can obtain the complete distribution from the one we have, because the complete distribution depends only on the square of the magnitude of the velocity, not upon the z -component. It must be something that is independent of direction, and there is only one function involved, the probability of different magnitudes. We have the distribution of the z -component, and therefore we can get the distribution of the other components from it. The result is that the probability is still proportional to $e^{-K.E./kT}$, but now the kinetic energy involves three parts, $mv_x^2/2$, $mv_y^2/2$, and $mv_z^2/2$, summed in the exponent. Or we can write it as a product:

$$f(v_x, v_y, v_z) dv_x dv_y dv_z \propto e^{-mv_x^2/2kT} \cdot e^{-mv_y^2/2kT} \cdot e^{-mv_z^2/2kT} dv_x dv_y dv_z. \quad (40.9)$$

You can see that this formula must be right because, first, it is a function only of v^2 , as required, and second, the probabilities of various values of v_z obtained by integrating over all v_x and v_y is just (40.7). But this one function (40.9) can do both those things!

40-5 The specific heats of gases

Now we shall look at some ways to test the theory, and to see how successful is the classical theory of gases. We saw earlier that if U is the internal energy of N molecules, then $PV = NkT = (\gamma - 1)U$ holds, sometimes, for some gases, maybe. If it is a monatomic gas, we know this is also equal to $\frac{2}{3}$ of the kinetic energy of the center-of-mass motion of the atoms. If it is a monatomic gas, then the kinetic energy is equal to the internal energy, and therefore $\gamma - 1 = \frac{2}{3}$. But suppose it is, say, a more complicated molecule, that can spin and vibrate, and let us suppose (it turns out to be true according to classical mechanics) that the energies of the internal motions are also proportional to kT . Then at a given temperature, in addition to kinetic energy kT , it has internal vibrational or rotational energy. So the total U includes not just the internal kinetic energy, but also the rotational energy, and we get a different value of γ . Technically, the best way to measure γ is by measuring the specific heat, which is the change in energy with temperature. We will return to that approach later. For our present purposes, we may suppose γ is found experimentally from the PV^γ curve for adiabatic compression.

Let us make a calculation of γ for some cases. First, for a monatomic gas U is the total energy, the same as the kinetic energy, and we know already that γ should be $\frac{5}{3}$. For a diatomic gas, we may take, as an example, oxygen, hydrogen iodide, hydrogen, etc., and suppose that the diatomic gas can be represented as two atoms held together by some kind of force like the one of Fig. 40-3. We may also suppose, and it turns out to be quite true, that at the temperatures that are of interest for the diatomic gas, the pairs of atoms tend strongly to be separated by r_0 , the distance of potential minimum. If this were not true, if the probability were not strongly varying enough to make the great majority sit near the bottom, we would have to remember that oxygen gas is a mixture of O_2 and single oxygen atoms in a nontrivial ratio. We know that there are, in fact, very few single oxygen atoms, which means that the potential energy minimum is very much greater in magnitude than kT , as we have seen. Since they are clustered strongly around r_0 , the only part of the curve that is needed is the part near the minimum, which may be approximated by a parabola. A parabolic potential implies a harmonic oscillator, and in fact, to an excellent approximation, the oxygen molecule can be represented as two atoms connected by a spring.

Now what is the total energy of this molecule at temperature T ? We know that for each of the two atoms, each of the kinetic energies should be $\frac{3}{2}kT$, so the kinetic energy of both of them is $\frac{3}{2}kT + \frac{3}{2}kT$. We can also put this in a different way: the same $\frac{3}{2}$ plus $\frac{3}{2}$ can also be looked at as kinetic energy of the center of mass

Table 40-1

Values of the specific heat ratio, γ ,
for various gases

Gas	T ($^{\circ}\text{C}$)	γ
He	-180	1.660
Kr	19	1.68
Ar	15	1.668
H ₂	100	1.404
O ₂	100	1.399
HI	100	1.40
Br ₂	300	1.32
I ₂	185	1.30
NH ₃	15	1.310
C ₂ H ₆	15	1.22

($\frac{3}{2}$), kinetic energy of rotation ($\frac{2}{2}$), and kinetic energy of vibration ($\frac{1}{2}$). We know that the kinetic energy of vibration is $\frac{1}{2}$, since there is just one dimension involved and each degree of freedom has $\frac{1}{2}kT$. Regarding the rotation, it can turn about either of two axes, so there are two independent motions. We assume that the atoms are some kind of points, and cannot spin about the line joining them; this is something to bear in mind, because if we get a disagreement, maybe that is where the trouble is. But we have one more thing, which is the *potential* energy of vibration; how much is that? In a harmonic oscillator the average kinetic energy and average potential energy are equal, and therefore the potential energy of vibration is $\frac{1}{2}kT$, also. The grand total of energy is $U = \frac{7}{2}kT$, or kT is $\frac{2}{7}U$ per atom. That means, then, that γ is $\frac{9}{7}$ instead of $\frac{5}{3}$, i.e., $\gamma = 1.286$.

We may compare these numbers with the relevant measured values shown in Table 40-1. Looking first at helium, which is a monatomic gas, we find very nearly $\frac{5}{3}$, and the error is probably experimental, although at such a low temperature there may be some forces between the atoms. Krypton and argon, both monatomic, agree also within the accuracy of the experiment.

We turn to the diatomic gases and find hydrogen with 1.404, which does not agree with the theory, 1.286. Oxygen, 1.399, is very similar, but again not in agreement. Hydrogen iodide again is similar at 1.40. It begins to look as though the right answer is 1.40, but it is not, because if we look further at bromine we see 1.32, and at iodine we see 1.30. Since 1.30 is reasonably close to 1.286, iodine may be said to agree rather well, but oxygen is far off. So here we have a dilemma. We have it right for one molecule, we do not have it right for another molecule, and we may need to be pretty ingenious in order to explain both.

Let us look further at a still more complicated molecule with large numbers of parts, for example, C₂H₆, which is ethane. It has eight different atoms, and they are all vibrating and rotating in various combinations, so the total amount of internal energy must be an enormous number of kT 's, at least $12kT$ for kinetic energy alone, and $\gamma - 1$ must be very close to zero, or γ almost exactly 1. In fact, it is lower, but 1.22 is not so much lower, and is higher than the $1\frac{1}{2}$ calculated from the kinetic energy alone, and it is just not understandable!

Furthermore, the whole mystery is deep, because the diatomic molecule cannot be made rigid by a limit. Even if we made the couplings stiffer indefinitely, although it might not vibrate much, it would nevertheless keep vibrating. The vibrational energy inside is still kT , since it does not depend on the strength of the coupling. But if we could imagine *absolute* rigidity, stopping all vibration to eliminate a variable, then we would get $U = \frac{5}{2}kT$ and $\gamma = 1.40$ for the diatomic case. This looks good for H₂ or O₂. On the other hand, we would still have problems, because γ for either hydrogen or oxygen varies with temperature! From the measured values shown in Fig. 40-6, we see that for H₂, γ varies from about 1.6 at -185°C to 1.3 at 2000°C . The variation is more substantial in the case of hydrogen than for oxygen, but nevertheless, even in oxygen, γ tends definitely to go up as we go down in temperature.

40-6 The failure of classical physics

So, all in all, we might say that we have some difficulty. We might try some force law other than a spring, but it turns out that anything else will only make γ higher. If we include more forms of energy, γ approaches unity more closely, contradicting the facts. All the classical theoretical things that one can think of will only make it worse. The fact is that there are electrons in each atom, and we know from their spectra that there are internal motions; each of the electrons should have at least $\frac{1}{2}kT$ of kinetic energy, and something for the potential energy, so when these are added in, γ gets still smaller. It is ridiculous. It is wrong.

The first great paper on the dynamical theory of gases was by Maxwell in 1859. On the basis of ideas we have been discussing, he was able accurately to explain a great many known relations, such as Boyle's law, the diffusion theory, the viscosity of gases, and things we shall talk about in the next chapter. He listed all these great successes in a final summary, and at the end he said, "Finally, by

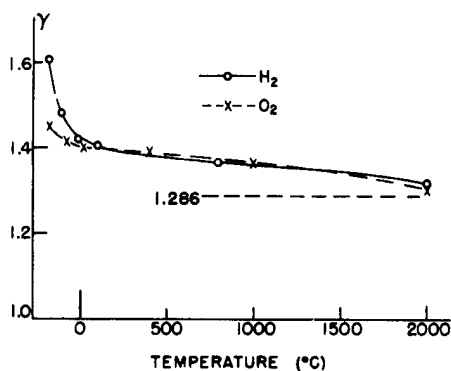


Fig. 40-6. Experimental values of γ as a function of temperature for hydrogen and oxygen. Classical theory predicts $\gamma = 1.286$, independent of temperature.

establishing a necessary relation between the motions of translation and rotation (he is talking about the $\frac{1}{2}kT$ theorem) of all particles not spherical, we proved that a system of such particles could not possibly satisfy the known relation between the two specific heats." He is referring to γ (which we shall see later is related to two ways of measuring specific heat), and he says we know we cannot get the right answer.

Ten years later, in a lecture, he said, "I have now put before you what I consider to be the greatest difficulty yet encountered by the molecular theory." These words represent the first discovery that the laws of classical physics were wrong. This was the first indication that there was something fundamentally impossible, because a rigorously proved theorem did not agree with experiment. About 1890, Jeans was to talk about this puzzle again. One often hears it said that physicists at the latter part of the nineteenth century thought they knew all the significant physical laws and that all they had to do was to calculate more decimal places. Someone may have said that once, and others copied it. But a thorough reading of the literature of the time shows they were all worrying about something. Jeans said about this puzzle that it is a very mysterious phenomenon, and it seems as though as the temperature falls, certain kinds of motions "freeze out."

If we could assume that the vibrational motion, say, did not exist at low temperature and did exist at high temperature, then we could imagine that a gas might exist at a temperature sufficiently low that vibrational motion does not occur, so $\gamma = 1.40$, or a higher temperature at which it begins to come in, so γ falls. The same might be argued for the rotation. If we can eliminate the rotation, say it "freezes out" at sufficiently low temperature, then we can understand the fact that the γ of hydrogen approaches 1.66 as we go down in temperature. How can we understand such a phenomenon? Of course that these motions "freeze out" cannot be understood by classical mechanics. It was only understood when quantum mechanics was discovered.

Without proof, we may state the results for statistical mechanics of the quantum-mechanical theory. We recall that according to quantum mechanics, a system which is bound by a potential, for the vibrations, for example, will have a discrete set of energy levels, i.e., states of different energy. Now the question is: how is statistical mechanics to be modified according to quantum-mechanical theory? It turns out, interestingly enough, that although most problems are more difficult in quantum mechanics than in classical mechanics, problems in statistical mechanics are much easier in quantum theory! The simple result we have in classical mechanics, that $n = n_0 e^{-\text{energy}/kT}$, becomes the following very important theorem: If the energies of the set of molecular states are called, say, $E_0, E_1, E_2, \dots, E_i, \dots$, then in thermal equilibrium the probability of finding a molecule in the particular state of having energy E_i is proportional to $e^{-E_i/kT}$. That gives the probability of being in various states. In other words, the relative chance, the probability, of being in state E_1 relative to the chance of being in state E_0 , is

$$\frac{P_1}{P_0} = \frac{e^{-E_1/kT}}{e^{-E_0/kT}}, \quad (40.10)$$

which, of course, is the same as

$$n_1 = n_0 e^{-(E_1 - E_0)/kT}, \quad (40.11)$$

since $P_1 = n_1/N$ and $P_0 = n_0/N$. So it is less likely to be in a higher energy state than in a lower one. The ratio of the number of atoms in the upper state to the number in the lower state is e raised to the power (minus the energy difference, over kT)—a very simple proposition.

Now it turns out that for a harmonic oscillator the energy levels are evenly spaced. Calling the lowest energy $E_0 = 0$ (it actually is not zero, it is a little different, but it does not matter if we shift all energies by a constant), the first one is then $E_1 = \hbar\omega$, and the second one is $2\hbar\omega$, and the third one is $3\hbar\omega$, and so on.

Now let us see what happens. We suppose we are studying the vibrations of a diatomic molecule, which we approximate as a harmonic oscillator. Let us ask

what is the relative chance of finding a molecule in state E_1 instead of in state E_0 . The answer is that the chance of finding it in state E_1 , relative to that of finding it in state E_0 , goes down as $e^{-\hbar\omega/kT}$. Now suppose that kT is much less than $\hbar\omega$, and we have a low-temperature circumstance. Then the probability of its being in state E_1 is extremely small. Practically all the atoms are in state E_0 . If we change the temperature but still keep it very small, then the chance of its being in state $E_1 = \hbar\omega$ remains infinitesimal—the energy of the oscillator remains nearly zero; it does not change with temperature so long as the temperature is much less than $\hbar\omega$. All oscillators are in the bottom state, and their motion is effectively “frozen”; *there is no contribution of it to the specific heat*. We can judge, then, from Table 40-1, that at 100°C, which is 373 degrees absolute, kT is much less than the vibrational energy in the oxygen or hydrogen molecules, but not so in the iodine molecule. The reason for the difference is that an iodine atom is very heavy, compared with hydrogen, and although the forces may be comparable in iodine and hydrogen, the iodine molecule is so heavy that the natural frequency of vibration is very low compared with the natural frequency of hydrogen. With $\hbar\omega$ higher than kT at room temperature for hydrogen, but lower for iodine, only the latter, iodine, exhibits the classical vibrational energy. As we increase the temperature of a gas, starting from a very low value of T , with the molecules almost all in their lowest state, they gradually begin to have an appreciable probability to be in the second state, and then in the next state, and so on. When the probability is appreciable for many states, the behavior of the gas approaches that given by classical physics, because the quantized states become nearly indistinguishable from a continuum of energies, and the system can have almost any energy. Thus, as the temperature rises, we should again get the results of classical physics, as indeed seems to be the case in Fig. 40-6. It is possible to show in the same way that the rotational states of atoms are also quantized, but the states are so much closer together that in ordinary circumstances kT is bigger than the spacing. Then many levels are excited, and the rotational kinetic energy in the system participates in the classical way. The one example where this is not quite true at room temperature is for hydrogen.

This is the first time that we have really deduced, by comparison with experiment, that there was something wrong with classical physics, and we have looked for a resolution of the difficulty in quantum mechanics in much the same way as it was done originally. It took 30 or 40 years before the next difficulty was discovered, and that had to do again with statistical mechanics, but this time the mechanics of a photon gas. That problem was solved by Planck, in the early years of this century.

The Brownian Movement

41-1 Equipartition of energy

The Brownian movement was discovered in 1827 by Robert Brown, a botanist. While he was studying microscopic life, he noticed little particles of plant pollens jiggling around in the liquid he was looking at in the microscope, and he was wise enough to realize that these were not living, but were just little pieces of dirt moving around in the water. In fact he helped to demonstrate that this had nothing to do with life by getting from the ground an old piece of quartz in which there was some water trapped. It must have been trapped for millions and millions of years, but inside he could see the same motion. What one sees is that very tiny particles are jiggling all the time.

This was later proved to be one of the effects of *molecular motion*, and we can understand it qualitatively by thinking of a great push ball on a playing field, seen from a great distance, with a lot of people underneath, all pushing the ball in various directions. We cannot see the people because we imagine that we are too far away, but we can see the ball, and we notice that it moves around rather irregularly. We also know, from the theorems that we have discussed in previous chapters, that the mean kinetic energy of a small particle suspended in a liquid or a gas will be $\frac{3}{2}kT$ even though it is very heavy compared with a molecule. If it is very heavy, that means that the speeds are relatively slow, but it turns out, actually, that the speed is not really so slow. In fact, we cannot see the speed of such a particle very easily because although the mean kinetic energy is $\frac{3}{2}kT$, which represents a speed of a millimeter or so per second for an object a micron or two in diameter, this is very hard to see even in a microscope, because the particle continuously reverses its direction and does not get anywhere. How far it does get we will discuss at the end of the present chapter. This problem was first solved by Einstein at the beginning of the present century.

Incidentally, when we say that the mean kinetic energy of this particle is $\frac{3}{2}kT$, we claim to have derived this result from the kinetic theory, that is, from Newton's laws. We shall find that we can derive all kinds of things—marvelous things—from the kinetic theory, and it is most interesting that we can apparently get so much from so little. Of course we do not mean that Newton's laws are “little”—they are enough to do it, really—what we mean is that *we* did not do very much. How do we get so much out? The answer is that we have been perpetually making a certain important assumption, which is that if a given system is in thermal equilibrium at some temperature, it will also be in thermal equilibrium with *anything else* at the same temperature. For instance, if we wanted to see how a particle would move if it was really colliding with water, we could imagine that there was a gas present, composed of another kind of particle, little fine pellets that (we suppose) do not interact with water, but only hit the particle with “hard” collisions. Suppose the particle has a prong sticking out of it; all our pellets have to do is hit the prong. We know all about this imaginary gas of pellets at temperature T —it is an ideal gas. Water is complicated, but an ideal gas is simple. Now, *our particle has to be in equilibrium with the gas of pellets*. Therefore, the mean motion of the particle must be what we get for gaseous collisions, because if it were not moving at the right speed relative to the water but, say, was moving faster, that would mean that the pellets would pick up energy from it and get hotter than the water. But we had started them at the same temperature, and we assume that if a thing is once in equilibrium, it stays in equilibrium—parts of it do not get hotter and other parts colder, spontaneously.

41-1 Equipartition of energy

41-2 Thermal equilibrium of radiation

41-3 Equipartition and the quantum oscillator

41-4 The random walk

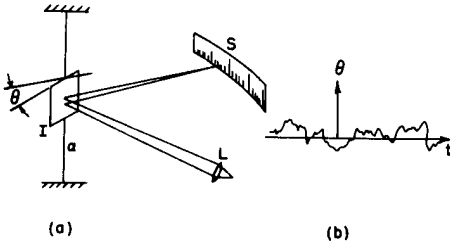


Fig. 41-1. (a) A sensitive light-beam galvanometer. Light from a source L is reflected from a small mirror onto a scale. (b) A schematic record of the reading of the scale as a function of the time.

This proposition is true and can be proved from the laws of mechanics, but the proof is very complicated and can be established only by using advanced mechanics. It is much easier to prove in quantum mechanics than it is in classical mechanics. It was proved first by Boltzmann, but for now we simply take it to be true, and then we can argue that our particle has to have $\frac{3}{2}kT$ of energy if it is hit with artificial pellets, so it also must have $\frac{3}{2}kT$ when it is being hit with water at the same temperature and we take away the pellets; so it is $\frac{3}{2}kT$. It is a strange line of argument, but perfectly valid.

In addition to the motion of colloidal particles for which the Brownian movement was first discovered, there are a number of other phenomena, both in the laboratory and in other situations, where one can see Brownian movement. If we are trying to build the most delicate possible equipment, say a very small mirror on a thin quartz fiber for a very sensitive ballistic galvanometer (Fig. 41-1), the mirror does not stay put, but jiggles all the time—all the time—so that when we shine a light on it and look at the position of the spot, we do not have a perfect instrument because the mirror is always jiggling. Why? Because the average kinetic energy of rotation of this mirror has to be, on the average, $\frac{1}{2}kT$.

What is the mean-square angle over which the mirror will wobble? Suppose we find the natural vibration period of the mirror by tapping on one side and seeing how long it takes to oscillate back and forth, and we also know the moment of inertia, I . We know the formula for the kinetic energy of rotation—it is given by Eq. (19.8): $T = \frac{1}{2}I\omega^2$. That is the kinetic energy, and the potential energy that goes with it will be proportional to the square of the angle—it is $V = \frac{1}{2}\alpha\theta^2$. But, if we know the period t_0 and calculate from that the natural frequency $\omega_0 = 2\pi/t_0$, then the potential energy is $V = \frac{1}{2}I\omega_0^2\theta^2$. Now we know that the average kinetic energy is $\frac{1}{2}kT$, but since it is a harmonic oscillator the average potential energy is also $\frac{1}{2}kT$. Thus

$$\frac{1}{2}I\omega_0^2\langle\theta^2\rangle = \frac{1}{2}kT,$$

or

$$\langle\theta^2\rangle = kT/I\omega_0^2. \quad (41.1)$$

In this way we can calculate the oscillations of a galvanometer mirror, and thereby find what the limitations of our instrument will be. If we want to have smaller oscillations, we have to cool the mirror. An interesting question is, *where* to cool it. This depends upon where it is getting its “kicks” from. If it is through the fiber, we cool it at the top—if the mirror is surrounded by a gas and is getting hit mostly by collisions in the gas, it is better to cool the gas. As a matter of fact, if we know where the *damping* of the oscillations comes from, it turns out that that is always the *source* of the fluctuations also, a point which we will come back to.

The same thing works, amazingly enough, in *electrical circuits*. Suppose that we are building a very sensitive, accurate amplifier for a definite frequency and have a resonant circuit (Fig. 41-2) in the input so as to make it very sensitive to this certain frequency, like a radio receiver, but a really good one. Suppose we wish to go down to the very lowest limit of things, so we take the voltage, say off the inductance, and send it into the rest of the amplifier. Of course, in any circuit like this, there is a certain amount of loss. It is not a perfect resonant circuit, but it is a very good one and there is a little resistance, say (we put the resistor in so we can see it, but it is supposed to be small). Now we would like to find out: How much does the voltage across the inductance fluctuate? *Answer*: We know that $\frac{1}{2}LI^2$ is the “kinetic energy”—the energy associated with a coil in a resonant circuit (Chapter 25). Therefore the mean value of $\frac{1}{2}LI^2$ is equal to $\frac{1}{2}kT$ —that tells us what the rms current is and we can find out what the rms voltage is from the rms current. For if we want the voltage across the inductance the formula is $\hat{V}_L = i\omega L\hat{I}$, and the mean absolute square voltage on the inductance is $\langle V_L^2 \rangle = L^2\omega_0^2\langle I^2 \rangle$, and putting in $\frac{1}{2}L\langle I^2 \rangle = \frac{1}{2}kT$, we obtain

$$\langle V_L^2 \rangle = L\omega_0^2 kT. \quad (41.2)$$

So now we can design circuits and tell when we are going to get what is called *Johnson noise*, the noise associated with thermal fluctuations!

Where do the fluctuations come from this time? They come again from the *resistor*—they come from the fact that the electrons in the resistor are jiggling around because they are in thermal equilibrium with the matter in the resistor, and they make fluctuations in the density of electrons. They thus make tiny electric fields which drive the resonant circuit.

Electrical engineers represent the answer in another way. Physically, the resistor is effectively the source of noise. However, we may replace the real circuit having an honest, true physical resistor which is making noise, by an artificial circuit which contains a little generator that is going to *represent* the noise, and now the resistor is otherwise ideal—no noise comes from it. All the noise is in the artificial generator. And so if we knew the characteristics of the noise generated by a resistor, if we had the formula for that, then we could calculate what the circuit is going to do in response to that noise. So, we need a formula for the noise fluctuations. Now the noise that is generated by the resistor is at all frequencies, since the resistor by itself is not resonant. Of course the resonant circuit only “listens” to the part that is near the right frequency, but the resistor has many different frequencies in it. We may describe how strong the generator is, as follows: The mean power that the resistor would absorb if it were connected directly across the noise generator would be $\langle E^2 \rangle / R$, if E were the voltage from the generator. But we would like to know in more detail how much power there is at every frequency. There is very little power in any one frequency; it is a distribution. Let $P(\omega) d\omega$ be the power that the generator would deliver in the frequency range $d\omega$ into the very same resistor. Then we can prove (we shall prove it for another case, but the mathematics is exactly the same) that the power comes out

$$P(\omega) d\omega = (2/\pi)kT d\omega, \quad (41.3)$$

and is *independent of the resistance* when put this way.

41-2 Thermal equilibrium of radiation

Now we go on to consider a still more advanced and interesting proposition that is as follows. Suppose we have a charged oscillator like those we were talking about when we were discussing light, let us say an electron oscillating up and down in an atom. If it oscillates up and down, it radiates light. Now suppose that this oscillator is in a very thin gas of other atoms, and that from time to time the atoms collide with it. Then in equilibrium, after a long time, this oscillator will pick up energy such that its kinetic energy of oscillation is $\frac{1}{2}kT$, and since it is a harmonic oscillator, its entire energy of motion will become kT . That is, of course, a wrong description so far, because the oscillator carries *electric charge*, and if it has an energy kT it is shaking up and down and *radiating light*. Therefore it is impossible to have equilibrium of real matter alone without the charges in it emitting light, and as light is emitted, energy flows away, the oscillator loses its kT as time goes on, and thus the whole gas which is colliding with the oscillator gradually cools off. And that is, of course, the way a hot stove cools on a cold night by radiating the light into the sky, because the atoms are jiggling their charge and they continually radiate, and slowly, because of this radiation, the jiggling motion slows down.

On the other hand, if we enclose the whole thing in a box so that the light does not go away to infinity, then we *can* eventually get thermal equilibrium. We may either put the gas in a box where we can say that there are other radiators in the box walls sending light back or, to take a nicer example, we may suppose the box has mirror walls. It is easier to think about that case. Thus we assume that all the radiation that goes out from the oscillator keeps running around in the box. Then, of course, it is true that the oscillator starts to radiate, but pretty soon it can maintain its kT of kinetic energy in spite of the fact that it is radiating, because it is being illuminated, we may say, by its own light reflected from the walls of the box. That is, after a while there is a great deal of light rushing around in the box, and although the oscillator is radiating some, the light comes back and returns some of the energy that was radiated.

We shall now determine how much light there must be in such a box at temperature T in order that the shining of the light on this oscillator will generate just enough energy to account for the light it radiated.

Let the gas atoms be very few and far between, so that we have an ideal oscillator with no resistance except radiation resistance. Then we consider that at thermal equilibrium the oscillator is doing two things at the same time. First, it has a mean energy kT , and we calculate how much radiation it emits. Second, this radiation should be exactly the amount that would result because of the fact that the light shining on the oscillator is scattered. Since there is nowhere else the energy can go, this effective radiation is really just scattered light from the light that is in there.

Thus we first calculate the energy that is radiated by the oscillator per second, if the oscillator has a certain energy. (We borrow from Chapter 32 on radiation resistance a number of equations without going back over their derivation.) The energy radiated per radian divided by the energy of the oscillator is called $1/Q$ (Eq. 32.8): $1/Q = (dW/dt)/\omega_0 W$. Using the quantity γ , the damping constant, this can also be written as $1/Q = \gamma/\omega_0$, where ω_0 is the natural frequency of the oscillator—if gamma is very small, Q is very large. The energy radiated per second is then

$$\frac{dW}{dt} = \frac{\omega_0 W}{Q} = \frac{\omega_0 W \gamma}{\omega_0} = \gamma W. \quad (41.4)$$

The energy radiated per second is thus simply gamma times the energy of the oscillator. Now the oscillator should have an average energy kT , so we see that gamma kT is the average amount of energy radiated per second:

$$\langle dW/dt \rangle = \gamma kT. \quad (41.5)$$

Now we only have to know what gamma is. Gamma is easily found from Eq. (32.12). It is

$$\gamma = \frac{\omega_0}{Q} = \frac{2}{3} \frac{r_0 \omega_0^2}{c}, \quad (41.6)$$

where $r_0 = e^2/mc^2$ is the classical electron radius, and we have set $\lambda = 2\pi c/\omega_0$.

Our final result for the average rate of radiation of light near the frequency ω_0 is therefore

$$\frac{dW}{dt} = \frac{2}{3} \frac{r_0 \omega_0^2 kT}{c}. \quad (41.7)$$

Next we ask how much light must be shining on the oscillator. It must be enough that the energy absorbed from the light (and thereupon scattered) is just exactly this much. In other words, the emitted light is accounted for as *scattered* light from the light that is shining on the oscillator in the cavity. So we must now calculate how much light is scattered from the oscillator if there is a certain amount—unknown—of radiation incident on it. Let $I(\omega) d\omega$ be the amount of light energy there is at the frequency ω , within a certain range $d\omega$ (because there is no light at *exactly* a certain frequency; it is spread all over the spectrum). So $I(\omega)$ is a certain *spectral distribution* which we are now going to find—it is the color of a furnace at temperature T that we see when we open the door and look in the hole. Now how much light is absorbed? We worked out the amount of radiation absorbed from a given incident light beam, and we calculated it in terms of a *cross section*. It is just as though we said that all of the light that falls on a certain cross section is absorbed. So the total amount that is re-radiated (scattered) is the incident intensity $I(\omega) d\omega$ multiplied by the cross section σ .

The formula for the cross section which we derived (Eq. 31.19) did not have the damping included. It is not hard to go through the derivation again and put in the resistance term which we neglected. If we do that, and calculate the cross section the same way, we get

$$\sigma_s = \frac{8\pi r_0^2}{3} \left(\frac{\omega^4}{(\omega^2 - \omega_0^2)^2 + \gamma^2 \omega^2} \right). \quad (41.8)$$

Now, as a function of frequency, σ_s is of significant size only for ω very near to the natural frequency ω_0 . (Remember that the Q for a radiating oscillator is about 10^8 .) The oscillator scatters very strongly when ω is equal to ω_0 , and very weakly for other values of ω . Therefore we can replace ω by ω_0 and $\omega^2 - \omega_0^2$ by $2\omega_0(\omega - \omega_0)$, and we get

$$\sigma_s = \frac{2\pi r_0^2 \omega_0^2}{3[(\omega - \omega_0)^2 + \gamma^2/4]} \quad (41.9)$$

Now the whole curve is localized near $\omega = \omega_0$. (We do not really have to make any approximations, but it is much easier to do the integrals if we simplify the equation a bit.) Now we multiply the intensity in a given frequency range by the cross section of scattering, to get the amount of energy scattered in the range $d\omega$. The *total* energy scattered is then the integral of this for all ω . Thus

$$\begin{aligned} \frac{dW_s}{dt} &= \int_0^\infty I(\omega) \sigma_s(\omega) d\omega \\ &= \int_0^\infty \frac{2\pi r_0^2 \omega_0^2 I(\omega) d\omega}{3[(\omega - \omega_0)^2 + \gamma^2/4]} \end{aligned} \quad (41.10)$$

Now we set $dW_s/dt = 3\gamma kT$. Why *three*? Because when we made our analysis of the cross section in Chapter 32, we assumed that the polarization was such that the light could drive the oscillator. If we had used an oscillator which could move only in one direction, and the light, say, was polarized in the wrong way, it would not give any scattering. So we must either average the cross section of an oscillator which can go only in one direction, over all directions of incidence and polarization of the light or, more easily, we can imagine an oscillator which will follow the field no matter which way the field is pointing. Such an oscillator, which can oscillate equally in three directions, would have $3kT$ average energy because there are 3 degrees of freedom in that oscillator. So we should use $3\gamma kT$ because of the 3 degrees of freedom.

Now we have to do the integral. Let us suppose that the unknown spectral distribution $I(\omega)$ of the light is a smooth curve and does not vary very much across the very narrow frequency region where σ_s is peaked (Fig. 41-3). Then the only significant contribution comes when ω is very close to ω_0 , within an amount gamma, which is very small. So therefore, although $I(\omega)$ may be an unknown and complicated function, the only place where it is important is near $\omega = \omega_0$, and there we may replace the smooth curve by a flat one—a “constant”—at the same height. In other words, we simply take $I(\omega)$ outside the integral sign and call it $I(\omega_0)$. We may also take the rest of the constants out in front of the integral, and what we have left is

$$\frac{2}{3} \pi r_0^2 \omega_0^2 I(\omega_0) \int_0^\infty \frac{d\omega}{(\omega - \omega_0)^2 + \gamma^2/4} = 3\gamma kT. \quad (41.11)$$

Now, the integral should go from 0 to ∞ , but 0 is so far from ω_0 that the curve is all finished by that time, so we go instead to minus ∞ —it makes no difference and it is much easier to do the integral. The integral is an inverse tangent function of the form $\int dx/(x^2 + a^2)$. If we look it up in a book we see that it is equal to π/a . So what it comes to for our case is $2\pi/\gamma$. Therefore we get, with some rearranging,

$$I(\omega_0) = \frac{9\gamma^2 kT}{4\pi^2 r_0^2 \omega_0^2}. \quad (41.12)$$

Then we substitute the formula (41.6) for gamma (do not worry about writing ω_0 ; since it is true of any ω_0 , we may just call it ω) and the formula for $I(\omega)$ then comes out

$$I(\omega) = \frac{\omega^2 kT}{\pi^2 c^2}. \quad (41.13)$$

And that gives us the distribution of light in a hot furnace. It is called the *black-*

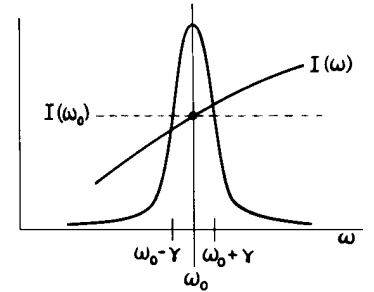


Fig. 41-3. The factors in the integrand (41.10). The peak is the resonance curve $1/(\omega - \omega_0)^2 + \gamma^2/4$. To a good approximation the factor $I(\omega)$ can be replaced by $I(\omega_0)$.

body radiation. Black, because the hole in the furnace that we look at is black when the temperature is zero.

Inside a closed box at temperature T , (41.13) is the distribution of energy of the radiation, according to classical theory. First, let us notice a remarkable feature of that expression. The charge of the oscillator, the mass of the oscillator, all properties specific to the oscillator, *cancel out*, because once we have reached equilibrium with one oscillator, we must be at equilibrium with any other oscillator of a different mass, or we will be in trouble. So this is an important kind of check on the proposition that equilibrium does not depend on what we are in equilibrium with, but *only on the temperature*. Now let us draw a picture of the $I(\omega)$ curve (Fig. 41-4). It tells us how much light we have at different frequencies.

The amount of intensity that there is in our box, per unit frequency range, goes, as we see, as the square of the frequency, which means that if we have a box at any temperature at all, and if we look at the x-rays that are coming out, there will be a lot of them!

Of course we know this is false. When we open the furnace and take a look at it, we do not burn our eyes out from x-rays at all. It is completely false. Furthermore, the *total energy* in the box, the total of all this intensity summed over all frequencies, would be the area under this infinite curve. Therefore, something is fundamentally, powerfully, and absolutely wrong.

Thus was the classical theory *absolutely incapable* of correctly describing the distribution of light from a blackbody, just as it was incapable of correctly describing the specific heats of gases. Physicists went back and forth over this derivation from many different points of view, and there is no escape. This is the prediction of classical physics. Equation (41.13) is called *Rayleigh's law*, and it is the prediction of classical physics, and is obviously absurd.

41-3 Equipartition and the quantum oscillator

The difficulty above was another part of the continual problem of classical physics, which started with the difficulty of the specific heat of gases, and now has been focused on the distribution of light in a blackbody. Now, of course, at the time that theoreticians studied this thing, there were also many *measurements* of the actual curve. And it turned out that the correct curve looked like the dashed curves in Fig. 41-4. That is, the x-rays were not there. If we lower the temperature, the whole curve goes down in proportion to T , according to the classical theory, but the observed curve also cuts off sooner at a lower temperature. Thus the low-frequency end of the curve is right, but the high-frequency end is wrong. Why? When Sir James Jeans was worrying about the specific heats of gases, he noted that motions which have high frequency are "frozen out" as the temperature goes too low. That is, if the temperature is too low, if the frequency is too high, the oscillators *do not have* kT of energy on the average. Now recall how our derivation of (41.13) worked: It all depends on the energy of an oscillator at thermal equilibrium. What the kT of (41.5) was, and what the same kT in (41.13) is, is the mean energy of a harmonic oscillator of frequency ω at temperature T . Classically, this is kT , but experimentally, no!—not when the temperature is too low or the oscillator frequency is too high. And so the reason that the curve falls off is the same reason that the specific heats of gases fail. It is easier to study the blackbody curve than it is the specific heats of gases, which are so complicated, therefore our attention is focused on determining the true blackbody curve, because this curve is a curve which correctly tells us, at every frequency, what the average energy of harmonic oscillators actually is as a function of temperature.

Planck studied this curve. He first determined the answer empirically, by fitting the observed curve with a nice function that fitted very well. Thus he had an empirical formula for the average energy of a harmonic oscillator as a function of frequency. In other words, he had the *right* formula instead of kT , and then by fiddling around he found a simple derivation for it which involved a very peculiar assumption. That assumption was that *the harmonic oscillator can take up energies only $\hbar\omega$ at a time*. The idea that they can have *any energy at all* is false. Of course, that was the beginning of the end of classical mechanics.

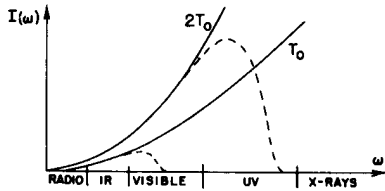


Fig. 41-4. The blackbody intensity distribution at two temperatures, according to classical physics (solid curves). The dashed curves show the actual distribution.

The very first correctly determined quantum-mechanical formula will now be derived. Suppose that the permitted energy levels of a harmonic oscillator were equally spaced at $\hbar\omega_0$ apart, so that the oscillator could take on only these different energies (Fig. 41-5). Planck made a somewhat more complicated argument than the one that is being given here, because that was the very beginning of quantum mechanics and he had to prove some things. But we are going to take it as a fact (which he demonstrated in this case) that the probability of occupying a level of energy E is $P(E) = \alpha e^{-E/kT}$. If we go along with that, we will obtain the right result.

Suppose now that we have a lot of oscillators, and each is a vibrator of frequency ω_0 . Some of these vibrators will be in the bottom quantum state, some will be in the next one, and so forth. What we would like to know is the average energy of all these oscillators. To find out, let us calculate the total energy of all the oscillators and divide by the number of oscillators. That will be the average energy per oscillator in thermal equilibrium, and will also be the energy that is in equilibrium with the blackbody radiation and that should go in Eq. (41.13) in place of kT . Thus we let N_0 be the number of oscillators that are in the ground state (the lowest energy state); N_1 the number of oscillators in the state E_1 ; N_2 the number that are in state E_2 ; and so on. According to the hypothesis (which we have not proved) that in quantum mechanics the law that replaced the probability $e^{-P.E./kT}$ or $e^{-K.E./kT}$ in classical mechanics is that the probability goes down as $e^{-\Delta E/kT}$, where ΔE is the excess energy, we shall assume that the number N_1 that are in the first state will be the number N_0 that are in the ground state, times $e^{-\hbar\omega/kT}$. Similarly, N_2 , the number of oscillators in the second state, is $N_2 = N_0 e^{-2\hbar\omega/kT}$. To simplify the algebra, let us call $e^{-\hbar\omega/kT} = x$. Then we simply have $N_1 = N_0 x$, $N_2 = N_0 x^2$, $\dots$, $N_n = N_0 x^n$.

The total energy of all the oscillators must first be worked out. If an oscillator is in the ground state, there is no energy. If it is in the first state, the energy is $\hbar\omega_0$, and there are N_1 of them. So $N_1 \hbar\omega$, or $\hbar\omega N_0 x$ is how much energy we get from those. Those that are in the second state have $2\hbar\omega_0$, and there are N_2 of them, so $N_2 \cdot 2\hbar\omega = 2\hbar\omega N_0 x^2$ is how much energy we get, and so on. Then we add it all together to get $E_{\text{tot}} = N_0 \hbar\omega (0 + x + 2x^2 + 3x^3 + \dots)$.

And now, how many oscillators are there? Of course, N_0 is the number that are in the ground state, N_1 in the first state, and so on, and we add them together: $N_{\text{tot}} = N_0 (1 + x + x^2 + x^3 + \dots)$. Thus the average energy is

$$\langle E \rangle = \frac{E_{\text{tot}}}{N_{\text{tot}}} = \frac{N_0 \hbar\omega (0 + x + 2x^2 + 3x^3 + \dots)}{N_0 (1 + x + x^2 + \dots)}. \quad (41.14)$$

Now the two sums which appear here we shall leave for the reader to play with and have some fun with. When we are all finished summing and substituting for x in the sum, we should get—if we make no mistakes in the sum—

$$\langle E \rangle = \frac{\hbar\omega}{e^{\hbar\omega/kT} - 1}. \quad (41.15)$$

This, then, was the first quantum-mechanical formula ever known, or ever discussed, and it was the beautiful culmination of decades of puzzlement. Maxwell knew that there was something wrong, and the problem was, what was *right*? Here is the quantitative answer of what is right instead of kT . This expression should, of course, approach kT as $\omega \rightarrow 0$ or as $T \rightarrow \infty$. See if you can prove that it does—learn how to do the mathematics.

This is the famous cutoff factor that Jeans was looking for, and if we use it instead of kT in (41.13), we obtain for the distribution of light in a black box

$$I(\omega) d\omega = \frac{\hbar\omega^3 d\omega}{\pi^2 c^2 (e^{\hbar\omega/kT} - 1)}. \quad (41.16)$$

We see that for a large ω , even though we have ω^3 in the numerator, there is an e raised to a tremendous power in the denominator, so the curve comes down again and does not “blow up”—we do not get ultraviolet light and x-rays where we do not expect them!

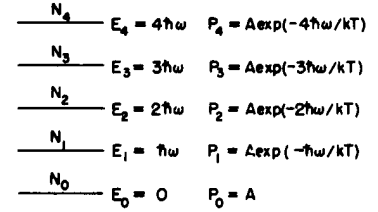


Fig. 41-5. The energy levels of a harmonic oscillator are equally spaced: $E_n = n\hbar\omega$.

One might complain that in our derivation of (41.16) we used the quantum theory for the energy levels of the harmonic oscillator, but the classical theory in determining the cross section σ_s . But the quantum theory of light interacting with a harmonic oscillator gives exactly the same result as that given by the classical theory. That, in fact, is why we were justified in spending so much time on our analysis of the index of refraction and the scattering of light, using a model of atoms like little oscillators—the quantum formulas are substantially the same.

Now let us return to the Johnson noise in a resistor. We have already remarked that the theory of this noise power is really the same theory as that of the classical blackbody distribution. In fact, rather amusingly, we have already said that if the resistance in a circuit were not a real resistance, but were an antenna (an antenna acts like a resistance because it radiates energy), a radiation resistance, it would be easy for us to calculate what the power would be. It would be just the power that runs into the antenna from the light that is all around, and we would get the same distribution, changed by only one or two factors. We can suppose that the resistor is a generator with an unknown power spectrum $P(\omega)$. The spectrum is determined by the fact that this same generator, connected to a resonant circuit of *any frequency*, as in Fig. 41-2(b), generates in the inductance a voltage of the magnitude given in Eq. (41.2). One is thus led to the same integral as in (41.10), and the same method works to give Eq. (41.3). For low temperatures the kT in (41.3) must of course be replaced by (41.15). The two theories (blackbody radiation and Johnson noise) are also closely related physically, for we may of course connect a resonant circuit to an *antenna*, so the resistance R is a pure *radiation resistance*. Since (41.2) does not depend on the physical origin of the resistance, we know the generator G for a real resistance and for radiation resistance is the same. What is the origin of the generated power $P(\omega)$ if the resistance R is only an ideal antenna in equilibrium with its environment at temperature T ? It is the radiation $I(\omega)$ in the space at temperature T which impinges on the antenna and, as “received signals,” makes an effective generator. Therefore one can deduce a direct relation of $P(\omega)$ and $I(\omega)$, leading then from (41.13) to (41.3).

All the things we have been talking about—the so-called Johnson noise and Planck’s distribution, and the correct theory of the Brownian movement which we are about to describe—are developments of the first decade or so of this century. Now with those points and that history in mind, we return to the Brownian movement.

41-4 The random walk

Let us consider how the position of a jiggling particle should change with time, for very long times compared with the time between “kicks.” Consider a little Brownian movement particle which is jiggling about because it is bombarded on all sides by irregularly jiggling water molecules. Query: After a given length of time, how far away is it likely to be from where it began? This problem was solved by Einstein and Smoluchowski. If we imagine that we divide the time into little intervals, let us say a hundredth of a second or so, then after the first hundredth of a second it moves here, and in the next hundredth it moves some more, in the next hundredth of a second it moves somewhere else, and so on. In terms of the rate of bombardment, a hundredth of a second is a very long time. The reader may easily verify that the number of collisions a single molecule of water receives in a second is about 10^{14} , so in a hundredth of a second it has 10^{12} collisions, which is a lot! Therefore, after a hundredth of a second it is not going to remember what happened before. In other words, the collisions are all *random*, so that one “step” is not related to the previous “step.” It is like the famous drunken sailor problem: the sailor comes out of the bar and takes a sequence of steps, but each step is chosen at an arbitrary angle, at random (Fig. 41-6). The question is: After a long time, where is the sailor? Of course we do not know! It is impossible to say. What do we mean—he is just somewhere more or less random. Well then, on the average, where is he? *On the average, how far away from the bar has he gone?* We have already answered this question, because once we were discussing the

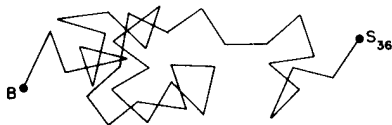


Fig. 41-6. A random walk of 36 steps of length l . How far is S_{36} from B ?
Ans: about $6l$ on the average.

superposition of light from a whole lot of different sources at different phases, and that meant adding a lot of arrows at different angles (Chapter 32). There we discovered that the mean square of the distance from one end to the other of the chain of random steps, which was the intensity of the light, is the sum of the intensities of the separate pieces. And so, by the same kind of mathematics, we can prove immediately that if $\mathbf{R}_N$ is the vector distance from the origin after N steps, the mean square of the distance from the origin is proportional to the number N of steps. That is, $\langle R_N^2 \rangle = NL^2$, where L is the length of each step. Since the number of steps is proportional to the time in our present problem, *the mean square distance is proportional to the time*:

$$\langle R^2 \rangle = \alpha t. \quad (41.17)$$

This does not mean that the *mean distance* is proportional to the time. If the mean distance were proportional to the time it would mean that the drifting is at a nice uniform velocity. The sailor *is* making some relatively sensible headway, but only such that his mean *square* distance is proportional to time. That is the characteristic of a random walk.

We may show very easily that in each successive step the square of the distance increases, on the average, by L^2 . For if we write $\mathbf{R}_N = \mathbf{R}_{N-1} + \mathbf{L}$, we find that $\mathbf{R}_N^2$ is

$$\mathbf{R}_N \cdot \mathbf{R}_N = R_N^2 = R_{N-1}^2 + 2\mathbf{R}_{N-1} \cdot \mathbf{L} + L^2,$$

and averaging over many trials, we have $\langle R_N^2 \rangle = \langle R_{N-1}^2 \rangle + L^2$, since $\langle \mathbf{R}_{N-1} \cdot \mathbf{L} \rangle = 0$. Thus, by induction,

$$R_N^2 = NL^2. \quad (41.18)$$

Now we would like to calculate the coefficient α in Eq. (41.17), and to do so we must add a feature. We are going to suppose that if we were to put a force on this particle (having nothing to do with the Brownian movement—we are taking a side issue for the moment), then it would react in the following way against the force. First, there would be inertia. Let m be the coefficient of inertia, the effective mass of the object (not necessarily the same as the real mass of the real particle, because the water has to move around the particle if we pull on it). Thus if we talk about motion in one direction, there is a term like $m(d^2x/dt^2)$ on one side. And next, we want also to assume that if we kept a steady pull on the object, there would be a drag on it from the fluid, proportional to its velocity. Besides the inertia of the fluid, there is a resistance to flow due to the viscosity and the complexity of the fluid. It is absolutely essential that there *be* some irreversible losses, something like resistance, in order that there be fluctuations. There is no way to produce the kT unless there are also losses. The source of the fluctuations is very closely related to these losses. What the mechanism of this drag is, we will discuss soon—we shall talk about forces that are proportional to the velocity and where they come from. But let us suppose for now that there is such a resistance. Then the formula for the motion under an external force, when we are pulling on it in a normal manner, is

$$m \frac{d^2x}{dt^2} + \mu \frac{dx}{dt} = F_{\text{ext}}. \quad (41.19)$$

The quantity μ can be determined directly from experiment. For example, we can watch the drop fall under gravity. Then we know that the force is mg , and μ is mg divided by the speed of fall the drop ultimately acquires. Or we could put the drop in a centrifuge and see how fast it sediments. Or if it is charged, we can put an electric field on it. So μ is a measurable thing, not an artificial thing, and it is known for many types of colloidal particles, etc.

Now let us use the same formula in the case where the force is not external, but is equal to the irregular forces of the Brownian movement. We shall then try to determine the mean square distance that the object goes. Instead of taking the distances in three dimensions, let us take just one dimension, and find the mean of x^2 , just to prepare ourselves. (Obviously the mean of x^2 is the same as the mean of y^2 is the same as the mean of z^2 , and therefore the mean square of the distance

is just 3 times what we are going to calculate.) The x -component of the irregular forces is, of course, just as irregular as any other component. What is the rate of change of x^2 ? It is $d(x^2)/dt = 2x(dx/dt)$, so what we have to find is the average of the position times the velocity. We shall show that this is a constant, and that therefore the mean square radius will increase proportionally to the time, and at what rate. Now if we multiply Eq. (41.19) by x , $mx(d^2x/dt^2) + \mu x(dx/dt) = xF_x$. We want the time average of $x(dx/dt)$, so let us take the average of the whole equation, and study the three terms. Now what about x times the force? If the particle happens to have gone a certain distance x , then, since the irregular force is *completely* irregular and does not know where the particle started from, the next impulse can be in any direction relative to x . If x is positive, there is no reason why the average force should also be in that direction. It is just as likely to be one way as the other. The bombardment forces are not driving it in a definite direction. So the average value of x times F is zero. On the other hand, for the term $mx(d^2x/dt^2)$ we will have to be a little fancy, and write this as

$$mx \frac{d^2x}{dt^2} = m \frac{d[x(dx/dt)]}{dt} - m \left(\frac{dx}{dt} \right)^2.$$

Thus we put in these two terms and take the average of both. So let us see how much x times the velocity should be. Now x times the velocity has a mean that does not change with time, because when it gets to some position it has no remembrance of where it was before, so things are no longer changing with time. So this quantity, on the average, is zero. We have left the quantity mv^2 , and that is the only thing we know: $mv^2/2$ has a mean value $\frac{1}{2}kT$. Therefore we find that

$$\left\langle mx \frac{d^2x}{dt^2} \right\rangle + \mu \left\langle x \frac{dx}{dt} \right\rangle = \langle xF_x \rangle$$

implies

$$-\langle mv^2 \rangle + \frac{\mu}{2} \frac{d}{dt} \langle x^2 \rangle = 0,$$

or

$$\frac{d\langle x^2 \rangle}{dt} = 2 \frac{kT}{\mu}. \quad (41.20)$$

Therefore the object has a mean square distance $\langle R^2 \rangle$, at the end of a certain amount of t , equal to

$$\langle R^2 \rangle = 6kT \frac{t}{\mu}. \quad (41.21)$$

And so we can actually determine *how far* the particles go! We first must determine how they react to a steady force, how fast they drift under a known force (to find μ), and then we can determine how far they go in their random motions. This equation was of considerable importance historically, because it was one of the first ways by which the constant k was determined. After all, we can measure μ , the time, how far the particles go, and we can take an average. The reason that the determination of k was important is that in the law $PV = RT$ for a mole, we know that R , which can also be measured, is equal to the number of atoms in a mole times k . A mole was originally defined as so and so many *grams* of oxygen—16 (now carbon is used), so the number of *atoms* in a mole was not known, originally. It is, of course, a very interesting and important problem. How big are atoms? How many are there? So one of the earliest determinations of the number of atoms was by the determination of how far a dirty little particle would move if we watched it patiently under a microscope for a certain length of time. And thus Boltzmann's constant k and the Avogadro number N_0 were determined because R had already been measured.

Applications of Kinetic Theory

42-1 Evaporation

In this chapter we shall discuss some further applications of kinetic theory. In the previous chapter we emphasized one particular aspect of kinetic theory, namely, that the average kinetic energy in any degree of freedom of a molecule or other object is $\frac{1}{2}kT$. The central feature of what we shall now discuss, on the other hand, is the fact that the probability of finding a particle in different places, per unit volume, varies as $e^{-\text{potential energy}/kT}$; we shall make a number of applications of this.

The phenomena which we want to study are relatively complicated: a liquid evaporating, or electrons in a metal coming out of the surface, or a chemical reaction in which there are a large number of atoms involved. In such cases it is no longer possible to make from the kinetic theory any simple and correct statements, because the situation is too complicated. Therefore, this chapter, except where otherwise emphasized, is quite inexact. The idea to be emphasized is only that we can understand, from the kinetic theory, *more or less* how things ought to behave. By using thermodynamic arguments, or some empirical measurements of certain critical quantities, we can get a more accurate representation of the phenomena.

However, it is very useful to know even only more or less why something behaves as it does, so that when the situation is a new one, or one that we have not yet started to analyze, we can say, more or less, what ought to happen. So this discussion is highly inaccurate but essentially right—right in idea, but a little bit simplified, let us say, in the specific details.

The first example that we shall consider is the evaporation of a liquid. Suppose we have a box with a large volume, partially filled with liquid in equilibrium and with the vapor at a certain temperature. We shall suppose that the molecules of the vapor are relatively far apart, and that inside the liquid, the molecules are packed close together. The problem is to find out how many molecules there are in the vapor phase, compared with the number there are in the liquid. How dense is the vapor at a given temperature, and how does it depend on the temperature?

Let us say that n equals the number of molecules per unit volume in the vapor. That number, of course, varies with the temperature. If we add heat, we get more evaporation. Now let another quantity, $1/V_a$, equal the number of atoms per unit volume in the liquid: We suppose that each molecule in the liquid occupies a certain volume, so that if there are more molecules of liquid, then all together they occupy a bigger volume. Thus if V_a is the volume occupied by one molecule, the number of molecules in a unit volume is a unit volume divided by the volume of each molecule. Furthermore, we suppose that there is a force of attraction between the molecules to hold them together in the liquid. Otherwise we cannot understand why it condenses. Thus suppose that there is such a force and that there is an energy of binding of the molecules in the liquid which is lost when they go into the vapor. That is, we are going to suppose that, in order to take a single molecule out of the liquid into the vapor, a certain amount of work W has to be done. There is a certain difference, W , in the energy of a molecule in the liquid from what it would have if it were in the vapor, because we have to pull it away from the other molecules which attract it.

Now we use the general principle that the number of atoms per unit volume in two different regions is $n_2/n_1 = e^{-(E_2-E_1)/kT}$. So the number n per unit volume in the vapor, divided by the number $1/V_a$ per unit volume in the liquid, is equal to

$$nV_a = e^{-W/kT}, \quad (42.1)$$

42-1 Evaporation

42-2 Thermionic emission

42-3 Thermal ionization

42-4 Chemical kinetics

42-5 Einstein's laws of radiation

because that is the general rule. It is like the atmosphere in equilibrium under gravity, where the gas at the bottom is denser than that at the top because of the work mgh needed to lift the gas molecules to the height h . In the liquid, the molecules are denser than in the vapor because we have to pull them out through the energy "hill" W , and the ratio of the densities is $e^{-W/kT}$.

This is what we wanted to deduce—that the vapor density varies as e to the minus some energy or other over kT . The factors in front are not really interesting to us, because in most cases the vapor density is very much lower than the liquid density. In those circumstances, where we are not near the critical point where they are almost the same, but where the vapor density is much lower than the liquid density, then the fact that n is very much less than $1/V_a$ is occasioned by the fact that W is very much greater than kT . So formulas such as (42.1) are interesting only when W is very much bigger than kT , because in those circumstances, since we are raising e to minus a tremendous amount, if we change T a little bit, that tremendous power changes a bit, and the change produced in the exponential factor is very much more important than any change that might occur in the factors out in front. Why should there be any changes in such factors as V_a ? Because ours was an approximate analysis. After all, there is not really a definite volume for each molecule; as we change the temperature, the volume V_a does not stay constant—the liquid expands. There are other little features like that, and so the actual situation is more complicated. There are slowly varying temperature-dependent factors all over the place. In fact, we might say that W itself varies slightly with temperature, because at a higher temperature, at a different molecular volume, there would be different average attractions, and so on. So, while we might think that if we have a formula in which everything varies in an unknown way with temperature then we have no formula at all, if we realize that the exponent W/kT is, in general, very large, we see that in the curve of the vapor density as a function of temperature most of the variation is occasioned by the exponential factor, and if we take W as a constant and the coefficient $1/V_a$ as nearly constant, it is a good approximation for short intervals along the curve. Most of the variation, in other words, is of the general nature $e^{-W/kT}$.

It turns out that there are many, many phenomena in nature which are characterized by having to borrow an energy from somewhere, and in which the central feature of the temperature variation is e to the minus the energy over kT . This is a useful fact only when the energy is large compared with kT , so that most of the variation is contained in the variation of the kT and not in the constant and in other factors.

Now let us consider another way of obtaining a somewhat similar result for the evaporation, but looking at it in more detail. To arrive at (42.1), we simply applied a rule which is valid at equilibrium, but in order to understand things better, there is no harm in trying to look at the details of what is going on. We may also describe what is going on in the following way: the molecules that are in the vapor continually bombard the surface of the liquid; when they hit it, they may bounce off or they may get stuck. There is an unknown factor for that—maybe 50–50, maybe 10 to 90—we do not know. Let us say they always get stuck—we can analyze it over again later on the assumption that they do not always get stuck. Then at a given moment there will be a certain number of atoms which are condensing onto the surface of the liquid. The number of condensing molecules, the number that arrive on a unit area, is the number n per unit volume times the velocity v . This velocity of the molecules is related to the temperature, because we know that $\frac{1}{2}mv^2$ is equal to $\frac{3}{2}kT$ on the average. So v is some kind of a mean velocity. Of course we should integrate over the angles and get some kind of an average, but it is roughly proportional to the root-mean-square velocity, within some factor. Thus

$$N_c = nv \quad (42.2)$$

is the number which arrive per unit area and are condensing.

At the same time, however, the atoms in the liquid are jiggling about, and from time to time one of them gets kicked out. Now we have to estimate how fast they get kicked out. The idea will be that at equilibrium the number that are kicked out per second and the number that arrive per second are equal.

How many get kicked out? In order to get kicked out, a particular molecule has to have acquired by accident an excess energy over its neighbors—a considerable excess energy, because it is attracted very strongly by the other molecules in the liquid. Ordinarily it does not leave because it is so strongly attracted, but in the collisions sometimes one of them gets an extra energy by accident. And the chance that it gets the extra energy W which it needs in our case is very small if $W \gg kT$. In fact, $e^{-W/kT}$ is the chance that an atom has picked up more than this much energy. That is the general principle in kinetic theory: in order to borrow an excess energy W over the average, the odds are e to the minus the energy that we have to borrow, over kT . Now suppose that some molecules have borrowed this energy. We now have to estimate how many leave the surface per second. Of course, just because a molecule has the necessary energy does not mean that it will actually evaporate, since it may be buried too deeply inside the liquid or, even if it is near the surface, it may be travelling in the wrong direction. The number that are going to leave a unit area per second is going to be something like this: the number of atoms there are near the surface, per unit area, divided by the time it takes one to escape, multiplied by the probability $e^{-W/kT}$ that they are ready to escape in the sense that they have enough energy.

We shall suppose that each molecule at the surface of the liquid occupies a certain cross-sectional area A . Then the number of molecules per unit area of liquid surface will be $1/A$. And now, how long does it take a molecule to escape? If the molecules have a certain average speed v , and have to move, say, one molecular diameter D , the thickness of the first layer, then the time it takes to get across that thickness is the time needed to escape, if the molecule has enough energy. The time will be D/v . Thus the number evaporating should be approximately

$$N_e = (1/A)(v/D)e^{-W/kT}. \quad (42.3)$$

Now the area of each atom times the thickness of the layer is approximately the same as the volume V_a occupied by a single atom. And so, in order to get equilibrium, we must have $N_e = N_c$, or

$$nv = (v/V_a)e^{-W/kT}. \quad (42.4)$$

We may cancel the v 's, since they are equal; even though one is the velocity of a molecule in the vapor and the other is the velocity of an evaporating molecule, these are the same, because we know their mean kinetic energy (in one direction) is $\frac{1}{2}kT$. But one may object, "No! No! These are the especially fast-moving ones; these are the ones that have picked up excess energy." Not really, because the moment they start to pull away from the liquid, they have to *lose* that excess energy against the potential energy. So, as they come to the surface they are slowed down to the velocity v ! It is the same as it was in our discussion of the distribution of molecular velocities in the atmosphere—at the bottom, the molecules had a certain distribution of energy. The ones that arrive at the top have the *same* distribution of energy, because the slow ones did not arrive at all, and the fast ones were slowed down. The molecules that are evaporating have the same distribution of energy as the ones inside—a rather remarkable fact. Anyway, it is useless to try to argue so closely about our formula because of other inaccuracies, such as the probability of bouncing back rather than entering the liquid, and so on. Thus we have a rough idea of the rate of evaporation and condensation, and we see, of course, that the vapor density n varies in the same way as before, but now we have understood it in some detail rather than just as an arbitrary formula.

This deeper understanding permits us to analyze some things. For example, suppose that we were to pump away the vapor at such a great rate that we removed the vapor as fast as it formed (if we had very good pumps and the liquid was evaporating very slowly), how fast would evaporation occur if we maintained a liquid temperature T ? Suppose that we have already experimentally measured the equilibrium vapor density, so that we know, at the given temperature, how many molecules per unit volume are in equilibrium with the liquid. Now we would like to know *how fast* it will evaporate. Even though we have used only a rough analysis so far as the evaporation part of it is concerned, the number of vapor

molecules *arriving* was not done so badly, aside from the unknown factor of reflection coefficient. So therefore we may use the fact that the number that are leaving, at equilibrium, is the same as the number that arrive. True, the vapor is being swept away and so the molecules are only coming out, but if the vapor were left alone, it would attain the equilibrium density at which the number that come back would equal the number that are evaporating. Therefore, we can easily see that the number that are coming off the surface per second is equal to the unknown reflection coefficient R times the number that would come down to the surface per second were the vapor still there, because that is how many would balance the evaporation at equilibrium:

$$N_e = nvR = (vR/V_a)e^{-W/kT}. \quad (42.5)$$

Of course, the number of molecules that hit the liquid from the vapor is easy to calculate, since we do not need to know as much about the forces as we do when we are worrying about how they get to escape through the liquid surface; it is much easier to make the argument the other way.

42-2 Thermionic emission

We may give another example of a very practical situation that is similar to the evaporation of a liquid—so similar that it is not worth making a separate analysis. It is essentially the same problem. In a radio tube there is a source of electrons, namely a heated tungsten filament, and a positively charged plate to attract the electrons. Any electron that escapes from the surface of the tungsten is immediately swept away to the plate. That is our ideal “pump,” which is “pumping” the electrons away all the time. Now the question is: How many electrons per second can we get out of a piece of tungsten, and how does that number vary with temperature? The answer to that problem is the same as (42.5), because it turns out that in a piece of metal, electrons are attracted to the ions, or to atoms, of the metal. They are attracted, if we may say it crudely, to the metal. In order to get an electron out of a piece of metal, it takes a certain amount of energy or work to pull it out. This work varies with the different kinds of metal. In fact, it varies even with the character of the surface of a given kind of metal, but the total work may be a few electron volts, which, incidentally, is typical of the energy involved in chemical reactions. We can remember the latter fact by remembering that the voltage in a chemical cell like a flashlight battery, which is produced by chemical reactions, is about one volt.

How can we find out how many electrons come out per second? It would be quite difficult to analyze the effects on the electrons going out; it is easier to analyze the situation the other way. So, we could start out by imagining that we did not draw the electrons away, and that the electrons were like a gas, and could come back to the metal. Then there would be a certain density of electrons at equilibrium which would, of course, be given by exactly the same formula as (42.1), where V_a is the volume per electron in the metal, roughly, and W is equal to $q_e\phi$, where ϕ is the so-called *work function*, or the voltage needed to pull an electron off the surface. This would tell us how many electrons would have to be in the surrounding space and striking the metal in order to balance the ones that are coming out. And thus it is easy to calculate how many are coming out if we sweep away all of them, because the number that are coming out is exactly equal to the number that would be going in with the above density of electron “vapor.” In other words, the answer is that the current of electricity that comes in per unit area is equal to the charge on each times the number that arrive per second per unit area, which is the number per unit volume times the velocity, as we have seen many times:

$$I = q_e nv = (q_e v R / V_a) e^{-q_e \phi / kT}. \quad (42.6)$$

Now one electron volt corresponds to kT at a temperature of 11,600 degrees. The filament of the tube may be operating at a temperature of, say, 1100 degrees, so the exponential factor is something like e^{-10} ; when we change the temperature a

little bit, the exponential factor changes a lot. Thus, again, the central feature of the formula is the $e^{-q_e\phi/kT}$. As a matter of fact, the factor in front is quite wrong—it turns out that the behavior of electrons in a metal is not correctly described by the classical theory, but by quantum mechanics, but this only changes the factor in front a little. Actually, no one has ever been able to get the thing straightened out very well, even though many people have used the high-class quantum-mechanical theory for their calculations. The big problem is, does W change slightly with temperature? If it does, one cannot distinguish a W changing slowly with temperature from a different coefficient in front. That is, if W changed linearly, say, with temperature, so that $W = W_0 + \alpha kT$, then we would have

$$e^{-W/kT} = e^{-(W_0 + \alpha kT)/kT} = e^{-\alpha} e^{-W_0/kT}.$$

Thus a linearly temperature-dependent W is equivalent to a shifted “constant.” It is really quite difficult and usually fruitless to try to obtain the coefficient in the front accurately.

42-3 Thermal ionization

Now we go on to another example of the same idea; always the same idea. This has to do with ionization. Suppose that in a gas we have a whole lot of atoms which are in the neutral state, say, but the gas is hot and the atoms can become ionized. We would like to know how many ions there are in a given circumstance if we have a certain density of atoms per unit volume at a certain temperature. Again we consider a box in which there are N atoms which can hold electrons. (If an electron has come off an atom, it is called an *ion*, and if the atom is neutral, we simply call it an atom.) Then suppose that, at any given moment, the number of neutral atoms is n_a , the number of ions is n_i , and the number of electrons is n_e , all per unit volume. The problem is: What is the relationship of these three numbers?

In the first place, we have two conditions or constraints on the numbers. For instance, as we vary different conditions, like the temperature and so on, $n_a + n_i$ would remain constant, because this would be simply the number N of atomic nuclei that are in the box. If we keep the number of nuclei per unit volume fixed, and change, say, the temperature, then as the ionization proceeded some atoms would turn to ions, but the total number of atoms plus ions would be unchanged. That is, $n_a + n_i = N$. Another condition is that if the entire gas is to be electrically neutral (and if we neglect double or triple ionization), that means that the number of ions is equal to the number of electrons at all times, or $n_i = n_e$. These are subsidiary equations that simply express the conservation of charge and the conservation of atoms.

These equations are true, and we ultimately will use them when we consider a real problem. But we want to obtain another relationship between the quantities. We can do this as follows. We again use the idea that it takes a certain amount of energy to lift the electron out of the atom, which we call the *ionization energy*, and we would write it as W , in order to make all of the formulas look the same. So we let W equal the energy needed to pull an electron out of an atom and make an ion. Now we again say that the number of free electrons per unit volume in the “vapor” is equal to the number of bound electrons per unit volume in the atoms, times e to the minus the energy difference between being bound and being free, over kT . That is the basic equation again. How can we write it? The number of free electrons per unit volume would, of course, be n_e , because that is the definition of n_e . Now what about the number of electrons per unit volume that are bound to atoms? The total number of places that we could put the electrons is apparently $n_a + n_i$, and we will suppose that when they are bound each one is bound within a certain volume V_a . So the total amount of volume which is available to electrons which would be bound is $(n_a + n_i)V_a$, so we might want to write our formula as

$$n_e = \frac{n_a}{(n_a + n_i)V_a} e^{-W/kT}.$$

The formula is wrong, however, in one essential feature, which is the following: when an electron is already on an atom, another electron cannot come to that volume anymore! In other words, all the volumes of all the possible sites are not really available for the one electron which is trying to make up its mind whether or not to be in the vapor or in the condensed position, because in this problem there is an extra feature that when one electron is where another electron is, it is not allowed to go—it is repelled. For that reason, it comes out that we should count only that part of the volume which is available for an electron to sit on or not. That is, those which are already occupied do not count in the total available volume, but the only volume which is allowed is that of the *ions*, where there are vacant places for the electron to go. Then, in those circumstances, we find that a nicer way to write our formula is

$$\frac{n_e n_i}{n_a} = \frac{1}{V_a} e^{-W/kT}. \quad (42.7)$$

This formula is called the *Saha ionization equation*. Now let us see if we can understand qualitatively why a formula like this is right, by arguing about the kinetic things that are happening.

First, every once in a while an electron comes to an ion and they combine to make an atom. And also, every once in a while, an atom gets into a collision and breaks up into an ion and an electron. Now those two rates must be equal. How fast do electrons and ions find each other? The rate is certainly increased if the number of electrons per unit volume is increased. It is also increased if the number of ions per unit volume is increased. That is, the total rate at which recombination is occurring is certainly proportional to the number of electrons times the number of ions. Now the total rate at which ionization is occurring due to collisions must be dependent linearly on how many atoms there are to ionize. And so the rates will balance when there is some relationship between the product $n_e n_i$ and the number of atoms, n_a . The fact that this relationship happens to be given by this particular formula, where W is the ionization energy, is of course a little bit more information, but we can easily understand that the formula would necessarily involve the concentrations of the electrons, ions, and atoms in the combination $n_e n_i / n_a$ to produce a constant independent of the n 's, and dependent only on temperature, the atomic cross sections, and other constant factors.

We may also note that, since the equation involves the numbers *per unit volume*, if we were to do two experiments with a given total number N of atoms plus ions, that is, a certain fixed number of nuclei, but using boxes with different volumes, the n 's would all be smaller in the larger box. But since the ratio $n_e n_i / n_a$ stays the same, the *total number* of electrons and ions must be greater in the larger box. To see this, suppose that there are N nuclei inside a box of volume V , and that a fraction f of them are ionized. Then $n_e = fN/V = n_i$, and $n_a = (1-f)N/V$. Then our equation becomes

$$\frac{f^2}{1-f} \frac{N}{V} = \frac{e^{-W/kT}}{V_a}. \quad (42.8)$$

In other words, if we take a smaller and smaller density of atoms, or make the volume of the container bigger and bigger, the fraction f of electrons and ions must increase. That ionization, just from “expansion” as the density goes down, is the reason why we believe that at very low densities, such as in the cold space between the stars, there may be ions present, even though we might not understand it from the point of view of the available energy. Although it takes many, many kT of energy to make them, there are ions present.

Why can there be ions present when there is so much space around, while if we increase the density, the ions tend to disappear? *Answer*: Consider an atom. Every once in a while, light, or another atom, or an ion, or whatever it is that maintains thermal equilibrium, strikes it. Very rarely, because it takes such a terrific amount of excess energy, an electron comes off and an ion is left. Now that electron, if the space is enormous, wanders and wanders and does not come near anything for years, perhaps. But once in a very great while, it does come back

to an ion and they combine to make an atom. So the rate at which electrons are coming out from the atoms is very slow. But if the volume is enormous, an electron which has escaped takes so long to find another ion to recombine with that its probability of recombination is very, very small; thus, in spite of the large excess energy needed, there may be a reasonable number of electrons.

42-4 Chemical kinetics

The same situation that we have just called “ionization” is also found in a chemical reaction. For instance, if two objects A and B combine into a compound AB , then if we think about it for a while we see that AB is what we have called an atom, B is what we call an electron, and A is what we call an ion. With these substitutions the equations of equilibrium are exactly the same in form:

$$\frac{n_A n_B}{n_{AB}} = c e^{-W/kT}. \quad (42.9)$$

This formula, of course, is not exact, since the “constant” c depends on how much volume is allowed for the A and B to combine, and so on, but by thermodynamic arguments one can identify what the meaning of the W in the exponential factor is, and it turns out that it is very close to the energy needed in the reaction.

Suppose that we tried to understand this formula as a result of collisions, much in the way that we understood the evaporation formula, by arguing about how many electrons came off and how many of them came back per unit time. Suppose that A and B combine in a collision every once in a while to form a compound AB . And suppose that the compound AB is a complicated molecule which jiggles around and is hit by other molecules, and from time to time it gets enough energy to explode and break up again into A and B .

Now it actually turns out, in chemical reactions, that if the atoms come together with too small an energy, even though energy may be released in the reaction $A + B \rightarrow AB$, the fact that A and B may touch each other does not necessarily make the reaction start. It usually is required that the collision be rather hard, in fact, to get the reaction to go at all—a “soft” collision between A and B may not do it, even though energy may be released in the process. So let us suppose that it is very common in chemical reactions that, in order for A and B to form AB , they cannot just hit each other, but they have to hit each other *with sufficient energy*. This energy is called the *activation energy*—the energy needed to “activate” the reaction. Call A^* the activation energy, the excess energy needed in a collision in order that the reaction may really occur. Then the rate R_f at which A and B produce AB would involve the number of atoms of A times the number of atoms of B , times the rate at which a single atom would strike a certain cross section σ_{AB} , times a factor $e^{-A^*/kT}$, which is the probability that they have enough energy:

$$R_f = n_A n_B v \sigma_{AB} e^{-A^*/kT}. \quad (42.10)$$

Now we have to find the opposite rate, R_r . There is a certain chance that AB will fly apart. In order to fly apart, it not only must have the energy W which it needs in order to get apart at all but, just as it was hard for A and B to come together, so there is a kind of hill that A and B have to climb over to get apart again; they must have not only enough energy just to get ready to pull apart, but a certain excess. It is like climbing a hill to get into a deep valley; they have to climb the hill coming in and they have to climb out of the valley and then over the hill coming back (Fig. 42-1). Thus the rate at which AB goes to A and B will be proportional to the number n_{AB} that are present, times $e^{-(W+A^*)/kT}$:

$$R_r = c' n_{AB} e^{-(W+A^*)/kT}. \quad (42.11)$$

The c' will involve the volume of atoms and the rate of collisions, which we can work out, as we did the case of evaporation, with areas and times and thicknesses; but we shall not do this. The main feature of interest to us is that when these two

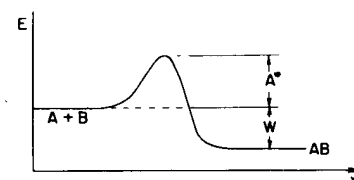


Fig. 42-1. The energy relationship for the reaction $A + B \rightarrow AB$.

rates are equal, the ratio of them is equal to unity. This tells us that $n_A n_B / n_{AB} = ce^{-W/kT}$, as before, where c involves the cross sections, velocities, and other factors independent of the n 's.

The interesting thing is that the rate of the reaction also varies as $e^{-\text{const}/kT}$, although the constant is not the same as that which governs the concentrations; the activation energy A^* is quite different from the energy W . W governs the proportions of A , B , and AB that we have in equilibrium, but if we want to know how fast $A + B$ goes to AB , that is not a question of equilibrium, and here a different energy, the activation energy, governs the rate of reaction through an exponential factor.

Furthermore, A^* is not a fundamental constant like W . Suppose that at the surface of the wall—or at some other place— A and B could temporarily stick there in such a way that they could combine more easily. In other words, we might find a “tunnel” through the hill, or perhaps a lower hill. By the conservation of energy, when we are all finished we have still made AB out of A and B , so the energy difference W will be quite independent of the way the reaction occurred, but the activation energy A^* will depend *very much* on the way the reaction occurs. This is why the rates of chemical reactions are very sensitive to outside conditions. We can change the rate by putting in a surface of a different kind, we can put it in a “different barrel” and it will go at a different rate, if it depends on the nature of the surface. Or if we put in a third kind of object it may change the rate very much; some things produce enormous changes in rate simply by changing the A^* a little bit—they are called *catalysts*. A reaction might practically not occur at all because A^* is too big at the given temperature, but when we put in this special stuff, the catalyst, then the reaction goes very fast indeed, because A^* is reduced.

Incidentally, there is some trouble with such a reaction, A plus B , making AB , because we cannot conserve both energy and momentum when we try to put two objects together to make one that is more stable. Therefore, we need at least a third object C , so the actual reaction is much more complicated. The forward rate would involve the product $n_A n_B n_C$, and it might seem that our formula is going wrong, but no! When we look at the rate at which AB goes the other way, we find that it also needs to collide with C , so there is an $n_{AB} n_C$ in the reverse rate; the n_C 's cancel out in the formula for the equilibrium concentrations. The law of equilibrium, (42.9), which we first wrote down is absolutely guaranteed to be true, no matter what the mechanism of the reaction may be!

42-5 Einstein's laws of radiation

We now turn to an interesting analogous situation having to do with the black-body radiation law. In the last chapter we worked out the distribution law for the radiation in a cavity the way Planck did, considering the radiation from an oscillator. The oscillator had to have a certain mean energy, and since it was oscillating, it would radiate and would keep pumping radiation into the cavity until it piled up enough radiation to balance the absorption and emission. In that way we found that the intensity of radiation at frequency ω was given by the formula

$$I(\omega) d\omega = \frac{\hbar \omega^3 d\omega}{\pi^2 c^2 (e^{\hbar \omega / kT} - 1)} . \quad (42.12)$$

This result involved the assumption that the oscillator which was generating the radiation had definite, equally spaced energy levels. We did not say that light had to be a photon or anything like that. There was no discussion about how, when an atom goes from one level to another, the energy must come out in one unit of energy, $\hbar \omega$, in the form of light. Planck's original idea was that the matter was quantized but not the light: material oscillators cannot take up just any energy, but have to take it in lumps. Furthermore, the trouble with the derivation is that it was partially classical. We calculated the rate of radiation from an oscillator according to classical physics; then we turned around and said, “No, this oscillator has a lot of energy levels.” So gradually, in order to find the right result, the completely quantum-mechanical result, there was a slow develop-

ment which culminated in the quantum mechanics of 1927. But in the meantime, there was an attempt by Einstein to convert Planck's viewpoint that only oscillators of matter were quantized, to the idea that light was really photons and could be considered in a certain way as particles with energy $\hbar\omega$. Furthermore, Bohr had pointed out that *any* system of atoms has energy levels, but they are not necessarily equally spaced like Planck's oscillator. And so it became necessary to rederive or at least rediscuss the radiation law from a more completely quantum-mechanical viewpoint.

Einstein assumed that Planck's final formula was right, and he used that formula to obtain some new information, previously unknown, about the interaction of radiation with matter. His discussion went as follows: Consider any two of the many energy levels of an atom, say the m th level and the n th level (Fig. 42-2). Now Einstein proposed that when such an atom has light of the right frequency shining on it, it can absorb that photon of light and make a transition from state n to state m , and that the probability that this occurs per second depends upon the two levels, of course, but is *proportional to how intense the light is* that is shining on it. Let us call the proportionality constant B_{nm} , merely to remind us that this is not a universal constant of nature, but depends on the particular pair of levels: some levels are easy to excite; some levels are hard to excite. Now what is the formula going to be for the rate of emission from m to n ? Einstein proposed that this must have two parts to it. First, even if there were no light present, there would be some chance that an atom in an excited state would fall to a lower state, emitting a photon; this we call *spontaneous emission*. It is analogous to the idea that an oscillator with a certain amount of energy, even in classical physics, does not keep that energy, but loses it by radiation. Thus the analog of spontaneous radiation of a classical system is that if the atom is in an excited state there is a certain probability A_{mn} , which depends on the levels again, for it to go down from m to n , and this probability is independent of whether light is shining on the atom or not. But then Einstein went further, and by comparison with the classical theory and by other arguments, concluded that emission was also influenced by the presence of light—that when light of the right frequency is shining on an atom, it has an increased rate of emitting a photon that is proportional to the intensity of the light, with a proportionality constant B_{mn} . Later, if we deduce that this coefficient is zero, then we will have found that Einstein was wrong. Of course we will find he was right.

Thus Einstein assumed that there are three kinds of processes: an absorption proportional to the intensity of light, an emission proportional to the intensity of light, called *induced emission* or sometimes *stimulated emission*, and a spontaneous emission independent of light.

Now suppose that we have, in equilibrium at temperature T , a certain number of atoms N_n in the state n and another number N_m in the state m . Then the total number of atoms that are going from n to m is the number that are in the state n times the rate per second that, if one is in n , it goes up to m . So we have a formula for the number that are going from n to m per second:

$$R_{n \rightarrow m} = N_n B_{nm} I(\omega). \quad (42.13)$$

The number that will go from m to n is expressed in the same manner, as the number N_m that are in m , times the chance per second that each one goes down to n . This time our expression is

$$R_{m \rightarrow n} = N_m [A_{mn} + B_{mn} I(\omega)]. \quad (42.14)$$

Now we shall suppose that in thermal equilibrium the number of atoms going up must equal the number coming down. That is one way, at least, in which the number will be sure to stay constant in each level.* So we take these two rates

* This is not the only way one can arrange to keep the numbers of atoms in the various levels constant, but it is the way it actually works. That every process must, in thermal equilibrium, be balanced by its exact opposite is called the *principle of detailed balancing*.

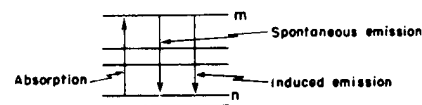


Fig. 42-2. Transitions between two energy levels of an atom.

to be equal at equilibrium. But we have one other piece of information: we know how large N_m is compared with N_n —the ratio of those two is $e^{-(E_m - E_n)/kT}$. Now Einstein assumed that the only light which is effective in making the transition from n to m is the light which has the frequency corresponding to the energy difference, so $E_m - E_n = \hbar\omega$ in all our formulas. Thus

$$N_m = N_n e^{-\hbar\omega/kT}. \quad (42.15)$$

Thus if we set the two rates equal: $N_n B_{nm} I(\omega) = N_m [A_{mn} + B_{mn} I(\omega)]$, and divide by N_m , we get

$$B_{nm} I(\omega) e^{\hbar\omega/kT} = A_{mn} + B_{mn} I(\omega). \quad (42.16)$$

From this equation, we can calculate $I(\omega)$. It is simply

$$I(\omega) = \frac{A_{mn}}{B_{nm} e^{\hbar\omega/kT} - B_{mn}}. \quad (42.17)$$

But Planck has already told us that the formula must be (42.12). Therefore we can deduce something: First, that B_{nm} must equal B_{mn} , since otherwise we cannot get the $(e^{\hbar\omega/kT} - 1)$. So Einstein discovered some things that he did not know how to calculate, namely *that the induced emission probability and the absorption probability must be equal*. This is interesting. And furthermore, in order for (42.17) and (42.12) to agree,

$$A_{mn}/B_{mn} \quad \text{must be} \quad \hbar\omega^3/\pi^2 c^2. \quad (42.18)$$

So if we know, for instance, the absorption rate for a given level, we can deduce the spontaneous emission rate and the induced emission rate, or any combination.

This is as far as Einstein or anyone else could go using such arguments. To actually compute the absolute spontaneous emission rate or the other rates for any specific atomic transition, of course, requires a knowledge of the machinery of the atom, called quantum electrodynamics, which was not discovered until eleven years later. This work of Einstein was done in 1916.

The possibility of induced emission has, today, found interesting applications. If there is light present, it will tend to induce the downward transition. The transition then adds its $\hbar\omega$ to the available light energy, if there were some atoms sitting in the upper state. Now we can arrange, by some nonthermal method, to have a gas in which the number in the state m is very much greater than the number in the state n . This is far out of equilibrium, and so is not given by the formula $e^{-\hbar\omega/kT}$, which is for equilibrium. We can even arrange it so that the number in the upper state is very large, while the number in the lower state is practically zero. Then light which has the frequency corresponding to the energy difference $E_m - E_n$ will not be strongly absorbed, because there are not many atoms in state n to absorb it. On the other hand, when that light is present, it will induce the emission from this upper state! So, if we had a lot of atoms in the upper state, there would be a sort of chain reaction, in which, the moment the atoms began to emit, more would be caused to emit, and the whole lot of them would dump down together. This is what is called a *laser*, or, in the case of the far infrared, a *maser*.

Various tricks can be used to obtain the atoms in state m . There may be higher levels to which the atoms can get if we shine in a strong beam of light of high frequency. From these high levels, they may trickle down, emitting various photons, until they all get stuck in the state m . If they tend to stay in the state m without emitting, the state is called *metastable*. And then they are all dumped down together by induced emissions. One more technical point—if we put this system in an ordinary box, it would radiate in so many different directions spontaneously, compared with the induced effect, that we would still be in trouble. But we can enhance the induced effect, increase its efficiency, by putting nearly perfect mirrors on each side of the box, so that the light which is emitted gets another chance, and another chance, and another chance, to induce more emission. Although the mirrors are almost one hundred percent reflecting, there is a slight amount of transmission of the mirror, and a little light gets out. In the end, of course, from the conservation of energy, all the light goes out in a nice uniform straight direction which makes the strong light beams that are possible today with lasers.

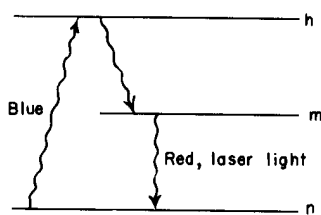


Fig. 42-3. By exciting, say by blue light, a higher state h , which may emit a photon leaving atoms in state m , the number in this state m becomes sufficiently large to start laser action.

Diffusion

43-1 Collisions between molecules

We have considered so far only the molecular motions in a gas which is in thermal equilibrium. We want now to discuss what happens when things are near, but not exactly in, equilibrium. In a situation far from equilibrium, things are extremely complicated, but in a situation very close to equilibrium we can easily work out what happens. To see what happens, we must, however, return to the kinetic theory. Statistical mechanics and thermodynamics deal with the equilibrium situation, but away from equilibrium we can only analyze what occurs atom by atom, so to speak.

As a simple example of a nonequilibrium circumstance, we shall consider the diffusion of ions in a gas. Suppose that in a gas there is a relatively small concentration of ions—electrically charged molecules. If we put an electric field on the gas, then each ion will have a force on it which is different from the forces on the neutral molecules of the gas. If there were no other molecules present, an ion would have a constant acceleration until it reached the wall of the container. But because of the presence of the other molecules, it cannot do that; its velocity increases only until it collides with a molecule and loses its momentum. It starts again to pick up more speed, but then it loses its momentum again. The net effect is that an ion works its way along an erratic path, but with a net motion in the direction of the electric force. We shall see that the ion has an average “drift” with a mean speed which is proportional to the electric field—the stronger the field, the faster it goes. While the field is on, and while the ion is moving along, it is, of course, *not* in thermal equilibrium, it is trying to get to equilibrium, which is to be sitting at the end of the container. By means of the kinetic theory we can compute the drift velocity.

It turns out that with our present mathematical abilities we cannot really compute *precisely* what will happen, but we can obtain approximate results which exhibit all the essential features. We can find out how things will vary with pressure, with temperature, and so on, but it will not be possible to get precisely the correct numerical factors in front of all the terms. We shall, therefore, in our derivations, not worry about the precise value of numerical factors. They can be obtained only by a very much more sophisticated mathematical treatment.

Before we consider what happens in nonequilibrium situations, we shall need to look a little closer at what goes on in a gas in thermal equilibrium. We shall need to know, for example, what the average time between successive collisions of a molecule is.

Any molecule experiences a sequence of collisions with other molecules—in a random way, of course. A particular molecule will, in a long period of time T , have a certain number, N , of hits. If we double the length of time, there will be twice as many hits. So the number of collisions is proportional to the time T . We would like to write it this way:

$$N = T/\tau \quad (43.1)$$

We have written the constant of proportionality as $1/\tau$, where τ will have the dimensions of a time. The constant τ is the average time between collisions. Suppose, for example, that in an hour there are 60 collisions; then τ is one minute. We would say that τ (one minute) is the *average time* between the collisions.

We may often wish to ask the following question: “What is the *chance* that a molecule will experience a collision during the next *small interval* of time dt ?” The answer, we may intuitively understand, is dt/τ . But let us try to make a more

43-1 Collisions between molecules

43-2 The mean free path

43-3 The drift speed

43-4 Ionic conductivity

43-5 Molecular diffusion

43-6 Thermal conductivity

convincing argument. Suppose that there were a very large number N of molecules. How many will have collisions in the next interval of time dt ? If there is equilibrium, nothing is changing *on the average* with time. So N molecules waiting the time dt will have the same number of collisions as *one* molecule waiting for the time $N dt$. That number we know is $N dt/\tau$. So the number of hits of N molecules is $N dt/\tau$ in a time dt , and the chance, or probability, of a hit for any one molecule is just $1/N$ as large, or $(1/N)(N dt/\tau) = dt/\tau$, as we guessed above. That is to say, the fraction of the molecules which will suffer a collision in the time dt is dt/τ . To take an example, if τ is one minute, then in one second the fraction of particles which will suffer collisions is $1/60$. What this means, of course, is that $1/60$ of the molecules happen to be close enough to what they are going to hit next that *their* collisions will occur in the next minute.

When we say that τ , the mean time between collisions, is one minute, we do not mean that all the collisions will occur at times separated by exactly one minute. A particular particle does not have a collision, wait one minute, and then have another collision. The times between successive collisions are quite variable. We will not need it for our later work here, but we may make a small diversion to answer the question: "What *are* the times between collisions?" We know that for the case above, the *average* time is one minute, but we might like to know, for example, what is the chance that we get no collision for *two* minutes?

We shall find the answer to the general question: "What is the probability that a molecule will go for a time t without having a collision?" At some arbitrary instant—that we call $t = 0$ —we begin to watch a particular molecule. What is the chance that it gets by until t without colliding with another molecule? To compute the probability, we observe what is happening to all N_0 molecules in a container. After we have waited a time t , some of them will have had collisions. We let $N(t)$ be the number that have *not* had collisions up to the time t . $N(t)$ is, of course, less than N_0 . We can find $N(t)$ because we know how it changes with time. If we know that $N(t)$ molecules have got by until t , then $N(t + dt)$, the number which get by until $t + dt$, is *less* than $N(t)$ by the number that have collisions in dt . The number that collide in dt we have written above in terms of the mean time τ as $dN = N(t) dt/\tau$. We have the equation

$$N(t + dt) = N(t) - N(t) \frac{dt}{\tau}. \quad (43.2)$$

The quantity on the left-hand side, $N(t + dt)$, can be written, according to the definitions of calculus, as $N(t) + (dN/dt) dt$. Making this substitution, Eq. (43.2) yields

$$\frac{dN(t)}{dt} = - \frac{N(t)}{\tau}. \quad (43.3)$$

The number that are being lost in the interval dt is proportional to the number that are present, and inversely proportional to the mean life τ . Equation (43.3) is easily integrated if we rewrite it as

$$\frac{dN(t)}{N(t)} = - \frac{dt}{\tau}. \quad (43.4)$$

Each side is a perfect differential, so the integral is

$$\ln N(t) = -t/\tau + (\text{a constant}), \quad (43.5)$$

which says the same thing as

$$N(t) = (\text{constant})e^{-t/\tau}. \quad (43.6)$$

We know that the constant must be just N_0 , the total number of molecules present, since all of them start at $t = 0$ to wait for their "next" collision. We can write our result as

$$N(t) = N_0 e^{-t/\tau}. \quad (43.7)$$

If we wish the *probability* of no collision, $P(t)$, we can get it by dividing $N(t)$ by N_0 , so

$$P(t) = e^{-t/\tau}. \quad (43.8)$$

Our result is: the probability that a particular molecule survives a time t without a collision is $e^{-t/\tau}$, where τ is the mean time between collisions. The probability starts out at 1 (or certainty) for $t = 0$, and gets less as t gets bigger and bigger. The probability that the molecule avoids a collision for a time equal to τ is $e^{-1} = 0.37 \dots$. The chance is less than one-half that it will have a greater than average time between collisions. That is all right, because there are enough molecules which go collision-free for times much *longer* than the mean time before colliding, so that the average time can still be τ .

We originally defined τ as the average time *between* collisions. The result we have obtained in Eq. (43.7) also says that the mean time from an *arbitrary* starting instant to the *next* collision is *also* τ . We can demonstrate this somewhat surprising fact in the following way. The number of molecules which experience their *next* collision in the interval dt at the time t after an arbitrarily chosen starting time is $N(t) dt/\tau$. Their "time until the next collision" is, of course, just t . The "average time until the next collision" is obtained in the usual way:

$$\text{Average time until the next collision} = \frac{1}{N_0} \int_0^\infty t \frac{N(t) dt}{\tau}.$$

Using $N(t)$ obtained in (43.7) and evaluating the integral, we find indeed that τ is the average time from *any* instant until the next collision.

43-2 The mean free path

Another way of describing the molecular collisions is to talk not about the *time* between collisions, but about *how far* the particle moves between collisions. If we say that the average time between collisions is τ , and that the molecules have a mean velocity v , we can expect that the average *distance* between collisions, which we shall call l , is just the product of τ and v . This distance between collisions is usually called the *mean free path*:

$$\text{Mean free path } l = \tau v. \quad (43.9)$$

In this chapter we shall be a little careless about *what kind of average* we mean in any particular case. The various possible averages—the mean, the root-mean-square, etc.—are all nearly equal and differ by factors which are near to one. Since a detailed analysis is required to obtain the correct numerical factors anyway, we need not worry about which average is required at any particular point. We may also warn the reader that the algebraic symbols we are using for some of the physical quantities (e.g., l for the mean free path) do not follow a generally accepted convention, mainly because there is no general agreement.

Just as the chance that a molecule will have a collision in a short time dt is equal to dt/τ , the chance that it will have a collision in going a distance dx is dx/l . Following the same line of argument used above, the reader can show that the probability that a molecule will go at least the distance x before having its next collision is $e^{-x/l}$.

The average distance a molecule goes before colliding with another molecule—the mean free path l —will depend on how many molecules there are around and on the "size" of the molecules, i.e., how big a target they represent. The effective "size" of a target in a collision we usually describe by a "collision cross section," the same idea that is used in nuclear physics, or in light-scattering problems.

Consider a moving particle which travels a distance dx through a gas which has n_0 scatterers (molecules) per unit volume (Fig. 43-1). If we look at each unit of area perpendicular to the direction of motion of our selected particle, we will find there $n_0 dx$ molecules. If each one presents an effective collision area or, as it is usually called, "collision cross section," σ_c , then the total area covered by the scatterers is $\sigma_c n_0 dx$.

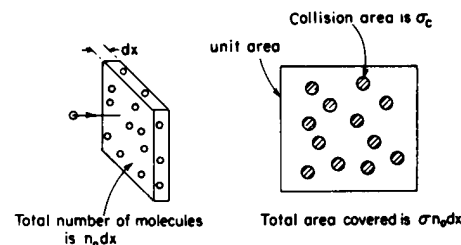


Fig. 43-1. Collision cross section.

By "collision cross section" we mean the area within which the center of our particle must be located if it is to collide with a particular molecule. If molecules were little spheres (a classical picture) we would expect that $\sigma_c = \pi(r_1 + r_2)^2$, where r_1 and r_2 are the radii of the two colliding objects. The chance that our particle will have a collision is the ratio of the area covered by scattering molecules to the total area, which we have taken to be one. So the probability of a collision in going a distance dx is just $\sigma_c n_0 dx$:

$$\text{Chance of a collision in } dx = \sigma_c n_0 dx. \quad (43.10)$$

We have seen above that the chance of a collision in dx can also be written in terms of the mean free path l as dx/l . Comparing this with (43.10), we can relate the mean free path to the collision cross section:

$$\frac{1}{l} = \sigma_c n_0, \quad (43.11)$$

which is easier to remember if we write it as

$$\sigma_c n_0 l = 1. \quad (43.12)$$

This formula can be thought of as saying that there should be one collision, on the average, when the particle goes through a distance l in which the scattering molecules *could* just cover the total area. In a cylindrical volume of length l and a base of unit area, there are $n_0 l$ scatterers; if each one has an area σ_c the total area covered is $n_0 l \sigma_c$, which is just one unit of area. The whole area is *not* covered, of course, because some molecules are partly hidden behind others. That is why some molecules go farther than l before having a collision. It is only *on the average* that the molecules have a collision by the time they go the distance l . From measurements of the mean free path l we can determine the scattering cross section σ_c , and compare the result with calculations based on a detailed theory of atomic structure. But that is a different subject! So we return to the problem of nonequilibrium states.

43-3 The drift speed

We want to describe what happens to a molecule, or several molecules, which are different in some way from the large majority of the molecules in a gas. We shall refer to the "majority" molecules as the "background" molecules, and we shall call the molecules which are different from the background molecules "special" molecules or, for short, the *S*-molecules. A molecule could be special for any number of reasons: It might be heavier than the background molecules. It might be a different chemical. It might have an electric charge—i.e., be an ion in a background of uncharged molecules. Because of their different masses or charges the *S*-molecules may have forces on them which are different from the forces on the background molecules. By considering what happens to these *S*-molecules we can understand the basic effects which come into play in a similar way in many different phenomena. To list a few: the diffusion of gases, electric currents in batteries, sedimentation, centrifugal separation, etc.

We begin by concentrating on the basic process: an *S*-molecule in a background gas is acted on by some specific force **F** (which might be, e.g., gravitational or electrical) and *in addition* by the not-so-specific forces due to collisions with the background molecules. We would like to describe the *general* behavior of the *S*-molecule. What happens to it, *in detail*, is that it darts around hither and yon as it collides over and over again with other molecules. But if we watch it carefully we see that it does make some net progress in the direction of the force **F**. We say that there is a *drift* superposed on its random motion. We would like to know what the speed of its drift is—its *drift velocity*—due to the force **F**.

If we start to observe an *S*-molecule at some instant we may expect that it is somewhere between two collisions. In addition to the velocity it was left with after its last collision it is picking up some velocity component due to the force **F**. In a

short time (on the average, in a time τ) it will experience a collision and start out on a new piece of its trajectory. It will have a new starting velocity, but the same acceleration from $\mathbf{F}$.

To keep things simple for the moment, we shall suppose that after each collision our S -molecule gets a completely “fresh” start. That is, that it keeps no remembrance of its past acceleration by $\mathbf{F}$. This might be a reasonable assumption if our S -molecule were much lighter than the background molecules, but it is certainly not valid in general. We shall discuss later an improved assumption.

For the moment, then, our assumption is that the S -molecule leaves each collision with a velocity which may be in any direction with equal likelihood. The starting velocity will take it equally in all directions and will not contribute to any net motion, so we shall not worry further about its initial velocity after a collision. In addition to its random motion, each S -molecule will have, at any moment, an additional velocity in the direction of the force $\mathbf{F}$, which it has picked up *since* its last collision. What is the *average* value of *this* part of the velocity? It is just the acceleration $\mathbf{F}/m$ (where m is the mass of the S -molecule) times the *average* time *since* the last collision. Now the average time *since* the *last* collision must be the same as the average time *until* the *next* collision, which we have called τ , above. The *average* velocity from $\mathbf{F}$, of course, is just what is called the drift velocity, so we have the relation

$$v_{\text{drift}} = \frac{F\tau}{m}. \quad (43.13)$$

This basic relation is the heart of our subject. There may be some complication in determining what τ is, but the basic process is defined by Eq. (43.13).

You will notice that the drift velocity is *proportional* to the force. There is, unfortunately, no generally used name for the constant of proportionality. Different names have been used for each different kind of force. If in an electrical problem the force is written as the charge times the electric field, $\mathbf{F} = q\mathbf{E}$, then the constant of proportionality between the velocity and the electric field $\mathbf{E}$ is usually called the “mobility.” In spite of the possibility of some confusion, we shall use the term *mobility* for the ratio of the drift velocity to the force for *any* force. We write

$$v_{\text{drift}} = \mu F \quad (43.14)$$

in general, and we shall call μ the mobility. We have from Eq. (43.13) that

$$\mu = \tau/m. \quad (43.15)$$

The mobility is proportional to the mean time between collisions (there are fewer collisions to slow it down) and inversely proportional to the mass (more inertia means less speed picked up between collisions).

To get the correct numerical coefficient in Eq. (43.13), which is correct as given, takes some care. Without intending to confuse, we should still point out that the arguments have a subtlety which can be appreciated only by a careful and detailed study. To illustrate that there are difficulties, in spite of appearances, we shall make over again the argument which led to Eq. (43.13) in a reasonable *but erroneous* way (and the way one will find in many textbooks!).

We might have said: The mean time between collisions is τ . After a collision the particle starts out with a random velocity, but it picks up an additional velocity between collisions, which is equal to the acceleration times the time. Since it takes the time τ to arrive at the *next* collision it gets there with the velocity $(F/m)\tau$. At the beginning of the collision it had zero velocity. So between the two collisions it has, on the average, a velocity one-half of the final velocity, so the mean drift velocity is $\frac{1}{2}F\tau/m$. (Wrong!) This result is wrong and the result in Eq. (43.13) is right, although the arguments may sound equally satisfactory. The reason the second result is wrong is somewhat subtle, and has to do with the following: The argument is made as though all collisions were separated by the mean time τ . The fact is that some times are shorter and others are longer than the mean. Short times occur *more often* but make *less* contribution to the drift velocity because they have less

chance “to really get going.” If one takes proper account of the *distribution* of free times between collisions, one can show that there should not be the factor $\frac{1}{2}$ that was obtained from the second argument. The error was made in trying to relate by a simple argument the *average final* velocity to the average velocity itself. This relationship is not simple, so it is best to concentrate on what is wanted: the average velocity itself. The first argument we gave determines the average velocity directly—and correctly! But we can perhaps see now why we shall not in general try to get all of the correct numerical coefficients in our elementary derivations!

We return now to our simplifying assumption that each collision knocks out all memory of the past motion—that a fresh start is made after each collision. Suppose our *S*-molecule is a heavy object in a background of lighter molecules. Then our *S*-molecule will not lose its “forward” momentum in each collision. It would take several collisions before its motion was “randomized” again. We should assume, instead, that at each collision—in each time τ on the average—it loses a certain fraction of its momentum. We shall not work out the details, but just state that the result is equivalent to replacing τ , the average collision time, by a new—and longer— τ which corresponds to the average “forgetting time,” i.e., the average time to forget its forward momentum. With such an interpretation of τ we can use our formula (43.15) for situations which are not quite as simple as we first assumed.

43-4 Ionic conductivity

We now apply our results to a special case. Suppose we have a gas in a vessel in which there are also some ions—atoms or molecules with a net electric charge. We show the situation schematically in Fig. 43-2. If two opposite walls of the container are metallic plates, we can connect them to the terminals of a battery and thereby produce an electric field in the gas. The electric field will result in a force on the ions, so they will begin to drift toward one or the other of the plates. An electric current will be induced, and the gas with its ions will behave like a resistor. By computing the ion flow from the drift velocity we can compute the resistance. We ask, specifically: How does the flow of electric current depend on the voltage difference V that we apply across the two plates?

We consider the case that our container is a rectangular box of length b and cross-sectional area A (Fig. 43-2). If the potential difference, or voltage, from one plate to the other is V , the electric field E between the plates is V/b . (The electric potential is the work done in carrying a unit charge from one plate to the other. The force on a unit charge is E . If E is the same everywhere between the plates, which is a good enough approximation for now, the work done on a unit charge is just Eb , so $V = Eb$.) The special force on an ion of the gas is qE , where q is the charge on the ion. The drift velocity of the ion is then μ times this force, or

$$v_{\text{drift}} = \mu F = \mu q E = \mu q \frac{V}{b}. \quad (43.16)$$

An electric current I is the flow of charge in a unit time. The electric current to one of the plates is given by the total charge of the ions which arrive at the plate in a unit of time. If the ions drift toward the plate with the velocity v_{drift} , then those which are within a distance $(v_{\text{drift}} \cdot T)$ will arrive at the plate in the time T . If there are n_i ions per unit volume, the number which reach the plate in the time T is $(n_i \cdot A \cdot v_{\text{drift}} \cdot T)$. Each ion carries the charge q , so we have that

$$\text{Charge collected in } T = q n_i A v_{\text{drift}} T. \quad (43.17)$$

The current I is the charge collected in T divided by T , so

$$I = q n_i A v_{\text{drift}}. \quad (43.18)$$

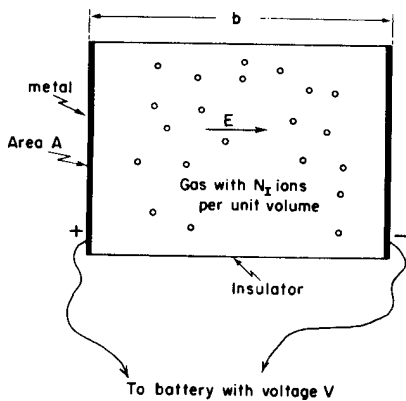


Fig. 43-2. Electric current from an ionized gas.

Substituting v_{drift} from (43.16), we have

$$I = \mu q^2 n_i \frac{A}{b} V. \quad (43.19)$$

We find that the current is proportional to the voltage, which is just the form of Ohm's law, and the resistance R is the inverse of the proportionality constant:

$$\frac{1}{R} = \mu q^2 n_i \frac{A}{b}. \quad (43.20)$$

We have a relation between the resistance and the molecular properties n_i , q , and μ , which depends in turn on m and τ . If we know n_i and q from atomic measurements, a measurement of R could be used to determine μ , and from μ also τ .

43-5 Molecular diffusion

We turn now to a different kind of problem, and a different kind of analysis: the theory of diffusion. Suppose that we have a container of gas in thermal equilibrium, and that we introduce a small amount of a different kind of gas at some place in the container. We shall call the original gas the "background" gas and the new one the "special" gas. The special gas will start to spread out through *the whole container*, but it will spread slowly because of the presence of the background gas. This slow spreading-out process is called *diffusion*. The diffusion is controlled mainly by the molecules of the special gas getting knocked about by the molecules of the background gas. After a large number of collisions, the special molecules end up spread out more or less evenly throughout the whole volume. We must be careful *not* to confuse diffusion of a gas with the gross transport that may occur due to convection currents. Most commonly, the mixing of two gases occurs by a combination of convection and diffusion. We are interested now only in the case that there are *no* "wind" currents. The gas is spreading only by molecular motions, by diffusion. We wish to compute how fast diffusion takes place.

We now compute the *net flow* of molecules of the "special" gas due to the molecular motions. There will be a net flow only when there is some nonuniform distribution of the molecules, otherwise all of the molecular motions would average to give no net flow. Let us consider first the flow in the x -direction. To find the flow, we consider an imaginary plane surface perpendicular to the x -axis and count the number of special molecules that cross this plane. To obtain the net flow, we must count as positive those molecules which cross in the direction of positive x and *subtract* from this number the number which cross in the negative x -direction. As we have seen many times, the number which cross a surface area in a time ΔT is given by the number which start the interval ΔT in a volume which extends the distance $v \Delta T$ from the plane. (Note that v , here, is the actual molecular velocity, not the drift velocity.)

We shall simplify our algebra by giving our surface one unit of area. Then the number of special molecules which pass from left to right (taking the $+x$ -direction to the right) is $n_- v \Delta T$, where n_- is the number of special molecules per unit volume to the left (within a factor of 2 or so, but we are ignoring such factors!). The number which cross from right to left is, similarly, $n_+ v \Delta T$, where n_+ is the number density of special molecules on the right-hand side of the plane. If we call the molecular current J , by which we mean the net flow of molecules per unit area per unit time, we have

$$J = \frac{n_- v \Delta T - n_+ v \Delta T}{\Delta T}, \quad (43.21)$$

or

$$J = (n_- - n_+)v. \quad (43.22)$$

What shall we use for n_- and n_+ ? When we say "the density on the left," how *far* to the left do we mean? We should choose the density at the place from which the molecules started their "flight," because the number which *start* such

trips is determined by the number present at that place. So by n_- we should mean the density a distance to the left equal to the mean free path l , and by n_+ , the density at the distance l to the right of our imaginary surface.

It is convenient to consider that the distribution of our special molecules in space is described by a continuous function of x , y , and z which we shall call n_a . By $n_a(x, y, z)$ we mean the number density of special molecules in a small volume element centered on (x, y, z) . In terms of n_a we can express the difference ($n_+ - n_-$) as

$$(n_+ - n_-) = \frac{dn_a}{dx} \Delta x = \frac{dn_a}{dx} \cdot 2l. \quad (43.23)$$

Substituting this result in Eq. (43.22) and neglecting the factor of 2, we get

$$J_x = -lv \frac{dn_a}{dx}. \quad (43.24)$$

We have found that the flow of special molecules is proportional to the derivative of the density, or to what is sometimes called the “gradient” of the density.

It is clear that we have made several rough approximations. Besides various factors of two we have left out, we have used v where we should have used v_x , and we have assumed that n_+ and n_- refer to places at the perpendicular distance l from our surface, whereas for those molecules which do not travel perpendicular to the surface element, l should correspond to the *slant* distance from the surface. All of these refinements can be made; the result of a more careful analysis shows that the right-hand side of Eq. (43.24) should be multiplied by $1/3$. So a better answer is

$$J_x = -\frac{lv}{3} \frac{dn_a}{dx}. \quad (43.25)$$

Similar equations can be written for the currents in the y - and z -directions.

The current J_x and the density gradient dn_a/dx can be measured by macroscopic observations. Their experimentally determined ratio is called the “diffusion coefficient,” D . That is,

$$J_x = -D \frac{dn_a}{dx}. \quad (43.26)$$

We have been able to show that for a gas we expect

$$D = \frac{1}{3}lv. \quad (43.27)$$

So far in this chapter we have considered two distinct processes: *mobility*, the drift of molecules due to “outside” forces; and *diffusion*, the spreading determined only by the internal forces, the random collisions. There is, however, a relation between them, since they both depend basically on the thermal motions, and the mean free path l appears in both calculations.

If, in Eq. (43.25), we substitute $l = v\tau$ and $\tau = \mu m$, we have

$$J_x = -\frac{1}{3}mv^2\mu \frac{dn_a}{dx}. \quad (43.28)$$

But mv^2 depends only on the temperature. We recall that

$$\frac{1}{2}mv^2 = \frac{3}{2}kT, \quad (43.29)$$

so

$$J_x = -\mu kT \frac{dn_a}{dx}. \quad (43.30)$$

We find that D , the *diffusion* coefficient, is just kT times μ , the *mobility* coefficient:

$$D = \mu kT. \quad (43.31)$$

And it turns out that the numerical coefficient in (43.31) is exactly right—no extra factors have to be thrown in to adjust for our rough assumptions. We can show,

in fact, that (43.31) must *always* be correct—even in complicated situations (for example, the case of a suspension in a liquid) where the details of our simple calculations would not apply at all.

To show that (43.31) must be correct in general, we shall derive it in a different way, using only our basic principles of statistical mechanics. Imagine a situation in which there is a gradient of “special” molecules, and we have a diffusion current proportional to the density gradient, according to Eq. (43.26). We now apply a force field in the x -direction, so that each special molecule feels the force F . According to the *definition* of the mobility μ there will be a drift velocity given by

$$v_{\text{drift}} = \mu F. \quad (43.32)$$

By our usual arguments, the *drift current* (the *net* number of molecules which pass a unit of area in a unit of time) will be

$$J_{\text{drift}} = n_a v_{\text{drift}}, \quad (43.33)$$

or

$$J_{\text{drift}} = n_a \mu F. \quad (43.34)$$

We now *adjust* the force F so that the drift current due to F just *balances* the diffusion, so that there is *no net flow* of our special molecules. We have $J_x + J_{\text{drift}} = 0$, or

$$D \frac{dn_a}{dx} = n_a \mu F. \quad (43.35)$$

Under the “balance” conditions we find a steady (with time) gradient of density given by

$$\frac{dn_a}{dx} = \frac{n_a \mu F}{D}. \quad (43.36)$$

But notice! We are describing an *equilibrium* condition, so our *equilibrium* laws of statistical mechanics apply. According to these laws the probability of finding a molecule at the coordinate x is proportional to $e^{-U/kT}$, where U is the potential energy. In terms of the number density n_a , this means that

$$n_a = n_0 e^{-U/kT}. \quad (43.37)$$

If we differentiate (43.37) with respect to x , we find

$$\frac{dn_a}{dx} = -n_0 e^{-U/kT} \cdot \frac{1}{kT} \frac{dU}{dx}, \quad (43.38)$$

or

$$\frac{dn_a}{dx} = -\frac{n_a}{kT} \frac{dU}{dx}. \quad (43.39)$$

In our situation, since the force F is in the x -direction, the potential energy U is just $-Fx$, and $-dU/dx = F$. Equation (43.39) then gives

$$\frac{dn_a}{dx} = \frac{n_a F}{kT}. \quad (43.40)$$

[This is just exactly Eq. (40.2), from which we deduced $e^{-U/kT}$ in the first place, so we have come in a circle]. Comparing (43.40) with (43.36), we get exactly Eq. (43.31). We have shown that Eq. (43.31), which gives the diffusion current in terms of the mobility, has the correct coefficient and is very generally true. Mobility and diffusion are intimately connected. This relation was first deduced by Einstein.

43-6 Thermal conductivity

The methods of the kinetic theory that we have been using above can be used also to compute the *thermal conductivity* of a gas. If the gas at the top of a container is hotter than the gas at the bottom, heat will flow from the top to the bottom. (We think of the top being hotter because otherwise convection currents would be

set up and the problem would no longer be one of heat *conduction*.) The transfer of heat from the hotter gas to the colder gas is by the diffusion of the “hot” molecules—those with more energy—downward and the diffusion of the “cold” molecules upward. To compute the flow of thermal energy we can ask about the energy carried downward across an element of area by the downward-moving molecules, and about the energy carried upward across the surface by the upward-moving molecules. The difference will give us the net downward flow of energy.

The thermal conductivity κ is defined as the ratio of the rate at which thermal energy is carried across a unit surface area, to the temperature gradient:

$$\frac{1}{A} \frac{dQ}{dt} = -\kappa \frac{dT}{dz}. \quad (43.41)$$

Since the details of the calculations are quite similar to those we have done above in considering the flow of electric current in an ionized gas, we shall leave it as an exercise for the reader to show that

$$\kappa = \frac{kn\bar{v}}{\gamma - 1}, \quad (43.42)$$

where $(\gamma - 1)kT$ is the average energy of a molecule at the temperature T .

If we use our relation $n\bar{\sigma}_c = 1$, the heat conductivity can be written as

$$\kappa = \frac{1}{\gamma - 1} \frac{kv}{\bar{\sigma}_c}. \quad (43.43)$$

We have a rather surprising result. We know that the average velocity of gas molecules depends on the temperature but *not on the density*. We expect $\bar{\sigma}_c$ to depend only on the *size* of the molecules. So our simple result says that the thermal conductivity κ (and therefore the *rate* of flow of heat in any particular circumstance) is independent of the *density* of the gas! The change in the number of “carriers” of energy with a change in density is just compensated by the larger distance the “carriers” can go between collisions.

One may ask: “Is the heat flow independent of the gas density in the limit as the density goes to zero? When there is no gas at all?” Certainly not! The formula (43.43) was derived, as were all the others in this chapter, under the assumption that the mean free path between collisions is much smaller than any of the dimensions of the container. Whenever the gas density is so low that a molecule has a fair chance of crossing from one wall of its container to the other without having a collision, none of the calculations of this chapter apply. We must in such cases go back to kinetic theory and calculate again the details of what will occur.

The Laws of Thermodynamics

44-1 Heat engines; the first law

So far we have been discussing the properties of matter from the atomic point of view, trying to understand roughly what will happen if we suppose that things are made of atoms obeying certain laws. However, there are a number of relationships among the properties of substances which can be worked out without consideration of the detailed structure of the materials. The determination of the relationships among the various properties of materials, without knowing their internal structure, is the subject of *thermodynamics*. Historically, thermodynamics was developed before an understanding of the internal structure of matter was achieved.

To give an example: we know from the kinetic theory that the pressure of a gas is caused by molecular bombardment, and we know that if we heat a gas, so that the bombardment increases, the pressure must increase. Conversely, if the piston in a container of the gas is moved inward against the force of bombardment, the energy of the molecules bombarding the piston will increase, and consequently the temperature will increase. So, on the one hand, if we increase the temperature at a given volume, we increase the pressure. On the other hand, if we compress the gas, we will find that the temperature will rise. From the kinetic theory, one can derive a quantitative relationship between these two effects, but instinctively one might guess that they are related in some necessary fashion which is independent of the details of the collisions.

Let us consider another example. Many people are familiar with this interesting property of rubber: If we take a rubber band and pull it, it gets warm. If one puts it between his lips, for example, and pulls it out, he can feel a distinct warming, and this warming is reversible in the sense that if he relaxes the rubber band quickly while it is between his lips, it is distinctly cooled. That means that when we stretch a rubber band it heats, and when we release the tension of the band it cools. Now our instincts might suggest that if we heated a band, it might pull: that the fact that pulling a band heats it might imply that heating a band should cause it to contract. And, in fact, if we apply a gas flame to a rubber band holding a weight, we will see that the band contracts abruptly (Fig. 44-1). So it is true that when we heat a rubber band it pulls, and this fact is definitely related to the fact that when we release the tension of it, it cools.

The internal machinery of rubber that causes these effects is quite complicated. We will describe it from a molecular point of view to some extent, although our main purpose in this chapter is to understand the relationship of these effects independently of the molecular model. Nevertheless, we can show from the molecular model that the effects are closely related. One way to understand the behavior of rubber is to recognize that this substance consists of an enormous tangle of long chains of molecules, a kind of "molecular spaghetti," with one extra complication: between the chains there are cross-links—like spaghetti that is sometimes welded together where it crosses another piece of spaghetti—a grand tangle. When we pull out such a tangle, some of the chains tend to line up along the direction of the pull. At the same time, the chains are in thermal motion, so they hit each other continually. It follows that such a chain, if stretched, would not by itself remain stretched, because it would be hit from the sides by the other chains and other molecules, and would tend to kink up again. So the real reason why a rubber band tends to contract is this: when one pulls it out, the chains are lengthwise, and the thermal agitations of the molecules on the sides of the chains tend

44-1 Heat engines; the first law

44-2 The second law

44-3 Reversible engines

44-4 The efficiency of an ideal engine

44-5 The thermodynamic temperature

44-6 Entropy

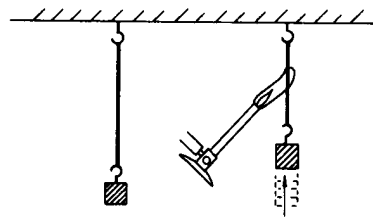


Fig. 44-1. The heated rubber band.

to kink the chains up, and make them shorten. One can then appreciate that if the chains are held stretched and the temperature is increased, so that the vigor of the bombardment on the sides of the chains is also increased, the chains tend to pull in, and they are able to pull a stronger weight when heated. If, after being stretched for a time, a rubber band is allowed to relax, each chain becomes soft, and the molecules striking it lose energy as they pound into the relaxing chain. So the temperature falls.

We have seen how these two process, contraction when heated and cooling during relaxation, can be related by the kinetic theory, but it would be a tremendous challenge to determine from the theory the precise relationship between the two. We would have to know how many collisions there were each second and what the chains look like, and we would have to take account of all kinds of other complications. The detailed mechanism is so complex that we cannot, by kinetic theory, really determine exactly what happens; still, a definite relation between the two effects we observe can be worked out without knowing anything about the internal machinery!

The whole subject of thermodynamics depends essentially upon the following kind of consideration: because a rubber band is "stronger" at higher temperatures than it is at lower temperatures, it ought to be possible to lift weights, and to move them around, and thus to do work with heat. In fact, we have already seen experimentally that a heated rubber band can lift a weight. The study of the way that one does work with heat is the beginning of the science of thermodynamics. Can we make an engine which uses the heating effect on a rubber band to do work? One can make a silly looking engine that does just this. It consists of a bicycle wheel in which all the spokes are rubber bands (Fig. 44-2). If one heats the rubber bands on one side of the wheel with a pair of heat lamps, they become "stronger" than the rubber bands on the other side. The center of gravity of the wheel will be pulled to one side, away from the bearing, so that the wheel turns. As it turns, cool rubber bands move toward the heat, and the heated bands move away from the heat and cool, so that the wheel turns slowly so long as the heat is applied. The efficiency of this engine is extremely low. Four hundred watts of power pour into the two lamps, but it is just possible to lift a fly with such an engine! An interesting question, however, is whether we can get heat to do the work in more efficient ways.

In fact, the science of thermodynamics began with an analysis, by the great engineer Sadi Carnot, of the problem of how to build the best and most efficient engine, and this constitutes one of the few famous cases in which engineering has contributed fundamentally to physical theory. Another example that comes to mind is the more recent analysis of information theory by Claude Shannon. These two analyses, incidentally, turn out to be closely related.

Now the way a steam engine ordinarily operates is that heat from a fire boils some water, and the steam so formed expands and pushes on a piston which makes a wheel go around. So the steam pushes the piston—what then? One has to finish the job: a stupid way to complete the cycle would be to let the steam escape into the air, for then one has to keep supplying water. It is cheaper—more efficient—to let the steam go into another box, where it is condensed by cool water, and then pump the water back into the boiler, so that it circulates continuously. Heat is thus supplied to the engine and converted into work. Now would it be better to use alcohol? What property should a substance have so that it makes the best possible engine? That was the question to which Carnot addressed himself, and one of the by-products was the discovery of the type of relationship that we have just explained above.

The results of thermodynamics are all contained implicitly in certain apparently simple statements called the *laws of thermodynamics*. At the time when Carnot lived, the first law of thermodynamics, the conservation of energy, was not known. Carnot's arguments were so carefully drawn, however, that they are valid even though the first law was not known in his time! Some time afterwards, Clausius made a simpler derivation that could be understood more easily than Carnot's very subtle reasoning. But it turned out that Clausius assumed, not the conserva-

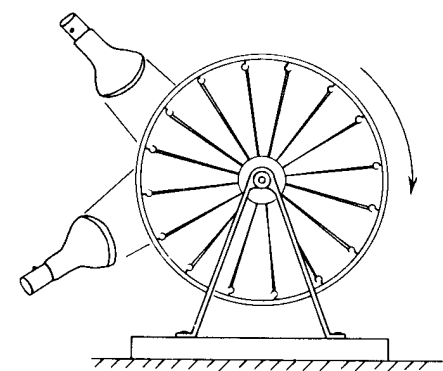


Fig. 44-2. The rubber-band heat engine.

tion of energy in general, but that *heat* was conserved according to the caloric theory, which was later shown to be false. So it has often been said that Carnot's logic was wrong. But his logic was quite correct. Only Clausius' simplified version, that everybody read, was incorrect.

The so-called second law of thermodynamics was thus discovered by Carnot before the first law! It would be interesting to give Carnot's argument that did not use the first law, but we shall not do so because we want to learn physics, not history. We shall use the first law from the start, in spite of the fact that a great deal can be done without it.

Let us begin by stating the first law, the conservation of energy: if one has a system and puts heat into it, and does work on it, then its energy is increased by the heat put in and the work done. We can write this as follows: The heat Q put into the system, plus the work W done on the system, is the increase in the energy U of the system; the latter energy is sometimes called the internal energy:

$$\text{Change in } U = Q + W. \quad (44.1)$$

The change in U can be represented as adding a little heat ΔQ and adding a little work ΔW :

$$\Delta U = \Delta Q + \Delta W, \quad (44.2)$$

which is a differential form of the same law. We know that very well, from an earlier chapter.

44-2 The second law

Now, what about the second law of thermodynamics? We know that if we do work against friction, say, the work lost to us is equal to the heat produced. If we do work in a room at temperature T , and we do the work slowly enough, the room temperature does not change much, and we have converted work into heat at a given temperature. What about the reverse possibility? Is it possible to convert the heat back into work at a given temperature? The second law of thermodynamics asserts that it is not. It would be very convenient to be able to convert heat into work merely by reversing a process like friction. If we consider only the conservation of energy, we might think that heat energy, such as that in the vibrational motions of molecules, might provide a goodly supply of useful energy. But Carnot assumed that it is impossible to extract the energy of heat at a single temperature. In other words, if the whole world were at the same temperature, one could not convert any of its heat energy into work: while the process of making work go into heat can take place at a given temperature, one cannot reverse it to get the work back again. Specifically, Carnot assumed that heat cannot be taken in at a certain temperature and converted into work *with no other change* in the system or the surroundings.

That last phrase is very important. Suppose we have a can of compressed air at a certain temperature, and we let the air expand. It can do work; it can make hammers go, for example. It cools off a little in the expansion, but if we had a big sea, like the ocean, at a given temperature—a heat reservoir—we could warm it up again. So we have taken the heat out of the sea, and we have done work with the compressed air. But Carnot was not wrong, because *we did not leave everything as it was*. If we recompress the air that we let expand, we will find we are doing extra work, and when we are finished we will discover that we not only got no work out of the system at temperature T , but we actually put some in. We must talk only about situations in which the *net result* of the whole process is to take heat away and convert it into work, just as the net result of the process of doing work against friction is to take work and convert it into heat. If we move in a circle, we can bring the system back precisely to its starting point, with the net result that we did work against friction and produced heat. Can we reverse the process? Turn a switch, so that everything goes backwards, so the friction does work against us, and cools the sea? According to Carnot: no! So let us suppose that this is impossible.

If it were possible it would mean, among other things, that we could take heat out of a cold body and put it into a hot body at no cost, as it were. Now we know it is natural that a hot thing can warm up a cool thing; if we simply put a hot body and a cold one together, and change nothing else, our experience assures us that it is not going to happen that the hot one gets hotter, and the cold one gets colder! But if we could obtain work by extracting the heat out of the ocean, say, or from anything else at a single temperature, then that work could be converted back into heat by friction at some other temperature. For instance, the other arm of a working machine could be rubbing something that is already hot. The net result would be to take heat from a “cold” body, the ocean, and to put it into a hot body. Now, the hypothesis of Carnot, the second law of thermodynamics, is sometimes stated as follows: heat cannot, of itself, flow from a cold to a hot object. But, as we have just seen, these two statements are equivalent: first, that one cannot devise a process whose only result is to convert heat to work at a single temperature, and second, that one cannot make heat flow by itself from a cold to a hot place. We shall mostly use the first form.

Carnot’s analysis of heat engines is quite similar to the argument that we gave about weight-lifting engines in our discussion of the conservation of energy in Chapter 4. In fact, that argument was patterned after Carnot’s argument about heat engines, and so the present treatment will sound very much the same.

Suppose we build a heat engine that has a “boiler” somewhere at a temperature T_1 . A certain heat Q_1 is taken from the boiler, the steam engine does some work W , and it then delivers some heat Q_2 into a “condenser” at another temperature T_2 (Fig. 44-3). Carnot did not say how much heat, because he did not know the first law, and he did not use the law that Q_2 was equal to Q_1 because he did not believe it. Although everybody thought that, according to the caloric theory, the heats Q_1 and Q_2 would have to be the same, Carnot did not say they were the same—that is part of the cleverness of his argument. If we do use the first law, we find that the heat delivered, Q_2 , is the heat Q_1 that was put in minus the work W that was done:

$$Q_2 = Q_1 - W. \quad (44.3)$$

(If we have some kind of cyclic process where water is pumped back into the boiler after it is condensed, we will say that we have heat Q_1 absorbed and work W done, during each cycle, for a certain amount of water that goes around the cycle.)

Now we shall build another engine, and see if we cannot get more work from the same amount of heat being delivered at the temperature T_1 , with the condenser still at the temperature T_2 . We shall use the same amount of heat Q_1 from the boiler, and we shall try to get more work than we did out of the steam engine, perhaps by using another fluid, such as alcohol.

44-3 Reversible engines

Now we must analyze our engines. One thing is clear: we will lose something if the engines contain devices in which there is friction. The best engine will be a frictionless engine. We assume, then, the same idealization that we did when we studied the conservation of energy; that is, a perfectly frictionless engine.

We must also consider the analog of frictionless motion, “frictionless” heat transfer. If we put a hot object at a high temperature against a cold object, so that the heat flows, then it is not possible to make that heat flow in a reverse direction by a very small change in the temperature of either object. But when we have a practically frictionless machine, if we push it with a little force one way, it goes that way, and if we push it with a little force the other way, it goes the other way. We need to find the analog of frictionless motion: heat transfer whose direction we can reverse with only a tiny change. If the difference in temperature is finite, that is impossible, but if one makes sure that heat flows always between two things at essentially the same temperature, with just an infinitesimal difference to make it flow in the desired direction, the flow is said to be reversible (Fig. 44-4). If we heat

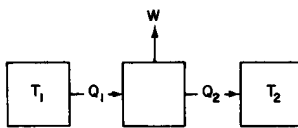


Fig. 44-3. Heat engine.

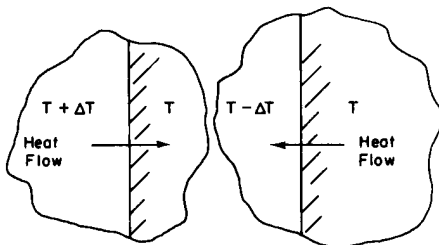


Fig. 44-4. Reversible heat transfer.

the object on the left a little, heat will flow to the right; if we cool it a little, heat will flow to the left. So we find that the ideal engine is a so-called *reversible* engine, in which every process is reversible in the sense that, by minor changes, infinitesimal changes, we can make the engine go in the opposite direction. That means that nowhere in the machine must there be any appreciable friction, and nowhere in the machine must there be any place where the heat of the reservoirs, or the flame of the boiler, is in direct contact with something definitely cooler or warmer.

Let us now consider an idealized engine in which all the processes are reversible. To show that such a thing is possible in principle, we will give an example of an engine cycle which may or may not be practical, but which is at least reversible, in the sense of Carnot's idea. Suppose that we have a gas in a cylinder equipped with a frictionless piston. The gas is not necessarily a perfect gas. The fluid does not even have to be a gas, but to be specific let us say we do have a perfect gas. Also, suppose that we have two heat pads, T_1 and T_2 —great big things that have definite temperatures, T_1 and T_2 . We will suppose in this case that T_1 is higher than T_2 . Let us first heat the gas and at the same time expand it, while it is in contact with the heat pad at T_1 . As we do this, pulling the piston out very slowly as the heat flows into the gas, we will make sure that the temperature of the gas never gets very far from T_1 . If we pull the piston out too fast, the temperature of the gas will fall too much below T_1 and then the process will not be quite reversible, but if we pull it out slowly enough, the temperature of the gas will never depart much from T_1 . On the other hand, if we push the piston back slowly, the temperature would be only infinitesimally higher than T_1 , and the heat would pour back. We see that such an isothermal (constant-temperature) expansion, done slowly and gently enough, is a reversible process.

To understand what we are doing, we shall use a plot (Fig. 44-6) of the pressure of the gas against its volume. As the gas expands, the pressure falls. The curve marked (1) tells us how the pressure and volume change if the temperature is kept fixed at the value T_1 . For an ideal gas this curve would be $PV = NkT_1$. During an isothermal expansion the pressure falls as the volume increases until we stop at the point b . At the same time, a certain heat Q_1 must flow into the gas from the reservoir, for if the gas were expanded without being in contact with the reservoir it would cool off, as we already know. Having completed the isothermal expansion, stopping at the point b , let us take the cylinder away from the reservoir and continue the expansion. This time we permit no heat to enter the cylinder. Again we perform the expansion slowly, so there is no reason why we cannot reverse it, and we again assume there is no friction. The gas continues to expand and the temperature falls, since there is no longer any heat entering the cylinder.

We let the gas expand, following the curve marked (2), until the temperature falls to T_2 , at the point marked c . This kind of expansion, made without adding heat, is called an *adiabatic* expansion. For an ideal gas, we already know that curve (2) has the form $PV^\gamma = \text{constant}$, where γ is a constant greater than 1, so that the adiabatic curve has a more negative slope than the isothermal curve. The gas cylinder has now reached the temperature T_2 , so that if we put it on the heat pad at temperature T_2 there will be no irreversible changes. Now we slowly compress the gas while it is in contact with the reservoir at T_2 , following the curve marked (3) (Fig. 44-5, Step 2). Because the cylinder is in contact with the reservoir, the temperature does not rise, but heat Q_2 flows from the cylinder into the reservoir at the temperature T_2 . Having compressed the gas isothermally along curve (3) to the point d , we remove the cylinder from the heat pad at temperature T_2 and compress it still further, without letting any heat flow out. The temperature will rise, and the pressure will follow the curve marked (4). If we carry out each step properly, we can return to the point a at temperature T_1 where we started, and repeat the cycle.

We see that on this diagram we have carried the gas around a complete cycle, and during one cycle we have put Q_1 in at temperature T_1 , and have removed Q_2 at temperature T_2 . Now the point is that this cycle is reversible, so that we could represent all the steps the other way around. We could have gone backwards instead of forwards: we could have started at point a , at temperature T_1 , expanded

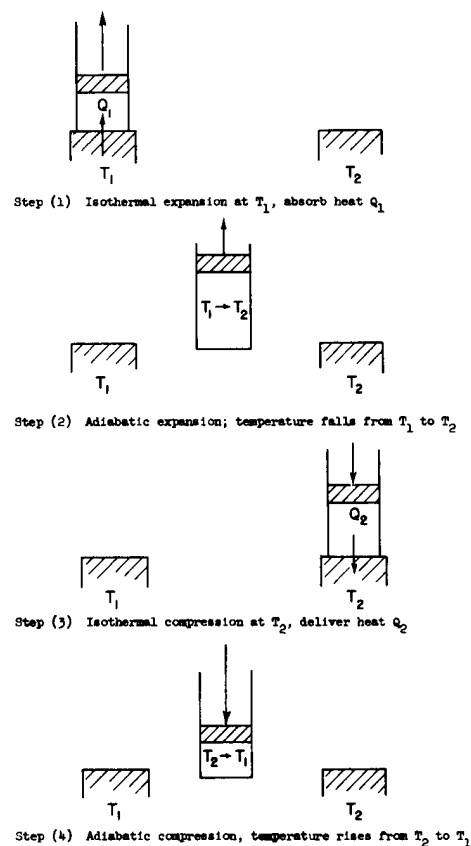


Fig. 44-5. Steps in Carnot cycle.

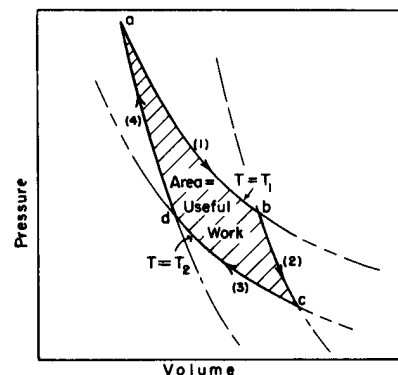


Fig. 44-6. The Carnot cycle.

along the curve (4), expanded further at the temperature T_2^* , absorbing heat Q_2 , and so on, going around the cycle backward. If we go around the cycle in one direction, we must do work on the gas; if we go in the other direction, the gas does work on us.

Incidentally, it is easy to find out what the total amount of work is, because the work during any expansion is the pressure times the change in volume, $\int P dV$. On this particular diagram, we have plotted P vertically and V horizontally. So if we call the vertical distance y and the horizontal distance x , this is $\int y dx$ —in other words, the area under the curve. So the area under each of the numbered curves is a measure of the work done by or on the gas in the corresponding step. It is easy to see that the net work done is the shaded area of the picture.

Now that we have given a single example of a reversible machine, we shall suppose that other such engines are also possible. Let us assume that we have a reversible engine A which takes Q_1 at T_1 , does work W , and delivers some heat at T_2 . Now let us assume we have any other engine B , made by man, already designed or not yet invented, made of rubber bands, steam, or whatever, reversible or not, which is designed so that it takes in the same amount of heat Q_1 at T_1 , and rejects the heat at the lower temperature T_2 (Fig. 44-7). Assume that engine B does some work, W' . Now we shall show that W' is not greater than W —that no engine can do more work than a reversible one. Why? Suppose that, indeed, W' were bigger than W . Then we could take the heat Q_1 out of the reservoir at T_1 , and with engine B we could do work W' and deliver some heat to the reservoir at T_2 ; we do not care how much. That done, we could save some of the work W' , which is supposed to be greater than W ; we could use a part of it, W , and save the remainder, $W' - W$, for useful work. With the work W we could run engine A backwards *because it is a reversible engine*. It will absorb some heat from the reservoir at T_2 and deliver Q_1 back to the reservoir at T_1 . After this double cycle, the net result would be that we would have put everything back the way it was before, and we would have done some excess work, namely $W' - W$, and *all* we would have done would be to extract energy from the reservoir at T_2 ! We were careful to restore the heat Q_1 to the reservoir at T_1 . So that reservoir can be small and “inside” our combined machine $A + B$, whose net effect is therefore to extract a net heat $W' - W$ from the reservoir at T_2 and convert it into work. But to obtain useful work from a reservoir at a single temperature *with no other changes* is impossible according to Carnot’s postulate; it cannot be done. Therefore no engine which absorbs a given amount of heat from a higher temperature T_1 and delivers it at the temperature T_2 can do more work than a reversible engine operating under the same temperature conditions.

Now suppose that engine B is also reversible. Then, of course, not only must W' be not greater than W , but now we can reverse the argument and show that W cannot be greater than W' . So, if both engines are reversible they must both do the same amount of work, and we thus come to Carnot’s brilliant conclusion: that if an engine is reversible, it makes no difference how it is designed, because the amount of work one will obtain if the engine absorbs a given amount of heat at temperature T_1 and delivers heat at some other temperature T_2 *does not depend on the design of the engine*. It is a property of the world, not a property of a particular engine.

If we could find out what the law is that determines how much work we obtain when we absorb the heat Q_1 at T_1 and deliver heat at T_2 , this quantity would be a universal thing, independent of the substance. Of course if we knew the properties of a particular substance, we could work it out and then say that all other substances must give the same amount of work in a reversible engine. That is the key idea, the clue by which we can find the relationship between how much, for instance, a rubber band contracts when we heat it, and how much it cools when we let it contract. Imagine that we put that rubber band in a reversible machine, and that we make it go around a reversible cycle. The net result, the total amount of work done, is that universal function, that great function which is independent of substance. So we see that a substance’s properties must be limited in a certain way; one

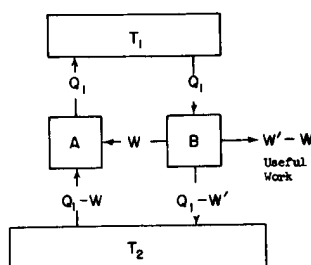


Fig. 44-7. Reversible engine A being driven backwards by engine B .

cannot make up anything he wants, or he would be able to invent a substance which he could use to produce more than the maximum allowable work when he carried it around a reversible cycle. This principle, this limitation, is the only real rule that comes out of the thermodynamics.

44-4 The efficiency of an ideal engine

Now we shall try to find the law which determines the work W as a function of Q_1 , T_1 , and T_2 . It is clear that W is proportional to Q_1 , for if we consider two reversible engines in parallel, both working together and both double engines, the combination is also a reversible engine. If each one absorbed heat Q_1 , the two together absorb $2Q_1$ and the work done is $2W$, and so on. So it is not unreasonable that W is proportional to Q_1 .

Now the next important step is to find this universal law. We can, and will, do so by studying a reversible engine with the one particular substance whose laws we know, a perfect gas. It is also possible to obtain the rule by a purely logical argument, using no particular substance at all. This is one of the very beautiful pieces of reasoning in physics and we are reluctant not to show it to you, so for those who would like to see it we shall discuss it in just a moment. But first we shall use the much less abstract and simpler method of direct calculation for a perfect gas.

We need only obtain formulas for Q_1 and Q_2 (for W is just $Q_1 - Q_2$), the heats exchanged with the reservoirs during the isothermal expansion or contraction. For example, how much heat Q_1 is absorbed from the reservoir at temperature T_1 during the isothermal expansion [marked (1) in Fig. 44-6] from point a , at pressure p_a , volume V_a , temperature T_1 , to point b with pressure p_b , volume V_b , and the same temperature T_1 ? For a perfect gas each molecule has an energy that depends only on the temperature, and since the temperature and the number of molecules are the same at a and at b , the internal energy is the same. *There is no change in U* ; all the work done by the gas,

$$W = \int_a^b p dV,$$

during the expansion is energy Q_1 taken from the reservoir. During the expansion, $pV = NkT_1$, or

$$p = \frac{NkT_1}{V}$$

or

$$Q_1 = \int_a^b p dV = \int_a^b NkT_1 \frac{dV}{V} \quad (44.4)$$

or

$$Q_1 = NkT_1 \ln \frac{V_b}{V_a}$$

is the heat taken from the reservoir at T_1 . In the same way, for the compression at T_2 [curve (3) of Fig. 44-6] the heat delivered to the reservoir at T_2 is

$$Q_2 = NkT_2 \ln \frac{V_c}{V_d}. \quad (44.5)$$

To finish our analysis we need only find a relation between V_c/V_d and V_b/V_a . This we do by noting that (2) is an adiabatic expansion from b to c , during which pV^γ is a constant. Since $pV = NkT$, we can write this as $(pV)V^{\gamma-1} = \text{const}$ or, in terms of T and V , as $TV^{\gamma-1} = \text{const}$, or

$$T_1 V_b^{\gamma-1} = T_2 V_c^{\gamma-1}. \quad (44.6)$$

Likewise, since (4), the expansion from d to a , is also adiabatic, we find

$$T_1 V_a^{\gamma-1} = T_2 V_d^{\gamma-1}. \quad (44.6a)$$

If we divide this equation by the previous one, we find that V_b/V_a must equal V_c/V_d , so the $\ln$'s in (44.4) and (44.5) are equal, and that

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}. \quad (44.7)$$

This is the relation we were seeking. Although proved for a perfect gas engine, we know it must be true *for any reversible engine at all*.

Now we shall see how this universal law could also be obtained by logical argument, without knowing the properties of any specific substances, as follows. Suppose that we have three engines and three temperatures, let us say T_1 , T_2 , and T_3 . Let one engine absorb heat Q_1 from the temperature T_1 and do a certain amount of work W_{13} , and let it deliver heat Q_3 to the temperature T_3 (Fig. 44-8). Let another engine run backwards between T_2 and T_3 . Suppose that we let the second engine be of such a size that it will absorb the same heat Q_3 , and deliver the heat Q_2 . We will have to put a certain amount of work, W_{32} , into it—negative because the engine is running backwards. When the first machine goes through a cycle, it absorbs heat Q_1 and delivers Q_3 at the temperature T_3 ; then the second machine takes the same heat Q_3 out of the reservoir at the temperature T_3 and delivers it into the reservoir at temperature T_2 . Therefore the net result of the two machines in tandem is to take the heat Q_1 from T_1 , and deliver Q_2 at T_2 . The two machines are thus equivalent to a third one, which absorbs Q_1 at T_1 , does work W_{12} , and delivers heat Q_2 at T_2 , because $W_{12} = W_{13} - W_{32}$, as one can immediately show from the first law, as follows:

$$W_{13} - W_{32} = (Q_1 - Q_3) - (Q_2 - Q_3) = Q_1 - Q_2 = W_{12}. \quad (44.8)$$

We can now obtain the laws which relate the efficiencies of the engines, because there clearly must be some kind of relationship between the efficiencies of engines running between the temperatures T_1 and T_3 , and between T_2 and T_3 , and between T_1 and T_2 .

We can make the argument very clear in the following way: We have just seen that we can always relate the heat absorbed at T_1 to the heat delivered at T_2 by finding the heat delivered at some other temperature T_3 . Therefore we can get all the engines' properties if we introduce a standard temperature, analyzing everything with that standard temperature. In other words, if we knew the efficiency of an engine running between a certain temperature T and a certain arbitrary standard temperature, then we could work out the efficiency for any other difference in temperature. Because we assume we are using only reversible engines, we can work from the initial temperature down to the standard temperature and back up to the final temperature again. We shall define the standard temperature arbitrarily as *one degree*. We shall also adopt a special symbol for the heat which is delivered at this standard temperature: we shall call it Q_s . In other words, when a reversible engine absorbs the heat Q at temperature T , it will deliver, at the unit temperature, a heat Q_s . *If one engine, absorbing heat Q_1 at T_1 , delivers the heat Q_s at one degree, and if an engine absorbing heat Q_2 at temperature T_2 will also deliver the same heat Q_s at one degree, then it follows that an engine which absorbs heat Q_1 at the temperature T_1 will deliver heat Q_2 if it runs between T_1 and T_2 , as we have already proved by considering engines running between three temperatures.* So all we really have to do is to find how much heat Q_1 we need to put in at the temperature T_1 in order to deliver a certain amount of heat Q_s at the unit temperature. If we discover that, we have everything. The heat Q , of course, is a function of the temperature T . It is easy to see that the heat must increase as the temperature increases, for we know that it takes work to run an engine backwards and deliver heat at a higher temperature. It is also easy to see that the heat Q_1 must be proportional to Q_s . So the great law is something like this: for a given amount of heat Q_s delivered at one degree from an engine running at temperature T degrees, the heat Q absorbed must be that amount Q_s times some increasing function of the temperature:

$$Q = Q_s f(T). \quad (44.9)$$

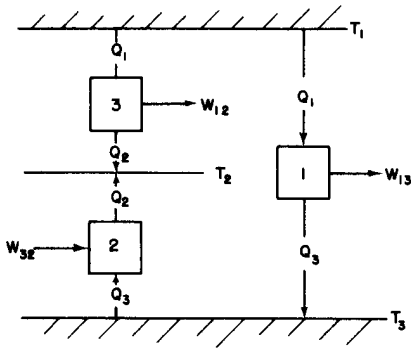


Fig. 44-8. Engines 1 and 2 together are equivalent to engine 3.

44-5 The thermodynamic temperature

At this stage we are not going to try to find the formula for the above increasing function of the temperature in terms of our familiar mercury temperature scale, but instead *we shall define temperature by a new scale*. At one time "the temperature" was defined arbitrarily by dividing the expansion of water into even degrees of a certain size. But when one then measures temperature with a mercury thermometer, one finds that the degrees are no longer even. But *now we can make a definition of temperature which is independent of any particular substance*. We can use that function $f(T)$, which does not depend on what device we use, because the efficiency of these reversible engines is independent of their working substances. Since the function we found is rising with temperature, we will define the *function itself* as the temperature, measured in units of the standard one-degree temperature, as follows:

$$Q = ST, \quad (44.10)$$

where

$$Q_S = S \cdot 1^\circ. \quad (44.11)$$

This means that we can tell how hot an object is by finding out how much heat is absorbed by a reversible engine working between the temperature of the object and the unit temperature (Fig. 44-9). If seven times more heat is taken out of a boiler than is delivered at a one-degree condenser, the temperature of the boiler will be called seven degrees, and so forth. So, by measuring how much heat is absorbed at different temperatures, we determine the temperature. The temperature defined in this way is called the *absolute thermodynamic temperature*, and it is independent of the substance. We shall use this definition exclusively from now on.*

Now we see that when we have two engines, one working between T_1 and one degree, the other working between T_2 and one degree, delivering the same heat at unit temperature, then the heats absorbed must be related by

$$\frac{Q_1}{T_1} = S = \frac{Q_2}{T_2}. \quad (44.12)$$

But that means that if we have a single engine running between T_1 and T_2 , then the result of the whole analysis, the grand finale, is that Q_1 is to T_1 as Q_2 is to T_2 , if the engine absorbs energy Q_1 at temperature T_1 and delivers heat Q_2 at temperature T_2 . Whenever the engine is reversible, this relationship between the heats must follow. That is all there is to it: that is the center of the universe of thermodynamics.

If this is all there is to thermodynamics, why is it considered such a difficult subject? In doing a problem involving a given mass of some substance, the condition of the substance at any moment can be described by telling what its temperature is and what its volume is. If we know the temperature and volume of a substance, and that the pressure is some function of the temperature and volume, then we know the internal energy. One could say, "I do not want to do it that way. Tell me the temperature and the pressure, and I will tell you the volume. I can think of the volume as a function of temperature and pressure, and the internal energy as a function of temperature and pressure, and so on." That is why thermodynamics is hard, because everyone uses a different approach. If we could only sit down once and decide on our variables, and stick to them, it would be fairly easy.

Now we start to make deductions. Just as $F \Rightarrow ma$ is the center of the universe in mechanics, and it goes on and on and on after that, in the same way the principle just found is all there is to thermodynamics. But can one make conclusions out of it?

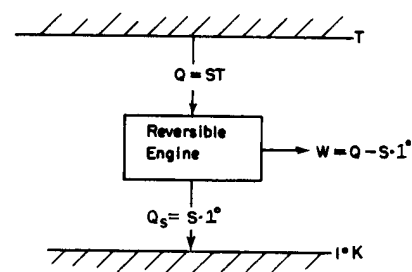


Fig. 44-9. Absolute thermodynamic temperature.

* We have previously defined our scale of temperature in a different way, namely by stating that the mean kinetic energy of a molecule in a perfect gas is proportional to the temperature, or that the perfect gas law says pV is proportional to T . Is this new definition equivalent? Yes, since the final result (44.7) derived from the gas law is the same as that derived here. We shall discuss this point again in the next chapter.

We begin. To obtain our first conclusion, we shall combine both laws, the law of conservation of energy and this law which relates the heats Q_2 and Q_1 , and we can easily obtain *the efficiency of a reversible engine*. From the first law, we have $W = Q_1 - Q_2$. According to our new principle,

$$Q_2 = \frac{T_2}{T_1} Q_1,$$

so the work becomes

$$W = Q_1 \left(1 - \frac{T_2}{T_1} \right) = Q_1 \frac{T_1 - T_2}{T_1}, \quad (44.13)$$

which tells us the efficiency of the engine—how much work we get out of so much heat. The efficiency of an engine is proportional to the difference in the temperatures between which the engine runs, divided by the higher temperature:

$$\text{Efficiency} = \frac{W}{Q_1} = \frac{T_1 - T_2}{T_1}. \quad (44.14)$$

The efficiency cannot be greater than unity and the absolute temperature cannot be less than zero, absolute zero. So, since T_2 must be positive, the efficiency is always less than unity. That is our first conclusion.

44-6 Entropy

Equation (44.7) or (44.12) can be interpreted in a special way. Working always with reversible engines, a heat Q_1 at temperature T_1 is “equivalent” to Q_2 at T_2 if $Q_1/T_1 = Q_2/T_2$, in the sense that as one is absorbed the other is delivered. This suggests that if we call Q/T something, we can say: in a reversible process as much Q/T is absorbed as is liberated; there is no gain or loss of Q/T . This Q/T is called *entropy*, and we say “there is no net change in entropy in a reversible cycle.” If T is 1° , then the entropy is $Q/1^\circ$ or, as we symbolized it, $Q_S/1^\circ = S$. Actually, S is the letter usually used for entropy, and it is numerically equal to the heat (which we have called Q_S) delivered to a 1° -reservoir (entropy is not itself a heat, it is heat divided by a temperature, hence it is measured in *joules per degree*).

Now it is interesting that besides the pressure, which is a function of the temperature and the volume, and the internal energy, which is a function of temperature and volume, we have found another quantity which is a function of the condition, i.e., the entropy of the substance. Let us try to explain how we compute it, and what we mean when we call it a “function of the condition.” Consider the system in two different conditions, much as we had in the experiment where we did the adiabatic and isothermal expansions. (Incidentally, there is no need that a heat engine have only two reservoirs, it could have three or four different temperatures at which it takes in and delivers heats, and so on.) We can move around on a pV diagram all over the place, and go from one condition to another. In other words, we could say the gas is in a certain condition a , and then it goes over to some other condition, b , and we will require that this transition, made from a to b , be reversible. Now suppose that all along the path from a to b we have little reservoirs at different temperatures, so that the heat dQ removed from the substance at each little step is delivered to each reservoir at the temperature corresponding to that point on the path. Then let us connect all these reservoirs, by reversible heat engines, to a single reservoir at the unit temperature. When we are finished carrying the substance from a to b , we shall bring all the reservoirs back to their original condition. Any heat dQ that has been absorbed from the substance at temperature T has now been converted by a reversible machine, and a certain amount of entropy dS has been delivered at the unit temperature as follows:

$$dS = dQ/T. \quad (44.15)$$

Let us compute the total amount of entropy which has been delivered. The entropy difference, or the entropy needed to go from a to b by this particular

reversible transformation, is the total entropy, the total of the entropy taken out of the little reservoirs, and delivered at the unit temperature:

$$S_b - S_a = \int_a^b \frac{dQ}{T}. \quad (44.16)$$

The question is, does the entropy difference depend upon the path taken? There is more than one way to go from a to b . Remember that in the Carnot cycle we could go from a to c in Fig. 44-6 by first expanding isothermally and then adiabatically; or we could first expand adiabatically and then isothermally. So the question is whether the entropy change which occurs when we go from a to b in Fig. 44-10 is the same on one route as it is on another. *It must be the same*, because if we went all the way around the cycle, going forward on one path and backward on another, we would have a reversible engine, and there would be no loss of heat to the reservoir at unit temperature. In a totally reversible cycle, no heat must be taken from the reservoir at the unit temperature, so the entropy needed to go from a to b is the same over one path as it is over another. It is *independent of path*, and depends only on the endpoints. We can, therefore, say that there is a certain function, which we call the entropy of the substance, that depends only on the condition, i.e., only on the volume and temperature.

We can find a function $S(V, T)$ which has the property that if we compute the change in entropy, as the substance is moved along any reversible path, in terms of the heat rejected at unit temperature, then

$$\Delta S = \int \frac{dQ}{T}, \quad (44.17)$$

where dQ is the heat removed from the substance at temperature T . This total entropy change is the difference between the entropy calculated at the initial and final points:

$$\Delta S = S(V_b, T_b) - S(V_a, T_a) = \int_a^b \frac{dQ}{T}. \quad (44.18)$$

This expression does not completely define the entropy, but rather only the *difference* of entropy between two different conditions. Only if we can evaluate the entropy for one special condition can we really define S absolutely.

For a long time it was believed that absolute entropy meant nothing—that only differences could be defined—but finally Nernst proposed what he called the *heat theorem*, which is also called the third law of thermodynamics. It is very simple. We will say what it is, but we will not explain why it is true. Nernst's postulate states simply that the entropy of any object at absolute zero is zero. We know of one case of T and V , namely $T = 0$, where S is zero; and so we can get the entropy at any other point.

To give an illustration of these ideas, let us calculate the entropy of a perfect gas. In an isothermal (and therefore reversible) expansion, $\int dQ/T$ is Q/T , since T is constant. Therefore (from 44.4) the change in entropy is

$$S(V_a, T) - S(V_b, T) = Nk \ln \frac{V_a}{V_b},$$

so $S(V, T) = Nk \ln V$ plus some function of T only. How does S depend on T ? We know that for a reversible adiabatic expansion, *no heat is exchanged*. Thus the entropy does not change even though V changes, provided that T changes also, such that $TV^{\gamma-1} = \text{constant}$. Can you see that this implies that

$$S(V, T) = Nk \left[\ln V + \frac{1}{\gamma - 1} \ln T \right] + a,$$

where a is some constant independent of *both* V and T ? [a is called the chemical constant. It depends on the gas in question, and may be determined experimentally from the Nernst theorem by measuring the heat liberated in cooling and condensing

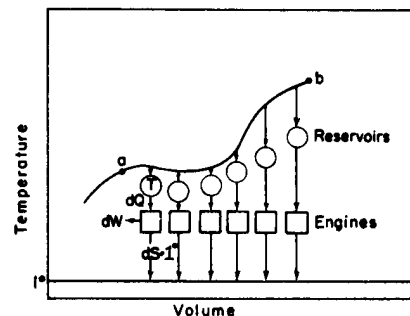


Fig. 44-10. Change in entropy during a reversible transformation.

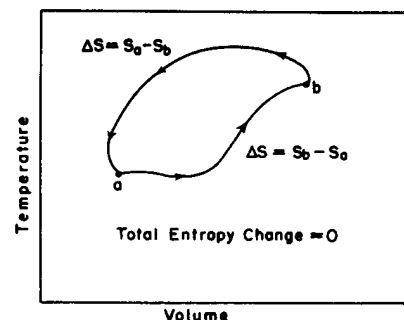


Fig. 44-11. Change in entropy in a completely reversible cycle.

the gas until it is brought to a solid (or for helium, a liquid) at 0° , by integrating $\int dQ/T$. It can also be determined theoretically by means of Planck's constant and quantum mechanics, but we shall not study it in this course.]

Now we shall remark on some of the properties of the entropy of things. We first remember that if we go along a reversible cycle from a to b , then the entropy of the substance will change by $S_b - S_a$. And we remember that as we go along the path, the entropy—the heat delivered at unit temperature—increases according to the rule $dS = dQ/T$, where dQ is the heat we remove from the substance when its temperature is T .

We already know that if we have a reversible cycle, the total entropy of everything is not changed, because the heat Q_1 absorbed at T_1 and the heat Q_2 delivered at T_2 correspond to equal and opposite changes in entropy, so that the net change in the entropy is zero. So for a reversible cycle there is no change in the entropy of anything, including the reservoirs. This rule may look like the conservation of energy again, but it is not; it applies only to reversible cycles. If we include irreversible cycles there is no law of conservation of entropy.

We shall give two examples. First, suppose that we do irreversible work on an object by friction, generating a heat Q on some object at temperature T . The entropy is increased by Q/T . The heat Q is equal to the work, and thus when we do a certain amount of work by friction against an object whose temperature is T , the entropy of the whole world increases by W/T .

Another example of irreversibility is this: If we put together two objects that are at different temperatures, say T_1 and T_2 , a certain amount of heat will flow from one to the other by itself. Suppose, for instance, we put a hot stone in cold water. Then when a certain heat ΔQ is transferred from T_1 to T_2 , how much does the entropy of the hot stone change? It decreases by $\Delta Q/T_1$. How much does the water entropy change? It increases by $\Delta Q/T_2$. The heat will, of course, flow only from the higher temperature T_1 to the lower temperature T_2 , so that ΔQ is positive if T_1 is greater than T_2 . So the change in entropy of the whole world is positive, and it is the difference of the two fractions:

$$\Delta S = \frac{\Delta Q}{T_2} - \frac{\Delta Q}{T_1}. \quad (44.19)$$

So the following proposition is true: in any process that is irreversible, the entropy of the whole world is increased. Only in reversible processes does the entropy remain constant. Since no process is absolutely reversible, there is always at least a small gain in the entropy; a reversible process is an idealization in which we have made the gain of entropy minimal.

Unfortunately, we are not going to enter into the field of thermodynamics very far. Our purpose is only to illustrate the principal ideas involved and the reasons why it is possible to make such arguments, but we will not use thermodynamics very much in this course. Thermodynamics is used very often by engineers and, particularly, by chemists. So we must learn our thermodynamics in practice in chemistry or engineering. Because it is not worth while duplicating everything, we shall just give some discussion of the origin of the theory, rather than much detail for special applications.

The two laws of thermodynamics are often stated this way:

First law: the energy of the universe is always constant.

Second law: the entropy of the universe is always increasing.

That is not a very good statement of the second law; it does not say, for example, that in a reversible cycle the entropy stays the same, and it does not say exactly what the entropy is. It is just a clever way of remembering the two laws, but it does not really tell us exactly where we stand. We have summarized the laws discussed in this chapter in Table 44-1. In the next chapter we shall apply these laws to discover the relationship between the heat generated in the expansion of a rubber band, and the extra tension when it is heated.

Table 44-1

Summary of the laws of thermodynamics

First law:

Heat put into a system + Work done on a system = Increase in internal energy of the system:

$$dQ + dW = dU.$$

Second law:

A process whose *only* net result is to take heat from a reservoir and convert it to work is impossible.

No heat engine taking heat Q_1 from T_1 and delivering heat Q_2 at T_2 can do more work than a reversible engine, for which

$$W = Q_1 - Q_2 = Q_1 \left(\frac{T_1 - T_2}{T_1} \right).$$

The entropy of a system is defined this way:

(a) If heat ΔQ is added reversibly to a system at temperature T , the increase in entropy of the system is $\Delta S = \Delta Q/T$.

(b) At $T = 0$, $S = 0$ (*third law*).

In a *reversible change*, the total entropy of all parts of the system (including reservoirs) does not change.

In *irreversible change*, the total entropy of the system always increases.

Illustrations of Thermodynamics

45-1 Internal energy

Thermodynamics is a rather difficult and complex subject when we come to apply it, and it is not appropriate for us to go very far into the applications in this course. The subject is of very great importance, of course, to engineers and chemists, and those who are interested in the subject can learn about the applications in physical chemistry or in engineering thermodynamics. There are also good reference books, such as Zemansky's *Heat and Thermodynamics*, where one can learn more about the subject. In the Encyclopedia Britannica, fourteenth edition, one can find excellent articles on thermodynamics and thermochemistry, and in the article on chemistry, the sections on physical chemistry, vaporization, liquefaction of gases, and so on.

The subject of thermodynamics is complicated because there are so many different ways of describing the same thing. If we wish to describe the behavior of a gas, we can say that the pressure depends on the temperature and on the volume, or we can say that the volume depends on the temperature and the pressure. Or with respect to the internal energy U , we might say that it depends on the temperature and volume, if those are the variables we have chosen—but we might also say that it depends on the temperature and the pressure, or the pressure and the volume, and so on. In the last chapter we discussed another function of temperature and volume, called the entropy S , and we can of course construct as many other functions of these variables as we like: $U - TS$ is a function of temperature and volume. So we have a large number of different quantities which can be functions of many different combinations of variables.

To keep the subject simple in this chapter, we shall decide at the start to use *temperature* and *volume* as the independent variables. Chemists use temperature and pressure, because they are easier to measure and control in chemical experiments, but we shall use temperature and volume throughout this chapter, except in one place where we shall see how to make the transformation into the chemists' system of variables.

We shall first, then, consider only one system of independent variables: temperature and volume. Secondly, we shall discuss only two dependent functions: the internal energy and the pressure. All the other functions can be derived from these, so it is not necessary to discuss them. With these limitations, thermodynamics is still a fairly difficult subject, but it is not quite so impossible!

First we shall review some mathematics. If a quantity is a function of two variables, the idea of the derivative of the quantity requires a little more careful thought than for the case where there is only one variable. What do we mean by the derivative of the pressure with respect to the temperature? The pressure change accompanying a change in the temperature depends partly, of course, on what happens to the *volume* while T is changing. We must specify the change in V before the concept of a derivative with respect to T has a precise meaning. We might ask, for example, for the rate of change of P with respect to T if V is held constant. This ratio is just the ordinary derivative that we usually write as dP/dT . We customarily use a special symbol, $\partial P/\partial T$, to remind us that P depends on another variable V as well as on T , and that this other variable is held constant. We shall not only use the symbol ∂ to call attention to the fact that the other variable is held constant, but we shall also write the variable that is held constant as a subscript, $(\partial P/\partial T)_V$. Since we have only two independent variables, this notation is redundant, but it will help us keep our wits about us in the thermodynamic jungle of partial derivatives.

45-1 Internal energy

45-2 Applications

45-3 The Clausius-Clapeyron equation

Let us suppose that the function $f(x, y)$ depends on the two independent variables x and y . By $(\partial f/\partial x)_y$ we mean simply the ordinary derivative, obtained in the usual way, if we treat y as a constant:

$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}.$$

Similarly, we define

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}.$$

For example, if $f(x, y) = x^2 + yx$, then $(\partial f/\partial x)_y = 2x + y$, and $(\partial f/\partial y)_x = x$. We can extend this idea to higher derivatives: $\partial^2 f/\partial y^2$ or $\partial^2 f/\partial y \partial x$. The latter symbol indicates that we first differentiate f with respect to x , treating y as a constant, then differentiate the result with respect to y , treating x as a constant. The actual order of differentiation is immaterial: $\partial^2 f/\partial x \partial y = \partial^2 f/\partial y \partial x$.

We will need to compute the change Δf in $f(x, y)$ when x changes to $x + \Delta x$ and y changes to $y + \Delta y$. We assume throughout the following that Δx and Δy are infinitesimally small:

$$\begin{aligned} \Delta f &= f(x + \Delta x, y + \Delta y) - f(x, y) \\ &= \underbrace{f(x + \Delta x, y + \Delta y) - f(x, y + \Delta y)}_{\Delta x \left(\frac{\partial f}{\partial x}\right)_y} + \underbrace{f(x, y + \Delta y) - f(x, y)}_{\Delta y \left(\frac{\partial f}{\partial y}\right)_x} \\ &= \Delta x \left(\frac{\partial f}{\partial x}\right)_y + \Delta y \left(\frac{\partial f}{\partial y}\right)_x. \end{aligned} \quad (45.1)$$

The last equation is the fundamental relation that expresses Δf in terms of Δx and Δy .

As an example of the use of this relation, let us calculate the change in the internal energy $U(T, V)$ when the temperature changes from T to $T + \Delta T$ and the volume changes from V to $V + \Delta V$. Using Eq. (45.1), we write

$$\Delta U = \Delta T \left(\frac{\partial U}{\partial T}\right)_V + \Delta V \left(\frac{\partial U}{\partial V}\right)_T. \quad (45.2)$$

In our last chapter we found another expression for the change ΔU in the internal energy when a quantity of heat ΔQ was added to the gas:

$$\Delta U = \Delta Q - P \Delta V. \quad (45.3)$$

In comparing Eqs. (45.2) and (45.3) one might at first be inclined to think that $P = (\partial U/\partial V)_T$, but this is not correct. To obtain the correct relation, let us first suppose that we add a quantity of heat ΔQ to the gas while keeping the volume constant, so that $\Delta V = 0$. With $\Delta V = 0$, Eq. (45.3) tells us that $\Delta U = \Delta Q$, and Eq. (45.2) tells us that $\Delta U = (\partial U/\partial T)_V \Delta T$, so that $(\partial U/\partial T)_V = \Delta Q/\Delta T$. The ratio $\Delta Q/\Delta T$, the amount of heat one must put into a substance in order to change its temperature by one degree with the volume held constant, is called the *specific heat at constant volume* and is designated by the symbol C_V . By this argument we have shown that

$$\left(\frac{\partial U}{\partial T}\right)_V = C_V. \quad (45.4)$$

Now let us again add a quantity of heat ΔQ to the gas, but this time we will hold T constant and allow the volume to change by ΔV . The analysis in this case is more complex, but we can calculate ΔU by the argument of Carnot, making use of the Carnot cycle we introduced in the last chapter.

The pressure-volume diagram for the Carnot cycle is shown in Fig. 45-1. As we have already shown, the total amount of work done by the gas in a reversible cycle is $\Delta Q(\Delta T/T)$, where ΔQ is the amount of heat energy added to the gas as it expands isothermally at temperature T from volume V to $V + \Delta V$, and $T - \Delta T$ is the final temperature reached by the gas as it expands adiabatically on the second leg of the cycle. Now we will show that this work done is also given by

the shaded area in Fig. 45-1. In any circumstances, the work done by the gas is $\int P dV$, and is positive when the gas expands and negative when the gas is compressed. If we plot P vs. V , the variation of P and V is represented by a curve which gives the value of P corresponding to a particular value of V . As the volume changes from one value to another, the work done by the gas, the integral $\int P dV$, is the area under the curve connecting the initial and final values of V . When we apply this idea to the Carnot cycle, we see that as we go around the cycle, paying attention to the sign of the work done by the gas, the net work done by the gas is just the shaded area in Fig. 45-1.

Now we want to evaluate the shaded area geometrically. The cycle we have used in Fig. 45-1 differs from that used in the previous chapter in that we now suppose that ΔT and ΔQ are infinitesimally small. We are working between adiabatic lines and isothermal lines that are very close together, and the figure described by the heavy lines in Fig. 45-1 will approach a parallelogram as the increments ΔT and ΔQ approach zero. The area of this parallelogram is just $\Delta V \Delta P$, where ΔV is the change in volume as energy ΔQ is added to the gas at constant temperature, and ΔP is the change in pressure as the temperature changes by ΔT at constant volume. One can easily show that the shaded area in Fig. 45-1 is given by $\Delta V \Delta P$ by recognizing that the shaded area is equal to the area enclosed by the dotted lines in Fig. 45-2, which in turn differs from the rectangle bounded by ΔP and ΔV only by the addition and subtraction of the equal triangular areas in Fig. 45-2.

Now let us summarize the results of the arguments we have developed so far:

$$\left. \begin{aligned} &\text{Work done by the gas} = \text{shaded area} = \Delta V \Delta P = \Delta Q \left(\frac{\Delta T}{T} \right) \\ \text{or} & \quad \frac{\Delta T}{T} \cdot (\text{heat needed to change } V \text{ by } \Delta V)_{\text{constant } T} \\ &= \Delta V \cdot (\text{change in } P \text{ when } T \text{ changes by } \Delta T)_{\text{constant } V} \\ \text{or} & \quad \frac{1}{\Delta V} \cdot (\text{heat needed to change } V \text{ by } \Delta V)_T = T \left(\frac{\partial P}{\partial T} \right)_V. \end{aligned} \right\} \quad (45.5)$$

Equation (45.5) expresses the essential result of Carnot's argument. The whole of thermodynamics can be deduced from Eq. (45.5) and the First Law, which is stated in Eq. (45.3). Equation (45.5) is essentially the Second Law, although it was originally deduced by Carnot in a slightly different form, since he did not use our definition of temperature.

Now we can proceed to calculate $(\partial U / \partial V)_T$. By how much would the internal energy U change if we changed the volume by ΔV ? First, U changes because heat is put in, and second, U changes because work is done. The heat put in is

$$\Delta Q = T \left(\frac{\partial P}{\partial T} \right)_V \Delta V,$$

according to Eq. (45.5), and the work done on the substance is $-P \Delta V$. Therefore the change ΔU in internal energy has two pieces:

$$\Delta U = T \left(\frac{\partial P}{\partial T} \right)_V \Delta V - P \Delta V. \quad (45.6)$$

Dividing both sides by ΔV , we find for the rate of change of U with V at constant T

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial P}{\partial T} \right)_V - P. \quad (45.7)$$

In our thermodynamics, in which T and V are the only variables and P and U are the only functions, Eqs. (45.3) and (45.7) are the basic equations from which all the result of the subject can be deduced.

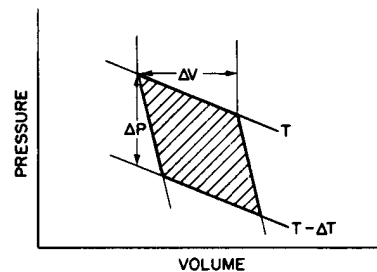


Fig. 45-1. Pressure-volume diagram for a Carnot cycle. The curves marked T and $T - \Delta T$ are isothermal lines; the steeper curves are adiabatic lines. ΔV is the volume change as heat ΔQ is added to the gas at constant temperature T . ΔP is the pressure change at constant volume as the gas temperature is changed from T to $T - \Delta T$.

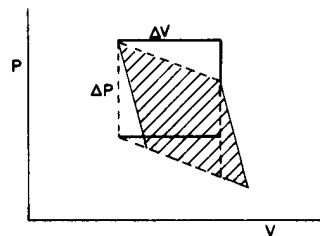


Fig. 45-2. Shaded area = area enclosed by dashed lines = area of rectangle = $\Delta P \Delta V$.

45-2 Applications

Now let us discuss the meaning of Eq. (45.7) and see why it answers the questions which we proposed in our last chapter. We considered the following problem: in kinetic theory it is obvious that an increase in temperature leads to an increase in pressure, because of the bombardments of the atoms on a piston. For the same physical reason, when we let the piston move back, heat is taken out of the gas and, in order to keep the temperature constant, heat will have to be put back in. The gas cools when it expands, and the pressure rises when it is heated. There must be some connection between these two phenomena, and this connection is given explicitly in Eq. (45.7). If we hold the volume fixed and increase the temperature, the pressure rises at a rate $(\partial P/\partial T)_V$. Related to that fact is this: if we increase the volume, the gas will cool unless we pour some heat in to maintain the temperature constant, and $(\partial U/\partial V)_T$ tells us the amount of heat needed to maintain the temperature. Equation (45.7) expresses the fundamental interrelationship between these two effects. That is what we promised we would find when we came to the laws of thermodynamics. Without knowing the internal mechanism of the gas, and knowing only that we cannot make perpetual motion of the second type, we can deduce the relationship between the amount of heat needed to maintain a constant temperature when the gas expands, and the pressure change when the gas is heated!

Now that we have the result we wanted for a gas, let us consider the rubber band. When we stretch a rubber band, we find that its temperature falls, and when we heat a rubber band, we find that it pulls itself in. What is the equation that gives the same relation for a rubber band as Eq. (45.3) gives for gas? For a rubber band the situation will be something like this: when heat ΔQ is put in, the internal energy is changed by ΔU and some work is done. The only difference will be that the work done by the rubber band is $-F \Delta L$ instead of $P \Delta V$, where F is the force on the band, and L is the length of the band. The force F is a function of temperature and of length of the band. Replacing $P \Delta V$ in Eq. (45.3) by $-F \Delta L$, we get

$$\Delta U = \Delta Q + F \Delta L \quad (45.8)$$

Comparing Eqs. (45.3) and (45.8), we see that the rubber band equation is obtained by a mere substitution of one letter for another. Furthermore, if we substitute L for V , and $-F$ for P , all of our discussion of the Carnot cycle applies to the rubber band. We can immediately deduce, for instance, that the heat ΔQ needed to change the length by ΔL is given by the analog to Eq. (45.5): $\Delta Q = -T(\partial F/\partial T)_L \Delta L$. This equation tells us that if we keep the length of a rubber band fixed and heat the band, we can calculate how much the force will increase in terms of the heat needed to keep the temperature constant when the band is stretched a little bit. So we see that the same equation applies to both gas and a rubber band. In fact, if one can write $\Delta U = \Delta Q + A \Delta B$, where A and B represent different quantities, force and length, pressure and volume, etc., one can apply the results obtained for a gas by substituting A and B for P and V . For example, consider the electric potential difference, or "voltage," E in a battery and the charge ΔZ that moves through the battery. We know that the work done in a reversible electric cell, like a storage battery, is $E \Delta Z$. (Since we include no $P \Delta V$ term in the work, we require that our battery maintain a constant volume.) Let us see what thermodynamics can tell us about the performance of a battery. If we substitute E for P and Z for V in Eq. (45.6), we obtain

$$\frac{\Delta U}{\Delta Z} = -T \left(\frac{\partial E}{\partial T} \right)_Z - E. \quad (45.9)$$

Equation (45.9) says that the internal energy U is changed when a charge ΔZ moves through the cell. Why is $\Delta U/\Delta Z$ not simply the voltage E of the battery? The answer is that a real battery gets warm when charge moves through the cell. The internal energy of the battery is changed, first, because the battery did some work on the outside circuit, and second, because the battery is heated. The re-

markable thing is that the second part can again be expressed in terms of the way in which the battery voltage changes with temperature. Incidentally, when the charge moves through the cell, chemical reactions occur, and Eq. (45.9) suggests a nifty way of measuring the amount of energy required to produce a chemical reaction. All we need do is construct a cell that works on the reaction, measure the voltage, and measure how much the voltage changes with temperature when we draw no charge from the battery!

Now we have assumed that the volume of the battery can be maintained constant, since we have omitted the $P\Delta V$ term when we set the work done by the battery equal to $E\Delta Z$. It turns out that it is technically quite difficult to keep the volume constant. It is much easier to keep the cell at constant atmospheric pressure. For that reason, the chemists do not like any of the equations we have written above: they prefer equations which describe performance under constant pressure. We chose at the beginning of this chapter to use V and T as independent variables. The chemists prefer P and T , and we will now consider how the results we have obtained so far can be transformed into the chemists' system of variables. Remember that in the following treatment confusion can easily set in because we are shifting gears from T and V to T and P .

We started in Eq. (45.3) with $\Delta U = \Delta Q - P\Delta V$; $P\Delta V$ may be replaced by $E\Delta Z$ or $A\Delta B$. If we could somehow replace the last term, $P\Delta V$, by $V\Delta P$, then we would have interchanged V and P , and the chemists would be happy. Well, a clever man noticed that the differential of the product PV is $d(PV) = PdV + VdP$, and if he added this equality to Eq. (45.3), he obtained

$$\begin{array}{rcl} \Delta(PV) & = & P\Delta V + V\Delta P \\ \Delta U & = & \Delta Q - P\Delta V \\ \hline \Delta(U + PV) & = & \Delta Q + V\Delta P \end{array}$$

In order that the result look like Eq. (45.3), we define $U + PV$ to be something new, called the *enthalpy*, H , and we write $\Delta H = \Delta Q + V\Delta P$.

Now we are ready to transform our results into chemists' language with the following rules: $U \rightarrow H$, $P \rightarrow -V$, $V \rightarrow P$. For example, the fundamental relationship that chemists would use instead of Eq. (45.7) is

$$\left(\frac{\partial H}{\partial P}\right)_T = T\left(\frac{\partial V}{\partial T}\right)_P - V.$$

It should now be clear how one transforms to the chemists' variables T and P . We now go back to our original variables: for the remainder of this chapter, T and V are the independent variables.

Now let us apply the results we have obtained to a number of physical situations. Consider first the ideal gas. From kinetic theory we know that the internal energy of a gas depends only on the motion of the molecules and the number of molecules. The internal energy depends on T , but not on V . If we change V , but keep T constant, U is not changed. Therefore $(\partial U/\partial V)_T = 0$, and Eq. (45.7) tells us that for an ideal gas

$$T\left(\frac{\partial P}{\partial T}\right)_V - P = 0. \quad (45.10)$$

Equation (45.10) is a differential equation that can tell us something about P . We take account of the partial derivatives in the following way: Since the partial derivative is at constant V , we will replace the partial derivative by an ordinary derivative and write explicitly, to remind us, "constant V ." Equation (45.10) then becomes

$$T \frac{\Delta P}{\Delta T} - P = 0; \quad \text{const } V, \quad (45.11)$$

which we can integrate to get

$$\begin{array}{ll} \ln P = \ln T + \text{const}; & \text{const } V, \\ P = \text{const} \times T; & \text{const } V. \end{array} \quad (45.12)$$

We know that for an ideal gas the pressure is equal to

$$P = \frac{RT}{V}, \quad (45.13)$$

which is consistent with (45.12), since V and R are constants. Why did we bother to go through this calculation if we already knew the results? Because we have been using *two independent definitions of temperature!* At one stage we assumed that the kinetic energy of the molecules was proportional to the temperature, an assumption that defines one scale of temperature which we will call the ideal gas scale. The T in Eq. (45.13) is based on the gas scale. We also call temperatures measured on the gas scale *kinetic* temperatures. Later, we defined the temperature in a second way which was completely independent of any substance. From arguments based on the Second Law we defined what we might call the “grand thermodynamic absolute temperature” T , the T that appears in Eq. (45.12). What we proved here is that the pressure of an ideal gas (defined as one for which the internal energy does not depend on the volume) is proportional to the grand thermodynamic absolute temperature. We also know that the pressure is proportional to the temperature measured on the gas scale. Therefore we can deduce that the kinetic temperature is proportional to the “grand thermodynamic absolute temperature.” That means, of course, that if we were sensible we could make two scales agree. In this instance, at least, the two scales *have* been chosen so that they coincide; the proportionality constant has been chosen to be 1. Most of the time man chooses trouble for himself, but in this case he made them equal!

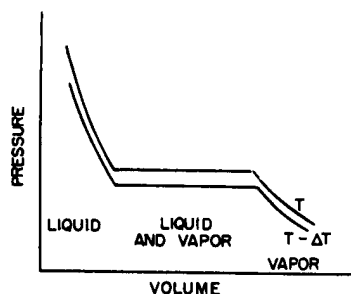


Fig. 45-3. Isothermal lines for a condensable vapor compressed in a cylinder. At the left, the substance is in the liquid phase. At the right, the substance is vaporized. In the center, both liquid and vapor are present in the cylinder.

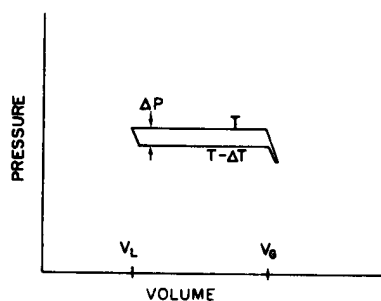


Fig. 45-4. Pressure-volume diagram for a Carnot cycle with a condensable vapor in the cylinder. At the left, the substance is in the liquid state. A quantity of heat L is added at temperature T to vaporize the liquid. The vapor expands adiabatically as T changes to $T - \Delta T$.

45-3 The Clausius-Clapeyron equation

The vaporization of a liquid is another application of the results we have derived. Suppose we have some liquid in a cylinder, such that we can compress it by pushing on the piston, and we ask ourselves, “If we keep the temperature constant, how does the pressure vary with volume?” In other words, we want to draw an isothermal line on the P - V diagram. The substance in the cylinder is not the ideal gas that we considered earlier; now it may be in the liquid or the vapor phase, or both may be present. If we apply sufficient pressure, the substance will condense to a liquid. Now if we squeeze still harder, the volume changes very little, and our isothermal line rises rapidly with decreasing volume, as shown at the left in Fig. 45-3.

If we increase the volume by pulling the piston out, the pressure drops until we reach the point at which the liquid starts to boil, and then vapor starts to form. If we pull the piston out farther, all that happens is that more liquid vaporizes. When there is part liquid and part vapor in the cylinder, the two phases are in equilibrium—liquid is evaporating and vapor is condensing at the same rate. If we make more room for the vapor, more vapor is needed to maintain the pressure, so a little more liquid evaporates, but the pressure remains constant. On the flat part of the curve in Fig. 45-3 the pressure does not change, and the value of the pressure here is called the *vapor pressure at temperature T* . As we continue to increase the volume, there comes a time when there is no more liquid to evaporate. At this juncture, if we expand the volume further, the pressure will fall as for an ordinary gas, as shown at the right of the P - V diagram. The lower curve in Fig. 45-3 is the isothermal line at a slightly lower temperature $T - \Delta T$. The pressure in the liquid phase is slightly reduced because liquid expands with an increase in temperature (for most substances, but not for water near the freezing point) and, of course, the vapor pressure is lower at the lower temperature.

We will now make a cycle out of the two isothermal lines by connecting them (say by adiabatic lines) at the ends of the flat sections, as shown in Fig. 45-4. The little jiggle in the lower right-hand corner of the figure will make little difference and we will neglect it. We are going to use the argument of Carnot, which tells us that the heat added to the substance in changing it from a liquid to a vapor is related to the work done by the substance as it goes around the cycle. Let

us call L the heat needed to vaporize the substance in the cylinder. As in the argument immediately preceding Eq. (45.5), we know that $L(\Delta T/T) = \text{work done by the substance}$. As before, the work done by the substance is the shaded area, which is approximately $\Delta P(V_G - V_L)$, where ΔP is the difference in vapor pressure at the two temperatures T and $T - \Delta T$, V_G is the volume of the gas, and V_L is the volume of the liquid, both volumes measured at the vapor pressure. Setting these two expressions for the area equal, we get $L\Delta T/T = \Delta P(V_G - V_L)$, or

$$\frac{L}{T(V_G - V_L)} = (\partial P_{\text{vap}}/\partial T). \quad (45.14)$$

Equation (45.14) gives the relationship between the rate of change of vapor pressure with temperature and the amount of heat required to evaporate the liquid. This relationship was deduced by Carnot, but it is called the Clausius-Clapeyron equation.

Now let us compare Eq. (45.14) with the results deduced from kinetic theory. Usually V_G is very much larger than V_L . So $V_G - V_L \approx V_G = RT/P$ per mole. If we further assume that L is a constant, independent of temperature—not a very good approximation—then we would have $\partial P/\partial T = L/(RT^2P)$. The solution of this differential equation is

$$P = \text{const } e^{-L/RT}. \quad (45.15)$$

Let us compare this with the pressure variation with temperature that we deduced earlier from kinetic theory. Kinetic theory indicated the possibility, at least roughly, that the number of molecules of vapor above a liquid would be

$$n = \left(\frac{1}{V_A}\right) e^{-(U_G - U_L)/RT}, \quad (45.16)$$

where $U_G - U_L$ is the internal energy per mole in the liquid minus the internal energy per mole in the gas, i.e., the energy needed to vaporize a mole of liquid. Equation (45.15) from thermodynamics and Eq. (45.16) from kinetic theory are very closely related because the pressure is nkT , but they are not exactly the same. However, they will turn out to be exactly the same if we assume $L - U_G = \text{const}$, instead of $L = \text{const}$. If we assume $L - U_G = \text{const}$, independent of temperature, then the argument leading to Eq. (45.15) will produce Eq. (45.16).

This comparison shows the advantages and disadvantages of thermodynamics over kinetic theory: First of all, Eq. (45.14) obtained by thermodynamics is exact, while Eq. (45.16) can only be approximated, for instance, if U is nearly constant, and if the model is right. Second, we may not understand correctly how the gas goes into the liquid; nevertheless, Eq. (45.14) is right, while (45.16) is only approximate. Third, although our treatment applies to a gas condensing into a liquid, the argument is true for any other change of state. For instance, the solid-to-liquid transition has the same kind of curve as that shown in Figs. 45-3 and 45-4. Introducing the latent heat for melting, M/mole , the formula analogous to Eq. (45.14) then is $(\partial P_{\text{melt}}/\partial T)_V = M/[T(V_{\text{liq}} - V_{\text{solid}})]$. Although we may not understand the kinetic theory of the melting process, we nevertheless have a correct equation. However, when we *can* understand the kinetic theory, we have another advantage. Equation (45.14) is only a differential relationship, and we have no way of obtaining the constants of integration. In the kinetic theory we can obtain the constants also if we have a good model that describes the phenomenon completely. So there are advantages and disadvantages to each. When knowledge is weak and the situation is complicated, thermodynamic relations are really the most powerful. When the situation is very simple and a theoretical analysis can be made, then it is better to try to get more information from theoretical analysis.

One more example: blackbody radiation. We have discussed a box containing radiation and nothing else. We have talked about the equilibrium between the oscillator and the radiation. We also found that the photons hitting the wall of the box would exert the pressure P , and we found $PV = U/3$, where U is the

total energy of all the photons and V is the volume of the box. If we substitute $U = 3PV$ in the basic Eq. (45.7), we find

$$\left(\frac{\partial U}{\partial V}\right)_T = 3P = T \left(\frac{\partial P}{\partial T}\right)_V - P. \quad (45.17)$$

Since the volume of our box is constant, we can replace $(\partial P/\partial T)_V$ by dP/dT to obtain an ordinary differential equation we can integrate: $\ln P = 4 \ln T + \text{const}$, or $P = \text{const} \times T^4$. The pressure of radiation varies as the fourth power of the temperature, and the energy content of the radiation, $U/V = P/3$, also varies as T^4 . It is usual to write $U/V = (4\sigma/c)T^4$, where c is the speed of light and σ is a constant. It is not possible to get σ from thermodynamics alone. Here is a good example of its power, and its limitations. To know that U/V goes as T^4 is a great deal, but to know how big U/V actually is at any temperature requires that we go into the kind of detail that only a complete theory can supply. For blackbody radiation we have such a theory and we can derive an expression for the constant σ in the following manner.

Let $I(\omega) d\omega$ be the intensity distribution, the energy flow through 1 m^2 in one second with frequency between ω and $\omega + d\omega$. The energy density distribution = energy/volume = $I(\omega) d\omega/c$ is

$$\begin{aligned} \frac{U}{V} &= \text{total energy density} \\ &= \int_{\omega=0}^{\infty} \text{energy density } \omega \text{ and } \omega + d\omega \\ &= \int_0^{\infty} \frac{I(\omega) d\omega}{c}. \end{aligned}$$

From our earlier discussions, we know that

$$I(\omega) = \frac{\hbar \omega^3}{\pi^2 c^2 (e^{\hbar \omega/kT} - 1)}.$$

Substituting this expression for $I(\omega)$ in our equation for U/V , we get

$$\frac{U}{V} = \frac{1}{\pi^2 c^3} \int_0^{\infty} \frac{\hbar \omega^3 d\omega}{e^{\hbar \omega/kT} - 1}.$$

If we substitute $x = \hbar \omega/kT$, the expression becomes

$$\frac{U}{V} = \frac{(kT)^4}{\hbar^3 \pi^2 c^3} \int_0^{\infty} \frac{x^3 dx}{e^x - 1}.$$

This integral is just some number that we can get, approximately, by drawing a curve and taking the area by counting squares. It is roughly 6.5. The mathematicians among us can show that the integral is exactly $\pi^4/15$.* Comparing this expression with $U/V = (4\sigma/c)T^4$, we find

$$\sigma = \frac{k^4 \pi^2}{60 \hbar^3 c^2} = 5.67 \times 10^{-8} \frac{\text{watts}}{(\text{meter})^2 (\text{degree})^4}.$$

* Since $(e^x - 1)^{-1} = e^{-x} + e^{-2x} + \dots$, the integral is

$$\sum_{n=1}^{\infty} \int_0^{\infty} e^{-nx} x^3 dx.$$

But $\int_0^{\infty} e^{-nx} dx = 1/n$, and differentiating with respect to n three times gives $\int_0^{\infty} x^3 e^{-nx} dx = 6/n^4$, so the integral is $6(1 + \frac{1}{16} + \frac{1}{81} + \dots)$ and a good estimate comes from adding the first few terms. In Chapter 50 we will find a way to show that the sum of the reciprocal fourth powers of the integers is, in fact, $\pi^4/90$.

If we make a small hole in our box, how much energy will flow per second through the hole of unit area? To go from energy density to energy flow, we multiply the energy density U/V by c . We also multiply by $\frac{1}{4}$, which arises as follows: first, a factor of $\frac{1}{2}$, because only the energy which is flowing *out* escapes; and second, another factor $\frac{1}{2}$, because energy which approaches the hole at an angle to the normal is less effective in getting through the hole by a cosine factor. The average value of the cosine is $\frac{1}{2}$. It is clear now why we write $U/V = (4\sigma/c)T^4$: so that we can ultimately say that the flux from a small hole is σT^4 per unit area.

Ratchet and pawl

46-1 How a ratchet works

In this chapter we discuss the ratchet and pawl, a very simple device which allows a shaft to turn only one way. The possibility of having something turn only one way requires some detailed and careful analysis, and there are some very interesting consequences.

The plan of the discussion came about in attempting to devise an elementary explanation, from the molecular or kinetic point of view, for the fact that there is a maximum amount of work which can be extracted from a heat engine. Of course we have seen the essence of Carnot's argument, but it would be nice to find an explanation which is elementary in the sense that we can see what is happening physically. Now, there are complicated mathematical demonstrations which follow from Newton's laws to demonstrate that we can get only a certain amount of work out when heat flows from one place to another, but there is great difficulty in converting this into an elementary demonstration. In short, we do not understand it, although we can follow the mathematics.

In Carnot's argument, the fact that more than a certain amount of work cannot be extracted in going from one temperature to another is deduced from another axiom, which is that if everything is at the same temperature, heat cannot be converted to work by means of a cyclic process. First, let us back up and try to see, in at least one elementary example, why this simpler statement is true.

Let us try to invent a device which will violate the Second Law of Thermodynamics, that is, a gadget which will generate work from a heat reservoir with everything at the same temperature. Let us say we have a box of gas at a certain temperature, and inside there is an axle with vanes in it. (See Fig. 46-1 but take $T_1 = T_2 = T$, say.) Because of the bombardments of gas molecules on the vane, the vane oscillates and jiggles. All we have to do is to hook onto the other end of the axle a wheel which can turn only one way—the ratchet and pawl. Then when the shaft tries to jiggle one way, it will not turn, and when it jiggles the other, it will turn. Then the wheel will slowly turn, and perhaps we might even tie a flea onto a string hanging from a drum on the shaft, and lift the flea! Now let us ask if this is possible. According to Carnot's hypothesis, it is impossible. But if we just look at it, we see, *prima facie*, that it seems quite possible. So we must look more closely. Indeed, if we look at the ratchet and pawl, we see a number of complications.

First, our idealized ratchet is as simple as possible, but even so, there is a pawl, and there must be a spring in the pawl. The pawl must return after coming off a tooth, so the spring is necessary.

Another feature of this ratchet and pawl, not shown in the figure, is quite essential. Suppose the device were made of perfectly elastic parts. After the pawl is lifted off the end of the tooth and is pushed back by the spring, it will bounce against the wheel and continue to bounce. Then, when another fluctuation came, the wheel could turn the other way, because the tooth could get underneath during the moment when the pawl was up! Therefore an essential part of the irreversibility of our wheel is a damping or deadening mechanism which stops the bouncing. When the damping happens, of course, the energy that was in the pawl goes into the wheel and shows up as heat. So, as it turns, the wheel will get hotter and hotter. To make the thing simpler, we can put a gas around the wheel to take up some of the heat. Anyway, let us say the gas keeps rising in temperature, along with the wheel. Will it go on forever? No! The pawl and wheel, both at some temperature

46-1 How a ratchet works

46-2 The ratchet as an engine

46-3 Reversibility in mechanics

46-4 Irreversibility

46-5 Order and entropy

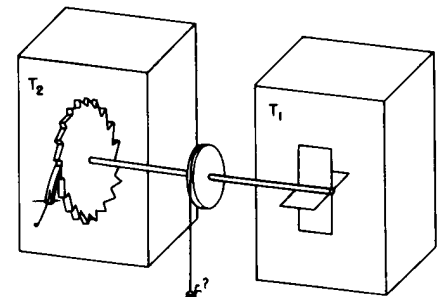


Fig. 46-1. The ratchet and pawl machine.

T , also have Brownian motion. This motion is such that, every once in a while, by accident, the pawl lifts itself up and over a tooth just at the moment when the Brownian motion on the vanes is trying to turn the axle backwards. And as things get hotter, this happens more often.

So, this is the reason this device does not work in perpetual motion. When the vanes get kicked, sometimes the pawl lifts up and goes over the end. But sometimes, when it tries to turn the other way, the pawl has already lifted due to the fluctuations of the motions on the wheel side, and the wheel goes back the other way! The net result is nothing. It is not hard to demonstrate that when the temperature on both sides is equal, there will be no net average motion of the wheel. Of course the wheel will do a lot of jiggling this way and that way, but it will not do what we would like, which is to turn just one way.

Let us look at the reason. It is necessary to do work against the spring in order to lift the pawl to the top of a tooth. Let us call this energy ϵ , and let θ be the angle between the teeth. The chance that the system can accumulate enough energy, ϵ , to get the pawl over the top of the tooth, is $e^{-\epsilon/kT}$. But the probability that the pawl will accidentally be up is also $e^{-\epsilon/kT}$. So the number of times that the pawl is up and the wheel can turn backwards freely is equal to the number of times that we have enough energy to turn it forward when the pawl is down. We thus get a "balance," and the wheel will not go around.

46-2 The ratchet as an engine

Let us now go further. Take the example where the temperature of the vanes is T_1 and the temperature of the wheel, or ratchet, is T_2 , and T_2 is less than T_1 . Because the wheel is cold and the fluctuations of the pawl are relatively infrequent, it will be very hard for the pawl to attain an energy ϵ . Because of the high temperature T_1 , the vanes will often attain the energy ϵ , so our gadget will go in one direction, as designed.

We would now like to see if it can lift weights. Onto the drum in the middle we tie a string, and put a weight, such as our flea, on the string. We let L be the torque due to the weight. If L is not too great, our machine will lift the weight because the Brownian fluctuations make it more likely to move in one direction than the other. We want to find how much weight it can lift, how fast it goes around, and so on.

First we consider a forward motion, the usual way one designs a ratchet to run. In order to make one step forward, how much energy has to be borrowed from the vane end? We must borrow an energy ϵ to lift the pawl. The wheel turns through an angle θ against a torque L , so we also need the energy $L\theta$. The total amount of energy that we have to borrow is thus $\epsilon + L\theta$. The probability that we get this energy is proportional to $e^{-(\epsilon+L\theta)/kT_1}$. Actually, it is not only a question of getting the energy, but we also would like to know the number of times per second it has this energy. The probability per second is proportional to $e^{-(\epsilon+L\theta)/kT_1}$, and we shall call the proportionality constant $1/\tau$. It will cancel out in the end anyway. When a forward step happens, the work done on the weight is $L\theta$. The energy taken from the vane is $\epsilon + L\theta$. The spring gets wound up with energy ϵ , then it goes clatter, clatter, bang, and this energy goes into heat. All the energy taken out goes to lift the weight and to drive the pawl, which then falls back and gives heat to the other side.

Now we look at the opposite case, which is backward motion. What happens here? To get the wheel to go backwards all we have to do is supply the energy to lift the pawl high enough so that the ratchet will slip. This is still energy ϵ . Our probability per second for the pawl to lift this high is now $(1/\tau)e^{-\epsilon/kT_2}$. Our proportionality constant is the same, but this time kT_2 shows up because of the different temperature. When this happens, the work is released because the wheel slips backward. It loses one notch, so it releases work $L\theta$. The energy taken from the ratchet system is ϵ , and the energy given to the gas at T_1 on the vane side is $L\theta + \epsilon$. It takes a little thinking to see the reason for that. Suppose the pawl has lifted itself up accidentally by a fluctuation. Then when it falls back and the spring

Table 46-1

Summary of operation of ratchet and pawl.

<i>Forward:</i>	Need energy	$\epsilon + L\theta$	from vane. $\therefore$ Rate = $\frac{1}{\tau} e^{-(L\theta + \epsilon)/kT_1}$
	Takes from vane	$L\theta + \epsilon$	
	Does work	$L\theta$	
	Gives to ratchet	ϵ	
<i>Backward:</i>	Needs energy	ϵ	for pawl. $\therefore$ Rate = $\frac{1}{\tau} e^{-\epsilon/kT_2}$
	Takes from ratchet	ϵ	} same as above with sign reversed.
	Releases work	$L\theta$	
	Gives to vane	$L\theta + \epsilon$	
If system is reversible, rates are equal, hence $\frac{\epsilon + L\theta}{T_1} = \frac{\epsilon}{T_2}$.			
$\frac{\text{Heat to ratchet}}{\text{Heat from vane}} = \frac{\epsilon}{L\theta + \epsilon}$.		Hence $\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$.	

pushes it down against the tooth, there is a force trying to turn the wheel, because the tooth is pushing on an inclined plane. This force is doing work, and so is the force due to the weights. So both together make up the total force, and all the energy which is slowly released appears at the vane end as heat. (Of course it must, by conservation of energy, but one must be careful to think the thing through!) We notice that all these energies are exactly the same, but reversed. So, depending upon which of these two rates is greater, the weight is either slowly lifted or slowly released. Of course, it is constantly jiggling around, going up for a while and down for a while, but we are talking about the average behavior.

Suppose that for a particular weight the rates happen to be equal. Then we add an infinitesimal weight to the string. The weight will slowly go down, and work will be done on the machine. Energy will be taken from the wheel and given to the vanes. If instead we take off a little bit of weight, then the imbalance is the other way. The weight is lifted, and heat is taken from the vane and put into the wheel. So we have the conditions of Carnot's reversible cycle, provided that the weight is just such that these two are equal. This condition is evidently that $(\epsilon + L\theta)/T_1 = \epsilon/T_2$. Let us say that the machine is slowly lifting the weight. Energy Q_1 is taken from the vanes and energy Q_2 is delivered to the wheel, and these energies are in the ratio $(\epsilon + L\theta)/\epsilon$. If we are lowering the weight, we also have $Q_1/Q_2 = (\epsilon + L\theta)/\epsilon$. Thus (Table 46-1) we have

$$Q_1/Q_2 = T_1/T_2.$$

Furthermore, the work we get out is to the energy taken from the vane as $L\theta$ is to $L\theta + \epsilon$, hence as $(T_1 - T_2)/T_1$. We see that our device cannot extract more work than this, operating reversibly. This is the result that we expected from Carnot's argument, and the main result of this lecture. However, we can use our device to understand a few other phenomena, even out of equilibrium, and therefore beyond the range of thermodynamics.

Let us now calculate *how fast* our one-way device would turn if everything were at the same temperature and we hung a weight on the drum. If we pull very, very hard, of course, there are all kinds of complications. The pawl slips over the ratchet, or the spring breaks, or something. But suppose we pull gently enough that everything works nicely. In those circumstances, the above analysis is right for the probability of the wheel going forward and backward, if we remember

that the two temperatures are equal. In each step an angle θ is obtained, so the angular velocity is θ times the probability of one of these jumps per second. It goes forward with probability $(1/\tau)e^{-(\epsilon+L\theta)/kT}$ and backward with probability $(1/\tau)e^{-\epsilon/kT}$, so that for the angular velocity we have

$$\begin{aligned}\omega &= (\theta/\tau)e^{-(\epsilon+L\theta)/kT} - e^{-\epsilon/kT} \\ &= (\theta/\tau)e^{-\epsilon/kT}(e^{-L\theta/kT} - 1).\end{aligned}\quad (46.1)$$

If we plot ω against L , we get the curve shown in Fig. 46-2. We see that it makes a great difference whether L is positive or negative. If L increases in the positive range, which happens when we try to drive the wheel backward, the backward velocity approaches a constant. As L becomes negative, ω really "takes off" forward, since e to a tremendous power is very great!

The angular velocity that was obtained from different forces is thus very unsymmetrical. Going one way it is easy: we get a lot of angular velocity for a little force. Going the other way, we can put on a lot of force, and yet the wheel hardly goes around.

We find the same thing in an *electrical rectifier*. Instead of the force, we have the electric field, and instead of the angular velocity, we have the electric current. In the case of a rectifier, the voltage is not proportional to resistance, and the situation is unsymmetrical. The same analysis that we made for the mechanical rectifier will also work for an electrical rectifier. In fact, the kind of formula we obtained above is typical of the current-carrying capacities of rectifiers as a function of their voltages.

Now let us take all the weights away, and look at the original machine. If T_2 were less than T_1 , the ratchet would go forward, as anybody would believe. But what is hard to believe, at first sight, is the opposite. If T_2 is greater than T_1 , the ratchet goes around the opposite way! A dynamic ratchet with lots of heat in it runs itself backwards, because the ratchet pawl is bouncing. If the pawl, for a moment, is on the incline somewhere, it pushes the inclined plane sideways. But it is *always* pushing on an inclined plane, because if it happens to lift up high enough to get past the point of a tooth, then the inclined plane slides by, and it comes down again on an inclined plane. So a hot ratchet and pawl is ideally built to go around in a direction exactly opposite to that for which it was originally designed!

In spite of all our cleverness of lopsided design, if the two temperatures are exactly equal there is no more propensity to turn one way than the other. The moment we look at it, it may be turning one way or the other, but in the long run, it gets nowhere. The fact that it gets nowhere is really the fundamental deep principle on which all of thermodynamics is based.

46-3 Reversibility in mechanics

What deeper mechanical principle tells us that, in the long run, if the temperature is kept the same everywhere, our gadget will turn neither to the right nor to the left? We evidently have a fundamental proposition that there is no way to design a machine which, left to itself, will be more likely to be turning one way than the other after a long enough time. We must try to see how this follows from the laws of mechanics.

The laws of mechanics go something like this: the mass times the acceleration is the force, and the force on each particle is some complicated function of the positions of all the other particles. There are other situations in which forces depend on velocity, such as in magnetism, but let us not consider that now. We take a simpler case, such as gravity, where forces depend only on position. Now suppose that we have solved our set of equations and we have a certain motion $x(t)$ for each particle. In a complicated enough system, the solutions are very complicated, and what happens with time turns out to be very surprising. If we write down any arrangement we please for the particles, we will see this arrangement actually occur if we wait long enough! If we follow our solution for a long

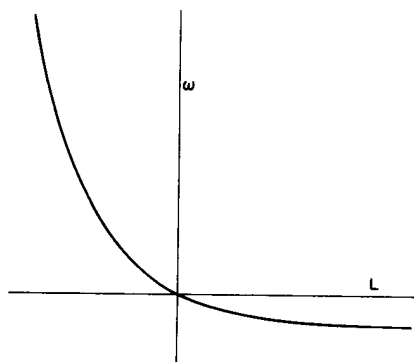


Fig. 46-2. Angular velocity of the ratchet as a function of torque.

enough time, it tries everything that it can do, so to speak. This is not absolutely necessary in the simplest devices, but when systems get complicated enough, with enough atoms, it happens. Now there is something else the solution can do. If we solve the equations of motion, we may get certain functions such as $t + t^2 + t^3$. We claim that another solution would be $-t + t^2 - t^3$. In other words, if we substitute $-t$ everywhere for t throughout the entire solution, we will once again get a solution of the same equation. This follows from the fact that if we substitute $-t$ for t in the original differential equation, nothing is changed, since only second derivatives with respect to t appear. This means that if we have a certain motion, then the exact opposite motion is also possible. In the complete confusion which comes if we wait long enough, it finds itself going one way sometimes, and it finds itself going the other way sometimes. There is nothing more beautiful about one of the motions than about the other. So it is impossible to design a machine which, in the long run, is more likely to be going one way than the other, if the machine is sufficiently complicated.

One might think up an example for which this is obviously untrue. If we take a wheel, for instance, and spin it in empty space, it will go the same way forever. So there are some conditions, like the conservation of angular momentum, which violate the above argument. This just requires that the argument be made with a little more care. Perhaps the walls take up the angular momentum, or something like that, so that we have no special conservation laws. Then, if the system is complicated enough, the argument is true. It is based on the fact that the laws of mechanics are reversible.

For historical interest, we would like to remark on a device invented by Maxwell, who first worked out the dynamical theory of gases. He supposed the following situation: We have two boxes of gas at the same temperature, with a little hole between them. At the hole sits a little demon (who may be a machine of course!). There is a door on the hole, which can be opened or closed by the demon. He watches the molecules coming from the left. Whenever he sees a fast molecule, he opens the door. When he sees a slow one, he leaves it closed. If we want him to be an extra special demon, he can have eyes at the back of his head, and do the opposite to the molecules from the other side. He lets the slow ones through to the left, and the fast through to the right. Pretty soon the left side will get cold and the right side hot. Then, are the ideas of thermodynamics violated because we could have such a demon?

It turns out, if we build a finite-sized demon, that the demon himself gets so warm that he cannot see very well after a while. The simplest possible demon, as an example, would be a trap door held over the hole by a spring. A fast molecule comes through, because it is able to lift the trap door. The slow molecule cannot get through, and bounces back. But this thing is nothing but our ratchet and pawl in another form, and ultimately the mechanism will heat up. If we assume that the specific heat of the demon is not infinite, it must heat up. It has but a finite number of internal gears and wheels, so it cannot get rid of the extra heat that it gets from observing the molecules. Soon it is shaking from Brownian motion so much that it cannot tell whether it is coming or going, much less whether the molecules are coming or going, so it does not work.

46-4 Irreversibility

Are all the laws of physics reversible? Evidently not! Just try to unscramble an egg! Run a moving picture backwards, and it takes only a few minutes for everybody to start laughing. The most natural characteristic of all phenomena is their obvious irreversibility.

Where does irreversibility come from? It does not come from Newton's laws. If we claim that the behavior of everything is ultimately to be understood in terms of the laws of physics, and if it also turns out that all the equations have the fantastic property that if we put $t = -t$ we have another solution, then every phenomenon is reversible. How then does it come about in nature on a large scale that things are not reversible? Obviously there must be some law, some

obscure but fundamental equation, perhaps in electricity, maybe in neutrino physics, in which it *does* matter which way time goes.

Let us discuss that question now. We already know one of those laws, which says that the entropy is always increasing. If we have a hot thing and a cold thing, the heat goes from hot to cold. So the law of entropy is one such law. But we expect to understand the law of entropy from the point of view of mechanics. In fact, we have just been successful in obtaining all the consequences of the argument that heat cannot flow backwards by itself from just mechanical arguments, and we thereby obtained an understanding of the Second Law. Apparently we can get irreversibility from reversible equations. But *was* it *only* a mechanical argument that we used? Let us look into it more closely.

Since our question has to do with the entropy, our problem is to try to find a microscopic description of entropy. If we say we have a certain amount of energy in something, like a gas, then we can get a microscopic picture of it, and say that every atom has a certain energy. All these energies added together give us the total energy. Similarly, maybe every atom has a certain entropy. If we add everything up, we would have the total entropy. It does not work so well, but let us see what happens.

As an example, we calculate the entropy difference between a gas at a certain temperature at one volume, and a gas at the same temperature at another volume. We remember, from Chapter 44, that we had, for the change in entropy,

$$\Delta S = \int \frac{dQ}{T}.$$

In the present case, the energy of the gas is the same before and after expansion, since the temperature does not change. So we have to add enough heat to equal the work done by the gas or, for each little change in volume,

$$dQ = P dV.$$

Putting this in for dQ , we get

$$\begin{aligned} \Delta S &= \int_{V_1}^{V_2} P \frac{dV}{T} = \int_{V_1}^{V_2} \frac{NkT}{V} \frac{dV}{T} \\ &= Nk \ln \frac{V_2}{V_1}, \end{aligned}$$

as we obtained in Chapter 44. For instance, if we expand the volume by a factor of 2, the entropy change is $Nk \ln 2$.

Let us now consider another interesting example. Suppose we have a box with a barrier in the middle. On one side is neon ("black" molecules), and on the other, argon ("white" molecules). Now we take out the barrier, and let them mix. How much has the entropy changed? It is possible to imagine that instead of the barrier we have a piston, with holes in it that let the whites through but not the blacks, and another kind of piston which is the other way around. If we move one piston to each end, we see that, for each gas, the problem is like the one we just solved. So we get an entropy change of $Nk \ln 2$, which means that the entropy has increased by $k \ln 2$ per molecule. The 2 has to do with the extra room that the molecule has, which is rather peculiar. It is not a property of the molecule itself, but of *how much room* the molecule has to run around in. This is a strange situation, where entropy increases but where everything has the same temperature and the same energy! The only thing that is changed is that the molecules are distributed differently.

We well know that if we just pull the barrier out, everything will get mixed up after a long time, due to the collisions, the jiggling, the banging, and so on. Every once in a while a white molecule goes toward a black, and a black one goes toward a white, and maybe they pass. Gradually the whites worm their way, by accident, across into the space of blacks, and the blacks worm their way, by accident, into the space of whites. If we wait long enough we get a mixture.

Clearly, this is an irreversible process in the real world, and ought to involve an increase in the entropy.

Here we have a simple example of an irreversible process which is completely composed of reversible events. Every time there is a collision between any two molecules, they go off in certain directions. If we took a moving picture of a collision in reverse, there would be nothing wrong with the picture. In fact, one kind of collision is just as likely as another. So the mixing is completely reversible, and yet it is irreversible. Everyone knows that if we started with white and with black, separated, we would get a mixture within a few minutes. If we sat and looked at it for several more minutes, it would not separate again but would stay mixed. So we have an irreversibility which is based on reversible situations. But we also see the *reason* now. We started with an arrangement which is, in some sense, *ordered*. Due to the chaos of the collisions, it becomes disordered. *It is the change from an ordered arrangement to a disordered arrangement which is the source of the irreversibility.*

It is true that if we took a motion picture of this, and showed it backwards, we would see it gradually become ordered. Someone would say, "That is against the laws of physics!" So we would run the film over again, and we would look at every collision. Every one would be perfect, and every one would be obeying the laws of physics. The reason, of course, is that every molecule's velocities are just right, so if the paths are all followed back, they get back to their original condition. But that is a very unlikely circumstance to have. If we start with the gas in no special arrangement, just whites and blacks, it will never get back.

46-5 Order and entropy

So we now have to talk about what we mean by disorder and what we mean by order. It is not a question of pleasant order or unpleasant disorder. What is different in our mixed and unmixed cases is the following. Suppose we divide the space into little volume elements. If we have white and black molecules, how many ways could we distribute them among the volume elements so that white is on one side, and black on the other? On the other hand, how many ways could we distribute them with no restriction on which goes where? Clearly, there are many more ways to arrange them in the latter case. We measure "disorder" by the number of ways that the insides can be arranged, so that from the outside it looks the same. *The logarithm of that number of ways is the entropy.* The number of ways in the separated case is less, so the entropy is less, or the "disorder" is less.

So with the above technical definition of disorder we can understand the proposition. First, the entropy measures the disorder. Second, the universe always goes from "order" to "disorder," so entropy always increases. Order is not order in the sense that we like the arrangement, but in the sense that the number of different ways we can hook it up, and still have it look the same from the outside, is relatively restricted. In the case where we reversed our motion picture of the gas mixing, there was not as much disorder as we thought. Every single atom had exactly the correct speed and direction to come out right! The entropy was not high after all, even though it appeared so.

What about the reversibility of the other physical laws? When we talked about the electric field which comes from an accelerating charge, it was said that we must take the retarded field. At a time t and at a distance r from the charge, we take the field due to the acceleration at a time $t - r/c$, not $t + r/c$. So it looks, at first, as if the law of electricity is not reversible. Very strangely, however, the laws we used come from a set of equations called Maxwell's equations, which are, in fact, reversible. Furthermore, it is possible to argue that if we were to use only the advanced field, the field due to the state of affairs at $t + r/c$, and do it absolutely consistently in a completely enclosed space, everything happens exactly the same way as if we use retarded fields! This apparent irreversibility in electricity, at least in an enclosure, is thus not an irreversibility at all. We have some feeling for that already, because we know that when we have an oscillating charge which generates fields which are bounced from the walls of an enclosure we ultimately

get to an equilibrium in which there is no one-sidedness. The retarded field approach is only a convenience in the method of solution.

So far as we know, all the fundamental laws of physics, like Newton's equations, are reversible. Then where does irreversibility come from? It comes from order going to disorder, but we do not understand this until we know the origin of the order. Why is it that the situations we find ourselves in every day are always out of equilibrium? One possible explanation is the following. Look again at our box of mixed white and black molecules. Now it is possible, if we wait long enough, by sheer, grossly improbable, but possible, accident, that the distribution of molecules gets to be mostly white on one side and mostly black on the other. After that, as times goes on and accidents continue, they get more mixed up again.

Thus one possible explanation of the high degree of order in the present-day world is that it is just a question of luck. Perhaps our universe happened to have had a fluctuation of some kind in the past, in which things got somewhat separated, and now they are running back together again. This kind of theory is not unsymmetrical, because we can ask what the separated gas looks like either a little in the future or a little in the past. In either case, we see a grey smear at the interface, because the molecules are mixing again. No matter which way we run time, the gas mixes. So this theory would say the irreversibility is just one of the accidents of life.

We would like to argue that this is not the case. Suppose we do not look at the whole box at once, but only at a piece of the box. Then, at a certain moment, suppose we discover a certain amount of order. In this little piece, white and black are separate. What should we deduce about the condition in places where we have not yet looked? If we really believe that the order arose from complete disorder by a fluctuation, we must surely take the most likely fluctuation which could produce it, and the most likely condition is *not* that the rest of it has also become disentangled! Therefore, from the hypothesis that the world is a fluctuation, all of the predictions are that if we look at a part of the world we have never seen before, we will find it mixed up, and not like the piece we just looked at. If our order were due to a fluctuation, we would not expect order anywhere but where we have just noticed it.

Now we assume the separation is because the past of the universe was really ordered. It is not due to a fluctuation, but the whole thing used to be white and black. This theory now predicts that there will be order in other places—the order is not due to a fluctuation, but due to a much higher ordering at the beginning of time. Then we would expect to find order in places where we have not yet looked.

The astronomers, for example, have only looked at some of the stars. Every day they turn their telescopes to other stars, and the new stars are doing the same thing as the other stars. We therefore conclude that the universe is *not* a fluctuation, and that the order is a memory of conditions when things started. This is not to say that we understand the logic of it. For some reason, the universe at one time had a very low entropy for its energy content, and since then the entropy has increased. So that is the way toward the future. That is the origin of all irreversibility, that is what makes the processes of growth and decay, that makes us remember the past and not the future, remember the things which are closer to that moment in the history of the universe when the order was higher than now, and why we are not able to remember things where the disorder is higher than now, which we call the future. So, as we commented in an earlier chapter, the entire universe is in a glass of wine, if we look at it closely enough. In this case the glass of wine is complex, because there is water and glass and light and everything else.

Another delight of our subject of physics is that even simple and idealized things, like the ratchet and pawl, work only because they are part of the universe. The ratchet and pawl works in only one direction because it has some ultimate contact with the rest of the universe. If the ratchet and pawl were in a box and isolated for some sufficient time, the wheel would be no more likely to go one way than the other. But because we pull up the shades and let the light out, because

we cool off on the earth and get heat from the sun, the ratchets and pawls that we make can turn one way. This one-wayness is interrelated with the fact that the ratchet is part of the universe. It is part of the universe not only in the sense that it obeys the physical laws of the universe, but its one-way behavior is tied to the one-way behavior of the entire universe. It cannot be completely understood until the mystery of the beginnings of the history of the universe are reduced still further from speculation to scientific understanding.

Sound. The wave equation

47-1 Waves

In this chapter we shall discuss the phenomenon of *waves*. This is a phenomenon which appears in many contexts throughout physics, and therefore our attention should be concentrated on it not only because of the particular example considered here, which is sound, but also because of the much wider application of the ideas in all branches of physics.

It was pointed out when we studied the harmonic oscillator that there are not only mechanical examples of oscillating systems but electrical ones as well. Waves are related to oscillating systems, except that wave oscillations appear not only as time-oscillations at one place, but propagate in space as well.

We have really already studied waves. When we studied light, in learning about the properties of waves in that subject, we paid particular attention to the interference in space of waves from several sources at different locations and all at the same frequency. There are two important wave phenomena that we have not yet discussed which occur in light, i.e., electromagnetic waves, as well as in any other form of waves. The first of these is the phenomenon of *interference in time* rather than interference in space. If we have two sources of sound which have slightly different frequencies and if we listen to both at the same time, then sometimes the waves come with the crests together and sometimes with the crest and trough together (see Fig. 47-1). The rising and falling of the sound that results is the phenomenon of *beats* or, in other words, of interference in time. The second phenomenon involves the wave patterns which result when the waves are confined within a given volume and reflect back and forth from walls.

These effects could have been discussed, of course, for the case of electromagnetic waves. The reason for not having done this is that by using one example we would not generate the feeling that we are actually learning about many different subjects at the same time. In order to emphasize the general applicability of waves beyond electrodynamics, we consider here a different example, in particular sound waves.

Other examples of waves are water waves consisting of long swells that we see coming in to the shore, or the smaller water waves consisting of surface tension ripples. As another example, there are two kinds of elastic waves in solids; a compressional (or longitudinal) wave in which the particles of the solid oscillate back and forth along the direction of propagation of the wave (sound waves in a gas are of this kind), and a transverse wave in which the particles of the solid oscillate in a direction perpendicular to the direction of propagation. Earthquake waves contain elastic waves of both kinds, generated by a motion at some place in the earth's crust.

Still another example of waves is found in modern physics. These are waves which give the probability amplitude of finding a particle at a given place—the “matter waves” which we have already discussed. Their frequency is proportional to the energy and their wave number is proportional to the momentum. They are the waves of quantum mechanics.

In this chapter we shall consider only waves for which the velocity is independent of the wavelength. This is, for example, the case for light in a vacuum. The speed of light is then the same for radiowaves, blue light, green light, or for any other wavelength. Because of this behavior, when we began to describe the wave phenomenon we did not notice at first that we had wave propagation. Instead, we said that if a charge is moved at one place, the electric field at a distance x was

47-1 Waves

47-2 The propagation of sound

47-3 The wave equation

47-4 Solutions of the wave equation

47-5 The speed of sound

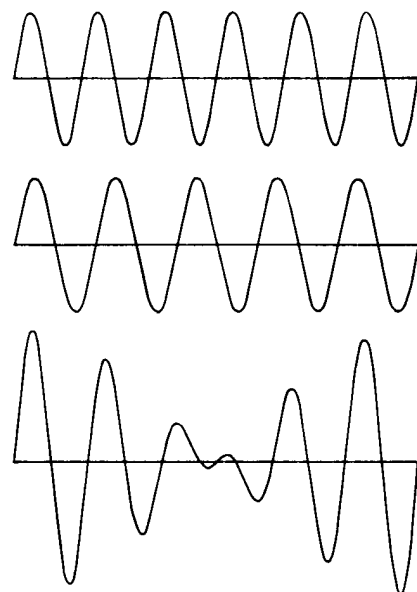


Fig. 47-1. Interference in time of two sound sources with slightly different frequencies, resulting in beats.

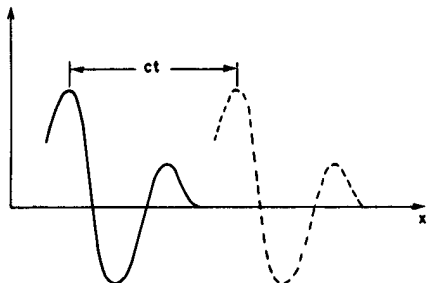


Fig. 47-2. The solid curve shows what the electric field might be like at some instant of time and the dashed curve shows what the electric field is at a time t later.

proportional to the acceleration, not at the time t , but at the earlier time $t - x/c$. Therefore if we were to picture the electric field in space at some instant of time, as in Fig. 47-2, the electric field at a time t later would have moved the distance ct , as indicated in the figure. Mathematically, we can say that in the one-dimensional example we are taking, the electric field is a function of $x - ct$. We see that at $t = 0$, it is some function of x . If we consider a later time, we need only to increase x somewhat to get the same value of the electric field. For example, if the maximum field occurred at $x = 3$ at time zero, then to find the new position of the maximum field at time t we need

$$x - ct = 3 \quad \text{or} \quad x = 3 + ct.$$

We see that this kind of function represents the propagation of a wave.

Such a function, $f(x - ct)$, then represents a wave. We may summarize this description of a wave by saying simply that

$$f(x - ct) = f(x + \Delta x - c(t + \Delta t)),$$

when $\Delta x = c \Delta t$. There is, of course, another possibility, i.e., that instead of a source to the left as indicated in Fig. 47-2, we have a source on the right, so that the wave propagates toward negative x . Then the wave would be described by $g(x + ct)$.

There is the additional possibility that more than one wave exists in space at the same time, and so the electric field is the sum of the two fields, each one propagating independently. This behavior of electric fields may be described by saying that if $f_1(x - ct)$ is a wave, and if $f_2(x - ct)$ is another wave, then their sum is also a wave. This is called the principle of superposition. The same principle is valid in sound.

We are familiar with the fact that if a sound is produced, we hear with complete fidelity the same sequence of sounds as was generated. If we had high frequencies travelling faster than low frequencies, a short, sharp noise would be heard as a succession of musical sounds. Similarly, if red light travelled faster than blue light, a flash of white light would be seen first as red, then as white, and finally as blue. We are familiar with the fact that this is not the case. Both sound and light travel with a speed in air which is very nearly independent of frequency. Examples of wave propagation for which this independence is not true will be considered in Chapter 48.

In the case of light (electromagnetic waves) we gave a rule which determined the electric field at a point as a result of the acceleration of a charge. One might expect now that what we should do is give a rule whereby some quality of the air, say the pressure, is determined at a given distance from a source in terms of the source motion, delayed by the travel time of the sound. In the case of light this procedure was acceptable because all that we knew was that a charge at one place exerts a force on another charge at another place. The details of propagation from the one place to the other were not absolutely essential. In the case of sound, however, we know that it propagates through the air between the source and the hearer, and it is certainly a natural question to ask what, at any given moment, the pressure of the air is. We would like, in addition, to know exactly how the air moves. In the case of electricity we could accept a rule, since we could say that we do not yet know the laws of electricity, but we cannot make the same remark with regard to sound. We would not be satisfied with a rule stating how the sound pressure moves through the air, because the process ought to be understandable as a consequence of the laws of mechanics. In short, sound is a branch of mechanics, and so it is to be understood in terms of Newton's laws. The propagation of sound from one place to another is merely a consequence of mechanics and the properties of gases, if it propagates in a gas, or of the properties of liquids or solids, if it propagates through such mediums. Later we shall derive the properties of light and its wave propagation in a similar way from the laws of electrodynamics.

47-2 The propagation of sound

We shall give a derivation of the properties of the propagation of sound *between* the source and the receiver as a consequence of Newton's laws, and we shall not consider the interaction with the source and the receiver. Ordinarily we emphasize a result rather than a particular derivation of it. In this chapter we take the opposite view. The point here, in a certain sense, is the derivation itself. This problem of explaining new phenomena in terms of old ones, when we know the laws of the old ones, is perhaps the greatest art of mathematical physics. The mathematical physicist has two problems: one is to find solutions, given the equations, and the other is to find the equations which describe a new phenomenon. The derivation here is an example of the second kind of problem.

We shall take the simplest example here—the propagation of sound in one dimension. To carry out such a derivation it is necessary first to have some kind of understanding of what is going on. Fundamentally what is involved is that if an object is moved at one place in the air, we observe that there is a disturbance which travels through the air. If we ask what kind of disturbance, we would say that we would expect that the motion of the object produces a change of pressure. Of course, if the object is moved gently, the air merely flows around it, but what we are concerned with is a rapid motion, so that there is not sufficient time for such a flow. Then, with the motion, the air is compressed and a change of pressure is produced which pushes on additional air. This air is in turn compressed, which leads again to an extra pressure, and a wave is propagated.

We now want to formulate such a process. We have to decide what variables we need. In our particular problem we would need to know how much the air has moved, so that the air *displacement* in the sound wave is certainly one relevant variable. In addition we would like to describe how the air *density* changes as it is displaced. The air *pressure* also changes, so this is another variable of interest. Then, of course, the air has a *velocity*, so that we shall have to describe the velocity of the air particles. The air particles also have *accelerations*—but as we list these many variables we soon realize that the velocity and acceleration would be known if we knew how the air *displacement* varies with time.

As we said, we shall consider the wave in one dimension. We can do this if we are sufficiently far from the source that what we call the *wavefronts* are very nearly planes. We thus make our argument simpler by taking the least complicated example. We shall then be able to say that the displacement, χ , depends only on x and t , and not on y and z . Therefore the description of the air is given by $\chi(x, t)$.

Is this description complete? It would appear to be far from complete, for we know none of the details of how the air molecules are moving. They are moving in all directions, and this state of affairs is certainly not described by means of this function $\chi(x, t)$. From the point of view of kinetic theory, if we have a higher density of molecules at one place and a lower density adjacent to that place, the molecules would move away from the region of higher density to the one of lower density, so as to equalize this difference. Apparently we would not get an oscillation and there would be no sound. What is necessary to get the sound wave is this situation: as the molecules rush out of the region of higher density and higher pressure, they give momentum to the molecules in the adjacent region of lower density. For sound to be generated, the regions over which the density and pressure change must be much larger than the distance the molecules travel before colliding with other molecules. This distance is the mean free path, and the distance between pressure crests and troughs must be much larger than this. Otherwise the molecules would move freely from the crest to the trough and immediately smear out the wave.

It is clear that we are going to describe the gas behavior on a scale large compared with the mean free path, and so the properties of the gas will not be described in terms of the individual molecules. The displacement, for example, will be the displacement of the center of mass of a small element of the gas, and the pressure or density will be the pressure or density in this region. We shall call the pressure P and the density ρ , and they will be functions of x and t . We must keep in mind that this description is an approximation which is valid only when these gas properties do not vary too rapidly with distance.

47-3 The wave equation

The physics of the phenomenon of sound waves thus involves three features:

- I. The gas moves and changes the density.
- II. The change in density corresponds to a change in pressure.
- III. Pressure inequalities generate gas motion.

Let us consider II first. For a gas, a liquid, or a solid, the pressure is some function of the density. Before the sound wave arrives, we have equilibrium, with a pressure P_0 and a corresponding density ρ_0 . A pressure P in the medium is connected to the density by some characteristic relation $P = f(\rho)$ and, in particular, the equilibrium pressure P_0 is given by $P_0 = f(\rho_0)$. The changes of pressure in sound from the equilibrium value are extremely small. A convenient unit for measuring pressure is the *bar*, where $1 \text{ bar} = 10^5 \text{ n/m}^2$. The pressure of 1 standard atmosphere is very nearly 1 bar: $1 \text{ atm} = 1.0133 \text{ bars}$. In sound we use a logarithmic scale of intensities since the sensitivity of the ear is roughly logarithmic. This scale is the decibel scale, in which the acoustic pressure level for the pressure amplitude P is defined as

$$I \text{ (acoustic pressure level)} = 20 \log_{10}(P/P_{\text{ref}}) \text{ in db}, \quad (47.1)$$

where the reference pressure $P_{\text{ref}} = 2 \times 10^{-10} \text{ bar}$. A pressure amplitude of $P = 10^3 P_{\text{ref}} = 2 \times 10^{-7} \text{ bar}^*$ corresponds to a moderately intense sound of 60 decibels. We see that the pressure changes in sound are extremely small compared with the equilibrium, or mean, pressure of 1 atm. The displacements and the density changes are correspondingly extremely small. In explosions we do not have such small changes; the excess pressures produced can be greater than 1 atm. These large pressure changes lead to new effects which we shall consider later. In sound we do not often consider acoustic intensity levels over 100 db; 120 db is a level which is painful to the ear. Therefore, for sound, if we write

$$P = P_0 + P_e, \quad \rho = \rho_0 + \rho_e, \quad (47.2)$$

we shall always have the pressure change P_e very small compared with P_0 and the density change ρ_e very small compared with ρ_0 . Then

$$P_0 + P_e = f(\rho_0 + \rho_e) = f(\rho_0) + \rho_e f'(\rho_0), \quad (47.3)$$

where $P_0 = f(\rho_0)$ and $f'(\rho_0)$ stands for the derivative of $f(\rho)$ evaluated at $\rho = \rho_0$. We can take the second step in this equality only because ρ_e is very small. We find in this way that the excess pressure P_e is proportional to the excess density ρ_e , and we may call the proportionality factor κ :

$$P_e = \kappa \rho_e, \quad \text{where } \kappa = f'(\rho_0) = (dP/d\rho)_0. \quad (\text{II}) \quad (47.4)$$

The relation we needed for II is this very simple one.

Let us now consider I. We shall suppose that the position of a portion of air undisturbed by the sound wave is x and the displacement at the time t due to the sound is $\chi(x, t)$, so that its new position is $x + \chi(x, t)$, as in Fig. 47-3. Now the undisturbed position of a nearby portion of air is $x + \Delta x$, and its new position is $x + \Delta x + \chi(x + \Delta x, t)$. We can now find the density changes in the following way. Since we are limiting ourselves to plane waves, we can take a unit area perpendicular to the x -direction, which is the direction of propagation of the sound wave. The amount of air, per unit area, in Δx is then $\rho_0 \Delta x$, where ρ_0 is the undisturbed, or equilibrium, air density. This air, when displaced by the sound wave, now lies between $x + \chi(x, t)$ and $x + \Delta x + \chi(x + \Delta x, t)$, so that we have the same matter in this interval that was in Δx when undisturbed. If ρ is the new density, then

$$\rho_0 \Delta x = \rho [x + \Delta x + \chi(x + \Delta x, t) - x - \chi(x, t)]. \quad (47.5)$$

* With this choice of P_{ref} , the P is not the peak pressure in the sound wave but the "root-mean-square" pressure, which is $1/(2)^{1/2}$ times the peak pressure.

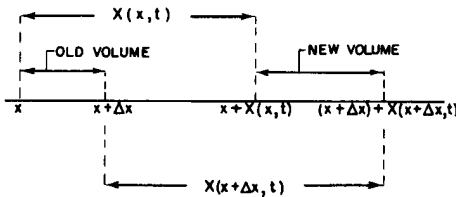


Fig. 47-3. The displacement of the air at x is $\chi(x, t)$, and at $x + \Delta x$ it is $\chi(x + \Delta x, t)$. The original volume of the air for a unit area of the plane wave is Δx ; the new volume is $\Delta x + \chi(x + \Delta x, t) - \chi(x, t)$.

Since Δx is small, we can write $\chi(x + \Delta x, t) - \chi(x, t) = (\partial\chi/\partial x) \Delta x$. This derivative is a partial derivative, since χ depends on the time as well as on x . Our equation then is

$$\rho_0 \Delta x = \rho \left(\frac{\partial\chi}{\partial x} \Delta x + \Delta x \right) \quad (47.6)$$

or

$$\rho_0 = (\rho_0 + \rho_e) \frac{\partial\chi}{\partial x} + \rho_0 + \rho_e. \quad (47.7)$$

Now in sound waves all changes are small so that ρ_e is small, χ is small, and $\partial\chi/\partial x$ is also small. Therefore in the relation that we have just found,

$$\rho_e = -\rho_0 \frac{\partial\chi}{\partial x} - \rho_e \frac{\partial\chi}{\partial x}, \quad (47.8)$$

we can neglect $\rho_e \partial\chi/\partial x$ compared with $\rho_0 \partial\chi/\partial x$. Thus we get the relation we needed for I:

$$\rho_e = -\rho_0 \frac{\partial\chi}{\partial x}. \quad (I) \quad (47.9)$$

This equation is what we would expect physically. If the displacements vary with x , then there will be density changes. The sign is also right: if the displacement χ increases with x , so that the air is stretched out, the density must go down.

We now need the third equation, which is the equation of the motion produced by the pressure. If we know the relation between the force and the pressure, we can then get the equation of motion. If we take a thin slab of air of length Δx and of unit area perpendicular to x , then the mass of air in this slab is $\rho_0 \Delta x$ and it has the acceleration $\partial^2\chi/\partial t^2$, so the mass times the acceleration for this slab of matter is $\rho_0 \Delta x (\partial^2\chi/\partial t^2)$. (It makes no difference for small Δx whether the acceleration $\partial^2\chi/\partial t^2$ is evaluated at an edge of the slab or at some intermediate position.) If now we find the force on this matter for a unit area perpendicular to x , it will then be equal to $\rho_0 \Delta x (\partial^2\chi/\partial t^2)$. We have the force in the $+x$ -direction, at x , of amount $P(x, t)$ per unit area, and we have the force in the opposite direction, at $x + \Delta x$, of amount $P(x + \Delta x, t)$ per unit area (Fig. 47-4):

$$P(x, t) - P(x + \Delta x, t) = -\frac{\partial P}{\partial x} \Delta x = -\frac{\partial P_e}{\partial x} \Delta x, \quad (47.10)$$

since Δx is small and since the only part of P which changes is the excess pressure P_e . We now have III:

$$\rho_0 \frac{\partial^2\chi}{\partial t^2} = -\frac{\partial P_e}{\partial x}, \quad (III) \quad (47.11)$$

and so we have enough equations to interconnect things and reduce down to one variable, say to χ . We can eliminate P_e from III by using II, so that we get

$$\rho_0 \frac{\partial^2\chi}{\partial t^2} = -\kappa \frac{\partial\rho_e}{\partial x}, \quad (47.12)$$

and then we can use I to eliminate ρ_e . In this way we find that ρ_0 cancels out and that we are left with

$$\frac{\partial^2\chi}{\partial t^2} = \kappa \frac{\partial^2\chi}{\partial x^2}. \quad (47.13)$$

We shall call $c_s^2 = \kappa$, so that we can write

$$\frac{\partial^2\chi}{\partial x^2} = \frac{1}{c_s^2} \frac{\partial^2\chi}{\partial t^2}. \quad (47.14)$$

This is the wave equation which describes the behavior of sound in matter.

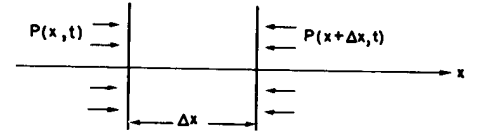


Fig. 47-4. The net force in the positive x -direction produced by the pressure acting on unit area perpendicular to x is $-(\partial P/\partial x) \Delta x$.

47-4 Solutions of the wave equation

We now can see whether this equation really does describe the essential properties of sound waves in matter. We want to deduce that a sound pulse, or disturbance, will move with a constant speed. We want to verify that two different pulses can move through each other—the principle of superposition. We also want to verify that sound can go either to the right or to the left. All these properties should be contained in this one equation.

We have remarked that any plane-wave disturbance which moves with a constant velocity v has the form $f(x - vt)$. Now we have to see whether $\chi(x, t) = f(x - vt)$ is a solution of the wave equation. When we calculate $\partial\chi/\partial x$, we get the derivative of the function, $\partial\chi/\partial x = f'(x - vt)$. Differentiating once more, we find

$$\frac{\partial^2 \chi}{\partial x^2} = f''(x - vt). \quad (47.15)$$

The differentiation of this same function with respect to t gives $-v$ times the derivative of the function, or $\partial\chi/\partial t = -vf'(x - vt)$, and the second time derivative is

$$\frac{\partial^2 \chi}{\partial t^2} = v^2 f''(x - vt). \quad (47.16)$$

It is evident that $f(x - vt)$ will satisfy the wave equation provided the wave velocity v is equal to c_s .

We find, therefore, from the *laws of mechanics* that any sound disturbance propagates with the velocity c_s , and in addition we find that

$$c_s = \kappa^{1/2} = (dP/d\rho)_0^{1/2},$$

and so we have related the wave velocity to a property of the medium.

If we consider a wave travelling in the opposite direction, so that $\chi(x, t) = g(x + vt)$, it is easy to see that such a disturbance also satisfies the wave equation. The only difference between such a wave and one travelling from left to right is in the sign of v , but whether we have $x + vt$ or $x - vt$ as the variable in the function does not affect the sign of $\partial^2 \chi / \partial t^2$, since it involves only v^2 . It follows that we have a solution for waves propagating in either direction with speed c_s .

An extremely interesting question is that of superposition. Suppose one solution of the wave equation has been found, say χ_1 . This means that the second derivative of χ_1 with respect to x is equal to $1/c_s^2$ times the second derivative of χ_1 with respect to t . Now any other solution χ_2 has this same property. If we superpose these two solutions, we have

$$\chi(x, t) = \chi_1(x, t) + \chi_2(x, t), \quad (47.17)$$

and we wish to verify that $\chi(x, t)$ is also a wave, i.e., that χ satisfies the wave equation. We can easily prove this result, since we have

$$\frac{\partial^2 \chi}{\partial x^2} = \frac{\partial^2 \chi_1}{\partial x^2} + \frac{\partial^2 \chi_2}{\partial x^2} \quad (47.18)$$

and, in addition,

$$\frac{\partial^2 \chi}{\partial t^2} = \frac{\partial^2 \chi_1}{\partial t^2} + \frac{\partial^2 \chi_2}{\partial t^2}. \quad (47.19)$$

It follows that $\partial^2 \chi / \partial x^2 = (1/c_s^2) \partial^2 \chi / \partial t^2$, so we have verified the principle of superposition. The proof of the principle of superposition follows from the fact that the wave equation is *linear* in χ .

We can now expect that a plane light wave propagating in the x -direction, polarized so that the electric field is in the y -direction, will satisfy the wave equation

$$\frac{\partial^2 E_y}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 E_y}{\partial t^2}, \quad (47.20)$$

where c is the speed of light. This wave equation is one of the consequences of Maxwell's equations. The equations of electrodynamics will lead to the wave equation for light just as the equations of mechanics lead to the wave equation for sound.

47-5 The speed of sound

Our deduction of the wave equation for sound has given us a *formula* which connects the wave speed with the rate of change of pressure with the density at the normal pressure:

$$c_s^2 = \left(\frac{dP}{d\rho} \right)_0. \quad (47.21)$$

In evaluating this rate of change, it is essential to know how the temperature varies. In a sound wave, we would expect that in the region of compression the temperature would be raised, and that in the region of rarefaction the temperature would be lowered. Newton was the first to calculate the rate of change of pressure with density, and he supposed that the temperature remained unchanged. He argued that the heat was conducted from one region to the other so rapidly that the temperature could not rise or fall. This argument gives the isothermal speed of sound, and it is wrong. The correct deduction was given later by Laplace, who put forward the opposite idea—that the pressure and temperature change adiabatically in a sound wave. The heat flow from the compressed region to the rarefied region is negligible so long as the wavelength is long compared with the mean free path. Under this condition the slight amount of heat flow in a sound wave does not affect the speed, although it gives a small absorption of the sound energy. We can expect correctly that this absorption increases as the wavelength approaches the mean free path, but these wavelengths are smaller by factors of about a million than the wavelengths of audible sound.

The actual variation of pressure with density in a sound wave is the one that allows no heat flow. This corresponds to the adiabatic variation, which we found to be $PV^\gamma = \text{const}$, where V was the volume. Since the density ρ varies inversely with V , the adiabatic connection between P and ρ is

$$P = \text{const } \rho^\gamma, \quad (47.22)$$

from which we get $dP/d\rho = \gamma P/\rho$. We then have for the speed of sound the relation

$$c_s^2 = \frac{\gamma P}{\rho}. \quad (47.23)$$

We can also write $c_s^2 = \gamma PV/\rho V$ and make use of the relation $PV = NkT$. Further, we see that ρV is the mass of gas, which can also be expressed as Nm , or as μ , where m is the mass of a molecule and μ is the molecular weight. In this way we find that

$$c_s^2 = \frac{\gamma kT}{m} = \frac{\gamma RT}{\mu}, \quad (47.24)$$

from which it is evident that the speed of sound depends only on the gas temperature and not on the pressure or the density. We also have observed that

$$kT = \frac{1}{3}m\langle v^2 \rangle, \quad (47.25)$$

where $\langle v^2 \rangle$ is the mean square of the speed of the molecules. It follows that $c_s^2 = (\gamma/3)\langle v^2 \rangle$, or

$$c_s = \left(\frac{\gamma}{3} \right)^{1/2} v_{\text{av}}. \quad (47.26)$$

This equation states that the speed of sound is some number which is roughly $1/(3)^{1/2}$ times some average speed, v_{av} , of the molecules (the square root of the

mean square velocity). In other words, the speed of sound is of the same order of magnitude as the speed of the molecules, and is actually somewhat less than this average speed.

Of course we could expect such a result, because a disturbance like a change in pressure is, after all, propagated by the motion of the molecules. However, such an argument does not tell us the precise propagation speed; it could have turned out that sound was carried primarily by the fastest molecules, or by the slowest molecules. It is reasonable and satisfying that the speed of sound is roughly $\frac{1}{2}$ of the average molecular speed v_{av} .

Beats

48-1 Adding two waves

Some time ago we discussed in considerable detail the properties of light waves and their interference—that is, the effects of the superposition of two waves from different sources. In all these analyses we assumed that the frequencies of the sources were all the same. In this chapter we shall discuss some of the phenomena which result from the interference of two sources which have *different* frequencies.

It is easy to guess what is going to happen. Proceeding in the same way as we have done previously, suppose we have two equal oscillating sources of the same frequency whose phases are so adjusted, say, that the signals arrive in phase at some point P . At that point, if it is light, the light is very strong; if it is sound, it is very loud; or if it is electrons, many of them arrive. On the other hand, if the arriving signals were 180° out of phase, we would get no signal at P , because the net amplitude there is then a minimum. Now suppose that someone twists the “phase knob” of one of the sources and changes the phase at P back and forth, say, first making it 0° and then 180° , and so on. Of course, we would then find variations in the net signal strength. Now we also see that if the phase of one source is slowly changing relative to that of the other in a gradual, uniform manner, starting at zero, going up to ten, twenty, thirty, forty degrees, and so on, then what we would measure at P would be a series of strong and weak “pulsations,” because when the phase shifts through 360° the amplitude returns to a maximum. Of course, to say that one source is shifting its phase relative to another at a uniform rate is the same as saying that the number of oscillations per second is slightly different for the two.

So we know the answer: if we have two sources at slightly different frequencies we should find, as a net result, an oscillation with a slowly pulsating intensity. That is all there really is to the subject!

It is very easy to formulate this result mathematically also. Suppose, for example, that we have two waves, and that we do not worry for the moment about all the spatial relations, but simply analyze what arrives at P . From one source, let us say, we would have $\cos \omega_1 t$, and from the other source, $\cos \omega_2 t$, where the two ω 's are not exactly the same. Of course the amplitudes may not be the same, either, but we can solve the general problem later; let us first take the case where the amplitudes are equal. Then the total amplitude at P is the sum of these two cosines. If we plot the amplitudes of the waves against the time, as in Fig. 48-1, we see that

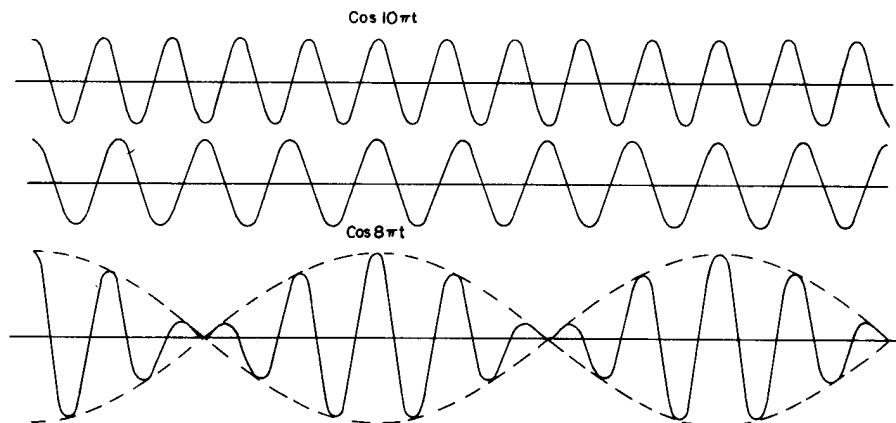


Fig. 48-1. The superposition of two cosine waves with frequencies in the ratio 8:10. The precise repetition of the pattern within each “beat” is not typical of the general case.

48-1 Adding two waves

48-2 Beat notes and modulation

48-3 Side bands

48-4 Localized wave trains

48-5 Probability amplitudes for particles

48-6 Waves in three dimensions

48-7 Normal modes

where the crests coincide we get a strong wave, and where a trough and crest coincide we get practically zero, and then when the crests coincide again we get a strong wave again.

Mathematically, we need only to add two cosines and rearrange the result somehow. There exist a number of useful relations among cosines which are not difficult to derive. Of course we know that

$$e^{i(a+b)} = e^{ia}e^{ib}, \quad (48.1)$$

and that e^{ia} has a real part, $\cos a$, and an imaginary part, $i \sin a$. If we take the real part of $e^{i(a+b)}$, we get $\cos(a+b)$. If we multiply out:

$$e^{ia}e^{ib} = (\cos a + i \sin a)(\cos b + i \sin b),$$

we get $\cos a \cos b - \sin a \sin b$, plus some imaginary parts. But we now need only the real part, so we have

$$\cos(a+b) = \cos a \cos b - \sin a \sin b. \quad (48.2)$$

Now if we change the sign of b , since the cosine does not change sign while the sine does, the same equation, for negative b , is

$$\cos(a-b) = \cos a \cos b + \sin a \sin b. \quad (48.3)$$

If we add these two equations together, we lose the sines and we learn that the product of two cosines is half the cosine of the sum, plus half the cosine of the difference:

$$\cos a \cos b = \frac{1}{2} \cos(a+b) + \frac{1}{2} \cos(a-b). \quad (48.4)$$

Now we can also reverse the formula and find a formula for $\cos \alpha + \cos \beta$ if we simply let $\alpha = a+b$ and $\beta = a-b$. That is, $a = \frac{1}{2}(\alpha + \beta)$ and $b = \frac{1}{2}(\alpha - \beta)$, so that

$$\cos \alpha + \cos \beta = 2 \cos \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta). \quad (48.5)$$

Now we can analyze our problem. The sum of $\cos \omega_1 t$ and $\cos \omega_2 t$ is

$$\cos \omega_1 t + \cos \omega_2 t = 2 \cos \frac{1}{2}(\omega_1 + \omega_2)t \cos \frac{1}{2}(\omega_1 - \omega_2)t. \quad (48.6)$$

Now let us suppose that the two frequencies are nearly the same, so that $\frac{1}{2}(\omega_1 + \omega_2)$ is the average frequency, and is more or less the same as either. But $\omega_1 - \omega_2$ is *much smaller* than ω_1 or ω_2 because, as we suppose, ω_1 and ω_2 are nearly equal. That means that we can represent the solution by saying that there is a high-frequency cosine wave more or less like the ones we started with, but that its "size" is slowly changing—its "size" is pulsating with a frequency which appears to be $\frac{1}{2}(\omega_1 - \omega_2)$. But is this the frequency at which the beats are heard? Although (48.6) says that the amplitude goes as $\cos \frac{1}{2}(\omega_1 - \omega_2)t$, what it is really telling us is that the high-frequency oscillations are contained between two opposed cosine curves (shown dotted in Fig. 48-1). On this basis one could say that the amplitude varies at the frequency $\frac{1}{2}(\omega_1 - \omega_2)$, but if we are talking about the *intensity* of the wave we must think of it as having twice this frequency. That is, the modulation of the amplitude, in the sense of the strength of its intensity, is at frequency $\omega_1 - \omega_2$, although the formula tells us that we multiply by a cosine wave at half that frequency. The technical basis for the difference is that the high frequency-wave has a little different phase relationship in the second half-cycle.

Ignoring this small complication, we may conclude that if we add two waves of frequency ω_1 and ω_2 , we will get a net resulting wave of average frequency $\frac{1}{2}(\omega_1 + \omega_2)$ which oscillates in strength with a frequency $\omega_1 - \omega_2$.

If the two amplitudes are different, we can do it all over again by multiplying the cosines by different amplitudes A_1 and A_2 , and do a lot of mathematics, rearranging, and so on, using equations like (48.2)–(48.5). However, there are other, easier ways of doing the same analysis. For example, we know that it is much

easier to work with exponentials than with sines and cosines and that we can represent $A_1 \cos \omega_1 t$ as the real part of $A_1 e^{i\omega_1 t}$. The other wave would similarly be the real part of $A_2 e^{i\omega_2 t}$. If we add the two, we get $A_1 e^{i\omega_1 t} + A_2 e^{i\omega_2 t}$. If we then factor out the average frequency, we have

$$A_1 e^{i\omega_1 t} + A_2 e^{i\omega_2 t} = e^{1/2 i(\omega_1 + \omega_2)t} [A_1 e^{1/2 i(\omega_1 - \omega_2)t} + A_2 e^{-1/2 i(\omega_1 - \omega_2)t}]. \quad (48.7)$$

Again we have the high-frequency wave with a modulation at the lower frequency.

48-2 Beat notes and modulation

If we are now asked for the intensity of the wave of Eq. (48.7), we can either take the absolute square of the left side, or of the right side. Let us take the left side. The intensity then is

$$I = A_1^2 + A_2^2 + 2A_1 A_2 \cos(\omega_1 - \omega_2)t. \quad (48.8)$$

We see that the intensity swells and falls at a frequency $\omega_1 - \omega_2$, varying between the limits $(A_1 + A_2)^2$ and $(A_1 - A_2)^2$. If $A_1 \neq A_2$, the minimum intensity is not zero.

One more way to represent this idea is by means of a drawing, like Fig. 48-2. We draw a vector of length A_1 , rotating at a frequency ω_1 , to represent one of the waves in the complex plane. We draw another vector of length A_2 , going around at a frequency ω_2 , to represent the second wave. If the two frequencies are exactly equal, their resultant is of fixed length as it keeps revolving, and we get a definite, fixed intensity from the two. But if the frequencies are slightly different, the two complex vectors go around at different speeds. Figure 48-3 shows what the situation looks like relative to the vector $A_1 e^{i\omega_1 t}$. We see that A_2 is turning slowly away from A_1 , and so the amplitude that we get by adding the two is first strong, and then, as it opens out, when it gets to the 180° relative position the resultant gets particularly weak, and so on. As the vectors go around, the amplitude of the sum vector gets bigger and smaller, and the intensity thus pulsates. It is a relatively simple idea, and there are many different ways of representing the same thing.

The effect is very easy to observe experimentally. In the case of acoustics, we may arrange two loudspeakers driven by two separate oscillators, one for each loudspeaker, so that they each make a tone. We thus receive one note from one source and a different note from the other source. If we make the frequencies exactly the same, the resulting effect will have a definite strength at a given space location. If we then de-tune them a little bit, we hear some variations in the intensity. The farther they are de-tuned, the more rapid are the variations of sound. The ear has some trouble following variations more rapid than ten or so per second.

We may also see the effect on an oscilloscope which simply displays the sum of the currents to the two speakers. If the frequency of pulsing is relatively low, we simply see a sinusoidal wave train whose amplitude pulsates, but as we make the pulsations more rapid we see the kind of wave shown in Fig. 48-1. As we go to greater frequency differences, the "bumps" move closer together. Also, if the amplitudes are not equal and we make one signal stronger than the other, then we get a wave whose amplitude does not ever become zero, just as we expect. Everything works the way it should, both acoustically and electrically.

The opposite phenomenon occurs too! In radio transmission using so-called *amplitude modulation* (AM), the sound is broadcast by the radio station as follows: the radio transmitter has an AC electric oscillation which is at a very high frequency, for example 800 kilocycles per second, in the broadcast band. If this *carrier signal* is turned on, the radio station emits a wave which is of uniform amplitude at 800,000 oscillations a second. The way the "information" is transmitted, the useless kind of information about what kind of car to buy, is that when somebody talks into a microphone the amplitude of the carrier signal is changed in step with the vibrations of sound entering the microphone.

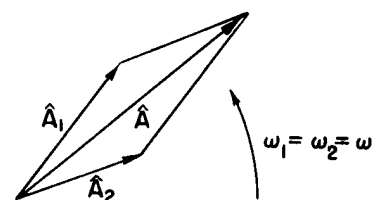


Fig. 48-2. The resultant of two complex vectors of equal frequency.

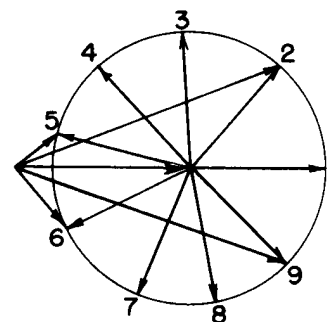


Fig. 48-3. The resultant of two complex vectors of unequal frequency, as seen in the rotating frame of reference of one vector. Nine successive positions of the slowly rotating vector are shown.

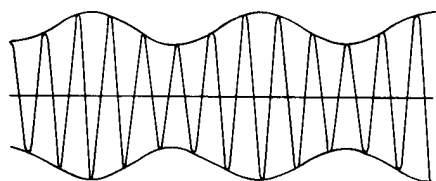


Fig. 48-4. A modulated carrier wave. In this schematic sketch, $\omega_c/\omega_m = 5$. In an actual radiowave, $\omega_c/\omega_m \sim 100$.

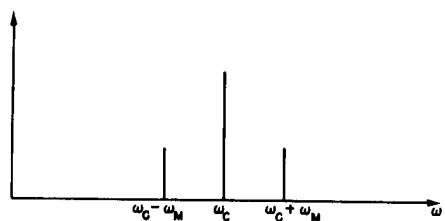


Fig. 48-5. The frequency spectrum of a carrier wave ω_c modulated by a single cosine wave ω_m .

If we take as the simplest mathematical case the situation where a soprano is singing a perfect note, with perfect sinusoidal oscillations of her vocal cords, then we get a signal whose strength is alternating as shown in Fig. 48-4. The audiofrequency alternation is then recovered in the receiver; we get rid of the carrier wave and just look at the envelope which represents the oscillations of the vocal cords, or the sound of the singer. The loudspeaker then makes corresponding vibrations at the same frequency in the air, and the listener is then essentially unable to tell the difference, so they say. Because of a number of distortions and other subtle effects, it is, in fact, possible to tell whether we are listening to a radio or to a real soprano; otherwise the idea is as indicated above.

48-3 Side bands

Mathematically, the modulated wave described above would be expressed as

$$S = (1 + b \cos \omega_m t) \cos \omega_c t, \quad (48.9)$$

where ω_c represents the frequency of the carrier and ω_m is the frequency of the audio tone. Again we use all those theorems about the cosines, or we can use $e^{i\theta}$; it makes no difference—it is easier with $e^{i\theta}$, but it is the same thing. We then get

$$S = \cos \omega_c t + \frac{1}{2}b \cos (\omega_c + \omega_m)t + \frac{1}{2}b \cos (\omega_c - \omega_m)t. \quad (48.10)$$

So, from another point of view, we can say that the output wave of the system consists of three waves added in superposition: first, the regular wave at the frequency ω_c , that is, at the carrier frequency, and then two new waves at two new frequencies. One is the carrier frequency plus the modulation frequency, and the other is the carrier frequency minus the modulation frequency. If, therefore, we make some kind of plot of the intensity being generated by the generator as a function of frequency, we would find a lot of intensity at the frequency of the carrier, naturally, but when a singer started to sing, we would suddenly also find intensity proportional to the strength of the singer, b^2 , at frequency $\omega_c + \omega_m$ and $\omega_c - \omega_m$, as shown in Fig. 48-5. These are called *side bands*; when there is a modulated signal from the transmitter, there are side bands. If there is more than one note at the same time, say ω_m and ω_m' , there are two instruments playing; or if there is any other complicated cosine wave, then, of course, we can see from the mathematics that we get some more waves that correspond to the frequencies $\omega_c \pm \omega_m'$.

Therefore, when there is a complicated modulation that can be represented as the sum of many cosines,* we find that the actual transmitter is transmitting over a range of frequencies, namely the carrier frequency plus or minus the maximum frequency that the modulation signal contains.

Although at first we might believe that a radio transmitter transmits only at the nominal frequency of the carrier, since there are big, superstable crystal oscillators in there, and everything is adjusted to be at precisely 800 kilocycles, the moment someone *announces* that they are at 800 kilocycles, he modulates the 800 kilocycles, and so they are no longer precisely at 800 kilocycles! Suppose that the amplifiers are so built that they are able to transmit over a good range of the ear's sensitivity (the ear can hear up to 20,000 cycles per second, but usually radio transmitters and receivers do not work beyond 10,000, so we do not hear the highest parts), then, when the man speaks, his voice may contain frequencies ranging up, say, to 10,000 cycles, so the transmitter is transmitting frequencies

* A slight side remark: In what circumstances can a curve be represented as a sum of a lot of cosines? *Answer:* In all ordinary circumstances, except for certain cases the mathematicians can dream up. Of course, the curve must have only one value at a given point, and it must not be a crazy curve which jumps an infinite number of times in an infinitesimal distance, or something like that. But aside from such restrictions any reasonable curve (one that a singer is going to be able to make by shaking her vocal cords) can always be compounded by adding cosine waves together.

which may range from 790 to 810 kilocycles per second. Now if there were another station at 795 kc/sec, there would be a lot of confusion. Also, if we made our receiver so sensitive that it picked up only 800, and did not pick up the 10 kilocycles on either side, we would not hear what the man was saying, because the information would be on these other frequencies! Therefore it is absolutely essential to keep the stations a certain distance apart, so that their side bands do not overlap and, also, the receiver must not be so selective that it does not permit reception of the side bands as well as of the main nominal frequency. In the case of sound, this problem does not really cause much trouble. We can hear over a ± 20 kc/sec range, and we have usually from 500 to 1500 kc/sec in the broadcast band, so there is plenty of room for lots of stations.

The television problem is more difficult. As the electron beam goes across the face of the picture tube, there are various little spots of light and dark. That "light" and "dark" is the "signal." Now ordinarily the beam scans over the whole picture, 500 lines, approximately, in a thirtieth of a second. Let us consider that the resolution of the picture vertically and horizontally is more or less the same, so that there are the same number of spots per inch along a scan line. We want to be able to distinguish dark from light, dark from light, dark from light, over, say, 500 lines. In order to be able to do this with cosine waves, the shortest wavelength needed thus corresponds to a wavelength, from maximum to maximum, of one 250th of the screen size. So we have $250 \times 500 \times 30$ pieces of information per second. The highest frequency that we are going to carry, therefore, is close to 4 megacycles per second. Actually, to keep the television stations apart, we have to use a little bit more than this, about 6 mc/sec; part of it is used to carry the sound signal, and other information. So, television channels are 6 megacycles per second wide. It certainly would not be possible to transmit TV on an 800 kc/sec carrier, since we cannot modulate at a higher frequency than the carrier.

At any rate, the television band starts at 54 megacycles. The first transmission channel, which is channel 2 (!), has a frequency range from 54 to 60 mc/sec, which is 6 mc/sec wide. "But," one might say, "we have just proved that there were side bands on both sides, and therefore it should be twice that wide." It turns out that the radio engineers are rather clever. If we analyze the modulation signal using not just cosine terms, but cosine and sine terms, to allow for phase differences, we then see that there is a definite, invariant relationship between the side band on the high-frequency side and the side band on the low-frequency side. What we mean is that there is no new information on that other side band. So what is done is to suppress one side band, and the receiver is wired inside such that the information which is missing is reconstituted by looking at the single side band and the carrier. Single side-band transmission is a clever scheme for decreasing the band widths needed to transmit information.

48-4 Localized wave trains

The next subject we shall discuss is the interference of waves in both space and time. Suppose that we have two waves travelling in space. We know, of course, that we can represent a wave travelling in space by $e^{i(\omega t - kx)}$. This might be, for example, the displacement in a sound wave. This is a solution of the wave equation provided that $\omega^2 = k^2 c^2$, where c is the speed of propagation of the wave. In this case we can write it as $e^{ik(x - ct)}$, which is of the general form $f(x - ct)$. Therefore this must be a wave which is travelling at this velocity, ω/k , and that is c and everything is all right.

Now we want to add two such waves together. Suppose we have a wave that is travelling with one frequency, and another wave travelling with another frequency. We leave to the reader to consider the case where the amplitudes are different; it makes no real difference. Thus we want to add $e^{i(\omega_1 t - k_1 x)} + e^{i(\omega_2 t - k_2 x)}$. We can add these by the same kind of mathematics we used when we added signal waves. Of course, if c is the same for both, this is easy, since it is the same as what we did before:

$$e^{i\omega_1(t-x/c)} + e^{i\omega_2(t-x/c)} = e^{i\omega_1 t'} + e^{i\omega_2 t'}, \quad (48.11)$$

except that $t' = t - x/c$ is the variable instead of t . So we get the same kind of modulations, naturally, but we see, of course, that those modulations are moving along with the wave. In other words, if we added two waves, but these waves were not just oscillating, but also moving in space, then the resultant wave would move along also, at the same speed.

Now we would like to generalize this to the case of waves in which the relationship between the frequency and the wave number k is not so simple. Example: material having an index of refraction. We have already studied the theory of the index of refraction in Chapter 31, where we found that we could write $k = n\omega/c$, where n is the index of refraction. As an interesting example, for x-rays we found that the index n is

$$n = 1 - \frac{Nq_e^2}{2\epsilon_0 m \omega^2}. \quad (48.12)$$

We actually derived a more complicated formula in Chapter 31, but this one is as good as any, as an example.

Incidentally, we know that even when ω and k are not linearly proportional, the ratio ω/k is certainly the speed of propagation for the particular frequency and wave number. We call this ratio the *phase velocity*; it is the speed at which the phase, or the nodes of a single wave, would move along:

$$v_p = \frac{\omega}{k}. \quad (48.13)$$

This phase velocity, for the case of x-rays in glass, is greater than the speed of light in vacuum (since n in 48.12 is less than 1), and that is a bit bothersome, because we do not think we can send signals faster than the speed of light!

What we are going to discuss now is the interference of two waves in which ω and k have a definite formula relating them. The above formula for n says that k is given as a definite function of ω . To be specific, in this particular problem, the formula for k in terms of ω is

$$k = \frac{\omega}{c} - \frac{a}{\omega c}, \quad (48.14)$$

where $a = Nq_e^2/2\epsilon_0 m$, a constant. At any rate, for each frequency there is a definite wave number, and we want to add two such waves together.

Let us do it just as we did in Eq. (48.7):

$$e^{i(\omega_1 t - k_1 x)} + e^{i(\omega_2 t - k_2 x)} = e^{1/2 i[(\omega_1 + \omega_2)t - (k_1 + k_2)x]} \times \{e^{1/2 i[(\omega_1 - \omega_2)t - (k_1 - k_2)x]} + e^{-1/2 i[(\omega_1 - \omega_2)t - (k_1 - k_2)x]}\}. \quad (48.15)$$

So we have a modulated wave again, a wave which travels with the mean frequency and the mean wave number, but whose strength is varying with a form which depends on the difference frequency and the difference wave number.

Now let us take the case that the difference between the two waves is relatively small. Let us suppose that we are adding two waves whose frequencies are nearly equal; then $(\omega_1 + \omega_2)/2$ is practically the same as either one of the ω 's, and similarly for $(k_1 + k_2)/2$. Thus the speed of the wave, the fast oscillations, the nodes, is still essentially ω/k . But look, the speed of propagation of the modulation is not the same! How much do we have to change x to account for a certain amount of t ? The speed of this modulation wave is the ratio

$$v_M = \frac{\omega_1 - \omega_2}{k_1 - k_2}. \quad (48.16)$$

The speed of modulation is sometimes called the *group velocity*. If we take the case that the difference in frequency is relatively small, and the difference in wave number is then also relatively small, then this expression approaches, in the limit,

$$v_g = \frac{d\omega}{dk}. \quad (48.17)$$

In other words, for the slowest modulation, the slowest beats, there is a definite speed at which they travel which is not the same as the phase speed of the waves—what a mysterious thing!

The group velocity is the derivative of ω with respect to k , and the phase velocity is ω/k .

Let us see if we can understand why. Consider two waves, again of slightly different wavelength, as in Fig. 48-1. They are out of phase, in phase, out of phase, and so on. Now these waves represent, really, the waves in space travelling with slightly different frequencies also. Now because the phase velocity, the velocity of the nodes of these two waves, is not precisely the same, something new happens. Suppose we ride along with one of the waves and look at the other one; if they both went at the same speed, then the other wave would stay right where it was relative to us, as we ride along on this crest. We ride on that crest and right opposite us we see a crest; if the two velocities are equal the crests stay on top of each other. But it is *not so* that the two velocities are really equal. There is only a small difference in frequency and therefore only a small difference in velocity, but because of that difference in velocity, as we ride along the other wave moves slowly forward, say, or behind, relative to our wave. So as time goes on, what happens to the node? If we move one wave train just a shade forward, the node moves forward (or backward) a considerable distance. That is, the sum of these two waves has an envelope, and as the waves travel along, the envelope rides on them at a different speed. The *group velocity* is the speed at which modulated signals would be transmitted.

If we made a signal, i.e., some kind of change in the wave that one could recognize when he listened to it, a kind of modulation, then that modulation would travel at the group velocity, provided that the modulations were relatively slow. (When they are fast, it is much more difficult to analyze.)

Now we may show (at long last), that the speed of propagation of x-rays in a block of carbon is *not* greater than the speed of light, although the phase velocity *is* greater than the speed of light. In order to do that, we must find $d\omega/dk$, which we get by differentiating (48.14): $dk/d\omega = 1/c + a/\omega^2 c$. The group velocity, therefore, is the reciprocal of this, namely,

$$v_g = \frac{c}{1 + a/\omega^2}, \quad (48.18)$$

which is smaller than c ! So although the phases can travel faster than the speed of light, the modulation signals travel slower, and that is the resolution of the apparent paradox! Of course, if we have the simple case that $\omega = kc$, then $d\omega/dk$ is also c . So when all the phases have the same velocity, naturally the group has the same velocity.

48-5 Probability amplitudes for particles

Let us now consider one more example of the phase velocity which is extremely interesting. It has to do with quantum mechanics. We know that the amplitude to find a particle at a place can, in some circumstances, vary in space and time, let us say in one dimension, in this manner:

$$\psi = Ae^{i(\omega t - kx)}, \quad (48.19)$$

where ω is the frequency, which is related to the classical idea of the energy through $E = \hbar\omega$, and k is the wave number, which is related to the momentum through $p = \hbar k$. We would say the particle had a definite momentum p if the wave number were exactly k , that is, a perfect wave which goes on with the same amplitude everywhere. Equation (48.19) gives the amplitude, and if we take the absolute square, we get the relative probability for finding the particle as a function of position and time. This is a *constant*, which means that the probability is the same to find a particle anywhere. Now suppose, instead, that we have a situation where we know that the particle is more likely to be at one place than at another. We would

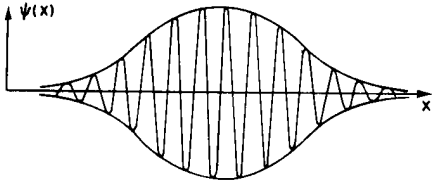


Fig. 48-6. A localized wave train.

represent such a situation by a wave which has a maximum and dies out on either side (Fig. 48-6). (It is not quite the same as a wave like (48.1) which has a series of maxima, but it is possible, by adding several waves of nearly the same ω and k together, to get rid of all but one maximum.)

Now in those circumstances, since the square of (48.19) represents the chance of finding a particle somewhere, we know that at a given instant the particle is most likely to be near the center of the "lump," where the amplitude of the wave is maximum. If now we wait a few moments, the waves will move, and after some time the "lump" will be somewhere else. If we knew that the particle originally was situated somewhere, classically, we would *expect* that it would later be elsewhere as a matter of fact, because it has a *speed*, after all, and a momentum. The quantum theory, then, will go into the correct classical theory for the relationship of momentum, energy, and velocity only if the group velocity, the velocity of the modulation, is equal to the velocity that we would obtain classically for a particle of the same momentum.

It is now necessary to demonstrate that this is, or is not, the case. According to the classical theory, the energy is related to the velocity through an equation like

$$E = \frac{mc^2}{\sqrt{1 - v^2/c^2}}. \quad (48.20)$$

Similarly, the momentum is

$$p = \frac{mv}{\sqrt{1 - v^2/c^2}}. \quad (48.21)$$

That is the classical theory, and as a consequence of the classical theory, by eliminating v , we can show that

$$E^2 - p^2c^2 = m^2c^4.$$

That is the four-dimensional grand result that we have talked and talked about, that $p_\mu p_\mu = m^2$; that is the relation between energy and momentum in the classical theory. Now that means, since these E 's and p 's are going to become ω 's and k 's, by substitution of $E = \hbar\omega$ and $p = \hbar k$, that for quantum mechanics it is necessary that

$$\frac{\hbar^2\omega^2}{c^2} - \hbar^2k^2 = m^2c^2. \quad (48.22)$$

This, then, is the relationship between the frequency and the wave number of a quantum-mechanical amplitude wave representing a particle of mass m . From this equation we can deduce that ω is

$$\omega = c\sqrt{k^2 + m^2c^2/\hbar^2}.$$

The phase velocity, ω/k , is here again faster than the speed of light!

Now let us look at the group velocity. The group velocity should be $d\omega/dk$, the speed at which the modulations move. We have to differentiate a square root, which is not very difficult. The derivative is

$$\frac{d\omega}{dk} = \frac{kc}{\sqrt{k^2 + m^2c^2/\hbar^2}}.$$

Now the square root is, after all, ω , so we could write this as $d\omega/dk = c^2k/\omega$. Further, k/ω is p/E , so

$$v_g = \frac{c^2p}{E}.$$

But from (48.20) and (48.21), $c^2p/E = v$, the velocity of the particle, according to classical mechanics. So we see that whereas the fundamental quantum-mechanical relationship $E = \hbar\omega$ and $p = \hbar k$, for the identification of ω and k with the classical E and p , only produces the equation $\omega^2 - k^2c^2 = m^2c^4/\hbar^2$, now we also understand the relationships (48.20) and (48.21) which connected E and p

to the velocity. Of course the group velocity must be the velocity of the particle if the interpretation is going to make any sense. If we think the particle is over here at one time, and then ten minutes later we think it is over there, as the quantum mechanics said, the distance traversed by the "lump," divided by the time interval, must be, classically, the velocity of the particle.

48-6 Waves in three dimensions

We shall now bring our discussion of waves to a close with a few general remarks about the wave equation. These remarks are intended to give some view of the future—not that we can understand everything exactly just now, but rather to see what things are going to look like when we study waves a little more. First of all, the wave equation for sound in one dimension was

$$\frac{\partial^2 \chi}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \chi}{\partial t^2},$$

where c is the speed of whatever the wave is—in the case of sound, it is the sound speed; in the case of light, it is the speed of light. We showed that for a sound wave the displacements would propagate themselves at a certain speed. But the excess pressure also propagates at a certain speed, and so does the excess density. So we should expect that the pressure would satisfy the same equation, as indeed it does. We shall leave it to the reader to prove that it does. *Hint:* ρ_e is proportional to the rate of change of χ with respect to x . Therefore if we differentiate the wave equation with respect to x , we will immediately discover that $\partial \chi / \partial x$ satisfies the same equation. That is to say, ρ_e satisfies the same equation. But P_e is proportional to ρ_e , and therefore P_e does too. So the pressure, the displacements, everything, satisfy the same wave equation.

Usually one sees the wave equation for sound written in terms of pressure instead of in terms of displacement, because the pressure is a scalar and has no direction. But the displacement is a vector and has direction, and it is thus easier to analyze the pressure.

The next matter we discuss has to do with the wave equation in three dimensions. We know that the sound wave solution in one dimension is $e^{i(\omega t - kx)}$, with $\omega = kc_s$, but we also know that in three dimensions a wave would be represented by $e^{i(\omega t - k_x x - k_y y - k_z z)}$, where, in this case, $\omega^2 = k^2 c_s^2$, which is, of course, $(k_x^2 + k_y^2 + k_z^2) c_s^2$. Now what we want to do is to guess what the correct wave equation in three dimensions is. Naturally, for the case of sound this can be deduced by going through the same dynamic argument in three dimensions that we made in one dimension. But we shall not do that; instead we just write down what comes out: the equation for the pressure (or displacement, or anything) is

$$\frac{\partial^2 P_e}{\partial x^2} + \frac{\partial^2 P_e}{\partial y^2} + \frac{\partial^2 P_e}{\partial z^2} = \frac{1}{c_s^2} \frac{\partial^2 P_e}{\partial t^2}. \quad (48.23)$$

That this is true can be verified by substituting in $e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}$. Clearly, every time we differentiate with respect to x , we multiply by $-ik_x$. If we differentiate twice, it is equivalent to multiplying by $-k_x^2$, so the first term would become $-k_x^2 P_e$, for that wave. Similarly, the second term becomes $-k_y^2 P_e$, and the third term becomes $-k_z^2 P_e$. On the right, we get $-(\omega^2/c_s^2) P_e$. Then, if we take away the P_e 's and change the sign, we see that the relationship between k and ω is the one that we want.

Working backwards again, we cannot resist writing down the grand equation which corresponds to the dispersion equation (48.22) for quantum-mechanical waves. If ϕ represents the amplitude for finding a particle at position x, y, z , at the time t , then the great equation of quantum mechanics for free particles is this:

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = \frac{m^2 c^2}{\hbar^2} \phi. \quad (48.24)$$

First of all, the relativity character of this expression is suggested by the appearance

of x , y , z and t in the nice combination relativity usually involves. Second, it is a wave equation which, if we try a plane wave, would produce as a consequence that $-k^2 + \omega^2/c^2 = m^2c^2/\hbar^2$, which is the right relationship for quantum mechanics. There is still another great thing contained in the wave equation: the fact that any superposition of waves is also a solution. So this equation contains all of the quantum mechanics and the relativity that we have been discussing so far, at least so long as it deals with a single particle in empty space with no external potentials or forces on it!

48-7 Normal modes

Now we turn to another example of the phenomenon of beats which is rather curious and a little different. Imagine two equal pendulums which have, between them, a rather weak spring connection. They are made as nearly as possible the same length. If we pull one aside and let go, it moves back and forth, and it pulls on the connecting spring as it moves back and forth, and so it really is a machine for generating a force which has the natural frequency of the other pendulum. Therefore, as a consequence of the theory of resonance, which we studied before, when we put a force on something at just the right frequency, it will drive it. So, sure enough, one pendulum moving back and forth drives the other. However, in this circumstance there is a new thing happening, because the total energy of the system is finite, so when one pendulum pours its energy into the other to drive it, it finds itself gradually losing energy, until, if the timing is just right along with the speed, it loses all its energy and is reduced to a stationary condition! Then, of course, it is the other pendulum ball that has all the energy and the first one which has none, and as time goes on we see that it works also in the opposite direction, and that the energy is passed back into the first ball; this is a very interesting and amusing phenomenon. We said, however, that this is related to the theory of beats, and we must now explain how we can analyze this motion from the point of view of the theory of beats.

We note that the motion of either of the two balls is an oscillation which has an amplitude which changes cyclically. Therefore the motion of one of the balls is presumably analyzable in a different way, in that it is the sum of two oscillations, present at the same time but having two slightly different frequencies. Therefore it ought to be possible to find two other motions in this system, and to claim that what we saw was a superposition of the two solutions, because this is of course a linear system. Indeed, it is easy to find two ways that we could start the motion, each one of which is a perfect, single-frequency motion—absolutely periodic. The motion that we started with before was not strictly periodic, since it did not last; soon one ball was passing energy to the other and so changing its amplitude; but there are ways of starting the motion so that nothing changes and, of course, as soon as we see it we understand why. For example, if we made both pendulums go together, then, since they are of the same length and the spring is not then doing anything, they will of course continue to swing like that for all time, assuming no friction and that everything is perfect. On the other hand, there is another possible motion which also has a definite frequency: that is, if we move the pendulums oppositely, pulling them aside exactly equal distances, then again they would be in absolutely periodic motion. We can appreciate that the spring just adds a little to the restoring force that the gravity supplies, that is all, and the system just keeps oscillating at a slightly higher frequency than in the first case. Why higher? Because the spring is pulling, in addition to the gravitation, and it makes the system a little “stiffer,” so that the frequency of this motion is just a shade higher than that of the other.

Thus this system has two ways in which it can oscillate with unchanging amplitude: it can either oscillate in a manner in which both pendulums go the same way and oscillate all the time at one frequency, or they could go in opposite directions at a slightly higher frequency.

Now the actual motion of the thing, because the system is linear, can be represented as a superposition of the two. (The subject of this chapter, remember,

is the effects of adding two motions with different frequencies.) So think what would happen if we combined these two solutions. If at $t = 0$ the two motions are started with equal amplitude and in the same phase, the sum of the two motions means that one ball, having been impressed one way by the first motion and the other way by the second motion, is at zero, while the other ball, having been displaced the same way in both motions, has a large amplitude. As time goes on, however, the two basic *motions* proceed independently, so the phase of one relative to the other is slowly shifting. That means, then, that after a sufficiently long time, when the time is enough that one motion could have gone “ $900\frac{1}{2}$ ” oscillations, while the other went only “900,” the relative phase would be just reversed with respect to what it was before. That is, the large-amplitude motion will have fallen to zero, and in the meantime, of course, the initially motionless ball will have attained full strength!

So we see that we could analyze this complicated motion either by the idea that there is a resonance and that one passes energy to the other, or else by the superposition of two constant-amplitude motions at two different frequencies.

Modes

49-1 The reflection of waves

This chapter will consider some of the remarkable phenomena which are a result of confining waves in some finite region. We will be led first to discover a few particular facts about vibrating strings, for example, and then the generalization of these facts will give us a principle which is probably the most far-reaching principle of mathematical physics.

Our first example of confining waves will be to confine a wave at one boundary. Let us take the simple example of a one-dimensional wave on a string. One could equally well consider sound in one dimension against a wall, or other situations of a similar nature, but the example of a string will be sufficient for our present purposes. Suppose that the string is held at one end, for example by fastening it to an "infinitely solid" wall. This can be expressed mathematically by saying that the displacement y of the string at the position $x = 0$ must be zero, because the end does not move. Now if it were not for the wall, we know that the general solution for the motion is the sum of two functions, $F(x - ct)$ and $G(x + ct)$, the first representing a wave travelling one way in the string, and the second a wave travelling the other way in the string:

$$y = F(x - ct) + G(x + ct) \quad (49.1)$$

is the general solution for any string. But we have next to satisfy the condition that the string does not move at one end. If we put $x = 0$ in Eq. (49.1) and examine y for any value of t , we get $y = F(-ct) + G(+ct)$. Now if this is to be zero for all times, it means that the function $G(ct)$ must be $-F(-ct)$. In other words, G of anything must be $-F$ of minus that same thing. If this result is put back into Eq. (49.1), we find that the solution for the problem is

$$y = F(x - ct) - F(-x - ct). \quad (49.2)$$

It is easy to check that we will get $y = 0$ if we set $x = 0$.

Figure 49-1 shows a wave travelling in the negative x -direction near $x = 0$, and a hypothetical wave travelling in the other direction reversed in sign and on the other side of the origin. We say hypothetical because, of course, there is no string to vibrate on that side of the origin. The total motion of the string is to be regarded as the sum of these two waves in the region of positive x . As they reach the origin, they will always cancel at $x = 0$, and finally the second (reflected) wave will be the only one to exist for positive x and it will, of course, be travelling in the opposite direction. These results are equivalent to the following statement: if a wave reaches the clamped end of a string, it will be reflected with a change in sign. Such a reflection can always be understood by imagining that what is coming to the end of the string comes out upside down from behind the wall. In short, if we assume that the string is infinite and that whenever we have a wave going one way we have another one going the other way with the stated symmetry, the displacement at $x = 0$ will always be zero and it would make no difference if we clamped the string there.

The next point to be discussed is the reflection of a periodic wave. Suppose that the wave represented by $F(x - ct)$ is a sine wave and has been reflected; then the reflected wave $-F(-x - ct)$ is also a sine wave of the same frequency, but travelling in the opposite direction. This situation can be most simply described by using the complex function notation: $F(x - ct) = e^{i\omega(t - x/c)}$ and

49-1 The reflection of waves

49-2 Confined waves, with natural frequencies

49-3 Modes in two dimensions

49-4 Coupled pendulums

49-5 Linear systems

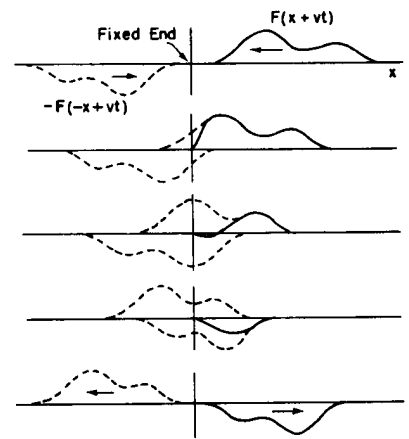


Fig. 49-1. Reflection of a wave at a fixed end as a superposition of two travelling waves.

$F(-x - ct) = e^{i\omega(t+x/c)}$. It can be seen that if these are substituted in (49.2) and if x is set equal to 0, then $y = 0$ for all values of t , so it satisfies the necessary condition. Because of the properties of exponentials, this can be written in a simpler form:

$$y = e^{i\omega t}(e^{-i\omega x/c} - e^{i\omega x/c}) = -2ie^{i\omega t} \sin(\omega x/c). \quad (49.3)$$

There is something interesting and new here, in that this solution tells us that if we look at any fixed x , the string oscillates at frequency ω . No matter where this point is, the frequency is the same! But there are some places, in particular wherever $\sin(\omega x/c) = 0$, where there is no displacement at all. Furthermore, if at any time t we take a snapshot of the vibrating string, the picture will be a sine wave. However, the displacement of this sine wave will depend upon the time t . From inspection of Eq. (49.3) we can see that the length of one cycle of the sine wave is equal to the wavelength of either of the superimposed waves:

$$\lambda = 2\pi c/\omega. \quad (49.4)$$

The points where there is no motion satisfy the condition $\sin(\omega x/c) = 0$, which means that $(\omega x/c) = 0, \pi, 2\pi, \dots, n\pi \dots$. These points are called *nodes*. Between any two successive nodes, every point moves up and down sinusoidally, but the pattern of motion stays fixed in space. This is the fundamental characteristic of what we call a *mode*. If one can find a pattern of motion which has the property that at any point the object moves perfectly sinusoidally, and that all points move at the same frequency (though some will move more than others), then we have what is called a mode.

49-2 Confined waves, with natural frequencies

The next interesting problem is to consider what happens if the string is held at both ends, say at $x = 0$ and $x = L$. We can begin with the idea of the reflection of waves, starting with some kind of a bump moving in one direction. As time goes on, we would expect the bump to get near one end, and as time goes still further it will become a kind of little wobble, because it is combining with the reversed-image bump which is coming from the other side. Finally the original bump will disappear and the image bump will move in the other direction to repeat the process at the other end. This problem has an easy solution, but an interesting question is whether we can have a sinusoidal motion (the solution just described is *periodic*, but of course it is not *sinusoidally* periodic). Let us try to put a sinusoidally periodic wave on a string. If the string is tied at one end, we know it must look like our earlier solution (49.3). If it is tied at the other end, it has to look the same at the other end. So the only possibility for periodic sinusoidal motion is that the sine wave must neatly fit into the string length. If it does not fit into the string length, then it is not a natural frequency at which the string can continue to oscillate. In short, if the string is started with a sine wave shape that just fits in, then it will continue to keep that perfect shape of a sine wave and will oscillate harmonically at some frequency.

Mathematically, we can write $\sin kx$ for the shape, where k is equal to the factor (ω/c) in Eqs. (49.3) and (49.4), and this function will be zero at $x = 0$. However, it must also be zero at the other end. The significance of this is that k is no longer arbitrary, as was the case for the half-open string. With the string closed at both ends, the only possibility is that $\sin(kL) = 0$, because this is the only condition that will keep both ends fixed. Now in order for a sine to be zero, the angle must be either 0, π , 2π , or some other integral multiple of π . The equation

$$kL = n\pi \quad (49.5)$$

will, therefore, give any one of the possible k 's, depending on what integer is put in. For each of the k 's there is a certain frequency ω , which, according to (49.3), is simply

$$\omega = kc = n\pi c/L. \quad (49.6)$$

So we have found the following: that a string has a property that it can have sinusoidal motions, *but only at certain frequencies*. This is the most important characteristic of confined waves. No matter how complicated the system is, it always turns out that there are some patterns of motion which have a perfect sinusoidal time dependence, but with frequencies that are a property of the particular system and the nature of its boundaries. In the case of the string we have many different possible frequencies, each one, by definition, corresponding to a mode, because a mode is a pattern of motion which repeats itself sinusoidally. Figure 49-2 shows the first three modes for a string. For the first mode the wavelength λ is $2L$. This can be seen if one continues the wave out to $x = 2L$ to obtain one complete cycle of the sine wave. The angular frequency ω is $2\pi c$ divided by the wavelength, in general, and in this case, since λ is $2L$, the frequency is $\pi c/L$, which is in agreement with (49.6) with $n = 1$. Let us call the first mode frequency ω_1 . Now the next mode shows two loops with one node in the middle. For this mode the wavelength, then, is simply L . The corresponding value of k is twice as great and the frequency is twice as large; it is $2\omega_1$. For the third mode it is $3\omega_1$, and so on. So all the different frequencies of the string are multiples, 1, 2, 3, 4, and so on, of the lowest frequency ω_1 .

Returning now to the general motion of the string, it turns out that any possible motion can always be analyzed by asserting that more than one mode is operating at the same time. In fact, for general motion an infinite number of modes must be excited at the same time. To get some idea of this, let us illustrate what happens when there are two modes oscillating at the same time: Suppose that we have the first mode oscillating as shown by the sequence of pictures in Fig. 49-3, which illustrates the deflection of the string for equally spaced time intervals extending through half a cycle of the lowest frequency.

Now, at the same time, we suppose that there is an oscillation of the second mode also. Figure 49-3 also shows a sequence of pictures of this mode, which at the start is 90° out of phase with the first mode. This means that at the start it has no displacement, but the two halves of the string have oppositely directed velocities. Now we recall a general principle relating to linear systems: if there are any two solutions, then their sum is also a solution. Therefore a third possible motion of the string would be a displacement obtained by adding the two solutions shown in Fig. 49-3. The result, also shown in the figure, begins to suggest the idea of a bump running back and forth between the ends of the string, although with only two modes we cannot make a very good picture of it; more modes are needed. This result is, in fact, a special case of a great principle for linear systems:

Any motion at all can be analyzed by assuming that it is the sum of the motions of all the different modes, combined with appropriate amplitudes and phases.

The importance of the principle derives from the fact that each mode is very simple—it is nothing but a sinusoidal motion in time. It is true that even the general motion of a string is not really very complicated, but there are other systems, for example the whipping of an airplane wing, in which the motion is much more complicated. Nevertheless, even with an airplane wing, we find there is a certain particular way of twisting which has one frequency and other ways of twisting that have other frequencies. If these modes can be found, then the complete motion can always be analyzed as a superposition of harmonic oscillations (except when the whipping is of such degree that the system can no longer be considered as linear).

49-3 Modes in two dimensions

The next example to be considered is the interesting situation of modes in two dimensions. Up to this point we have talked only about one-dimensional situations—a stretched string or sound waves in a tube. Ultimately we should consider three dimensions, but an easier step will be that to two dimensions. Consider for definiteness a rectangular rubber drumhead which is confined so as to have no displacement anywhere on the rectangular edge, and let the dimensions of the rectangle

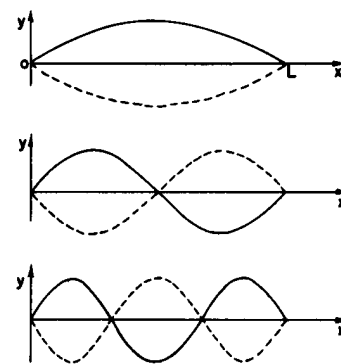


Fig. 49-2. The first three modes of a vibrating string.

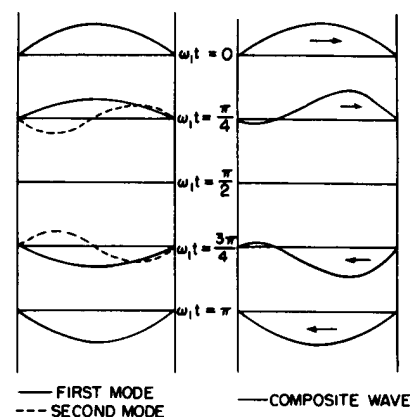


Fig. 49-3. Two modes combine to give a travelling wave.

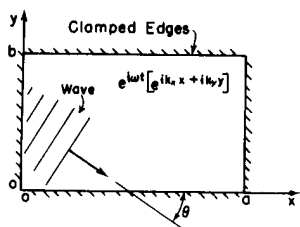


Fig. 49-4. Vibrating rectangular plate.

be a and b , as shown in Fig. 49-4. Now the question is, what are the characteristics of the possible motion? We can start with the same procedure used for the string. If we had no confinement at all, we would expect waves travelling along with some kind of wave motion. For example, $(e^{i\omega t})(e^{-ik_x x + ik_y y})$ would represent a sine wave travelling in some direction which depends on the relative values of k_x and k_y . Now how can we make the x -axis, that is, the line $y = 0$, a node? Using the ideas developed for the one-dimensional string, we can imagine another wave represented by the complex function $(-e^{i\omega t})(e^{-ik_x x - ik_y y})$. The superposition of these waves will give zero displacement at $y = 0$ regardless of the values of x and t . (Although these functions are defined for negative y where there is no drumhead to vibrate, this can be ignored, since the displacement is truly zero at $y = 0$.) In this case we can look upon the second function as the reflected wave.

However, we want a nodal line at $y = b$ as well as at $y = 0$. How do we do that? The solution is related to something we did when studying reflection from crystals. These waves which cancel each other at $y = 0$ will do the same at $y = b$ only if $2b \sin \theta$ is an integral multiple of λ , where θ is the angle shown in Fig. 49-4:

$$m\lambda = 2b \sin \theta, \quad m = 0, 1, 2, \dots \quad (49.7)$$

Now in the same way we can make the y -axis a nodal line by adding two more functions $-(e^{i\omega t})(e^{+ik_x x + ik_y y})$ and $+(e^{i\omega t})(e^{+ik_x x - ik_y y})$, each representing a reflection of one of the other two waves from the $x = 0$ line. The condition for a nodal line at $x = a$ is similar to the one for $y = b$. It is that $2a \cos \theta$ must also be an integral multiple of λ :

$$n\lambda = 2a \cos \theta. \quad (49.8)$$

Then the final result is that the waves bouncing about in the box produce a standing-wave pattern, that is, a definite mode.

So we must satisfy the above two conditions if we are to have a mode. Let us first find the wavelength. This can be obtained by eliminating the angle θ from (49.7) and (49.8) to obtain the wavelength in terms of a , b , n and m . The easiest way to do that is to divide both sides of the respective equations by $2b$ and $2a$, square them, and add the two equations together. The result is $\sin^2 \theta + \cos^2 \theta = 1 = (n\lambda/2a)^2 + (m\lambda/2b)^2$, which can be solved for λ :

$$\frac{1}{\lambda^2} = \frac{n^2}{4a^2} + \frac{m^2}{4b^2}. \quad (49.9)$$

In this way we have determined the wavelength in terms of two integers, and from the wavelength we immediately get the frequency ω , because, as we know, the frequency is equal to $2\pi c$ divided by the wavelength.

This result is interesting and important enough that we should deduce it by a purely mathematical analysis instead of by an argument about the reflections. Let us represent the vibration by a superposition of four waves chosen so that the four lines $x = 0$, $x = a$, $y = 0$, and $y = b$ are all nodes. In addition we shall require that all waves have the same frequency, so that the resulting motion will represent a mode. From our earlier treatment of light reflection we know that $(e^{i\omega t})(e^{-ik_x x + ik_y y})$ represents a wave travelling in the direction indicated in Fig. 49-4. Equation (49.6), that is, $k = \omega/c$, still holds, provided

$$k^2 = k_x^2 + k_y^2. \quad (49.10)$$

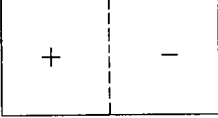
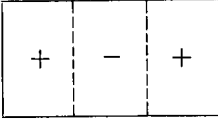
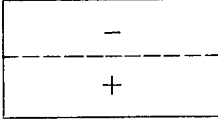
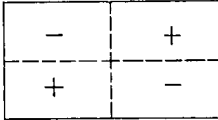
It is clear from the figure that $k_x = k \cos \theta$ and $k_y = k \sin \theta$.

Now our equation for the displacement, say ϕ , of the rectangular drumhead takes on the grand form

$$\phi = [e^{i\omega t}][e^{(-ik_x x + ik_y y)} - e^{(+ik_x x + ik_y y)} - e^{(-ik_x x - ik_y y)} + e^{(+ik_x x - ik_y y)}]. \quad (49.11a)$$

Although this looks rather a mess, the sum of these things now is not very hard.

Table 49-1

Mode shape	m	n	$(\omega/\omega_0)^2$	ω/ω_0
	1	1	1.25	1.12
	1	2	2.00	1.41
	1	3	3.25	1.80
	2	1	4.25	2.06
	2	2	5.00	2.24

The exponentials can be combined to give sine functions, so that the displacement turns out to be

$$\phi = [-4 \sin k_x x \sin k_y y][e^{i\omega t}]. \quad (49.11b)$$

In other words, it is a sinusoidal oscillation, all right, with a pattern that is also sinusoidal in both the x - and the y -direction. Our boundary conditions are of course satisfied at $x = 0$ and $y = 0$. We also want ϕ to be zero when $x = a$ and when $y = b$. Therefore we have to put in two other conditions: $k_x a$ must be an integral multiple of π , and $k_y b$ must be another integral multiple of π . Since we have seen that $k_x = k \cos \theta$ and $k_y = k \sin \theta$, we immediately get equations (49.7) and (49.8) and from these the final result (49.9).

Now let us take as an example a rectangle whose width is twice the height. If we take $a = 2b$ and use Eqs. (49.4) and (49.9), we can calculate the frequencies of all of the modes:

$$\omega^2 = \left(\frac{\pi c}{b}\right)^2 \frac{4m^2 + n^2}{4}. \quad (49.12)$$

Table 49-1 lists a few of the simple modes and also shows their shape in a qualitative way.

The most important point to be emphasized about this particular case is that the frequencies are not multiples of each other, nor are they multiples of any number. The idea that the natural frequencies are harmonically related is not generally true. It is not true for a system of more than one dimension, nor is it true for one-dimensional systems which are more complicated than a string with uniform density and tension. A simple example of the latter is a hanging chain in which the tension is higher at the top than at the bottom. If such a chain is set in harmonic oscillation, there are various modes and frequencies, but the frequencies are not simple multiples of any number, nor are the mode shapes sinusoidal.

The modes of more complicated systems are still more elaborate. For example, inside the mouth we have a cavity above the vocal cords, and by moving the tongue and the lips, and so forth, we make an open-ended pipe or a closed-ended pipe of different diameters and shapes; it is a terribly complicated resonator, but

it is a resonator nevertheless. Now when one talks with the vocal cords, they are made to produce some kind of tone. The tone is rather complicated and there are many sounds coming out, but the cavity of the mouth further modifies that tone because of the various resonant frequencies of the cavity. For instance, a singer can sing various vowels, a, or o, or oo, and so forth, at the same pitch, but they sound different because the various harmonics are in resonance in this cavity to different degrees. The very great importance of the resonant frequencies of a cavity in modifying the voice sounds can be demonstrated by a simple experiment. Since the speed of sound goes as the reciprocal of the square root of the density, the speed of sound may be varied by using different gases. If one uses helium instead of air, so that the density is lower, the speed of sound is much higher, and all the frequencies of a cavity will be raised. Consequently if one fills one's lungs with helium before speaking, the character of his voice will be drastically altered even though the vocal cords may still be vibrating at the same frequency.

49-4 Coupled pendulums

Finally we should emphasize that not only do modes exist for complicated continuous systems, but also for very simple mechanical systems. A good example is the system of two coupled pendulums discussed in the preceding chapter. In that chapter it was shown that the motion could be analyzed as a superposition of two harmonic motions with different frequencies. So even this system can be analyzed in terms of harmonic motions or modes. The string has an infinite number of modes and the two-dimensional surface also has an infinite number of modes. In a sense it is a double infinity, if we know how to count infinities. But a simple mechanical thing which has only two degrees of freedom, and requires only two variables to describe it, has only two modes.

Let us make a mathematical analysis of these two modes for the case where the pendulums are of equal length. Let the displacement of one be x , and the displacement of the other be y , as shown in Fig. 49-5. Without a spring, the force on the first mass is proportional to the displacement of that mass, because of gravity. There would be, if there were no spring, a certain natural frequency ω_0 for this one alone. The equation of motion without a spring would be

$$m \frac{d^2 x}{dt^2} = -m\omega_0^2 x. \quad (49.13)$$

The other pendulum would swing in the same way if there were no spring. In addition to the force of restoration due to gravitation, there is an additional force pulling the first mass. That force depends upon the excess distance of x over y and is proportional to that difference, so it is some constant which depends on the geometry, times $(x - y)$. The same force in reverse sense acts on the second mass. The equations of motion that have to be solved are therefore

$$m \frac{d^2 x}{dt^2} = -m\omega_0^2 x - k(x - y), \quad m \frac{d^2 y}{dt^2} = -m\omega_0^2 y - k(y - x). \quad (49.14)$$

In order to find a motion in which both of the masses move at the same frequency, we must determine how much each mass moves. In other words, pendulum x and pendulum y will oscillate at the same frequency, but their amplitudes must have certain values, A and B , whose relation is fixed. Let us try this solution:

$$x = Ae^{i\omega t}, \quad y = Be^{i\omega t}. \quad (49.15)$$

If these are substituted in Eqs. (49.14) and similar terms are collected, the results are

$$\begin{aligned} \left(\omega^2 - \omega_0^2 - \frac{k}{m} \right) A &= -\frac{k}{m} B, \\ \left(\omega^2 - \omega_0^2 - \frac{k}{m} \right) B &= -\frac{k}{m} A. \end{aligned} \quad (49.16)$$

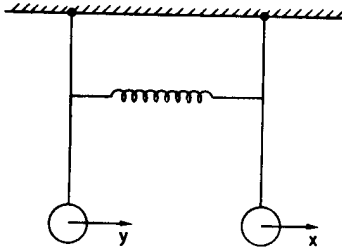


Fig. 49-5. Two coupled pendulums.

The equations as written have had the common factor $e^{i\omega t}$ removed and have been divided by m .

Now we see that we have two equations for what looks like two unknowns. But there really are not *two* unknowns, because the whole size of the motion is something that we cannot determine from these equations. The above equations can determine only the *ratio* of A to B , but they must both give the same ratio. The necessity for both of these equations to be consistent is a requirement that the frequency be something very special.

In this particular case this can be worked out rather easily. If the two equations are multiplied together, the result is

$$\left(\omega^2 - \omega_0^2 - \frac{k}{m}\right)^2 AB = \left(\frac{k}{m}\right)^2 AB. \quad (49.17)$$

The term AB can be removed from both sides unless A and B are zero, which means there is no motion at all. If there is motion, then the other terms must be equal, giving a quadratic equation to solve. The result is that there are two possible frequencies:

$$\omega_1^2 = \omega_0^2, \quad \omega_2^2 = \omega_0^2 + \frac{2k}{m}. \quad (49.18)$$

Furthermore, if these values of frequency are substituted back into Eq. (49.16), we find that for the first frequency $A = B$, and for the second frequency $A = -B$. These are the “mode shapes,” as can be readily verified by experiment.

It is clear that in the first mode, where $A = B$, the spring is never stretched, and both masses oscillate at the frequency ω_0 , as though the spring were absent. In the other solution, where $A = -B$, the spring contributes a restoring force and raises the frequency. A more interesting case results if the pendulums have different lengths. The analysis is very similar to that given above, and is left as an exercise for the reader.

49-5 Linear systems

Now let us summarize the ideas discussed above, which are all aspects of what is probably the most general and wonderful principle of mathematical physics. If we have a linear system whose character is independent of the time, then the motion does not have to have any particular simplicity, and in fact may be exceedingly complex, but there are very special motions, usually a series of special motions, in which the whole pattern of motion varies exponentially with the time. For the vibrating systems that we are talking about now, the exponential is imaginary, and instead of saying “exponentially” we might prefer to say “sinusoidally” with time. However, one can be more general and say that the motions will vary exponentially with the time in very special modes, with very special shapes. The most general motion of the system can always be represented as a superposition of motions involving each of the different exponentials.

This is worth stating again for the case of sinusoidal motion: a linear system need not be moving in a purely sinusoidal motion, i.e., at a definite single frequency, but no matter how it does move, this motion can be represented as a superposition of pure sinusoidal motions. The frequency of each of these motions is a characteristic of the system, and the pattern or waveform of each motion is also a characteristic of the system. The general motion in any such system can be characterized by giving the strength and the phase of each of these modes, and adding them all together. Another way of saying this is that any linear vibrating system is equivalent to a set of independent harmonic oscillators, with the natural frequencies corresponding to the modes.

We conclude this chapter by remarking on the connection of modes with quantum mechanics. In quantum mechanics the vibrating object, or the thing that varies in space, is the amplitude of a probability function that gives the probability of finding an electron, or system of electrons, in a given configuration. This amplitude function can vary in space and time, and satisfies, in fact, a linear equation.

But in quantum mechanics there is a transformation, in that what we call frequency of the probability amplitude is equal, in the classical idea, to energy. Therefore we can translate the principle stated above to this case by taking the word *frequency* and replacing it with *energy*. It becomes something like this: a quantum-mechanical system, for example an atom, need not have a definite energy, just as a simple mechanical system does not have to have a definite frequency; but no matter how the system behaves, its behavior can always be represented as a superposition of states of definite energy. The energy of each state is a characteristic of the atom, and so is the pattern of amplitude which determines the probability of finding particles in different places. The general motion can be described by giving the amplitude of each of these different energy states. This is the origin of energy levels in quantum mechanics. Since quantum mechanics is represented by waves, in the circumstance in which the electron does not have enough energy to ultimately escape from the proton, they are *confined waves*. Like the confined waves of a string, there are definite frequencies for the solution of the wave equation for quantum mechanics. The quantum-mechanical interpretation is that these are definite *energies*. Therefore a quantum-mechanical system, because it is represented by waves, can have definite states of fixed energy; examples are the energy levels of various atoms.

Harmonics

50-1 Musical tones

Pythagoras is said to have discovered the fact that two similar strings under the same tension and differing only in length, when sounded together give an effect that is pleasant to the ear *if* the lengths of the strings are in the ratio of two small integers. If the lengths are as one is to two, they then correspond to the octave in music. If the lengths are as two is to three, they correspond to the interval between *C* and *G*, which is called a fifth. These intervals are generally accepted as “pleasant” sounding chords.

Pythagoras was so impressed by this discovery that he made it the basis of a school—Pythagoreans they were called—which held mystic beliefs in the great powers of numbers. It was believed that something similar would be found out about the planets—or “spheres.” We sometimes hear the expression: “the music of the spheres.” The idea was that there would be some numerical relationships between the orbits of the planets or between other things in nature. People usually think that this is just a kind of superstition held by the Greeks. But is it so different from our own scientific interest in quantitative relationships? Pythagoras’ discovery was the first example, outside geometry, of any numerical relationship in nature. It must have been very surprising to suddenly discover that there was a *fact* of nature that involved a simple numerical relationship. Simple measurements of lengths gave a prediction about something which had no apparent connection to geometry—the production of pleasant sounds. This discovery led to the extension that perhaps a good tool for understanding nature would be arithmetic and mathematical analysis. The results of modern science justify that point of view.

Pythagoras could only have made his discovery by making an experimental observation. Yet this important aspect does not seem to have impressed him. If it had, physics might have had a much earlier start. (It is always easy to look back at what someone else has done and to decide what he *should* have done!)

We might remark on a third aspect of this very interesting discovery: that the discovery had to do with two notes that *sound pleasant* to the ear. We may question whether we are any better off than Pythagoras in understanding *why* only certain sounds are pleasant to our ear. The general theory of aesthetics is probably no further advanced now than in the time of Pythagoras. In this one discovery of the Greeks, there are the three aspects: experiment, mathematical relationships, and aesthetics. Physics has made great progress on only the first two parts. This chapter will deal with our present-day understanding of the discovery of Pythagoras.

Among the sounds that we hear, there is one kind that we call *noise*. Noise corresponds to a sort of irregular vibration of the eardrum that is produced by the irregular vibration of some object in the neighborhood. If we make a diagram to indicate the pressure of the air on the eardrum (and, therefore, the displacement of the drum) as a function of time, the graph which corresponds to a noise might look like that shown in Fig. 50-1(a). (Such a noise might correspond roughly to the sound of a stamped foot.) The sound of *music* has a different character. Music is characterized by the presence of more-or-less *sustained tones*—or musical “notes.” (Musical instruments may make noises as well!) The tone may last for a relatively short time, as when a key is pressed on a piano, or it may be sustained almost indefinitely, as when a flute player holds a long note.

What is the special character of a musical note from the point of view of the pressure in the air? A musical note differs from a noise in that there is a periodicity in its graph. There is some uneven shape to the variation of the air pressure with

50-1 Musical tones

50-2 The Fourier series

50-3 Quality and consonance

50-4 The Fourier coefficients

50-5 The energy theorem

50-6 Nonlinear responses

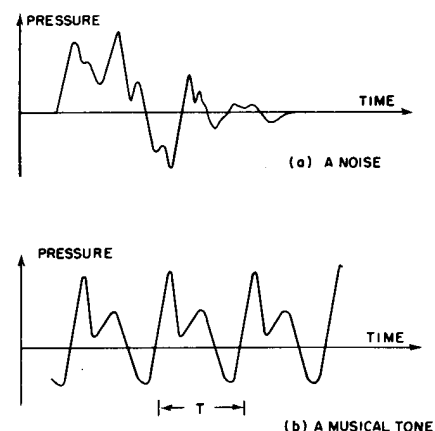


Fig. 50-1. Pressure as a function of time for (a) a noise, and (b) a musical tone.

time, and the shape repeats itself over and over again. An example of a pressure-time function that would correspond to a musical note is shown in Fig. 50-1(b).

Musicians will usually speak of a musical tone in terms of three characteristics: the loudness, the pitch, and the "quality." The "loudness" is found to correspond to the magnitude of the pressure changes. The "pitch" corresponds to the period of time for one repetition of the basic pressure function. ("Low" notes have longer periods than "high" notes.) The "quality" of a tone has to do with the differences we may still be able to hear between two notes of the same loudness and pitch. An oboe, a violin, or a soprano are still distinguishable even when they sound notes of the same pitch. The quality has to do with the structure of the repeating pattern.

Let us consider, for a moment, the sound produced by a vibrating string. If we pluck the string, by pulling it to one side and releasing it, the subsequent motion will be determined by the motions of the waves we have produced. We know that these waves will travel in both directions, and will be reflected at the ends. They will slosh back and forth for a long time. No matter how complicated the wave is, however, it will repeat itself. The period of repetition is just the time T required for the wave to travel two full lengths of the string. For that is just the time required for any wave, once started, to reflect off each end and return to its starting position, and be proceeding in the original direction. The time is the same for waves which start out in either direction. Each point on the string will, then, return to its starting position after one period, and again one period later, etc. The sound wave produced must also have the same repetition. We see why a plucked string produces a musical tone.

50-2 The Fourier series

We have discussed in the preceding chapter another way of looking at the motion of a vibrating system. We have seen that a string has various natural modes of oscillation, and that any particular kind of vibration that may be set up by the starting conditions can be thought of as a combination—in suitable proportions—of several of the natural modes, oscillating together. For a string we found that the normal modes of oscillation had the frequencies ω_0 , $2\omega_0$, $3\omega_0$, . . . The most general motion of a plucked string, therefore, is composed of the sum of a sinusoidal oscillation at the fundamental frequency ω_0 , another at the second harmonic frequency $2\omega_0$, another at the third harmonic $3\omega_0$, etc. Now the fundamental mode repeats itself every period $T_1 = 2\pi/\omega_0$. The second harmonic mode repeats itself every $T_2 = 2\pi/2\omega_0$. It *also* repeats itself every $T_1 = 2T_2$, after *two* of its periods. Similarly, the third harmonic mode repeats itself after a time T_1 which is 3 of its periods. We see again why a plucked string repeats its whole pattern with a periodicity of T_1 . It produces a musical tone.

We have been talking about the motion of the string. But the *sound*, which is the motion of the air, is produced by the motion of the string, so its vibrations too must be composed of the same harmonics—though we are no longer thinking about the normal modes of the air. Also, the relative strength of the harmonics may be different in the air than in the string, particularly if the string is "coupled" to the air via a sounding board. The efficiency of the coupling to the air is different for different harmonics.

If we let $f(t)$ represent the air pressure as a function of time for a musical tone [such as that in Fig. 50-1(b)], then we expect that $f(t)$ can be written as the sum of a number of simple harmonic functions of time—like $\cos \omega t$ —for each of the various harmonic frequencies. If the period of the vibration is T , the fundamental angular frequency will be $\omega = 2\pi/T$, and the harmonics will be 2ω , 3ω , etc.

There is one slight complication. For each frequency we may expect that the starting phases will not necessarily be the same for all frequencies. We should, therefore, use functions like $\cos(\omega t + \phi)$. It is, however, simpler to use instead both the sine and cosine functions for *each* frequency. We recall that

$$\cos(\omega t + \phi) = (\cos \phi \cos \omega t - \sin \phi \sin \omega t) \quad (50.1)$$

and since ϕ is a constant, *any* sinusoidal oscillation at the frequency ω can be written as the sum of a term with $\cos \omega t$ and another term with $\sin \omega t$.

We conclude, then, that *any* function $f(t)$ that is periodic with the period T can be written mathematically as

$$\begin{aligned} f(t) = & a_0 \\ & + a_1 \cos \omega t + b_1 \sin \omega t \\ & + a_2 \cos 2\omega t + b_2 \sin 2\omega t \\ & + a_3 \cos 3\omega t + b_3 \sin 3\omega t \\ & + \dots + \dots \end{aligned} \quad (50.2)$$

where $\omega = 2\pi/T$ and the a 's and b 's are numerical constants which tell us how much of each component oscillation is present in the oscillation $f(t)$. We have added the "zero-frequency" term a_0 so that our formula will be completely general, although it is usually zero for a musical tone. It represents a shift of the average value (that is, the "zero" level) of the sound pressure. With it our formula can take care of any case. The equality of Eq. (50.2) is represented schematically in Fig. 50-2. (The amplitudes, a_n and b_n , of the harmonic functions must be suitably chosen. They are shown schematically and without any particular scale in the figure.) The series (50.2) is called the *Fourier series* for $f(t)$.

We have said that *any* periodic function can be made up in this way. We should correct that and say that any sound wave, or any function we ordinarily encounter in physics, can be made up of such a sum. The mathematicians can invent functions which cannot be made up of simple harmonic functions—for instance, a function that has a "reverse twist" so that it has two values for some values of t ! We need not worry about such functions here.

50-3 Quality and consonance

Now we are able to describe what it is that determines the "quality" of a musical tone. It is the relative amounts of the various harmonics—the values of the a 's and b 's. A tone with only the first harmonic is a "pure" tone. A tone with many strong harmonics is a "rich" tone. A violin produces a different proportion of harmonics than does an oboe.

We can "manufacture" various musical tones if we connect several "oscillators" to a loudspeaker. (An oscillator usually produces a nearly pure simple harmonic function.) We should choose the frequencies of the oscillators to be ω , 2ω , 3ω , etc. Then by adjusting the volume control on each oscillator, we can add in any amount we wish of each harmonic—thereby producing tones of different quality. An electric organ works in much this way. The "keys" select the frequency of the fundamental oscillator and the "stops" are switches that control the relative proportions of the harmonics. By throwing these switches, the organ can be made to sound like a flute, or an oboe, or a violin.

It is interesting that to produce such "artificial" tones we need only one oscillator for each frequency—we do not need separate oscillators for the sine and cosine components. The ear is not very sensitive to the relative phases of the harmonics. It pays attention mainly to the *total* of the sine and cosine parts of each frequency. Our analysis is more accurate than is necessary to explain the *subjective* aspect of music. The response of a microphone or other physical instrument does depend on the phases, however, and our complete analysis may be needed to treat such cases.

The "quality" of a spoken sound also determines the vowel sounds that we recognize in speech. The shape of the mouth determines the frequencies of the natural modes of vibration of the air in the mouth. Some of these modes are set into vibration by the sound waves from the vocal chords. In this way, the amplitudes of some of the harmonics of the sound are increased with respect to others. When we change the shape of our mouth, harmonics of different frequencies are given preference. These effects account for the difference between an "e-e-e" sound and an "a-a-a" sound.

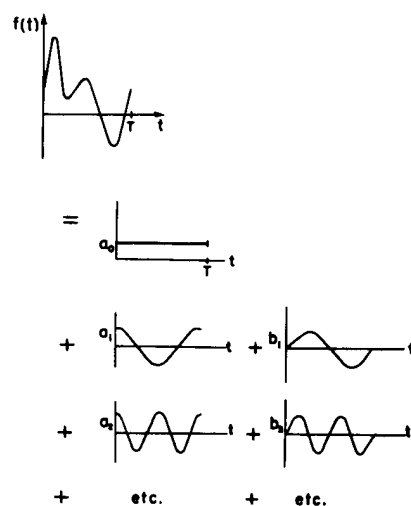


Fig. 50-2. Any periodic function $f(t)$ is equal to a sum of simple harmonic functions.

We all know that a particular vowel sound—say “e-e-e”—still “sounds like” the same vowel whether we say (or sing) it at a high or a low pitch. From the mechanism we describe, we would expect that *particular* frequencies are emphasized when we shape our mouth for an “e-e-e,” and that they *do not* change as we change the pitch of our voice. So the relation of the important harmonics to the fundamental—that is, the “quality”—changes as we change pitch. Apparently the mechanism by which we recognize speech is not based on specific harmonic relationships.

What should we say now about Pythagoras’ discovery? We understand that two similar strings with lengths in the ratio of 2 to 3 will have fundamental frequencies in the ratio 3 to 2. But why should they “sound pleasant” together? Perhaps we should take our clue from the frequencies of the harmonics. The second harmonic of the lower shorter string will have the *same* frequency as the third harmonic of the longer string. (It is easy to show—or to believe—that a plucked string produces strongly the several lowest harmonics.)

Perhaps we should make the following rules. Notes sound consonant when they have harmonics with the same frequency. Notes sound dissonant if their upper harmonics have frequencies near to each other but far enough apart that there are rapid beats between the two. Why beats do not sound pleasant, and why unison of the upper harmonics does sound pleasant, is something that we do not know how to define or describe. We cannot say from this knowledge of what *sounds* good, what ought, for example, to *smell* good. In other words, our understanding of it is not anything more general than the statement that when they are in unison they sound good. It does not permit us to deduce anything more than the properties of concordance in music.

It is easy to check on the harmonic relationships we have described by some simple experiments with a piano. Let us label the 3 successive C’s near the middle of the keyboard by C, C’, and C’’, and the G’s just above by G, G’, and G’’. Then the fundamentals will have relative frequencies as follows:

$$\begin{array}{ll} \text{C} - 2 & \text{G} - 3 \\ \text{C}' - 4 & \text{G}' - 6 \\ \text{C}'' - 8 & \text{G}'' - 12 \end{array}$$

These harmonic relationships can be demonstrated in the following way: Suppose we press C’ *slowly*—so that it does not sound but we cause the damper to be lifted. If we then sound C, it will produce its own fundamental *and* some second harmonic. The second harmonic will set the strings of C’ into vibration. If we now release C (keeping C’ pressed) the damper will stop the vibration of the C strings, and we can hear (softly) the note C’ as it dies away. In a similar way, the third harmonic of C can cause a vibration of G’. Or the sixth of C (now getting much weaker) can set up a vibration in the fundamental of G’.

A somewhat different result is obtained if we press G quietly and then sound C’. The third harmonic of C’ will correspond to the fourth harmonic of G, so *only* the fourth harmonic of G will be excited. We can hear (if we listen closely) the sound of G’’, which is two octaves above the G we have pressed! It is easy to think up many more combinations for this game.

We may remark in passing that the major scale can be defined just by the condition that the three major chords (F-A-C); (C-E-G); and (G-B-D) *each* represent tone sequences with the frequency ratio (4: 5: 6). These ratios—plus the fact that an octave (C-C’, B-B’, etc.) has the ratio 1: 2—determine the whole scale for the “ideal” case, or for what is called “just intonation.” Keyboard instruments like the piano are *not* usually tuned in this manner, but a little “fudging” is done so that the frequencies are *approximately* correct for all possible starting tones. For this tuning, which is called “tempered,” the octave (still 1: 2) is divided into 12 equal intervals for which the frequency ratio is $(2)^{1/12}$. A fifth no longer has the frequency ratio 3/2, but $2^{7/12} = 1.499$, which is apparently close enough for most ears.

We have stated a rule for consonance in terms of the coincidence of harmonics. Is this coincidence perhaps the *reason* that two notes are consonant? One worker has claimed that two *pure* tones—tones carefully manufactured to be free of harmonics—do not give the *sensations* of consonance or dissonance as the relative frequencies are placed at or near the expected ratios. (Such experiments are difficult because it is difficult to manufacture pure tones, for reasons that we shall see later.) We cannot still be certain whether the ear is matching harmonics or doing arithmetic when we decide that we like a sound.

50-4 The Fourier coefficients

Let us return now to the idea that any note—that is, a *periodic* sound—can be represented by a suitable combination of harmonics. We would like to show how we can find out what amount of each harmonic is required. It is, of course, easy to compute $f(t)$, using Eq. (50.2), if we are *given* all the coefficients a and b . The question now is, if we are given $f(t)$ how can we know what the coefficients of the various harmonic terms should be? (It is easy to make a cake from a recipe; but can we write down the recipe if we are given a cake?)

Fourier discovered that it was not really very difficult. The term a_0 is certainly easy. We have already said that it is just the average value of $f(t)$ over one period (from $t = 0$ to $t = T$). We can easily see that this is indeed so. The average value of a sine or cosine function over one period is zero. Over two, or three, or any whole number of periods, it is also zero. So the average value of all of the terms on the right-hand side of Eq. (50.2) is zero, except for a_0 . (Recall that we must choose $\omega = 2\pi/T$.)

Now the average of a sum is the sum of the averages. So the average of $f(t)$ is just the average of a_0 . But a_0 is a *constant*, so its average is just the same as its value. Recalling the definition of an average, we have

$$a_0 = \frac{1}{T} \int_0^T f(t) dt. \quad (50.3)$$

The other coefficients are only a little more difficult. To find them we can use a trick discovered by Fourier. Suppose we multiply both sides of Eq. (50.2) by some harmonic function—say by $\cos 7\omega t$. We have then

$$\begin{aligned} f(t) \cdot \cos 7\omega t &= a_0 \cdot \cos 7\omega t \\ &+ a_1 \cos \omega t \cdot \cos 7\omega t + b_1 \sin \omega t \cdot \cos 7\omega t \\ &+ a_2 \cos 2\omega t \cdot \cos 7\omega t + b_2 \sin 2\omega t \cdot \cos 7\omega t \\ &+ \cdots \qquad \qquad \qquad + \cdots \\ &+ a_7 \cos 7\omega t \cdot \cos 7\omega t + b_7 \sin 7\omega t \cdot \cos 7\omega t \\ &+ \cdots \qquad \qquad \qquad + \cdots \end{aligned} \quad (50.4)$$

Now let us average both sides. The average of $a_0 \cos 7\omega t$ over the time T is proportional to the average of a cosine over 7 whole periods. But that is just zero. The average of *almost all* of the rest of the terms is *also* zero. Let us look at the a_1 term. We know, in general, that

$$\cos A \cos B = \frac{1}{2} \cos (A + B) + \frac{1}{2} \cos (A - B). \quad (50.5)$$

The a_1 term becomes

$$\frac{1}{2} a_1 (\cos 8\omega t + \cos 6\omega t). \quad (50.6)$$

We thus have two cosine terms, one with 8 full periods in T and the other with 6. *They both average to zero.* The average of the a_1 term is therefore zero.

For the a_2 term, we would find $a_2 \cos 9\omega t$ and $a_2 \cos 5\omega t$, each of which also averages to zero. For the a_9 term, we would find $\cos 16\omega t$ and $\cos (-2\omega t)$. But $\cos (-2\omega t)$ is the same as $\cos 2\omega t$, so both of these have zero averages. It is clear

that *all* of the a terms will have a zero average *except* one. And that one is the a_7 term. For this one we have

$$\frac{1}{2}a_7(\cos 14\omega t + \cos 0). \quad (50.7)$$

The cosine of zero is one, and its average, of course, is one. So we have the result that the average of all of the a terms of Eq. (50.4) equals $\frac{1}{2}a_7$.

The b terms are even easier. When we multiply by any cosine term like $\cos n\omega t$, we can show by the same method that *all* of the b terms have the average value zero.

We see that Fourier's "trick" has acted like a sieve. When we multiply by $\cos 7\omega t$ and average, all terms drop out except a_7 , and we find that

$$\text{Average } [f(t) \cdot \cos 7\omega t] = a_7/2, \quad (50.8)$$

or

$$a_7 = \frac{2}{T} \int_0^T f(t) \cdot \cos 7\omega t \, dt. \quad (50.9)$$

We shall leave it for the reader to show that the coefficient b_7 can be obtained by multiplying Eq. (50.2) by $\sin 7\omega t$ and averaging both sides. The result is

$$b_7 = \frac{2}{T} \int_0^T f(t) \cdot \sin 7\omega t \, dt. \quad (50.10)$$

Now what is true for 7 we expect is true for any integer. So we can summarize our proof and result in the following more elegant mathematical form. If m and n are integers other than zero, and if $\omega = 2\pi/T$, then

$$\text{I. } \int_0^T \sin n\omega t \cos m\omega t \, dt = 0. \quad (50.11)$$

$$\text{II. } \int_0^T \cos n\omega t \cos m\omega t \, dt = \begin{cases} 0 & \text{if } n \neq m. \\ T/2 & \text{if } n = m. \end{cases} \quad (50.12)$$

$$\text{III. } \int_0^T \sin n\omega t \sin m\omega t \, dt = \begin{cases} 0 & \text{if } n \neq m. \\ T/2 & \text{if } n = m. \end{cases}$$

$$\text{IV. } f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t + \sum_{n=1}^{\infty} b_n \sin n\omega t. \quad (50.13)$$

$$\text{V. } a_0 = \frac{1}{T} \int_0^T f(t) \, dt. \quad (50.14)$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cdot \cos n\omega t \, dt. \quad (50.15)$$

$$b_n = \frac{2}{T} \int_0^T f(t) \cdot \sin n\omega t \, dt. \quad (50.16)$$

In earlier chapters it was convenient to use the exponential notation for representing simple harmonic motion. Instead of $\cos \omega t$ we used $\text{Re } e^{i\omega t}$, the real part of the exponential function. We have used cosine and sine functions in this chapter because it made the derivations perhaps a little clearer. Our final result of Eq. (50.13) can, however, be written in the compact form

$$f(t) = \text{Re } \sum_{n=0}^{\infty} \hat{a}_n e^{in\omega t}, \quad (50.17)$$

where $\hat{a}_n$ is the complex number $a_n - ib_n$ (with $b_0 = 0$). If we wish to use the same notation throughout, we can write also

$$a_n = \frac{2}{T} \int_0^T f(t) e^{-in\omega t} \, dt \quad (n \geq 1). \quad (50.18)$$

We now know how to “analyze” a periodic wave into its harmonic components. The procedure is called *Fourier analysis*, and the separate terms are called Fourier components. We have *not* shown, however, that once we find all of the Fourier components and add them together, we do indeed get back our $f(t)$. The mathematicians have shown, for a wide class of functions, in fact for all that are of interest to physicists, that if we can do the integrals we will get back $f(t)$. There is one minor exception. If the function $f(t)$ is discontinuous, i.e., if it jumps suddenly from one value to another, the Fourier sum will give a value at the breakpoint halfway between the upper and lower values at the discontinuity. So if we have the strange function $f(t) = 0, 0 \leq t < t_0$, and $f(t) = 1$ for $t_0 \leq t \leq T$, the Fourier sum will give the right value everywhere *except* at t_0 , where it will have the value $\frac{1}{2}$ instead of 1. It is rather unphysical anyway to insist that a function should be zero *up to* t_0 , but 1 *right at* t_0 . So perhaps we should make the “rule” for physicists that any discontinuous function (which can only be a simplification of a *real* physical function) should be defined with halfway values at the discontinuities. Then any such function—with any finite number of such jumps—as well as all other physically interesting functions, are given correctly by the Fourier sum.

As an exercise, we suggest that the reader determine the Fourier series for the function shown in Fig. 50-3. Since the function cannot be written in an explicit algebraic form, you will not be able to do the integrals from zero to T in the usual way. The integrals are easy, however, if we separate them into two parts: the integral from zero to $T/2$ (over which $f(t) = 1$) and the integral from $T/2$ to T (over which $f(t) = -1$). The result should be

$$f(t) = \frac{4}{\pi} (\sin \omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{5} \sin 5\omega t + \dots), \quad (50.19)$$

where $\omega = 2\pi/T$. We thus find that our square wave (with the particular phase chosen) has only odd harmonics, and their amplitudes are in inverse proportion to their frequencies.

Let us check that Eq. (50.19) does indeed give us back $f(t)$ for some value of t . Let us choose $t = T/4$, or $\omega t = \pi/2$. We have

$$f(t) = \frac{4}{\pi} \left(\sin \frac{\pi}{2} + \frac{1}{3} \sin \frac{3\pi}{2} + \frac{1}{5} \sin \frac{5\pi}{2} + \dots \right) \quad (50.20)$$

$$= \frac{4}{\pi} \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right) \quad (50.21)$$

The series* has the value $\pi/4$, and we find that $f(t) = 1$.

50-5 The energy theorem

The energy in a wave is proportional to the square of its amplitude. For a wave of complex shape, the energy in one period will be proportional to $\int_0^T f^2(t) dt$. We can also relate this energy to the Fourier coefficients. We write

$$\int_0^T f^2(t) dt = \int_0^T \left[a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t + \sum_{n=1}^{\infty} b_n \sin n\omega t \right]^2 dt. \quad (50.22)$$

When we expand the square of the bracketed term we will get all possible cross terms, such as $a_5 \cos 5\omega t \cdot b_7 \cos 7\omega t$. We have shown above, however, [Eqs. (50.11) and (50.12)] that the integrals of all such terms over one period is zero.

* The series can be evaluated in the following way. First we remark that $\int_0^x [dx/(1+x^2)] = \tan^{-1} x$. Second, we expand the integrand in a series $1/(1+x^2) = 1 - x^2 + x^4 - x^6 + \dots$. We integrate the series term by term (from zero to x) to obtain $\tan^{-1} x = x - x^3/3 + x^5/5 - x^7/7 + \dots$. Setting $x = 1$, we have the stated result, since $\tan^{-1} 1 = \pi/4$.

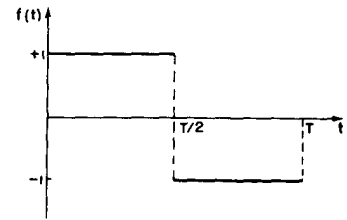


Fig. 50-3. Square-wave function.
 $f(t) = +1$ for $0 < t < T/2$,
 $f(t) = -1$ for $T/2 < t < T$.

We have left only the square terms like $a_5^2 \cos^2 5\omega t$. The integral of any cosine squared or sine squared over one period is equal to $T/2$, so we get

$$\begin{aligned} \int_0^T f^2(t) dt &= Ta_0^2 + \frac{T}{2} (a_1^2 + a_2^2 + \cdots + b_1^2 + b_2^2 + \cdots) \\ &= Ta_0^2 + \frac{T}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2). \end{aligned} \quad (50.23)$$

This equation is called the “energy theorem,” and says that the total energy in a wave is just the sum of the energies in all of the Fourier components. For example, applying this theorem to the series (50.19), since $[f(t)]^2 = 1$ we get

$$T = \frac{T}{2} \cdot \left(\frac{4}{\pi}\right)^2 \left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} \cdots\right),$$

so we learn that the sum of the squares of the reciprocals of the odd integers is $\pi^2/8$. In a similar way, by first obtaining the Fourier series for the function and using the energy theorem, we can prove that $1 + 1/2^4 + 1/3^4 + \cdots$ is $\pi^4/90$, a result we needed in Chapter 45.

50-6 Nonlinear responses

Finally, in the theory of harmonics there is an important phenomenon which should be remarked upon because of its practical importance—that of nonlinear effects. In all the systems that we have been considering so far, we have supposed that everything was linear, that the responses to forces, say the displacements or the accelerations, were always proportional to the forces. Or that the currents in the circuits were proportional to the voltages, and so on. We now wish to consider cases where there is not a strict proportionality. We think, at the moment, of some device in which the response, which we will call x_{out} at the time t , is determined by the input x_{in} at the time t . For example, x_{in} might be the force and x_{out} might be the displacement. Or x_{in} might be the current and x_{out} the voltage. If the device is linear, we would have

$$x_{out}(t) = Kx_{in}(t), \quad (50.24)$$

where K is a constant independent of t and of x_{in} . Suppose, however, that the device is nearly, but not exactly, linear, so that we can write

$$x_{out}(t) = K[x_{in}(t) + \epsilon x_{in}^2(t)], \quad (50.25)$$

where ϵ is small in comparison with unity. Such linear and nonlinear responses are shown in the graphs of Fig. 50-4.

Nonlinear responses have several important practical consequences. We shall discuss some of them now. First we consider what happens if we apply a pure tone at the input. We let $x_{in} = \cos \omega t$. If we plot x_{out} as a function of time we get the solid curve shown in Fig. 50-5. The dashed curve gives, for comparison, the response of a linear system. We see that the output is no longer a cosine function. It is more peaked at the top and flatter at the bottom. We say that the output is *distorted*. We know, however, that such a wave is no longer a pure tone, that it will have harmonics. We can find what the harmonics are. Using $x_{in} = \cos \omega t$ with Eq. (50.25), we have

$$x_{out} = K(\cos \omega t + \epsilon \cos^2 \omega t). \quad (50.26)$$

From the equality $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$, we have

$$x_{out} = K \left(\cos \omega t + \frac{\epsilon}{2} + \frac{\epsilon}{2} \cos 2\omega t \right). \quad (50.27)$$

The output has not only a component at the fundamental frequency, that was present at the input, but also has some of its second harmonic. There has also

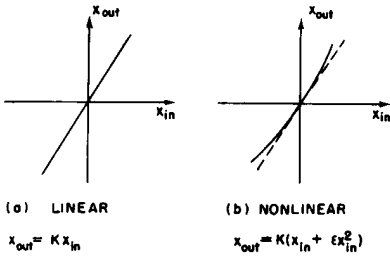


Fig. 50-4. Linear and nonlinear responses.

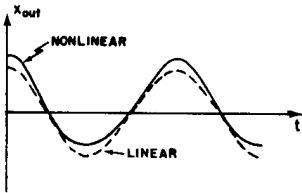


Fig. 50-5. The response of a nonlinear device to the input $\cos \omega t$. A linear response is shown for comparison.

appeared at the output a constant term $K(\epsilon/2)$, which corresponds to the shift of the average value, shown in Fig. 50-5. The process of producing a shift of the average value is called *rectification*.

A nonlinear response will rectify and will produce harmonics of the frequencies at its input. Although the nonlinearity we assumed produced only second harmonics, nonlinearities of higher order—those which have terms like x_{in}^3 and x_{in}^4 , for example—will produce harmonics higher than the second.

Another effect which results from a nonlinear response is *modulation*. If our input function contains two (or more) pure tones, the output will have not only their harmonics, but still other frequency components. Let $x_{in} = A \cos \omega_1 t + B \cos \omega_2 t$, where now ω_1 and ω_2 are *not* intended to be in a harmonic relation. In addition to the linear term (which is K times the input) we shall have a component in the output given by

$$x_{out} = K\epsilon(A \cos \omega_1 t + B \cos \omega_2 t)^2 \quad (50.28)$$

$$= K\epsilon(A^2 \cos^2 \omega_1 t + B^2 \cos^2 \omega_2 t + 2AB \cos \omega_1 t \cos \omega_2 t). \quad (50.29)$$

The first two terms in the parentheses of Eq. (50.29) are just those which gave the constant terms and second harmonic terms we found above. The last term is new.

We can look at this new “cross term” $AB \cos \omega_1 t \cos \omega_2 t$ in two ways. First, if the two frequencies are widely different (for example, if ω_1 is much greater than ω_2) we can consider that the cross term represents a cosine oscillation of varying amplitude. That is, we can think of the factors in this way:

$$AB \cos \omega_1 t \cos \omega_2 t = C(t) \cos \omega_1 t, \quad (50.30)$$

with

$$C(t) = AB \cos \omega_2 t. \quad (50.31)$$

We say that the amplitude of $\cos \omega_1 t$ is *modulated* with the frequency ω_2 .

Alternatively, we can write the cross term in another way:

$$AB \cos \omega_1 t \cos \omega_2 t = \frac{AB}{2} [\cos (\omega_1 + \omega_2)t + \cos (\omega_1 - \omega_2)t]. \quad (50.32)$$

We would now say that two *new* components have been produced, one at the *sum* frequency $(\omega_1 + \omega_2)$, another at the *difference* frequency $(\omega_1 - \omega_2)$.

We have two different, but equivalent, ways of looking at the same result. In the special case that $\omega_1 \gg \omega_2$, we can relate these two different views by remarking that since $(\omega_1 + \omega_2)$ and $(\omega_1 - \omega_2)$ are near to each other we would expect to observe beats between them. But these beats have just the effect of *modulating* the amplitude of the *average* frequency ω_1 by one-half the difference frequency $2\omega_2$. We see, then, why the two descriptions are equivalent.

In summary, we have found that a nonlinear response produces several effects: rectification, generation of harmonics, and modulation, or the generation of components with sum and difference frequencies.

We should notice that all these effects (Eq. 50.29) are proportional not only to the nonlinearity coefficient ϵ , but also to the product of two amplitudes—either A^2 , B^2 , or AB . We expect these effects to be much more important for *strong* signals than for weak ones.

The effects we have been describing have many practical applications. First, with regard to sound, it is believed that the ear is nonlinear. This is believed to account for the fact that with loud sounds we have the sensation that we *hear* harmonics and also sum and difference frequencies even if the sound waves contain only pure tones.

The components which are used in sound-reproducing equipment—amplifiers, loudspeakers, etc.—always have some nonlinearity. They produce distortions in the sound—they generate harmonics, etc.—which were not present in the original sound. These new components are heard by the ear and are apparently objectionable. It is for this reason that “Hi-Fi” equipment is designed to be as linear as

possible. (Why the nonlinearities of the *ear* are *not* “objectionable” in the same way, or how we even know that the nonlinearity is in the *loudspeaker* rather than in the *ear* is not clear!)

Nonlinearities are quite *necessary*, and are, in fact, intentionally made large in certain parts of radio transmitting and receiving equipment. In an AM transmitter the “voice” signal (with frequencies of some kilocycles per second) is combined with the “carrier” signal (with a frequency of some megacycles per second) in a nonlinear circuit called a *modulator*, to produce the modulated oscillation that is transmitted. In the receiver, the components of the received signal are fed to a nonlinear circuit which combines the sum and difference frequencies of the modulated carrier to generate again the voice signal.

When we discussed the transmission of light, we assumed that the induced oscillations of charges were proportional to the electric field of the light—that the response was linear. That is indeed a very good approximation. It is only within the last few years that light sources have been devised (lasers) which produce an intensity of light strong enough so that nonlinear effects can be observed. It is now possible to generate harmonics of light frequencies. When a strong red light passes through a piece of glass, a little bit of blue light—second harmonic—comes out!

Waves

51-1 Bow waves

Although we have finished our quantitative analyses of waves, this added chapter on the subject is intended to give some appreciation, qualitatively, for various phenomena that are associated with waves, which are too complicated to analyze in detail here. Since we have been dealing with waves for several chapters, more properly the subject might be called "some of the more complex phenomena associated with waves."

The first topic to be discussed concerns the effects that are produced by a source of waves which is moving faster than the wave velocity, or the phase velocity. Let us first consider waves that have a definite velocity, like sound and light. If we have a source of sound which is moving faster than the speed of sound, then something like this happens: Suppose at a given moment a sound wave is generated from the source at point x_1 in Fig. 51-1; then, in the next moment, as the source moves to x_2 , the wave from x_1 expands by a radius r_1 smaller than the distance that the source moves; and, of course, another wave starts from x_2 . When the sound source has moved still farther, to x_3 , and a wave is starting there, the wave from x_2 has now expanded to r_2 , and the one from x_1 has expanded to r_3 . Of course the thing is done continuously, not in steps, and therefore, we have a series of wave circles with a common tangent line which goes through the center of the source. We see that instead of a source generating spherical waves, as it would if it were standing still, it generates a wavefront which forms a cone in three dimensions, or a pair of lines in two dimensions. The angle of the cone is very easy to figure out. In a given amount of time the source moves a distance, say $x_3 - x_1$, proportional to v , the velocity of the source. In the meantime the wavefront has moved out a distance r_3 , proportional to c_w , the speed of the wave. Therefore it is clear that the half-angle of opening has a sine equal to the ratio of the speed of the waves, divided by the speed of the source, and this sine has a solution only if c_w is less than v , or the speed of the object is faster than the speed of the wave:

$$\sin \theta = \frac{c_w}{v}. \quad (51.1)$$

Incidentally, although we implied that it is necessary to have a *source* of sound, it turns out, very interestingly, that once the object is moving faster than the speed of sound, it will *make* sound. That is, it is not necessary that it have a certain tone vibrational character. Any object moving through a medium faster than the speed at which the medium carries waves will generate waves on each side, automatically, just from the motion itself. This is simple in the case of sound, but it also occurs in the case of light. At first one might think nothing can move faster than the speed of light. However, light in glass has a phase velocity less than the speed of light in a vacuum, and it is possible to shoot a charged particle of very high energy through a block of glass such that the particle velocity is close to the speed of light in a vacuum, while the speed of light in the glass may be only $\frac{2}{3}$ the speed of light in the vacuum. A particle moving faster than the speed of light in the medium will produce a conical wave of light with its apex at the source, like the wave wake from a boat (which is from the same effect, as a matter of fact). By measuring the cone angle, we can determine the speed of the particle. This is used technically to determine the speeds of particles as one of the methods of determining their energy in high-energy research. The direction of the light is all that needs to be measured.

51-1 Bow waves

51-2 Shock waves

51-3 Waves in solids

51-4 Surface waves

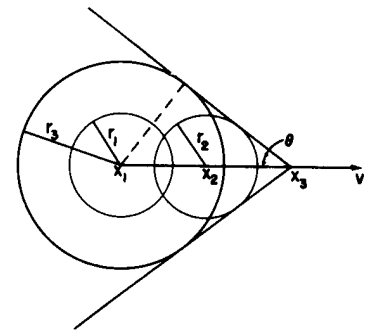


Fig. 51-1. The shock wave front lies on a cone with apex at the source and half-angle $\theta = \sin^{-1} v/c_w$.

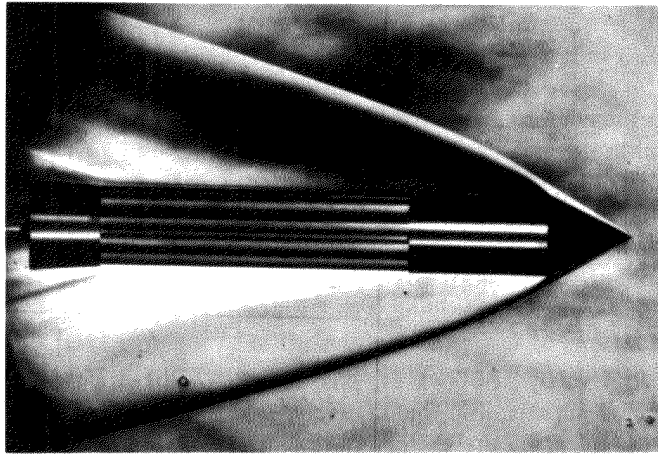


Fig. 51-2. A shock wave induced in a gas by a projectile moving faster than sound.

This light is sometimes called Čerenkov radiation, because it was first observed by Čerenkov. How intense this light should be was analyzed theoretically by Frank and Tamm. The 1958 Nobel Prize for physics was awarded jointly to all three for this work.

The corresponding circumstances in the case of sound are illustrated in Fig. 51-3, which is a photograph of an object moving through a gas at a speed greater than the speed of sound. The changes in pressure produce a change in refractive index, and with a suitable optical system the edges of the waves can be made visible. We see that the object moving faster than the speed of sound does, indeed, produce a conical wave. But closer inspection reveals that the surface is actually curved. It is straight asymptotically, but it is curved near the apex, and we have now to discuss how that can be, which brings us to the second topic of this chapter.

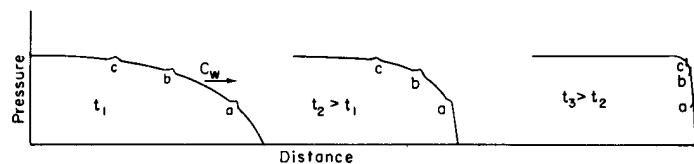


Fig. 51-3. Wavefront "snapshots" at successive instants in time.

51-2 Shock waves

Wave speed often depends on the amplitude, and in the case of sound the speed depends upon the amplitude in the following way. An object moving through the air has to move the air out of the way, so the disturbance produced in this case is some kind of a pressure step, with the pressure higher behind the wavefront than in the undisturbed region not yet reached by the wave (running along at the normal speed, say). But the air that is left behind, after the wavefront passes, has been compressed adiabatically, and therefore the temperature is increased. Now the speed of sound increases with the temperature, so the speed in the region behind the jump is faster than in the air in front. That means that any other disturbance that is made behind this step, say by a continuous pushing of the body, or any other disturbance, will ride faster than the front, the speed increasing with higher pressure. Figure 51-3 illustrates the situation, with some little bumps of pressure added to the pressure contour to aid visualization. We see that the higher pressure regions at the rear overtake the front as time goes on, until ultimately the compressional wave develops a sharp front. If the strength is very high, "ultimately" means right away; if it is rather weak, it takes a long time; it may be, in fact, that the sound is spreading and dying out before it has time to do this.

The sounds we make in talking are extremely weak relative to the atmospheric pressure—only 1 part in a million or so. But for pressure changes of the order of 1 atmosphere, the wave velocity increases by about twenty percent, and the wavefront sharpens up at a correspondingly high rate. In nature nothing happens *infinitely* rapidly, presumably, and what we call a "sharp" front has, actually, a very slight thickness; it is not infinitely steep. The distances over which it is varying are of the

order of one mean free path, in which the theory of the wave equation begins to fail because we did not consider the structure of the gas.

Now, referring again to Fig. 51-2, we see that the curvature can be understood if we appreciate that the pressures near the apex are higher than they are farther back, and so the angle θ is greater. That is, the curve is the result of the fact that the speed depends upon the strength of the wave. Therefore the wave from an atomic bomb explosion travels much faster than the speed of sound for a while, until it gets so far out that it is weakened to such an extent from spreading that the pressure bump is small compared with atmospheric pressure. The speed of the bump then approaches the speed of sound in the gas into which it is going. (Incidentally, it always turns out that the speed of the shock is higher than the speed of sound in the gas ahead, but is lower than the speed of sound in the gas behind. That is, impulses from the back will arrive at the front, but the front rides into the medium in which it is going faster than the normal speed of signals. So one cannot tell, acoustically, that the shock is coming until it is too late. The light from the bomb arrives first, but one cannot tell that the shock is coming until it arrives, because there is no sound signal coming ahead of it.)

This is a very interesting phenomenon, this piling up of waves, and the main point on which it depends is that after a wave is present, the speed of the resulting wave should be higher. Another example of the same phenomenon is the following. Consider water flowing in a long channel with finite width and finite depth. If a piston, or a wall across the channel, is moved along the channel fast enough, water piles up, like snow before a snow plow. Now suppose the situation is as shown in Fig. 51-4, with a sudden step in water height somewhere in the channel. It can be demonstrated that long waves in a channel travel faster in deeper water than they do in shallow water. Therefore any new bumps or irregularities in energy supplied by the piston run off forward and pile up at the front. Again, ultimately what we have is just water with a sharp front, theoretically. However, as Fig. 51-4 shows, there are complications. Pictured is a wave coming up a channel; the piston is at the far right end of the channel. At first it might have appeared like a well-behaved wave, as one might expect, but farther along the channel, it has become sharper and sharper until the events pictured occurred. There is a terrible churning at the surface, as the pieces of water fall down, but it is essentially a very sharp rise with no disturbance of the water ahead.

Actually water is much more complicated than sound. However, just to illustrate a point, we will try to analyze the speed of such a so-called *bore*, in a channel. The point here is not that this is of any basic importance for our purposes—it is not a great generalization—it is only to illustrate that the laws of mechanics that we already know are capable of explaining the phenomenon.

Imagine, for a moment, that the water does look something like Fig. 51-5(a), that water at the higher height h_2 is moving with a velocity v , and that the front is moving with velocity u into undisturbed water which is at height h_1 . We would like to determine the speed at which the front moves. In a time Δt a vertical plane initially at x_1 moves a distance $v \Delta t$ to x_2 , while the front of the wave has moved $u \Delta t$.

Now we apply the equations of conservation of matter and momentum. First, the former: Per unit channel width, we see that the amount $h_2 v \Delta t$ of matter that has moved past x_1 (shown shaded) is compensated by the other shaded region, which amounts to $(h_2 - h_1)u \Delta t$. So, dividing by Δt , $vh_2 = u(h_2 - h_1)$. That does not yet give us enough, because although we have h_2 and h_1 , we do not know either u or v ; we are trying to get both of them.

Now the next step is to use conservation of momentum. We have not discussed the problems of water pressure, or anything in hydrodynamics, but it is clear anyway that the pressure of water at a given depth is just enough to hold up the column of water above it. Therefore the pressure of water is equal to ρ , the density of water, times g , times the depth below the surface. Since the pressure increases linearly with depth, the average pressure over the plane at x_1 , say, is $\frac{1}{2}\rho gh_2$, which is also the average force per unit width and per unit height pushing the plane toward x_2 . So we multiply by another h_2 to get the total force which is acting on the water



Figure 51-4

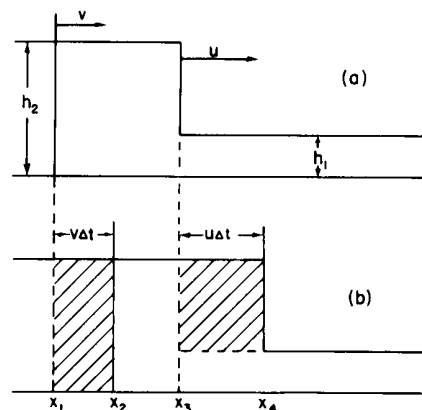


Fig. 51-5. Two cross sections of a bore in a channel, with (b) at an interval Δt later than (a).

pushing from the left. On the other hand, there is pressure in the water on the right also, exerting an opposite force on the region in question, which is, by the same kind of analysis, $\frac{1}{2}\rho gh_1^2$. Now we must balance the forces against the rate of change of the momentum. Thus we have to figure out how much more momentum there is in situation (b) in Fig. 51-5 than there was in (a). We see that the additional mass that has acquired the speed v is just $\rho h_2 u \Delta t - \rho h_2 v \Delta t$ (per unit width), and multiplying this by v gives the additional momentum to be equated to the impulse $F \Delta t$:

$$(\rho h_2 u \Delta t - \rho h_2 v \Delta t)v = (\frac{1}{2}\rho gh_2^2 - \frac{1}{2}\rho gh_1^2) \Delta t.$$

If we eliminate v from this equation by substituting $vh_2 = u(h_2 - h_1)$, already found, and simplify, we get finally that $u^2 = gh_2(h_1 + h_2)/2h_1$.

If the height difference is very small, so that h_1 and h_2 are nearly equal, this says that the velocity $= \sqrt{gh}$. As we will see later, that is only true provided the wavelength of the wave is longer than the depth of the channel.

We could also do the analogous thing for sound waves—including the conservation of internal energy, not the conservation of entropy, because the shock is irreversible. In fact, if one checks the conservation of energy in the bore problem, one finds that energy is not conserved. If the height difference is small, it is almost perfectly conserved, but as soon as the height difference becomes very appreciable, there is a net loss of energy. This is manifested as the falling water and the churning shown in Fig. 51-4.

In shock waves there is a corresponding apparent loss of energy, from the point of view of adiabatic reactions. The energy in the sound wave, behind the shock, goes into heating of the gas after shock passes, corresponding to churning of the water in the bore. In working it out, three equations for the sound case turn out to be necessary for solution, and the temperature behind the shock is not the same as the temperature in front, as we have seen.

If we try to make a bore that is upside down ($h_2 < h_1$), then we find that the energy loss per second is negative. Since energy is not available from anywhere, that bore cannot then maintain itself; it is unstable. If we were to start a wave of that sort, it would flatten out, because the speed dependence on height that resulted in sharpening in the case we discussed would now have the opposite effect.

51-3 Waves in solids

The next kind of waves to be discussed are the more complicated waves in solids. We have already discussed sound waves in gas and in liquid, and there is a direct analog to a sound wave in a solid. If a sudden push is applied to a solid, it is compressed. It resists the compression, and a wave analogous to sound is started. However there is another kind of wave that is possible in a solid, and which is not possible in a fluid. If a solid is distorted by pushing it sideways (called *shearing*), then it tries to pull itself back. That is by definition what distinguishes a solid from a liquid: if we distort a liquid (internally), hold it a minute so that it calms down, and then let go, it will stay that way, but if we take a solid and push it, like shearing a piece of "Jello," and let it go, it flies back and starts a *shear* wave, travelling in the same way the compressions travel. In all cases, the shear wave speed is less than the speed of longitudinal waves. The shear waves are somewhat more analogous, so far as their polarizations are concerned, to light waves. Sound has no polarization, it is just a pressure wave. Light has a characteristic orientation perpendicular to its direction of travel.

In a solid, the waves are of both kinds. First, there is a compression wave, analogous to sound, that runs at one speed. If the solid is not crystalline, then a shear wave polarized in any direction will propagate at a characteristic speed. (Of course all solids are crystalline, but if we use a block made up of microcrystals of all orientations, the crystal anisotropies average out.)

Another interesting question concerning sound waves is the following: What happens if the wavelength in a solid gets shorter, and shorter, and shorter? How short can it get? It is interesting that it cannot get any shorter than the space

between the atoms, because if there is supposed to be a wave in which one point goes up and the next down, etc., the shortest possible wavelength is clearly the atom spacing. In terms of the modes of oscillation, we say that there are longitudinal modes, and transverse modes, long wave modes, short wave modes. As we consider wavelengths comparable to the spacing between the atoms, then the speeds are no longer constant; there is a dispersion effect where the velocity is not independent of the wave number. But, ultimately, the highest mode of transverse waves would be that in which every atom is doing the opposite of neighboring atoms.

Now from the point of view of atoms, the situation is like the two pendulums that we were talking about, for which there are two modes, one in which they both go together, and the other in which they go apart. It is possible to analyze the solid waves another way, in terms of a system of coupled harmonic oscillators, like an enormous number of pendulums, with the highest mode such that they oscillate oppositely, and lower modes with different relationships of the timing.

The shortest wavelengths are so short that they are not usually available technically. However they are of great interest because, in the theory of thermodynamics of a solid, the heat properties of a solid, for example specific heats, can be analyzed in terms of the properties of the short sound waves. Going to the extreme of sound waves of ever shorter wavelength, one necessarily comes to the individual motions of the atoms; the two things are the same ultimately.

A very interesting example of sound waves in a solid, both longitudinal and transverse, are the waves that are in the solid earth. Who makes the noises we do not know, but inside the earth, from time to time, there are earthquakes—some rock slides past some other rock. That is like a little noise. So waves like sound waves start out from such a source very much longer in wavelength than one usually considers in sound waves, but still they are sound waves, and they travel around in the earth. The earth is not homogeneous, however, and the properties of pressure, density, compressibility, and so on, change with depth, and therefore the speed varies with depth. Then the waves do not travel in straight lines—there is a kind of index of refraction and they go in curves. The longitudinal waves and the transverse waves have different speeds, so there are different solutions for the different speeds. Therefore if we place a seismograph at some location and watch the way the thing jiggles after there has been an earthquake somewhere else, then we do not just get an irregular jiggling. We might get a jiggling, and a quieting down, and then another jiggling—what happens depends upon the location. If it were close enough, we would first receive longitudinal waves from the disturbance, and then, a few moments later, transverse waves, because they travel more slowly. By measuring the time difference between the two, we can tell how far away the earthquake is, if we know enough about the speeds and composition of the interior regions involved.

An example of the behavior pattern of waves in the earth is shown in Fig. 51-6. The two kinds of waves are represented by different symbols. If there were an earthquake at the place marked "source," the transverse waves and longitudinal waves would arrive at different times at the station by the most direct routes, and there would also be reflections at discontinuities, resulting in other paths and times. It turns out that there is a core in the earth which does not carry transverse waves. If the station is opposite the source, transverse waves still arrive, but the timing is not right. What happens is that the transverse wave comes to the core, and whenever the transverse waves come to a surface which is oblique, between two materials, two new waves are generated, one transverse and one longitudinal. But inside the core of the earth, a transverse wave is not propagated (or at least, there is no evidence for it, only for a longitudinal wave); it comes out again in both forms and comes to the station.

It is from the behavior of these earthquake waves that it has been determined that transverse waves cannot be propagated within the inner circle. This means that the center of the earth is liquid in the sense that it cannot propagate transverse waves. The only way we know what is inside the earth is by studying earthquakes. So, by using a large number of observations of many earthquakes at different

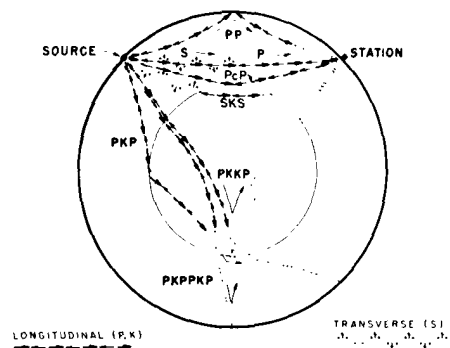


Fig. 51-6. Schematic of the earth, showing paths of longitudinal and transverse sound waves.

stations, the details have been worked out—the speed, the curves, etc. are all known. We know what the speeds of various kinds of waves are at every depth. Knowing that, therefore, it is possible to figure out what the normal modes of the earth are, because we know the speed of propagation of sound waves—in other words, the elastic properties of both kinds of waves at every depth. Suppose the earth were distorted into an ellipsoid and let go. It is just a matter of superposing waves travelling around in the ellipsoid to determine the period and shapes in a free mode. We have figured out that if there is a disturbance, there are a lot of modes, from the lowest, which is ellipsoidal, to higher modes with more structure.

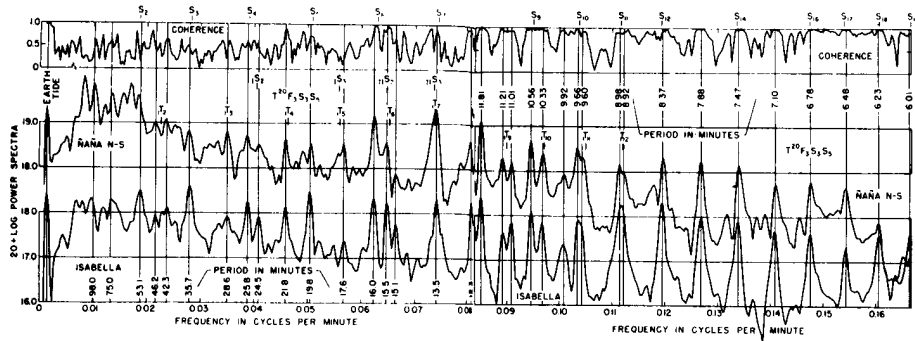


Fig. 51-7. Power versus frequency as detected at seismographs in Naña, Peru, and Isabella, California. The coherence is a measure of the coupling between the stations. [From Benioff, Press and Smith, *J. Geoph. Research* **66**, 605 (1961)].

The Chilean earthquake of May 1960 made a loud enough “noise” that the signals went around the earth many times, and new seismographs of great delicacy were made just in time to determine the frequencies of the fundamental modes of the earth and to compare them with the values that were calculated from the theory of sound with the known velocities, as measured from the independent earthquakes. The result of this experiment is illustrated in Fig. 51-7, which is a plot of the strength of the signal versus the frequency of its oscillation (a *Fourier analysis*). Note that at certain particular frequencies there is much more being received than at other frequencies; there are very definite maxima. These are the natural frequencies of the earth, because these are the main frequencies at which the earth can oscillate. In other words, if the entire motion of the earth is made up of many different modes, we would expect to obtain, for each station, irregular bumpings which indicate a superposition of many frequencies. If we analyze this in terms of frequencies, we should be able to find the characteristic frequencies of the earth. The vertical dark lines in the figure are the calculated frequencies, and we find a remarkable agreement, an agreement due to the fact that the theory of sound is right for the inside of the earth.

A very curious point is revealed in Fig. 51-8, which shows a very careful measurement, with better resolution of the lowest mode, the ellipsoidal mode of the earth. Note that it is not a single maximum, but a double one, 54.7 minutes and 53.1 minutes—slightly different. The reason for the two different frequencies was not known at the time that it was measured, although it may have been found in the meantime. There are at least two possible explanations: One would be that there may be asymmetry in the earth’s distribution, which would result in two similar modes. Another possibility, which is even more interesting, is this: Imagine the waves going around the earth in two directions from the source. The speeds will not be equal because of effects of the rotation of the earth in the equations of motion, which have not been taken into account in making the analysis. Motion in a rotating system is modified by Coriolis forces, and these may cause the observed splitting.

Regarding the method by which these quakes have been analyzed, what is obtained on the seismograph is not a curve of amplitude as a function of frequency, but displacement as a function of time, always a very irregular tracing. To find the amount of all the different sine waves for all different frequencies, we know that the trick is to multiply the data by a sine wave of a given frequency and integrate, i.e., average it, and in the average all other frequencies disappear. The figures were thus plots of the integrals found when the data were multiplied by sine waves of different cycles per minute, and integrated.

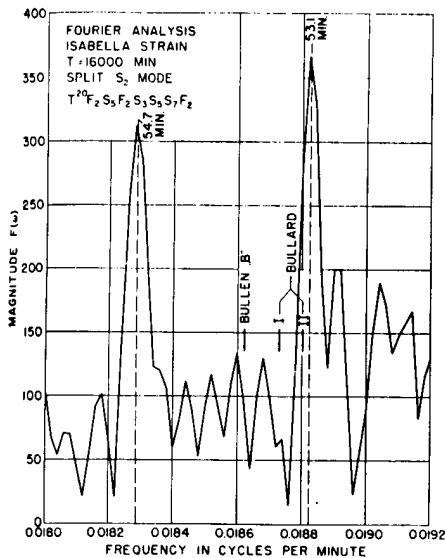
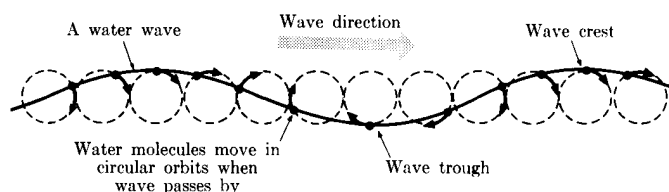


Fig. 51-8. High-resolution analysis of one of the seismograph records, showing spectral doublet.

51-4 Surface waves

Now, the next waves of interest, that are easily seen by everyone and which are usually used as an example of waves in elementary courses, are water waves. As we shall soon see, they are the worst possible example, because they are in no respects like sound and light; they have all the complications that waves can have. Let us start with long water waves in deep water. If the ocean is considered infinitely deep and a disturbance is made on the surface, waves are generated. All kinds of irregular motions occur, but the sinusoidal type motion, with a very small disturbance, might look like the common smooth ocean waves coming in toward the shore. Now with such a wave, the water, of course, on the average, is standing still, but the wave moves. What is the motion, is it transverse or longitudinal? It must be neither; it is not transverse, nor is it longitudinal. Although the water at a given place is alternately trough or hill, it cannot simply be moving up and down, by the conservation of water. That is, if it goes down, where is the water going to go? The water is essentially incompressible. The speed of compression of waves—that is, sound in the water—is much, much higher, and we are not considering that now. Since water is incompressible on this scale, as a hill comes down the water must move away from the region. What actually happens is that particles of water near the surface move approximately in circles. When smooth swells are coming, a person floating in a tire can look at a nearby object and see it going in a circle. So it is a mixture of longitudinal and transverse, to add to the confusion. At greater depths in the water the motions are smaller circles until, reasonably far down, there is nothing left of the motion (Fig. 51-9).

Fig. 51-9. Deep-water waves are formed from particles moving in circles. Note the systematic phase shifts from circle to circle. How would a floating object move?



To find the velocity of such waves is an interesting problem: it must be some combination of the density of the water, the acceleration of gravity, which is the restoring force that makes the waves, and possibly of the wavelength and of the depth. If we take the case where the depth goes to infinity, it will no longer depend on the depth. Whatever formula we are going to get for the velocity of the phases of the waves must combine the various factors to make the proper dimensions, and if we try this in various ways, we find only one way to combine the density, g , and λ in order to make a velocity, namely, $\sqrt{g\lambda}$, which does not include the density at all. Actually, this formula for the phase velocity is not exactly right, but a complete analysis of the dynamics, which we will not go into, shows that the factors are as we have them, except for $\sqrt{2\pi}$:

$$v_{\text{phase}} = \sqrt{g\lambda/2\pi} \text{ (for gravity waves).}$$

It is interesting that the long waves go faster than the short waves. Thus if a boat makes waves far out, because there is some sports-car driver in a motorboat travelling by, then after a while the waves come to shore with slow sloshings at first and then more and more rapid sloshings, because the first waves that come are long. The waves get shorter and shorter as the time goes on, because the velocities go as the square root of the wavelength.

One may object, "That is not right, we must look at the *group* velocity in order to figure it out!" Of course that is true. The formula for the phase velocity does not tell us what is going to arrive first; what tells us is the group velocity. So we have to work out the group velocity, and it is left as a problem to show it to be one-half of the phase velocity, assuming that the velocity goes as the square root of the wavelength, which is all that is needed. The group velocity also goes as the square root of the wavelength. How can the group velocity go half as fast as the phase? If one looks at the bunch of waves that are made by a boat travelling

along, following a particular crest, he finds that it moves forward in the group and gradually gets weaker and dies out in the front, and mystically and mysteriously a weak one in the back works its way forward and gets stronger. In short, the waves are moving through the group while the group is only moving at half the speed that the waves are moving.

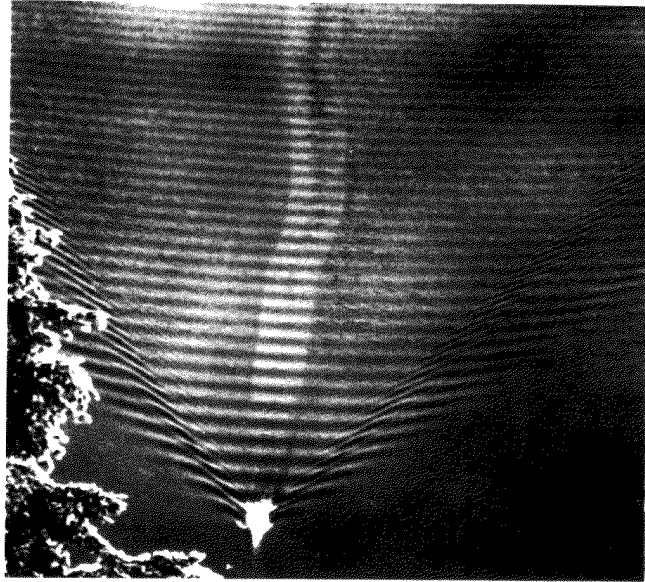


Fig. 51-10. The wake of a boat.

Because the group velocities and phase velocities are not equal, then the waves that are produced by an object moving through are no longer simply a cone, but it is much more interesting. We can see that in Fig. 51-10, which shows the waves produced by an object moving through the water. Note that it is quite different than what we would have for sound, in which the velocity is independent of wavelength, where we would have wavefronts only along the cone, travelling outward. Instead of that, we have waves in the back with fronts moving parallel to the motion of the boat, and then we have little waves on the sides at other angles. This entire pattern of waves can, with ingenuity, be analyzed by knowing only this: that the phase velocity is proportional to the square root of the wavelength. The trick is that the pattern of waves is stationary relative to the (constant-velocity) boat; any other pattern would get lost from the boat.

The water waves that we have been considering so far were long waves in which the force of restoration is due to gravitation. But when waves get very short in the water, the main restoring force is capillary attraction, i.e., the energy of the surface, the surface tension. For surface tension waves, it turns out that the phase velocity is

$$v_{\text{phase}} = \sqrt{2\pi T / \lambda \rho} \text{ (for ripples),}$$

where T is the surface tension and ρ the density. It is the exact opposite: the phase velocity is *higher*, the shorter the wavelength, when the wavelength gets very small. When we have both gravity and capillary action, as we always do, we get the combination of these two together:

$$v_{\text{phase}} = \sqrt{Tk/\rho + g/k},$$

where $k = 2\pi/\lambda$ is the wave number. So the velocity of the waves of water is really quite complicated. The phase velocity as a function of the wavelength is shown in Fig. 51-11; for very short waves it is fast, for very long waves it is fast, and there is a minimum speed at which the waves can go. The group velocity can be calculated from the formula: it goes to $\frac{3}{2}$ the phase velocity for ripples and $\frac{1}{2}$ the phase velocity for gravity waves. To the left of the minimum the group velocity is higher than the phase velocity; to the right, the group velocity is less than the

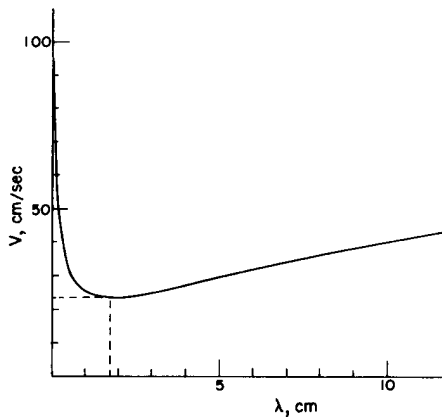


Fig. 51-11. Phase velocity vs. wavelength for water.

phase velocity. There are a number of interesting phenomena associated with these facts. In the first place, since the group velocity is increasing so rapidly as the wavelength goes down, if we make a disturbance there will be a slowest end of the disturbance going at the minimum speed with the corresponding wavelength, and then in front, going at higher speed, will be a short wave and a very long wave. It is very hard to see the long ones, but it is easy to see the short ones in a water tank.

So we see that the ripples often used to illustrate simple waves are quite interesting and complicated; they do not have a sharp wavefront at all, as is the case for simple waves like sound and light. The main wave has little ripples which run out ahead. A sharp disturbance in the water does not produce a sharp wave because of the dispersion. First come the very fine waves. Incidentally, if an object moves through the water at a certain speed, a rather complicated pattern results, because all the different waves are going at different speeds. One can demonstrate this with a tray of water and see that the fastest ones are the fine capillary waves. There are slowest waves, of a certain kind, which go behind. By inclining the bottom, one sees that where the depth is lower, the speed is lower. If a wave comes in at an angle to the line of maximum slope, it bends and tends to follow that line. In this way one can show various things, and we conclude that waves are more complicated in water than in air.

The speed of long waves in water with circulatory motions is slower when the depth is less, faster in deep water. Thus as water comes toward a beach where the depth lessens, the waves go slower. But where the water is deeper, the waves are faster, so we get the effects of shock waves. This time, since the wave is not so simple, the shocks are much more contorted, and the wave over-curves itself, in the familiar way shown in Fig. 51-12. This is what happens when waves come into the shore, and the real complexities in nature are well revealed in such a circumstance. No one has yet been able to figure out what shape the wave should take as it breaks. It is easy enough when the waves are small, but when one gets large and breaks, then it is much more complicated.



Fig. 51-12. A water wave.

An interesting feature about capillary waves can be seen in the disturbances made by an object moving through the water. From the point of view of the object itself, the water is flowing past, and the waves which ultimately sit around it are always the waves which have just the right speed to stay still with the object in the water. Similarly, around an object in a stream, with the stream flowing by, the pattern of waves is stationary, and at just the right wavelengths to go at the same speed as the water going by. But if the group velocity is less than the phase velocity, then the disturbances propagate out backwards in the stream, because the group velocity is not quite enough to keep up with the stream. If the group velocity is faster than the velocity of the phase, the pattern of waves will appear in front of the object. If one looks closely at objects in a stream, one can see that there are little ripples in front and long “slurps” in the back.

Another interesting feature of this sort can be observed in pouring liquids. If milk is poured fast enough out of a bottle, for instance, a large number of lines can be seen crossing both ways in the outgoing stream. They are waves starting from the disturbance at the edges and running out, much like the waves about an object in a stream. There are effects from both sides which produce the crossed pattern.

We have investigated some of the interesting properties of waves and the various complications of dependence of phase velocity on wavelength, the speed of the waves on depth, and so forth, that produce the really complex, and therefore interesting, phenomena of nature.

Symmetry in Physical Laws

52-1 Symmetry operations

The subject of this chapter is what we may call *symmetry in physical laws*. We have already discussed certain features of symmetry in physical laws in connection with vector analysis (Chapter 11), the theory of relativity (Chapter 16), and rotation (Chapter 20).

Why should we be concerned with symmetry? In the first place, symmetry is fascinating to the human mind, and everyone likes objects or patterns that are in some way symmetrical. It is an interesting fact that nature often exhibits certain kinds of symmetry in the objects we find in the world around us. Perhaps the most symmetrical object imaginable is a sphere, and nature is full of spheres—stars, planets, water droplets in clouds. The crystals found in rocks exhibit many different kinds of symmetry, the study of which tells us some important things about the structure of solids. Even the animal and vegetable worlds show some degree of symmetry, although the symmetry of a flower or of a bee is not as perfect or as fundamental as is that of a crystal.

But our main concern here is not with the fact that the *objects* of nature are often symmetrical. Rather, we wish to examine some of the even more remarkable symmetries of the universe—the symmetries that exist in the *basic laws themselves* which govern the operation of the physical world.

First, what *is* symmetry? How can a physical *law* be “symmetrical”? The problem of defining symmetry is an interesting one and we have already noted that Weyl gave a good definition, the substance of which is that a thing is symmetrical if there is something we can do to it so that after we have done it, it looks the same as it did before. For example, a symmetrical vase is of such a kind that if we reflect or turn it, it will look the same as it did before. The question we wish to consider here is what we can do to physical phenomena, or to a physical situation in an experiment, and yet leave the result the same. A list of the known operations under which various physical phenomena remain invariant is shown in Table 52-1.

52-2 Symmetry in space and time

The first thing we might try to do, for example, is to *translate* the phenomenon in space. If we do an experiment in a certain region, and then build another apparatus at another place in space (or move the original one over) then, whatever went on in one apparatus, in a certain order in time, will occur in the same way if we have arranged the same condition, with all due attention to the restrictions that we mentioned before: that all of those features of the environment which make it not behave the same way have also been moved over—we talked about how to define how much we should include in those circumstances, and we shall not go into those details again.

In the same way, we also believe today that *displacement in time* will have no effect on physical laws. (That is, *as far as we know today*—all of these things are as far as we know today!) That means that if we build a certain apparatus and start it at a certain time, say on Thursday at 10:00 a.m., and then build the same apparatus and start it, say, three days later in the same condition, the two apparatuses will go through the same motions in exactly the same way as a function of time no matter what the starting time, provided again, of course, that the relevant features of the environment are also modified appropriately in *time*. That symmetry means,

52-1 Symmetry operations

52-2 Symmetry in space and time

52-3 Symmetry and conservation laws

52-4 Mirror reflections

52-5 Polar and axial vectors

52-6 Which hand is right?

52-7 Parity is not conserved!

52-8 Antimatter

52-9 Broken symmetries

Table 52-1

Symmetry Operations

Translation in space
Translation in time
Rotation through a fixed angle
Uniform velocity in a straight line (Lorentz transformation)
Reversal of time
Reflection of space
Interchange of identical atoms or identical particles
Quantum-mechanical phase
Matter-antimatter (charge conjugation)

of course, that if one bought General Motors stock three months ago, the same thing would happen to it if he bought it now!

We have to watch out for geographical differences too, for there are, of course, variations in the characteristics of the earth's surface. So, for example, if we measure the magnetic field in a certain region and move the apparatus to some other region, it may not work in precisely the same way because the magnetic field is different, but we say that is because the magnetic field is associated with the earth. We can imagine that if we move the whole earth and the equipment, it would make no difference in the operation of the apparatus.

Another thing that we discussed in considerable detail was rotation in space: if we turn an apparatus at an angle it works just as well, provided we turn everything else that is relevant along with it. In fact, we discussed the problem of symmetry under rotation in space in some detail in Chapter 11, and we invented a mathematical system called *vector analysis* to handle it as neatly as possible.

On a more advanced level we had another symmetry—the symmetry under uniform velocity in a straight line. That is to say—a rather remarkable effect—that if we have a piece of apparatus working a certain way and then take the same apparatus and put it in a car, and move the whole car, plus all the relevant surroundings, at a uniform velocity in a straight line, then so far as the phenomena inside the car are concerned there is no difference: all the laws of physics appear the same. We even know how to express this more technically, and that is that the mathematical equations of the physical laws must be unchanged under a *Lorentz transformation*. As a matter of fact, it was a study of the relativity problem that concentrated physicists' attention most sharply on symmetry in physical laws.

Now the above-mentioned symmetries have all been of a geometrical nature, time and space being more or less the same, but there are other symmetries of a different kind. For example, there is a symmetry which describes the fact that we can replace one atom by another of the same kind; to put it differently, there *are* atoms of the same kind. It is possible to find groups of atoms such that if we change a pair around, it makes no difference—the atoms are identical. Whatever one atom of oxygen of a certain type will do, another atom of oxygen of that type will do. One may say, "That is ridiculous, that is the *definition* of equal types!" That may be merely the definition, but then we still do not know whether there *are* any "atoms of the same type"; the *fact* is that there are many, many atoms of the same type. Thus it does mean something to say that it makes no difference if we replace one atom by another of the same type. The so-called elementary particles of which the atoms are made are also identical particles in the above sense—all electrons are the same; all protons are the same; all positive pions are the same; and so on.

After such a long list of things that can be done without changing the phenomena, one might think we could do practically anything; so let us give some examples to the contrary, just to see the difference. Suppose that we ask: "Are the physical laws symmetrical under a change of scale?" Suppose we build a certain piece of apparatus, and then build another apparatus five times bigger in every part, will it work exactly the same way? The answer is, in this case, *no*! The wavelength of light emitted, for example, by the atoms inside one box of sodium atoms and the wavelength of light emitted by a gas of sodium atoms five times in volume is not five times longer, but is, in fact, exactly the same as the other. So the ratio of the wavelength to the size of the emitter will change.

Another example: we see in the newspaper, every once in a while pictures of a great cathedral made with little matchsticks—a tremendous work of art by some retired fellow who keeps gluing matchsticks together. It is much more elaborate and wonderful than any real cathedral. If we imagine that this wooden cathedral were actually built on the scale of a real cathedral, we see where the trouble is; it would not last—the whole thing would collapse because of the fact that scaled-up matchsticks are just not strong enough. "Yes," one might say, "but we also know that when there is an influence from the outside, it also must be changed in proportion!" We are talking about the ability of the object to withstand gravitation. So what we should do is first to take the model cathedral of real matchsticks and

the real earth, and then we know it is stable. Then we should take the larger cathedral and take a bigger earth. But then it is even worse, because the gravitation is increased still more!

Today, of course, we understand the fact that phenomena depend on the scale on the grounds that matter is atomic in nature, and certainly if we built an apparatus that was so small there were only five atoms in it, it would clearly be something we could not scale up and down arbitrarily. The scale of an individual atom is not at all arbitrary—it is quite definite.

The fact that the laws of physics are not unchanged under a change of scale was discovered by Galileo. He realized that the strengths of materials were not in exactly the right proportion to their sizes, and he illustrated this property that we were just discussing, about the cathedral of matchsticks, by drawing two bones, the bone of one dog, in the right proportion for holding up his weight, and the imaginary bone of a “super dog” that would be, say, ten or a hundred times bigger—that bone was a big, solid thing with quite different proportions. We do not know whether he ever carried the argument quite to the conclusion that the laws of nature must have a definite scale, but he was so impressed with this discovery that he considered it to be as important as the discovery of the laws of motion, because he published them both in the same volume, called “On Two New Sciences.”

Another example in which the laws are not symmetrical, that we know quite well, is this: a system in rotation at a uniform angular velocity does not give the same apparent laws as one that is not rotating. If we make an experiment and then put everything in a space ship and have the space ship spinning in empty space, all alone at a constant angular velocity, the apparatus will not work the same way because, as we know, things inside the equipment will be thrown to the outside, and so on, by the centrifugal or coriolis forces, etc. In fact, we can tell that the earth is rotating by using a so-called Foucault pendulum, without looking outside.

Next we mention a very interesting symmetry which is obviously false, i.e., *reversibility in time*. The physical laws apparently cannot be reversible in time, because, as we know, all obvious phenomena are irreversible on a large scale: “The moving finger writes, and having writ, moves on.” So far as we can tell, this irreversibility is due to the very large number of particles involved, and if we could see the individual molecules, we would not be able to discern whether the machinery was working forward or backwards. To make it more precise: we build a small apparatus in which we know what all the atoms are doing, in which we can watch them jiggling. Now we build another apparatus like it, but which starts its motion in the final condition of the other one, with all the velocities precisely reversed. *It will then go through the same motions, but exactly in reverse.* Putting it another way: if we take a motion picture, with sufficient detail, of all the inner works of a piece of material and shine it on a screen and run it backwards, no physicist will be able to say, “That is against the laws of physics, that is doing something wrong!” If we do not see all the details, of course, the situation will be perfectly clear. If we see the egg splattering on the sidewalk and the shell cracking open, and so on, then we will surely say, “That is irreversible, because if we run the moving picture backwards the egg will all collect together and the shell will go back together, and that is obviously ridiculous!” But if we look at the individual atoms themselves, the laws look completely reversible. This is, of course, a much harder discovery to have made, but apparently it is true that the fundamental physical laws, on a microscopic and fundamental level, are completely reversible in time!

52-3 Symmetry and conservation laws

The symmetries of the physical laws are very interesting at this level, but they turn out, in the end, to be even more interesting and exciting when we come to quantum mechanics. For a reason which we cannot make clear at the level of the present discussion—a fact that most physicists still find somewhat staggering, a most profound and beautiful thing, is that, in quantum mechanics, *for each of the rules of symmetry there is a corresponding conservation law*; there is a definite

connection between the laws of conservation and the symmetries of physical laws. We can only state this at present, without any attempt at explanation.

The fact, for example, that the laws are symmetrical for translation in space when we add the principles of quantum mechanics, turns out to mean that *momentum is conserved*.

That the laws are symmetrical under translation in time means, in quantum mechanics, that *energy is conserved*.

Invariance under rotation through a fixed angle in space corresponds to the *conservation of angular momentum*. These connections are very interesting and beautiful things, among the most beautiful and profound things in physics.

Incidentally, there are a number of symmetries which appear in quantum mechanics which have no classical analog, which have no method of description in classical physics. One of these is as follows: If ψ is the amplitude for some process or other, we know that the absolute square of ψ is the probability that the process will occur. Now if someone else were to make his calculations, not with this ψ , but with a ψ' which differs merely by a change in phase (let Δ be some constant, and multiply $e^{i\Delta}$ times the old ψ), the absolute square of ψ' , which is the probability of the event, is then equal to the absolute square of ψ :

$$\psi' = \psi e^{i\Delta}; \quad |\psi'|^2 = |\psi|^2. \quad (52.1)$$

Therefore the physical laws are unchanged if the phase of the wave function is shifted by an arbitrary constant. That is another symmetry. Physical laws must be of such a nature that a shift in the quantum-mechanical phase makes no difference. As we have just mentioned, in quantum mechanics there is a conservation law for every symmetry. The conservation law which is connected with the quantum-mechanical phase seems to be the *conservation of electrical charge*. This is altogether a very interesting business!

52-4 Mirror reflections

Now the next question, which is going to concern us for most of the rest of this chapter, is the question of symmetry under *reflection in space*. The problem is this: Are the physical laws symmetrical under reflection? We may put it this way: Suppose we build a piece of equipment, let us say a clock, with lots of wheels and hands and numbers; it ticks, it works, and it has things wound up inside. We look at the clock in the mirror. How it *looks* in the mirror is not the question. But let us actually *build* another clock which is exactly the same as the first clock looks in the mirror—every time there is a screw with a right-hand thread in one, we use a screw with a left-hand thread in the corresponding place of the other; where one is marked “2” on the face, we mark a “5” on the face of the other; each coiled spring is twisted one way in one clock and the other way in the mirror-image clock; when we are all finished, we have two clocks, both physical, which bear to each other the relation of an object and its mirror image, although they are both actual, material objects, we emphasize. Now the question is: If the two clocks are started in the same condition, the springs wound to corresponding tightnesses, will the two clocks tick and go around, forever after, as exact mirror images? (This is a physical question, not a philosophical question.) Our intuition about the laws of physics would suggest that they *would*.

We would suspect that, at least in the case of these clocks, reflection in space is one of the symmetries of physical laws, that if we change everything from “right” to “left” and leave it otherwise the same, we cannot tell the difference. Let us, then, suppose for a moment that this is true. If it is true, then it would be impossible to distinguish “right” and “left” by any physical phenomenon, just as it is, for example, impossible to define a particular absolute velocity by a physical phenomenon. So it should be impossible, by any physical phenomenon, to define absolutely what we mean by “right” as opposed to “left,” because the physical laws should be symmetrical.

Of course, the world does not *have* to be symmetrical. For example, using what we may call “geography,” surely “right” can be defined. For instance, we stand

in New Orleans and look at Chicago, and Florida is to our right (when our feet are on the ground!). So we can define “right” and “left” by geography. Of course, the actual situation in any system does not have to have the symmetry that we are talking about; it is a question of whether the *laws* are symmetrical—in other words, whether it is *against the physical laws* to have a sphere like the earth with “left-handed dirt” on it and a person like ourselves standing looking at a city like Chicago from a place like New Orleans, but with everything the other way around, so Florida is on the other side. It clearly seems not impossible, not against the physical laws, to have everything changed left for right.

Another point is that our definition of “right” should not depend on history. An easy way to distinguish right from left is to go to a machine shop and pick up a screw at random. The odds are it has a right-hand thread—not necessarily, but it is much more likely to have a right-hand thread than a left-hand one. This is a question of history or convention, or the way things happen to be, and is again not a question of fundamental laws. As we can well appreciate, everyone could have started out making left-handed screws!

So we must try to find some phenomenon in which “right hand” is involved fundamentally. The next possibility we discuss is the fact that polarized light rotates its plane of polarization as it goes through, say, sugar water. As we saw in Chapter 33, it rotates, let us say, to the right in a certain sugar solution. That is a way of defining “right-hand,” because we may dissolve some sugar in the water and then the polarization goes to the right. But sugar has come from living things, and if we try to make the sugar artificially, then we discover that it *does not* rotate the plane of polarization! But if we then take that same sugar which is made artificially and which does not rotate the plane of polarization, and put bacteria in it (they eat some of the sugar) and then filter out the bacteria, we find that we still have sugar left (almost half as much as we had before), and this time it does rotate the plane of polarization, but *the other way*! It seems very confusing, but is easily explained.

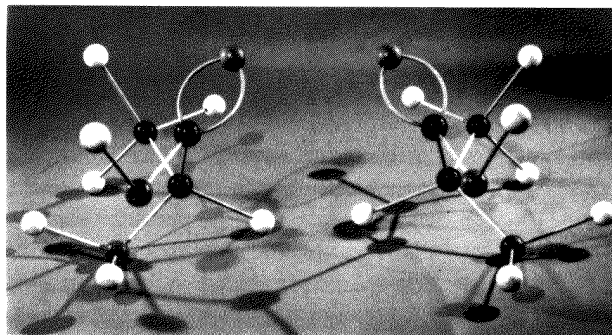


Fig. 52-1. (a) L-alanine (left), and (b) D-alanine (right).

Take another example: One of the substances which is common to all living creatures and that is fundamental to life is protein. Proteins consist of chains of amino acids. Figure 52-1 shows a model of an amino acid that comes out of a protein. This amino acid is called alanine, and the molecular arrangement would look like that in Fig. 52-1(a) if it came out of a protein of a real living thing. On the other hand, if we try to make alanine from carbon dioxide, ethane, and ammonia (and we *can* make it, it is not a complicated molecule), we discover that we are making equal amounts of this molecule and the one shown in Fig. 52-1(b)! The first molecule, the one that comes from the living thing, is called *L-alanine*. The other one, which is the same chemically, in that it has the same kinds of atoms and the same connections of the atoms, is a “right-hand” molecule, compared with the “left-hand” L-alanine, and it is called *D-alanine*. The interesting thing is that when we make alanine at home in a laboratory from simple gases, we get an equal mixture of both kinds. However, the only thing that life uses is L-alanine. (This is not exactly true. Here and there in living creatures there is a special use for D-alanine, but it is very rare. All proteins use L-alanine exclusively.) Now if we make both kinds, and we feed the mixture to some animal which likes to “eat,” or use up, alanine, it cannot use D-alanine, so it only uses the L-alanine; that is what happened to our sugar—after the bacteria eat the sugar that works

well for them, only the “wrong” kind is left! (Left-handed sugar tastes sweet, but not the same as right-handed sugar.)

So it looks as though the phenomena of life permit a distinction between “right” and “left,” or chemistry permits a distinction, because the two molecules are chemically different. But no, it does not! So far as physical measurements can be made, such as of energy, the rates of chemical reactions, and so on, the two kinds work exactly the same way if we make everything else in a mirror image too. One molecule will rotate light to the right, and the other will rotate it to the left in precisely the same amount, through the same amount of fluid. Thus, so far as physics is concerned, these two amino acids are equally satisfactory. So far as we understand things today, the fundamentals of the Schrödinger equation have it that the two molecules should behave in exactly corresponding ways, so that one is to the right as the other is to the left. Nevertheless, in life it is all one way!

It is presumed that the reason for this is the following. Let us suppose, for example, that life is somehow at one moment in a certain condition in which all the proteins in some creatures have left-handed amino acids, and all the enzymes are lopsided—every substance in the living creature is lopsided—it is not symmetrical. So when the digestive enzymes try to change the chemicals in the food from one kind to another, one kind of chemical “fits” into the enzyme, but the other kind does not (like Cinderella and the slipper, except that it is a “left foot” that we are testing). So far as we know, in principle, we could build a frog, for example, in which every molecule is reversed, everything is like the “left-hand” mirror image of a real frog; we have a left-hand frog. This left-hand frog would go on all right for a while, but he would find nothing to eat, because if he swallows a fly, his enzymes are not built to digest it. The fly has the wrong “kind” of amino acids (unless we give him a left-hand fly). So as far as we know, the chemical and life processes would continue in the same manner if everything were reversed.

If life is entirely a physical and chemical phenomenon, then we can understand that the proteins are all made in the same corkscrew only from the idea that at the very beginning some living molecules, by accident, got started and a few won. Somewhere, once, one organic molecule was lopsided in a certain way, and from this particular thing the “right” happened to evolve in our particular geography; a particular historical accident was one-sided, and ever since then the lopsidedness has propagated itself. Once having arrived at the state that it is in now, of course, it will always continue—all the enzymes digest the right things, manufacture the right things: when the carbon dioxide and the water vapor, and so on, go in the plant leaves, the enzymes that make the sugars make them lopsided because the enzymes are lopsided. If any new kind of virus or living thing were to originate at a later time, it would survive only if it could “eat” the kind of living matter already present. Thus it, too, must be of the same kind.

There is no conservation of the number of right-handed molecules. Once started, we could keep increasing the number of right-handed molecules. So the presumption is, then, that the phenomena in the case of life do not show a lack of symmetry in physical laws, but do show, on the contrary, the universal nature and the commonness of ultimate origin of all creatures on earth, in the sense described above.

52-5 Polar and axial vectors

Now we go further. We observe that in physics there are a lot of other places where we have “right” and “left” hand rules. As a matter of fact, when we learned about vector analysis we learned about the right-hand rules we have to use in order to get the angular momentum, torque, magnetic field, and so on, to come out right. The force on a charge moving in a magnetic field, for example, is $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$. In a given situation, in which we know $\mathbf{F}$, $\mathbf{v}$, and $\mathbf{B}$, isn’t that equation enough to define right-handedness? As a matter of fact, if we go back and look at where the vectors came from, we know that the “right-hand rule” was merely a convention; it was a trick. The original quantities, like the angular momenta and the angular velocities, and things of this kind, were not really vectors at all! They are all

somehow associated with a certain plane, and it is just because there are three dimensions in space that we can associate the quantity with a direction perpendicular to that plane. Of the two possible directions, we chose the “right-hand” direction.

So if the laws of physics are symmetrical, we should find that if some demon were to sneak into all the physics laboratories and replace the word “right” for “left” in every book in which “right-hand rules” are given, and instead we were to use all “left-hand rules,” uniformly, then it should make no difference whatever in the physical laws.

Let us give an illustration. There are two kinds of vectors. There are “honest” vectors, for example a step $\Delta \mathbf{r}$ in space. If in our apparatus there is a piece here and something else there, then in a mirror apparatus there will be the image piece and the image something else, and if we draw a vector from the “piece” to the “something else,” one vector is the mirror image of the other (Fig. 52-2). The vector arrow changes its head, just as the whole space turns inside out; such a vector we call a *polar vector*.

But the other kind of vector, which has to do with rotations, is of a different nature. For example, suppose that in three dimensions something is rotating as shown in Fig. 52-3. Then if we look at it in a mirror, it will be rotating as indicated, namely, as the mirror image of the original rotation. Now we have agreed to represent the mirror rotation by the same rule, it is a “vector” which, on reflection, does *not* change about as the polar vector does, but is reversed relative to the polar vectors and to the geometry of the space; such a vector is called an *axial vector*.

Now if the law of reflection symmetry is right in physics, then it must be true that the equations must be so designed that if we change the sign of each axial vector and each cross-product of vectors, which would be what corresponds to reflection, nothing will happen. For instance, when we write a formula which says that the angular momentum is $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, that equation is all right, because if we change to a left-hand coordinate system, we change the sign of $\mathbf{L}$, but $\mathbf{p}$ and $\mathbf{r}$ do not change; the cross-product sign is changed, since we must change from a right-hand rule to a left-hand rule. As another example, we know that the force on a charge moving in a magnetic field is $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$, but if we change from a right- to a left-handed system, since $\mathbf{F}$ and $\mathbf{v}$ are known to be polar vectors the sign change required by the cross-product must be cancelled by a sign change in $\mathbf{B}$, which means that $\mathbf{B}$ must be an axial vector. In other words, if we make such a reflection, $\mathbf{B}$ must go to $-\mathbf{B}$. So if we change our coordinates from right to left, we must also change the poles of magnets from north to south.

Let us see how that works in an example. Suppose that we have two magnets, as in Fig. 52-4. One is a magnet with the coils going around a certain way, and with current in a given direction. The other magnet looks like the reflection of the first magnet in a mirror—the coil will wind the other way, everything that happens inside the coil is exactly reversed, and the current goes as shown. Now, from the laws for the production of magnetic fields, which we do not know yet officially, but which we most likely learned in high school, it turns out that the magnetic field is as shown in the figure. In one case the pole is a south magnetic pole, while in the other magnet the current is going the other way and the magnetic field is reversed—it is a north magnetic pole. So we see that when we go from right to left we must indeed change from north to south!

Never mind changing north to south; these too are mere conventions. Let us talk about *phenomena*. Suppose, now, that we have an electron moving through one field, going into the page. Then, if we use the formula for the force, $\mathbf{v} \times \mathbf{B}$ (remember the charge is minus), we find that the electron will deviate in the indicated direction according to the physical law. So the phenomenon is that we have a coil with a current going in a specified sense and an electron curves in a certain way—that is the physics—never mind how we label everything.

Now let us do the same experiment with a mirror: we send an electron through in a corresponding direction and now the force is reversed, if we calculate it from the same rule, and that is very good because the corresponding *motions* are then mirror images!



Fig. 52-2. A step in space and its mirror image.

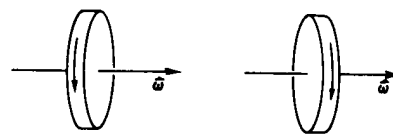


Fig. 52-3. A rotating wheel and its mirror image. Note that the angular velocity “vector” is not reversed in direction.

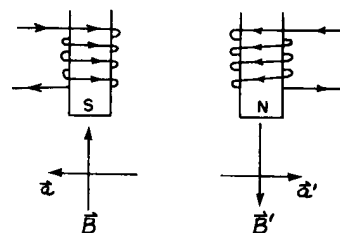


Fig. 52-4. A magnet and its mirror image.

52-6 Which hand is right?

So the fact of the matter is that in studying any phenomenon there are always two right-hand rules, or an even number of them, and the net result is that the phenomena always look symmetrical. In short, therefore, we cannot tell right from left if we also are not able to tell north from south. However, it may seem that we *can* tell the north pole of a magnet. The north pole of a compass needle, for example, is one that points to the north. But of course that is again a local property that has to do with geography of the earth; that is just like talking about in which direction is Chicago, so it does not count. If we have seen compass needles, we may have noticed that the north-seeking pole is a sort of bluish color. But that is just due to the man who painted the magnet. These are all local, conventional criteria.

However, if a magnet were to have the property that if we looked at it closely enough we would see small hairs growing on its north pole but not on its south pole, if that were the general rule, or if there were *any* unique way to distinguish the north from the south pole of a magnet, then we could tell which of the two cases we actually had, and *that would be the end of the law of reflection symmetry*.

To illustrate the whole problem still more clearly, imagine that we were talking to a Martian, or someone very far away, by telephone. We are not allowed to send him any actual samples to inspect; for instance, if we could send light, we could send him right-hand circularly polarized light and say, "That is right-hand light—just watch the way it is going." But we cannot *give* him anything, we can only talk to him. He is far away, or in some strange location, and he cannot see anything we can see. For instance, we cannot say, "Look at Ursa major; now see how those stars are arranged. What we mean by 'right' is . . ." We are only allowed to telephone him.

Now we want to tell him all about us. Of course, first we start defining numbers, and say, "Tick, tick, *two*, tick, tick, tick, *three* . . .," so that gradually he can understand a couple of words, and so on. After a while we may become very familiar with this fellow, and he says, "What do you guys look like?" We start to describe ourselves, and say, "Well, we are six feet tall." He says, "Wait a minute, what is six feet?" Is it possible to tell him what six feet is? Certainly! We say, "You know about the diameter of hydrogen atoms—we are 17,000,000,000 hydrogen atoms high!" That is possible because physical laws are not variant under change of scale, and therefore we *can* define an absolute length. And so we define the size of the body, and tell him what the general shape is—it has prongs with five bumps sticking out on the ends, and so on, and he follows us along, and we finish describing how we look on the outside, presumably without encountering any particular difficulties. He is even making a model of us as we go along. He says, "My, you are certainly very handsome fellows; now what is on the inside?" So we start to describe the various organs on the inside, and we come to the heart, and we carefully describe the shape of it, and say, "Now put the heart on the left side." He says, "Duhhh—the left side?" Now our problem is to describe to him which side the heart goes on without his ever seeing anything that we see, and without our ever sending any sample to him of what we mean by "right"—no standard right-handed object. Can we do it?

52-7 Parity is not conserved!

It turns out that the laws of gravitation, the laws of electricity and magnetism, nuclear forces, all satisfy the principle of reflection symmetry, so these laws, or anything derived from them, cannot be used. But associated with the many particles that are found in nature there is a phenomenon called *beta decay*, or *weak decay*. One of the examples of weak decay, in connection with a particle discovered in about 1954, posed a strange puzzle. There was a certain charged particle which disintegrated into three π -mesons, as shown schematically in Fig. 52-5. This particle was called, for a while, a τ -meson. Now in Fig. 52-5 we also see another particle which disintegrates into *two* mesons; one must be neutral, from the con-

servation of charge. This particle was called a θ -meson. So on the one hand we have a particle called a τ , which disintegrates into three π -mesons, and a θ , which disintegrates into two π -mesons. Now it was soon discovered that the τ and the θ are almost equal in mass; in fact, within the experimental error, they are equal. Next, the length of time it took for them to disintegrate into three π 's and two π 's was found to be almost exactly the same; they live the same length of time. Next, whenever they were made, they were made in the same proportions, say, 14 percent τ 's to 86 percent θ 's.

Anyone in his right mind realizes immediately that they must be the same particle, that we merely produce an object which has two different ways of disintegrating—not two different particles. This object that can disintegrate in two different ways has, therefore, the same lifetime and the same production ratio (because this is simply the ratio of the odds with which it disintegrates into these two kinds).

However, it was possible to prove (and we cannot here explain at all *how*), from the principle of reflection symmetry in quantum mechanics, that it was *impossible* to have these both come from the same particle—the same particle *could not* disintegrate in both of these ways. The conservation law corresponding to the principle of reflection symmetry is something which has no classical analog, and so this kind of quantum-mechanical conservation was called the *conservation of parity*. So, it was a result of the conservation of parity or, more precisely, from the symmetry of the quantum-mechanical equations of the weak decays under reflection, that the same particle could not go into both, so it must be some kind of coincidence of masses, lifetimes, and so on. But the more it was studied, the more remarkable the coincidence, and the suspicion gradually grew that possibly the deep law of the reflection symmetry of nature may be false.

As a result of this apparent failure, the physicists Lee and Yang suggested that other experiments be done in related decays to try to test whether the law was correct in other cases. The first such experiment was carried out by Miss Wu from Columbia, and was done as follows. Using a very strong magnet at a very low temperature, it turns out that a certain isotope of cobalt, which disintegrates by emitting an electron, is magnetic, and if the temperature is low enough that the thermal oscillations do not jiggle the atomic magnets about too much, they line up in the magnetic field. So the cobalt atoms will all line up in this strong field. They then disintegrate, emitting an electron, and it was discovered that when the atoms were lined up in a field whose $\mathbf{B}$ vector points upward, most of the electrons were emitted in a downward direction.

If one is not really “hep” to the world, such a remark does not sound like anything of significance, but if one appreciates the problems and interesting things in the world, then he sees that it is a most dramatic discovery: When we put cobalt atoms in an extremely strong magnetic field, more disintegration electrons go down than up. Therefore if we were to put it in a corresponding experiment in a “mirror,” in which the cobalt atoms would be lined up in the opposite direction, they would spit their electrons *up*, not *down*; the action is *unsymmetrical*. *The magnet has grown hairs!* The south pole of a magnet is of such a kind that the electrons in a β -disintegration tend to go away from it; that distinguishes, in a physical way, the north pole from the south pole.

After this, a lot of other experiments were done: the disintegration of the π into μ and ν ; μ into an electron and two neutrinos; nowadays, the Λ into proton and π ; disintegration of Σ 's; and many other disintegrations. In fact, in almost all cases where it could be expected, all have been found *not* to obey reflection symmetry! Fundamentally, the law of reflection symmetry, at this level in physics, is incorrect.

In short, we can tell a Martian where to put the heart: we say, “Listen, build yourself a magnet, and put the coils in, and put the current on, and then take some cobalt and lower the temperature. Arrange the experiment so the electrons go from the foot to the head, then the direction in which the current goes through the coils is the direction that goes in on what we call the right and comes out on the left.” So it is possible to define right and left, now, by doing an experiment of this kind.

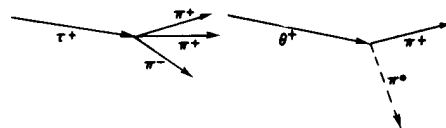


Fig. 52-5. A schematic diagram of the disintegration of a τ^+ and a θ^+ particle.

There are a lot of other features that were predicted. For example, it turns out that the spin, the angular momentum, of the cobalt nucleus before disintegration is 5 units of $\hbar$, and after disintegration it is 4 units. The electron carries spin angular momentum, and there is also a neutrino involved. It is easy to see from this that the electron must carry its spin angular momentum aligned along its direction of motion, the neutrino likewise. So it looks as though the electron is spinning to the left, and that was also checked. In fact, it was checked right here at Caltech by Boehm and Wapstra, that the electrons spin mostly to the left. (There were some other experiments that gave the opposite answer, but they were wrong!)

The next problem, of course, was to find the law of the failure of parity conservation. What is the rule that tells us how strong the failure is going to be? The rule is this: it occurs only in these very slow reactions, called weak decays, and when it occurs, the rule is that the particles which carry spin, like the electron, neutrino, and so on, come out with a spin tending to the left. That is a lopsided rule; it connects a polar vector velocity and an axial vector angular momentum, and says that the angular momentum is more likely to be opposite to the velocity than along it.

Now that is the rule, but today we do not really understand the whys and wherefores of it. *Why* is this the right rule, what is the fundamental reason for it, and how is it connected to anything else? At the moment we have been so shocked by the fact that this thing is unsymmetrical that we have not been able to recover enough to understand what it means with regard to all the other rules. However, the subject is interesting, modern, and still unsolved, so it seems appropriate that we discuss some of the questions associated with it.

52-8 Antimatter

The first thing to do when one of the symmetries is lost is to immediately go back over the list of known or assumed symmetries and ask whether any of the others are lost. Now we did not mention one operation on our list, which must necessarily be questioned, and that is the relation between matter and antimatter. Dirac predicted that in addition to electrons there must be another particle, called the positron (discovered at Caltech by Anderson), that is necessarily related to the electron. All the properties of these two particles obey certain rules of correspondence: the energies are equal; the masses are equal; the charges are reversed; but, more important than anything, the two of them, when they come together, can annihilate each other and liberate their entire mass in the form of energy, say γ -rays. The positron is called an *antiparticle* to the electron, and these are the characteristics of a particle and its antiparticle. It was clear from Dirac's argument that all the rest of the particles in the world should also have corresponding antiparticles. For instance, for the proton there should be an antiproton, which is now symbolized by a $\bar{p}$. The $\bar{p}$ would have a negative electrical charge and the same mass as a proton, and so on. The most important feature, however, is that a proton and an antiproton coming together can annihilate each other. The reason we emphasize this is that people do not understand it when we say there is a neutron and also an antineutron, because they say, "A neutron is neutral, so how *can* it have the opposite charge?" The rule of the "anti" is not just that it has the opposite charge, it has a certain set of properties, the whole lot of which are opposite. The antineutron is distinguished from the neutron in this way: if we bring two neutrons together, they just stay as two neutrons, but if we bring a neutron and an antineutron together, they annihilate each other with a great explosion of energy being liberated, with various π -mesons, γ -rays, and whatnot.

Now if we have antineutrons, antiprotons, and antielectrons, we can make antiatoms, in principle. They have not been made yet, but it is possible in principle. For instance, a hydrogen atom has a proton in the center with an electron going around outside. Now imagine that somewhere we can make an antiproton with a positron going around, would it go around? Well, first of all, the antiproton is electrically negative and the antielectron is electrically positive, so they attract each other in a corresponding manner—the masses are all the same; everything is the

same. It is one of the principles of the symmetry of physics, the equations seem to show, that if a clock, say, were made of matter on one hand, and then we made the same clock of antimatter, it would run in this way. (Of course, if we put the clocks together, they would annihilate each other, but that is different.)

An immediate question then arises. We can build, out of matter, two clocks, one which is "left-hand" and one which is "right-hand." For example, we could build a clock which is not built in a simple way, but has cobalt and magnets and electron detectors which detect the presence of β -decay electrons and count them. Each time one is counted, the second hand moves over. Then the mirror clock, receiving fewer electrons, will not run at the same rate. So evidently we can make two clocks such that the left-hand clock does not agree with the right-hand one. Let us make, out of matter, a clock which we call the standard or right-hand clock. Now let us make, also out of matter, a clock which we call the left-hand clock. We have just discovered that, in general, these two will *not* run the same way; prior to that famous physical discovery, it was thought that they would. Now it was also supposed that matter and antimatter were equivalent. That is, if we made an antimatter clock, right-hand, the same shape, then it would run the same as the right-hand matter clock, and if we made the same clock to the left it would run the same. In other words, in the beginning it was believed that *all four* of these clocks were the same; now of course we know that the right-hand and left-hand matter are not the same. Presumably, therefore, the right-handed antimatter and the left-handed antimatter are not the same.

So the obvious question is, which goes with which, if either? In other words, does the right-handed matter behave the same way as the right-handed antimatter? Or does the right-handed matter behave the same as the left-handed antimatter? β -decay experiments, using positron decay instead of electron decay, indicate that this is the interconnection: matter to the "right" works the same way as antimatter to the "left."

Therefore, at long last, it is really true that right and left symmetry is still maintained! If we made a left-hand clock, but made it out of the other kind of matter, antimatter instead of matter, it would run in the same way. So what has happened is that instead of having two independent rules in our list of symmetries, two of these rules go together to make a new rule, which says that matter to the right is symmetrical with antimatter to the left.

So if our Martian is made of antimatter and we give him instructions to make this "right" handed model like us, it will, of course, come out the other way around. What would happen when, after much conversation back and forth, we each have taught the other to make space ships and we meet halfway in empty space? We have instructed each other on our traditions, and so forth, and the two of us come rushing out to shake hands. Well, if he puts out his left hand, watch out!

52-9 Broken symmetries

The next question is, what can we make out of laws which are *nearly* symmetrical? The marvelous thing about it all is that for such a wide range of important, strong phenomena—nuclear forces, electrical phenomena, and even weak ones like gravitation—over a tremendous range of physics, all the laws for these seem to be symmetrical. On the other hand, this little extra piece says, "No, the laws are not symmetrical!" How is it that nature can be almost symmetrical, but not perfectly symmetrical? What shall we make of this? First, do we have any other examples? The answer is, we do, in fact, have a few other examples. For instance, the nuclear part of the force between proton and proton, between neutron and neutron, and between neutron and proton, is all exactly the same—there is a symmetry for nuclear forces, a new one, that we can interchange neutron and proton—but it evidently is not a general symmetry, for the electrical repulsion between two protons at a distance does not exist for neutrons. So it is not generally true that we can *always* replace a proton with a neutron, but only to a good approximation. Why *good*? Because the nuclear forces are much stronger than the

electrical forces. So this is an “almost” symmetry also. So we do have examples in other things.

We have, in our minds, a tendency to accept symmetry as some kind of perfection. In fact it is like the old idea of the Greeks that circles were perfect, and it was rather horrible to believe that the planetary orbits were not circles, but only nearly circles. The difference between being a circle and being nearly a circle is not a small difference, it is a fundamental change so far as the mind is concerned. There is a sign of perfection and symmetry in a circle that is not there the moment the circle is slightly off—that is the end of it—it is no longer symmetrical. Then the question is why it is only *nearly* a circle—that is a much more difficult question. The actual motion of the planets, in general, should be ellipses, but during the ages, because of tidal forces, and so on, they have been made almost symmetrical. Now the question is whether we have a similar problem here. The problem from the point of view of the circles is if they were perfect circles there would be nothing to explain, that is clearly simple. But since they are only nearly circles, there is a lot to explain, and the result turned out to be a big dynamical problem, and now our problem is to explain why they are nearly symmetrical by looking at tidal forces and so on.

So our problem is to explain where symmetry comes from. Why is nature so nearly symmetrical? No one has any idea why. The only thing we might suggest is something like this: There is a gate in Japan, a gate in Neiko, which is sometimes called by the Japanese the most beautiful gate in all Japan; it was built in a time when there was great influence from Chinese art. This gate is very elaborate, with lots of gables and beautiful carving and lots of columns and dragon heads and princes carved into the pillars, and so on. But when one looks closely he sees that in the elaborate and complex design along one of the pillars, one of the small design elements is carved upside down; otherwise the thing is completely symmetrical. If one asks why this is, the story is that it was carved upside down so that the gods will not be jealous of the perfection of man. So they purposely put an error in there, so that the gods would not be jealous and get angry with human beings.

We might like to turn the idea around and think that the true explanation of the near symmetry of nature is this: that God made the laws only nearly symmetrical so that we should not be jealous of His perfection!

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The Feynman

LECTURES ON PHYSICS

MAINLY ELECTROMAGNETISM AND MATTER

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Feynman's Preface

These are the lectures in physics that I gave last year and the year before to the freshman and sophomore classes at Caltech. The lectures are, of course, not verbatim—they have been edited, sometimes extensively and sometimes less so. The lectures form only part of the complete course. The whole group of 180 students gathered in a big lecture room twice a week to hear these lectures and then they broke up into small groups of 15 to 20 students in recitation sections under the guidance of a teaching assistant. In addition, there was a laboratory session once a week.

The special problem we tried to get at with these lectures was to maintain the interest of the very enthusiastic and rather smart students coming out of the high schools and into Caltech. They have heard a lot about how interesting and exciting physics is—the theory of relativity, quantum mechanics, and other modern ideas. By the end of two years of our previous course, many would be very discouraged because there were really very few grand, new, modern ideas presented to them. They were made to study inclined planes, electrostatics, and so forth, and after two years it was quite stultifying. The problem was whether or not we could make a course which would save the more advanced and excited student by maintaining his enthusiasm.

The lectures here are not in any way meant to be a survey course, but are very serious. I thought to address them to the most intelligent in the class and to make sure, if possible, that even the most intelligent student was unable to completely encompass everything that was in the lectures—by putting in suggestions of applications of the ideas and concepts in various directions outside the main line of attack. For this reason, though, I tried very hard to make all the statements as accurate as possible, to point out in every case where the equations and ideas fitted into the body of physics, and how—when they learned more—things would be modified. I also felt that for such students it is important to indicate what it is that they should—if they are sufficiently clever—be able to understand by deduction from what has been said before, and what is being put in as something new. When new ideas came in, I would try either to deduce them if they were deducible, or to explain that it *was* a new idea which hadn't any basis in terms of things they had already learned and which was not supposed to be provable—but was just added in.

At the start of these lectures, I assumed that the students knew something when they came out of high school—such things as geometrical optics, simple chemistry ideas, and so on. I also didn't see that there was any reason to make the lectures

in a definite order, in the sense that I would not be allowed to mention something until I was ready to discuss it in detail. There was a great deal of mention of things to come, without complete discussions. These more complete discussions would come later when the preparation became more advanced. Examples are the discussions of inductance, and of energy levels, which are at first brought in in a very qualitative way and are later developed more completely.

At the same time that I was aiming at the more active student, I also wanted to take care of the fellow for whom the extra fireworks and side applications are merely disquieting and who cannot be expected to learn most of the material in the lecture at all. For such students I wanted there to be at least a central core or backbone of material which he *could* get. Even if he didn't understand everything in a lecture, I hoped he wouldn't get nervous. I didn't expect him to understand everything, but only the central and most direct features. It takes, of course, a certain intelligence on his part to see which are the central theorems and central ideas, and which are the more advanced side issues and applications which he may understand only in later years.

In giving these lectures there was one serious difficulty: in the way the course was given, there wasn't any feedback from the students to the lecturer to indicate how well the lectures were going over. This is indeed a very serious difficulty, and I don't know how good the lectures really are. The whole thing was essentially an experiment. And if I did it again I wouldn't do it the same way—I hope I *don't* have to do it again! I think, though, that things worked out—so far as the physics is concerned—quite satisfactorily in the first year.

In the second year I was not so satisfied. In the first part of the course, dealing with electricity and magnetism, I couldn't think of any really unique or different way of doing it—of any way that would be particularly more exciting than the usual way of presenting it. So I don't think I did very much in the lectures on electricity and magnetism. At the end of the second year I had originally intended to go on, after the electricity and magnetism, by giving some more lectures on the properties of materials, but mainly to take up things like fundamental modes, solutions of the diffusion equation, vibrating systems, orthogonal functions, . . . developing the first stages of what are usually called “the mathematical methods of physics.” In retrospect, I think that if I were doing it again I would go back to that original idea. But since it was not planned that I would be giving these lectures again, it was suggested that it might be a good idea to try to give an introduction to the quantum mechanics—what you will find in Volume III.

It is perfectly clear that students who will major in physics can wait until their third year for quantum mechanics. On the other hand, the argument was made that many of the students in our course study physics as a background for their primary interest in other fields. And the usual way of dealing with quantum mechanics makes that subject almost unavailable for the great majority of students because they have to take so long to learn it. Yet, in its real applications—especially in its more complex applications, such as in electrical engineering and chemistry—the full machinery of the differential equation approach is not actually used. So I tried to describe the principles of quantum mechanics in a way which wouldn't require that one first know the mathematics of partial differential equations. Even for a physicist I think that is an interesting thing to try to do—to present quantum mechanics in this reverse fashion—for several reasons which may be apparent in the lectures themselves. However, I think that the experiment in the quantum mechanics part was not completely successful—in large part because I really did not have enough time at the end (I should, for instance, have had three or four more lectures in order to deal more completely with such matters as energy bands and the spatial dependence of amplitudes). Also, I had never presented the subject this way before, so the lack of feedback was particularly serious. I now believe the quantum mechanics should be given at a later time. Maybe I'll have a chance to do it again someday. Then I'll do it right.

The reason there are no lectures on how to solve problems is because there were recitation sections. Although I did put in three lectures in the first year on how to solve problems, they are not included here. Also there was a lecture on inertial

guidance which certainly belongs after the lecture on rotating systems, but which was, unfortunately, omitted. The fifth and sixth lectures are actually due to Matthew Sands, as I was out of town.

The question, of course, is how well this experiment has succeeded. My own point of view—which, however, does not seem to be shared by most of the people who worked with the students—is pessimistic. I don't think I did very well by the students. When I look at the way the majority of the students handled the problems on the examinations, I think that the system is a failure. Of course, my friends point out to me that there were one or two dozen students who—very surprisingly—understood almost everything in all of the lectures, and who were quite active in working with the material and worrying about the many points in an excited and interested way. These people have now, I believe, a first-rate background in physics—and they are, after all, the ones I was trying to get at. But then, “The power of instruction is seldom of much efficacy except in those happy dispositions where it is almost superfluous.” (Gibbons)

Still, I didn't want to leave any student completely behind, as perhaps I did. I think one way we could help the students more would be by putting more hard work into developing a set of problems which would elucidate some of the ideas in the lectures. Problems give a good opportunity to fill out the material of the lectures and make more realistic, more complete, and more settled in the mind the ideas that have been exposed.

I think, however, that there isn't any solution to this problem of education other than to realize that the best teaching can be done only when there is a direct individual relationship between a student and a good teacher—a situation in which the student discusses the ideas, thinks about the things, and talks about the things. It's impossible to learn very much by simply sitting in a lecture, or even by simply doing problems that are assigned. But in our modern times we have so many students to teach that we have to try to find some substitute for the ideal. Perhaps my lectures can make some contribution. Perhaps in some small place where there are individual teachers and students, they may get some inspiration or some ideas from the lectures. Perhaps they will have fun thinking them through—or going on to develop some of the ideas further.

RICHARD P. FEYNMAN

June, 1963

Foreword

For some forty years Richard P. Feynman focussed his curiosity on the mysterious workings of the physical world, and bent his intellect to searching out the order in its chaos. Now, he has given two years of his ability and his energy to his Lectures on Physics for beginning students. For them he has distilled the essence of his knowledge, and has created in terms they can hope to grasp a picture of the physicist's universe. To his lectures he has brought the brilliance and clarity of his thought, the originality and vitality of his approach, and the contagious enthusiasm of his delivery. It was a joy to behold.

The first year's lectures formed the basis for the first volume of this set of books. We have tried in this the second volume to make some kind of a record of a part of the second year's lectures—which were given to the sophomore class during the 1962–1963 academic year. The rest of the second year's lectures will make up Volume III.

Of the second year of lectures, the first two-thirds were devoted to a fairly complete treatment of the physics of electricity and magnetism. Its presentation was intended to serve a dual purpose. We hoped, first, to give the students a complete view of one of the great chapters of physics—from the early gropings of Franklin, through the great synthesis of Maxwell, on to the Lorentz electron theory of material properties, and ending with the still unsolved dilemmas of the electromagnetic self-energy. And we hoped, second, by introducing at the outset the calculus of vector fields, to give a solid introduction to the mathematics of field theories. To emphasize the general utility of the mathematical methods, related subjects from other parts of physics were sometimes analyzed together with their electric counterparts. We continually tried to drive home the generality of the mathematics. (“The same equations have the same solutions.”) And we emphasized this point by the kinds of exercises and examinations we gave with the course.

Following the electromagnetism there are two chapters each on elasticity and fluid flow. In the first chapter of each pair, the elementary and practical aspects are treated. The second chapter on each subject attempts to give an overview of the whole complex range of phenomena which the subject can lead to. These four chapters can well be omitted without serious loss, since they are not at all a necessary preparation for Volume III.

The last quarter, approximately, of the second year was dedicated to an introduction to quantum mechanics. This material has been put into the third volume.

In this record of the Feynman Lectures we wished to do more than provide a transcription of what was said. We hoped to make the written version as clear an exposition as possible of the ideas on which the original lectures were based. For some of the lectures this could be done by making only minor adjustments of the wording in the original transcript. For others of the lectures a major reworking and rearrangement of the material was required. Sometimes we felt we should add some new material to improve the clarity or balance of the presentation. Throughout the process we benefitted from the continual help and advice of Professor Feynman.

The translation of over 1,000,000 spoken words into a coherent text on a tight schedule is a formidable task, particularly when it is accompanied by the

other onerous burdens which come with the introduction of a new course—preparing for recitation sections, and meeting students, designing exercises and examinations, and grading them, and so on. Many hands—and heads—were involved. In some instances we have, I believe, been able to render a faithful image—or a tenderly retouched portrait—of the original Feynman. In other instances we have fallen far short of this ideal. Our successes are owed to all those who helped. The failures, we regret.

As explained in detail in the Foreword to Volume I, these lectures were but one aspect of a program initiated and supervised by the Physics Course Revision Committee (R. B. Leighton, Chairman, H. V. Neher, and M. Sands) at the California Institute of Technology, and supported financially by the Ford Foundation. In addition, the following people helped with one aspect or another of the preparation of textual material for this second volume: T. K. Caughey, M. L. Clayton, J. B. Curcio, J. B. Hartle, T. W. H. Harvey, M. H. Israel, W. J. Karzas, R. W. Kavanagh, R. B. Leighton, J. Mathews, M. S. Plesset, F. L. Warren, W. Whaling, C. H. Wilts, and B. Zimmerman. Others contributed indirectly through their work on the course: J. Blue, G. F. Chapline, M. J. Clauser, R. Dolen, H. H. Hill, and A. M. Title. Professor Gerry Neugebauer contributed in all aspects of our task with a diligence and devotion far beyond the dictates of duty.

The story of physics you find here would, however, not have been, except for the extraordinary ability and industry of Richard P. Feynman.

MATTHEW SANDS

March, 1964

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INDEX

Electromagnetism

1-1 Electrical forces

Consider a force like gravitation which varies predominantly inversely as the square of the distance, but which is about a *billion-billion-billion-billion* times stronger. And with another difference. There are two kinds of "matter," which we can call positive and negative. Like kinds repel and unlike kinds attract—unlike gravity where there is only attraction. What would happen?

A bunch of positives would repel with an enormous force and spread out in all directions. A bunch of negatives would do the same. But an evenly mixed bunch of positives and negatives would do something completely different. The opposite pieces would be pulled together by the enormous attractions. The net result would be that the terrific forces would balance themselves out almost perfectly, by forming tight, fine mixtures of the positive and the negative, and between two separate bunches of such mixtures there would be practically no attraction or repulsion at all.

There is such a force: the electrical force. And all matter is a mixture of positive protons and negative electrons which are attracting and repelling with this great force. So perfect is the balance, however, that when you stand near someone else you don't feel any force at all. If there were even a little bit of unbalance you would know it. If you were standing at arm's length from someone and each of you had *one percent* more electrons than protons, the repelling force would be incredible. How great? Enough to lift the Empire State Building? No! To lift Mount Everest? No! The repulsion would be enough to lift a "weight" equal to that of the entire earth!

With such enormous forces so perfectly balanced in this intimate mixture, it is not hard to understand that matter, trying to keep its positive and negative charges in the finest balance, can have a great stiffness and strength. The Empire State Building, for example, swings only eight feet in the wind because the electrical forces hold every electron and proton more or less in its proper place. On the other hand, if we look at matter on a scale small enough that we see only a few atoms, any small piece will not, usually, have an equal number of positive and negative charges, and so there will be strong residual electrical forces. Even when there are equal numbers of both charges in two neighboring small pieces, there may still be large net electrical forces because the forces between individual charges vary inversely as the square of the distance. A net force can arise if a negative charge of one piece is closer to the positive than to the negative charges of the other piece. The attractive forces can then be larger than the repulsive ones and there can be a net attraction between two small pieces with no excess charges. The force that holds the atoms together, and the chemical forces that hold molecules together, are really electrical forces acting in regions where the balance of charge is not perfect, or where the distances are very small.

You know, of course, that atoms are made with positive protons in the nucleus and with electrons outside. You may ask: "If this electrical force is so terrific, why don't the protons and electrons just get on top of each other? If they want to be in an intimate mixture, why isn't it still more intimate?" The answer has to do with the quantum effects. If we try to confine our electrons in a region that is very close to the protons, then according to the uncertainty principle they must have some mean square momentum which is larger the more we try to confine them. It is this motion, required by the laws of quantum mechanics, that keeps the electrical attraction from bringing the charges any closer together.

1-1 Electrical forces

1-2 Electric and magnetic fields

1-3 Characteristics of vector fields

1-4 The laws of electromagnetism

1-5 What are the fields?

1-6 Electromagnetism in science and technology

Review: Chapter 12, Vol. I, Characteristics of Force

**Lower case Greek letters
and commonly used capitals**

α	alpha
β	beta
γ Γ	gamma
δ Δ	delta
ϵ	epsilon
ζ	zeta
η	eta
θ Θ	theta
ι	iota
κ	kappa
λ Λ	lambda
μ	mu
ν	nu
ξ Ξ	xi (ksi)
$\omicron$	omicron
π Π	pi
ρ	rho
σ Σ	sigma
τ	tau
υ Υ	upsilon
ϕ Φ	phi
χ	chi (khi)
ψ Ψ	psi
ω Ω	omega

There is another question: "What holds the nucleus together"? In a nucleus there are several protons, all of which are positive. Why don't they push themselves apart? It turns out that in nuclei there are, in addition to electrical forces, nonelectrical forces, called nuclear forces, which are greater than the electrical forces and which are able to hold the protons together in spite of the electrical repulsion. The nuclear forces, however, have a short range—their force falls off much more rapidly than $1/r^2$. And this has an important consequence. If a nucleus has too many protons in it, it gets too big, and it will not stay together. An example is uranium, with 92 protons. The nuclear forces act mainly between each proton (or neutron) and its nearest neighbor, while the electrical forces act over larger distances, giving a repulsion between each proton and all of the others in the nucleus. The more protons in a nucleus, the stronger is the electrical repulsion, until, as in the case of uranium, the balance is so delicate that the nucleus is almost ready to fly apart from the repulsive electrical force. If such a nucleus is just "tapped" lightly (as can be done by sending in a slow neutron), it breaks into two pieces, each with positive charge, and these pieces fly apart by electrical repulsion. The energy which is liberated is the energy of the atomic bomb. This energy is usually called "nuclear" energy, but it is really "electrical" energy released when electrical forces have overcome the attractive nuclear forces.

We may ask, finally, what holds a negatively charged electron together (since it has no nuclear forces). If an electron is all made of one kind of substance, each part should repel the other parts. Why, then, doesn't it fly apart? But does the electron have "parts"? Perhaps we should say that the electron is just a point and that electrical forces only act between *different* point charges, so that the electron does not act upon itself. Perhaps. All we can say is that the question of what holds the electron together has produced many difficulties in the attempts to form a complete theory of electromagnetism. The question has never been answered. We will entertain ourselves by discussing this subject some more in later chapters.

As we have seen, we should expect that it is a combination of electrical forces and quantum-mechanical effects that will determine the detailed structure of materials in bulk, and, therefore, their properties. Some materials are hard, some are soft. Some are electrical "conductors"—because their electrons are free to move about; others are "insulators"—because their electrons are held tightly to individual atoms. We shall consider later how some of these properties come about, but that is a very complicated subject, so we will begin by looking at the electrical forces only in simple situations. We begin by treating only the laws of electricity—including magnetism, which is really a part of the same subject.

We have said that the electrical force, like a gravitational force, decreases inversely as the square of the distance between charges. This relationship is called Coulomb's law. But it is not precisely true when charges are moving—the electrical forces depend also on the motions of the charges in a complicated way. One part of the force between moving charges we call the *magnetic* force. It is really one aspect of an electrical effect. That is why we call the subject "electromagnetism."

There is an important general principle that makes it possible to treat electromagnetic forces in a relatively simple way. We find, from experiment, that the force that acts on a particular charge—no matter how many other charges there are or how they are moving—depends only on the position of that particular charge, on the velocity of the charge, and on the amount of charge. We can write the force F on a charge q moving with a velocity v as

$$F = q(E + v \times B). \quad (1.1)$$

We call E the *electric field* and B the *magnetic field* at the location of the charge. The important thing is that the electrical forces from all the other charges in the universe can be summarized by giving just these two vectors. Their values will depend on *where* the charge is, and may change with *time*. Furthermore, if we replace that charge with another charge, the force on the new charge will be just in proportion to the amount of charge so long as all the rest of the charges in the

world do not change their positions or motions. (In real situations, of course, each charge produces forces on all other charges in the neighborhood and may cause these other charges to move, and so in some cases the fields *can* change if we replace our particular charge by another.)

We know from Vol. I how to find the motion of a particle if we know the force on it. Equation (1.1) can be combined with the equation of motion to give

$$\frac{d}{dt} \left[\frac{mv}{(1 - v^2/c^2)^{1/2}} \right] = \mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (1.2)$$

So if $\mathbf{E}$ and $\mathbf{B}$ are given, we can find the motions. Now we need to know how the $\mathbf{E}$'s and $\mathbf{B}$'s are produced.

One of the most important simplifying principles about the way the fields are produced is this: Suppose a number of charges moving in some manner would produce a field $\mathbf{E}_1$, and another set of charges would produce $\mathbf{E}_2$. If both sets of charges are in place at the same time (keeping the same locations and motions they had when considered separately), then the field produced is just the sum

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2. \quad (1.3)$$

This fact is called *the principle of superposition* of fields. It holds also for magnetic fields.

This principle means that if we know the law for the electric and magnetic fields produced by a *single* charge moving in an arbitrary way, then all the laws of electrodynamics are complete. If we want to know the force on charge A we need only calculate the $\mathbf{E}$ and $\mathbf{B}$ produced by each of the charges B, C, D , etc., and then add the $\mathbf{E}$'s and $\mathbf{B}$'s from all the charges to find the fields, and from them the forces acting on charge A . If it had only turned out that the field produced by a single charge was simple, this would be the neatest way to describe the laws of electrodynamics. We have already given a description of this law (Chapter 28, Vol. I) and it is, unfortunately, rather complicated.

It turns out that the form in which the laws of electrodynamics are simplest are not what you might expect. It is *not* simplest to give a formula for the force that one charge produces on another. It is true that when charges are standing still the Coulomb force law is simple, but when charges are moving about the relations are complicated by delays in time and by the effects of acceleration, among others. As a result, we do not wish to present electrodynamics only through the force laws between charges; we find it more convenient to consider another point of view—a point of view in which the laws of electrodynamics appear to be the most easily manageable.

1-2 Electric and magnetic fields

First, we must extend, somewhat, our ideas of the electric and magnetic vectors, $\mathbf{E}$ and $\mathbf{B}$. We have defined them in terms of the forces that are felt by a charge. We wish now to speak of electric and magnetic fields *at a point* even when there is no charge present. We are saying, in effect, that since there are forces “acting on” the charge, there is still “something” there when the charge is removed. If a charge located at the point (x, y, z) at the time t feels the force $\mathbf{F}$ given by Eq. (1.1) we associate the vectors $\mathbf{E}$ and $\mathbf{B}$ with *the point* in space (x, y, z) . We may think of $\mathbf{E}(x, y, z, t)$ and $\mathbf{B}(x, y, z, t)$ as giving the forces that *would be* experienced at the time t by a charge located at (x, y, z) , *with the condition* that placing the charge there *did not disturb* the positions or motions of all the other charges responsible for the fields.

Following this idea, we associate with *every* point (x, y, z) in space two vectors $\mathbf{E}$ and $\mathbf{B}$, which may be changing with time. The electric and magnetic fields are, then, viewed as *vector functions* of x, y, z , and t . Since a vector is specified by its components, each of the fields $\mathbf{E}(x, y, z, t)$ and $\mathbf{B}(x, y, z, t)$ represent three mathematical functions of x, y, z , and t .

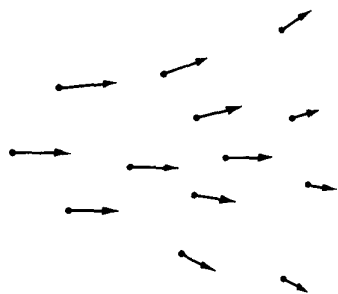


Fig. 1-1. A vector field may be represented by drawing a set of arrows whose magnitudes and directions indicate the values of the vector field at the points from which the arrows are drawn.

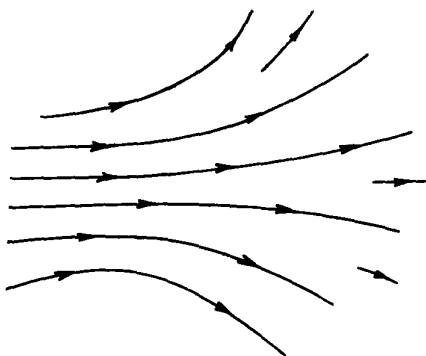


Fig. 1-2. A vector field can be represented by drawing lines which are tangent to the direction of the field vector at each point, and by drawing the density of lines proportional to the magnitude of the field vector.

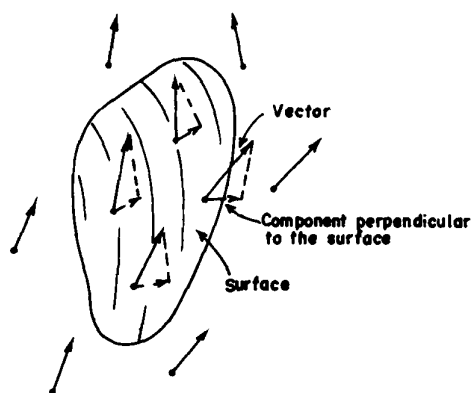


Fig. 1-3. The flux of a vector field through a surface is defined as the average value of the normal component of the vector times the area of the surface.

It is precisely because E (or B) can be specified at every point in space that it is called a "field." A "field" is any physical quantity which takes on different values at different points in space. Temperature, for example, is a field—in this case a scalar field, which we write as $T(x, y, z)$. The temperature could also vary in time, and we would say the temperature field is time-dependent, and write $T(x, y, z, t)$. Another example is the "velocity field" of a flowing liquid. We write $v(x, y, z, t)$ for the velocity of the liquid at each point in space at the time t . It is a vector field.

Returning to the electromagnetic fields—although they are produced by charges according to complicated formulas, they have the following important characteristic: the relationships between the values of the fields at *one point* and the values at a *nearby point* are very simple. With only a few such relationships in the form of differential equations we can describe the fields completely. It is in terms of such equations that the laws of electrodynamics are most simply written.

There have been various inventions to help the mind visualize the behavior of fields. The most correct is also the most abstract: we simply consider the fields as mathematical functions of position and time. We can also attempt to get a mental picture of the field by drawing vectors at many points in space, each of which gives the field strength and direction at that point. Such a representation is shown in Fig. 1-1. We can go further, however, and draw lines which are everywhere tangent to the vectors—which, so to speak, follow the arrows and keep track of the direction of the field. When we do this we lose track of the *lengths* of the vectors, but we can keep track of the strength of the field by drawing the lines far apart when the field is weak and close together when it is strong. We adopt the convention that the *number of lines per unit area* at right angles to the lines is proportional to the *field strength*. This is, of course, only an approximation, and it will require, in general, that new lines sometimes start up in order to keep the number up to the strength of the field. The field of Fig. 1-1 is represented by field lines in Fig. 1-2.

1-3 Characteristics of vector fields

There are two mathematically important properties of a vector field which we will use in our description of the laws of electricity from the field point of view. Suppose we imagine a closed surface of some kind and ask whether we are losing "something" from the inside; that is, does the field have a quality of "outflow"? For instance, for a velocity field we might ask whether the velocity is always outward on the surface or, more generally, whether more fluid flows out (per unit time) than comes in. We call the net amount of fluid going out through the surface per unit time the "flux of velocity" through the surface. The flow through an element of a surface is just equal to the component of the velocity perpendicular to the surface times the area of the surface. For an arbitrary closed surface, the *net outward flow*—or *flux*—is the average outward normal component of the velocity, times the area of the surface:

$$\text{Flux} = (\text{average normal component}) \cdot (\text{surface area}). \quad (1.4)$$

In the case of an electric field, we can mathematically define something analogous to an outflow, and we again call it the flux, but of course it is not the flow of any substance, because the electric field is not the velocity of anything. It turns out, however, that the mathematical quantity which is the average normal component of the field still has a useful significance. We speak, then, of the *electric flux*—also defined by Eq. (1.4). Finally, it is also useful to speak of the flux not only through a completely closed surface, but through any bounded surface. As before, the flux through such a surface is defined as the average normal component of a vector times the area of the surface. These ideas are illustrated in Fig. 1-3.

There is a second property of a vector field that has to do with a line, rather than a surface. Suppose again that we think of a velocity field that describes the flow of a liquid. We might ask this interesting question: Is the liquid circulating?

By that we mean: Is there a net rotational motion around some loop? Suppose that we instantaneously freeze the liquid everywhere except inside of a tube which is of uniform bore, and which goes in a loop that closes back on itself as in Fig. 1-4. Outside of the tube the liquid stops moving, but inside the tube it may keep on moving because of the momentum in the trapped liquid—that is, if there is more momentum heading one way around the tube than the other. We define a quantity called the *circulation* as the resulting speed of the liquid in the tube times its circumference. We can again extend our ideas and define the “circulation” for any vector field (even when there isn’t anything moving). For any vector field the *circulation around any imagined closed curve* is defined as the average tangential component of the vector (in a consistent sense) multiplied by the circumference of the loop (Fig. 1-5).

$$\text{Circulation} = (\text{average tangential component}) \cdot (\text{distance around}). \quad (1.5)$$

You will see that this definition does indeed give a number which is proportional to the circulation velocity in the quickly frozen tube described above.

With just these two ideas—flux and circulation—we can describe all the laws of electricity and magnetism at once. You may not understand the significance of the laws right away, but they will give you some idea of the way the physics of electromagnetism will be ultimately described.

1-4 The laws of electromagnetism

The first law of electromagnetism describes the flux of the electric field:

$$\text{The flux of } \mathbf{E} \text{ through any closed surface} = \frac{\text{the net charge inside}}{\epsilon_0}, \quad (1.6)$$

where ϵ_0 is a convenient constant. (The constant ϵ_0 is usually read as “epsilon-zero” or “epsilon-naught”.) If there are no charges inside the surface, even though there are charges nearby outside the surface, the *average* normal component of $\mathbf{E}$ is zero, so there is no net flux through the surface. To show the power of this type of statement, we can show that Eq. (1.6) is the same as Coulomb’s law, provided only that we also add the idea that the field from a single charge is spherically symmetric. For a point charge, we draw a sphere around the charge. Then the average normal component is just the value of the magnitude of $\mathbf{E}$ at any point, since the field must be directed radially and have the same strength for all points on the sphere. Our rule now says that the field at the surface of the sphere, times the area of the sphere—that is, the outgoing flux—is proportional to the charge inside. If we were to make the radius of the sphere bigger, the area would increase as the square of the radius. The average normal component of the electric field times that area must still be equal to the same charge inside, and so the field must decrease as the square of the distance—we get an “inverse square” field.

If we have an arbitrary curve in space and measure the circulation of the electric field around the curve, we will find that it is not, in general, zero (although it is for the Coulomb field). Rather, for electricity there is a second law that states: for any surface S (not closed) whose edge is the curve C ,

$$\text{Circulation of } \mathbf{E} \text{ around } C = \frac{d}{dt} (\text{flux of } \mathbf{B} \text{ through } S). \quad (1.7)$$

We can complete the laws of the electromagnetic field by writing two corresponding equations for the magnetic field $\mathbf{B}$.

$$\text{Flux of } \mathbf{B} \text{ through any closed surface} = 0. \quad (1.8)$$

For a surface S bounded by the curve C ,

$$c^2(\text{circulation of } \mathbf{B} \text{ around } C) = \frac{d}{dt} (\text{flux of } \mathbf{E} \text{ through } S) + \frac{\text{flux of electric current through } S}{\epsilon_0}. \quad (1.9)$$

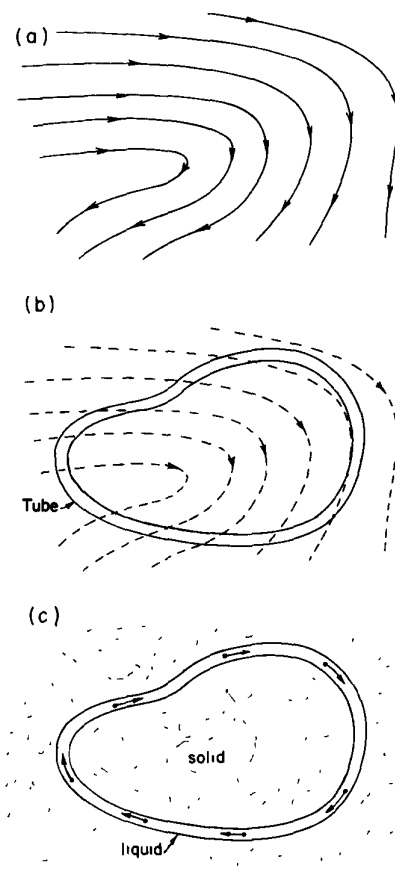


Fig. 1-4. (a) The velocity field in a liquid. Imagine a tube of uniform cross section that follows an arbitrary closed curve as in (b). If the liquid were suddenly frozen everywhere except inside the tube, the liquid in the tube would circulate as shown in (c).

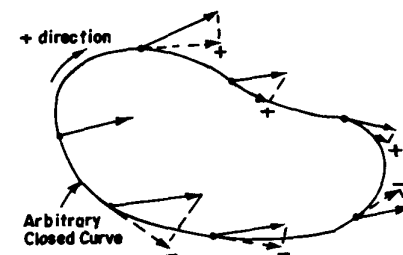


Fig. 1-5. The circulation of a vector field is the average tangential component of the vector (in a consistent sense) times the circumference of the loop.

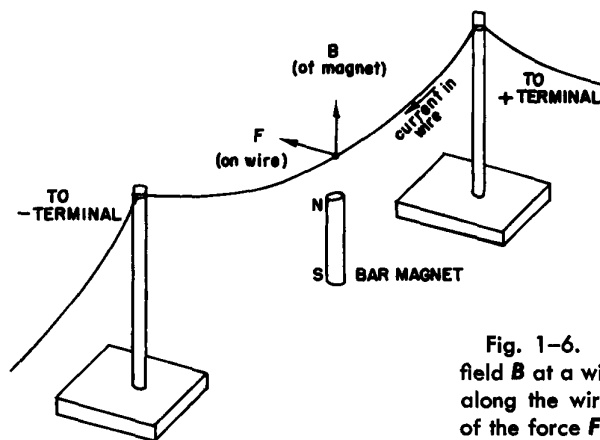


Fig. 1-6. A bar magnet gives a field B at a wire. When there is a current along the wire, the wire moves because of the force $F = qv \times B$.

The constant c^2 that appears in Eq. (1.9) is the square of the velocity of light. It appears because magnetism is in reality a relativistic effect of electricity. The constant ϵ_0 has been stuck in to make the units of electric current come out in a convenient way.

Equations (1.6) through (1.9), together with Eq. (1.1), are all the laws of electrodynamics*. As you remember, the laws of Newton were very simple to write down, but they had a lot of complicated consequences and it took us a long time to learn about them all. These laws are not nearly as simple to write down, which means that the consequences are going to be more elaborate and it will take us quite a lot of time to figure them all out.

We can illustrate some of the laws of electrodynamics by a series of small experiments which show qualitatively the interrelationships of electric and magnetic fields. You have experienced the first term of Eq. (1.1) when combing your hair, so we won't show that one. The second part of Eq. (1.1) can be demonstrated by passing a current through a wire which hangs above a bar magnet, as shown in Fig. 1-6. The wire will move when a current is turned on because of the force $F = qv \times B$. When a current exists, the charges inside the wire are moving, so they have a velocity v , and the magnetic field from the magnet exerts a force on them, which results in pushing the wire sideways.

When the wire is pushed to the left, we would expect that the magnet must feel a push to the right. (Otherwise we could put the whole thing on a wagon and have a propulsion system that didn't conserve momentum!) Although the force is too small to make movement of the bar magnet visible, a more sensitively supported magnet, like a compass needle, will show the movement.

How does the wire push on the magnet? The current in the wire produces a magnetic field of its own that exerts forces on the magnet. According to the last

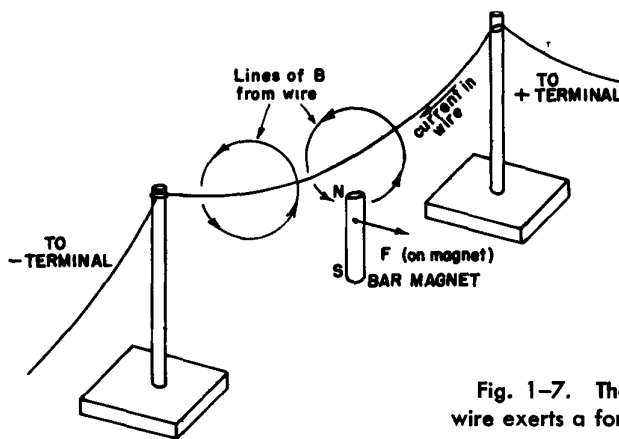


Fig. 1-7. The magnetic field of the wire exerts a force on the magnet.

* We need only to add a remark about some conventions for the *sign* of the circulation.

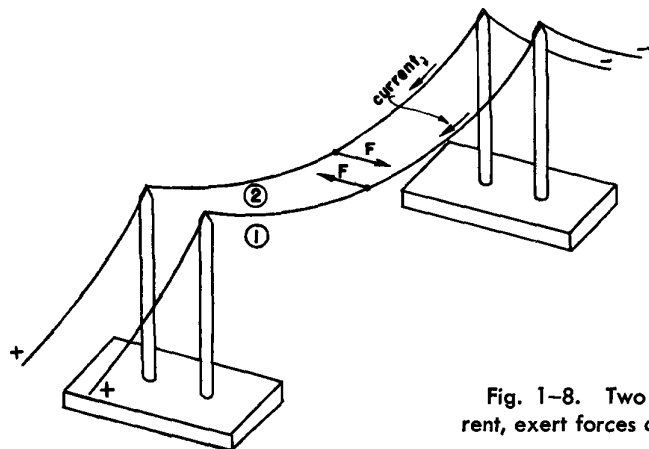


Fig. 1-8. Two wires, carrying current, exert forces on each other.

term in Eq. (1.9), a current must have a *circulation* of B —in this case, the lines of B are loops around the wire, as shown in Fig. 1-7. This B -field is responsible for the force on the magnet.

Equation (1.9) tells us that for a fixed current through the wire the circulation of B is the same for *any* curve that surrounds the wire. For curves—say circles—that are farther away from the wire, the circumference is larger, so the tangential component of B must decrease. You can see that we would, in fact, expect B to decrease linearly with the distance from a long straight wire.

Now, we have said that a current through a wire produces a magnetic field, and that when there is a magnetic field present there is a force on a wire carrying a current. Then we should also expect that if we make a magnetic field with a current in one wire, it should exert a force on another wire which also carries a current. This can be shown by using two hanging wires as shown in Fig. 1-8. When the currents are in the same direction, the two wires attract, but when the currents are opposite, they repel.

In short, electrical currents, as well as magnets, make magnetic fields. But wait, what is a magnet, anyway? If magnetic fields are produced by moving charges, is it not possible that the magnetic field from a piece of iron is really the result of currents? It appears to be so. We can replace the bar magnet of our experiment with a coil of wire, as shown in Fig. 1-9. When a current is passed through the coil—as well as through the straight wire above it—we observe a motion of the wire exactly as before, when we had a magnet instead of a coil. In other words, the current in the coil imitates a magnet. It appears, then, that a piece of iron acts as though it contains a perpetual circulating current. We can, in fact, understand magnets in terms of permanent currents in the atoms of the iron. The force on the magnet in Fig. 1-7 is due to the second term in Eq. (1.1).

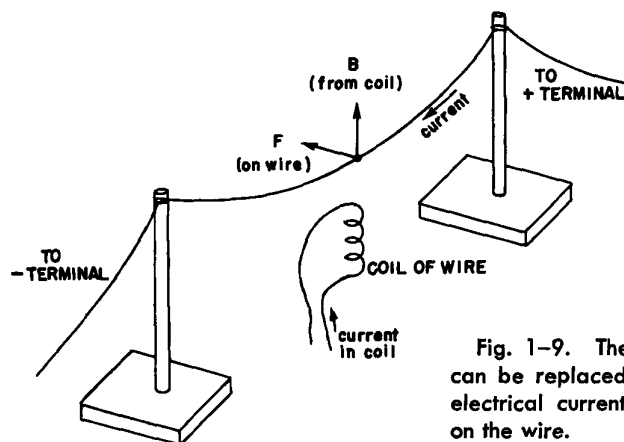
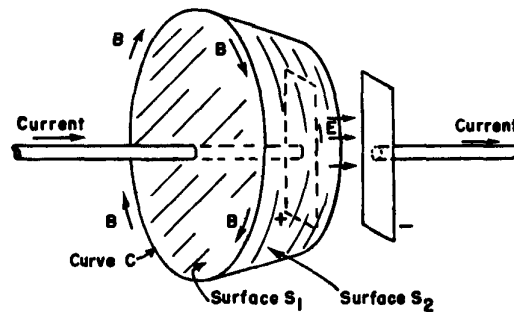


Fig. 1-9. The bar magnet of Fig. 1-6 can be replaced by a coil carrying an electrical current. A similar force acts on the wire.

Where do the currents come from? One possibility would be from the motion of the electrons in atomic orbits. Actually, that is not the case for iron, although it is for some materials. In addition to moving around in an atom, an electron also spins about on its own axis—something like the spin of the earth—and it is the current from this spin that gives the magnetic field in iron. (We say “something like the spin of the earth” because the question is so deep in quantum mechanics that the classical ideas do not really describe things too well.) In most substances, some electrons spin one way and some spin the other, so the magnetism cancels out, but in iron—for a mysterious reason which we will discuss later—many of the electrons are spinning with their axes lined up, and that is the source of the magnetism.

Since the fields of magnets are from currents, we do not have to add any extra term to Eqs. (1.8) or (1.9) to take care of magnets. We just take *all* currents, including the circulating currents of the spinning electrons, and then the law is right. You should also notice that Eq. (1.8) says that there are no magnetic “charges” analogous to the electrical charges appearing on the right side of Eq. (1.6). None has been found.

Fig. 1-10. The circulation of B around the curve C is given either by the current passing through the surface S_1 , or by the rate of change of the flux of E through the surface S_2 .



The first term on the right-hand side of Eq. (1.9) was discovered theoretically by Maxwell and is of great importance. It says that changing *electric* fields produce magnetic effects. In fact, without this term the equation would not make sense, because without it there could be no currents in circuits that are not complete loops. But such currents do exist, as we can see in the following example. Imagine a capacitor made of two flat plates. It is being charged by a current that flows toward one plate and away from the other, as shown in Fig. 1-10. We draw a curve C around one of the wires and fill it in with a surface which crosses the wire, as shown by the surface S_1 in the figure. According to Eq. (1.9), the circulation of B around C is given by the current in the wire (times c^2). But what if we fill in the curve with a *different* surface S_2 , which is shaped like a bowl and passes between the plates of the capacitor, staying always away from the wire? There is certainly no current through this surface. But, surely, just changing the location of an imaginary surface is not going to change a real magnetic field! The circulation of B must be what it was before. The first term on the right-hand side of Eq. (1.9) does, indeed, combine with the second term to give the same result for the two surfaces S_1 and S_2 . For S_2 the circulation of B is given in terms of the rate of change of the flux of E between the plates of the capacitor. And it works out that the changing E is related to the current in just the way required for Eq. (1.9) to be correct. Maxwell saw that it was needed, and he was the first to write the complete equation.

With the setup shown in Fig. 1-6 we can demonstrate another of the laws of electromagnetism. We disconnect the ends of the hanging wire from the battery and connect them to a galvanometer which tells us when there is a current through the wire. When we *push* the wire sideways through the magnetic field of the magnet, we observe a current. Such an effect is again just another consequence of Eq. (1.1)—the electrons in the wire feel the force $F = qv \times B$. The electrons have a sideways velocity because they move with the wire. This v with a vertical B from the magnet results in a force on the electrons directed *along* the wire, which starts the electrons moving toward the galvanometer.

Suppose, however, that we leave the wire alone and move the magnet. We guess from relativity that it should make no difference, and indeed, we observe a similar current in the galvanometer. How does the magnetic field produce forces on charges at rest? According to Eq. (1.1) there must be an electric field. A moving magnet must make an electric field. How that happens is said quantitatively by Eq. (1.7). This equation describes many phenomena of great practical interest, such as those that occur in electric generators and transformers.

The most remarkable consequence of our equations is that the combination of Eq. (1.7) and Eq. (1.9) contains the explanation of the radiation of electromagnetic effects over large distances. The reason is roughly something like this: suppose that somewhere we have a magnetic field which is increasing because, say, a current is turned on suddenly in a wire. Then by Eq. (1.7) there must be a circulation of an electric field. As the electric field builds up to produce its circulation, then according to Eq. (1.9) a magnetic circulation will be generated. But the building up of *this* magnetic field will produce a new circulation of the electric field, and so on. In this way fields work their way through space without the need of charges or currents except at their source. That is the way we *see* each other! It is all in the equations of the electromagnetic fields.

1-5 What are the fields?

We now make a few remarks on our way of looking at this subject. You may be saying: "All this business of fluxes and circulations is pretty abstract. There are electric fields at every point in space; then there are these 'laws.' But what is *actually* happening? Why can't you explain it, for instance, by whatever it *is* that goes between the charges." Well, it depends on your prejudices. Many physicists used to say that direct action with nothing in between was inconceivable. (How could they find an idea inconceivable when it had already been conceived?) They would say: "Look, the only forces we know are the direct action of one piece of matter on another. It is impossible that there can be a force with nothing to transmit it." But what really happens when we study the "direct action" of one piece of matter right against another? We discover that it is not one piece right against the other; they are slightly separated, and there are electrical forces acting on a tiny scale. Thus we find that we are going to explain so-called direct-contact action in terms of the picture for electrical forces. It is certainly not sensible to try to insist that an electrical force has to look like the old, familiar, muscular push or pull, when it will turn out that the muscular pushes and pulls are going to be interpreted as electrical forces! The only sensible question is what is the *most convenient* way to look at electrical effects. Some people prefer to represent them as the interaction at a distance of charges, and to use a complicated law. Others love the field lines. They draw field lines all the time, and feel that writing E 's and B 's is too abstract. The field lines, however, are only a crude way of describing a field, and it is very difficult to give the correct, quantitative laws directly in terms of field lines. Also, the ideas of the field lines do not contain the deepest principle of electrodynamics, which is the superposition principle. Even though we know how the field lines look for one set of charges and what the field lines look like for another set of charges, we don't get any idea about what the field line patterns will look like when both sets are present together. From the mathematical standpoint, on the other hand, superposition is easy—we simply add the two vectors. The field lines have some advantage in giving a vivid picture, but they also have some disadvantages. The direct interaction way of thinking has great advantages when thinking of electrical charges at rest, but has great disadvantages when dealing with charges in rapid motion.

The best way is to use the abstract field idea. That it is abstract is unfortunate, but necessary. The attempts to try to represent the electric field as the motion of some kind of gear wheels, or in terms of lines, or of stresses in some kind of material have used up more effort of physicists than it would have taken simply to get the right answers about electrodynamics. It is interesting that the correct equations for the behavior of light in crystals were worked out by McCullough in 1843. But

people said to him: "Yes, but there is no real material whose mechanical properties could possibly satisfy those equations, and since light is an oscillation that must vibrate in *something*, we cannot believe this abstract equation business." If people had been more open-minded, they might have believed in the right equations for the behavior of light a lot earlier than they did.

In the case of the magnetic field we can make the following point: Suppose that you finally succeeded in making up a picture of the magnetic field in terms of some kind of lines or of gear wheels running through space. Then you try to explain what happens to two charges moving in space, both at the same speed and parallel to each other. Because they are moving, they will behave like two currents and will have a magnetic field associated with them (like the currents in the wires of Fig. 1-8). An observer who was riding along with the two charges, however, would see both charges as stationary, and would say that there is *no* magnetic field. The "gear wheels" or "lines" disappear when you ride along with the object! All we have done is to invent a *new* problem. How can the gear wheels disappear?! The people who draw field lines are in a similar difficulty. Not only is it not possible to say whether the field lines move or do not move with charges—they may disappear completely in certain coordinate frames

What we are saying, then, is that magnetism is really a relativistic effect. In the case of the two charges we just considered, travelling parallel to each other, we would expect to have to make relativistic corrections to their motion, with terms of order v^2/c^2 . These corrections must correspond to the magnetic force. But what about the force between the two wires in our experiment (Fig. 1-8). There the magnetic force is the *whole* force. It didn't look like a "relativistic correction." Also, if we estimate the velocities of the electrons in the wire (you can do this yourself), we find that their average speed along the wire is about 0.01 centimeter per second. So v^2/c^2 is about 10^{-25} . Surely a negligible "correction." But no! Although the magnetic force is, in this case, 10^{-25} of the "normal" electrical force between the moving electrons, remember that the "normal" electrical forces have disappeared because of the almost perfect balancing out—because the wires have the same number of protons as electrons. The balance is much more precise than one part in 10^{25} , and the small relativistic term which we call the magnetic force is the only term left. It becomes the dominant term.

It is the near-perfect cancellation of electrical effects which allowed relativity effects (that is, magnetism) to be studied and the correct equations—to order v^2/c^2 —to be discovered, even though physicists didn't *know* that's what was happening. And that is why, when relativity was discovered, the electromagnetic laws didn't need to be changed. They—unlike mechanics—were already correct to a precision of v^2/c^2 .

1-6 Electromagnetism in science and technology

Let us end this chapter by pointing out that among the many phenomena studied by the Greeks there were two very strange ones: that if you rubbed a piece of amber you could lift up little pieces of papyrus, and that there was a strange rock from the island of Magnesia which attracted iron. It is amazing to think that these were the only phenomena known to the Greeks in which the effects of electricity or magnetism were apparent. The reason that these were the only phenomena that appeared is due primarily to the fantastic precision of the balancing of charges that we mentioned earlier. Study by scientists who came after the Greeks uncovered one new phenomena after another that were really some aspect of these amber and/or lodestone effects. Now we realize that the phenomena of chemical interaction and, ultimately, of life itself are to be understood in terms of electromagnetism.

At the same time that an understanding of the subject of electromagnetism was being developed, technical possibilities that defied the imagination of the people that came before were appearing: it became possible to signal by telegraph over long distances, and to talk to another person miles away without any connections between, and to run huge power systems—a great water wheel, connected by

filaments over hundreds of miles to another engine that turns in response to the master wheel—many thousands of branching filaments—ten thousand engines in ten thousand places running the machines of industries and homes—all turning because of the knowledge of the laws of electromagnetism.

Today we are applying even more subtle effects. The electrical forces, enormous as they are, can also be very tiny, and we can control them and use them in very many ways. So delicate are our instruments that we can tell what a man is doing by the way he affects the electrons in a thin metal rod hundreds of miles away. All we need to do is to use the rod as an antenna for a television receiver!

From a long view of the history of mankind—seen from, say, ten thousand years from now—there can be little doubt that the most significant event of the 19th century will be judged as Maxwell's discovery of the laws of electrodynamics. The American Civil War will pale into provincial insignificance in comparison with this important scientific event of the same decade.

Differential Calculus of Vector Fields

2-1 Understanding physics

The physicist needs a facility in looking at problems from several points of view. The exact analysis of real physical problems is usually quite complicated, and any particular physical situation may be too complicated to analyze directly by solving the differential equation. But one can still get a very good idea of the behavior of a system if one has some feel for the character of the solution in different circumstances. Ideas such as the field lines, capacitance, resistance, and inductance are, for such purposes, very useful. So we will spend much of our time analyzing them. In this way we will get a feel as to what should happen in different electromagnetic situations. On the other hand, none of the heuristic models, such as field lines, is really adequate and accurate for all situations. There is only one precise way of presenting the laws, and that is by means of differential equations. They have the advantage of being fundamental and, so far as we know, precise. If you have learned the differential equations you can always go back to them. There is nothing to unlearn.

It will take you some time to understand what should happen in different circumstances. You will have to solve the equations. Each time you solve the equations, you will learn something about the character of the solutions. To keep these solutions in mind, it will be useful also to study their meaning in terms of field lines and of other concepts. This is the way you will really “understand” the equations. That is the difference between mathematics and physics. Mathematicians, or people who have very mathematical minds, are often led astray when “studying” physics because they lose sight of the physics. They say: “Look, these differential equations—the Maxwell equations—are all there is to electrodynamics; it is admitted by the physicists that there is nothing which is not contained in the equations. The equations are complicated, but after all they are only mathematical equations and if I understand them mathematically inside out, I will understand the physics inside out.” Only it doesn’t work that way. Mathematicians who study physics with that point of view—and there have been many of them—usually make little contribution to physics and, in fact, little to mathematics. They fail because the actual physical situations in the real world are so complicated that it is necessary to have a much broader understanding of the equations.

What it means really to understand an equation—that is, in more than a strictly mathematical sense—was described by Dirac. He said: “I understand what an equation means if I have a way of figuring out the characteristics of its solution without actually solving it.” So if we have a way of knowing what should happen in given circumstances without actually solving the equations, then we “understand” the equations, as applied to these circumstances. A physical understanding is a completely unmathematical, imprecise, and inexact thing, but absolutely necessary for a physicist.

Ordinarily, a course like this is given by developing gradually the physical ideas—by starting with simple situations and going on to more and more complicated situations. This requires that you continuously forget things you previously learned—things that are true in certain situations, but which are not true in general. For example, the “law” that the electrical force depends on the square of the distance is not *always* true. We prefer the opposite approach. We prefer to take first the *complete* laws, and then to step back and apply them to simple situations, developing the physical ideas as we go along. And that is what we are going to do.

2-1 Understanding physics

2-2 Scalar and vector fields— T and h

2-3 Derivatives of fields—the gradient

2-4 The operator ∇

2-5 Operations with ∇

2-6 The differential equation of heat flow

2-7 Second derivatives of vector fields

2-8 Pitfalls

Review: Chapter 11, Vol. I, Vectors

Our approach is completely opposite to the historical approach in which one develops the subject in terms of the experiments by which the information was obtained. But the subject of physics has been developed over the past 200 years by some very ingenious people, and as we have only a limited time to acquire our knowledge, we cannot possibly cover everything they did. Unfortunately one of the things that we shall have a tendency to lose in these lectures is the historical, experimental development. It is hoped that in the laboratory some of this lack can be corrected. You can also fill in what we must leave out by reading the Encyclopedia Britannica, which has excellent historical articles on electricity and on other parts of physics. You will also find historical information in many textbooks on electricity and magnetism.

2-2 Scalar and vector fields— T and h

We begin now with the abstract, mathematical view of the theory of electricity and magnetism. The ultimate idea is to explain the meaning of the laws given in Chapter 1. But to do this we must first explain a new and peculiar notation that we want to use. So let us forget electromagnetism for the moment and discuss the mathematics of vector fields. It is of very great importance, not only for electromagnetism, but for all kinds of physical circumstances. Just as ordinary differential and integral calculus is so important to all branches of physics, so also is the differential calculus of vectors. We turn to that subject.

Listed below are a few facts from the algebra of vectors. It is assumed that you already know them.

$$\mathbf{A} \cdot \mathbf{B} = \text{scalar} = A_x B_x + A_y B_y + A_z B_z \quad (2.1)$$

$$\mathbf{A} \times \mathbf{B} = \text{vector} \quad (2.2)$$

$$(\mathbf{A} \times \mathbf{B})_z = A_x B_y - A_y B_x$$

$$(\mathbf{A} \times \mathbf{B})_x = A_y B_z - A_z B_y$$

$$(\mathbf{A} \times \mathbf{B})_y = A_z B_x - A_x B_z$$

$$\mathbf{A} \times \mathbf{A} = 0 \quad (2.3)$$

$$\mathbf{A} \cdot (\mathbf{A} \times \mathbf{B}) = 0 \quad (2.4)$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C} \quad (2.5)$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \quad (2.6)$$

Also we will want to use the two following equalities from the calculus:

$$\Delta f(x, y, z) = \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y + \frac{\partial f}{\partial z} \Delta z, \quad (2.7)$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}. \quad (2.8)$$

The first equation (2.7) is, of course, true only in the limit that Δx , Δy , and Δz go toward zero.

The simplest possible physical field is a scalar field. By a field, you remember, we mean a quantity which depends upon position in space. By a *scalar field* we merely mean a field which is characterized at each point by a single number—a scalar. Of course the number may change in time, but we need not worry about that for the moment. We will talk about what the field looks like at a given instant. As an example of a scalar field, consider a solid block of material which has been heated at some places and cooled at others, so that the temperature of the body varies from point to point in a complicated way. Then the temperature will be a function of x , y , and z , the position in space measured in a rectangular coordinate system. Temperature is a scalar field.

Writing vectors by hand.

Some people use

$\vec{E}$ or $\vec{E}$ or just $\bar{E}$.

Others prefer

$\underline{E}$.

We like the following way:

A B C D E F G

H I J K L M N

O P Q R S T U

V W X Y Z

Small letters are harder:

a b c d e f g

h i j k l m n

o p q r s t u

v w x y z

You can invent your own.

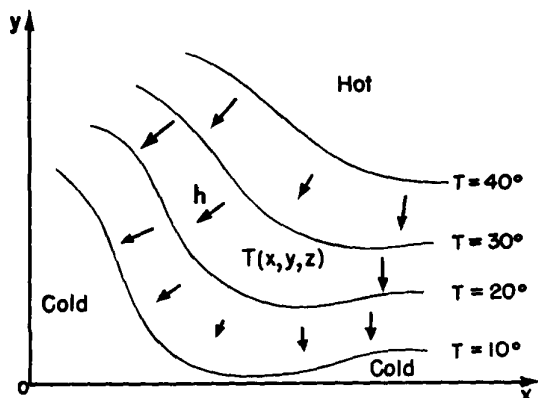


Fig. 2-1. Temperature T is an example of a scalar field. With each point (x, y, z) in space there is associated a number $T(x, y, z)$. All points on the surface marked $T = 20^\circ$ (shown as a curve at $z = 0$) are at the same temperature. The arrows are samples of the heat flow vector h .

One way of thinking about scalar fields is to imagine “contours” which are imaginary surfaces drawn through all points for which the field has the same value, just as contour lines on a map connect points with the same height. For a temperature field the contours are called “isothermal surfaces” or isotherms. Figure 2-1 illustrates a temperature field and shows the dependence of T on x and y when $z = 0$. Several isotherms are drawn.

There are also vector fields. The idea is very simple. A vector is given for each point in space. The vector varies from point to point. As an example, consider a rotating body. The velocity of the material of the body at any point is a vector which is a function of position (Fig. 2-2). As a second example, consider the flow of heat in a block of material. If the temperature in the block is high at one place and low at another, there will be a flow of heat from the hotter places to the colder. The heat will be flowing in different directions in different parts of the block. The heat flow is a directional quantity which we call h . Its magnitude is a measure of how much heat is flowing. Examples of the heat flow vector are also shown in Fig. 2-1.

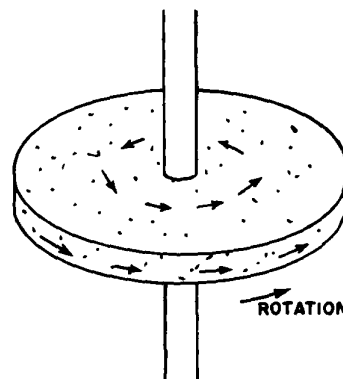


Fig. 2-2. The velocity of the atoms in a rotating object is an example of a vector field.

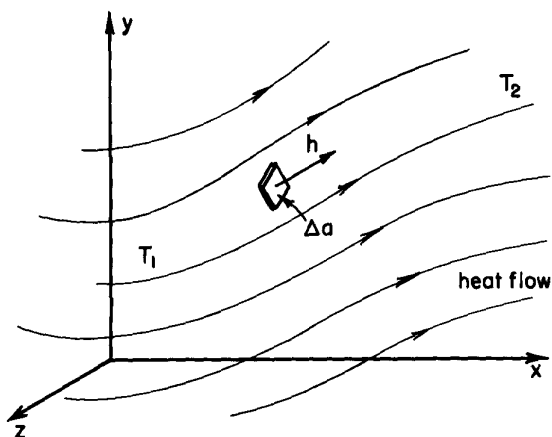


Fig. 2-3. Heat flow is a vector field. The vector h points along the direction of the flow. Its magnitude is the energy transported per unit time across a surface element oriented perpendicular to the flow, divided by the area of the surface element.

Let's make a more precise definition of h : The magnitude of the vector heat flow at a point is the amount of thermal energy that passes, per unit time and per unit area, through an infinitesimal surface element, at right angles to the direction of flow. The vector points in the direction of flow (see Fig. 2-3). In symbols: If ΔJ is the thermal energy that passes per unit time through the surface element Δa , then

$$h = \frac{\Delta J}{\Delta a} e_f, \quad (2.9)$$

where e_f is a unit vector in the direction of flow.

The vector h can be defined in another way—in terms of its components. We ask how much heat flows through a small surface at *any* angle with respect to the flow. In Fig. 2-4 we show a small surface Δa_2 inclined with respect to Δa_1 , which is perpendicular to the flow. The unit vector n is normal to the surface Δa_2 . The

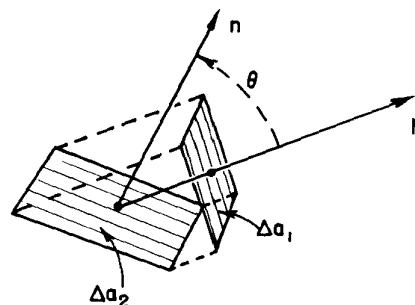


Fig. 2-4. The heat flow through Δa_2 is the same as through Δa_1 .

angle θ between $\mathbf{n}$ and $\mathbf{h}$ is the same as the angle between the surfaces (since $\mathbf{h}$ is normal to Δa_1). Now what is the heat flow *per unit area* through Δa_2 ? The flow through Δa_2 is the same as through Δa_1 ; only the areas are different. In fact, $\Delta a_1 = \Delta a_2 \cos \theta$. The heat flow through Δa_2 is

$$\frac{\Delta J}{\Delta a_2} = \frac{\Delta J}{\Delta a_1} \cos \theta = \mathbf{h} \cdot \mathbf{n}. \quad (2.10)$$

We interpret this equation: the heat flow (per unit time and per unit area) through *any* surface element whose unit normal is $\mathbf{n}$, is given by $\mathbf{h} \cdot \mathbf{n}$. Equally, we could say: the component of the heat flow perpendicular to the surface element Δa_2 is $\mathbf{h} \cdot \mathbf{n}$. We can, if we wish, consider that these statements *define* $\mathbf{h}$. We will be applying the same ideas to other vector fields.

2-3 Derivatives of fields—the gradient

When fields vary in time, we can describe the variation by giving their derivatives with respect to t . We want to describe the variations with position in a similar way, because we are interested in the relationship between, say, the temperature in one place and the temperature at a nearby place. How shall we take the derivative of the temperature with respect to position? Do we differentiate the temperature with respect to x ? Or with respect to y , or z ?

Useful physical laws do not depend upon the orientation of the coordinate system. They should, therefore, be written in a form in which either both sides are scalars or both sides are vectors. What is the derivative of a scalar field, say $\partial T / \partial x$? Is it a scalar, or a vector, or what? It is neither a scalar nor a vector, as you can easily appreciate, because if we took a different x -axis, $\partial T / \partial x$ would certainly be different. But notice: We have three possible derivatives: $\partial T / \partial x$, $\partial T / \partial y$, and $\partial T / \partial z$. Since there are three kinds of derivatives and we know that it takes three numbers to form a vector, perhaps these three derivatives are the components of a vector:

$$\left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right) \stackrel{?}{=} \text{a vector}. \quad (2.11)$$

Of course it is not generally true that *any* three numbers form a vector. It is true only if, when we rotate the coordinate system, the components of the vector transform among themselves in the correct way. So it is necessary to analyze how these derivatives are changed by a rotation of the coordinate system. We shall show that (2.11) is indeed a vector. The derivatives do transform in the correct way when the coordinate system is rotated.

We can see this in several ways. One way is to ask a question whose answer is independent of the coordinate system, and try to express the answer in an “invariant” form. For instance, if $S = \mathbf{A} \cdot \mathbf{B}$, and if $\mathbf{A}$ and $\mathbf{B}$ are vectors, we know—because we proved it in Chapter 11 of Vol. I—that S is a scalar. We *know* that S is a scalar without investigating whether it changes with changes in coordinate systems. It *can't*, because it's a dot product of two vectors. Similarly, if we *know* that $\mathbf{A}$ is a vector, and we have three numbers B_1 , B_2 , and B_3 , and we find out that

$$A_x B_1 + A_y B_2 + A_z B_3 = S, \quad (2.12)$$

where S is the same for any coordinate system, then it *must* be that the three numbers B_1 , B_2 , B_3 are the components B_x , B_y , B_z of some vector $\mathbf{B}$.

Now let's think of the temperature field. Suppose we take two points P_1 and P_2 , separated by the small interval ΔR . The temperature at P_1 is T_1 and at P_2 is T_2 , and the difference $\Delta T = T_2 - T_1$. The temperatures at these real, physical points certainly do not depend on what axis we choose for measuring the coordinates. In particular, ΔT is a number independent of the coordinate system. It is a scalar.

If we choose some convenient set of axes, we could write $T_1 = T(x, y, z)$ and $T_2 = T(x + \Delta x, y + \Delta y, z + \Delta z)$, where Δx , Δy , and Δz are the components of the vector $\Delta \mathbf{R}$ (Fig. 2-5). Remembering Eq. (2.7), we can write

$$\Delta T = \frac{\partial T}{\partial x} \Delta x + \frac{\partial T}{\partial y} \Delta y + \frac{\partial T}{\partial z} \Delta z. \quad (2.13)$$

The left side of Eq. (2.13) is a scalar. The right side is the sum of three products with Δx , Δy , and Δz , which are the components of a vector. It follows that the three numbers

$$\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z}$$

are also the x -, y -, and z -components of a vector. We write this new vector with the symbol ∇T . The symbol ∇ (called “del”) is an upside-down Δ , and is supposed to remind us of differentiation. People read ∇T in various ways: “del- T ,” or “gradient of T ,” or “grad T ,”

$$\text{grad } T = \nabla T = \left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right)^*. \quad (2.14)$$

Using this notation, we can rewrite Eq. (2.13) in the more compact form

$$\Delta T = \nabla T \cdot \Delta \mathbf{R}. \quad (2.15)$$

In words, this equation says that the difference in temperature between two nearby points is the dot product of the gradient of T and the vector displacement between the points. The form of Eq. (2.15) also illustrates clearly our proof above that ∇T is indeed a vector.

Perhaps you are still not convinced? Let’s prove it in a different way. (Although if you look carefully, you may be able to see that it’s really the same proof in a longer-winded form!) We shall show that the components of ∇T transform in just the same way that components of $\mathbf{R}$ do. If they do, ∇T is a vector according to our original definition of a vector in Chapter 11 of Vol. I. We take a new coordinate system x' , y' , z' , and in this new system we calculate $\partial T / \partial x'$, $\partial T / \partial y'$, and $\partial T / \partial z'$. To make things a little simpler, we let $z = z'$, so that we can forget about the z -coordinate. (You can check out the more general case for yourself.)

We take an $x'y'$ -system rotated an angle θ with respect to the xy -system, as in Fig. 2-6(a). For a point (x, y) the coordinates in the prime system are

$$x' = x \cos \theta + y \sin \theta, \quad (2.16)$$

$$y' = -x \sin \theta + y \cos \theta. \quad (2.17)$$

Or, solving for x and y ,

$$x = x' \cos \theta - y' \sin \theta, \quad (2.18)$$

$$y = x' \sin \theta + y' \cos \theta. \quad (2.19)$$

If any pair of numbers transforms with these equations in the same way that x and y do, they are the components of a vector.

Now let’s look at the difference in temperature between the two nearby points P_1 and P_2 , chosen as in Fig. 2-6(b). If we calculate with the x - and y -coordinates, we would write

$$\Delta T = \frac{\partial T}{\partial x} \Delta x \quad (2.20)$$

—since Δy is zero.

* In our notation, the expression (a, b, c) represents a vector with components a , b , and c . If you like to use the unit vectors i, j , and k , you may write

$$\nabla T = i \frac{\partial T}{\partial x} + j \frac{\partial T}{\partial y} + k \frac{\partial T}{\partial z}.$$

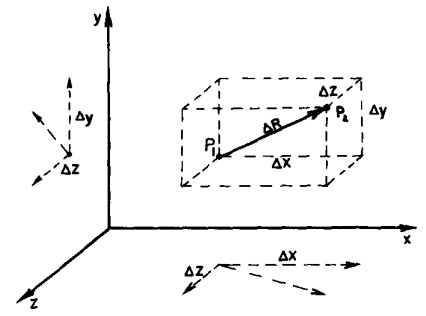


Fig. 2-5. The vector $\Delta \mathbf{R}$, whose components are Δx , Δy , and Δz .

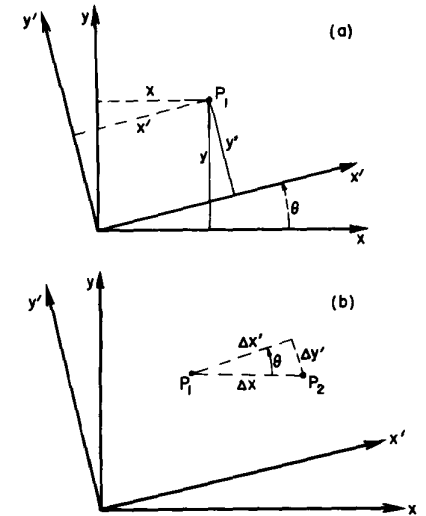


Fig. 2-6. (a) Transformation to a rotated coordinate system. (b) Special case of an interval $\Delta \mathbf{R}$ parallel to the x -axis.

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$$\nabla T = i \frac{\partial T}{\partial x} + j \frac{\partial T}{\partial y} + k \frac{\partial T}{\partial z}.$$

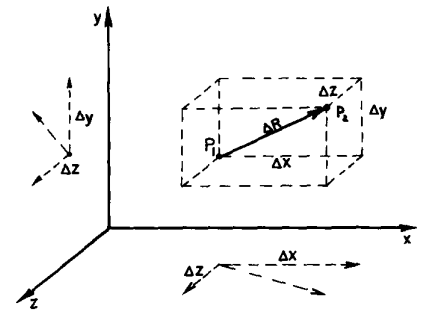


Fig. 2-5. The vector $\Delta \mathbf{R}$, whose components are Δx , Δy , and Δz .

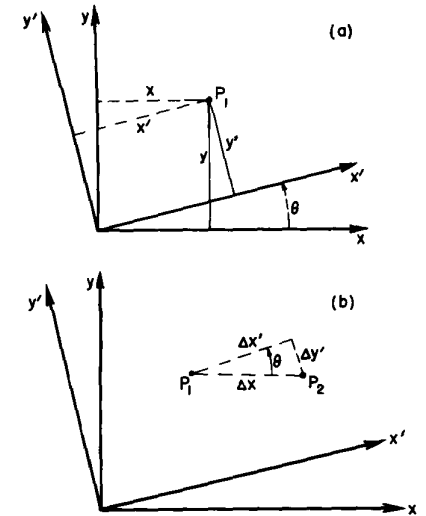


Fig. 2-6. (a) Transformation to a rotated coordinate system. (b) Special case of an interval $\Delta \mathbf{R}$ parallel to the x -axis.

What would a computation in the prime system give? We would have written

$$\Delta T = \frac{\partial T}{\partial x'} \Delta x' + \frac{\partial T}{\partial y'} \Delta y'. \quad (2.21)$$

Looking at Fig. 2-6(b), we see that

$$\Delta x' = \Delta x \cos \theta \quad (2.22)$$

and

$$\Delta y' = -\Delta x \sin \theta, \quad (2.23)$$

since Δy is negative when Δx is positive. Substituting these in Eq. (2.21), we find that

$$\Delta T = \frac{\partial T}{\partial x'} \Delta x \cos \theta - \frac{\partial T}{\partial y'} \Delta x \sin \theta \quad (2.24)$$

$$= \left(\frac{\partial T}{\partial x'} \cos \theta - \frac{\partial T}{\partial y'} \sin \theta \right) \Delta x. \quad (2.25)$$

Comparing Eq. (2.25) with (2.20), we see that

$$\frac{\partial T}{\partial x} = \frac{\partial T}{\partial x'} \cos \theta - \frac{\partial T}{\partial y'} \sin \theta. \quad (2.26)$$

This equation says that $\partial T/\partial x$ is obtained from $\partial T/\partial x'$ and $\partial T/\partial y'$, just as x is obtained from x' and y' in Eq. (2.18). So $\partial T/\partial x$ is the x -component of a vector. The same kind of arguments would show that $\partial T/\partial y$ and $\partial T/\partial z$ are y - and z -components. So ∇T is definitely a vector. It is a vector field derived from the scalar field T .

2-4 The operator ∇

Now we can do something that is extremely amusing and ingenious—and characteristic of the things that make mathematics beautiful. The argument that $\text{grad } T$, or ∇T , is a vector did not depend upon *what* scalar field we were differentiating. All the arguments would go the same if T were replaced by *any scalar field*. Since the transformation equations are the same no matter what we differentiate, we could just as well omit the T and replace Eq. (2.26) by the operator equation

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial x'} \cos \theta - \frac{\partial}{\partial y'} \sin \theta. \quad (2.27)$$

We leave the operators, as Jeans said, “hungry for something to differentiate.”

Since the differential operators themselves transform as the components of a vector should, we can call them components of a *vector operator*. We can write

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right), \quad (2.28)$$

which means, of course,

$$\nabla_x = \frac{\partial}{\partial x}, \quad \nabla_y = \frac{\partial}{\partial y}, \quad \nabla_z = \frac{\partial}{\partial z}. \quad (2.29)$$

We have abstracted the gradient away from the T —that is the wonderful idea.

You must always remember, of course, that ∇ is an operator. Alone, it means nothing. If ∇ by itself means nothing, what does it mean if we multiply it by a scalar—say T —to get the product $T\nabla$? (One can always multiply a vector by a scalar.) It still does not mean anything. Its x -component is

$$T \frac{\partial}{\partial x}, \quad (2.30)$$

which is not a number, but is still some kind of operator. However, according to the algebra of vectors we would still call $T\nabla$ a vector.

Now let's multiply ∇ by a scalar on the other side, so that we have the product (∇T) . In ordinary algebra

$$TA = AT, \quad (2.31)$$

but we have to remember that operator algebra is a little different from ordinary vector algebra. With operators we must always keep the sequence right, so that the operations make proper sense. You will have no difficulty if you just remember that the operator ∇ obeys the same convention as the derivative notation. What is to be differentiated must be placed on the right of the ∇ . The order is important.

Keeping in mind this problem of order, we understand that $T\nabla$ is an operator, but the product ∇T is no longer a hungry operator; the operator is completely satisfied. It is indeed a physical vector having a meaning. It represents the spatial rate of change of T . The x -component of ∇T is how fast T changes in the x -direction. What is the direction of the vector ∇T ? We know that the rate of change of T in any direction is the component of ∇T in that direction (see Eq. 2.15). It follows that the direction of ∇T is that in which it has the largest possible component—in other words, the direction in which T changes the fastest. The gradient of T has the direction of the steepest uphill slope (in T).

2-5 Operations with ∇

Can we do any other algebra with the vector operator ∇ ? Let us try combining it with a vector. We can combine two vectors by making a dot product. We could make the products

$$(\text{a vector}) \cdot \nabla, \quad \text{or} \quad \nabla \cdot (\text{a vector}).$$

The first one doesn't mean anything yet, because it is still an operator. What it might ultimately mean would depend on what it is made to operate on. The second product is some scalar field. ($A \cdot B$ is always a scalar.)

Let's try the dot product of ∇ with a vector field we know, say $\mathbf{h}$. We write out the components:

$$\nabla \cdot \mathbf{h} = \nabla_x h_x + \nabla_y h_y + \nabla_z h_z \quad (2.32)$$

or

$$\nabla \cdot \mathbf{h} = \frac{\partial h_x}{\partial x} + \frac{\partial h_y}{\partial y} + \frac{\partial h_z}{\partial z}. \quad (2.33)$$

The sum is invariant under a coordinate transformation. If we were to choose a different system (indicated by primes), we would have*

$$\nabla' \cdot \mathbf{h} = \frac{\partial h_{x'}}{\partial x'} + \frac{\partial h_{y'}}{\partial y'} + \frac{\partial h_{z'}}{\partial z'}, \quad (2.34)$$

which is the *same* number as would be gotten from Eq. (2.33), even though it looks different. That is,

$$\nabla' \cdot \mathbf{h} = \nabla \cdot \mathbf{h} \quad (2.35)$$

for every point in space. So $\nabla \cdot \mathbf{h}$ is a scalar field, which must represent some physical quantity. You should realize that the combination of derivatives in $\nabla \cdot \mathbf{h}$ is rather special. There are all sorts of other combinations like $\partial h_y / \partial x$, which are neither scalars nor components of vectors.

The scalar quantity $\nabla \cdot (\text{a vector})$ is extremely useful in physics. It has been given the name the *divergence*. For example,

$$\nabla \cdot \mathbf{h} = \text{div } \mathbf{h} = \text{"divergence of } \mathbf{h} \text{"} \quad (2.36)$$

As we did for ∇T , we can ascribe a physical significance to $\nabla \cdot \mathbf{h}$. We shall, however, postpone that until later.

* We think of $\mathbf{h}$ as a *physical* quantity that depends on position in space, and not strictly as a mathematical function of three variables. When $\mathbf{h}$ is "differentiated" with respect to x , y , and z , or with respect to x' , y' , and z' , the mathematical expression for $\mathbf{h}$ must first be expressed as a function of the appropriate variables.

First, we wish to see what else we can cook up with the vector operator ∇ . What about a cross product? We must expect that

$$\nabla \times \mathbf{h} = \text{a vector.} \quad (2.37)$$

It is a vector whose components we can write by the usual rule for cross products (see Eq. 2.2):

$$(\nabla \times \mathbf{h})_z = \nabla_x h_y - \nabla_y h_x = \frac{\partial h_y}{\partial x} - \frac{\partial h_x}{\partial y}. \quad (2.38)$$

Similarly,

$$(\nabla \times \mathbf{h})_x = \nabla_y h_z - \nabla_z h_y = \frac{\partial h_z}{\partial y} - \frac{\partial h_y}{\partial z} \quad (2.39)$$

and

$$(\nabla \times \mathbf{h})_y = \nabla_z h_x - \nabla_x h_z = \frac{\partial h_x}{\partial z} - \frac{\partial h_z}{\partial x}. \quad (2.40)$$

The combination $\nabla \times \mathbf{h}$ is called “the *curl* of $\mathbf{h}$.” The reason for the name and the physical meaning of the combination will be discussed later.

Summarizing, we have three kinds of combinations with ∇ :

$$\nabla T = \text{grad } T = \text{a vector,}$$

$$\nabla \cdot \mathbf{h} = \text{div } \mathbf{h} = \text{a scalar,}$$

$$\nabla \times \mathbf{h} = \text{curl } \mathbf{h} = \text{a vector.}$$

Using these combinations, we can write about the spatial variations of fields in a convenient way—in a way that is general, in that it doesn’t depend on any particular set of axes.

As an example of the use of our vector differential operator ∇ , we write a set of vector equations which contain the same laws of electromagnetism that we gave in words in Chapter 1. They are called Maxwell’s equations.

Maxwell’s Equations

$$\begin{aligned} (1) \quad \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \\ (2) \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ (3) \quad \nabla \cdot \mathbf{B} &= 0 \\ (4) \quad c^2 \nabla \times \mathbf{B} &= \frac{\partial \mathbf{E}}{\partial t} + \frac{\mathbf{j}}{\epsilon_0} \end{aligned} \quad (2.41)$$

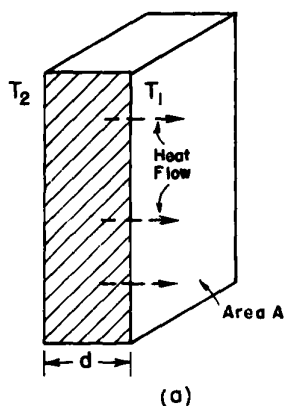
where ρ (rho), the “electric charge density,” is the amount of charge per unit volume, and $\mathbf{j}$, the “electric current density,” is the rate at which charge flows through a unit area per second. These four equations contain the complete classical theory of the electromagnetic field. You see what an elegantly simple form we can get with our new notation!

2-6 The differential equation of heat flow

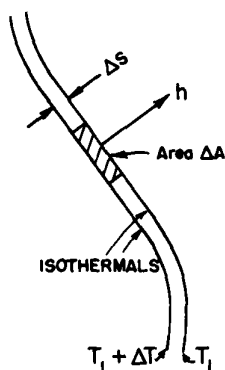
Let us give another example of a law of physics written in vector notation. The law is not a precise one, but for many metals and a number of other substances that conduct heat it is quite accurate. You know that if you take a slab of material and heat one face to temperature T_2 and cool the other to a different temperature T_1 , the heat will flow through the material from T_2 to T_1 [Fig. 2-7(a)]. The heat flow is proportional to the area A of the faces, and to the temperature difference. It is also inversely proportional to d , the distance between the plates. (For a given temperature difference, the thinner the slab the greater the heat flow.) Letting J be the thermal energy that passes per unit time through the slab, we write

$$J = \kappa(T_2 - T_1) \frac{A}{d}. \quad (2.42)$$

The constant of proportionality κ (kappa) is called the *thermal conductivity*.



(a)



(b)

Fig. 2-7. (a) Heat flow through a slab. (b) An infinitesimal slab parallel to an isothermal surface in a large block.

What will happen in a more complicated case? Say in an odd-shaped block of material in which the temperature varies in peculiar ways? Suppose we look at a tiny piece of the block and imagine a slab like that of Fig. 2-7(a) on a miniature scale. We orient the faces parallel to the isothermal surfaces, as in Fig. 2-7(b), so that Eq. (2.42) is correct for the small slab.

If the area of the small slab is ΔA , the heat flow per unit time is

$$\Delta J = \kappa \Delta T \frac{\Delta A}{\Delta s}, \quad (2.43)$$

where Δs is the thickness of the slab. Now $\Delta J/\Delta A$ we have defined earlier as the magnitude of $\mathbf{h}$, whose direction is the heat flow. The heat flow will be from $T_1 + \Delta T$ toward T_1 , and so it will be perpendicular to the isotherms, as drawn in Fig. 2-7(b). Also, $\Delta T/\Delta s$ is just the rate of change of T with position. And since the position change is perpendicular to the isotherms, our $\Delta T/\Delta s$ is the maximum rate of change. It is, therefore, just the magnitude of ∇T . Now since the direction of ∇T is opposite to that of $\mathbf{h}$, we can write (2.43) as a vector equation:

$$\mathbf{h} = -\kappa \nabla T. \quad (2.44)$$

(The minus sign is necessary because heat flows “downhill” in temperature.) Equation (2.44) is the differential equation of heat conduction in bulk materials. You see that it is a proper vector equation. Each side is a vector if κ is just a number. It is the generalization to arbitrary cases of the special relation (2.42) for rectangular slabs. Later we should learn to write all sorts of elementary physics relations like (2.42) in the more sophisticated vector notation. This notation is useful not only because it makes the equations *look* simpler. It also shows most clearly the *physical content* of the equations without reference to any arbitrarily chosen coordinate system.

2-7 Second derivatives of vector fields

So far we have had only first derivatives. Why not second derivatives? We could have several combinations:

$$\begin{aligned} (a) \quad & \nabla \cdot (\nabla T) \\ (b) \quad & \nabla \times (\nabla T) \\ (c) \quad & \nabla(\nabla \cdot \mathbf{h}) \\ (d) \quad & \nabla \cdot (\nabla \times \mathbf{h}) \\ (e) \quad & \nabla \times (\nabla \times \mathbf{h}) \end{aligned} \quad (2.45)$$

You can check that these are all the possible combinations.

Let's look first at the second one, (b). It has the same form as

$$\mathbf{A} \times (\mathbf{A}T) = (\mathbf{A} \times \mathbf{A})T = 0,$$

since $\mathbf{A} \times \mathbf{A}$ is always zero. So we should have

$$\text{curl}(\text{grad } T) = \nabla \times (\nabla T) = 0. \quad (2.46)$$

We can see how this equation comes about if we go through once with the components:

$$\begin{aligned} [\nabla \times (\nabla T)]_z &= \nabla_x(\nabla T)_y - \nabla_y(\nabla T)_x \\ &= \frac{\partial}{\partial x} \left(\frac{\partial T}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial x} \right), \end{aligned} \quad (2.47)$$

which is zero (by Eq. 2.8). It goes the same for the other components. So $\nabla \times (\nabla T) = 0$, for any temperature distribution—in fact, for *any* scalar function.

Now let us take another example. Let us see whether we can find another zero. The dot product of a vector with a cross product which contains that vector is zero:

$$\mathbf{A} \cdot (\mathbf{A} \times \mathbf{B}) = 0. \quad (2.48)$$

because $\mathbf{A} \times \mathbf{B}$ is perpendicular to $\mathbf{A}$, and so has no components in the direction $\mathbf{A}$. The same combination appears in (d) of (2.45), so we have

$$\nabla \cdot (\nabla \times \mathbf{h}) = \text{div}(\text{curl } \mathbf{h}) = 0. \quad (2.49)$$

Again, it is easy to show that it is zero by carrying through the operations with components.

Now we are going to state two mathematical theorems that we will not prove. They are very interesting and useful theorems for physicists to know.

In a physical problem we frequently find that the curl of some quantity—say of the vector field $\mathbf{A}$ —is zero. Now we have seen (Eq. 2.46) that the curl of a gradient is zero, which is easy to remember because of the way the vectors work. It could certainly be, then, that $\mathbf{A}$ is the gradient of some quantity, because then its curl would necessarily be zero. The interesting theorem is that if the curl $\mathbf{A}$ is zero, then $\mathbf{A}$ is *always* the gradient of *something*—there is some scalar field ψ (psi) such that $\mathbf{A}$ is equal to $\text{grad } \psi$. In other words, we have the

THEOREM:

$$\begin{array}{ll} \text{If} & \nabla \times \mathbf{A} = 0 \\ \text{there is a} & \psi \\ \text{such that} & \mathbf{A} = \nabla \psi. \end{array} \quad (2.50)$$

There is a similar theorem if the divergence of $\mathbf{A}$ is zero. We have seen in Eq. (2.49) that the divergence of a curl of something is always zero. If you come across a vector field $\mathbf{D}$ for which $\text{div } \mathbf{D}$ is zero, then you can conclude that $\mathbf{D}$ is the curl of some vector field $\mathbf{C}$.

THEOREM:

$$\begin{array}{ll} \text{If} & \nabla \cdot \mathbf{D} = 0 \\ \text{there is a} & \mathbf{C} \\ \text{such that} & \mathbf{D} = \nabla \times \mathbf{C}. \end{array} \quad (2.51)$$

In looking at the possible combinations of two ∇ operators, we have found that two of them always give zero. Now we look at the ones that are *not* zero. Take the combination $\nabla \cdot (\nabla T)$, which was first on our list. It is not, in general, zero. We write out the components:

$$\nabla T = \nabla_x T + \nabla_y T + \nabla_z T.$$

Then

$$\begin{aligned} \nabla \cdot (\nabla T) &= \nabla_x(\nabla_x T) + \nabla_y(\nabla_y T) + \nabla_z(\nabla_z T) \\ &= \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}, \end{aligned} \quad (2.52)$$

which would, in general, come out to be some number. It is a scalar field.

You see that we do not need to keep the parentheses, but can write, without any chance of confusion,

$$\nabla \cdot (\nabla T) = \nabla \cdot \nabla T = (\nabla \cdot \nabla) T = \nabla^2 T. \quad (2.53)$$

We look at ∇^2 as a new operator. It is a scalar operator. Because it appears often in physics, it has been given a special name—the *Laplacian*.

$$\text{Laplacian} = \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}. \quad (2.54)$$

Since the Laplacian is a scalar operator, we may operate with it on a vector—by which we mean the same operation on each component in rectangular coordinates:

$$\nabla^2 \mathbf{h} = (\nabla^2 h_x, \nabla^2 h_y, \nabla^2 h_z).$$

Let's look at one more possibility: $\nabla \times (\nabla \times \mathbf{h})$, which was (e) in the list (2.45). Now the curl of the curl can be written differently if we use the vector equality (2.6):

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}). \quad (2.55)$$

In order to use this formula, we should replace $\mathbf{A}$ and $\mathbf{B}$ by the operator ∇ and put $\mathbf{C} = \mathbf{h}$. If we do that, we get

$$\nabla \times (\nabla \times \mathbf{h}) = \nabla(\nabla \cdot \mathbf{h}) - \mathbf{h}(\nabla \cdot \nabla) \dots ???$$

Wait a minute! Something is wrong. The first two terms are vectors all right (the operators are satisfied), but the last term doesn't come out to anything. It's still an operator. The trouble is that we haven't been careful enough about keeping the order of our terms straight. If you look again at Eq. (2.55), however, you see that we could equally well have written it as

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}. \quad (2.56)$$

The order of terms looks better. Now let's make our substitution in (2.56). We get

$$\nabla \times (\nabla \times \mathbf{h}) = \nabla(\nabla \cdot \mathbf{h}) - (\nabla \cdot \nabla)\mathbf{h}. \quad (2.57)$$

This form looks all right. It is, in fact, correct, as you can verify by computing the components. The last term is the Laplacian, so we can equally well write

$$\nabla \times (\nabla \times \mathbf{h}) = \nabla(\nabla \cdot \mathbf{h}) - \nabla^2 \mathbf{h}. \quad (2.58)$$

We have had something to say about all of the combinations in our list of double ∇ 's, except for (c), $\nabla(\nabla \cdot \mathbf{h})$. It is a possible vector field, but there is nothing special to say about it. It's just some vector field which may occasionally come up.

It will be convenient to have a table of our conclusions:

- (a) $\nabla \cdot (\nabla T) = \nabla^2 T =$ a scalar field
- (b) $\nabla \times (\nabla T) = 0$
- (c) $\nabla(\nabla \cdot \mathbf{h}) =$ a vector field
- (d) $\nabla \cdot (\nabla \times \mathbf{h}) = 0$
- (e) $\nabla \times (\nabla \times \mathbf{h}) = \nabla(\nabla \cdot \mathbf{h}) - \nabla^2 \mathbf{h}$
- (f) $(\nabla \cdot \nabla)\mathbf{h} = \nabla^2 \mathbf{h} =$ a vector field

You may notice that we haven't tried to invent a new vector operator ($\nabla \times \nabla$). Do you see why?

2-8 Pitfalls

We have been applying our knowledge of ordinary vector algebra to the algebra of the operator ∇ . We have to be careful, though, because it is possible to go astray. There are two pitfalls which we will mention, although they will not come up in this course. What would you say about the following expression, that involves the two scalar functions ψ and ϕ (phi):

$$(\nabla\psi) \times (\nabla\phi)?$$

You might want to say: it must be zero because it's just like

$$(\mathbf{A}\mathbf{a}) \times (\mathbf{A}\mathbf{b}),$$

which is zero because the cross product of two *equal* vectors $A \times A$ is always zero. But in our example the two operators ∇ are not equal! The first one operates on one function, ψ ; the other operates on a different function, ϕ . So although we represent them by the same symbol ∇ , they must be considered as different operators. Clearly, the direction of $\nabla\psi$ depends on the function ψ , so it is not likely to be parallel to $\nabla\phi$.

$$(\nabla\psi) \times (\nabla\phi) \neq 0 \quad (\text{generally}).$$

Fortunately, we won't have to use such expressions. (What we have said doesn't change the fact that $\nabla \times \nabla\psi = 0$ for any scalar field, because here both ∇ 's operate on the same function.)

Pitfall number two (which, again, we need not get into in our course) is the following: The rules that we have outlined here are simple and nice when we use rectangular coordinates. For example, if we have $\nabla^2 h$ and we want the x -component, it is

$$(\nabla^2 h)_x = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) h_x = \nabla^2 h_x. \quad (2.60)$$

The same expression would *not* work if we were to ask for the *radial* component of $\nabla^2 h$. The radial component of $\nabla^2 h$ is not equal to $\nabla^2 h_r$. The reason is that when we are dealing with the algebra of vectors, the directions of the vectors are all quite definite. But when we are dealing with vector fields, their directions are different at different places. If we try to describe a vector field in, say, polar coordinates, what we call the "radial" direction varies from point to point. So we can get into a lot of trouble when we start to differentiate the components. For example, even for a *constant* vector field, the radial component changes from point to point.

It is usually safest and simplest just to stick to rectangular coordinates and avoid trouble, but there is one exception worth mentioning: Since the Laplacian ∇^2 , is a scalar, we can write it in any coordinate system we want to (for example, in polar coordinates). But since it is a differential operator, we should use it only on vectors whose components are in a fixed direction—that means rectangular coordinates. So we shall express all of our vector fields in terms of their x -, y -, and z -components when we write our vector differential equations out in components.

Vector Integral Calculus

3-1 Vector integrals; the line integral of $\nabla\psi$

We found in Chapter 2 that there were various ways of taking derivatives of fields. Some gave vector fields; some gave scalar fields. Although we developed many different formulas, everything in Chapter 2 could be summarized in one rule: the operators $\partial/\partial x$, $\partial/\partial y$, and $\partial/\partial z$ are the three components of a vector operator ∇ . We would now like to get some understanding of the significance of the derivatives of fields. We will then have a better feeling for what a vector field equation means.

We have already discussed the meaning of the gradient operation (∇ on a scalar). Now we turn to the meanings of the divergence and curl operations. The interpretation of these quantities is best done in terms of certain vector integrals and equations relating such integrals. These equations cannot, unfortunately, be obtained from vector algebra by some easy substitution, so you will just have to learn them as something new. Of these integral formulas, one is practically trivial, but the other two are not. We will derive them and explain their implications. The equations we shall study are really mathematical theorems. They will be useful not only for interpreting the meaning and the content of the divergence and the curl, but also in working out general physical theories. These mathematical theorems are, for the theory of fields, what the theorem of the conservation of energy is to the mechanics of particles. General theorems like these are important for a deeper understanding of physics. You will find, though, that they are not very useful for solving problems—except in the simplest cases. It is delightful, however, that in the beginning of our subject there will be many simple problems which can be solved with the three integral formulas we are going to treat. We will see, however, as the problems get harder, that we can no longer use these simple methods.

We take up first an integral formula involving the gradient. The relation contains a very simple idea: Since the gradient represents the rate of change of a field quantity, if we integrate that rate of change, we should get the total change. Suppose we have the scalar field $\psi(x, y, z)$. At any two points (1) and (2), the function ψ will have the values $\psi(1)$ and $\psi(2)$, respectively. [We use a convenient notation, in which (2) represents the point (x_2, y_2, z_2) and $\psi(2)$ means the same thing as $\psi(x_2, y_2, z_2)$.] If Γ (gamma) is any curve joining (1) and (2), as in Fig. 3-1, the following relation is true:

THEOREM 1.

$$\psi(2) - \psi(1) = \int_{\text{along } \Gamma}^{(2)} (\nabla\psi) \cdot ds. \quad (3.1)$$

The integral is a *line integral*, from (1) to (2) along the curve Γ , of the dot product of $\nabla\psi$ —a vector—with ds —another vector which is an infinitesimal line element of the curve Γ (directed away from (1) and toward (2)).

First, we should review what we mean by a line integral. Consider a scalar function $f(x, y, z)$, and the curve Γ joining two points (1) and (2). We mark off the curve at a number of points and join these points by straight-line segments, as shown in Fig. 3-2. Each segment has the length Δs_i , where i is an index that runs 1, 2, 3, . . . By the line integral

$$\int_{\text{along } \Gamma}^{(2)} f ds$$

3-1 Vector integrals; the line integral of $\nabla\psi$

3-2 The flux of a vector field

3-3 The flux from a cube; Gauss' theorem

3-4 Heat conduction; the diffusion equation

3-5 The circulation of a vector field

3-6 The circulation around a square; Stokes' theorem

3-7 Curl-free and divergence-free fields

3-8 Summary

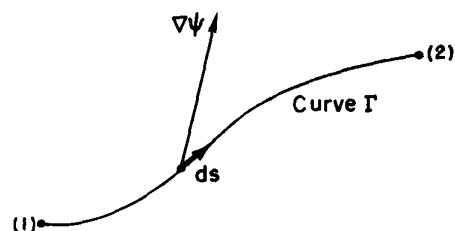


Fig. 3-1. The terms used in Eq. (3.1). The vector $\nabla\psi$ is evaluated at the line element ds .

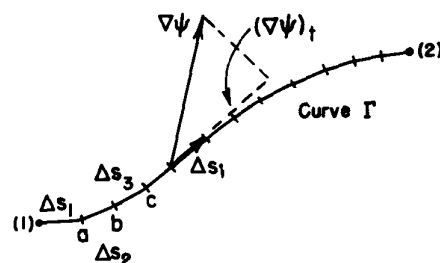


Fig. 3-2. The line integral is the limit of a sum.

we mean the limit of the sum

$$\sum f_i \Delta s_i,$$

where f_i is the value of the function at the i th segment. The limiting value is what the sum approaches as we add more and more segments (in a sensible way, so that the largest $\Delta s_i \rightarrow 0$).

The integral in our theorem, Eq. (3.1), means the same thing, although it looks a little different. Instead of f , we have another scalar—the component of $\nabla\psi$ in the direction of Δs . If we write $(\nabla\psi)_t$ for this tangential component, it is clear that

$$(\nabla\psi)_t \Delta s = (\nabla\psi) \cdot \Delta s. \quad (3.2)$$

The integral in Eq. (3.1) means the sum of such terms.

Now let's see why Eq. (3.1) is true. In Chapter 1, we showed that the component of $\nabla\psi$ along a small displacement ΔR was the rate of change of ψ in the direction of ΔR . Consider the line segment Δs from (1) to point a in Fig. 3-2. According to our definition,

$$\Delta\psi_1 = \psi(a) - \psi(1) = (\nabla\psi)_1 \cdot \Delta s_1. \quad (3.3)$$

Also, we have

$$\psi(b) - \psi(a) = (\nabla\psi)_2 \cdot \Delta s_2, \quad (3.4)$$

where, of course, $(\nabla\psi)_1$ means the gradient evaluated at the segment Δs_1 , and $(\nabla\psi)_2$, the gradient evaluated at Δs_2 . If we add Eqs. (3.3) and (3.4), we get

$$\psi(b) - \psi(1) = (\nabla\psi)_1 \cdot \Delta s_1 + (\nabla\psi)_2 \cdot \Delta s_2. \quad (3.5)$$

You can see that if we keep adding such terms, we get the result

$$\psi(2) - \psi(1) = \sum (\nabla\psi)_i \cdot \Delta s_i. \quad (3.6)$$

The left-hand side doesn't depend on how we choose our intervals—if (1) and (2) are kept always the same—so we can take the limit of the right-hand side. We have therefore proved Eq. (3.1).

You can see from our proof that just as the equality doesn't depend on how the points $a, b, c, \dots$, are chosen, similarly it doesn't depend on what we choose for the curve Γ to join (1) and (2). Our theorem is correct for *any* curve from (1) to (2).

One remark on notation: You will see that there is no confusion if we write, for convenience,

$$(\nabla\psi) \cdot ds = \nabla\psi \cdot ds. \quad (3.7)$$

With this notation, our theorem is

$$\text{THEOREM 1.} \quad \psi(2) - \psi(1) = \int_{\text{any curve from (1) to (2)}}^{(2)} \nabla\psi \cdot ds. \quad (3.8)$$

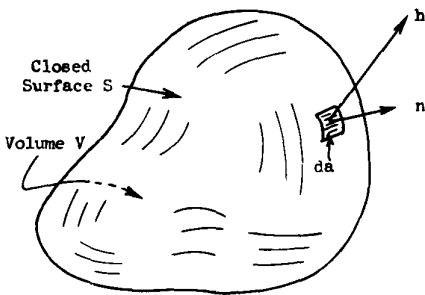


Fig. 3-3. The closed surface S defines the volume V . The unit vector n is the outward normal to the surface element da , and h is the heat-flow vector at the surface element.

3-2 The flux of a vector field

Before we consider our next integral theorem—a theorem about the divergence—we would like to study a certain idea which has an easily understood physical significance in the case of heat flow. We have defined the vector h , which represents the heat that flows through a unit area in a unit time. Suppose that inside a block of material we have some closed surface S which encloses the volume V (Fig. 3-3). We would like to find out how much heat is flowing out of this volume. We can, of course, find it by calculating the total heat flow out of the surface S .

We write da for the area of an element of the surface. The symbol stands for a two-dimensional differential. If, for instance, the area happened to be in the xy -plane we would have

$$da = dx dy.$$

Later we shall have integrals over volume and for these it is convenient to consider a differential volume that is a little cube. So when we write dV we mean

$$dV = dx dy dz.$$

Some people like to write d^2a instead of da to remind themselves that it is kind of a second-order quantity. They would also write d^3V instead of dV . We will use the simpler notation, and assume that you can remember that an area has two dimensions and a volume has three.

The heat flow out through the surface element da is the area times the component of $\mathbf{h}$ perpendicular to da . We have already defined $\mathbf{n}$ as a unit vector pointing outward at right angles to the surface (Fig. 3-3). The component of $\mathbf{h}$ that we want is

$$h_n = \mathbf{h} \cdot \mathbf{n}. \quad (3.9)$$

The heat flow out through da is then

$$\mathbf{h} \cdot \mathbf{n} da. \quad (3.10)$$

To get the total heat flow through any surface we sum the contributions from all the elements of the surface. In other words, we integrate (3.10) over the whole surface:

$$\text{Total heat flow outward through } S = \int_S \mathbf{h} \cdot \mathbf{n} da. \quad (3.11)$$

We are also going to call this surface integral “the flux of $\mathbf{h}$ through the surface.” Originally the word flux meant flow, so that the surface integral just means the flow of $\mathbf{h}$ through the surface. We may think: $\mathbf{h}$ is the “current density” of heat flow and the surface integral of it is the total heat current directed out of the surface; that is, the thermal energy per unit time (joules per second).

We would like to generalize this idea to the case where the vector does not represent the flow of anything; for instance, it might be the electric field. We can certainly still integrate the normal component of the electric field over an area if we wish. Although it is not the flow of anything, we still call it the “flux.” We say

$$\text{Flux of } \mathbf{E} \text{ through the surface } S = \int_S \mathbf{E} \cdot \mathbf{n} da. \quad (3.12)$$

We generalize the word “flux” to mean the “surface integral of the normal component” of a vector. We will also use the same definition even when the surface considered is not a closed one, as it is here.

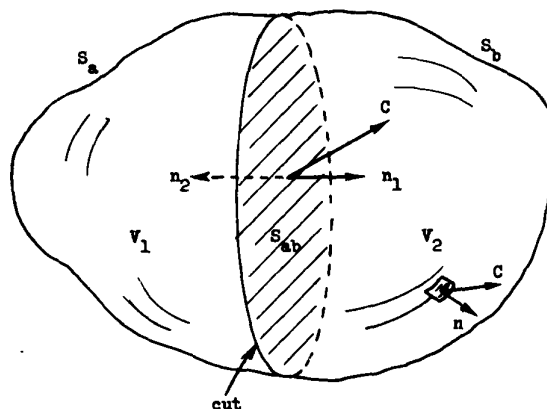
Returning to the special case of heat flow, let us take a situation in which *heat is conserved*. For example, imagine some material in which after an initial heating no further heat energy is generated or absorbed. Then, if there is a net heat flow out of a closed surface, the heat content of the volume inside must decrease. So, in circumstances in which heat would be conserved, we say that

$$\int_S \mathbf{h} \cdot \mathbf{n} da = - \frac{dQ}{dt}, \quad (3.13)$$

where Q is the heat inside the surface. The heat flux out of S is equal to minus the rate of change with respect to time of the total heat Q inside of S . This interpretation is possible because we are speaking of heat flow and also because we supposed that the heat was conserved. We could not, of course, speak of the total heat inside the volume if heat were being generated there.

Now we shall point out an interesting fact about the flux of any vector. You may think of the heat flow vector if you wish, but what we say will be true for any vector field $\mathbf{C}$. Imagine that we have a closed surface S that encloses the volume V . We now separate the volume into two parts by some kind of a “cut,” as in Fig. 3-4. Now we have two closed surfaces and volumes. The volume V_1 is enclosed in the surface S_1 , which is made up of part of the original surface S_a and of the surface of the cut, S_{ab} . The volume V_2 is enclosed by S_2 , which is made up of the rest of the original surface S_b and closed off by the cut S_{ab} . Now consider the

Fig. 3-4. A volume V contained inside the surface S is divided into two pieces by a "cut" at the surface S_{ab} . We now have the volume V_1 enclosed in the surface $S_1 = S_a + S_{ab}$ and the volume V_2 enclosed in the surface $S_2 = S_b + S_{ab}$.



following question: Suppose we calculate the flux out through surface S_1 and add to it the flux through surface S_2 . Does the sum equal the flux through the whole surface that we started with? The answer is yes. The flux through the part of the surfaces S_{ab} common to both S_1 and S_2 just exactly cancels out. For the flux of the vector C out of V_1 , we can write

$$\text{Flux through } S_1 = \int_{S_a} C \cdot n \, da + \int_{S_{ab}} C \cdot n_1 \, da, \quad (3.14)$$

and for the flux out of V_2 ,

$$\text{Flux through } S_2 = \int_{S_b} C \cdot n \, da + \int_{S_{ab}} C \cdot n_2 \, da. \quad (3.15)$$

Note that in the second integral we have written n_1 for the outward normal for S_{ab} when it belongs to S_1 , and n_2 when it belongs to S_2 , as shown in Fig. 3-4. Clearly, $n_1 = -n_2$, so that

$$\int_{S_{ab}} C \cdot n_1 \, da = - \int_{S_{ab}} C \cdot n_2 \, da. \quad (3.16)$$

If we now add Eqs. (3.14) and (3.15), we see that the sum of the fluxes through S_1 and S_2 is just the sum of two integrals which, taken together, give the flux through the original surface $S = S_a + S_b$.

We see that the flux through the complete outer surface S can be considered as the sum of the fluxes from the two pieces into which the volume was broken. We can similarly subdivide again—say by cutting V_1 into two pieces. You see that the same arguments apply. So for *any* way of dividing the original volume, it must be generally true that the flux through the outer surface, which is the original integral, is equal to a sum of the fluxes out of all the little interior pieces.

3-3 The flux from a cube; Gauss' theorem

We now take the special case of a small cube* and find an interesting formula for the flux out of it. Consider a cube whose edges are lined up with the axes as in Fig. 3-5. Let us suppose that the coordinates of the corner nearest the origin are x, y, z . Let Δx be the length of the cube in the x -direction, Δy be the length in the y -direction, and Δz be the length in the z -direction. We wish to find the flux of a vector field C through the surface of the cube. We shall do this by making a sum of the fluxes through each of the six faces. First, consider the face marked 1 in the figure. The flux *outward* on this face is the negative of the x -component of C , integrated over the area of the face. This flux is

$$- \int C_x \, dy \, dz.$$

Since we are considering a *small* cube, we can approximate this integral by the

* The following development applies equally well to any rectangular parallelepiped.

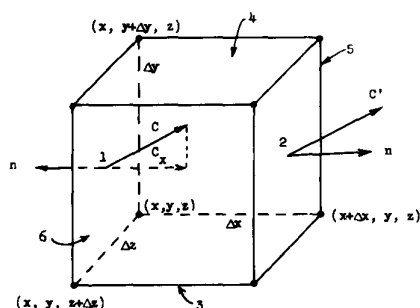


Fig. 3-5. Computation of the flux of C out of a small cube.

value of C_x at the center of the face—which we call the point (1)—multiplied by the area of the face, $\Delta y \Delta z$:

$$\text{Flux out of 1} = -C_x(1) \Delta y \Delta z.$$

Similarly, for the flux out of face 2, we write

$$\text{Flux out of 2} = C_x(2) \Delta y \Delta z.$$

Now $C_x(1)$ and $C_x(2)$ are, in general, slightly different. If Δx is small enough, we can write

$$C_x(2) = C_x(1) + \frac{\partial C_x}{\partial x} \Delta x.$$

There are, of course, more terms, but they will involve $(\Delta x)^2$ and higher powers, and so will be negligible if we consider only the limit of small Δx . So the flux through face 2 is

$$\text{Flux out of 2} = \left[C_x(1) + \frac{\partial C_x}{\partial x} \Delta x \right] \Delta y \Delta z.$$

Summing the fluxes for faces 1 and 2, we get

$$\text{Flux out of 1 and 2} = \frac{\partial C_x}{\partial x} \Delta x \Delta y \Delta z.$$

The derivative should really be evaluated at the center of face 1; that is, at $[x, y + (\Delta y/2), z + (\Delta z/2)]$. But in the limit of an infinitesimal cube, we make a negligible error if we evaluate it at the corner (x, y, z) .

Applying the same reasoning to each of the other pairs of faces, we have

$$\text{Flux out of 3 and 4} = \frac{\partial C_y}{\partial y} \Delta x \Delta y \Delta z$$

and

$$\text{Flux out of 5 and 6} = \frac{\partial C_z}{\partial z} \Delta x \Delta y \Delta z.$$

The total flux through all the faces is the sum of these terms. We find that

$$\int_{\text{cube}} \mathbf{C} \cdot \mathbf{n} \, da = \left(\frac{\partial C_x}{\partial x} + \frac{\partial C_y}{\partial y} + \frac{\partial C_z}{\partial z} \right) \Delta x \Delta y \Delta z,$$

and the sum of the derivatives is just $\nabla \cdot \mathbf{C}$. Also, $\Delta x \Delta y \Delta z = \Delta V$, the volume of the cube. So we can say that *for an infinitesimal cube*

$$\int_{\text{surface}} \mathbf{C} \cdot \mathbf{n} \, da = (\nabla \cdot \mathbf{C}) \Delta V. \quad (3.17)$$

We have shown that the outward flux from the surface of an infinitesimal cube is equal to the divergence of the vector multiplied by the volume of the cube. We now see the “meaning” of the divergence of a vector. The divergence of a vector at the point P is the flux—the outgoing “flow” of $\mathbf{C}$ —*per unit volume*, in the neighborhood of P .

We have connected the divergence of $\mathbf{C}$ to the flux of $\mathbf{C}$ out of each infinitesimal volume. For any finite volume we can use the fact we proved above—that the total flux from a volume is the sum of the fluxes out of each part. We can, that is, integrate the divergence over the entire volume. This gives us the theorem that the integral of the normal component of any vector over any closed surface can also be written as the integral of the divergence of the vector over the volume enclosed by the surface. This theorem is named after Gauss.

GAUSS’ THEOREM.

$$\int_S \mathbf{C} \cdot \mathbf{n} \, da = \int_V \nabla \cdot \mathbf{C} \, dV, \quad (3.18)$$

where S is any closed surface and V is the volume inside it.

3-4 Heat conduction; the diffusion equation

Let's consider an example of the use of this theorem, just to get familiar with it. Suppose we take again the case of heat flow in, say, a metal. Suppose we have a simple situation in which all the heat has been previously put in and the body is just cooling off. There are no sources of heat, so that heat is conserved. Then how much heat is there inside some chosen volume at any time? It must be *decreasing* by just the amount that flows out of the surface of the volume. If our volume is a little cube, we would write, following Eq. (3.17),

$$\text{Heat out} = \int_{\text{cube}} \mathbf{h} \cdot \mathbf{n} \, da = \nabla \cdot \mathbf{h} \, \Delta V. \quad (3.19)$$

But this must equal the rate of loss of the heat inside the cube. If q is the heat per unit volume, the heat in the cube is $q \, \Delta V$, and the rate of *loss* is

$$- \frac{d}{dt} (q \, \Delta V) = - \frac{dq}{dt} \, \Delta V. \quad (3.20)$$

Comparing (3.19) and (3.20), we see that

$$- \frac{dq}{dt} = \nabla \cdot \mathbf{h}. \quad (3.21)$$

Take careful note of the form of this equation; the form appears often in physics. It expresses a conservation law—here the conservation of heat. We have expressed the same physical fact in another way in Eq. (3.13). Here we have the *differential* form of a conservation equation, while Eq. (3.13) is the *integral* form.

We have obtained Eq. (3.21) by applying Eq. (3.13) to an infinitesimal cube. We can also go the other way. For a big volume V bounded by S , Gauss' law says that

$$\int_S \mathbf{h} \cdot \mathbf{n} \, da = \int_V \nabla \cdot \mathbf{h} \, dV. \quad (3.22)$$

Using (3.21), the integral on the right-hand side is found to be just $-dQ/dt$, and again we have Eq. (3.13).

Now let's consider a different case. Imagine that we have a block of material and that inside it there is a very tiny hole in which some chemical reaction is taking place and generating heat. Or we could imagine that there are some wires running into a tiny resistor that is being heated by an electric current. We shall suppose that the heat is generated practically at a point, and let W represent the energy liberated per second at that point. We shall suppose that in the rest of the volume heat is conserved, and that the heat generation has been going on for a long time—so that now the temperature is no longer changing anywhere. The problem is: What does the heat vector $\mathbf{h}$ look like at various places in the metal? How much heat flow is there at each point?

We know that if we integrate the normal component of $\mathbf{h}$ over a closed surface that encloses the source, we will always get W . All the heat that is being generated at the point source must flow out through the surface, since we have supposed that the flow is steady. We have the difficult problem of finding a vector field which, when integrated over any surface, always gives W . We can, however, find the field rather easily by taking a somewhat special surface. We take a sphere of radius R , centered at the source, and assume that the heat flow is radial (Fig. 3-6). Our intuition tells us that $\mathbf{h}$ should be radial if the block of material is large and we don't get too close to the edges, and it should also have the same magnitude at all points on the sphere. You see that we are adding a certain amount of guesswork—usually called “physical intuition”—to our mathematics in order to find the answer.

When $\mathbf{h}$ is radial and spherically symmetric, the integral of the normal component of $\mathbf{h}$ over the area is very simple, because the normal component is just

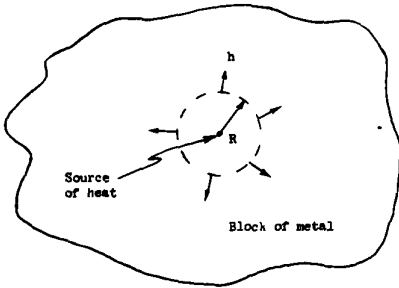


Fig. 3-6. In the region near a point source of heat, the heat flow is radially outward.

the magnitude of $\mathbf{h}$ and is constant. The area over which we integrate is $4\pi R^2$. We have then that

$$\int_S \mathbf{h} \cdot \mathbf{n} \, da = h \cdot 4\pi R^2 \quad (3.23)$$

(where h is the magnitude of $\mathbf{h}$). This integral should equal W , the rate at which heat is produced at the source. We get

$$h = \frac{W}{4\pi R^2},$$

or

$$\mathbf{h} = \frac{W}{4\pi R^2} \mathbf{e}_r, \quad (3.24)$$

where, as usual, $\mathbf{e}_r$ represents a unit vector in the radial direction. Our result says that $\mathbf{h}$ is proportional to W and varies inversely as the square of the distance from the source.

The result we have just obtained applies to the heat flow in the vicinity of a point source of heat. Let's now try to find the equations that hold in the most general kind of heat flow, keeping only the condition that heat is conserved. We will be dealing only with what happens at places outside of any sources or absorbers of heat.

The differential equation for the conduction of heat was derived in Chapter 2. According to Eq. (2.44),

$$\mathbf{h} = -\kappa \nabla T. \quad (3.25)$$

(Remember that this relationship is an approximate one, but fairly good for some materials like metals.) It is applicable, of course, only in regions of the material where there is no generation or absorption of heat. We derived above another relation, Eq. (3.21), that holds when heat is conserved. If we combine that equation with (3.25), we get

$$-\frac{dq}{dt} = \nabla \cdot \mathbf{h} = -\nabla \cdot (\kappa \nabla T),$$

or

$$\frac{dq}{dt} = \kappa \nabla \cdot \nabla T = \kappa \nabla^2 T, \quad (3.26)$$

if κ is a constant. You remember that q is the amount of heat in a unit volume and $\nabla \cdot \nabla = \nabla^2$ is the Laplacian operator

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

If we now make one more assumption we can obtain a very interesting equation. We assume that the temperature of the material is proportional to the heat content per unit volume—that is, that the material has a definite specific heat. When this assumption is valid (as it often is), we can write

$$\Delta q = c_v \Delta T$$

or

$$\frac{dq}{dt} = c_v \frac{dT}{dt}. \quad (3.27)$$

The rate of change of heat is proportional to the rate of change of temperature. The constant or proportionality c_v is, here, the specific heat per unit *volume* of the material. Using Eq. (3.27) with (3.26), we get

$$\frac{dT}{dt} = \frac{\kappa}{c_v} \nabla^2 T. \quad (3.28)$$

We find that the *time* rate of change of T —at every point—is proportional to the Laplacian of T , which is the second derivative of its spatial dependence. We have a differential equation—in x , y , z , and t —for the temperature T .

The differential equation (3.28) is called the *heat diffusion equation*. It is often written as

$$\frac{dT}{dt} = D \nabla^2 T, \quad (3.29)$$

where D is called the *diffusion constant*, and is here equal to κ/c_v .

The diffusion equation appears in many physical problems—in the diffusion of gases, in the diffusion of neutrons, and in others. We have already discussed the physics of some of these phenomena in Chapter 43 of Vol. I. Now you have the complete equation that describes diffusion in the most general possible situation. At some later time we will take up ways of solving the diffusion equation to find how the temperature varies in particular cases. We turn back now to consider other theorems about vector fields.

3-5 The circulation of a vector field

We wish now to look at the curl in somewhat the same way we looked at the divergence. We obtained Gauss' theorem by considering the integral over a surface, although it was not obvious at the beginning that we were going to be dealing with the divergence. How did we know that we were supposed to integrate over a surface in order to get the divergence? It was not at all clear that this would be the result. And so with an apparent equal lack of justification, we shall calculate something else about a vector and show that it is related to the curl. This time we calculate what is called the circulation of a vector field. If $\mathbf{C}$ is any vector field, we take its component along a curved line and take the integral of this component all the way around a complete loop. The integral is called the *circulation* of the vector field around the loop. We have already considered a line integral of $\nabla\psi$ earlier in this chapter. Now we do the same kind of thing for *any* vector field $\mathbf{C}$.

Let Γ be any closed loop in space—imaginary, of course. An example is given in Fig. 3-7. The line integral of the tangential component of $\mathbf{C}$ around the loop is written as

$$\oint_{\Gamma} C_t ds = \oint_{\Gamma} \mathbf{C} \cdot d\mathbf{s}. \quad (3.30)$$

You should note that the integral is taken all the way around, not from one point to another as we did before. The little circle on the integral sign is to remind us that the integral is to be taken all the way around. This integral is called the circulation of the vector field around the curve Γ . The name came originally from considering the circulation of a liquid. But the name—like flux—has been extended to apply to any field even when there is no material “circulating.”

Playing the same kind of game we did with the flux, we can show that the circulation around a loop is the sum of the circulations around two partial loops. Suppose we break up our curve of Fig. 3-7 into two loops, by joining two points (1) and (2) on the original curve by some line that cuts across as shown in Fig. 3-8. There are now two loops, Γ_1 and Γ_2 . Γ_1 is made up of Γ_a , which is that part of the original curve to the left of (1) and (2), plus Γ_{ab} , the “short cut.” Γ_2 is made up of the rest of the original curve plus the short cut.

The circulation around Γ_1 is the sum of an integral along Γ_a and along Γ_{ab} . Similarly, the circulation around Γ_2 is the sum of two parts, one along Γ_b and the other along Γ_{ab} . The integral along Γ_{ab} will have, for the curve Γ_2 , the opposite sign from what it has for Γ_1 , because the direction of travel is opposite—we must take both our line integrals with the same “sense” of rotation.

Following the same kind of argument we used before, you can see that the sum of the two circulations will give just the line integral around the original curve Γ . The parts due to Γ_{ab} cancel. The circulation around the one part plus the circulation around the second part equals the circulation about the outer line. We can continue the process of cutting the original loop into any number of smaller loops. When we add the circulations of the smaller loops, there is always a cancellation of the parts on their adjacent portions, so that the sum is equivalent to the circulation around the original single loop.

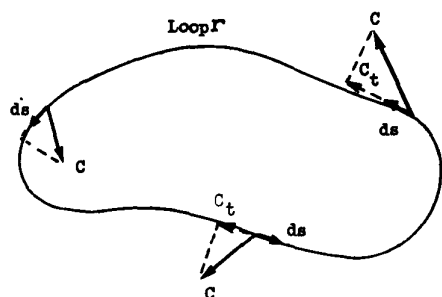


Fig. 3-7. The circulation of $\mathbf{C}$ around the curve Γ is the line integral of C_t , the tangential component of $\mathbf{C}$.

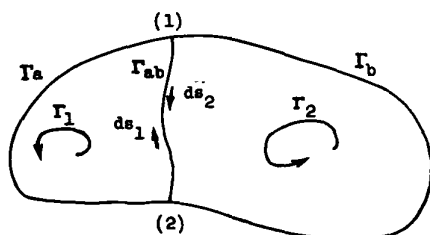


Fig. 3-8. The circulation around the whole loop is the sum of the circulations around the two loops: $\Gamma_1 = \Gamma_a + \Gamma_{ab}$ and $\Gamma_2 = \Gamma_b + \Gamma_{ab}$.

Now let us suppose that the original loop is the boundary of some surface. There are, of course, an infinite number of surfaces which all have the original loops as the boundary. Our results will not, however, depend on which surface we choose. First, we break our original loop into a number of small loops that all lie on the surface we have chosen, as in Fig. 3-9. No matter what the shape of the surface, if we choose our small loops small enough, we can assume that each of the small loops will enclose an area which is essentially flat. Also, we can choose our small loops so that each is very nearly a square. Now we can calculate the circulation around the big loop Γ by finding the circulations around all of the little squares and then taking their sum.

3-6 The circulation around a square; Stokes' theorem

How shall we find the circulation for each little square? One question is, how is the square oriented in space? We could easily make the calculation if it had a special orientation. For example, if it were in one of the coordinate planes. Since we have not assumed anything as yet about the orientation of the coordinate axes, we can just as well choose the axes so that the one little square we are concentrating on at the moment lies in the xy -plane, as in Fig. 3-10. If our result is expressed in vector notation, we can say that it will be the same no matter what the particular orientation of the plane.

We want now to find the circulation of the field C around our little square. It will be easy to do the line integral if we make the square small enough that the vector C doesn't change much along any one side of the square. (The assumption is better the smaller the square, so we are really talking about infinitesimal squares.) Starting at the point (x, y) —the lower left corner of the figure—we go around in the direction indicated by the arrows. Along the first side—marked (1)—the tangential component is $C_x(1)$ and the distance is Δx . The first part of the integral is $C_x(1) \Delta x$. Along the second leg, we get $C_y(2) \Delta y$. Along the third, we get $-C_x(3) \Delta x$, and along the fourth, $-C_y(4) \Delta y$. The minus signs are required because we want the tangential component in the direction of travel. The whole line integral is then

$$\oint C \cdot ds = C_x(1) \Delta x + C_y(2) \Delta y - C_x(3) \Delta x - C_y(4) \Delta y. \quad (3.31)$$

Now let's look at the first and third pieces. Together they are

$$[C_x(1) - C_x(3)] \Delta x. \quad (3.32)$$

You might think that to our approximation the difference is zero. That is true to the first approximation. We can be more accurate, however, and take into account the rate of change of C_x . If we do, we may write

$$C_x(3) = C_x(1) + \frac{\partial C_x}{\partial y} \Delta y. \quad (3.33)$$

If we included the next approximation, it would involve terms in $(\Delta y)^2$, but since we will ultimately think of the limit as $\Delta y \rightarrow 0$, such terms can be neglected. Putting (3.33) together with (3.32), we find that

$$[C_x(1) - C_x(3)] \Delta y = -\frac{\partial C_x}{\partial y} \Delta x \Delta y. \quad (3.34)$$

The derivative can, to our approximation, be evaluated at (x, y) .

Similarly, for the other two terms in the circulation, we may write

$$C_y(2) \Delta y - C_y(4) \Delta y = \frac{\partial C_y}{\partial x} \Delta x \Delta y. \quad (3.35)$$

The circulation around our square is then

$$\left(\frac{\partial C_y}{\partial x} - \frac{\partial C_x}{\partial y} \right) \Delta x \Delta y, \quad (3.36)$$

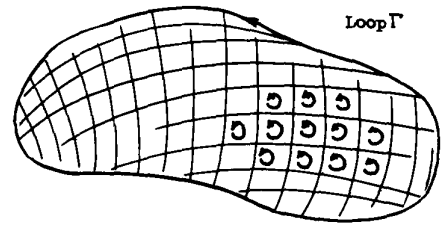


Fig. 3-9. Some surface bounded by the loop Γ is chosen. The surface is divided into a number of small areas, each approximately a square. The circulation around Γ is the sum of the circulations around the little loops.

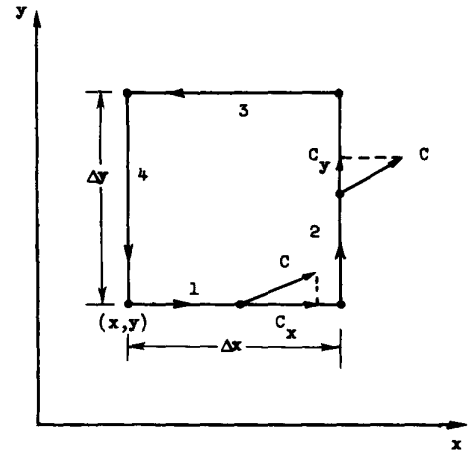


Fig. 3-10. Computing the circulation of C around a small square.

which is interesting, because the two terms in the parentheses are just the z -component of the curl. Also, we note that $\Delta x \Delta y$ is the area of our square. So we can write our circulation (3.36) as

$$(\nabla \times \mathbf{C})_z \Delta a.$$

But the z -component really means the component *normal* to the surface element. We can, therefore, write the circulation around a differential square in an invariant vector form:

$$\oint \mathbf{C} \cdot d\mathbf{s} = (\nabla \times \mathbf{C})_n \Delta a = (\nabla \times \mathbf{C}) \cdot \mathbf{n} \Delta a. \quad (3.37)$$

Our result is: the circulation of any vector $\mathbf{C}$ around an infinitesimal square is the component of the curl of $\mathbf{C}$ normal to the surface, times the area of the square.

The circulation around any loop Γ can now be easily related to the curl of the vector field. We fill in the loop with any convenient surface S , as in Fig. 3-11, and add the circulations around a set of infinitesimal squares in this surface. The sum can be written as an integral. Our result is a very useful theorem called Stokes' theorem (after Mr. Stokes).

STOKES' THEOREM.

$$\oint_{\Gamma} \mathbf{C} \cdot d\mathbf{s} = \int_S (\nabla \times \mathbf{C})_n da, \quad (3.38)$$

where S is any surface bounded by Γ .

We must now speak about a convention of signs. In Fig. 3-10 the z -axis would point *toward* you in a "usual"—that is, "right-handed"—system of axes. When we took our line integral with a "positive" sense of rotation, we found that the circulation was equal to the z -component of $\nabla \times \mathbf{C}$. If we had gone around the other way, we would have gotten the opposite sign. Now how shall we know, in general, what direction to choose for the positive direction of the "normal" component of $\nabla \times \mathbf{C}$? The "positive" normal must always be related to the sense of rotation, as in Fig. 3-10. It is indicated for the general case in Fig. 3-11.

One way of remembering the relationship is by the "right-hand rule." If you make the fingers of your *right* hand go around the curve Γ , with the fingertips pointed in the direction of the positive sense of $d\mathbf{s}$, then your thumb points in the direction of the *positive* normal to the surface S .

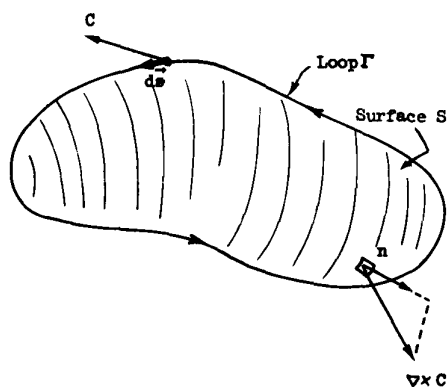


Fig. 3-11. The circulation of $\mathbf{C}$ around Γ is the surface integral of the normal component of $\nabla \times \mathbf{C}$.

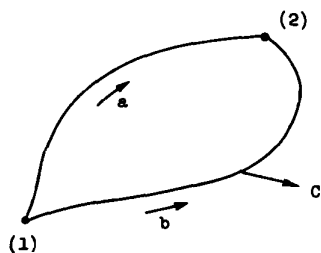


Fig. 3-12. If $\nabla \times \mathbf{C}$ is zero, the circulation around the closed curve Γ is zero. The line integral of $\mathbf{C} \cdot d\mathbf{s}$ from (1) to (2) along a must be the same as the line integral along b .

3-7 Curl-free and divergence-free fields

We would like, now, to consider some consequences of our new theorems. Take first the case of a vector whose curl is *everywhere* zero. Then Stokes' theorem says that the circulation around any loop is zero. Now if we choose two points (1) and (2) on a closed curve (Fig. 3-12), it follows that the line integral of the tangential component from (1) to (2) is independent of which of the two possible paths is taken. We can conclude that the integral from (1) to (2) can depend only on the location of these points—that is to say, it is some function of position only. The same logic was used in Chapter 14 of Vol. I, where we proved that if the integral around a closed loop of some quantity is always zero, then that integral can be represented as the difference of a function of the position of the two ends. This fact allowed us to invent the idea of a potential. We proved, furthermore, that the vector field was the gradient of this potential function (see Eq. 14.13 of Vol. I).

It follows that any vector field whose curl is zero is equal to the gradient of some scalar function. That is, if $\nabla \times \mathbf{C} = 0$, everywhere, there is some ψ (psi) for which $\mathbf{C} = \nabla\psi$ —a useful idea. We can, if we wish, describe this special kind of vector field by means of a scalar field.

Let's show something else. Suppose we have *any* scalar field ϕ (phi). If we take its gradient, $\nabla\phi$, the integral of this vector around any closed loop must be zero. Its line integral from point (1) to point (2) is $[\phi(2) - \phi(1)]$. If (1) and (2)